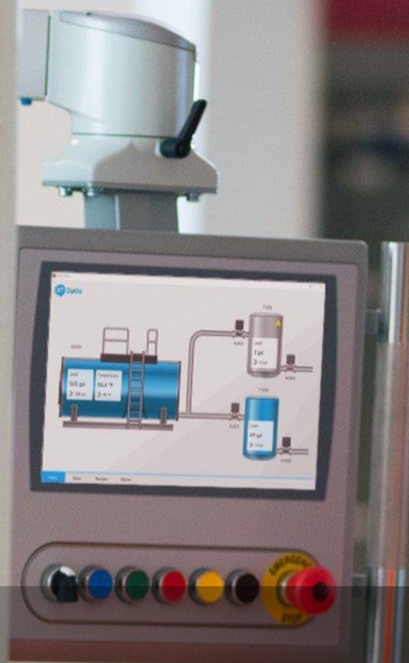




expanding human possibility®



PUBLIC



FactoryTalk Optix Technical Training

ASEM • APPLICATION SUPPORT ENGINEERS • 05/2026

Table of contents



Table of contents – Pre requirements

- (Pre-requirement) Getting started
- (Pre-requirement) Be familiar with FactoryTalk Optix Studio
- (Pre-requirement) How to Create a simple HMI project in few steps using wizards
- (Pre-requirement) WebIDE
- (Pre-requirement) Project versions and components
- (Pre-requirements) FactoryTalk Optix Entitlements

Table of contents – Basic topics

- (Base) Define communication drivers and tags
- (Base) Develop the user interface
- (Base) Dynamic links
- (Base) Work with reusable objects
- (Base) Template Library
- (Base) Project Templates
- (Base) UI Objects Protection
- (Base) Animations
- (Base) Events
- (Base) DataStores
- (Base) DataLoggers and EventLoggers
- (Base) Represent data by Trends
- (Base) OPC-UA Client
- (Base) OPC-UA Server

Table of contents – Basic topics

- (Base) Transfer and Execute the project
- (Base) Manage alarms
- (Base) Use legacy recipes
- (Base) Configure Users and Groups
- (Base) Configure Reports
- (Base) Set Locale, Languages and Measurement System
- (Base) Develop using version control and collaboration

Table of contents – Advanced topics

- (Advanced) Define client-server architectures via OPC-UA
- (Advanced) Domain authentication
- (Advanced) Define UI Session and Session variables
- (Advanced) Work with NetLogic scripts
- (Advanced) Debugging NetLogic scripts
- (Advanced) Use Retentivity
- (Advanced) MQTT
- (Advanced) REST API
- (Advanced) Optix Runtime on Docker containers
- (Advanced) OPC-UA NodeSet
- (Advanced) OPC-UA custom behaviors
- (Advanced) Custom keyboards
- (Advanced) Recipes

Table of contents - Appendix

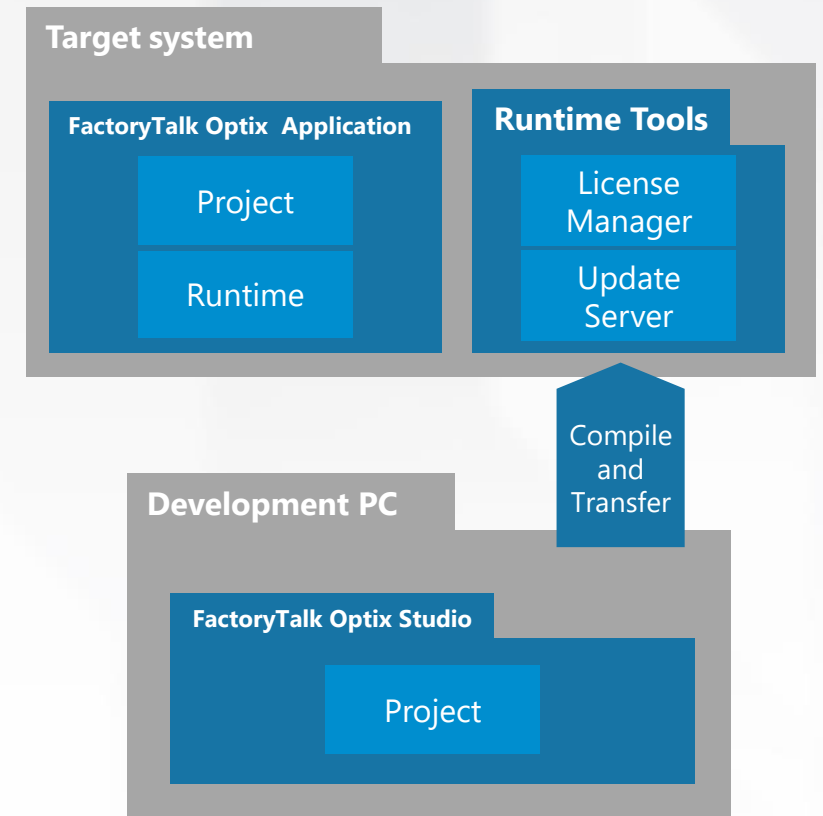
- [Appendix: See the light about OPC-UA Security](#)
- [Appendix: Generating OPC-UA certificates](#)
- [Appendix: Using OPC-UA certificates](#)
- [Appendix: Returning values from NetLogic methods](#)
- [Appendix: Connect to an External SQL Database via ODBC connector](#)
- [Appendix: Connect to an External InfluxDB instance](#)
- [Appendix: Additional resources](#)

Getting Started



Architecture

- **FactoryTalk Optix Studio** is the Integrated Development Environment (IDE) used to design and deploy HMI and IIOT applications
- **FactoryTalk Optix Application** is the compiled and "portable" application that combines the Project code and the Runtime executable for the Target (HMI or iPC)
- **Runtime Tools** is the set of tools that includes:
 - **Update Server**: Software to allow the deployment of applications from Studio to the target
 - **License Manager**: Software to manage the Runtime entitlements on the target device



FactoryTalk Optix key features

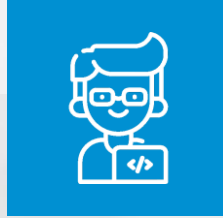


EXTENSIBILITY options

Natively OPC UA

Libraries

Scripting



DESIGN options

Object-oriented

Software As
A Service

Version Control



GRAPHIC & DEPLOYMENT options

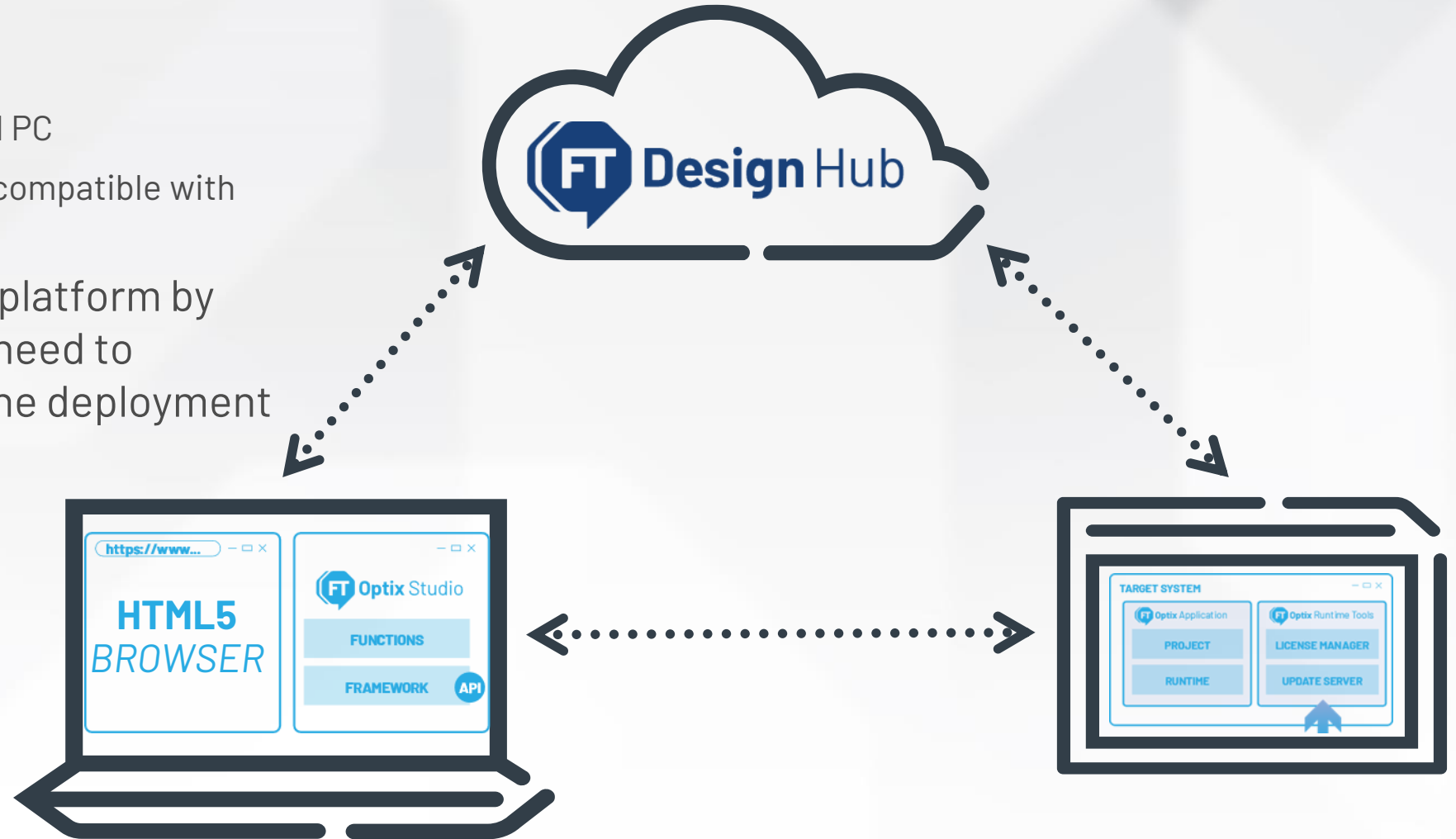
Cross-platform

Universal UI

Audit

FactoryTalk Optix Studio

- Available both
 - On premise: as a setup on local PC
 - As a service: a cloud instance compatible with any HTML5 browser
- All applications are cross-platform by design, the user does not need to choose or know which is the deployment target



FactoryTalk Optix Studio flavours

STANDARD

Installed on local PC

Design and Deploy application
from your PC

No screens, tags, alarms or other
Runtime features limitations

FREE



PRO

STANDARD capabilities plus...

Use the **Version Control** System
(both on project and libraries)
to enable multi-user **collaboration**

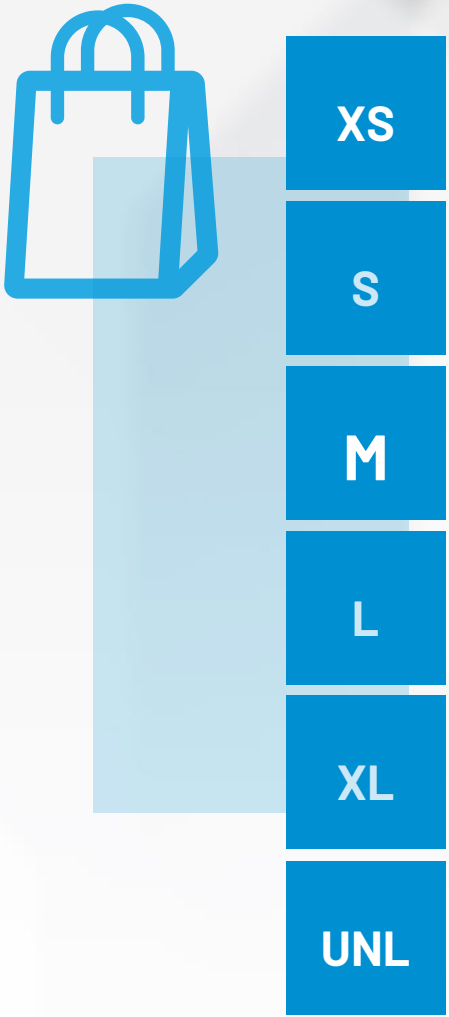
Design application using
web-based Studio

Annual subscription
per User

FactoryTalk Optix Runtime – Feature tokens overview

- Every **feature** used by the Runtime has a cost in **Tokens**
- License **size** defines the number of Tokens available

SIZE	RUNTIME FOR IPC
XS	5
S	8
M	11
L	15
XL	21
UNL	Unlimited



FactoryTalk Optix Runtime – Feature tokens example

SMALL HMI	
Native UI	1
Web UI (up to 1 clients)	1
Alarms + Alarms History	1
Data logger	1
Recipes	1
1x Comm Driver (multiple stations)	2
TOTAL	7



MEDIUM HMI	
Native UI	1
Web UI (up to 3 clients)	2
Alarms + Alarms History	1
Data logger	1
Retentivity	1
Recipes	1
OPC UA Server (1 client)	1
1x Comm Driver (multiple station)	2
TOTAL	10



Ready to start with FactoryTalk Optix?

- Just need to create a «My Rockwell» account then...
- Optix Studio STANDARD can be downloaded for free from <https://compatibility.rockwellautomation.com/>
- Optix Studio PRO is available as 90 Days Trial at <https://home.cloud.rockwellautomation.com>

Be familiar with FactoryTalk Optix Studio



Development environment: new project wizard

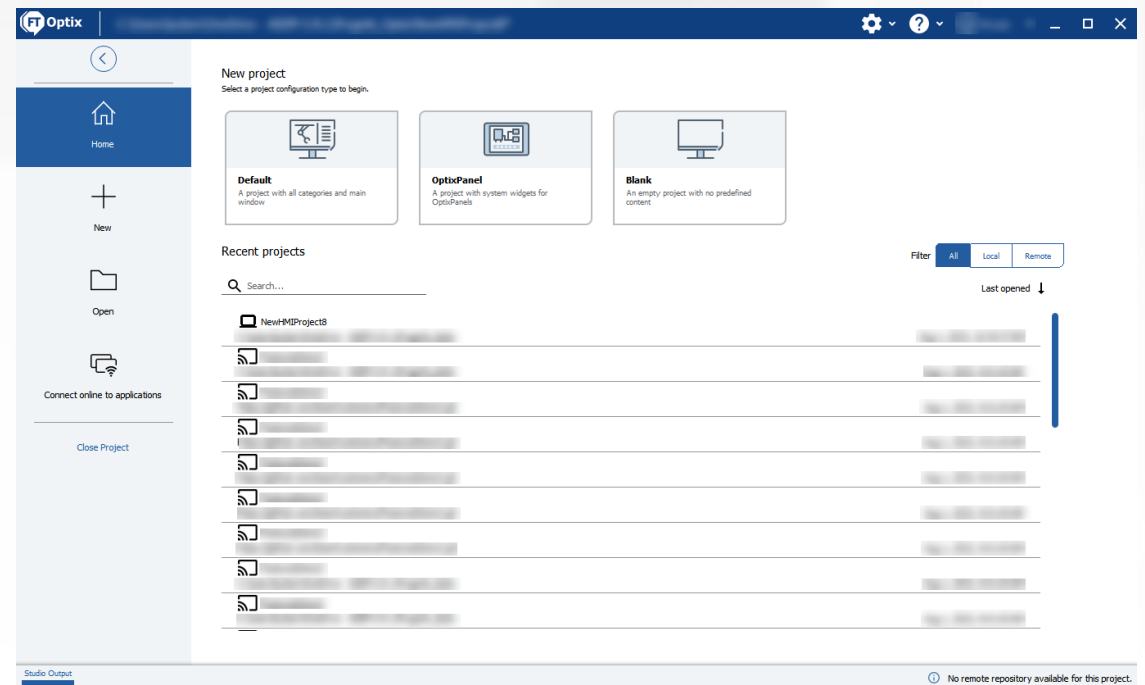
Choose between

- Default: project with default folders/resource structure (recommended)
- OptixPanel: project with Optix Panel's dedicated widgets (network, screen settings, etc)
- Blank: empty project, without folders/resources structure (only if «advanced mode» is enabled)

Important: this selection does not limit in the deployment capabilities or available features

Project configuration

- Name, Location
- Use version control: allow the project versioning and collaboration between developers (GIT)
- Encrypt secrets: passwords are encrypted via a private key stored in the FactoryTalk HUB domain
- Default screens resolution (can be changed later)



Development environment: new project wizard

Choose between

- Default: project with default folders/resource structure (recommended)
- OptixPanel: project with Optix Panel's dedicated widgets (network, screen settings, etc)
- Blank: empty project, without folders/resources structure (only if «advanced mode» is enabled)

Important: this selection does not limit in the deployment capabilities or available features

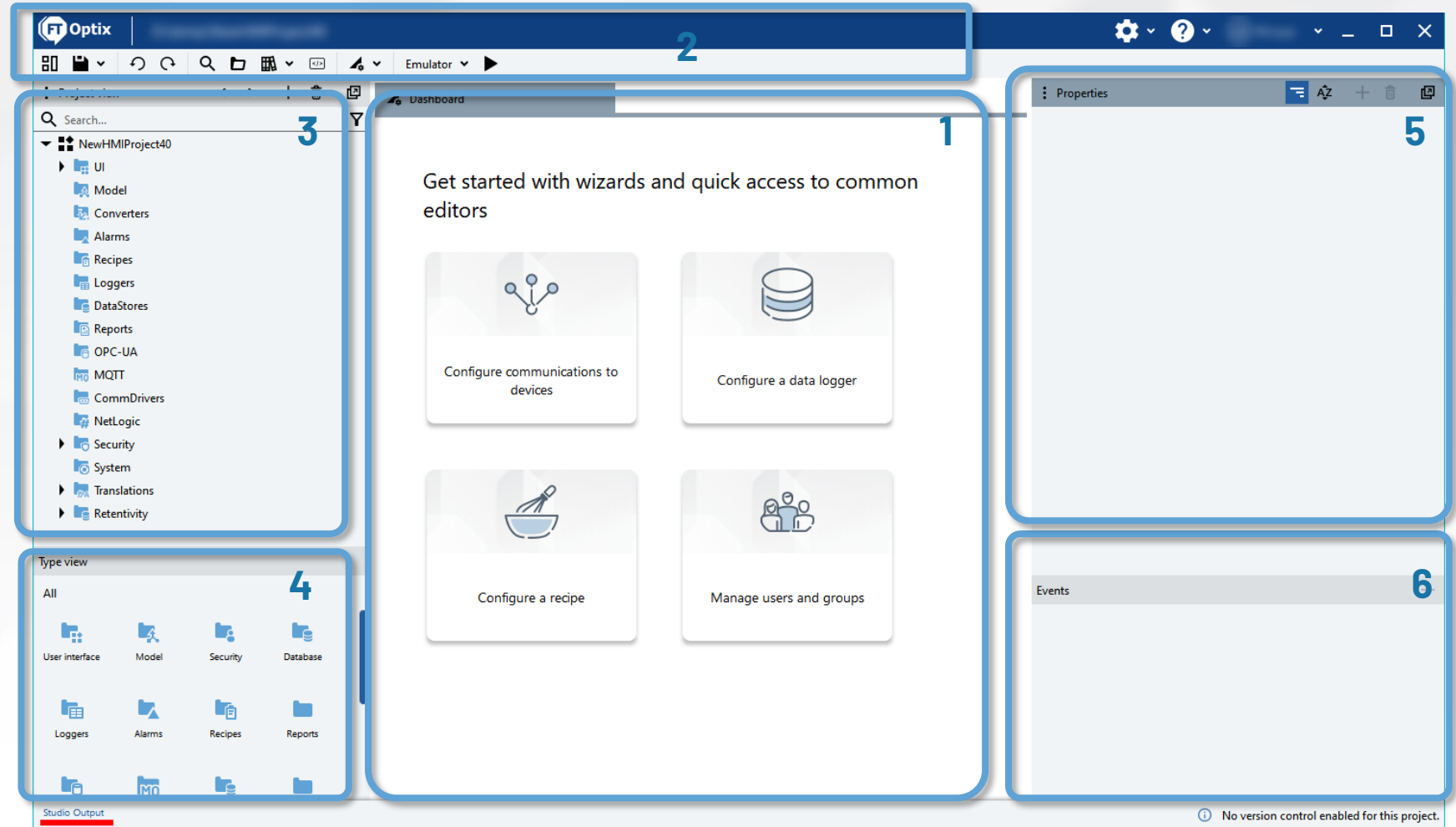
Project configuration

- Name, Location
- Use version control: allow the project versioning and collaboration between developers (GIT)
- Encrypt secrets: passwords are encrypted via a private key stored in the FactoryTalk HUB domain
- Default screens resolution (can be changed later)

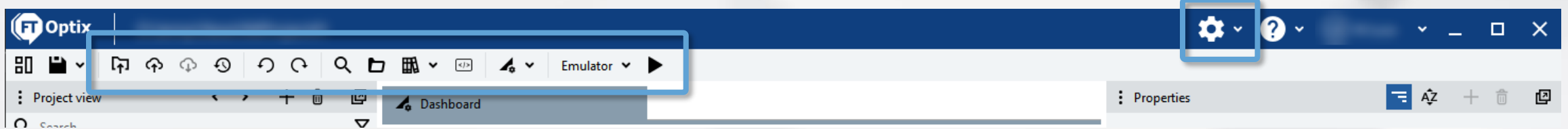
The screenshot shows the 'New project' wizard in the Optix application. The interface is divided into a left sidebar and a main content area. The sidebar contains icons for 'Home', 'New' (highlighted), 'Open', and 'Connect online to applications'. The main content area has a title bar with 'Optix' and standard window controls. Below the title bar, there's a 'New project' section with the instruction 'Select a project configuration type to begin.' Three options are presented: 'Default' (A project with all categories and main window), 'OptixPanel' (A project with system widgets for OptixPanels), and 'Blank' (An empty project with no predefined content). Below these options, there are input fields for 'Name' (containing 'NewHMIProject9') and 'Location' (with a 'Browse' button). Checkboxes for 'Use Version Control' and 'Encrypt secrets' are both checked. A 'Set main window size' section allows selecting a resolution from a list: 'PC Runtime 1366x768' (selected), 'PC Runtime 1920x1080', 'OptixPanel Compact 4.3" W (480x272)', 'OptixPanel Compact 7" W (800x480)', 'OptixPanel Standard 7" W (800x480)', and 'OptixPanel Standard 10.1" W (1280x800)'. To the right, 'Resolution' settings show 'Width' as 1366 px and 'Height' as 768 px. At the bottom right, there are 'Create' and 'Cancel' buttons. A 'Studio Output' tab is visible at the very bottom.

Development environment

1. Wizards
2. Main Toolbar
3. Project View panel
4. Types View panel
5. Properties panel
6. Events panel



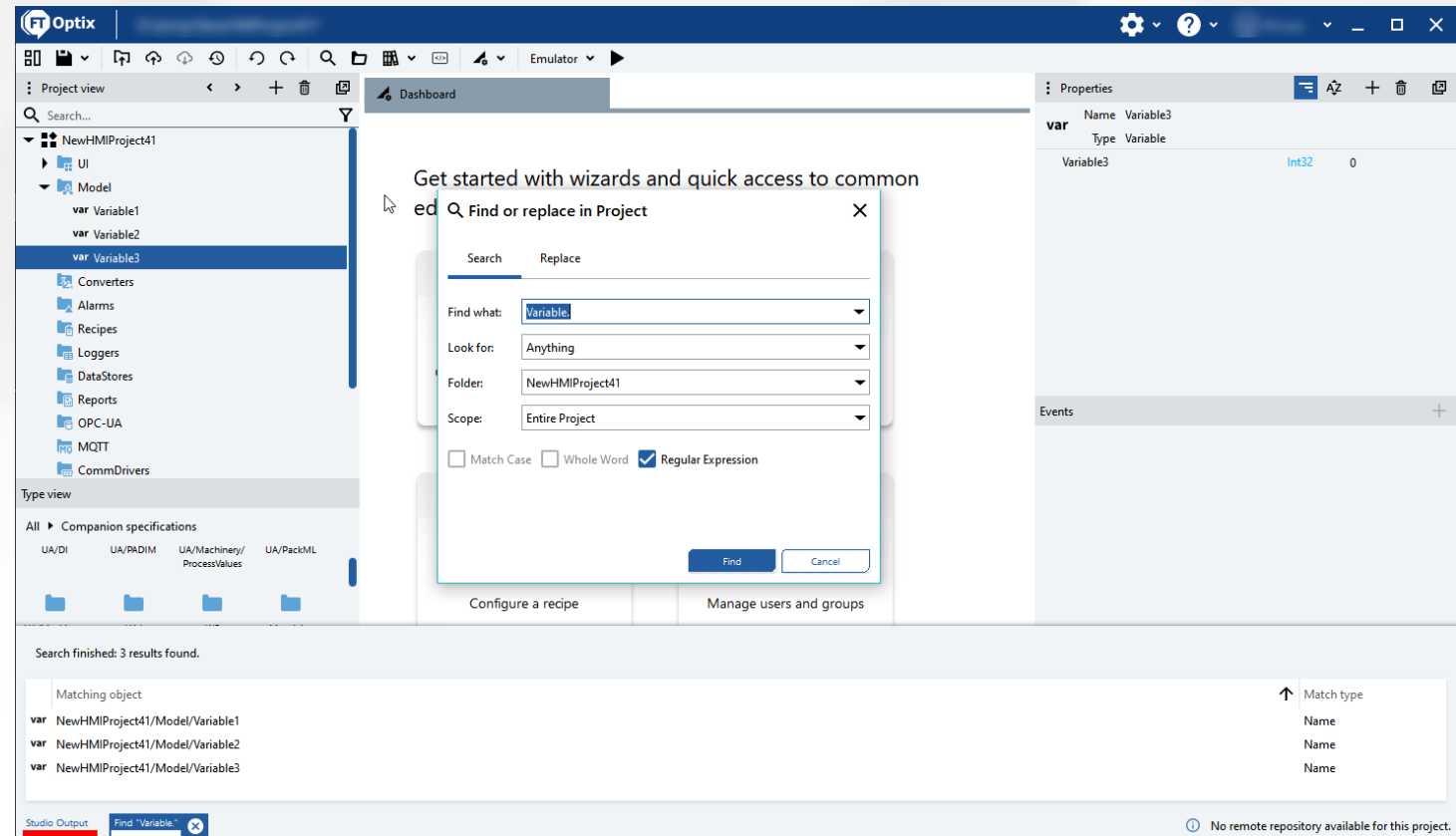
Main toolbar



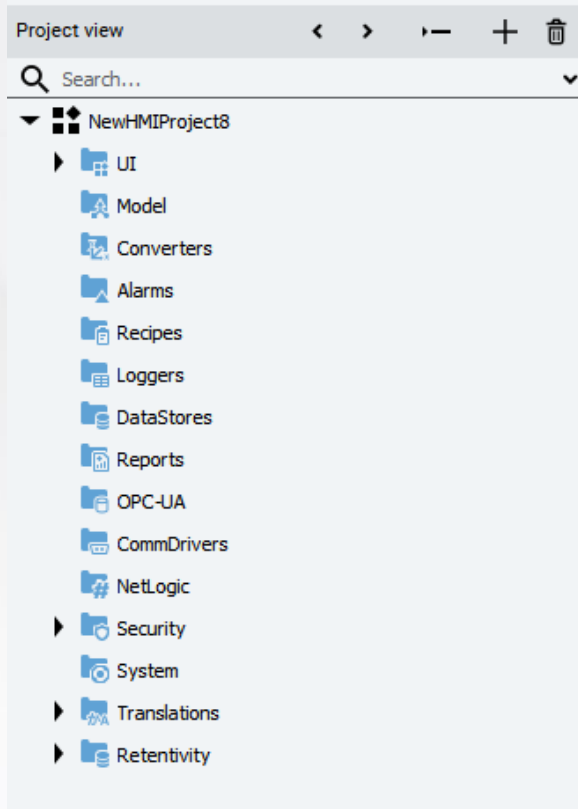
- **Collaboration:** Save & Commit/Push/Pull/History
- **Undo/Redo**
- **Global search**
- **Project Files**
- **Libraries** (%UserProfile%\Documents\Rockwell Automation\FactoryTalk Optix\Libraries)
- **NetSolution**
- **Wizards**
- **Deploy options:** Run the project within Emulator or transfer and run on a Target
- **Studio Options**

Main toolbar

- **Global Search:** helps finding elements anywhere in the project
 - Supports RegEx (using POSIX syntax, e.g: «.*» to search for any characters)
 - Can search in NetLogic, project tree and values
 - Output is returned in the a dedicated tab of the IDE
 - Supports replacing text and values



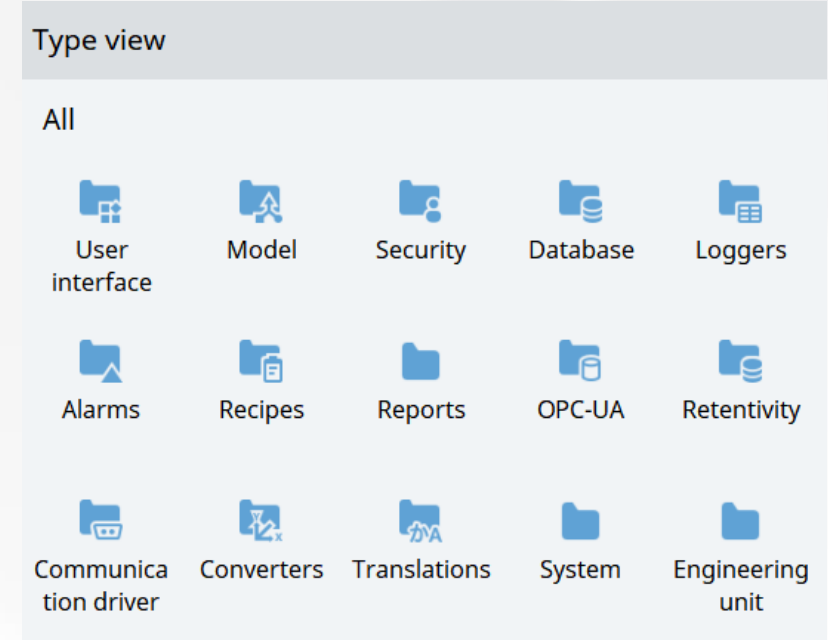
Project view



- The Project view shows, in a **typical tree architecture**, the project folders
- Folder structure constitutes the foundation of a project and are organized by type and functionality
- The Project view tree supports:
 - Context commands with the mouse (right-click)
 - Multiple selections
 - Drag&Drop
 - Copy/Paste
 - Search and filter
 - Detach
- Please, do not move or rename the default folders

Type view

- Collection of **Native Types**, grouped in folders according to their purpose:
 - **Basic User Control**: to develop the User Interface
 - **Optix Types**: all the objects to be added to the project folders
 - **Companion specifications**: set of standard and custom types definition which are used to share information between different manufacturers (PackML, DI, TMC, etc)
- Object from Type view can be added into the project Tree by Drag&Drop
- Can be also the repository of **Custom Types**



Properties panel

- This pane shows properties of the selected node
 - Project
 - Resource
 - Objects
- Depending on the type of property, the editing can be done through:
 - Direct editing by typing
 - by Selecting the pencil icon 
 - Drop-down menu
 - by Selecting the DynamicLink icon 
- Can edit common properties of Multiple selections

Properties

Name

MyProject

Type

Project folder

Locales

en-US

Translation fallback locales

en-US

Branching enabled

False

Measurement systems map

Default mapping

Authentication mode

Model only

Default user folder

Users

▼ Password policy

Maximum password age

0

Enforce password history

1

Minimum password age

0

Minimum password length

8

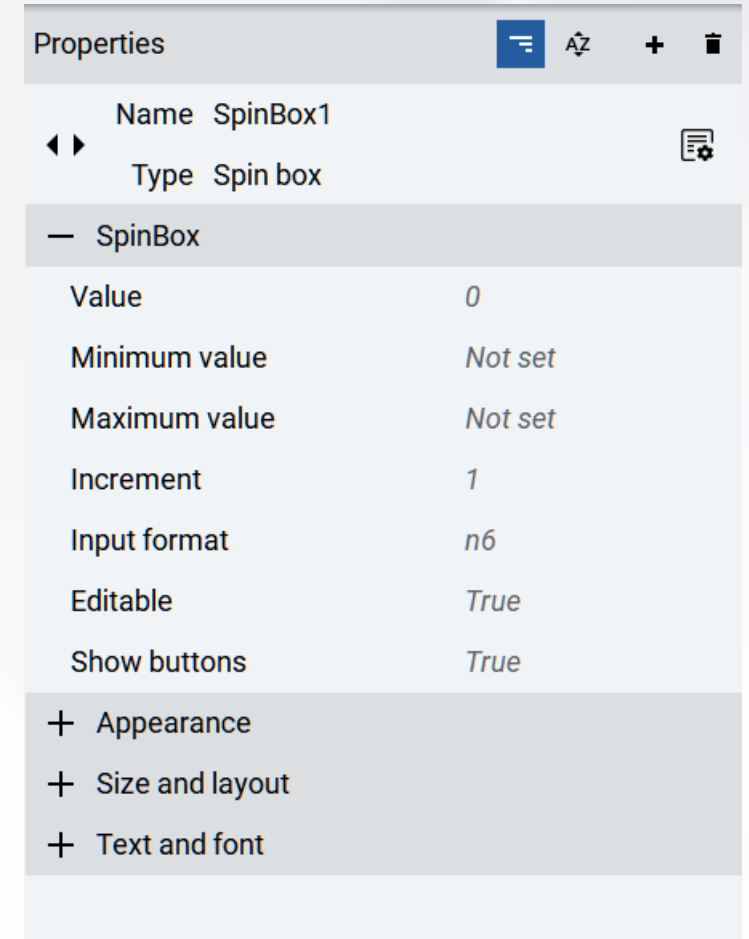
Properties panel

- This pane shows properties of the selected node
 - Project
 - Resource
 - Objects
- Depending on the type of property, the editing can be done through:
 - Direct editing by typing
 - by Selecting the pencil icon 
 - Drop-down menu
 - by Selecting the DynamicLink icon 
- Can edit common properties of Multiple selections

Properties	
Name Screen1 (type)	
Type Panel	
Visible	True
Enabled	True
Opacity	100
Rotation	0
Hit test visible	False
— Size and layout	
Horizontal alignment	Left
Vertical alignment	Top
Width	300
Height	300
Left margin	0
Top margin	0

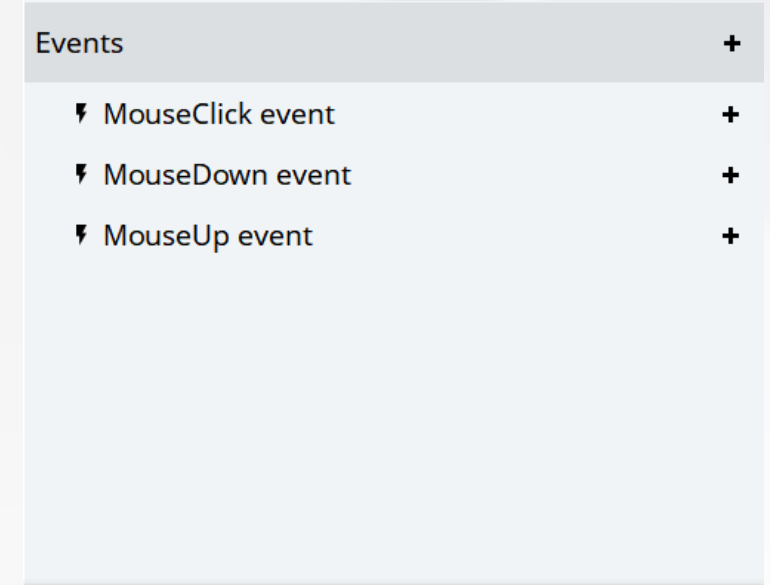
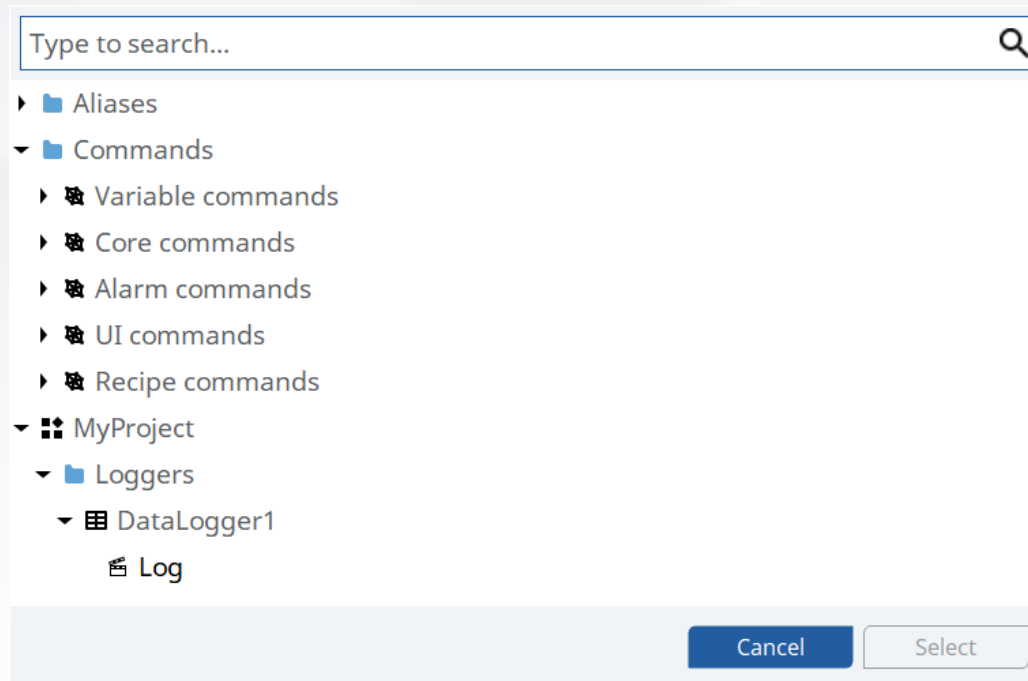
Properties panel

- This pane shows properties of the selected node
 - Project
 - Resource
 - Objects
- Depending on the type of property, the editing can be done through:
 - Direct editing by typing
 - by Selecting the pencil icon 
 - Drop-down menu
 - by Selecting the DynamicLink icon 
- Can edit common properties of Multiple selections



Events panel

- Used to configure Commands or Methods to be executed when events are triggered by the selected object
- Different objects have different events

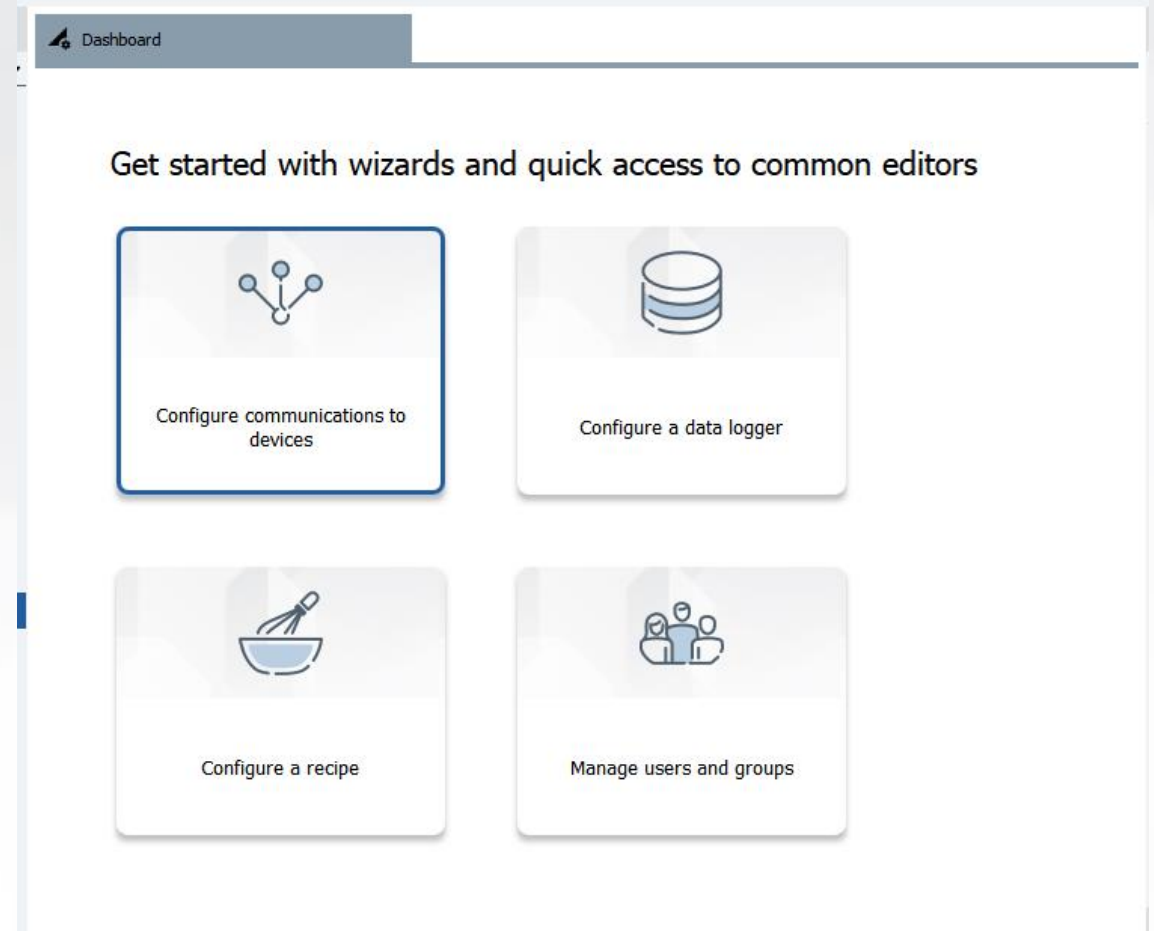


How to create a simple HMI project in few steps



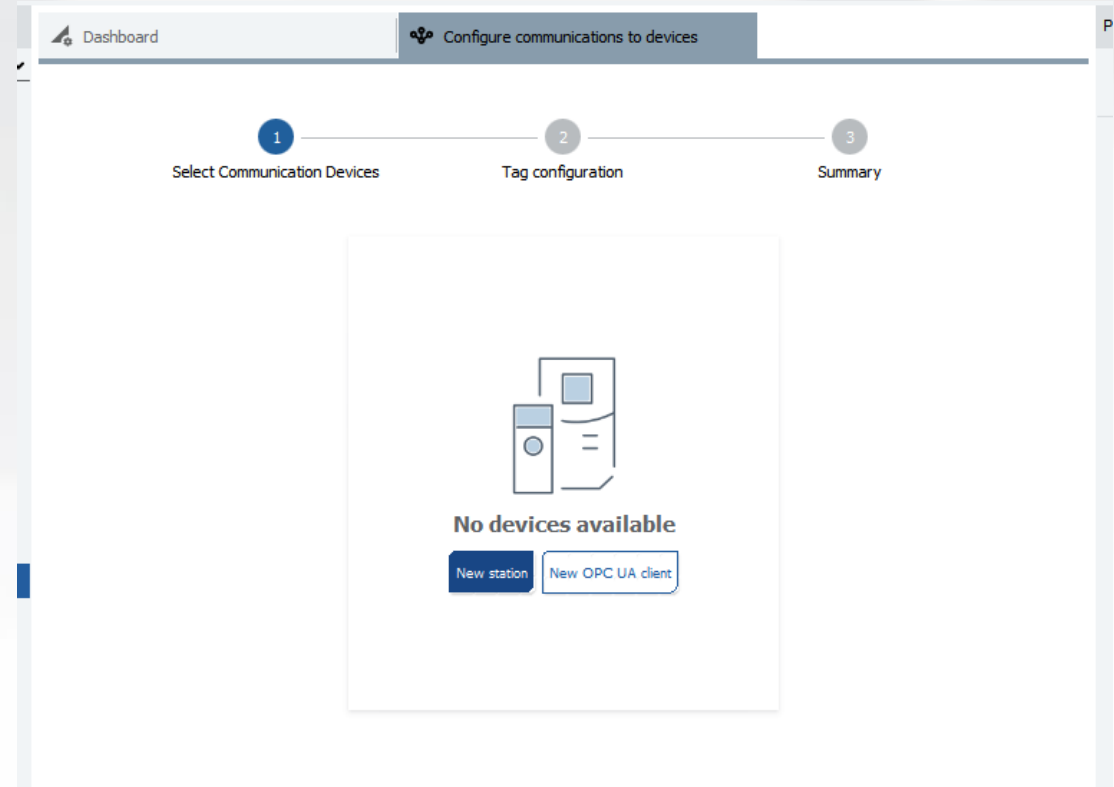
Create a simple HMI project in few steps

1. Configure connected device



Create a simple HMI project in few steps

1. Configure connected device



Create a simple HMI project in few steps

1. Configure connected device

The screenshot displays the 'Configure communications to devices' step in the FactoryTalk HMI configuration software. The interface includes a top navigation bar with 'Dashboard' and 'Configure communications to devices' tabs. A progress indicator at the top shows three steps: 1. Select Communication Devices (active), 2. Tag configuration, and 3. Summary. The main content area is titled 'Select protocol' and features two radio buttons: 'CommDriver' (selected) and 'OPC UA Client'. Below this is a section titled 'New station' with a 'Name' field containing 'RAEtherNet_IPStation1'. A 'Type' section lists various communication protocols with radio buttons, where 'RA EtherNet/IP Station' is selected. The list of types includes: CODESYS Station, MELSEC FX3U Station, MELSEC Q station, Modbus Station, OMRON EtherNet/IP station, OMRON Fins Station, RA EtherNet/IP Station, S7TCP Station, S7 TIA PROFINET station, and TwinCAT station.

Dashboard | Configure communications to devices

1 Select Communication Devices | 2 Tag configuration | 3 Summary

Select protocol

☒ CommDriver ☐ OPC UA Client

New station

Name

RAEtherNet_IPStation1

Type

☐ CODESYS Station

☐ MELSEC FX3U Station

☐ MELSEC Q station

☐ Modbus Station

☐ OMRON EtherNet/IP station

☐ OMRON Fins Station

☒ RA EtherNet/IP Station

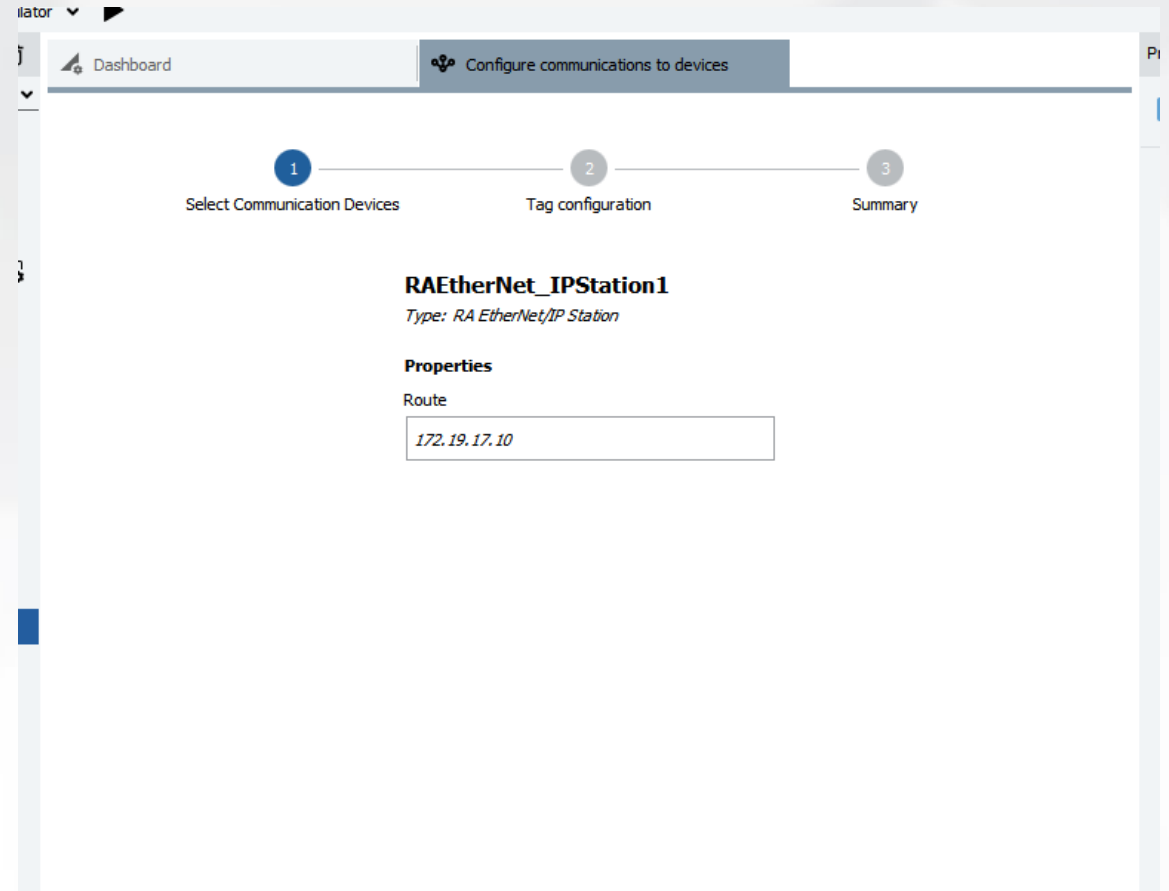
☐ S7TCP Station

☐ S7 TIA PROFINET station

☐ TwinCAT station

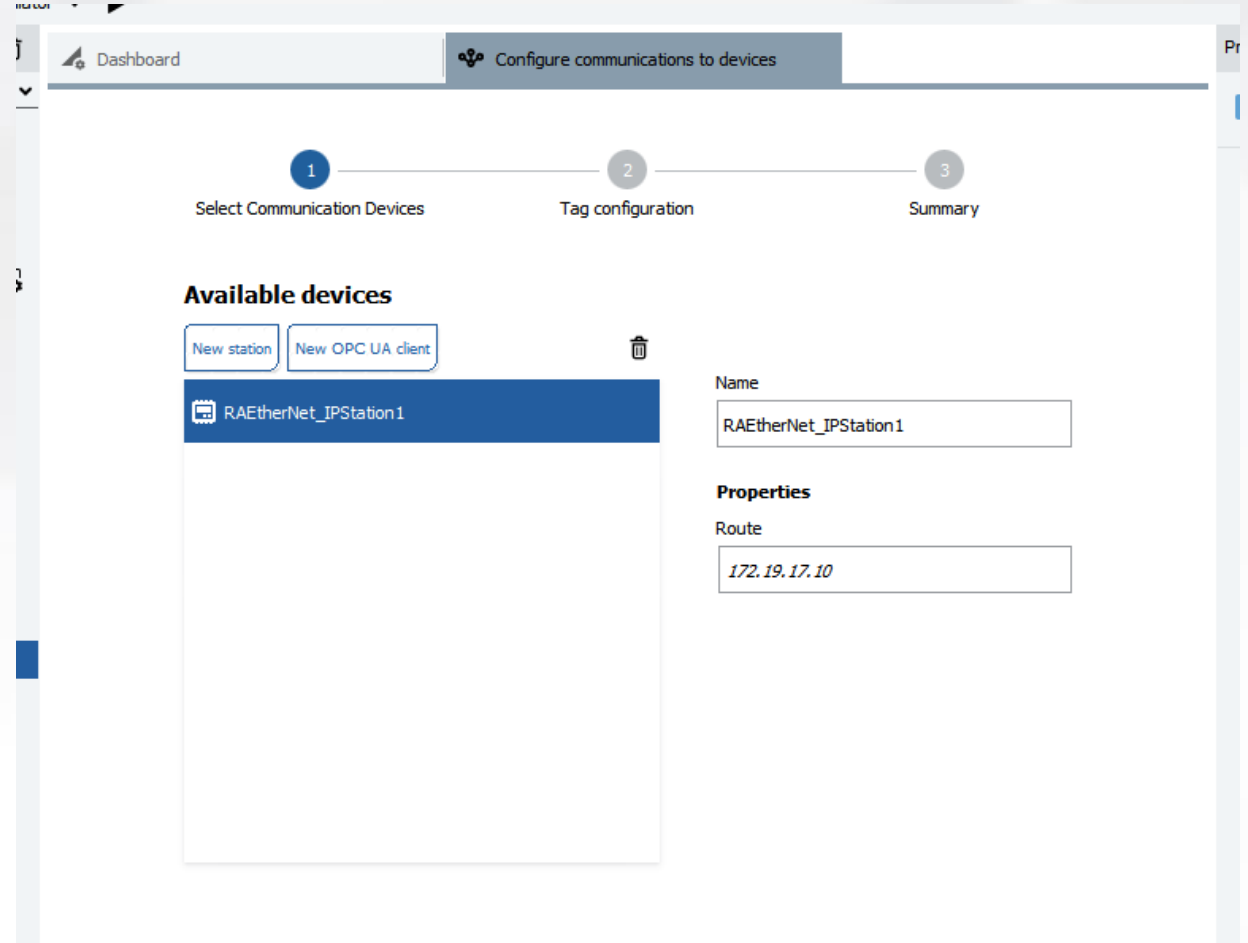
Create a simple HMI project in few steps

1. Configure connected device



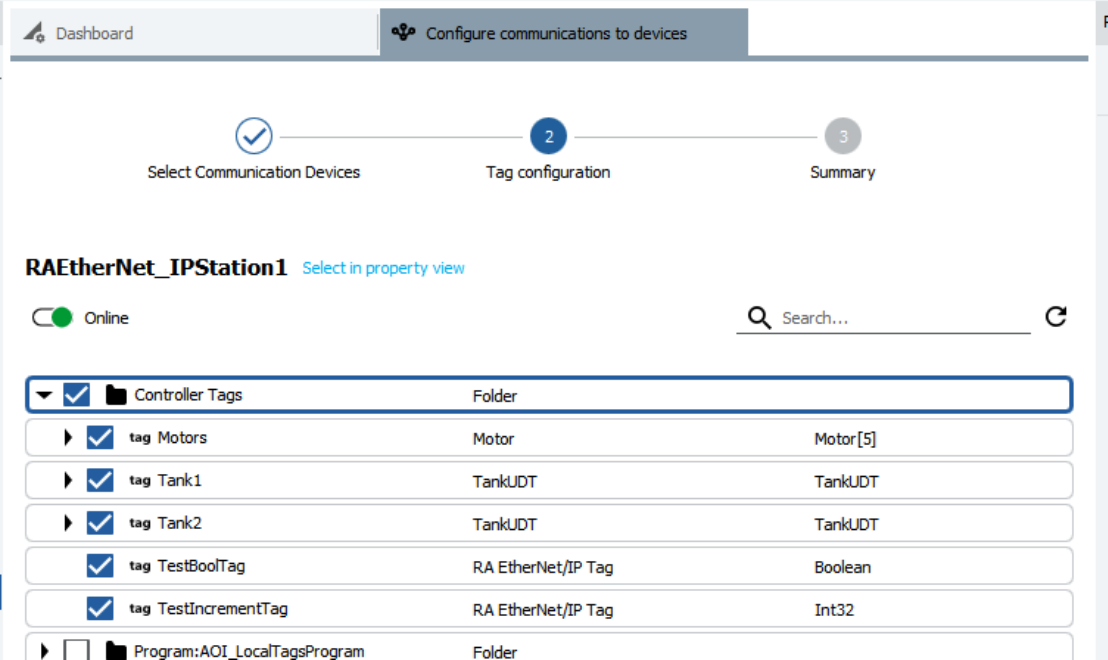
Create a simple HMI project in few steps

1. Configure connected device



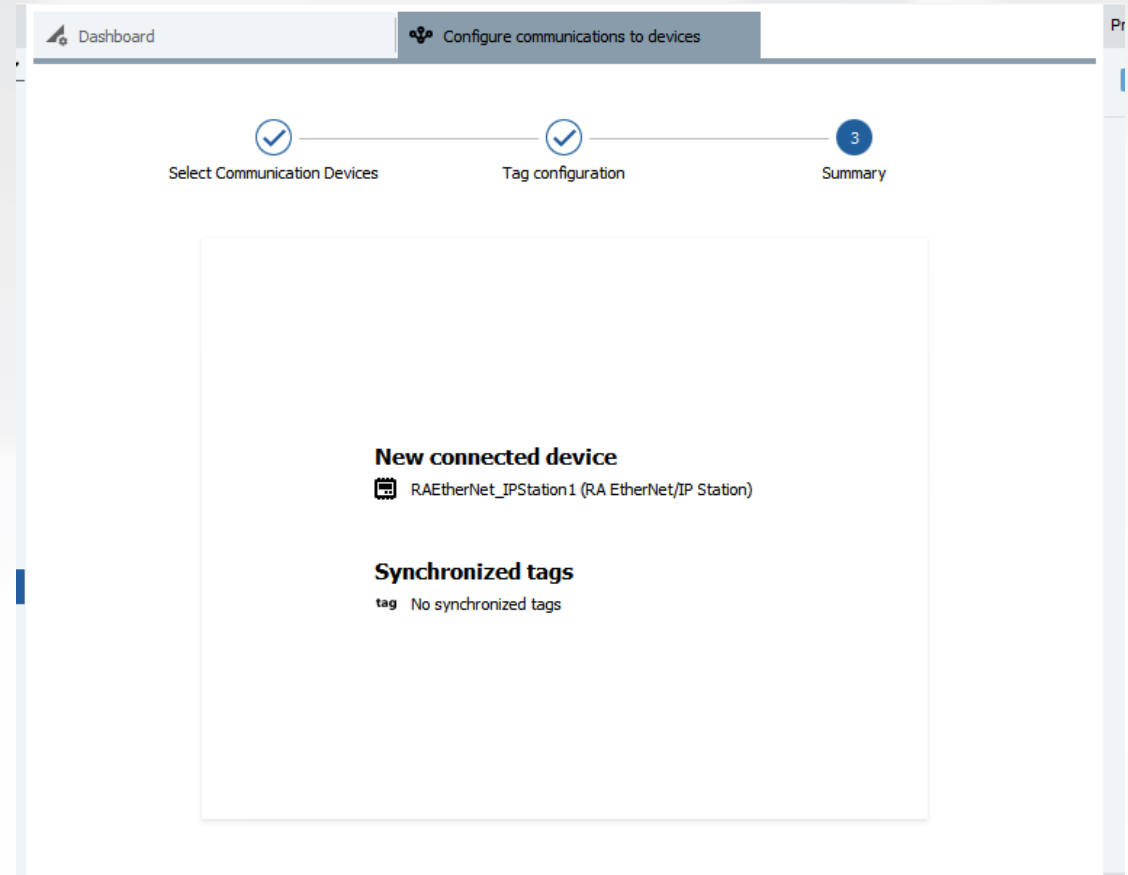
Create a simple HMI project in few steps

1. Configure connected device



Create a simple HMI project in few steps

1. Configure connected device



Create a simple HMI project in few steps

2. Configure a datalogger

Let's begin building your user interface



I want to configure
connected devices



I want to configure a data
logger



I want to configure a recipe



I want to manage users and
groups

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

1

2

3

4

Select data logger

Select database

Select variables

Summary



No data loggers

New data loggers

Exit

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

1

2

3

4

Select data logger

Select database

Select variables

Summary

New data logger

Name

DataLogger1

Properties

Sampling mode

Periodic

Sampling period

0000:00:01.000

Exit

Back

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

1

Select data logger

2

Select database

3

Select variables

4

Summary

Available data loggers

New data logger

DataLogger1

Name

DataLogger1

Properties

Sampling mode

Periodic

Sampling period

0000:00:01.000

✔ Data logger DataLogger1 successfully created

✕

Exit

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

1

2

3

4

Select data logger

Select database

Select variables

Summary



No databases

New database

Exit

Back

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

✓

2

3

4

Select data logger

Select database

Select variables

Summary

New database

Name

EmbeddedDatabase1

Type

☒ Embedded

☐ ODBC connection

Exit

Back

Next

Create a simple HMI project in few steps

2. Configure a datalogger

- In memory: data are put in RAM, content of the DB is lost if the application is restarted

Configure data Logger

1

2

3

4

Select data logger

Select database

Select variables

Summary

EmbeddedDatabase1
Type: Embedded database

Properties
In memory

Exit

Back

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

✓

2

3

4

Select data loggerSelect databaseSelect variablesSummary

Available databases

New database

EmbeddedDatabase1

Name

EmbeddedDatabase1

Type: Embedded database

Properties

In memory

False

✓ Database EmbeddedDatabase1 successfully added

Exit

Back

Next

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

✓

Select data logger

✓

Select database

3

Select variables

4

Summary

Select variables

Type to search...

Q

↺

▼

Tank2

▶

☐ tag ValveInStatus

▶

☐ tag ValveInCmd

▶

☐ tag ValveOutStatus

▶

☐ tag ValveOutCmd

▶

☒ tag Level

▶

☐ tag SetPoint

▼

Tank1

▶

☐ tag ValveInStatus

▶

☐ tag ValveInCmd

▶

☐ tag ValveOutStatus

▶

☐ tag ValveOutCmd

▶

☒ tag Level

Exit

Back

Next

PUBLIC • Copyright ©2024 Rockwell Automation, Inc.

48

 **FactoryTalk**
by ROCKWELL AUTOMATION

Create a simple HMI project in few steps

2. Configure a datalogger

Configure data Logger

✓

✓

✓

4

Select data loggerSelect databaseSelect variablesSummary

Data logger
DataLogger1

Database
EmbeddedDatabase1

Logged variables
var 2 variables configured

✓ Data logger DataLogger1 successfully configured

×

Configure another data logger

Exit

Create a simple HMI project in few steps

3. Configure the user interface

Let's begin building your user interface



I want to configure connected devices



I want to configure a data logger

Preview



Configure my UI displays starting with main window



I want to manage users and groups

Create a simple HMI project in few steps

3. Configure the user interface

Preview

1

Configure UI

Select screen resolution, orientation and template

2

Cusotmize

Select widgets, add title and style sheets

3

Preview & finish

Review and publish

What screen resolution do you want for your main window?

Responsive behavior is supported for scalability

3440 x 1440

1920 x 1080

1366 x 768

1024 x 768

Custom

Width

1920

px

Height

1080

px

Orientation

What navigation style do you want?

Templates are editable

Navigation 1

Navigation 2

Tab bar 1

Tab bar 2

Navigation 3

Navigation 4

Navigation 5

Logo

Tab

April 15, 2022 10:31 AM

Logo

Tab

April 15, 2022 10:31 AM

Logo

Tab

April 15, 2022 10:31 AM

Logo

Tab

April 15, 2022 10:31 AM

How many parent pages would you like to create?

Three children pages are included with each parent page.

Number of pages

-

000

+

How many tabs would you like to create?

Tabs are editable

Number of tabs

-

000

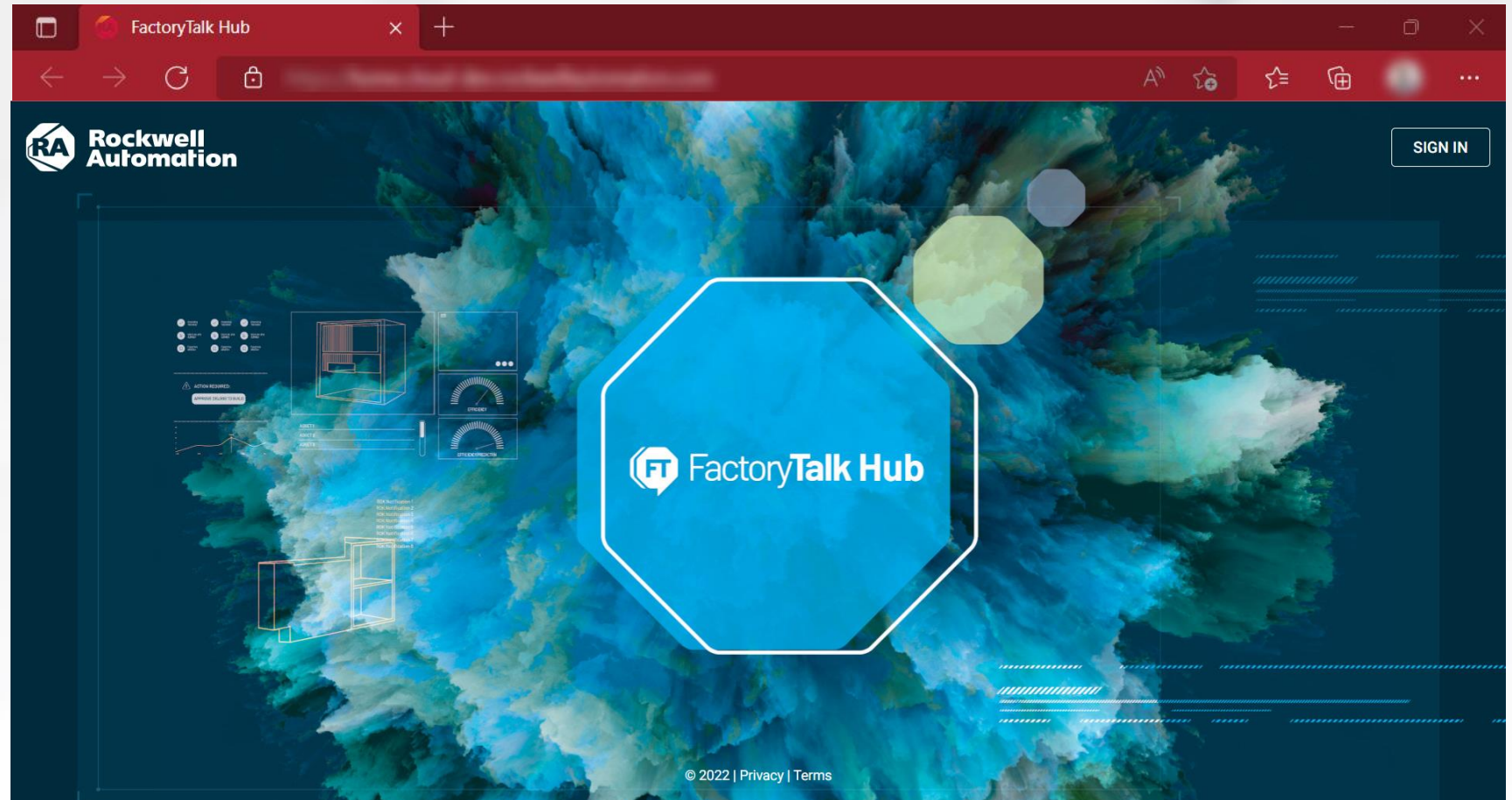
+

Web IDE



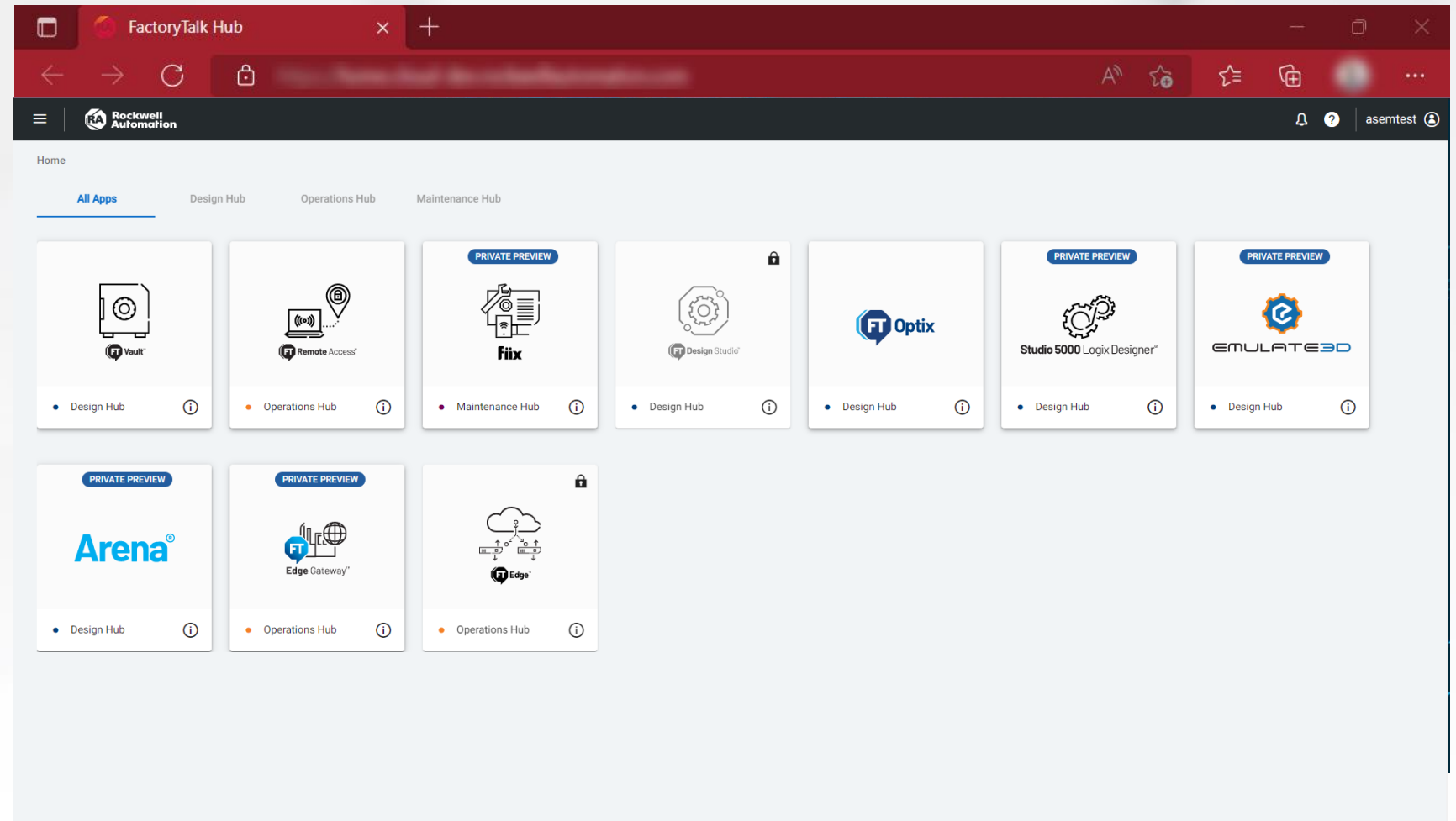
Web IDE using FactoryTalk Hub

- Sign in to FactoryTalk Hub
- Click on the FactoryTalk Optix tile
- Select which build to launch on Web IDE or continue with the last used version



Web IDE using FactoryTalk Hub

- Sign in to FactoryTalk Hub
- Click on the FactoryTalk Optix tile
- Select which build to launch on Web IDE or continue with the last used version



Web IDE using FactoryTalk Hub

- Sign in to FactoryTalk Hub
- Click on the FactoryTalk Optix tile
- Select which build to launch on Web IDE or continue with the last used version

FactoryTalk Hub

Releases Entitlements

Luca Bennati

Continue with the last used version
beta, 1.0.2.151

FT Optix Release 1.0.1.193
Download or open the latest Release build of FT Optix.

Open Web IDE Download to desktop Changelog

Previous Releases

Date created	Version	
14/9/2022	1.0.1.192	Open Web IDE Download to desktop
13/9/2022	1.0.1.191	Open Web IDE Download to desktop
9/9/2022	1.0.1.190	Open Web IDE Download to desktop
7/9/2022	1.0.1.188	Open Web IDE Download to desktop
6/9/2022	1.0.1.187	Open Web IDE Download to desktop

FT Optix Beta 1.0.2.151
Download or open the latest Beta build of FT Optix.

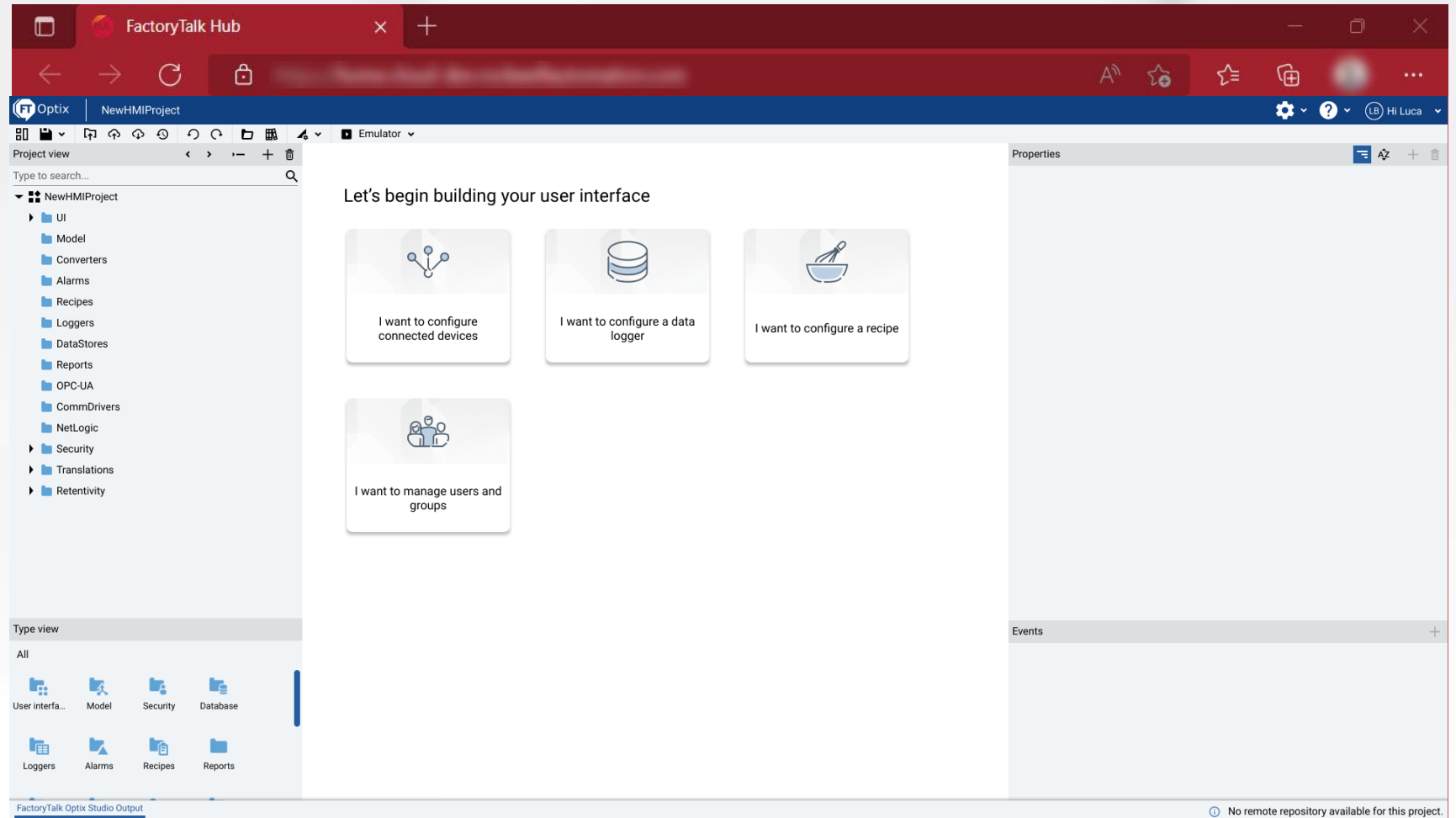
Open Web IDE Download to desktop Changelog

Previous Beta

Date created	Version	
22/9/2022	1.0.2.142	Open Web IDE Download to desktop
21/9/2022	1.0.2.122	Open Web IDE Download to desktop
20/9/2022	1.0.2.111	Open Web IDE Download to desktop
19/9/2022	1.0.2.94	Open Web IDE Download to desktop
15/9/2022	1.0.2.77	Open Web IDE Download to desktop

Web IDE using FactoryTalk Hub

- Sign in to FactoryTalk Hub
- Click on the FactoryTalk Optix tile
- Select which build to launch on Web IDE or continue with the last used version



Project versions and components



Project compatibility and backup

- **Upgrade:**

- Projects created with an old version of Studio **can be opened with later versions**
- Studio automatically updates the project and all its modules to the current version

- **Downgrade:**

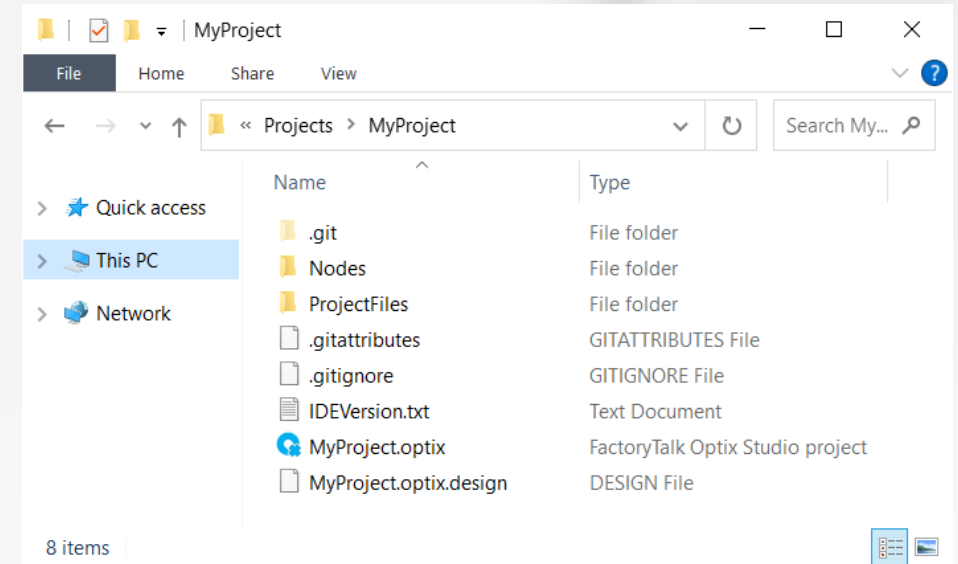
- Projects created with newer versions of Studio **cannot be opened with older versions**

- **Backups:** %localappdata%\Rockwell Automation\FactoryTalk Optix\FTOptixStudio\Backups\

- Automatic: contain the last three backups, performed when saving the project
- Upgrade: performed when updating the project to make it compatible with a newer version
- Recovery: performed when a corrupt project is opened, and repair is attempted

Project structure

- Contents of the project folder
 - **.optix** file: information on the project's elements
 - **.optix.design** file: information on accessory elements only required at design time.
- **Nodes** folder: contains a YAML file for each node that describes its related information model
- **ProjectFiles** folder: containing all files used in the project, including images, fonts and documents.
 - NetSolution subfolder contains all files used to create customized C# scripts
 - PKI subfolder: Public Key Infrastructure folder, used to manage security certificates for OPC-UA

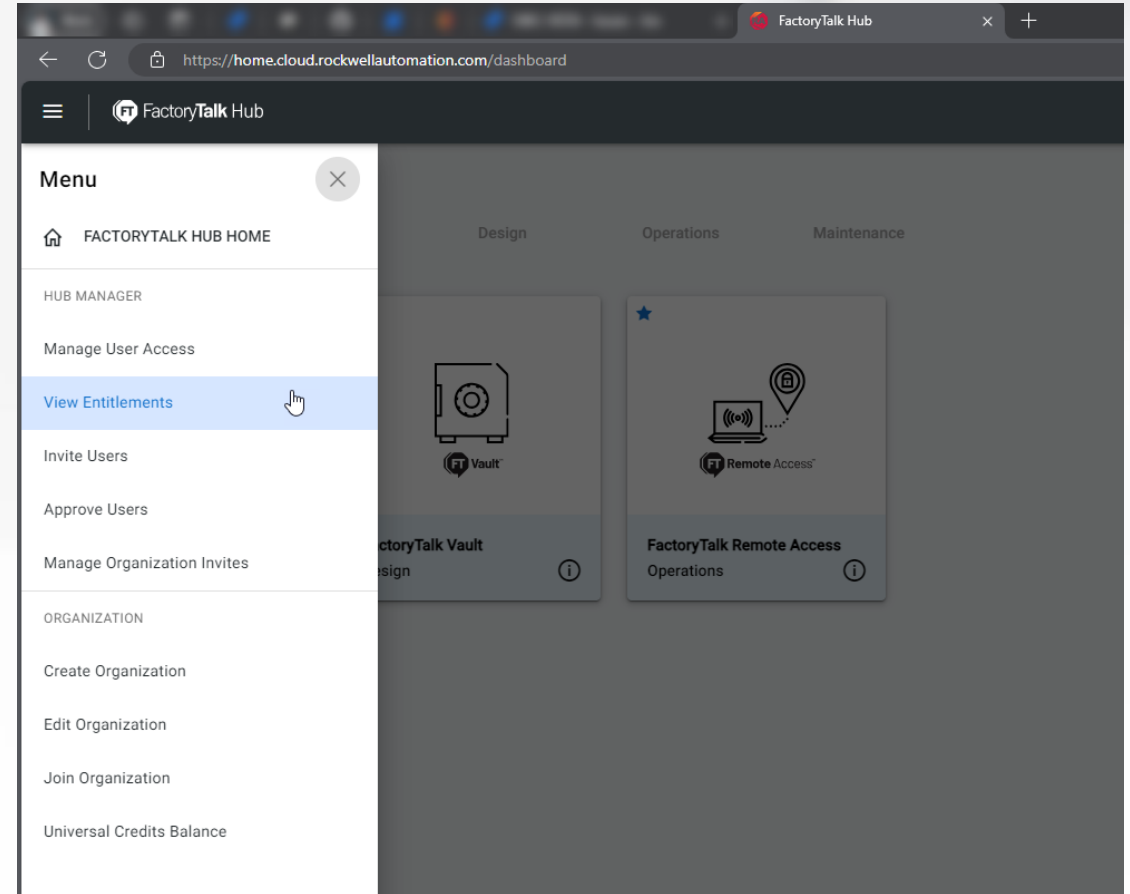


Factory Talk Optix Entitlements



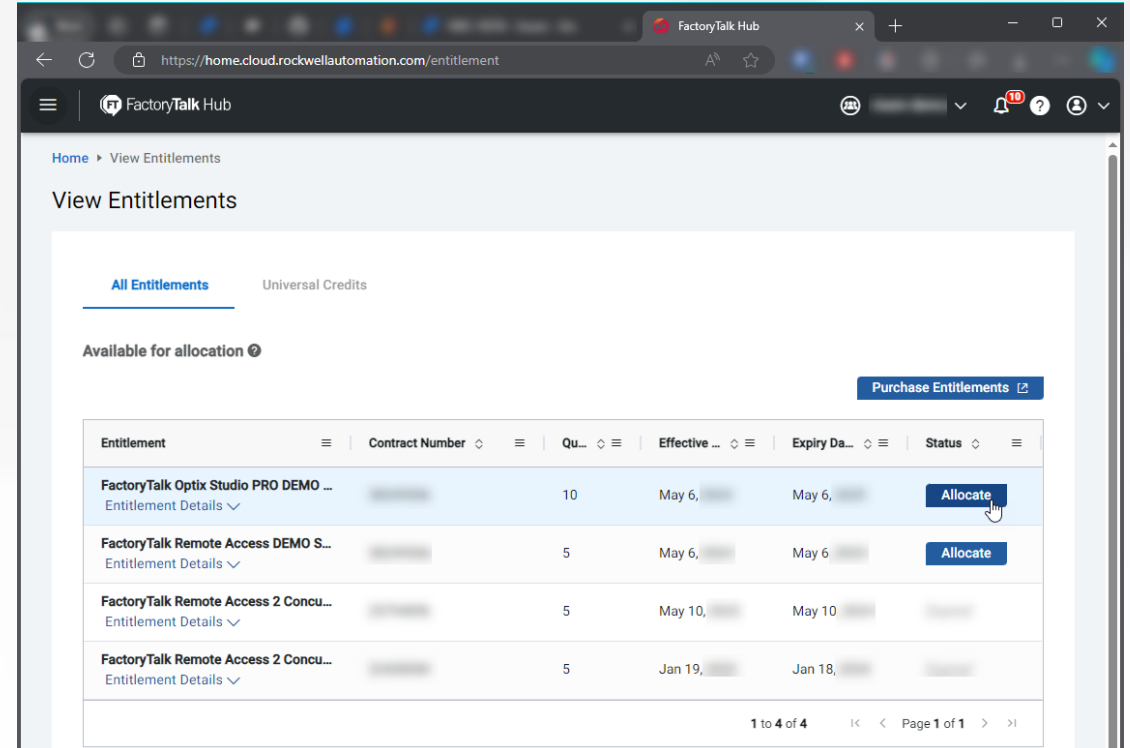
FactoryTalk Optix Entitlements

- Licenses are called «Entitlements»
- Entitlements are assigned to the user that bought the license by accessing the «Entitlements» menu of FactoryTalk Hub
- Entitlements have to be assigned to a FactoryTalk Optix organization



FactoryTalk Optix Entitlements

- Licenses are called «Entitlements»
- Entitlements are assigned to the user that bought the license by accessing the «Entitlements» menu of FactoryTalk Hub
- Entitlements have to be assigned to a FactoryTalk Optix organization



FactoryTalk Optix Entitlements

- Licenses are called «Entitlements»
- Entitlements are assigned to the user that bought the license by accessing the «Entitlements» menu of FactoryTalk Hub
- Entitlements have to be assigned to a FactoryTalk Optix organization

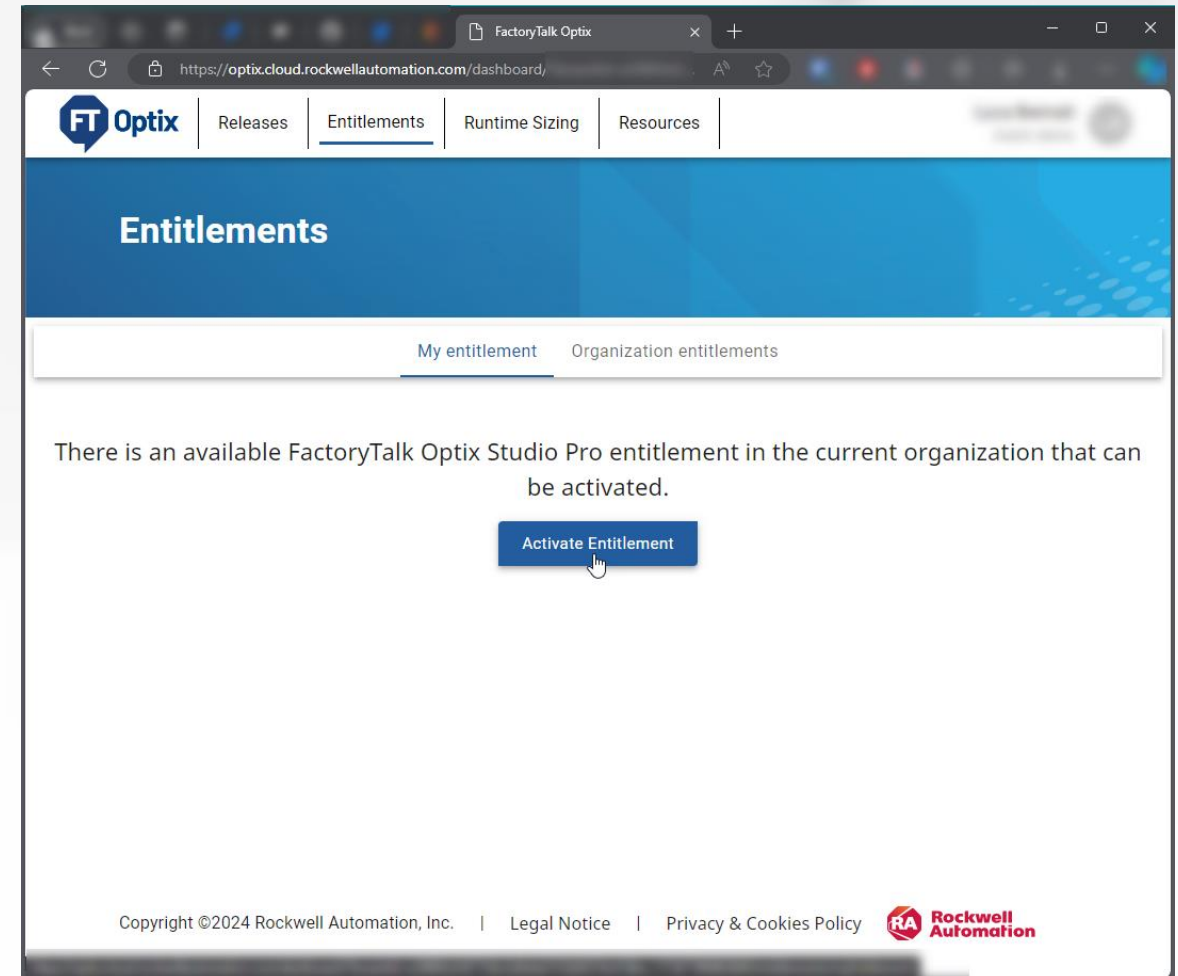
The screenshot shows a web application interface with a modal dialog titled "Allocate Entitlement". The dialog contains the following fields and options:

- Entitlement Details**
 - FactoryTalk Optix Studio PRO** (with a blurred ID)
 - Catalog ID: [blurred]
 - Contract Number: [blurred]
 - Organization ①: [blurred]
- Select Allocation Quantity**
 - ☐ **Allocate Full Quantity**
10
 - ☒ **Allocate Partial Quantity**
Enter custom quantity: [up/down arrows]
- Buttons**
 - Review Allocation** (highlighted with a mouse cursor)
 - Cancel**

The background shows a table with columns: Contract Number, Allocated, Consumed, and Effect. The first row is "Studio PRO DEMO 8x5 support" with Contract Number "23015565", Allocated "1", and Consumed "0".

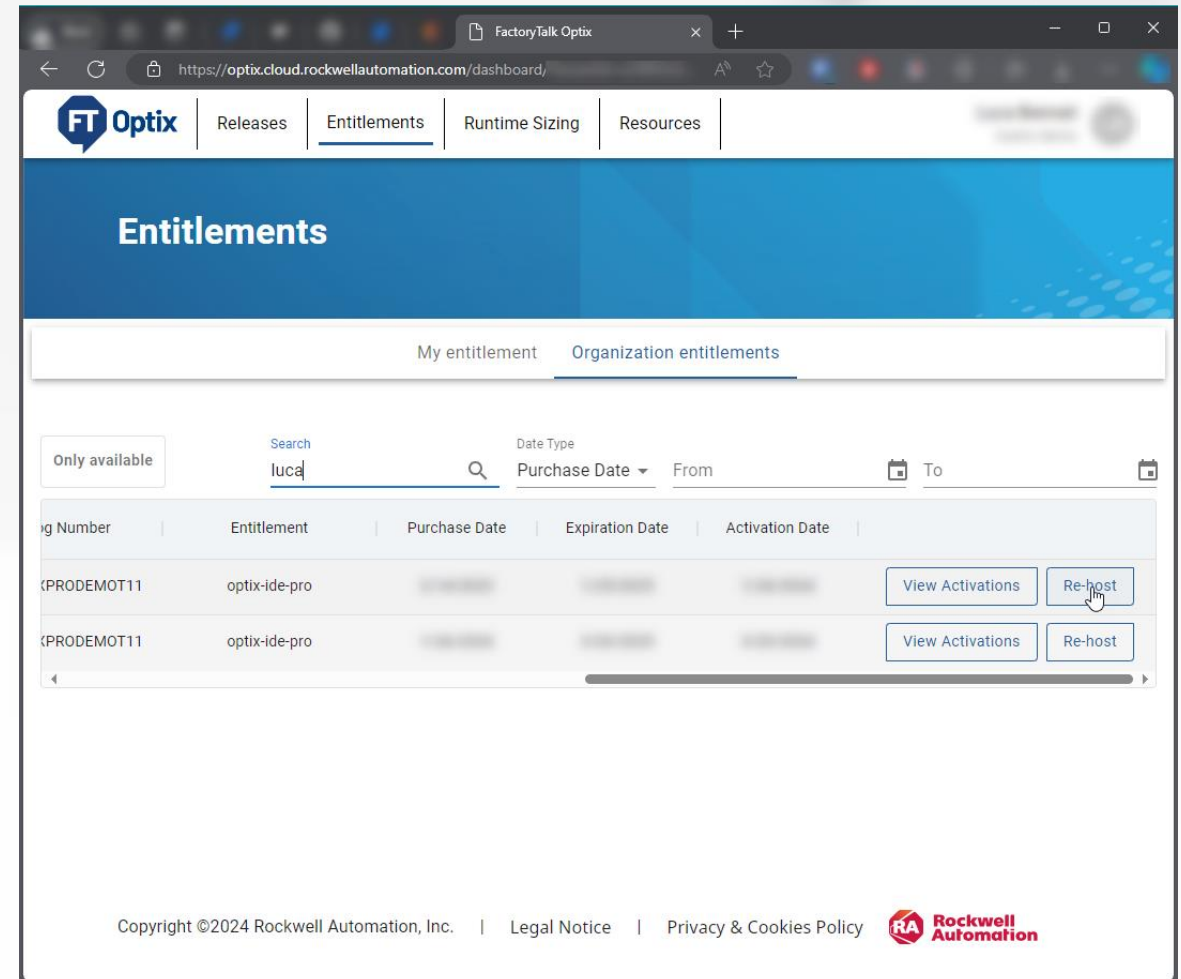
FactoryTalk Optix Studio Entitlements

- Users of an organization can self-assign any available Studio entitlement from FactoryTalk Optix web page
- Studio Entitlements can be released once they are not needed anymore and used by a different user in the same organization either from FactoryTalk Optix Studio or the web dashboard
- Studio Entitlements can only be rehosted by the user that has is using the license or by an administrator of the organization



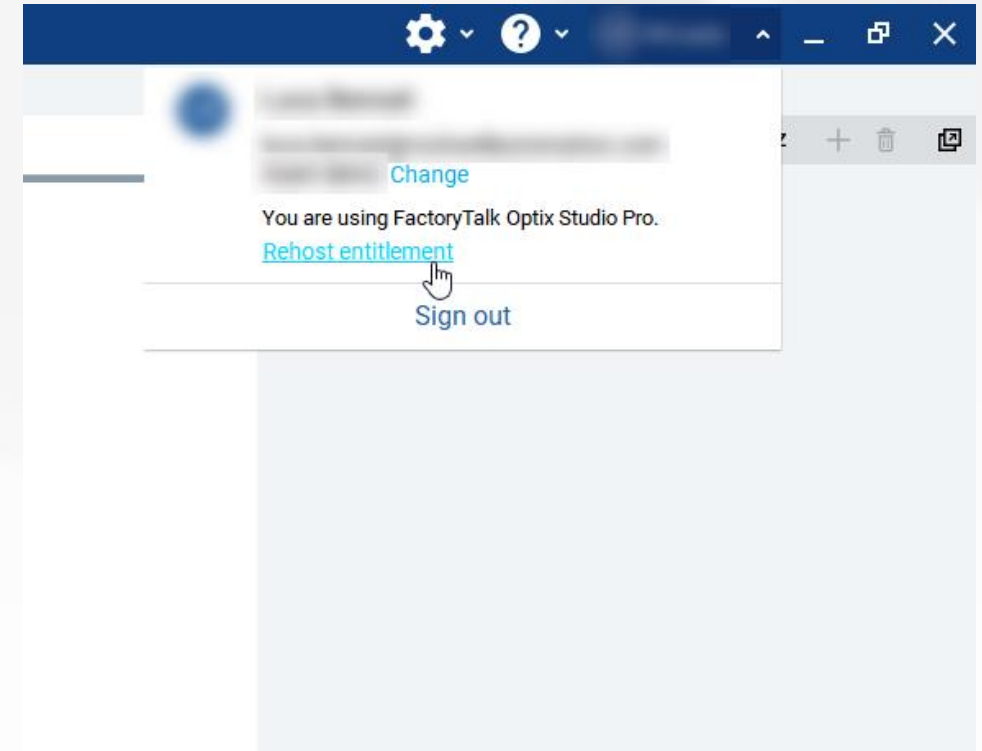
FactoryTalk Optix Studio Entitlements

- Users of an organization can self-assign any available Studio entitlement from FactoryTalk Optix web page
- Studio Entitlements can be released once they are not needed anymore and used by a different user in the same organization either from FactoryTalk Optix Studio or the web dashboard
- Studio Entitlements can only be rehosted by the user that has is using the license or by an administrator of the organization



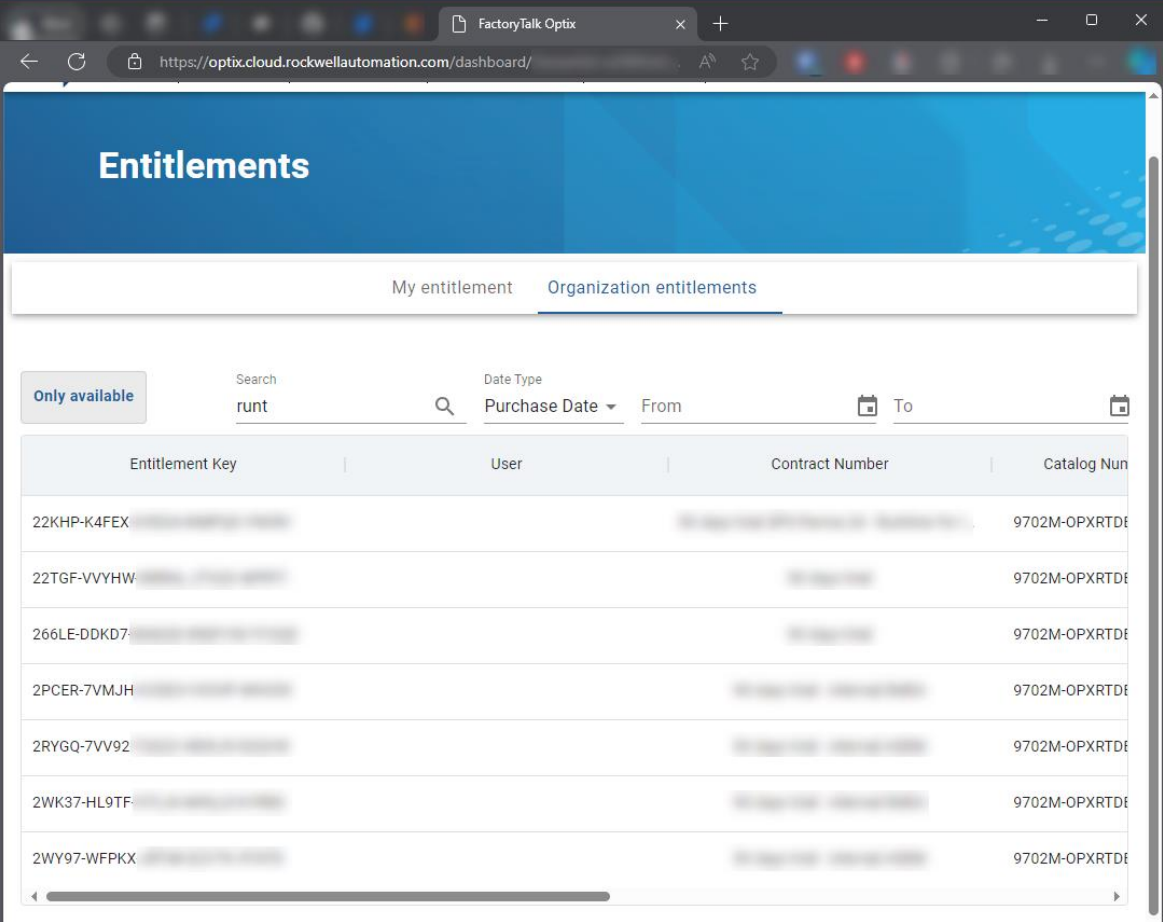
FactoryTalk Optix Studio Entitlements

- Users of an organization can self-assign any available Studio entitlement from FactoryTalk Optix web page
- Studio Entitlements can be released once they are not needed anymore and used by a different user in the same organization either from FactoryTalk Optix Studio or the web dashboard
- Studio Entitlements can only be rehosted by the user that has is using the license or by an administrator of the organization



FactoryTalk Optix Runtime Entitlements

- Embedded devices comes with a preinstalled entitlements (size varies depending on the device)
- «Open» devices (like iPC) should be manually activated using the License Manager which is installed with the Runtime Tools
- Runtime licenses can be upgraded by buying a bigger size
 - Runtime licenses upgrade on Embedded devices (like OptixPanels) may be limited to specific sizes



Entitlement Key	User	Contract Number	Catalog Nun
22KHP-K4FEX			9702M-OPXRTD
22TGF-VVYHW			9702M-OPXRTD
266LE-DDKD7			9702M-OPXRTD
2PCER-7VMJH			9702M-OPXRTD
2RYGQ-7VV92			9702M-OPXRTD
2WK37-HL9TF			9702M-OPXRTD
2WY97-WFPKX			9702M-OPXRTD

FactoryTalk Optix Entitlements

- A dedicated tutorial is available

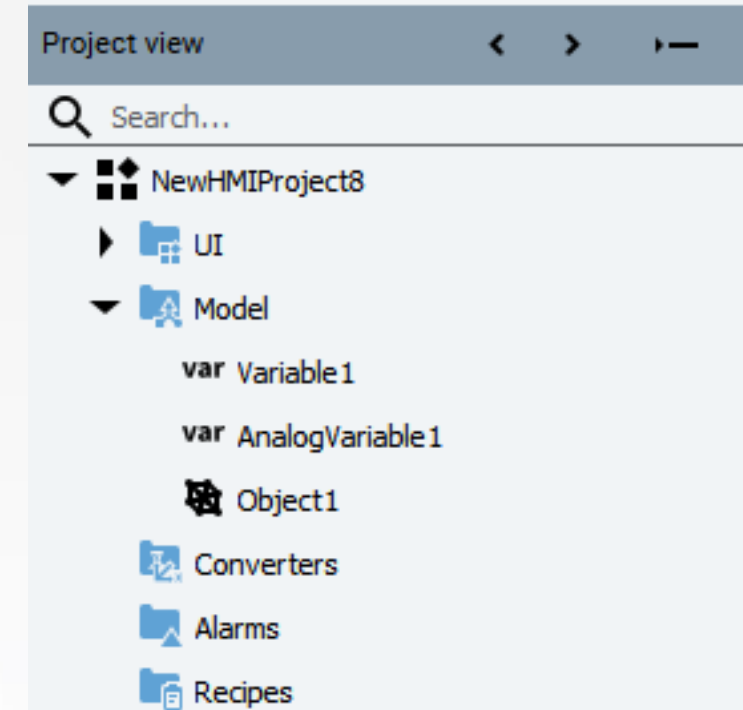


Variables and tags



Model's variables

- What are the purposes of these Tags?
 - as "**internal variables**" used only on the HMI project (such as parameters or global variables)
 - as "**bridge**" between the Tag connected to the Comm Driver and the User Interface (such as using same UI for different PLC vendors)
- Under the "Model" node, it's possible to define different types of tags:
 - **Variable**: Standard Tag with desired data type
 - **Analog Variable**: Tag with Engineering Unit (used in combination with Measurement System)
 - **Object**: Structured Tag (a.k.a. User Defined Type)



Communication drivers

- Used to:
 - communicate with a PLC exchanging Tag values
 - import Tags from the PLC application (if supported by the protocol)
- To configure the communication,
 1. Add a Communication Driver object that represents the communication protocol. Some drivers (like Modbus) have a choice between serial and eth.
 2. Add one (or more) Station object that represents PLC and configure communication parameters

Communication driver	Online tags import	Offline tags import
Modbus driver		
MELSEC FX3U Driver		✓
S7TCP driver	✓*	✓
OMRON EtherNet/IP driver		✓
MELSEC Q driver		✓
S7 TIA PROFINET driver	✓	✓
OMRON Fins Driver		✓
CODESYS Driver	✓	✓
TwinCAT driver	✓	✓
Serial port		
RA EtherNet/IP Driver	✓	✓
Micro Controller Driver	✓	

Communication drivers

- Some drivers also supports runtime import of PLC Tags
 - A custom logic must be added to the project (NetLogic examples are available on GitHub)
 - Only supported for:
 - RA Ethernet/IP
 - Twincat
 - Siemens S7 TCP

```
[FTOptix.NetLogic.ExportMethod]
public static void ImportFromRAEthernetIP()
{
    Log.Info("RuntimeTagsImport.ImportFromRAEthernetIP", "Starting runtime tags import");

    var station = Project.Current.Get<FTOptix.RAEtherNetIP.Station>("CommDrivers/RAEtherNet_IPDriver1/RAEtherNet_IPStation1");

    // Call Browse method to retrieve data from PLC
    station.Browse(out var plcItems, out var prototypes);

    Log.Info("RuntimeTagsImport.ImportFromRAEthernetIP", "Fetched " + plcItems.Length + " tags structures from the PLC");

    // Filter tag to import by tag name
    var tagInstances = FilterTagsToImport(new List<string>() { "Controller Tags/Tank1/Level", "Controller Tags/Tank2/Level" }, plcItems);

    Log.Info("RuntimeTagsImport.ImportFromRAEthernetIP", "Importing " + tagInstances.Length + " tags");

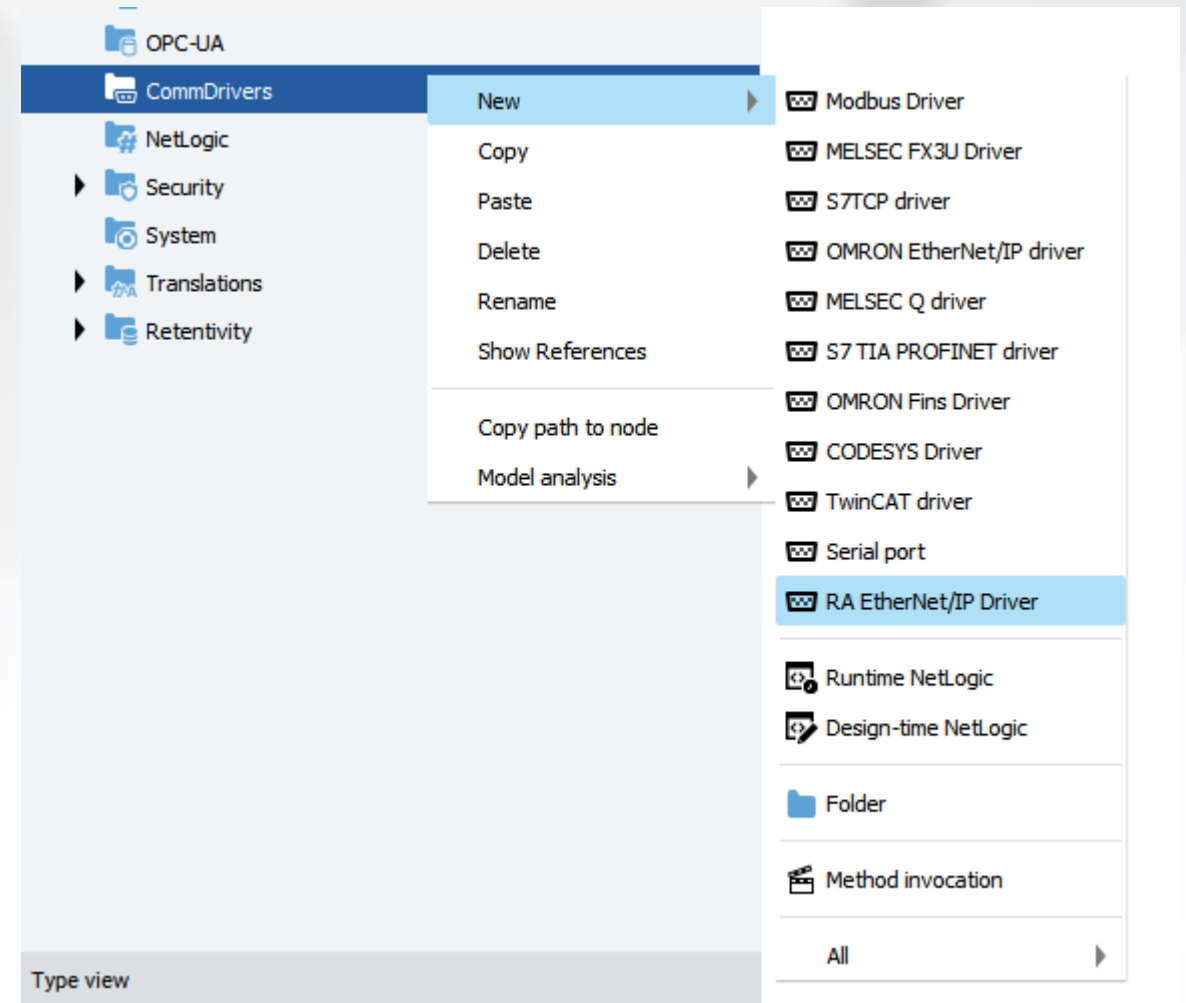
    if (tagInstances.Length == 0)
    {
        Log.Warning("RuntimeTagsImport.ImportFromRAEthernetIP", "No tags to import. Returning.");
        return;
    }

    //Filter tag to import by prototypeId
    // var tagInstances = GetTagsOfType("MachineParameter", plcItems);

    station.Import(tagInstances, prototypes);
}
```

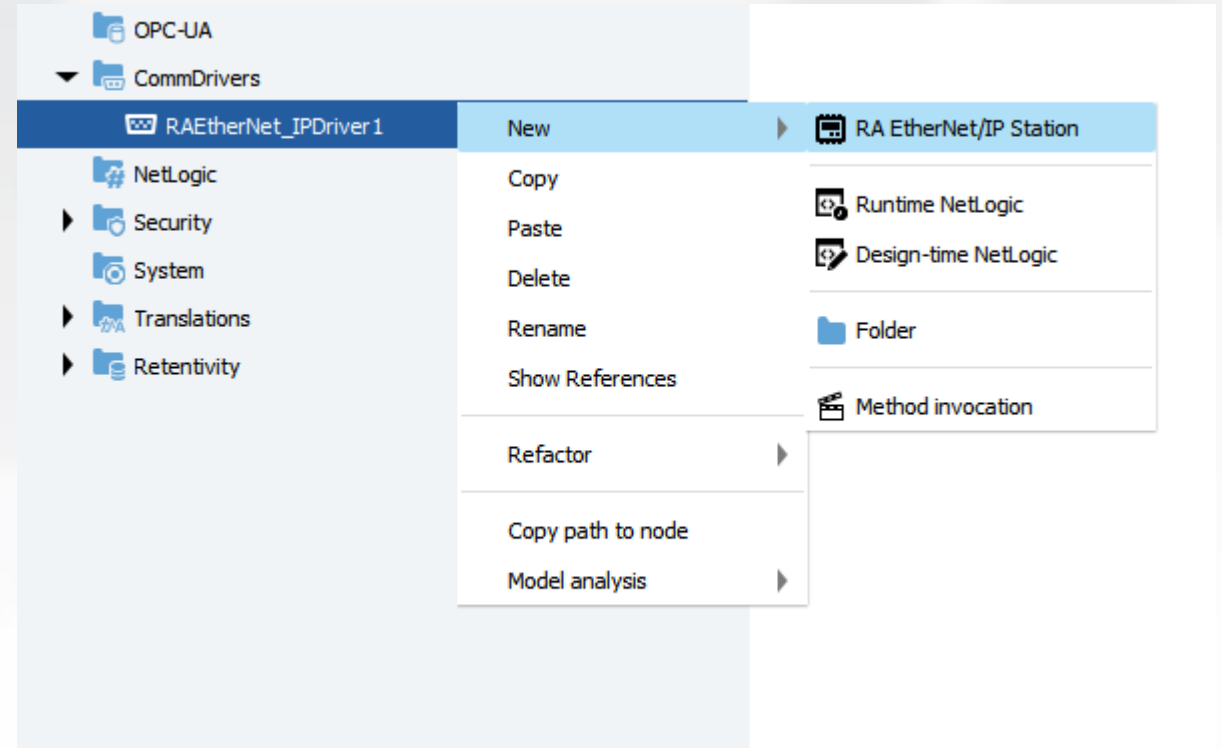
Communication drivers

- Used to:
 - communicate with a PLC exchanging Tag values
 - import Tags from the PLC application (if supported by the protocol)
- To configure the communication,
 1. Add a Communication Driver object that represents the communication protocol. Some drivers (like Modbus) have a choice between serial and eth.
 2. Add one (or more) Station object that represents PLC and configure communication parameters



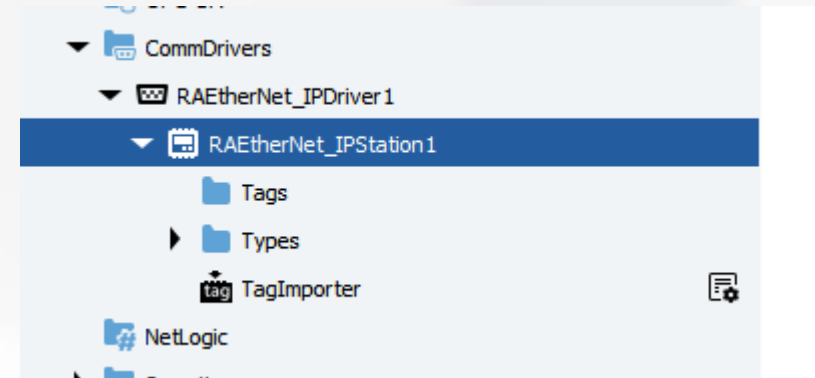
Communication drivers

- Used to:
 - communicate with a PLC exchanging Tag values
 - import Tags from the PLC application (if supported by the protocol)
- To configure the communication,
 1. Add a Communication Driver object that represents the communication protocol. Some drivers (like Modbus) have a choice between serial and eth.
 2. Add one (or more) Station object that represents PLC and configure communication parameters



Communication drivers

- Used to:
 - communicate with a PLC exchanging Tag values
 - import Tags from the PLC application (if supported by the protocol)
- To configure the communication,
 1. Add a Communication Driver object that represents the communication protocol. Some drivers (like Modbus) have a choice between serial and eth.
 2. Add one (or more) Station object that represents PLC and configure communication parameters

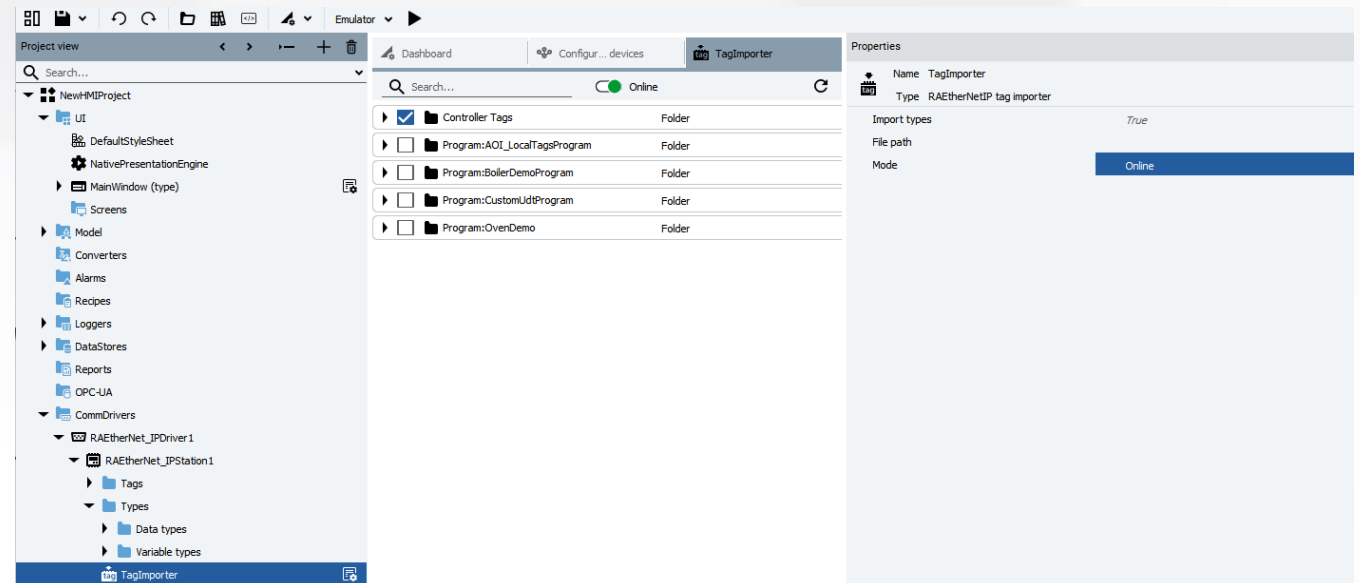


Add tags exchanged with plc: tag importer

- Tag Importer
 - *Offline*: from an export file generated by PLC programming software
 - *Online*: reading tags directly from an online PLC (if supported by the comm driver)

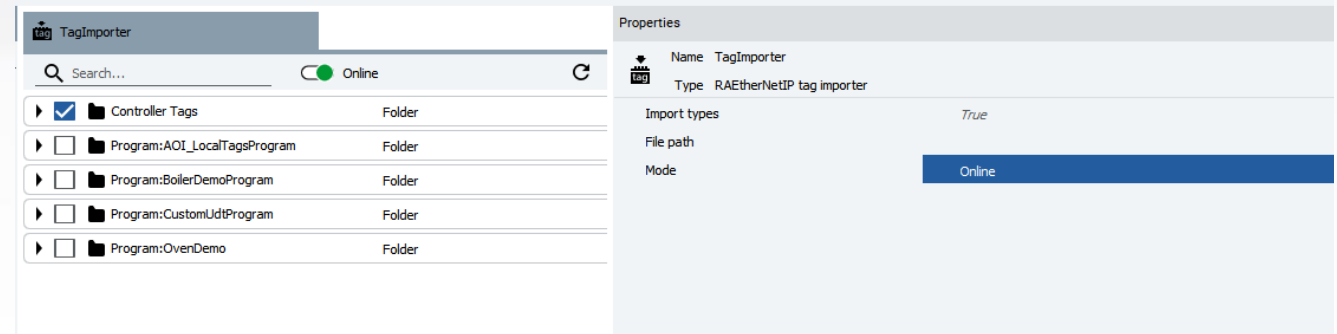
- Remind!

- the number of tags exchanged with PLC are not counted by the license manager



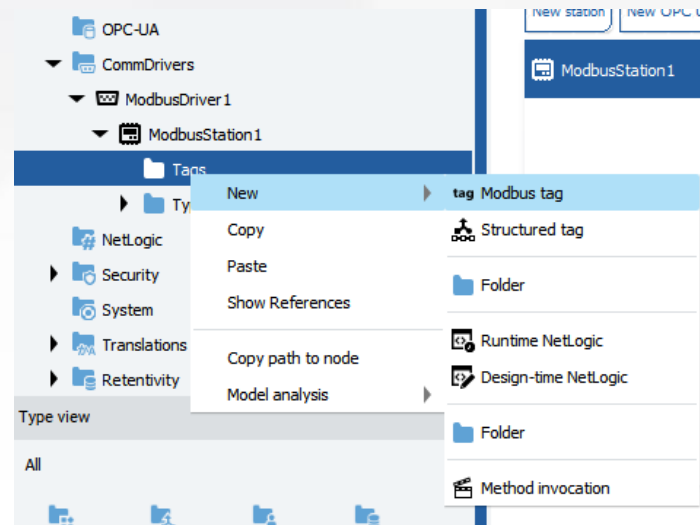
Add tags exchanged with plc: tag importer

- Tag Importer
 - *Offline*: from an export file generated by PLC programming software
 - *Online*: reading tags directly from an online PLC (if supported by the comm driver)
- Remind!
 - the number of tags exchanged with PLC are not counted by the license manager



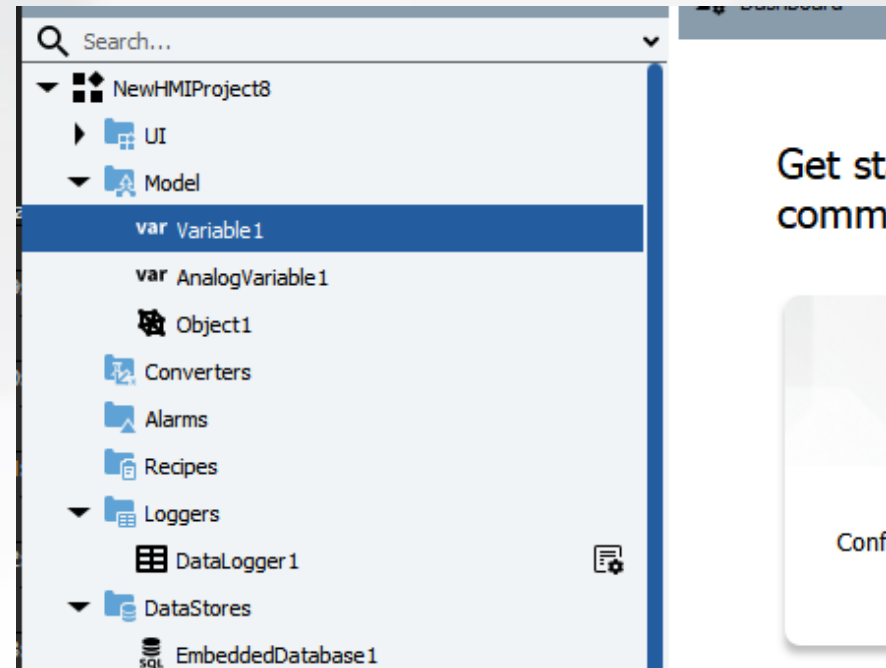
Add tags exchanged with plc: manually

- for Communication Drivers not using symbolic names (i.e. Modbus TCP)
- tags can be created “manually”
 - Right-click on the Tags node and select the "New Variable" command
 - on the Properties panel, configure the tag



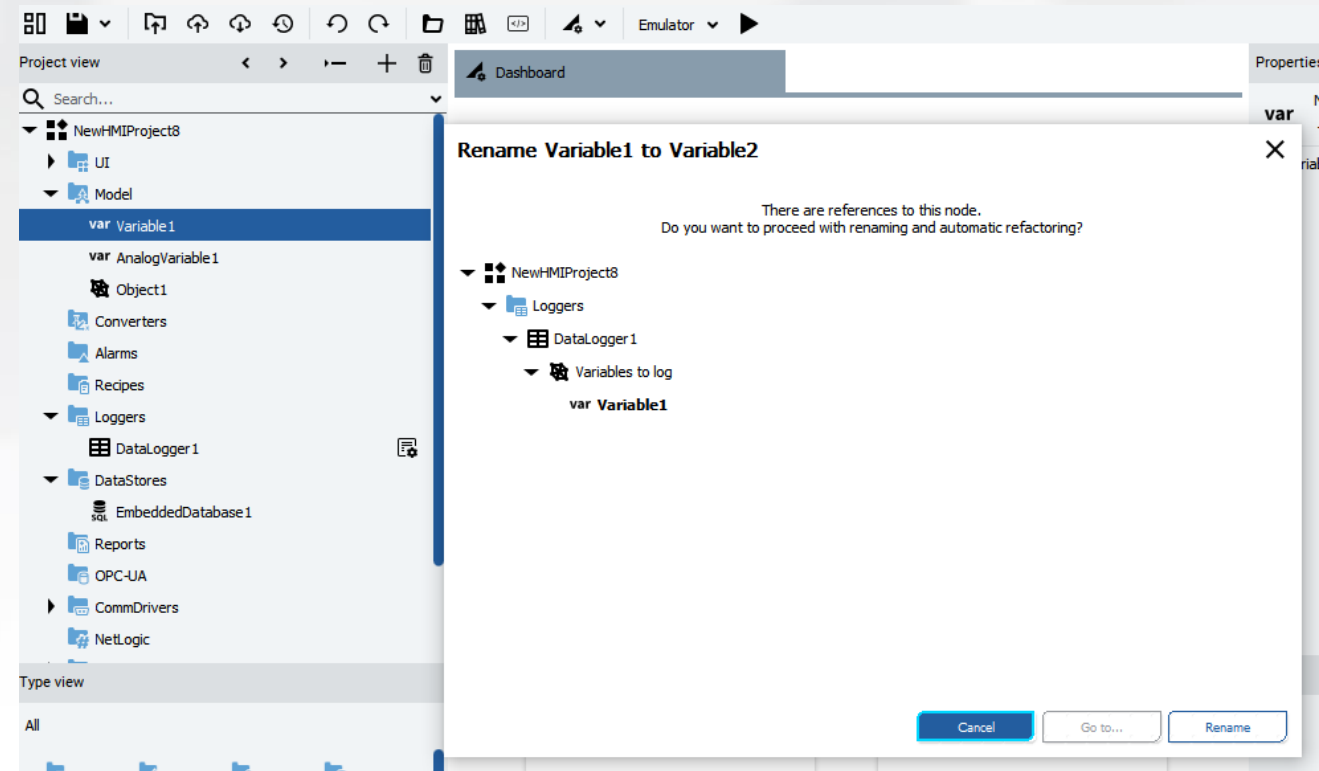
Tag rename

- Tags can be renamed
 - Applies to Model's Tags
 - Applies to Tags exchanged with PLC created manually and also imported!
 - If the Tag is referenced by a resource that requires refactoring (example: a datalogger), a popup will appear automatically
 - If it has been imported, the Tag will be marked as "Renamed"



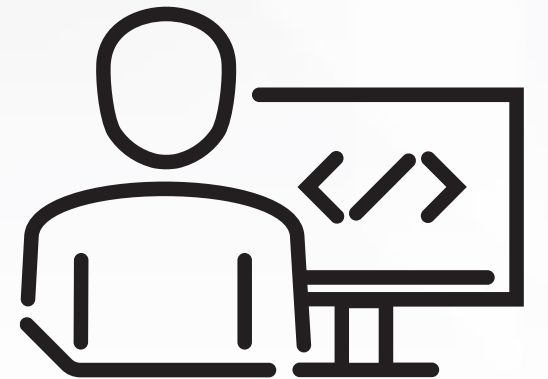
Tag rename

- Tags can be renamed
 - Applies to Model's Tags
 - Applies to Tags exchanged with PLC created manually and also imported!
 - If the Tag is referenced by a resource that requires refactoring (example: a datalogger), a popup will appear automatically
 - If it has been imported, the Tag will be marked as "Renamed"



Hands-on session

- Create some Model tags
- Add the RA Ethernet/IP communication driver
- Import tags from FactoryTalk Logix Echo

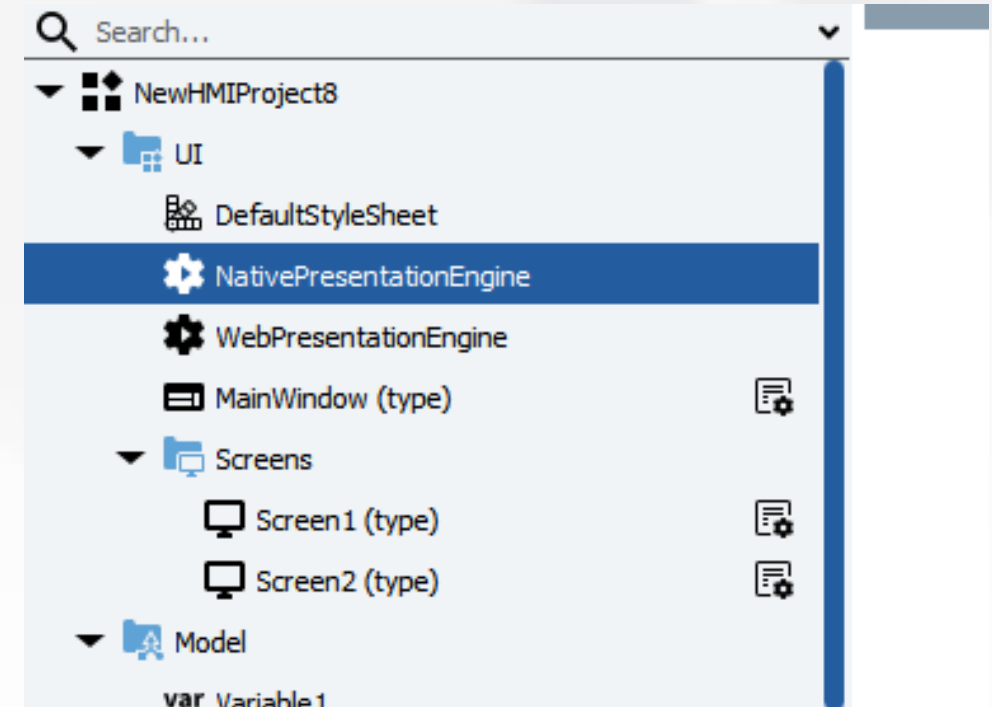


Develop the user interface



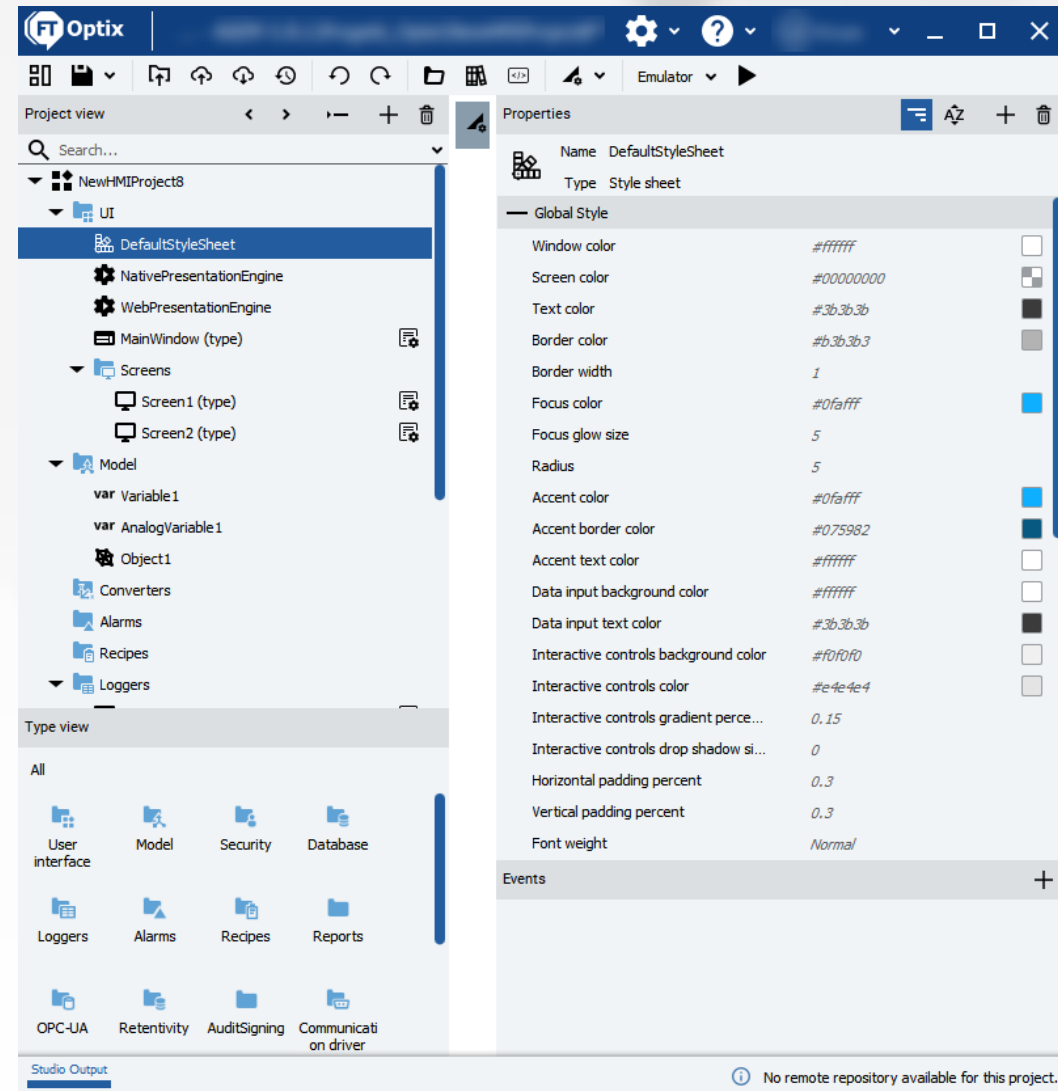
User interface (UI)

- Default Style Sheet
 - makes it possible to globally set some style properties of all graphical objects in the project
- Presentation Engine
 - **Native** Presentation Engine it's the object that create the user interface of the Application
 - **Web** Presentation Engine, it's the object that create the user interface for Web access
- Main Window
 - This object is the root container of graphical objects



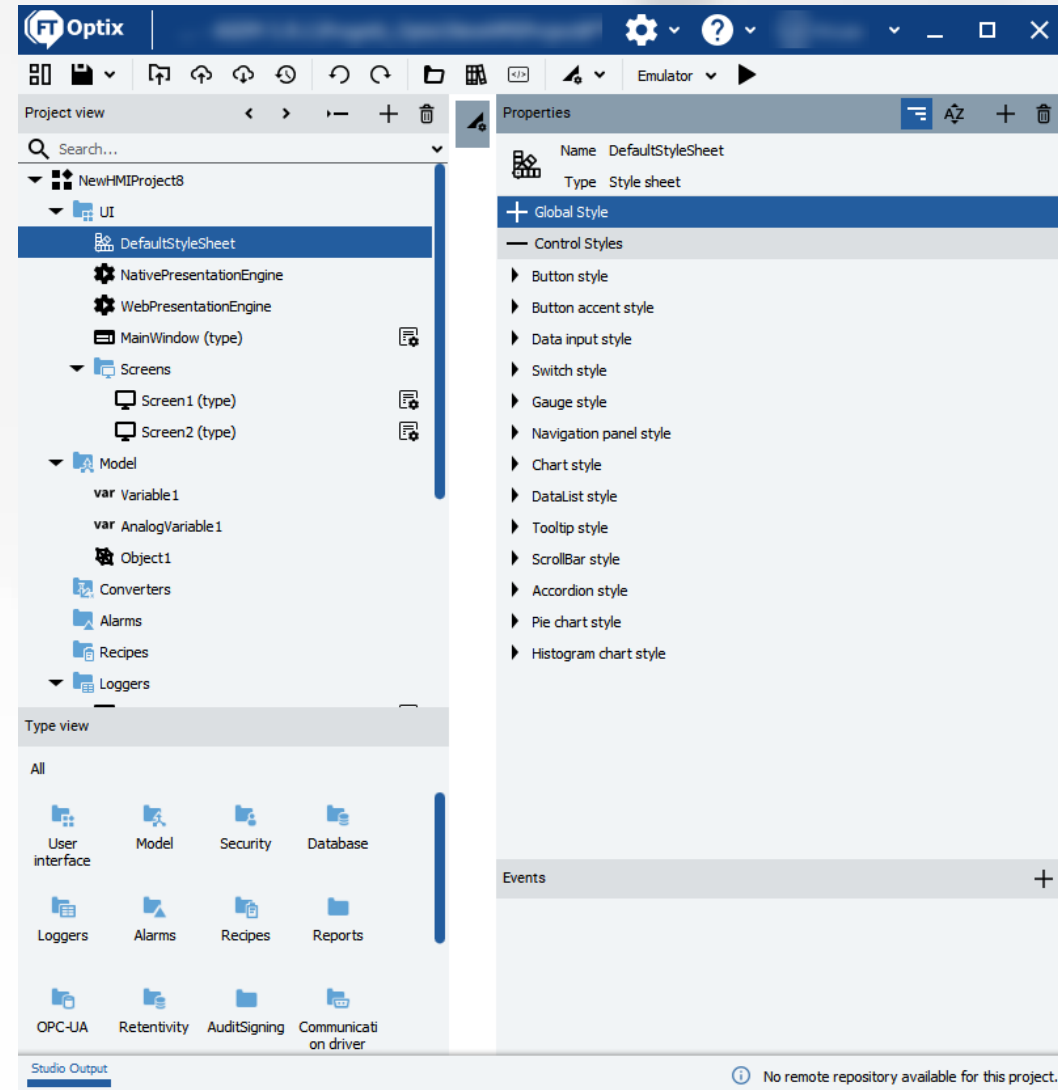
Default style sheet

- Properties
 - Global Style: style properties in common with all objects (like radius)
 - Control Styles: style properties for a specific type of object (Button, Switch...)
- Multiple style sheets
 - into the same project can be defined multiple style sheets for example, to allow to switch between a light/dark interface at Runtime
- Template Libraries
 - contain some pre-defined and "ready to use" Style sheets



Default style sheet

- Properties
 - Global Style: style properties in common with all objects (like radius)
 - Control Styles: style properties for a specific type of object (Button, Switch...)
- Multiple style sheets
 - into the same project can be defined multiple style sheets for example, to allow to switch between a light/dark interface at Runtime
- Template Libraries
 - contain some pre-defined and "ready to use" Style sheets



Native presentation engine

- Properties

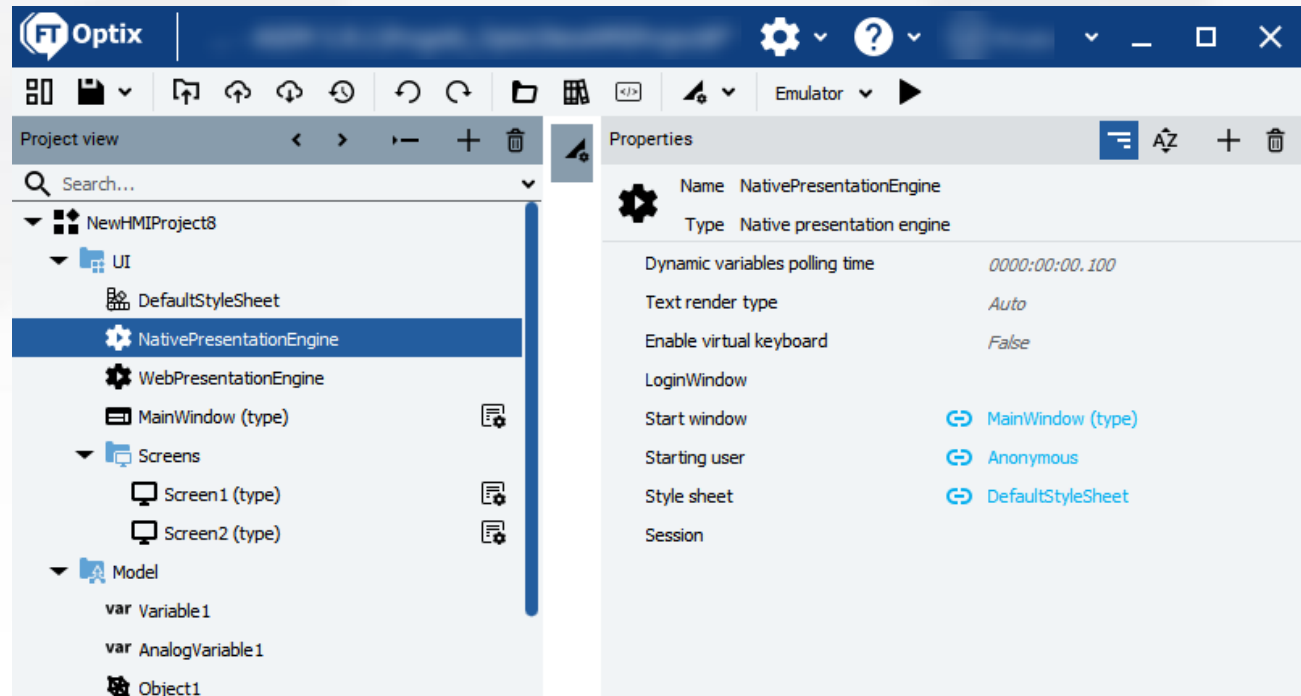
- Dynamic Variables polling time: used to refresh Tags used in Presentation Engine.
NOTE: this will also set the polling time used by comm drivers (for active tags)

- Start Window

- Starting User

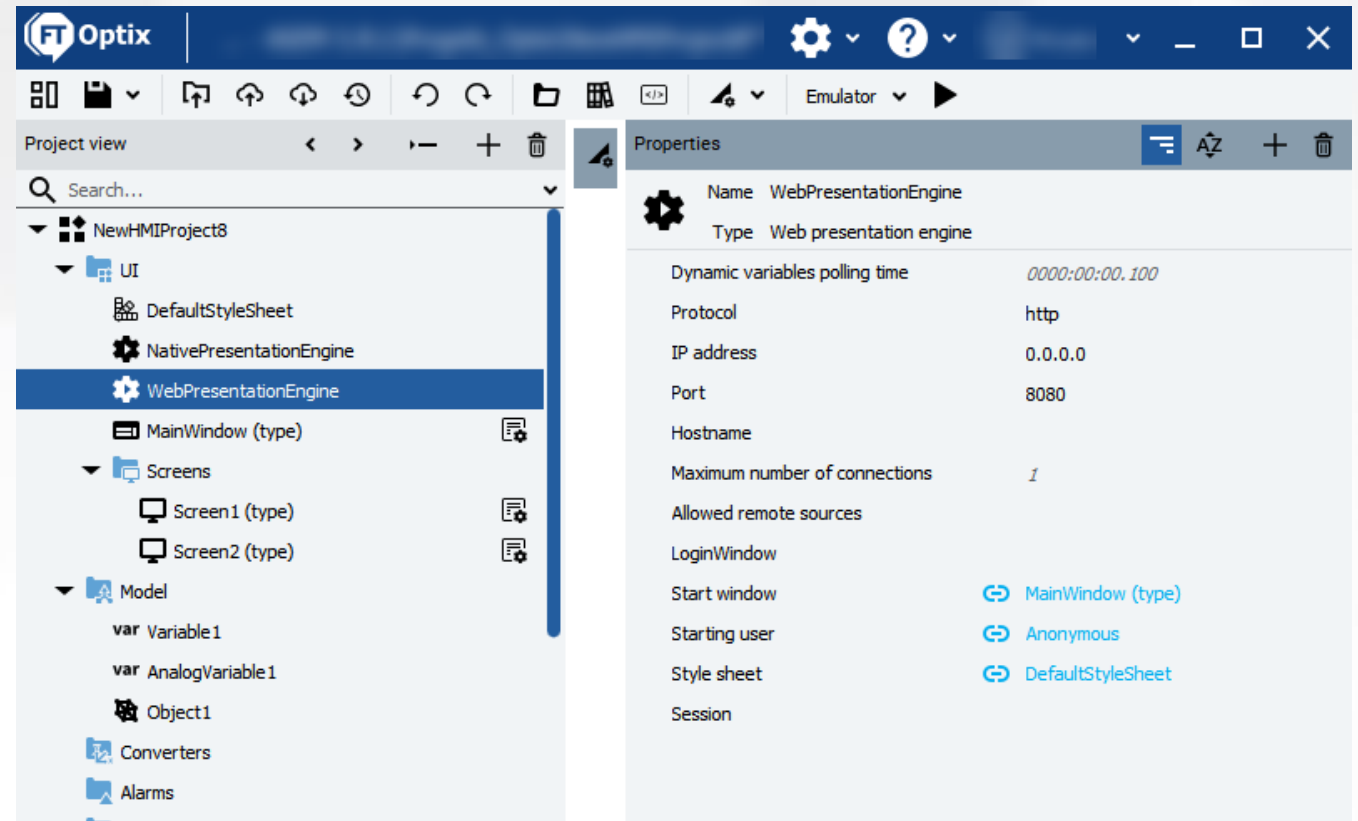
- Style Sheet

- Enable Virtual Keyboard



Web presentation engine

- Allow publishing the UI through the integrated Web Server
- Properties:
 - Start Window, Starting User, Style Sheet...
 - **Protocol:** http/https
 - **IP Address:** this will be the IP address used by Web Server
0.0.0.0 = listen on all addresses
 - **Port:** define the listening port
 - Hostname
 - Maximum Number of connections
 - Allowed remote sources: external sources that can be embedded (websites)

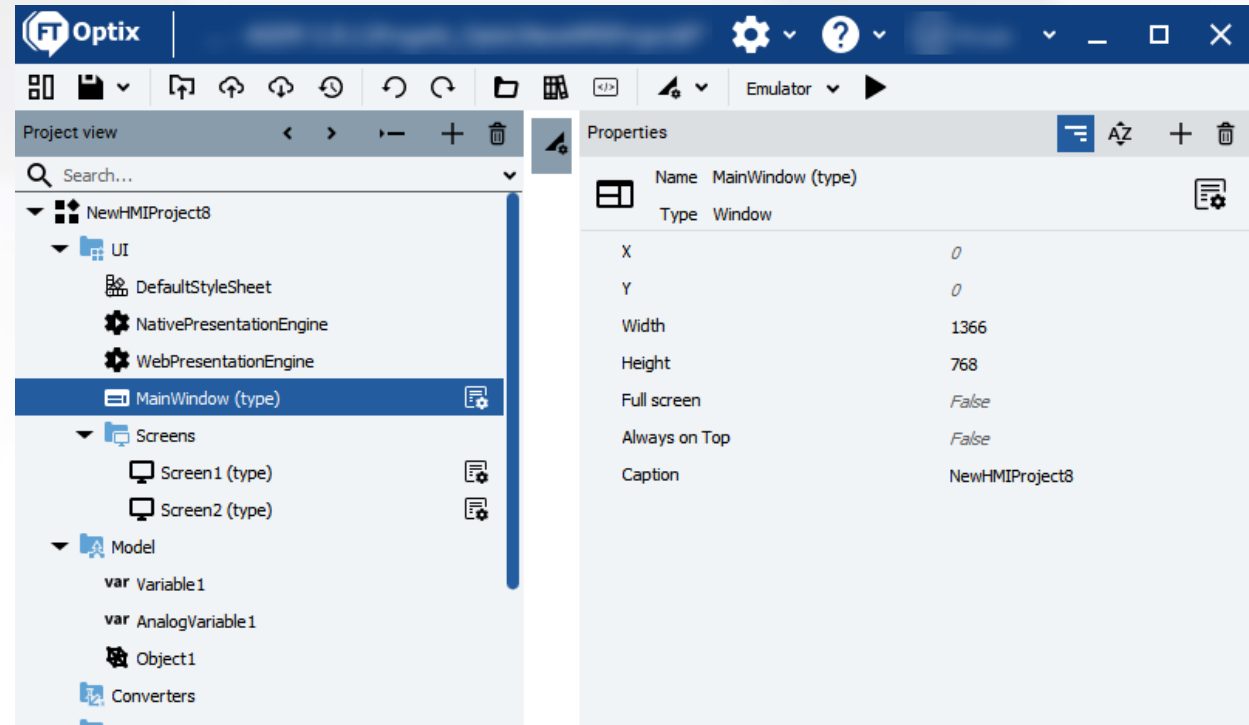


Main window

- Define the window where all graphical objects are showed
 - width/height: defines (manually) the size of the entire User Interface
 - Full Screen: set the size of UI equal to the current screen resolution hiding the title bar
 - Always on Top: force the window to be always on top of other applications
 - Caption: title of the window

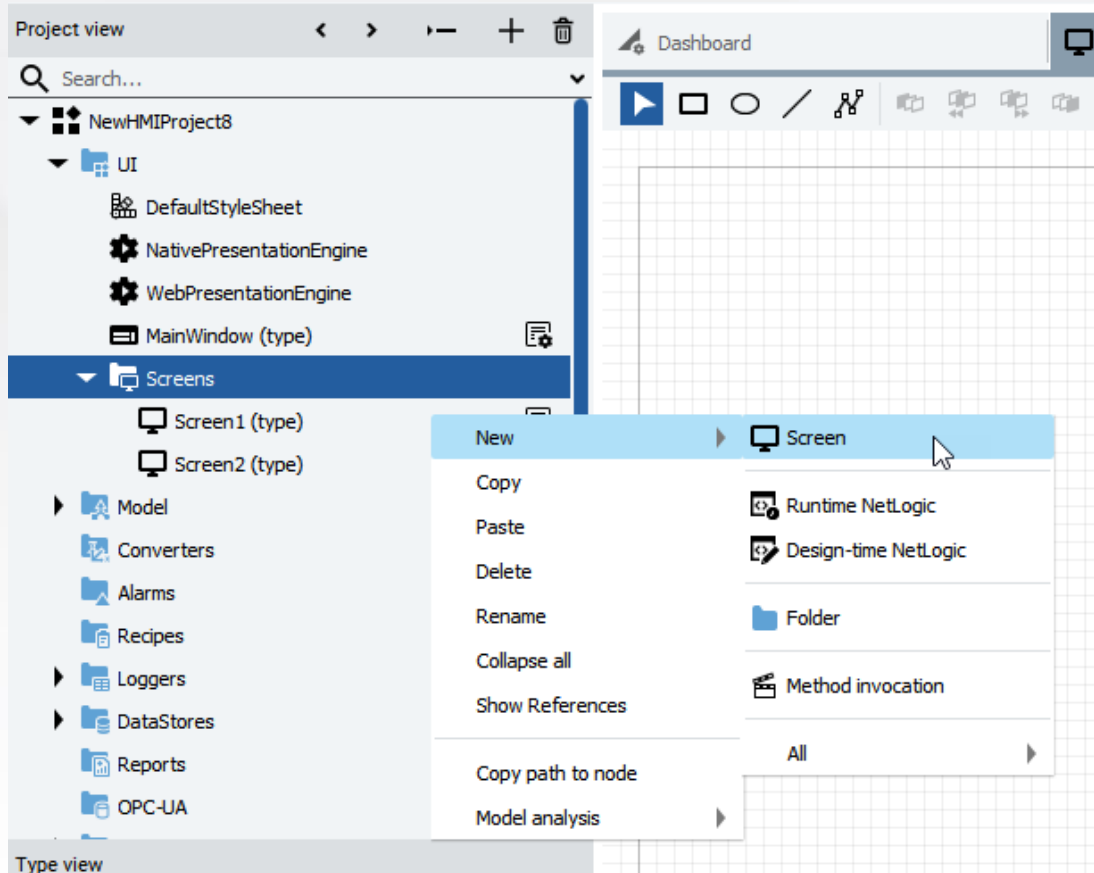
- NOTE:

- **Application is a single-window application**
- Switching between a page to another will not close/open a window, but will only change its content (containers)



Screens

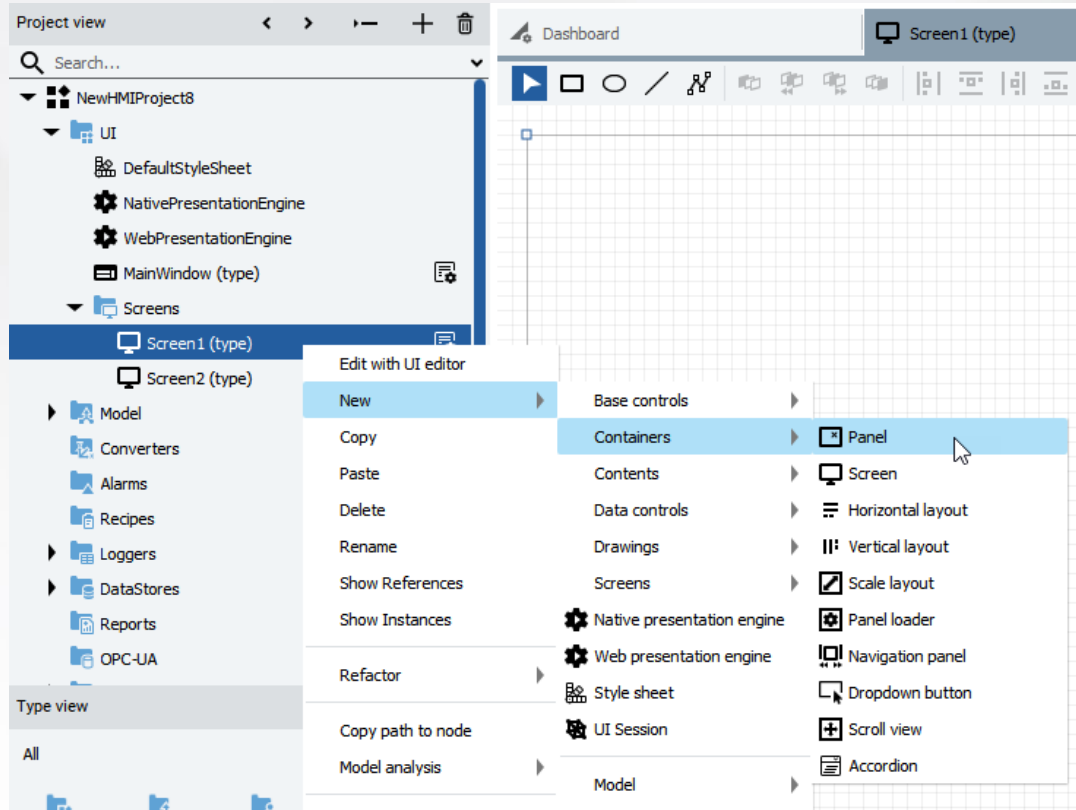
- Screens are the pages that are loaded to the running applications



- Screens sizes are stretched to the parent container (PanelLoader or NavigationPanel)
- Background can be customized with StyleSheet

Panels

- Panel is a Container that aggregates graphical objects (i.e. widgets)



- create a new Panel:
 - Right-click on the Page
 - select New > Containers > Panel
- Panels are transparent only
 - Can be used to group elements to trigger specific behaviors on all child components (like visibility, enablement, etc)

Navigation

- Navigation can take place using two different ways

1. Panel Loader

- Container to display panels or screens
- "Change panel" method allow to load a different panel

2. Navigation Panel

- Container that automatically organizes Panels into navigable tabs.
- Just Drag&Drop Screens into the object
- Tabs can be hidden or disabled

The screenshot displays the FactoryTalk software interface. The main workspace shows a 'MainWindow' with a toolbar at the top. Below the toolbar, there are three tabs labeled 'Panel1', 'Panel2', and 'Panel3'. The 'Panel1' tab is currently selected. To the right of the main workspace is a 'Properties' panel. The 'Properties' panel shows the following information:

- Name:** PanelLoader1
- Type:** Panel loader
- Panel loader section:**
 - Panel:** Panel1 (type) (with a link icon)
 - Panel alias node:**
 - Enter animation:** None
 - Exit animation:** None
- Appearance section:**
 - Visible:** True
 - Enabled:** True
 - Opacity:** 100
 - Rotation:** 0
- Size and layout section:** (indicated by a plus sign)

Navigation

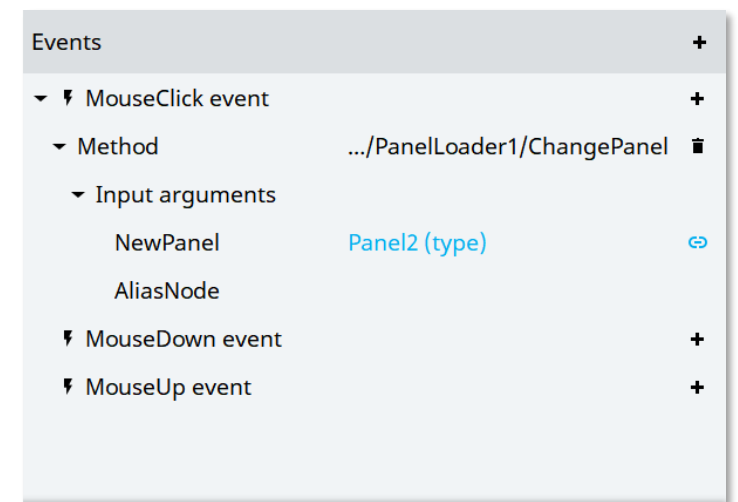
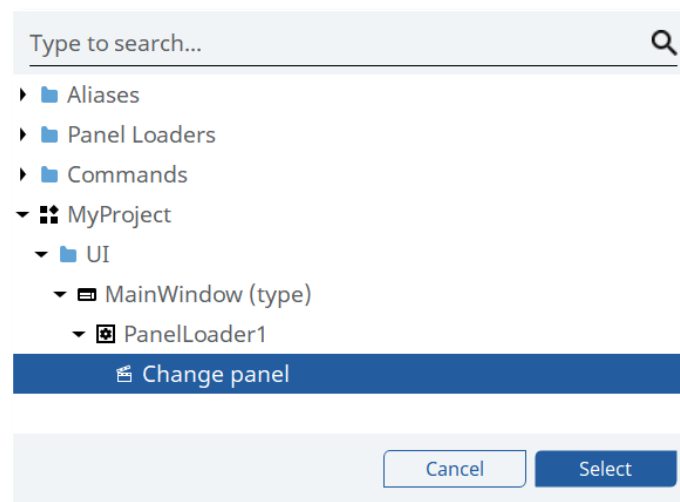
- Navigation can take place using two different ways

1. Panel Loader

- Container to display panels or screens
- "Change panel" method allow to load a different panel

2. Navigation Panel

- Container that automatically organizes Panels into navigable tabs.
- Just Drag&Drop Screens into the object
- Tabs can be hidden or disabled



Navigation

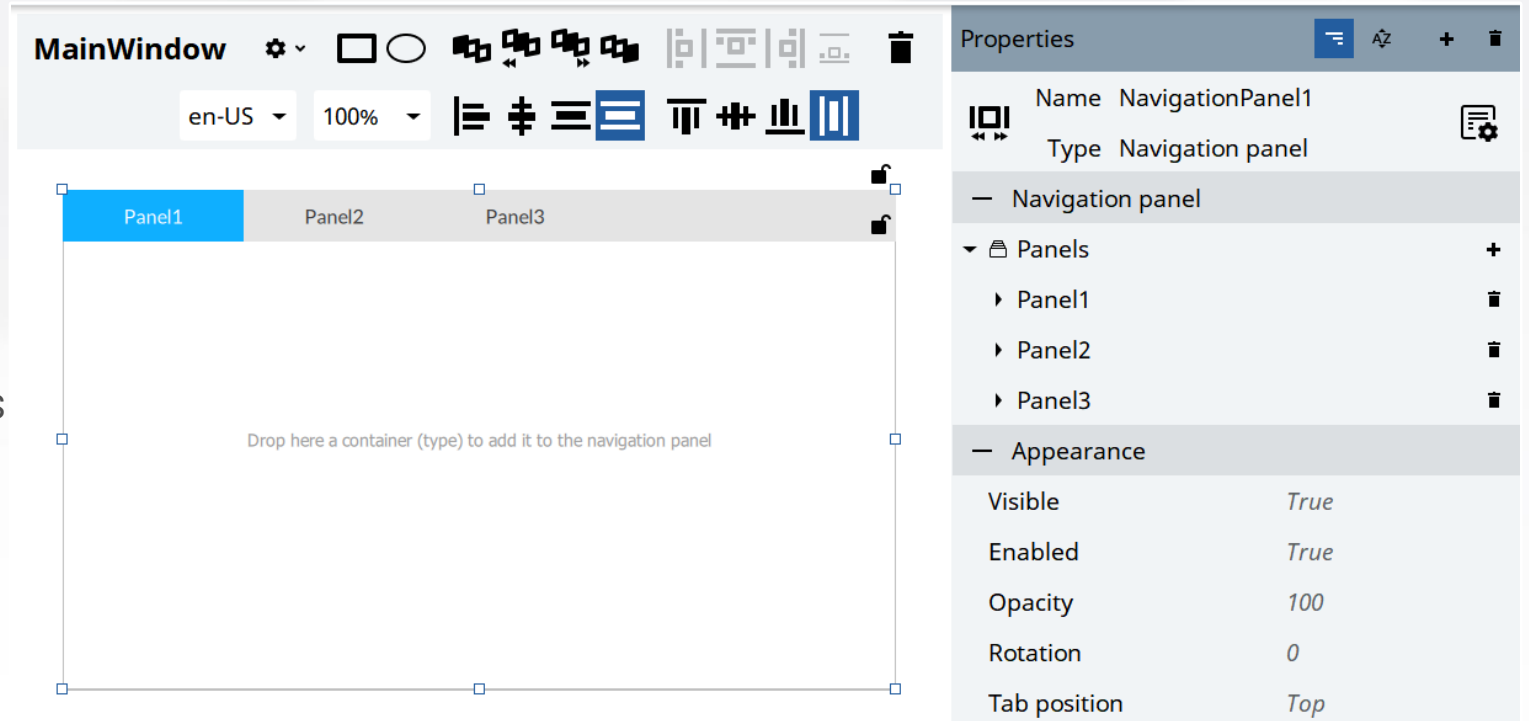
- Navigation can take place using two different ways

1. Panel Loader

- Container to display panels or screens
- "Change panel" method allow to load a different panel

2. Navigation Panel

- Container that automatically organizes Panels into navigable tabs.
- Just Drag&Drop Screens into the object
- Tabs can be hidden or disabled



Responsive design

- This can be achieved by using the **relative positioning** of objects or **scaling** them to be bigger or smaller

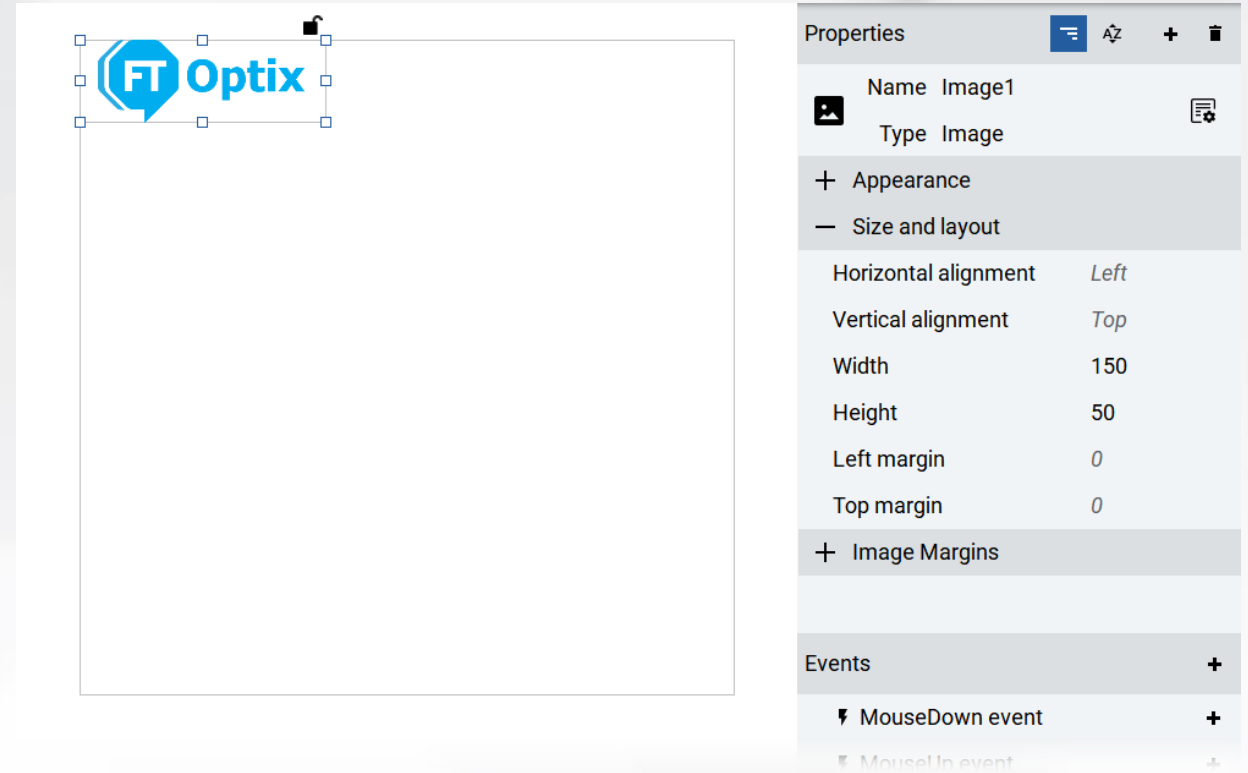
1. **Alignment** *: manage the reference

- Left/Right and Top/Bottom
- Center
- Stretch

2. **Margins**: manage the distance from the reference

- Horizontal Layout and Vertical Layout can be used for structured layouts

* Remember: to set Alignment on objects and also on all its containers: Panel, Navigation Panel...



Responsive design

- This can be achieved by using the **relative positioning** of objects or **scaling** them to be bigger or smaller

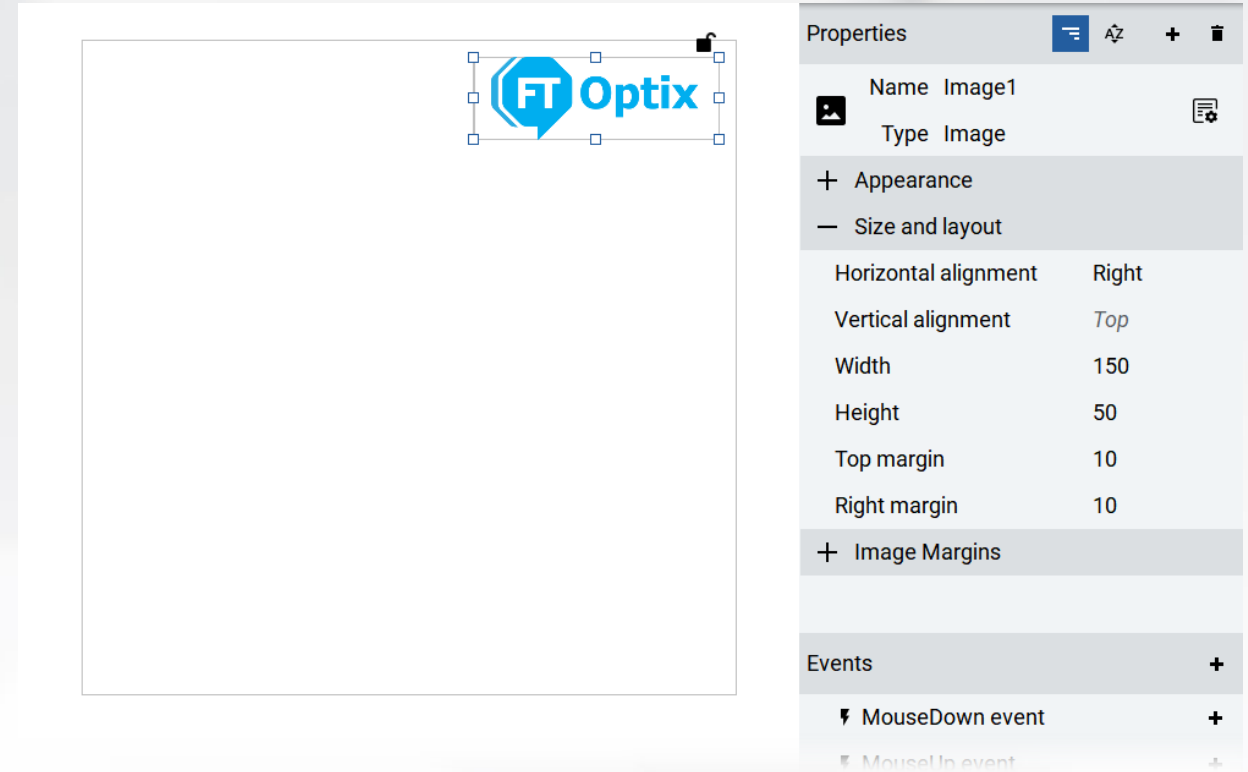
1. Alignment*: manage the reference

- Left/Right and Top/Bottom
- Center
- Stretch

2. Margins: manage the distance from the reference

- Horizontal Layout and Vertical Layout can be used for structured layouts

* Remember: to set Alignment on objects and also on all its containers: Panel, Navigation Panel...



Responsive design

- This can be achieved by using the **relative positioning** of objects or **scaling** them to be bigger or smaller

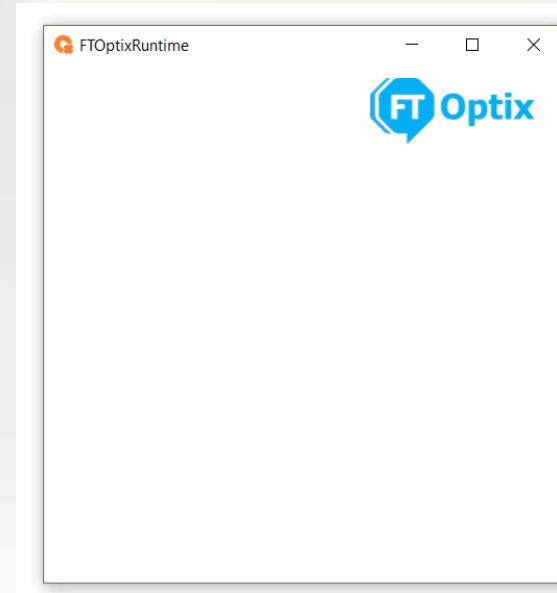
1. **Alignment** *: manage the reference

- Left/Right and Top/Bottom
- Center
- Stretch

2. **Margins**: manage the distance from the reference

- Horizontal Layout and Vertical Layout can be used for structured layouts

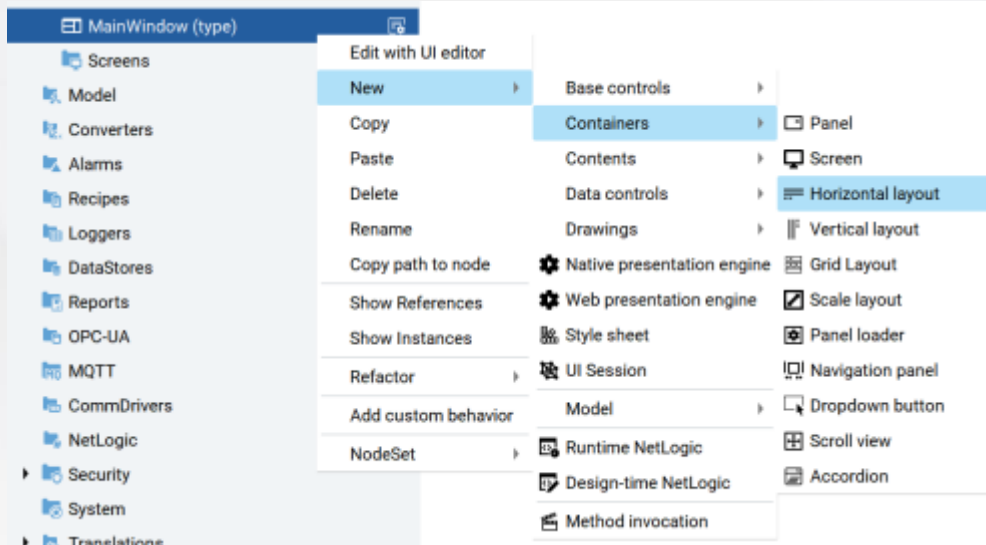
* Remember: to set Alignment on objects and also on all its containers: Panel, Navigation Panel...



MouseUp event

Horizontal Layout

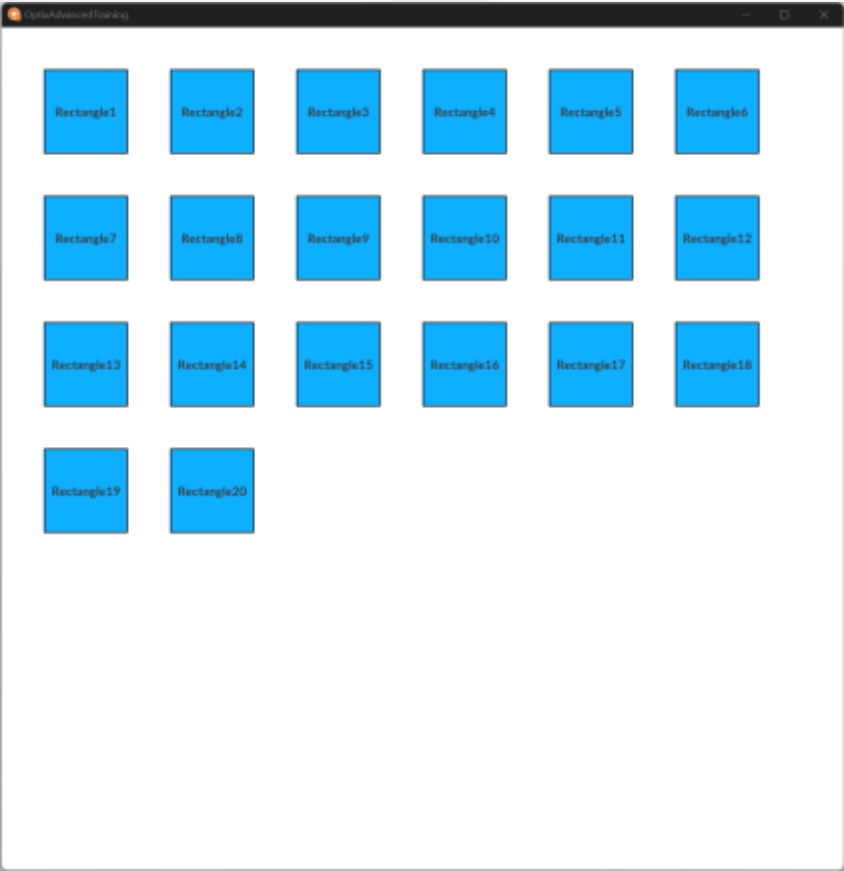
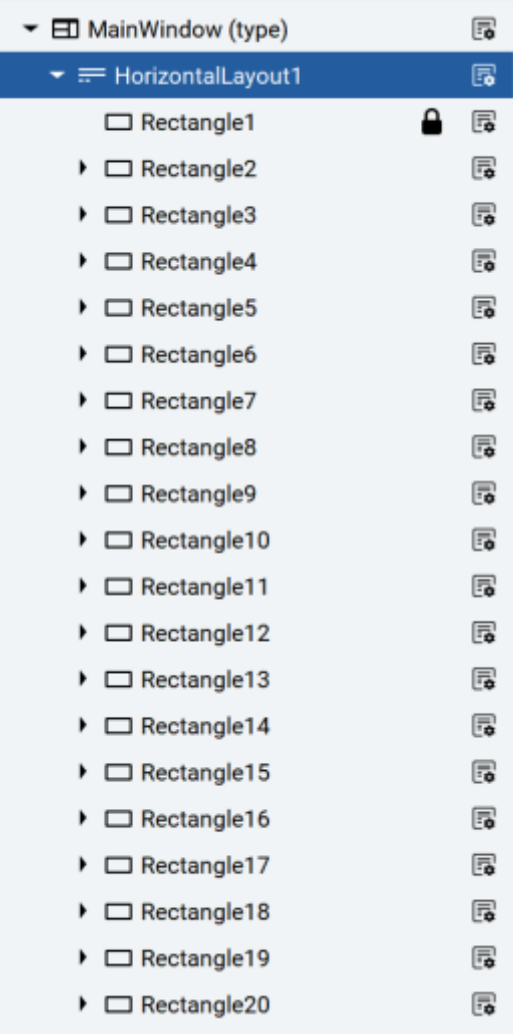
- Automatically lay out content horizontally inside a container



Content alignment	<i>Left aligned</i>
Wrap	<i>False</i>
Horizontal gap	<i>0</i>
Vertical gap	<i>0</i>

- Content alignment: Where to place content inside the container (Left aligned, Center aligned, Right aligned)
- Wrap: Start a new row when content exceeds the horizontal bounds of the container
- Horizontal gap: How much horizontal space between content
- Vertical gap: How much vertical space between content

Horizontal Layout



Horizontal Layout

Content alignment	<i>Left aligned</i>
Wrap	True
Horizontal gap	50
Vertical gap	50

Appearance

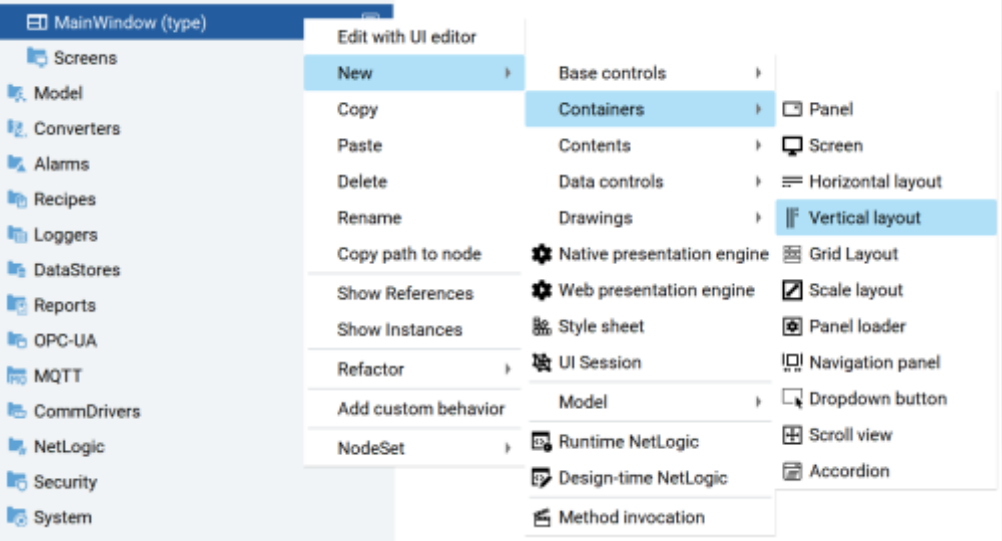
Visible	<i>True</i>
Enabled	<i>True</i>
Opacity	<i>100</i>
Rotation	<i>0</i>
Move target	<i>None</i>

Size and layout

Horizontal alignment	Stretch
Vertical alignment	Stretch
Left margin	50
Top margin	50
Right margin	50
Bottom margin	50

Vertical Layout

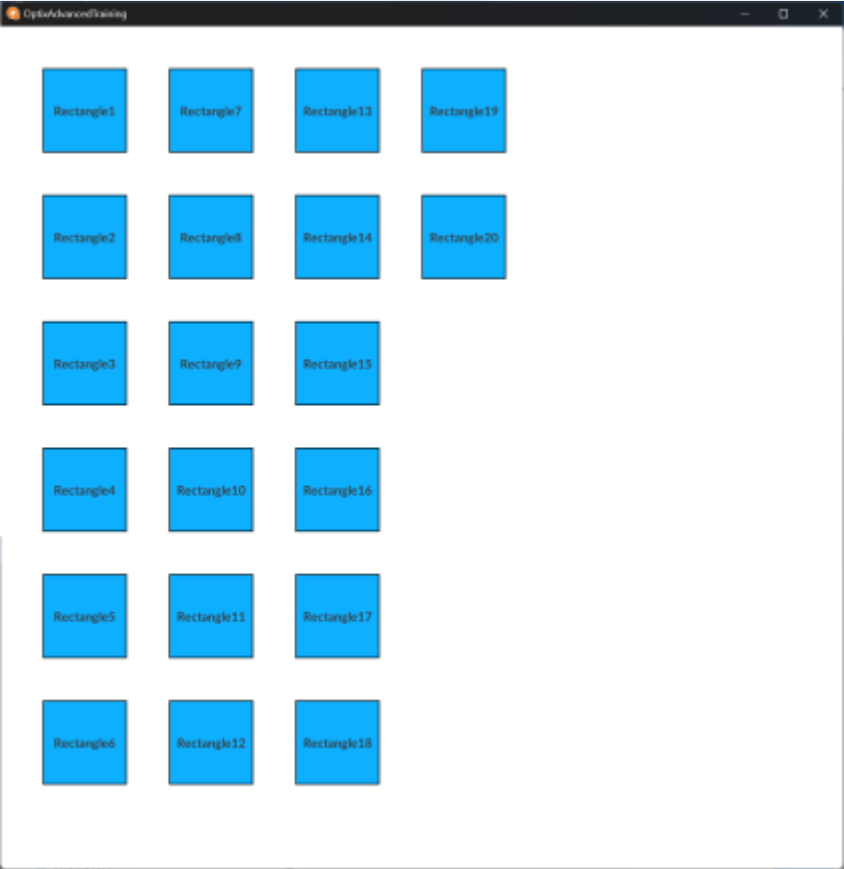
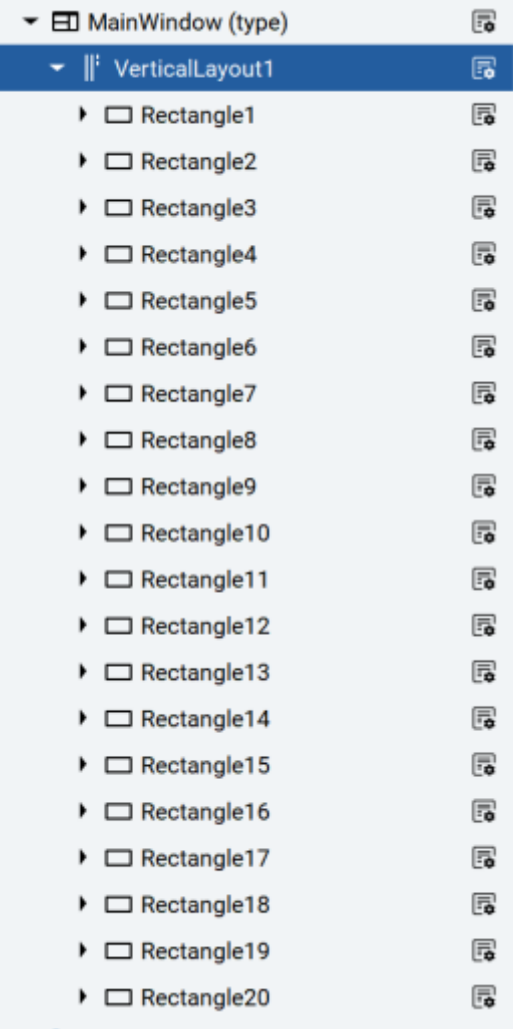
- Automatically lay out content vertically inside a container



Content alignment	<i>Top aligned</i>
Wrap	<i>False</i>
Horizontal gap	<i>0</i>
Vertical gap	<i>0</i>

- Content alignment: Where to place content inside the container (Left aligned, Center aligned, Right aligned)
- Wrap: Start a new column when content exceeds the vertical bounds of the container
- Horizontal gap: How much horizontal space between content
- Vertical gap: How much vertical space between content

Vertical Layout



Horizontal Layout

Content alignment	Top aligned
Wrap	True
Horizontal gap	50
Vertical gap	50

Appearance

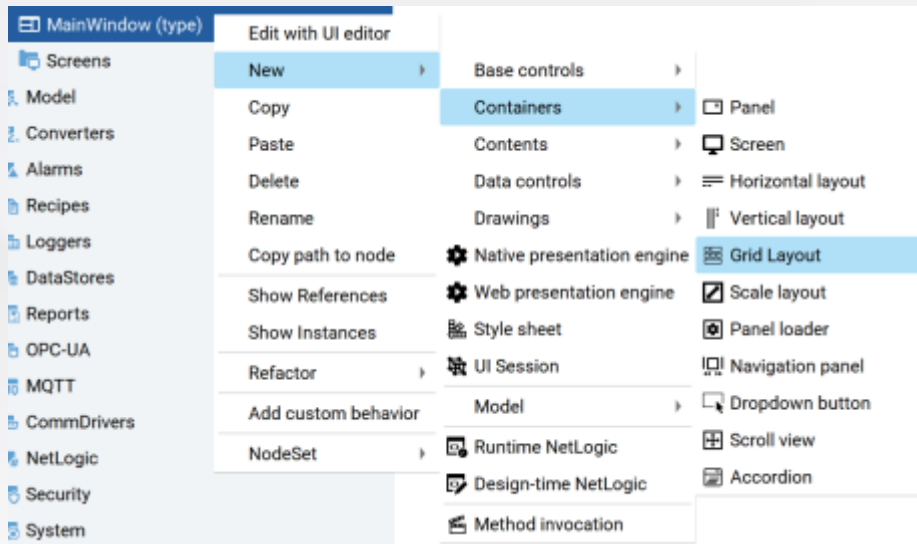
Visible	True
Enabled	True
Opacity	100
Rotation	0
Move target	None

Size and layout

Horizontal alignment	Stretch
Vertical alignment	Stretch
Left margin	50
Top margin	50
Right margin	50
Bottom margin	50

Grid Layout

- Automatically lay out content based on configured grid pattern



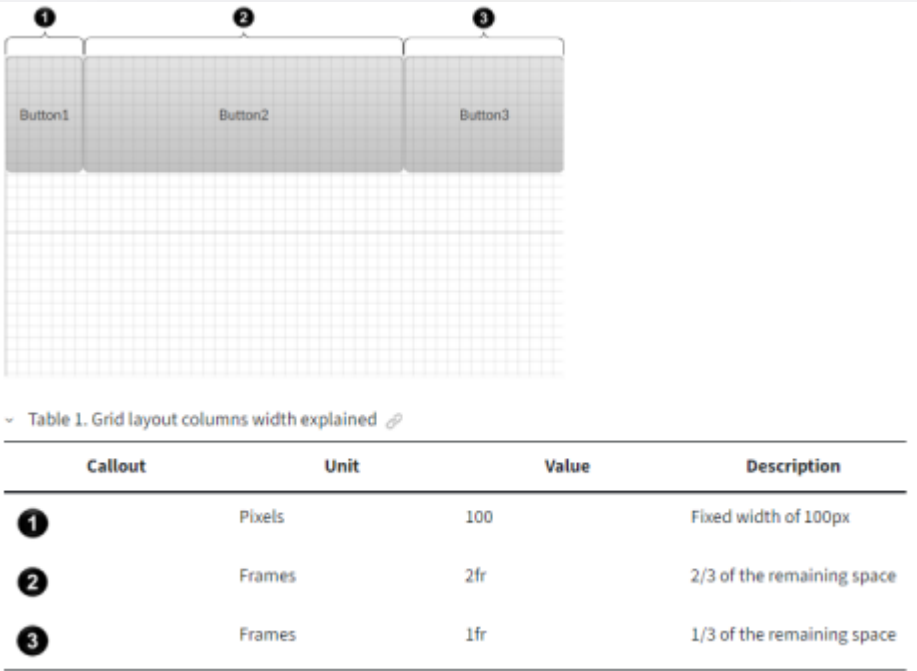
— Grid Layout

Columns	<i>1fr;2fr;3fr</i>
Rows	<i>1fr;2fr;3fr</i>
Horizontal gap	<i>0</i>
Vertical gap	<i>0</i>
Items Horizontal Alignment	<i>Stretch</i>
Items Vertical Alignment	<i>Stretch</i>

- Columns: Specify column width in units of static pixels or relative frame units (1fr = 1 unit, 2fr = 2 frame units)
- Rows: Specify column width in units of static pixels or relative frame units (1fr = 1 unit, 2fr = 2 frame units)
- Horizontal gap: Space between columns
- Vertical gap: Space between rows
- Items Horizontal Alignment: Column content alignment
- Items Vertical Alignment: Row content alignment

Grid Layout

- Grid Pattern Configuration

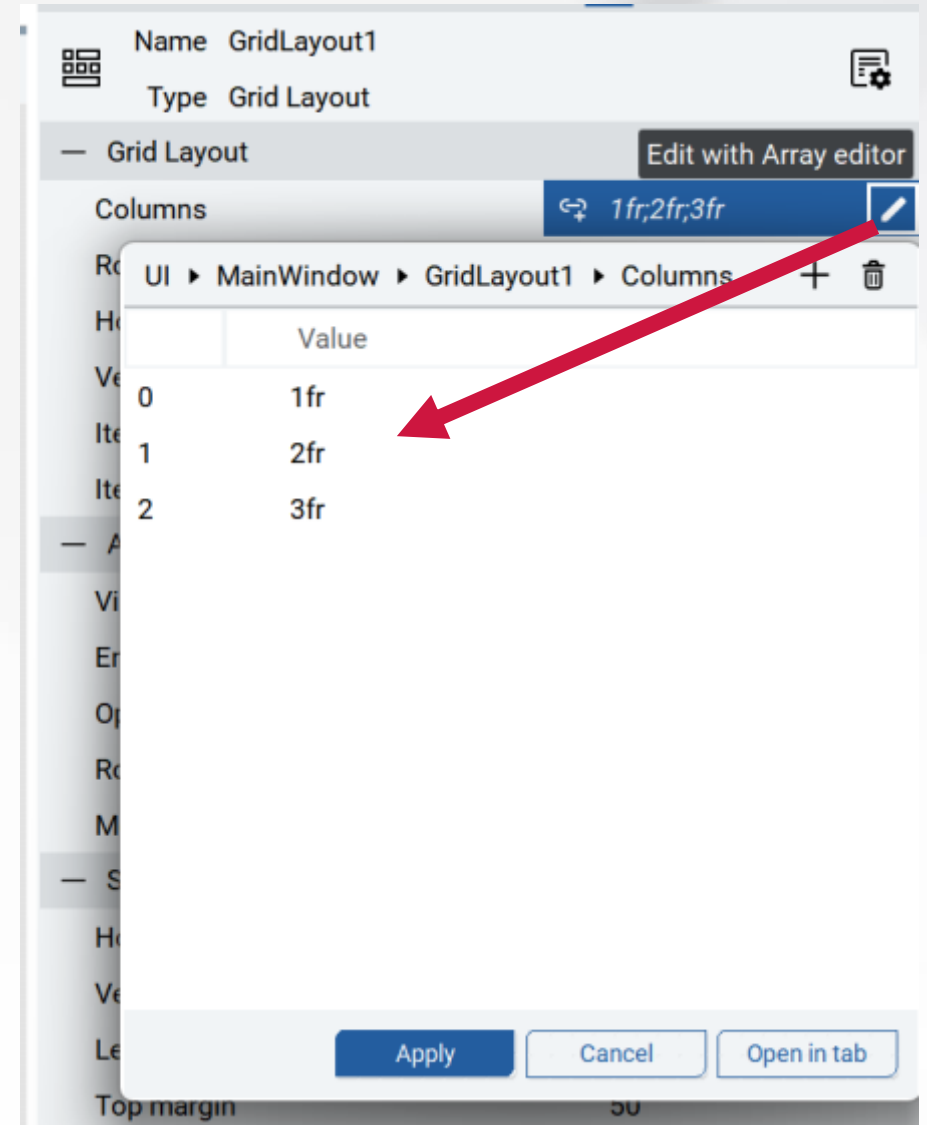


- Columns and Rows are arrays of values dictating content behaviour
- Column and Row array values can be static pixels or dynamic at runtime using xfr notation
- Column and Row array values are separated by semicolon
- Example:
 - 3x3 grid
 - Column 1 = 1/6 of Grid Layout Width
 - Column 2 = 1/3 of Grid Layout Width
 - Column 3 = 1/2 of Grid Layout Width
 - Row 1 = 1/6 of Grid Layout Height
 - Row 2 = 1/3 of Grid Layout Height
 - Row 3 = 1/2 of Grid Layout Height

— Grid Layout	
Columns	1fr;2fr;3fr
Rows	1fr;2fr;3fr

Grid Layout

- Grid Pattern Configuration
 - Add / Delete Columns and Rows using the Edit (Pencil) button



Grid Layout

▼ MainWindow (type)

▼ GridLayout1

▶ Rectangle1

▶ Rectangle2

▶ Rectangle3

▶ Rectangle4

▶ Rectangle5

▶ Rectangle6

▶ Rectangle7

▶ Rectangle8

▶ Rectangle9

Name GridLayout1

Type Grid Layout

— Grid Layout

Columns1fr;2fr;3fr

Rows1fr;2fr;3fr

Horizontal gap50

Vertical gap50

Items Horizontal AlignmentStretch

Items Vertical AlignmentStretch

— Appearance

VisibleTrue

EnabledTrue

Opacity100

Rotation0

Move targetNone

— Size and layout

Horizontal alignmentStretch

Vertical alignmentStretch

Left margin50

Top margin50

Right margin50

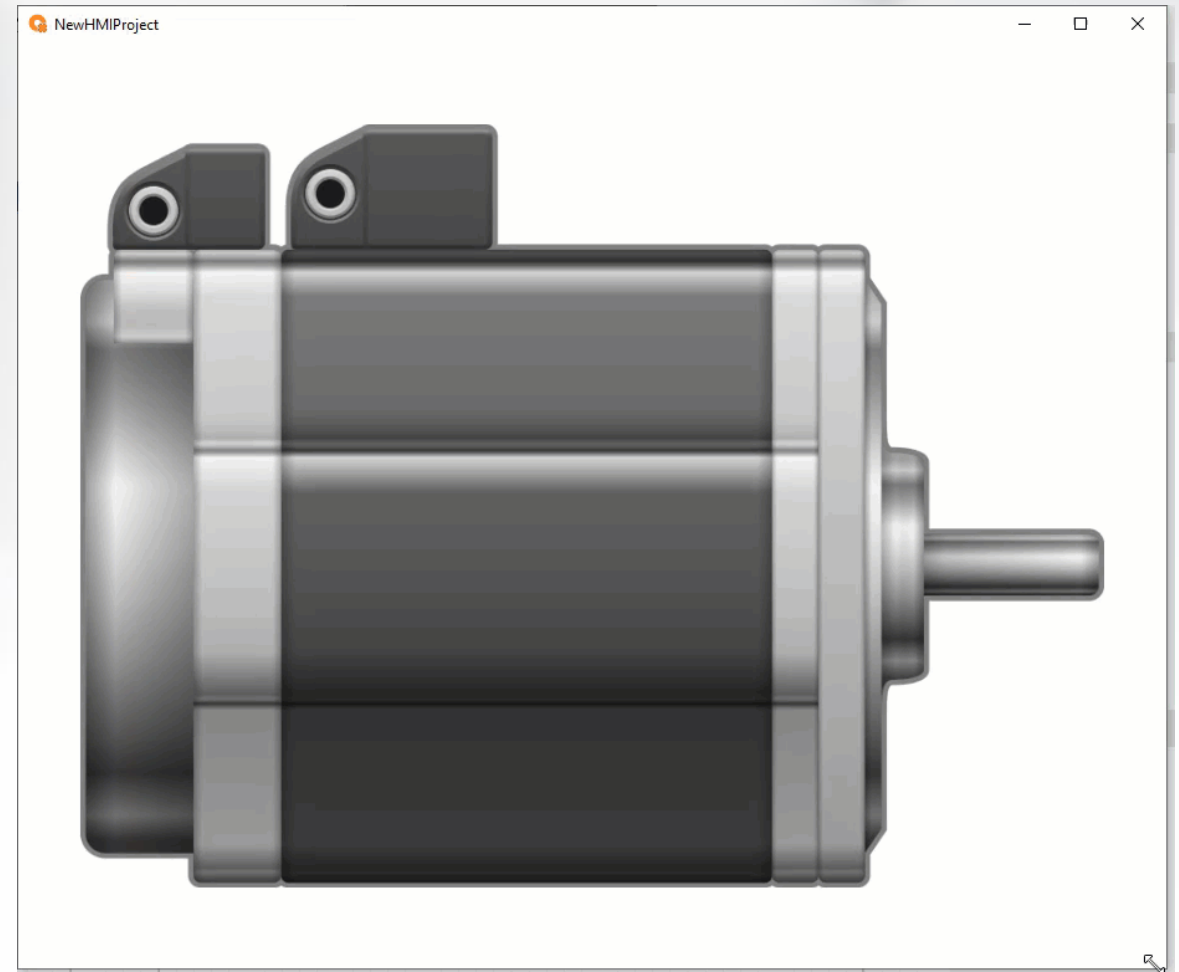
Bottom margin50

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 108

FactoryTalk
by ROCKWELL AUTOMATION

ScaleLayout

- Objects can be adapted to the size of the container using a **ScaleLayout**
- Scaling ratio can be set to:
 - Fit: original ratio is maintained, empty bars can appear around the element if the container's ratio is different from the object's ratio
 - Stretch: object is stretch to the container size, ratio is ignored, graphics may get distorted



Scale Layout

Adjust the Width and Height of the logo

Width

< 100 >

Height

< 100 >

Scale Layout Width & Height

Fill Mode: Fit

[100px]



[100px]

Rectangle Width & Height

Fill Mode: Stretch

[100px]



[100px]

Adjust the Width and Height of the logo

Width

< 250 >

Height

< 250 >

Scale Layout Width & Height

Fill Mode: Fit

[250px]



[250px]

Rectangle Width & Height

Fill Mode: Stretch

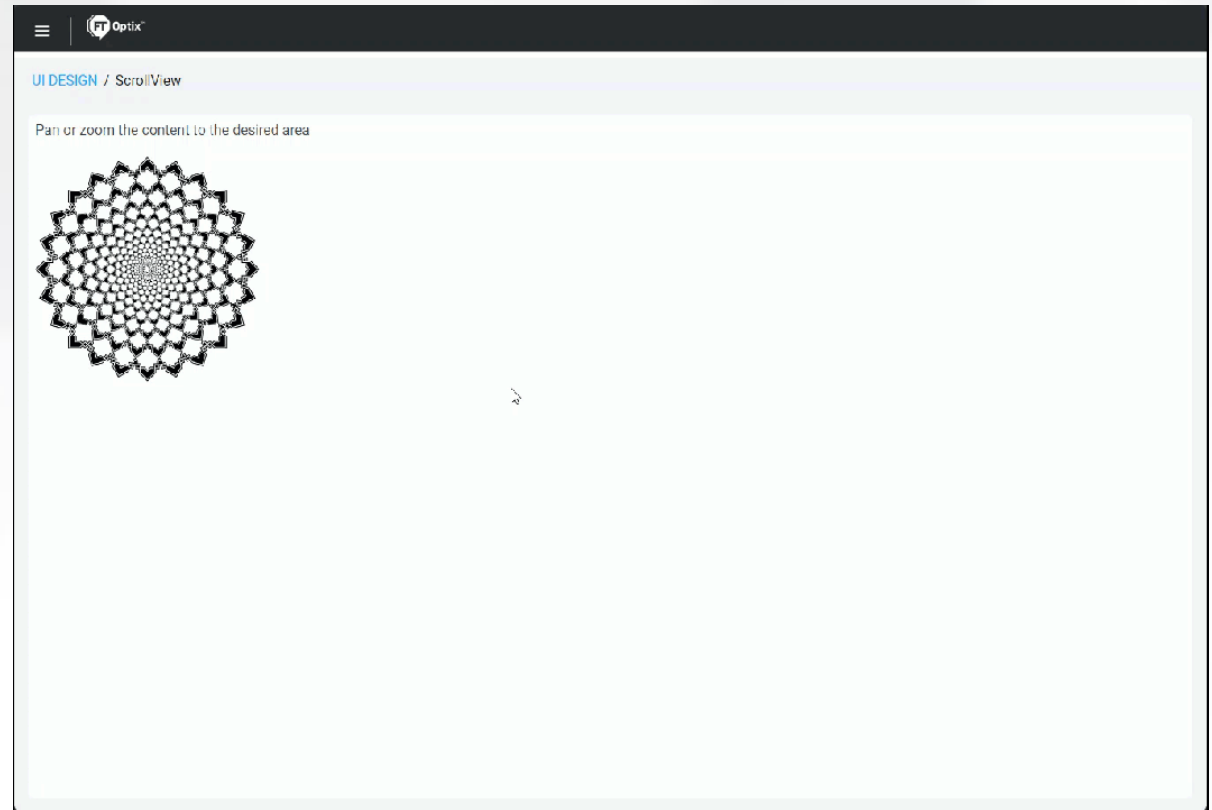
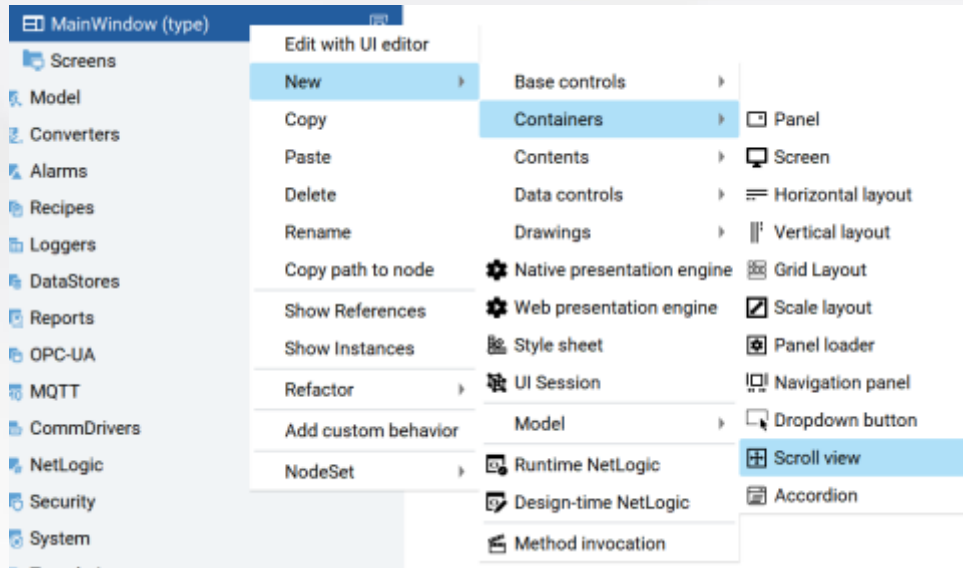
[250px]



[250px]

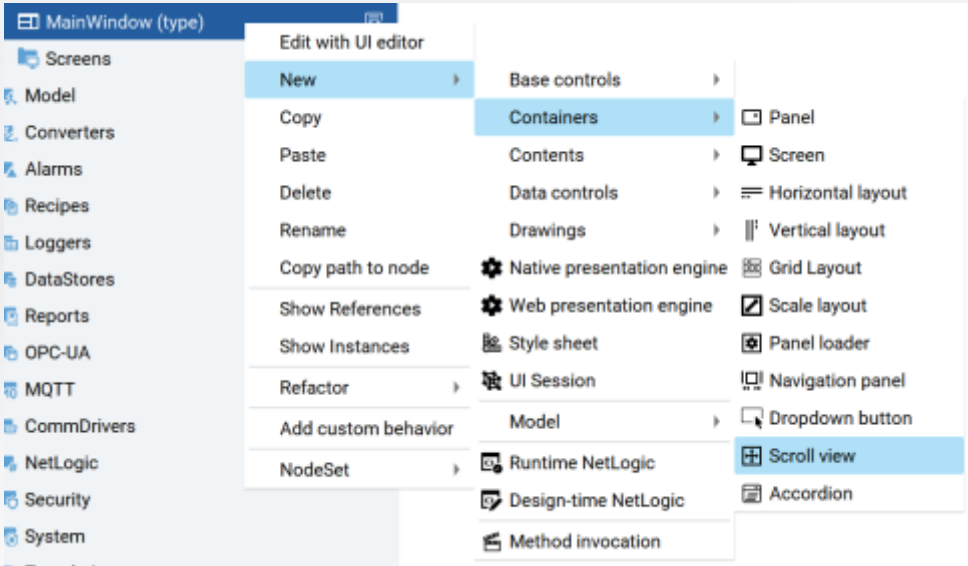
Scroll View

- Pan and Zoom on content
 - Zoom supported with Ctrl+Mouse Scroll or Multi-touch pinch



Scroll View

- Pan and Zoom on content



— ScrollView	
Enables zoom	<i>False</i>
Maximum zoom in	<i>100</i>
Minimum zoom out	<i>100</i>
Scroll axes enabled	<i>Both</i>

- Enables zoom: Turn on / off zoom capabilities
- Maximum / Minimum zoom in / out: Set zoom boundaries as a percentage
- Scroll axes enabled: Control scroll directions available (X Axis, Y Axis, Both)

Scroll View + Horizontal Layout (Wrapped)

▼	MainWindow (type)	
▼	ScrollView1	
▼	HorizontalLayout1	
▶	Rectangle1	
▶	Rectangle2	
▶	Rectangle3	
▶	Rectangle4	
▶	Rectangle5	
▶	Rectangle6	
▶	Rectangle7	
▶	Rectangle8	
▶	Rectangle9	
▶	Rectangle10	
▶	Rectangle11	
▶	Rectangle12	
▶	Rectangle13	
▶	Rectangle14	
▶	Rectangle15	
▶	Rectangle16	
▶	Rectangle17	
▶	Rectangle18	
▶	Rectangle19	
▶	Rectangle20	

+	Name	ScrollView1	
	Type	Scroll view	
—	ScrollView		
	Enables zoom	False	
	Maximum zoom in	100	
	Minimum zoom out	100	
	Scroll axes enabled	Both	
—	Appearance		
	Visible	True	
	Enabled	True	
	Opacity	100	
	Rotation	0	
	ScrollBar style	Default	
—	Size and layout		
	Horizontal alignment	Stretch	
	Vertical alignment	Stretch	
	Left margin	0	
	Top margin	0	
	Right margin	0	
	Bottom margin	0	

≡	Name	HorizontalLayout1	
	Type	Horizontal layout	
—	Horizontal Layout		
	Content alignment	Left aligned	
	Wrap	True	
	Horizontal gap	50	
	Vertical gap	50	
—	Appearance		
	Visible	True	
	Enabled	True	
	Opacity	100	
	Rotation	0	
	Move target	None	
—	Size and layout		
	Horizontal alignment	Stretch	
	Vertical alignment	Top	
	Height	Auto	
	Left margin	50	
	Top margin	50	
	Right margin	50	

Scroll View + Horizontal Layout (Wrapped)

▼  MainWindow (type)

▼  ScrollView1

▼  HorizontalLayout1

▶  Rectangle1

▶  Rectangle2

▶  Rectangle3

▶  Rectangle4

▶  Rectangle5

▶  Rectangle6

▶  Rectangle7

▶  Rectangle8

▶  Rectangle9

▶  Rectangle10

▶  Rectangle11

▶  Rectangle12

▶  Rectangle13

▶  Rectangle14

▶  Rectangle15

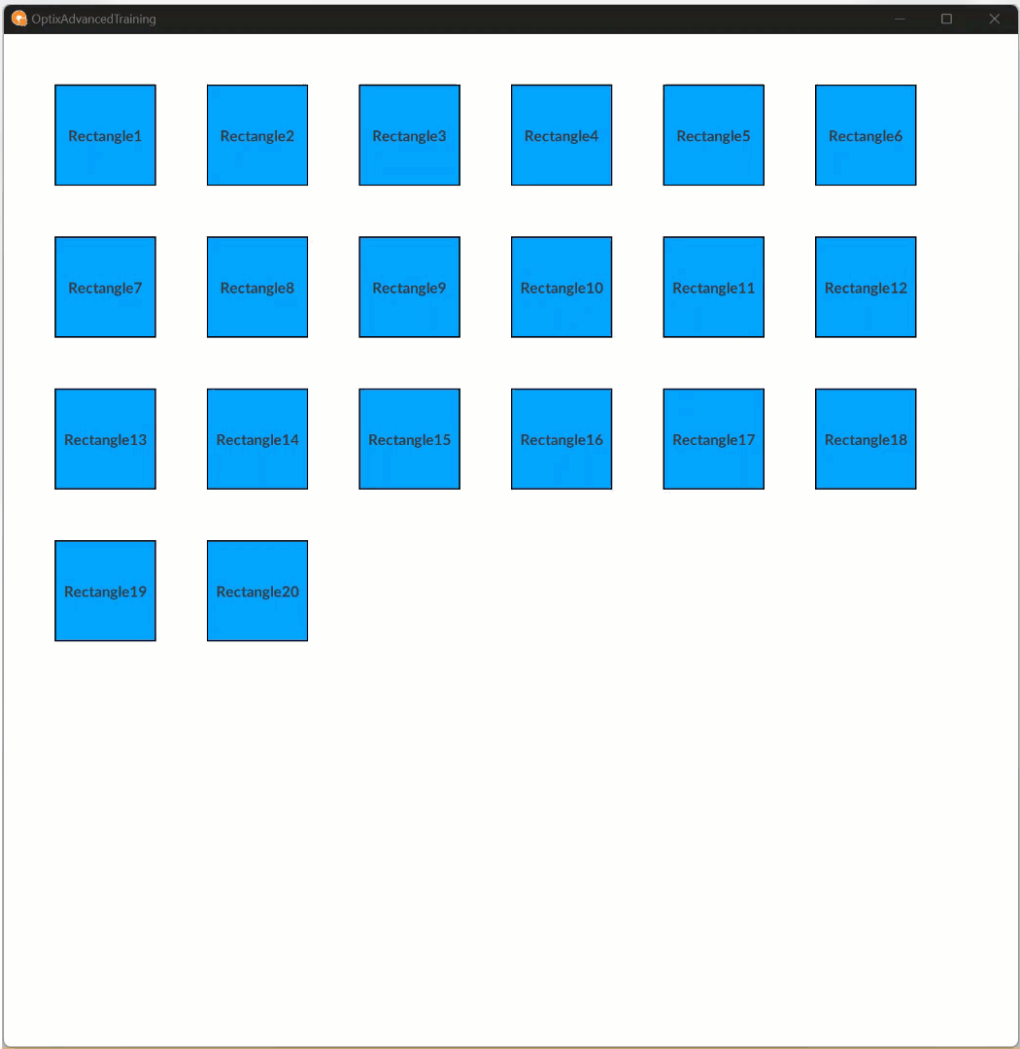
▶  Rectangle16

▶  Rectangle17

▶  Rectangle18

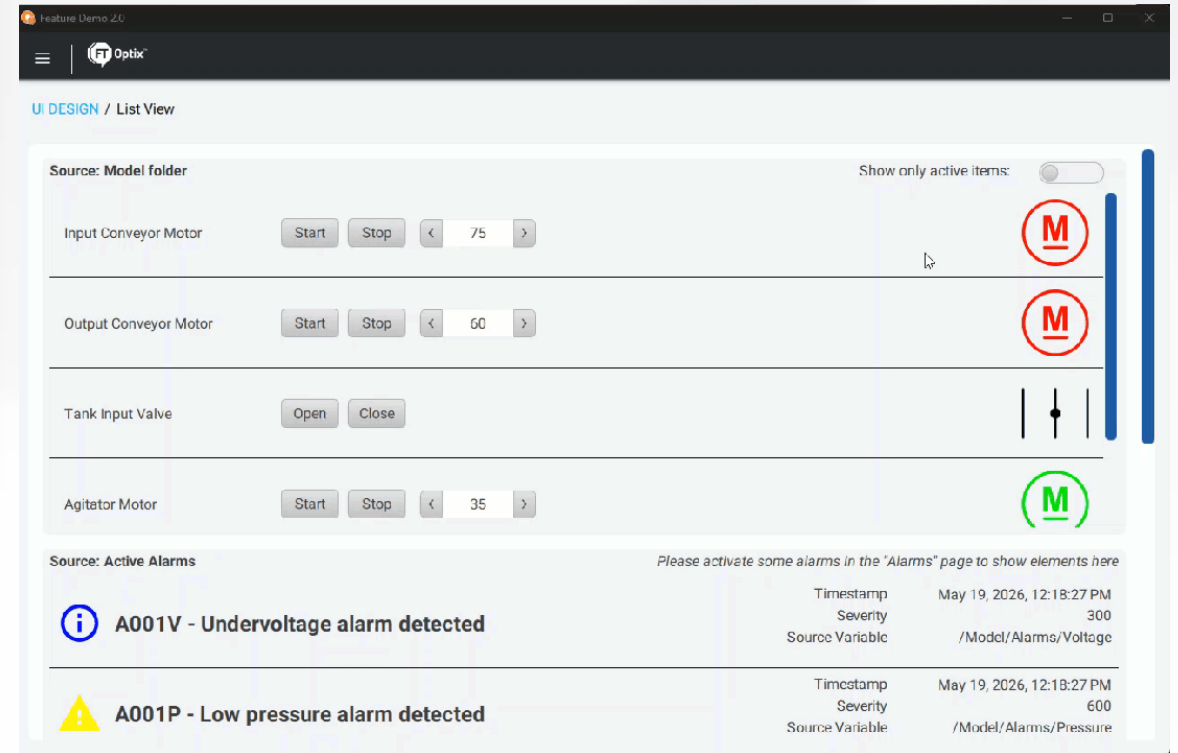
▶  Rectangle19

▶  Rectangle20



ListView

- Special container designed to enumerate content of the element configured in Model
 - Very useful to create custom lists or grids
 - Each object type is represented with a dedicated graphical container (ListViewRow)
- Designed to be extremely memory-efficient
 - Rows are “virtualized” meaning only displayed elements are loaded in RAM
 - When an element scrolls outside the screen, it is disposed
- Can be used with multiple sources
 - Model
 - Database
 - Alarms
- Supports queries



List View

- Configuration:

1. Configure Model source

- Model folder to show items from the project
- Database to return content of tables
- LocalizedAlarms to display live alarms list

2. Configure query (if needed)

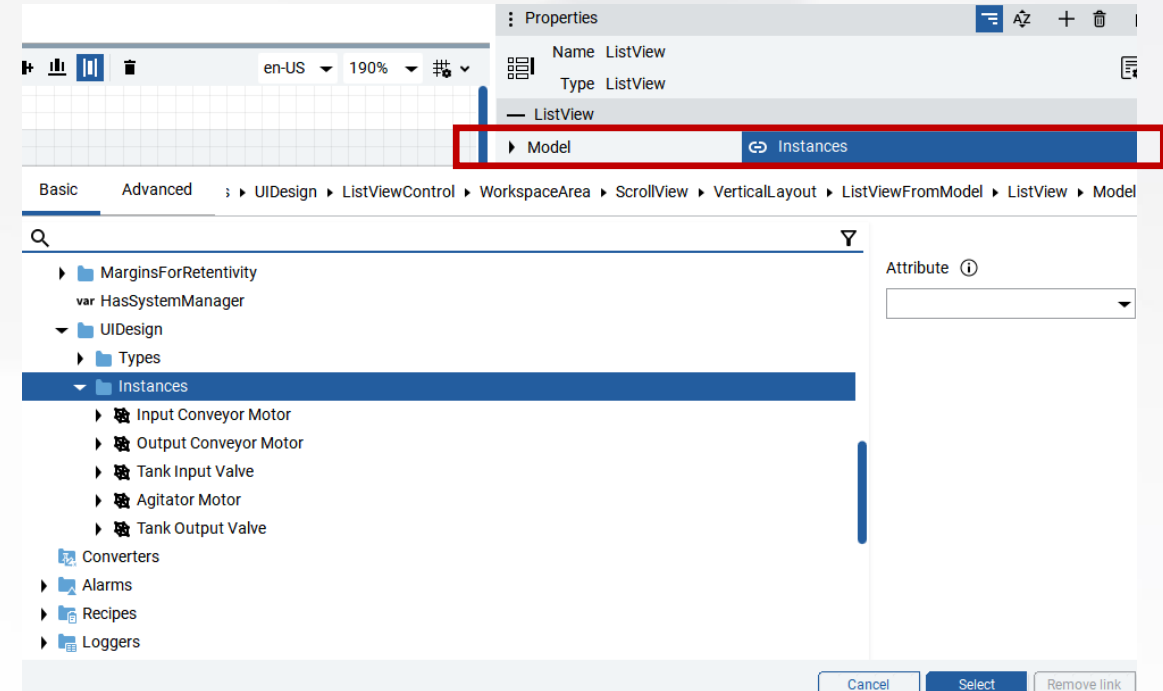
3. Create a ListViewRow per each object type

- Configure Kind property based on the Model content:
 - Object type for Model elements (MyMotorType, ValveUDT, etc)
 - Variable for database rows
 - AlarmCondition for live alarms

4. Add TypeSelectors for each object type

5. Configure a refresh button (if needed)

- Only when retrieving data from a Database



List View

- Configuration:

1. Configure Model source

- Model folder to show items from the project
- Database to return content of tables
- LocalizedAlarms to display live alarms list

2. Configure query (if needed)

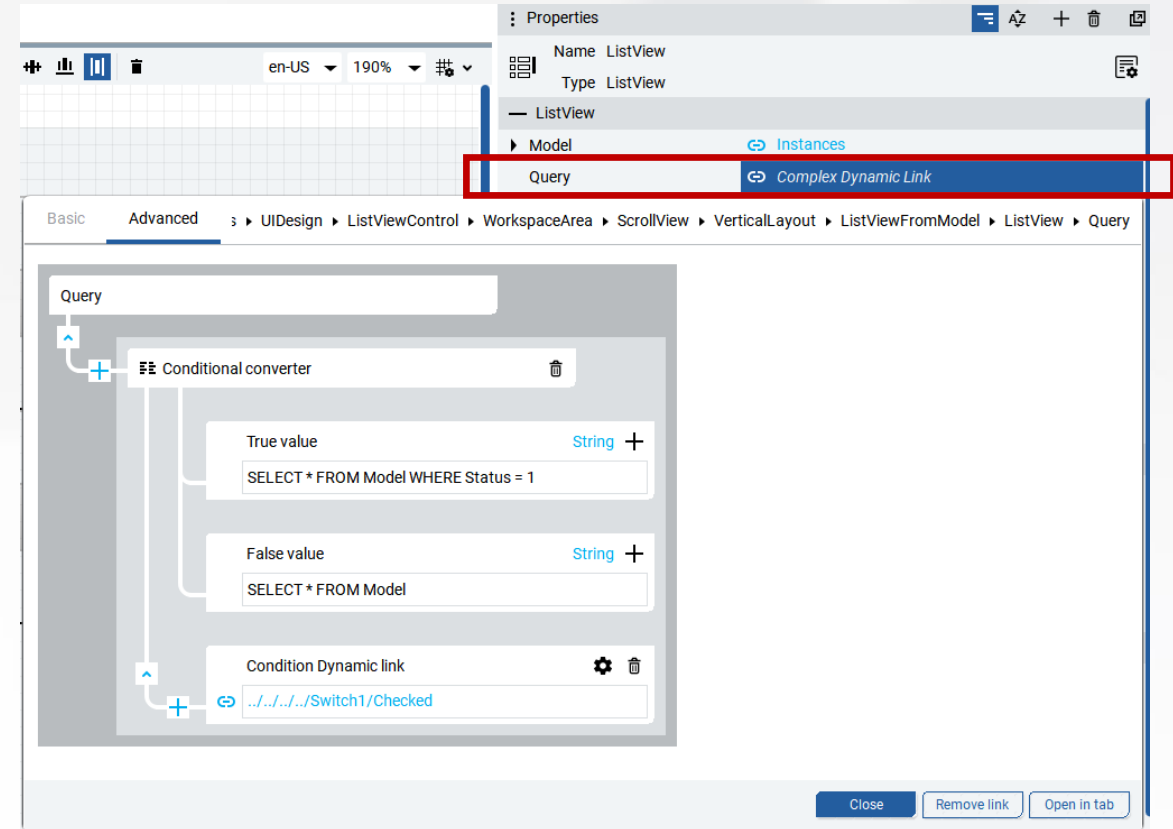
3. Create a ListViewRow per each object type

- Configure Kind property based on the Model content:
 - Object type for Model elements (MyMotorType, ValveUDT, etc)
 - Variable for database rows
 - AlarmCondition for live alarms

4. Add TypeSelectors for each object type

5. Configure a refresh button (if needed)

- Only when retrieving data from a Database



List View

- Configuration:

1. Configure Model source

- Model folder to show items from the project
- Database to return content of tables
- LocalizedAlarms to display live alarms list

2. Configure query (if needed)

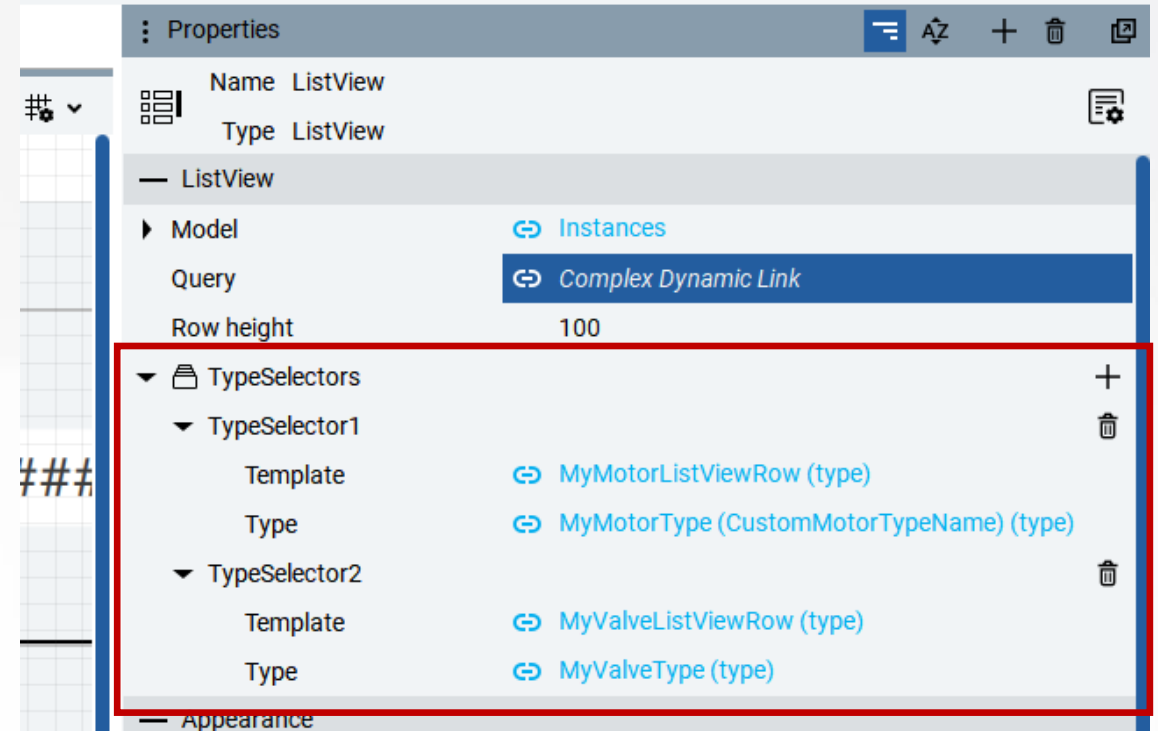
3. Create a ListViewRow per each object type

- Configure Kind property based on the Model content:
 - Object type for Model elements (MyMotorType, ValveUDT, etc)
 - Variable for database rows
 - AlarmCondition for live alarms

4. Add TypeSelectors for each object type

5. Configure a refresh button (if needed)

- Only when retrieving data from a Database



List View

The screenshot displays the FactoryTalk Optix UI Designer interface. The left pane shows the Project view with a tree structure. The right pane shows the List View configuration for 'Source: Model folder'.

Project view (Left Pane):

- Search...
- DateAndTimeFormatEnum
- NumericFormatEnum
- BinaryFormatEnum
- UpperLowerFormatEnum
- ZoomModeEnum
- MarginsForRetentivity
- var HasSystemManager
- UI Design
 - Types
 - Instances**
 - MyMotorType1 (Input Conveyor Motor)
 - MyMotorType2 (Output Conveyor Motor)
 - MyValveType1 (Tank Input Valve)
 - MyMotorType3 (Agitator Motor)
 - MyValveType2 (Tank Output Valve)
- Converters
- Alarms
- Recipes
- Loggers
- DataStores
- Reports

List View Configuration (Right Pane):

- Source: Model folder
- Show only active items: ☐
- Input Conveyor Motor: Start Stop < 80 > (M)
- Output Conveyor Motor: Start Stop < 54 > (M)
- Tank Input Valve: Open Close (Valve symbol)
- Agitator Motor: Start Stop < 35 > (M)

Red arrows indicate the mapping from the 'Instances' folder in the Project view to the List View items:

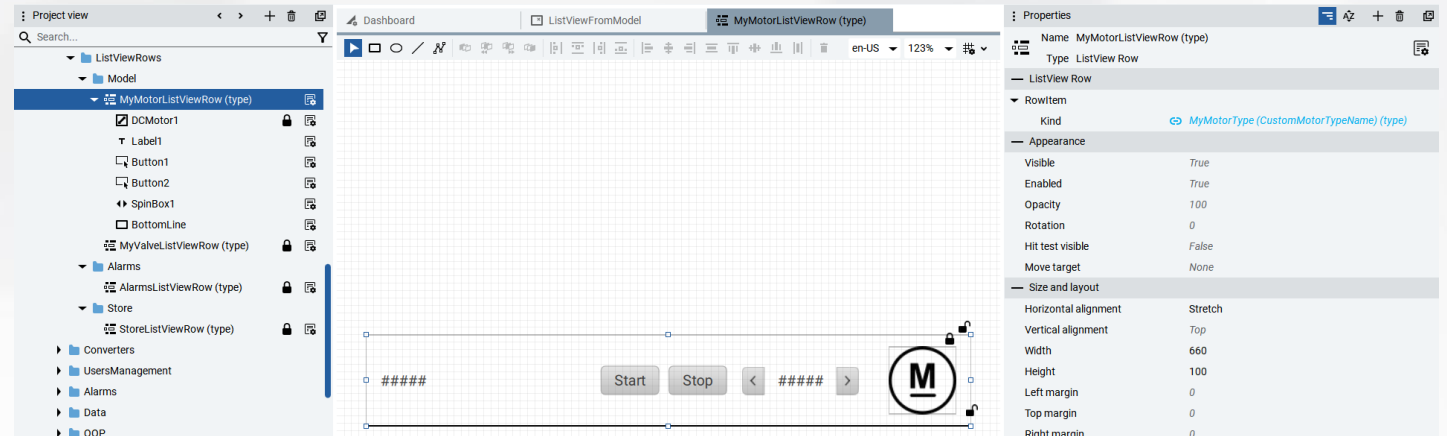
- MyMotorType1 (Input Conveyor Motor) to Input Conveyor Motor
- MyMotorType2 (Output Conveyor Motor) to Output Conveyor Motor
- MyValveType1 (Tank Input Valve) to Tank Input Valve
- MyMotorType3 (Agitator Motor) to Agitator Motor

Listview: things to know

- All ListViewRow items must have same height
- ListViewRow must have a known and valid height
 - Auto, stretched, or other dynamic behaviors are not supported
- Every parameter which is not linked to a persisted object is reset as soon as the element is scrolled outside the screen
 - Example: CheckBox selection, SpinBox values, etc
- ORDER BY clause is not supported when displaying project nodes
- Does not support recursion
 - Cannot navigate nested folders or objects
- If there is no ListViewRow template defined for a given object type, a blank row is displayed
- Does not support change runtime in visibility of ListViewRows
- Horizontal scrolling is not supported

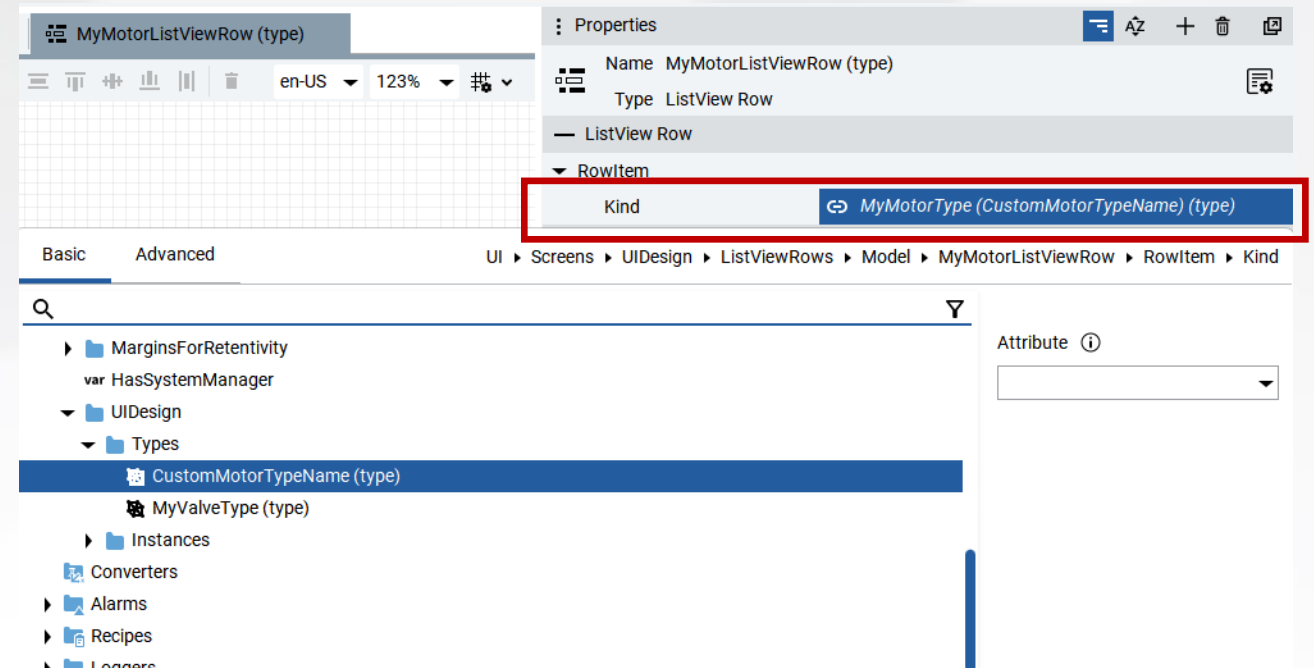
ListViewRow

- Atomic element of a ListView
 - Used to interact with the content being displayed in a ListView
- Each ListViewRow refers to a specific object type
- A ListViewRow does not support multiple aliases



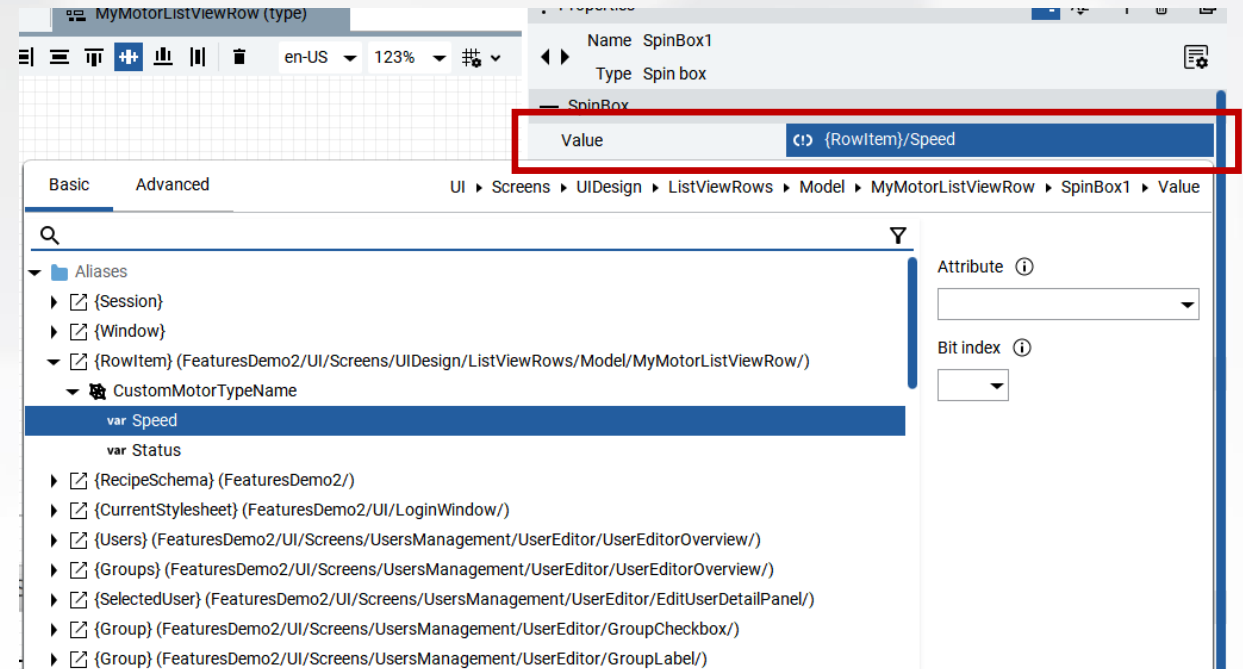
ListViewRow

- Configuration:
 1. Specify the object type in the Kind property
 2. Configure UI controls to interact with the given object type
 3. Assign the ListViewRow to the ListView (as seen before)



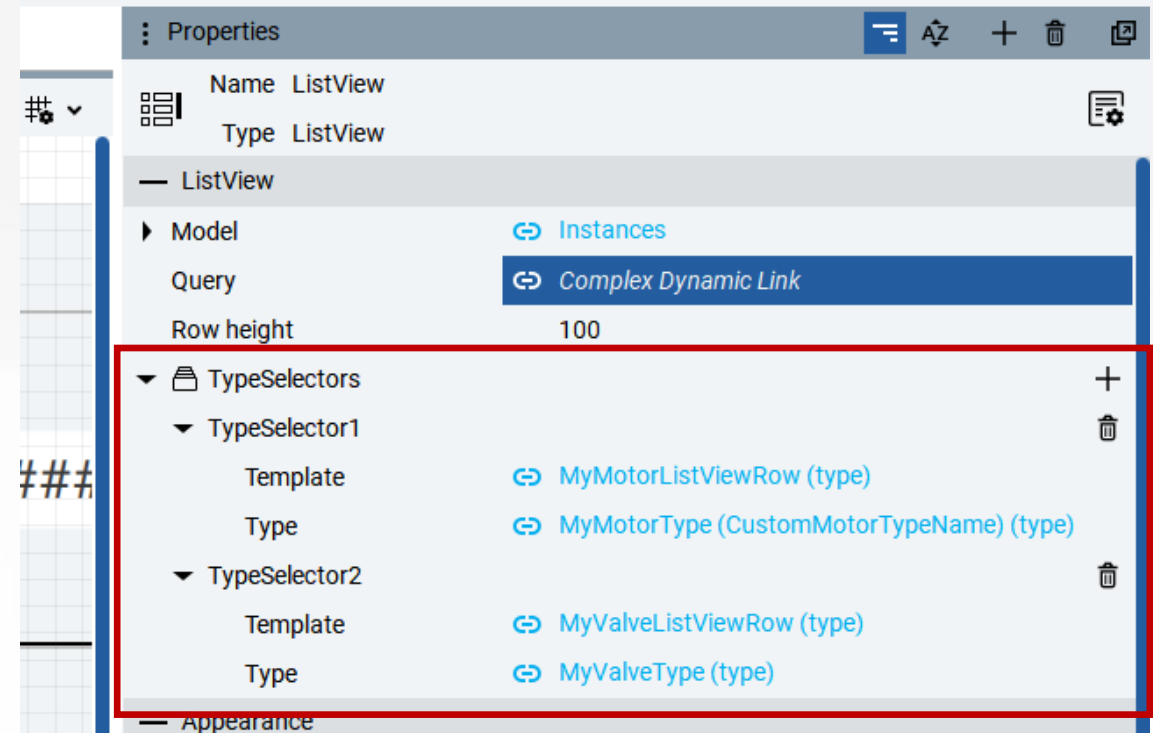
ListViewRow

- Configuration:
 1. Specify the object type in the Kind property
 2. Configure UI controls to interact with the given object type
 3. Assign the ListViewRow to the ListView (as seen before)



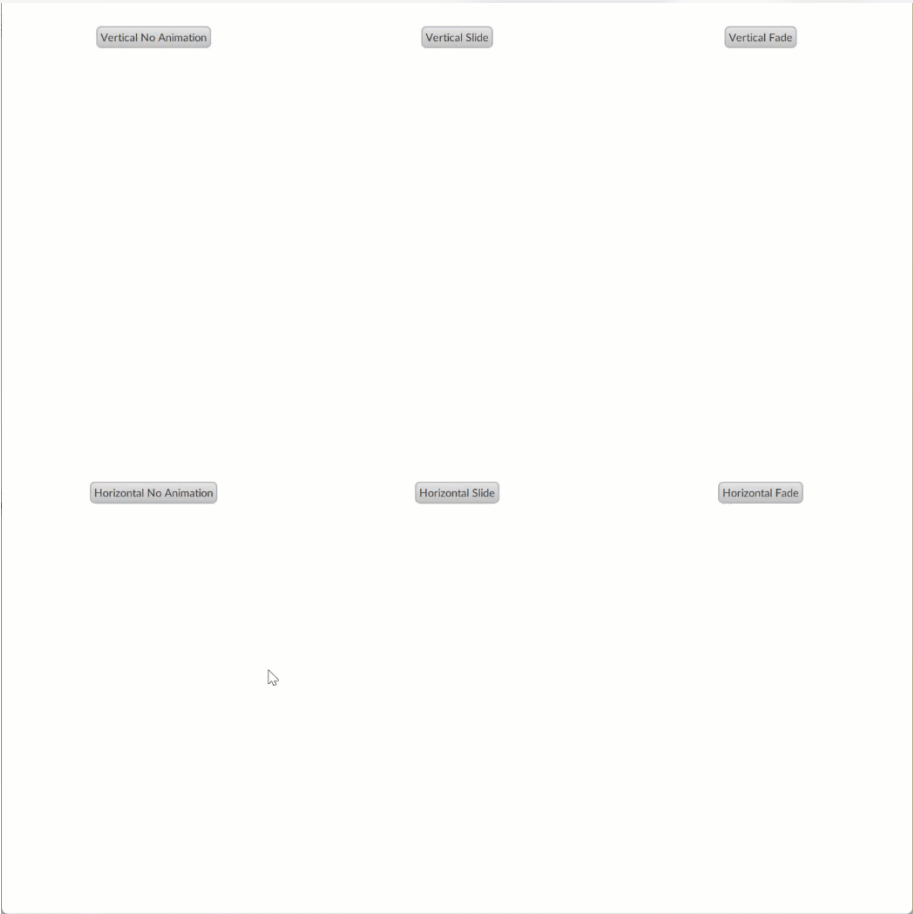
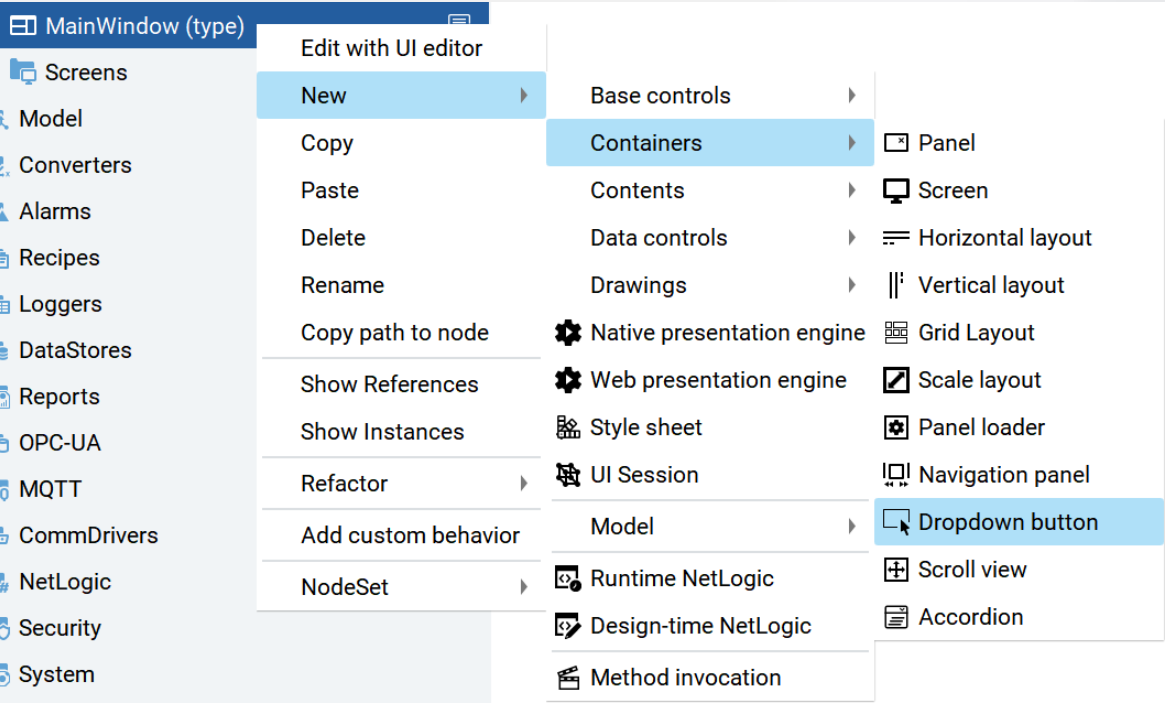
ListViewRow

- Configuration:
 1. Specify the object type in the Kind property
 2. Configure UI controls to interact with the given object type
 3. Assign the ListViewRow to the ListView (as seen before)



Dropdown button

- Present another container where clicked (i.e. Drop down menus or Faceplate-like functionality)



Dropdown button

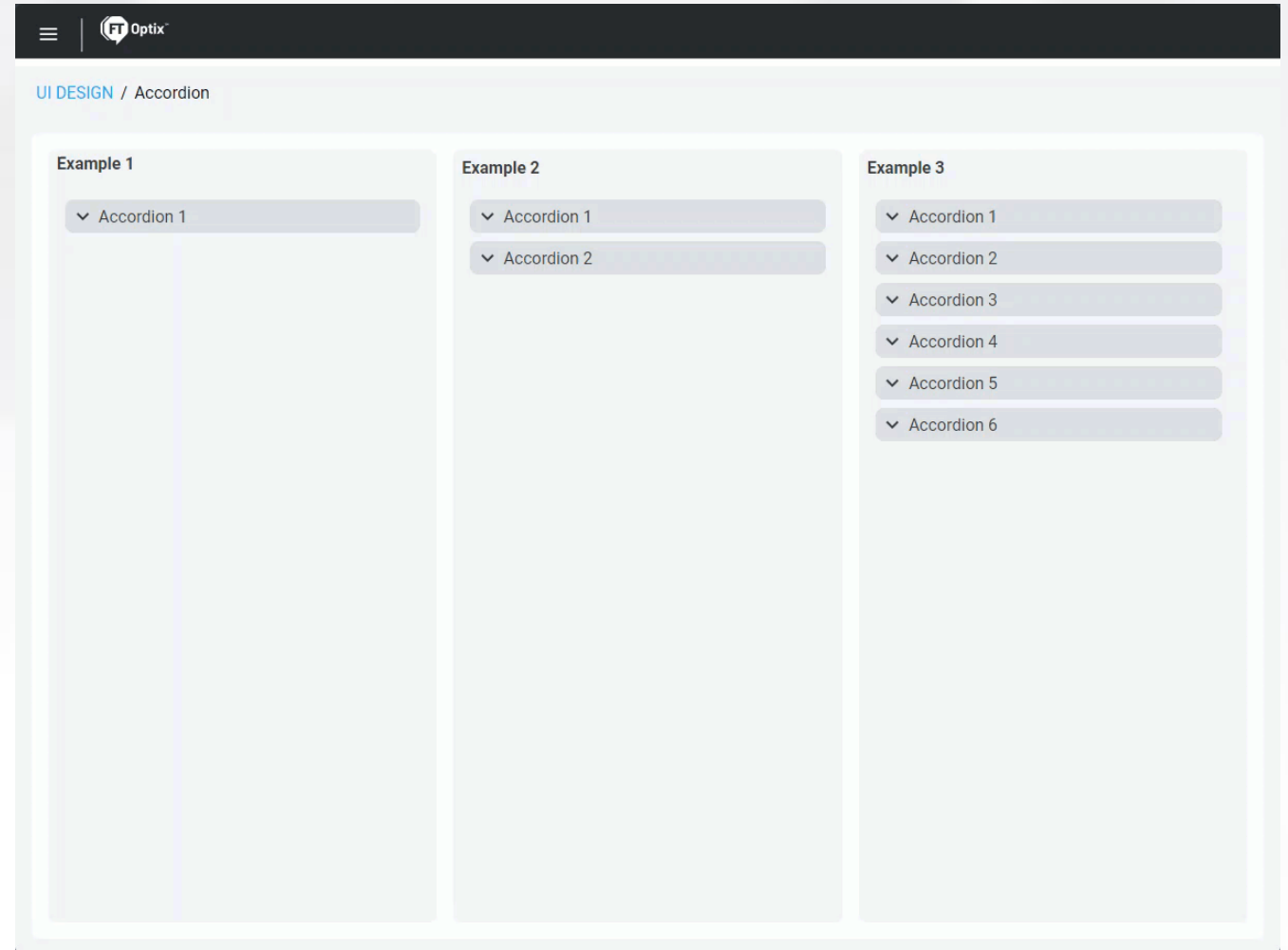
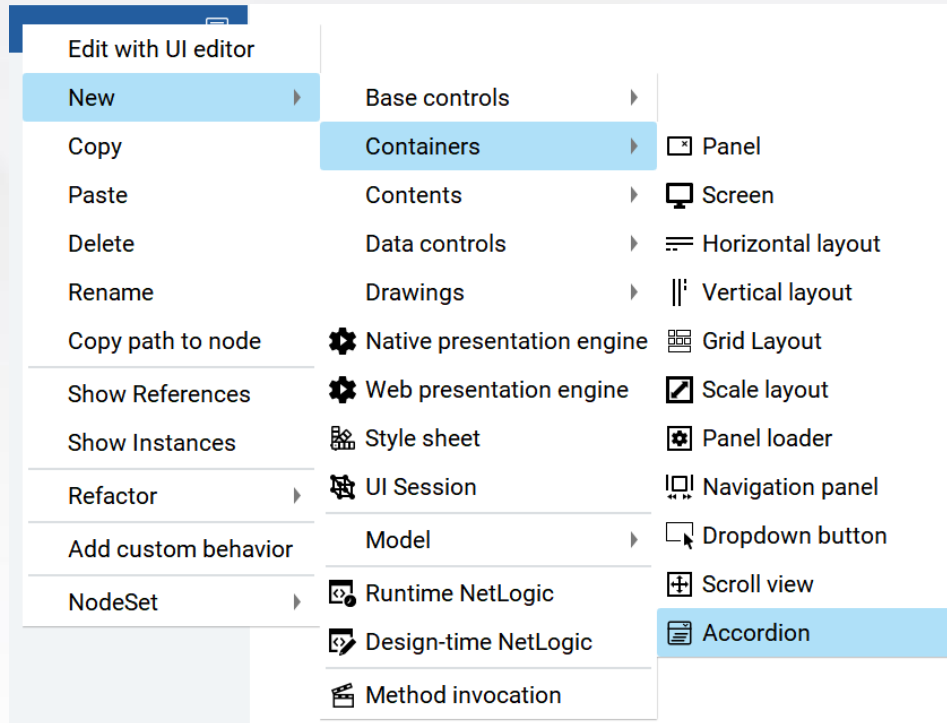
- Present another container where clicked (i.e. Drop down menus or Faceplate-like functionality)

— DropDown Button	
Panel	↗ Faceplate (type)
Panel alias node	
Direction	<i>Vertical</i>
Animation	<i>None</i>
Close on click inside	<i>False</i>
Close on click outside	<i>True</i>

- Panel: Panel to load on click
- Panel alias node: Aliases to pass to Panel
- Direction: Vertical or Horizontal
- Animation: None, Slide, Fade
- Close on Click inside / outside: Close panel behavior

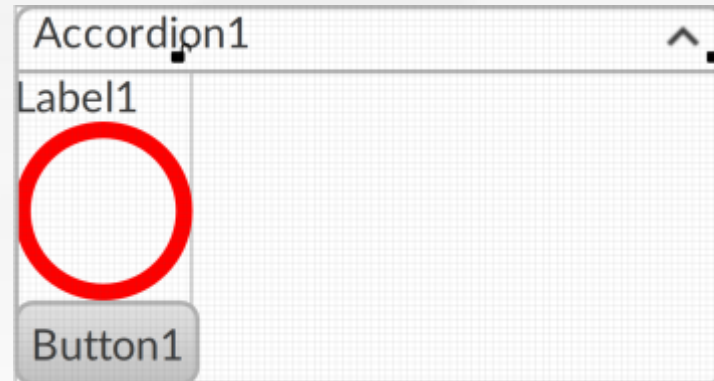
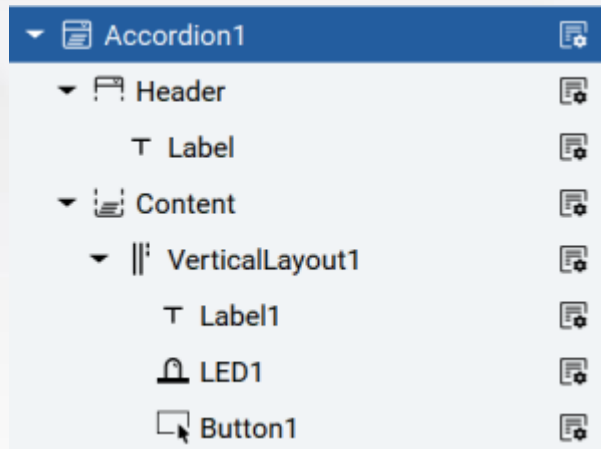
Accordion

- Configurable Dropdown button with nesting capabilities



Accordion

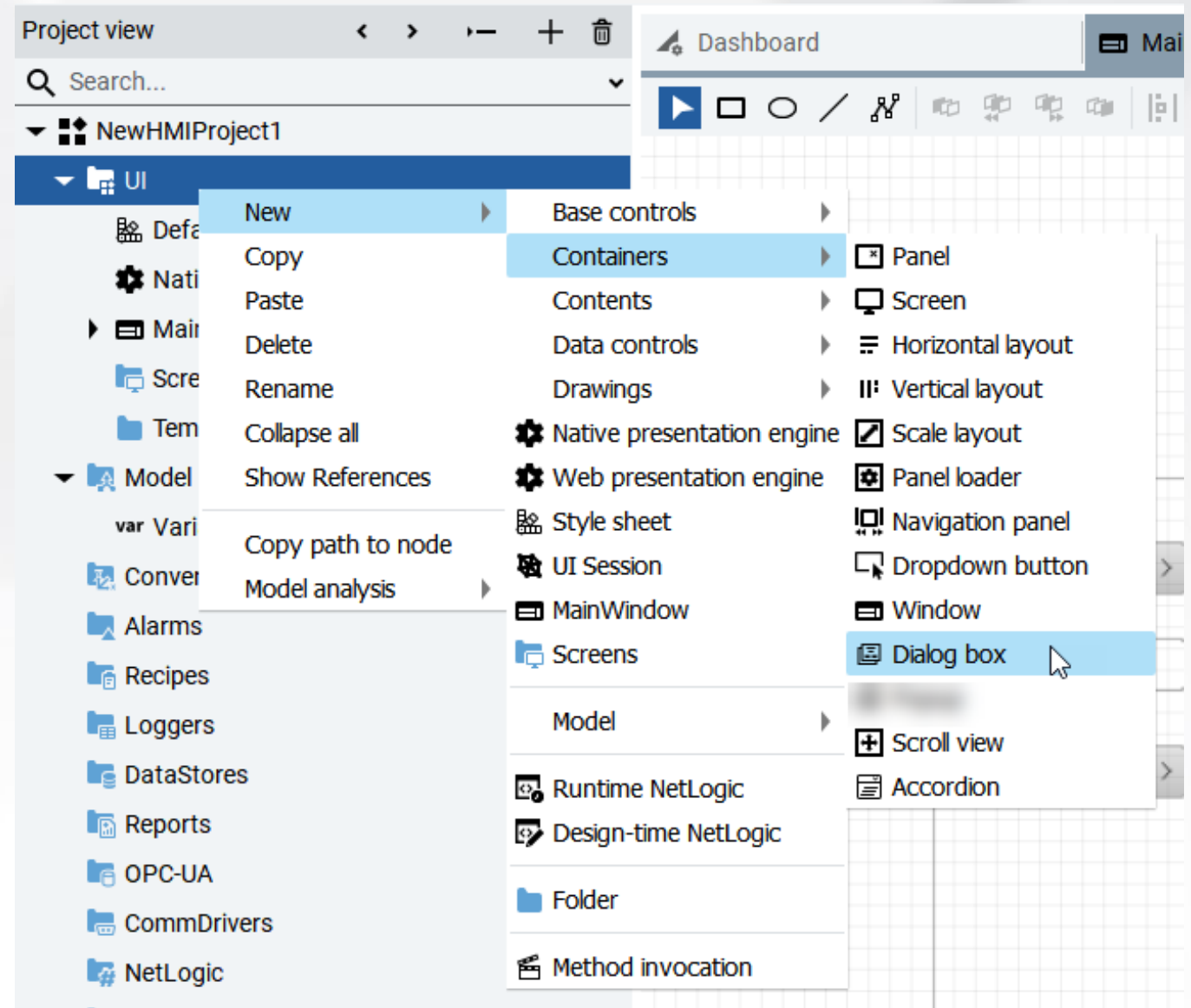
- Configurable Dropdown button with nesting capabilities



— Accordion	
Expanded	False

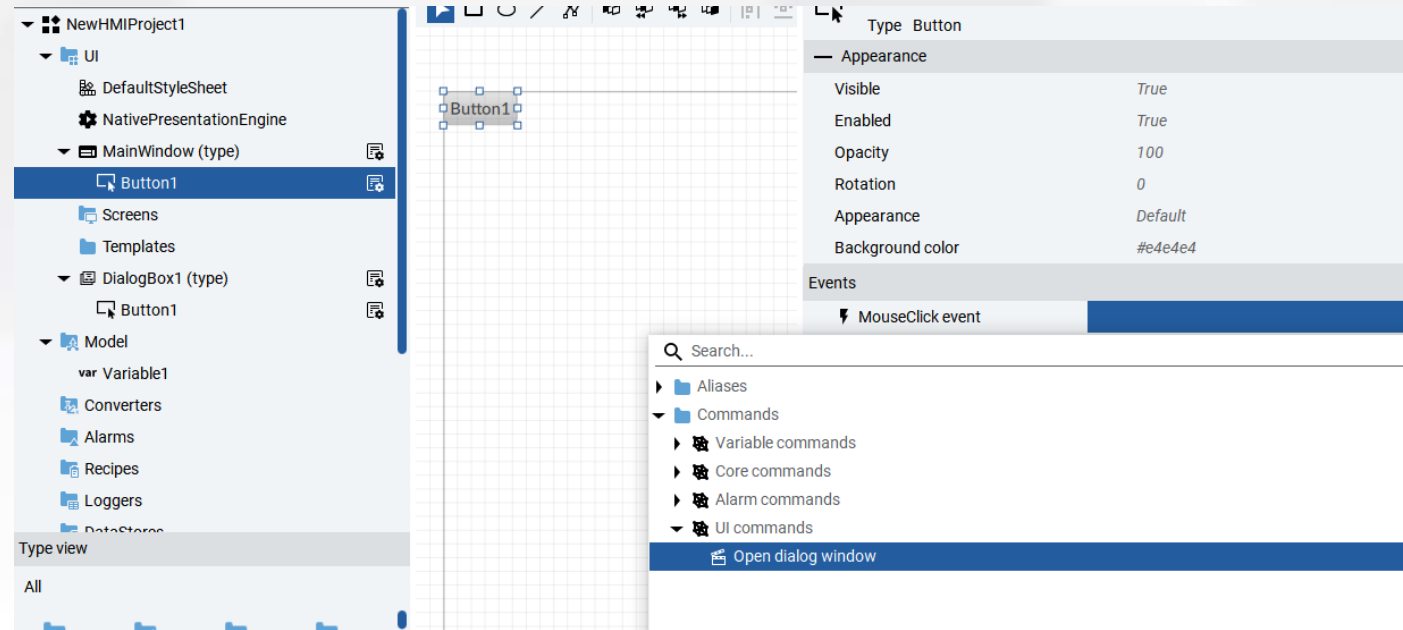
DialogBox

- Popup displays in Optix are called «DialogBox»
- DialogBox must be a type
- DialogBoxes are transparent by default
- **Note:** DialogBox does not open a new window on the device, it is a container on the highest Z-level of the MainWindow
- The «Close» method is exposed by the DialogBox itself



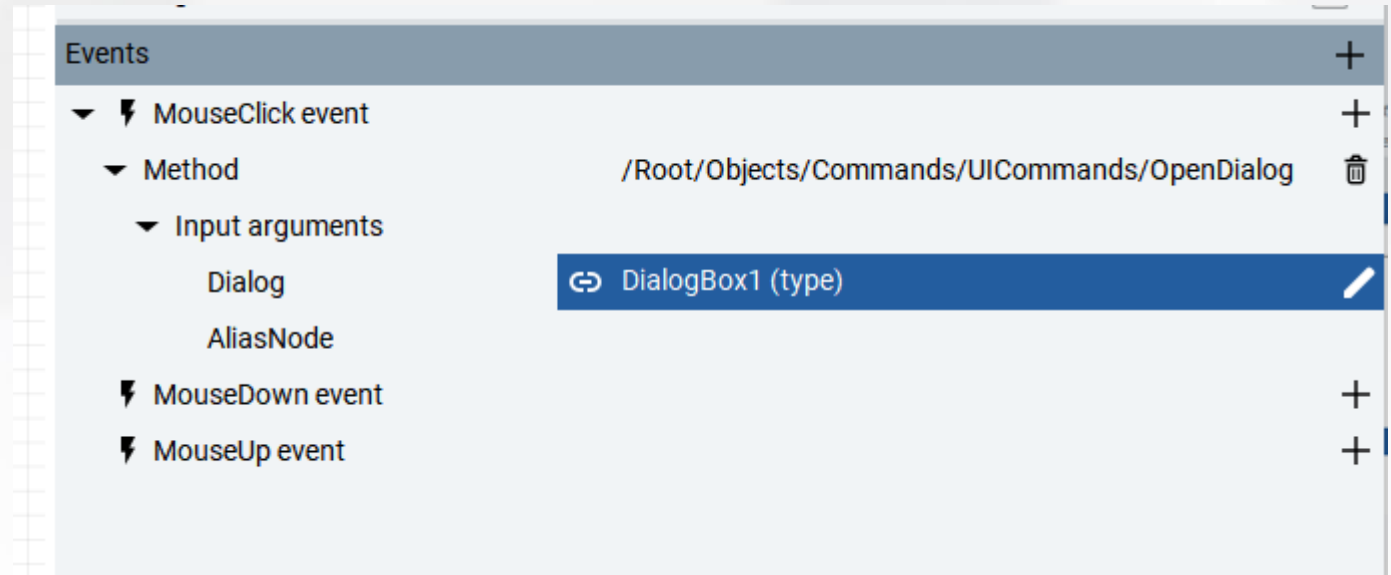
DialogBox

- Popup displays in Optix are called «DialogBox»
- DialogBox must be a type
- DialogBoxes are transparent by default
- **Note:** DialogBox does not open a new window on the device, it is a container on the highest Z-level of the MainWindow
- The «Close» method is exposed by the DialogBox itself



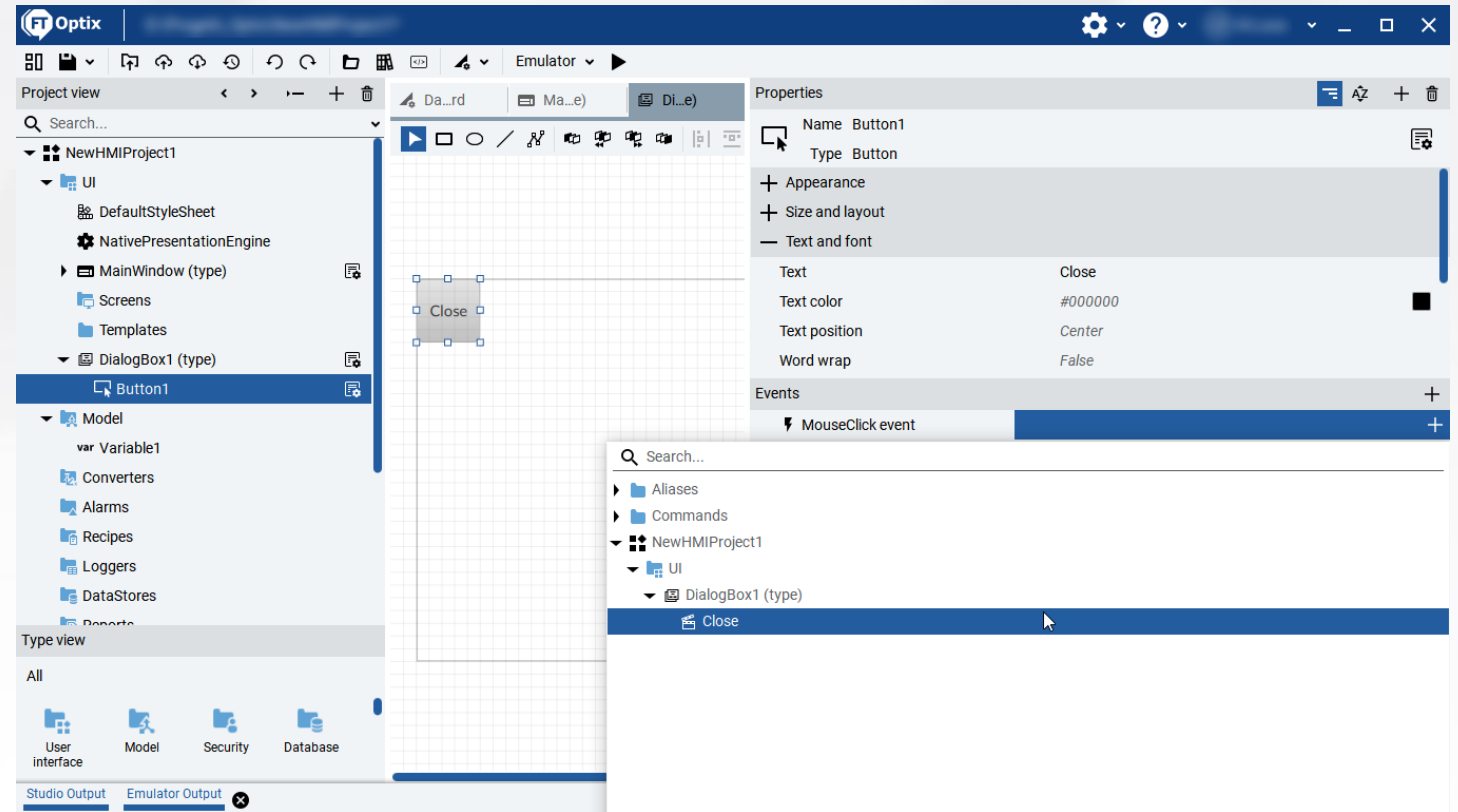
DialogBox

- Popup displays in Optix are called «DialogBox»
- DialogBox must be a type
- DialogBoxes are transparent by default
- **Note:** DialogBox does not open a new window on the device, it is a container on the highest Z-level of the MainWindow
- The «Close» method is exposed by the DialogBox itself



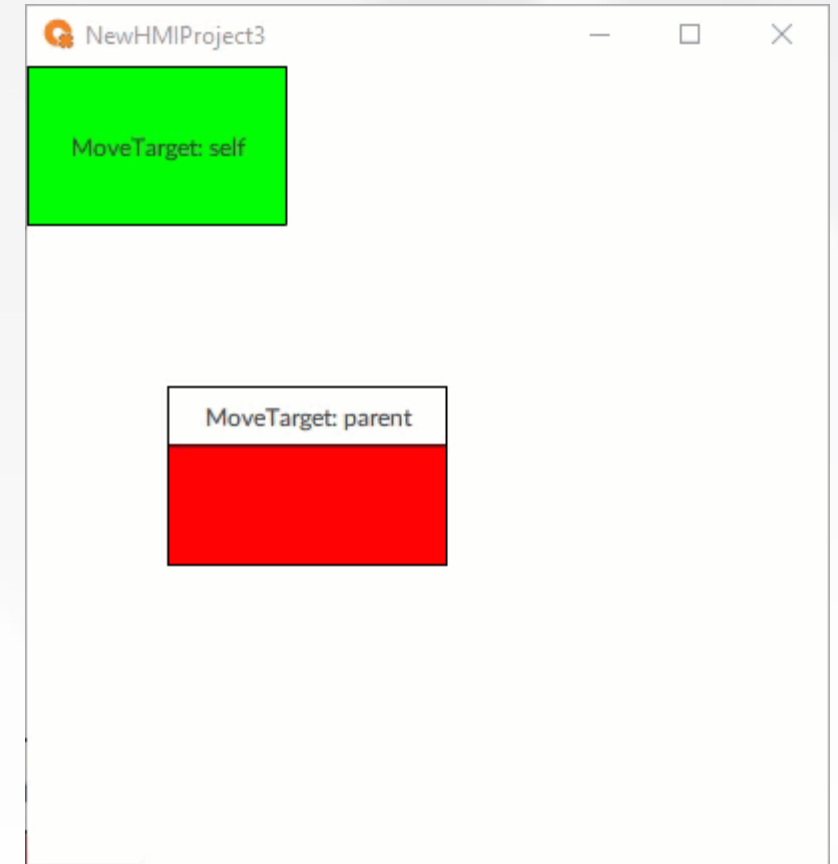
DialogBox

- Popup displays in Optix are called «DialogBox»
- DialogBox must be a type
- DialogBoxes are transparent by default
- **Note:** DialogBox does not open a new window on the device, it is a container on the highest Z-level of the MainWindow
- The «Close» method is exposed by the DialogBox itself



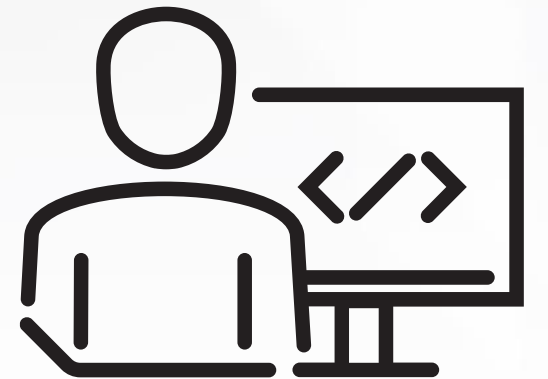
Move Target

- Elements can be moved on the screen with the MoveTarget property
 - Self -> Dragging the object will move the object itself (and its children)
 - Parent -> When selecting any of the parent objects, dragging the control will move the whole parent and all children (e.g: dragging the title bar moves the whole DialogBox)



Hands-on session

- Create a simple HMI with two or three screens and a Navigation Panel
- Add a web presentation engine and connect to the emulator using a browser
- Try to make the graphics responsive

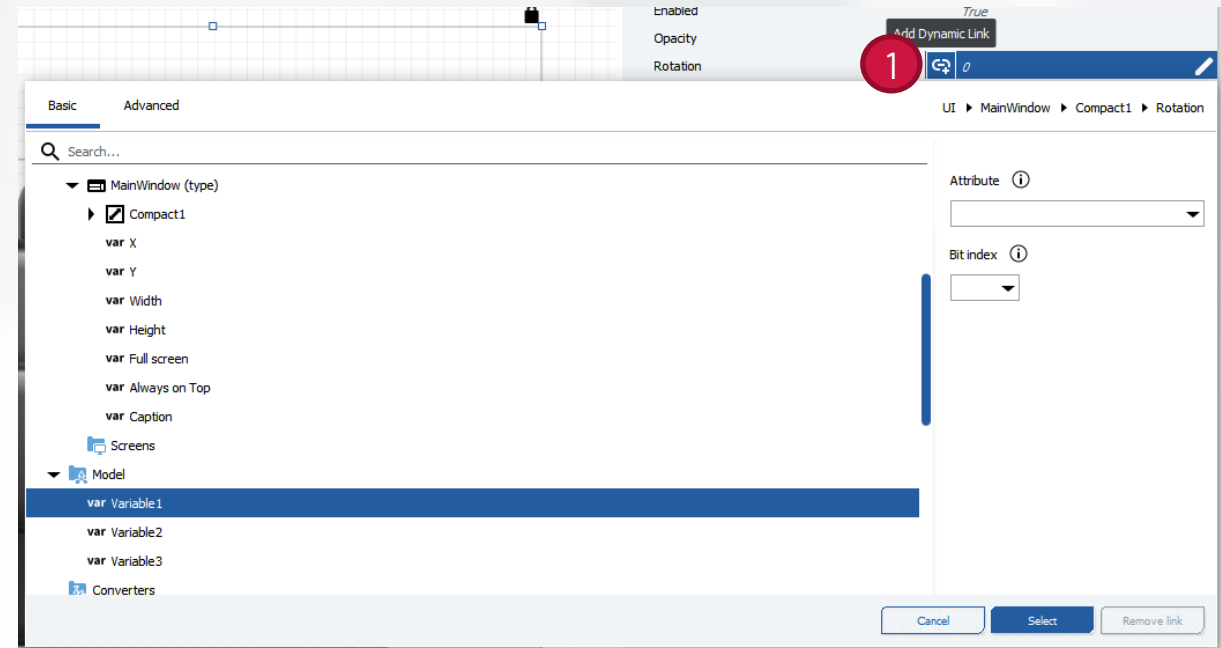


Dynamic links



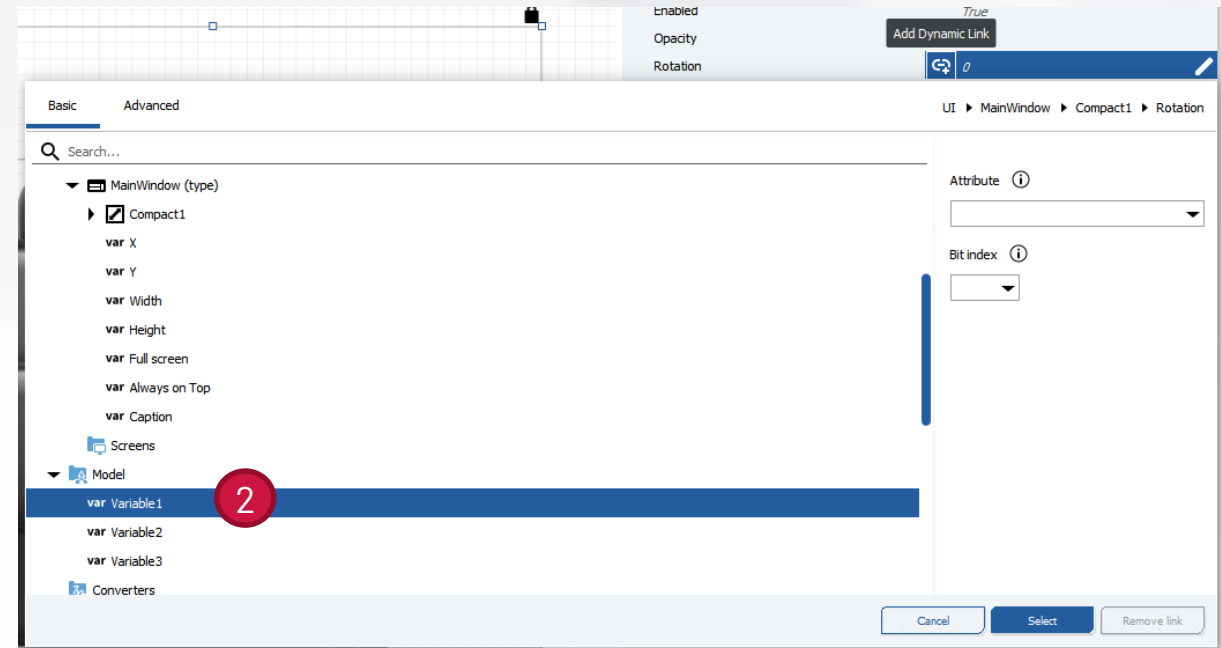
Dynamic links

- Dynamic Link is a connection between two nodes
 - Property connected to a Variable
absolute link = "[Variable1](#)"
 - Property connected to a Property of the same object
relative link = "[../Enabled](#)"
 - Property connected to a Property of another object
relative link = "[../../Switch1/Checked](#)"
- Can be created in different ways
 - Manual selection using the dynamic link window
 - Dragging the variable on a property
 - Dragging the variable on the graphical object
in this case, Studio sets the dynamic link
on the most suitable property, based on heuristics.



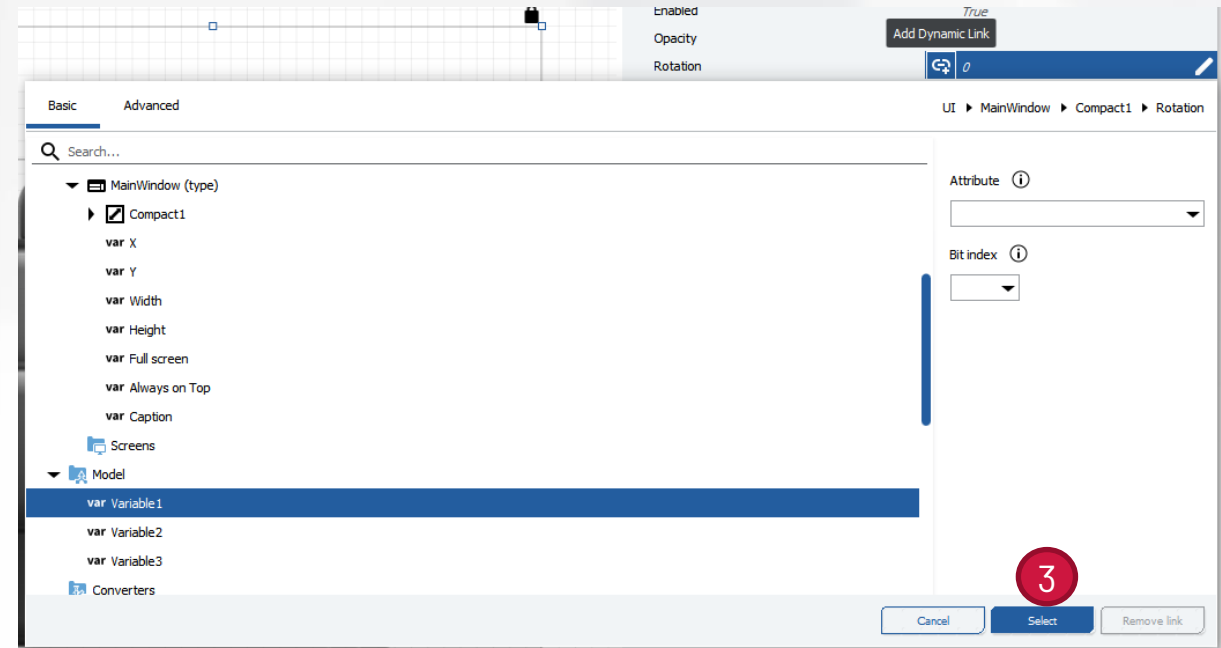
Dynamic links

- Dynamic Link is a connection between two nodes
 - Property connected to a Variable
absolute link = "[Variable1](#)"
 - Property connected to a Property of the same object
relative link = "[../Enabled](#)"
 - Property connected to a Property of another object
relative link = "[../../Switch1/Checked](#)"
- Can be created in different ways
 - Manual selection using the dynamic link window
 - Dragging the variable on a property
 - Dragging the variable on the graphical object
in this case, Studio sets the dynamic link
on the most suitable property, based on heuristics.



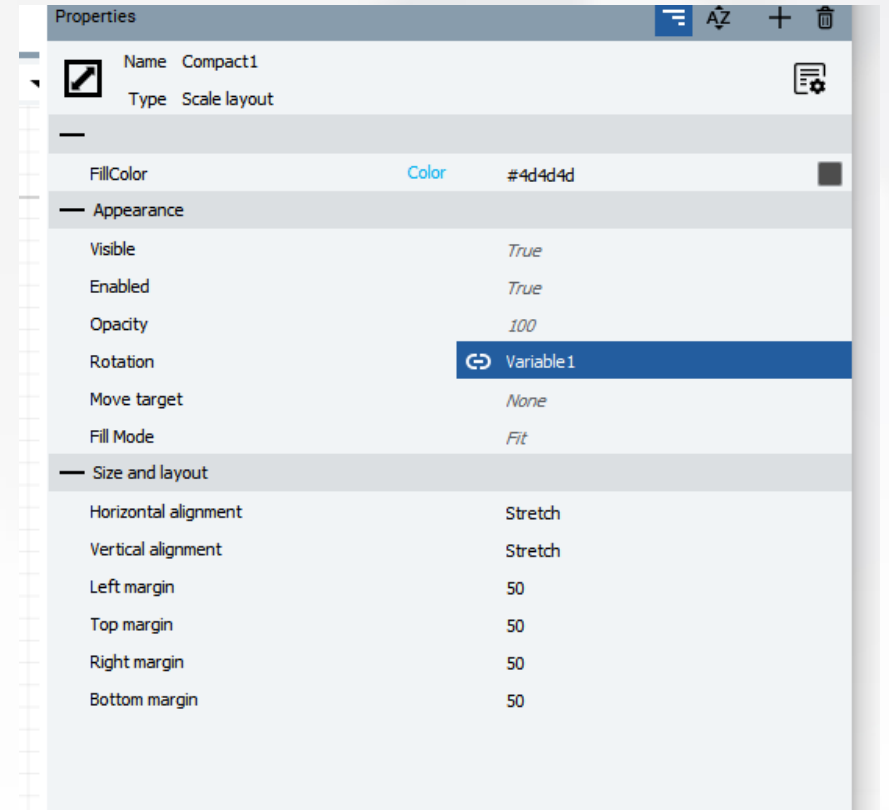
Dynamic links

- Dynamic Link is a connection between two nodes
 - Property connected to a Variable
absolute link = "[Variable1](#)"
 - Property connected to a Property of the same object
relative link = "[../Enabled](#)"
 - Property connected to a Property of another object
relative link = "[../../Switch1/Checked](#)"
- Can be created in different ways
 - Manual selection using the dynamic link window
 - Dragging the variable on a property
 - Dragging the variable on the graphical object
in this case, Studio sets the dynamic link
on the most suitable property, based on heuristics.



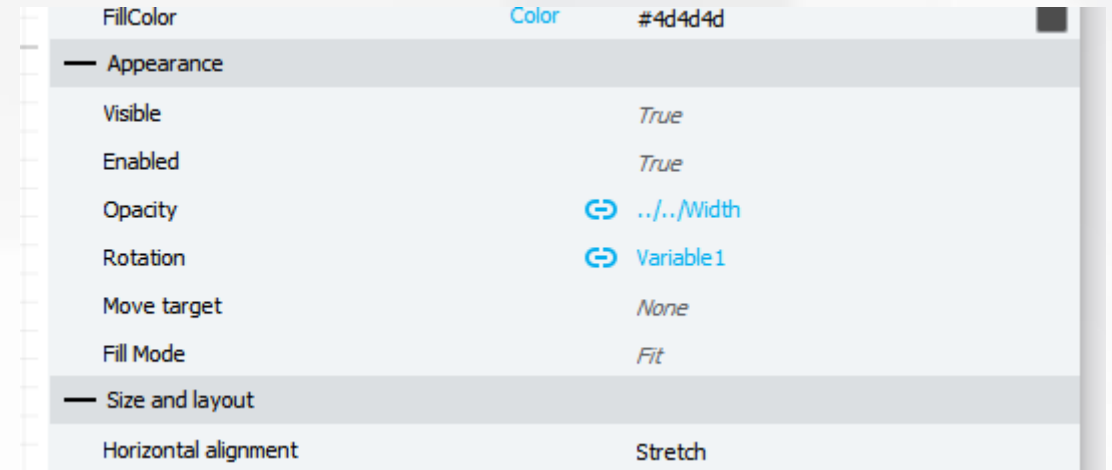
Dynamic links

- Dynamic Link is a connection between two nodes
 - Property connected to a Variable
absolute link = "[Variable1](#)"
 - Property connected to a Property of the same object
relative link = "[../Enabled](#)"
 - Property connected to a Property of another object
relative link = "[../Switch1/Checked](#)"
- Can be created in different ways
 - Manual selection using the dynamic link window
 - Dragging the variable on a property
 - Dragging the variable on the graphical object
in this case, Studio sets the dynamic link
on the most suitable property, based on heuristics.



Dynamic links

- Dynamic Link is a connection between two nodes
 - Property connected to a Variable
absolute link = "[Variable1](#)"
 - Property connected to a Property of the same object
relative link = "[../Enabled](#)"
 - Property connected to a Property of another object
relative link = "[../Switch1/Checked](#)"
- Can be created in different ways
 - Manual selection using the dynamic link window
 - Dragging the variable on a property
 - Dragging the variable on the graphical object
in this case, Studio sets the dynamic link
on the most suitable property, based on heuristics.



FillColor	Color	#4d4d4d
— Appearance		
Visible		True
Enabled		True
Opacity		../Width
Rotation		Variable1
Move target		None
Fill Mode		Fit
— Size and layout		
Horizontal alignment		Stretch

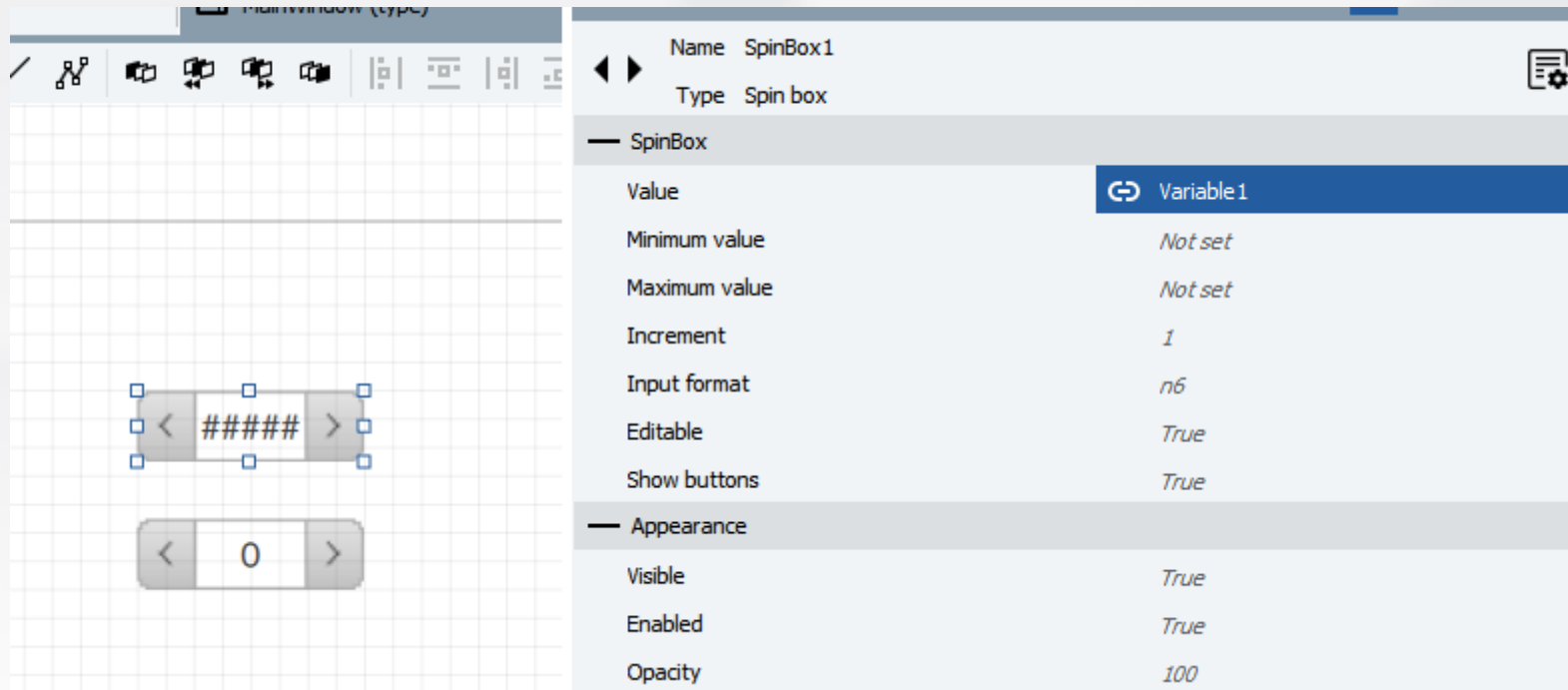
Dynamic links indicators

- Dynamic links can be
 - **Valid**: solid blue indicator, target variable is reachable
 - **Unknown**: blue exclamation mark, the target variable is not reachable at DesignTime (an Alias for example) but can be good at Runtime
 - **Unresolved**: the target variable does not exist, DynamicLink is most likely broken

Show buttons	True
— Appearance	
Visible	 ../../Switch1/Checked
Enabled	True
Opacity	100
Rotation	0
+ Size and layout	
— Text and font	
Text color	 {Alias1}\Name
Font family	Default
Font size	 <Unresolved>
Font weight	Normal
Font italic	False
Events	

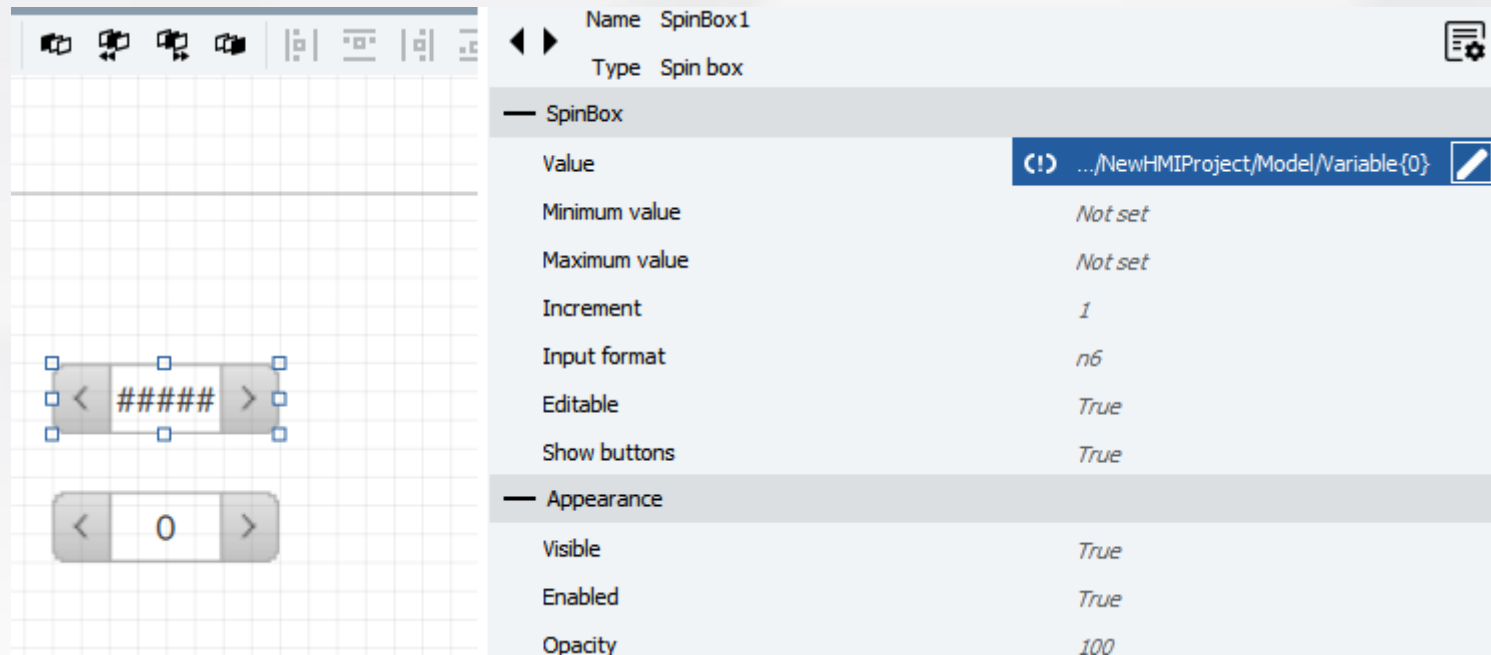
Advanced dynamic links: indexing

- Variables can be indexed by using {0} placeholder



Advanced dynamic links: indexing

- Variables can be indexed by using {0} placeholder

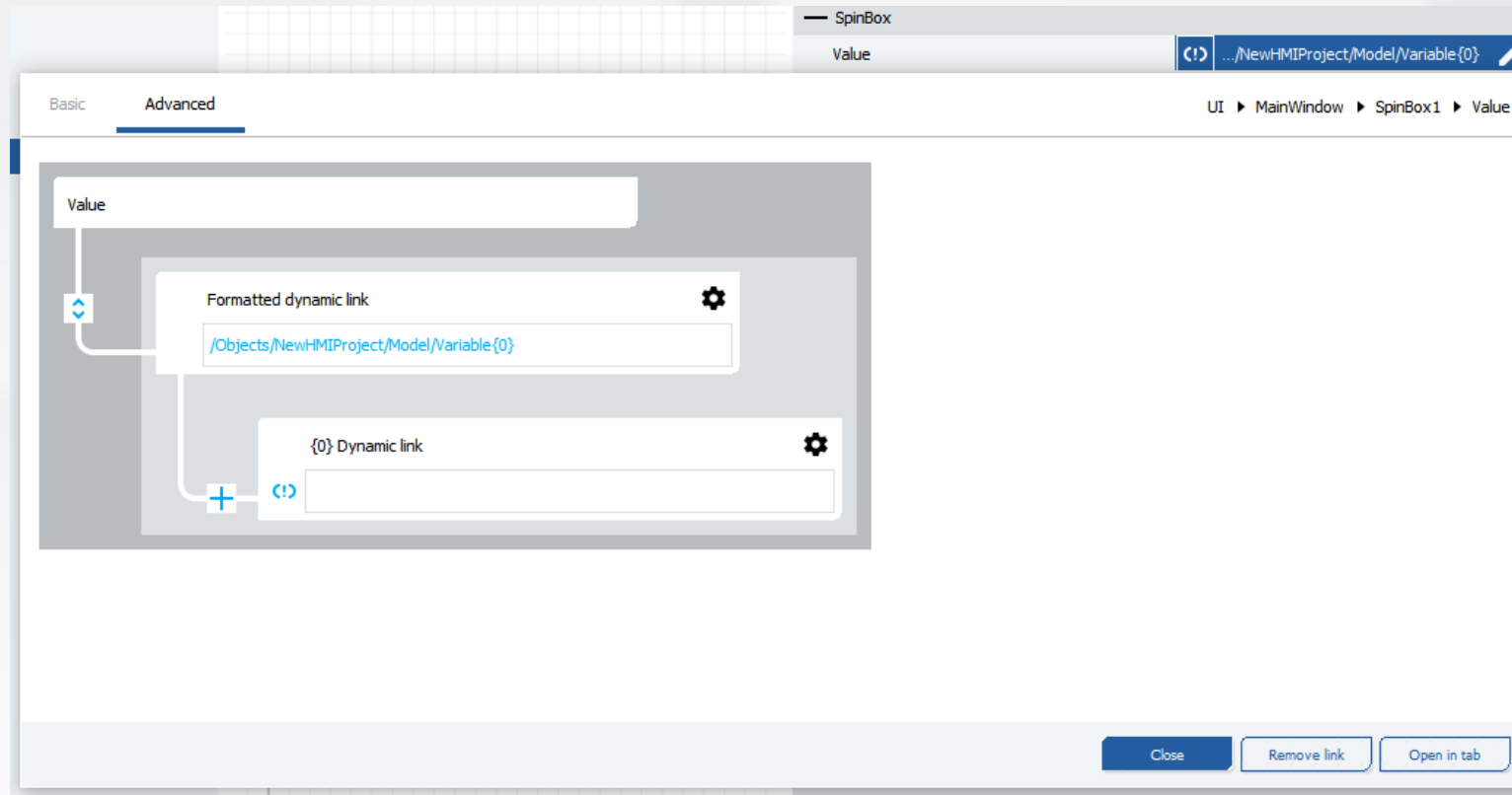


The screenshot displays the Rockwell Studio software interface. On the left, a grid-based workspace contains two SpinBox controls. The top one is a template with five hash symbols (#####) and the bottom one shows the number 0. On the right, the Properties window for 'SpinBox1' is open. It shows the 'Spin box' type and various configuration options.

Property	Value
Name	SpinBox1
Type	Spin box
SpinBox	
Value	.../NewHMIPProject/Model/Variable{0}
Minimum value	Not set
Maximum value	Not set
Increment	1
Input format	n6
Editable	True
Show buttons	True
Appearance	
Visible	True
Enabled	True
Opacity	100

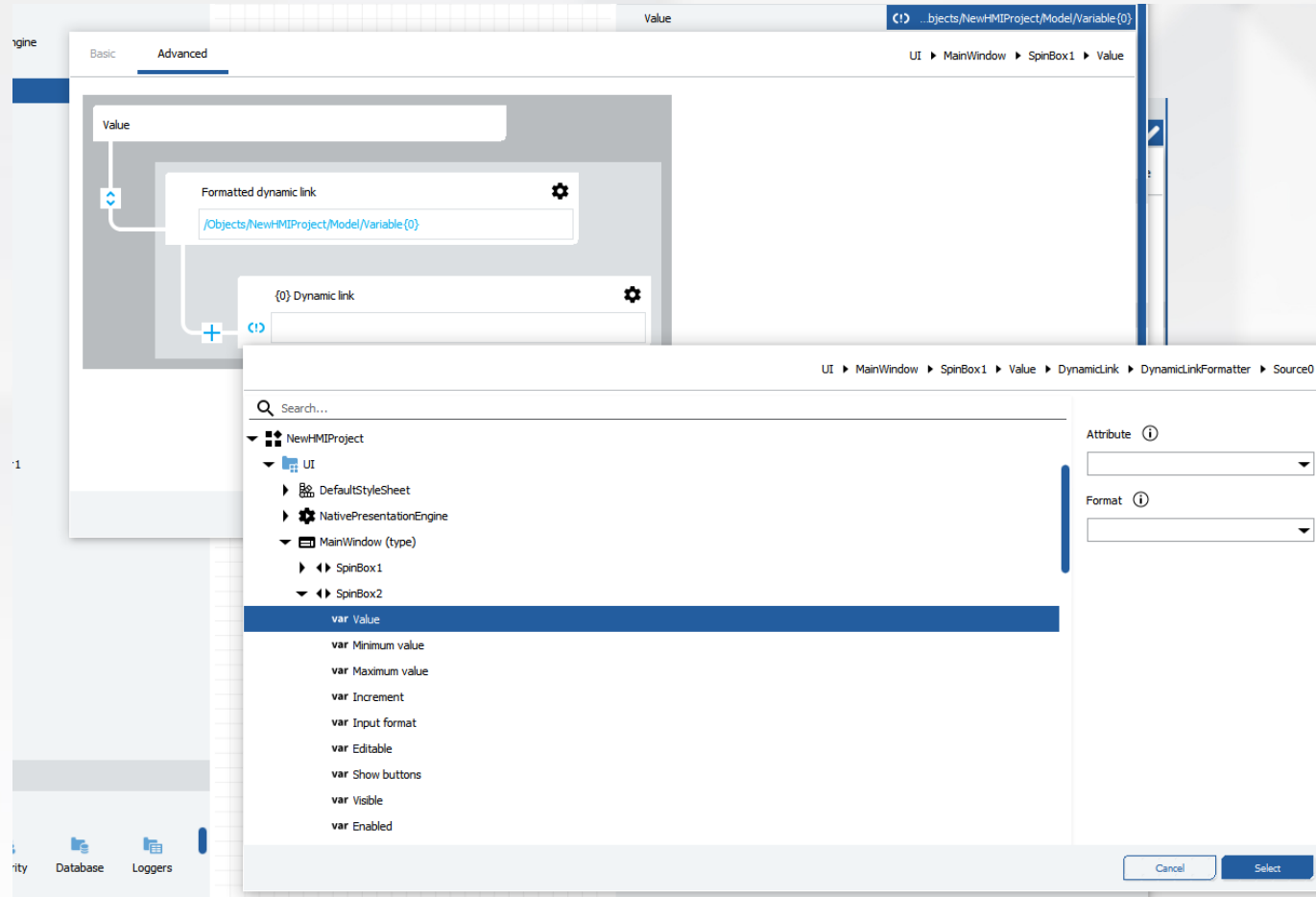
Advanced dynamic links: indexing

- Variables can be indexed by using {0} placeholder



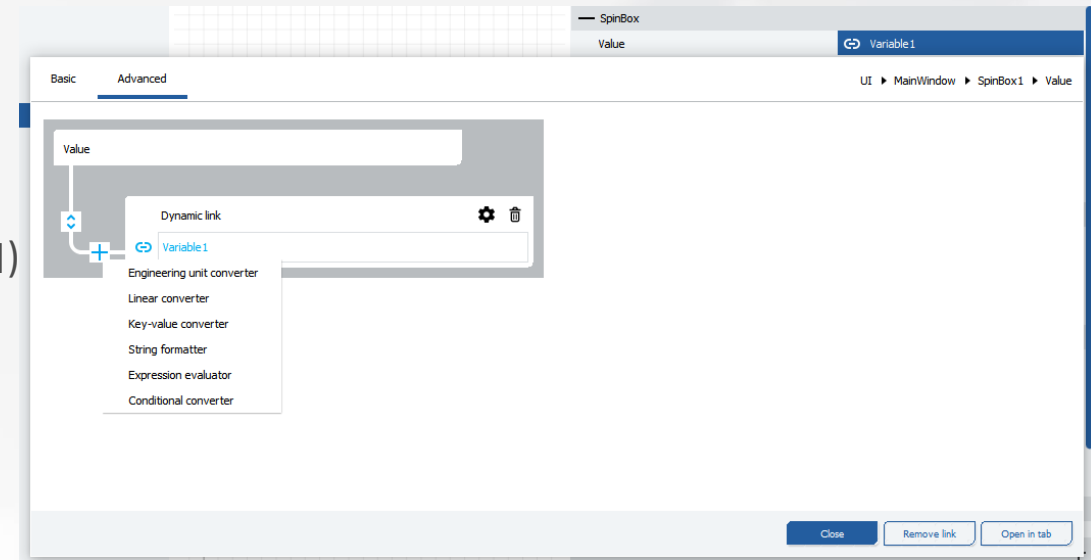
Advanced dynamic links: indexing

- Variables can be indexed by using {0} placeholder



Advanced dynamic links: converters

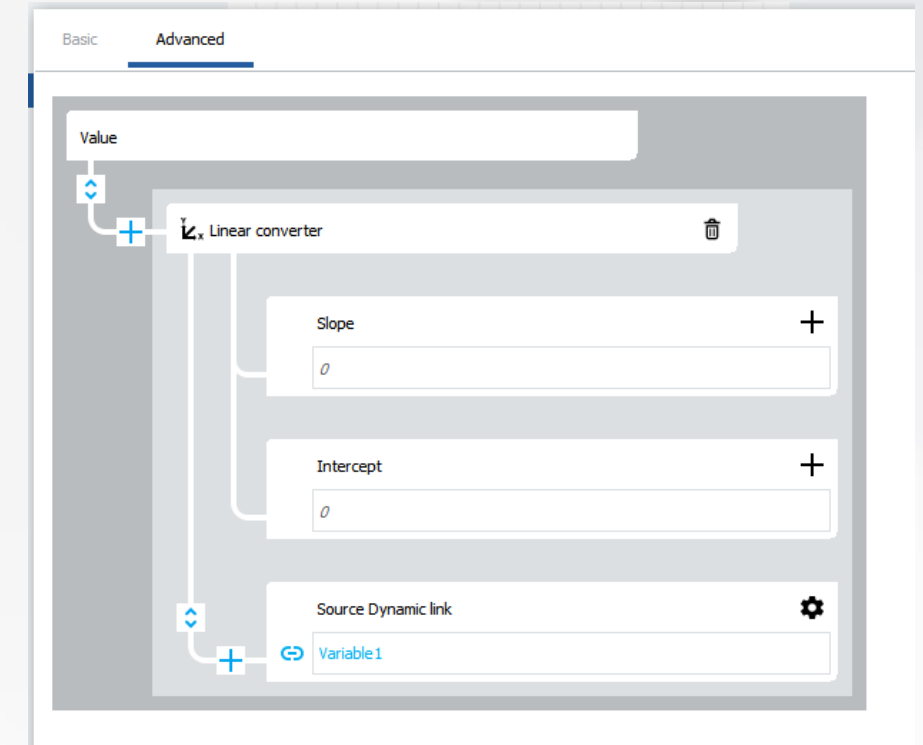
- By [Advanced...] button on Dynamic Link window, the value of a Property can be set through a Converter
- **Engineering Unit converter**: it's a linear conversion defined by Raw range (PLC) and Scaled range (HMI)
- **Linear converter**: it's a linear conversion defined by equation $y = mx + q$
- **Key-Value converter**^{*}: converts the value of the source variable according to a table of key-value pairs.
- **String formatter**: modify the format of a Numeric, Date/Time, String, SQL Query
- **Expression evaluator**: performs arithmetic calculations with one or more input values
- **Conditional converter**: writes two different values according to a boolean Condition DynamicLink



^{*} Key-Value converter table can be exported to csv file and imported back, using a Template Library script

Advanced dynamic links: converters

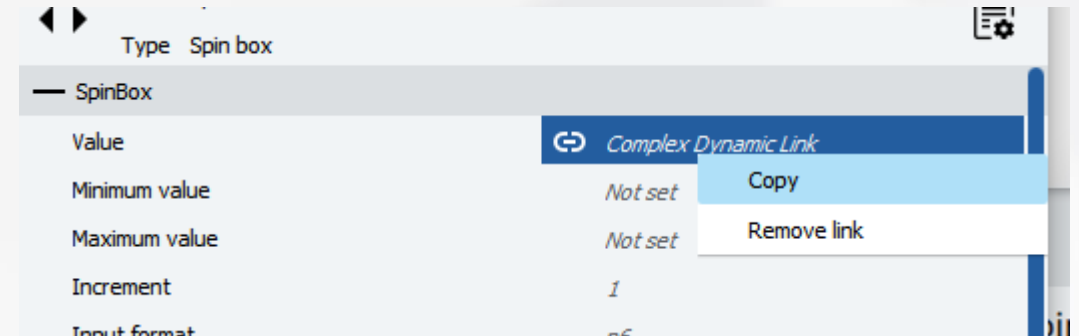
- By [Advanced...] button on Dynamic Link window, the value of a Property can be set through a Converter
- **Engineering Unit converter**: it's a linear conversion defined by Raw range (PLC) and Scaled range (HMI)
- **Linear converter**: it's a linear conversion defined by equation $y = mx + q$
- **Key-Value converter**^{*}: converts the value of the source variable according to a table of key-value pairs.
- **String formatter**: modify the format of a Numeric, Date/Time, String, SQL Query
- **Expression evaluator**: performs arithmetic calculations with one or more input values
- **Conditional converter**: writes two different values according to a boolean Condition DynamicLink



^{*} Key-Value converter table can be exported to csv file and imported back, using a Template Library script

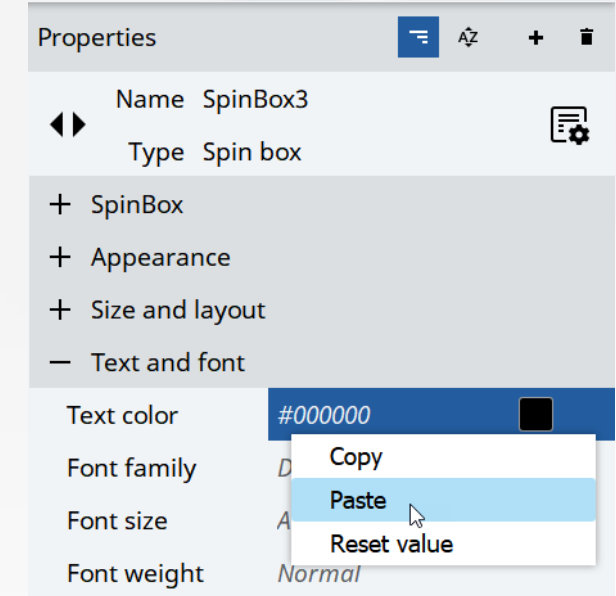
Advanced dynamic links: reusable converters

- Converters can be reused
 - Using **copy/paste**
 1. Copy the converter from the property with "*Complex Dynamic Link*"
 2. Paste to another property
 - Creating a **Converter as a Type**
 1. Create a Converter into Converters node
 2. The Converter will be automatically listed on the Advanced Dynamic Link configuration



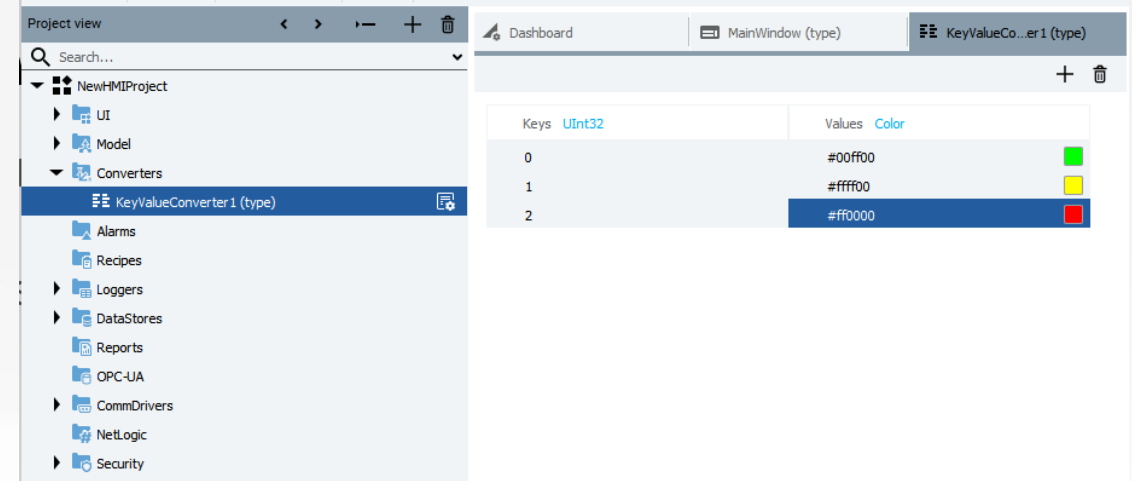
Advanced dynamic links: reusable converters

- Converters can be reused
 - Using **copy/paste**
 1. Copy the converter from the property with "*Complex Dynamic Link*"
 2. Paste to another property
 - Creating a **Converter as a Type**
 1. Create a Converter into Converters node
 2. The Converter will be automatically listed on the Advanced Dynamic Link configuration



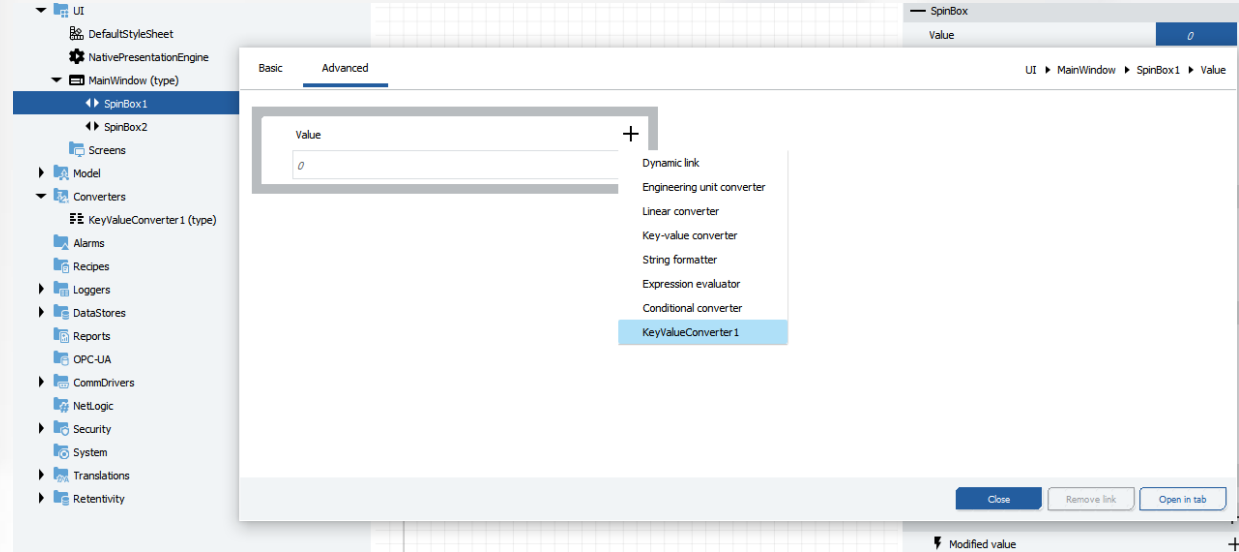
Advanced dynamic links: reusable converters

- Converters can be reused
- Using **copy/paste**
 1. Copy the converter from the property with "*Complex Dynamic Link*"
 2. Paste to another property
- Creating a **Converter as a Type**
 1. Create a Converter into Converters node
 2. The Converter will be automatically listed on the Advanced Dynamic Link configuration

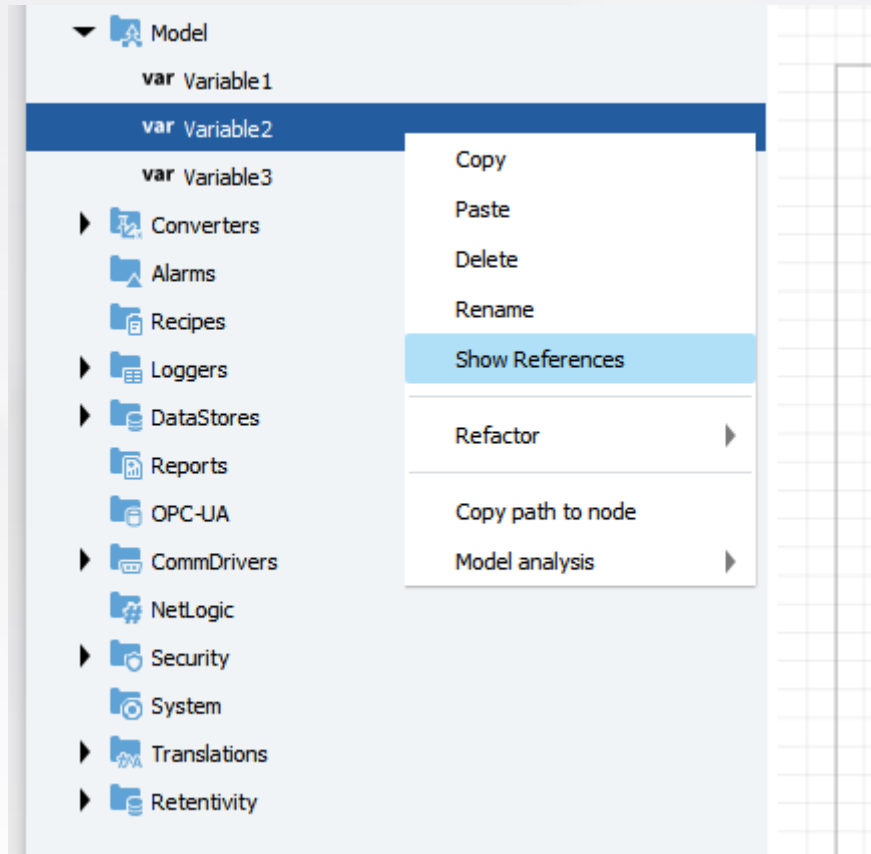


Advanced dynamic links: reusable converters

- Converters can be reused
 - Using **copy/paste**
 1. Copy the converter from the property with "*Complex Dynamic Link*"
 2. Paste to another property
 - Creating a **Converter as a Type**
 1. Create a Converter into Converters node
 2. The Converter will be automatically listed on the Advanced Dynamic Link configuration



Show reference



- Answer the question:
"who refers to this object ?"
- Can be used mainly for Tags, but also, for other objects like
 - Datastore,
 - Scripts,
 - ...

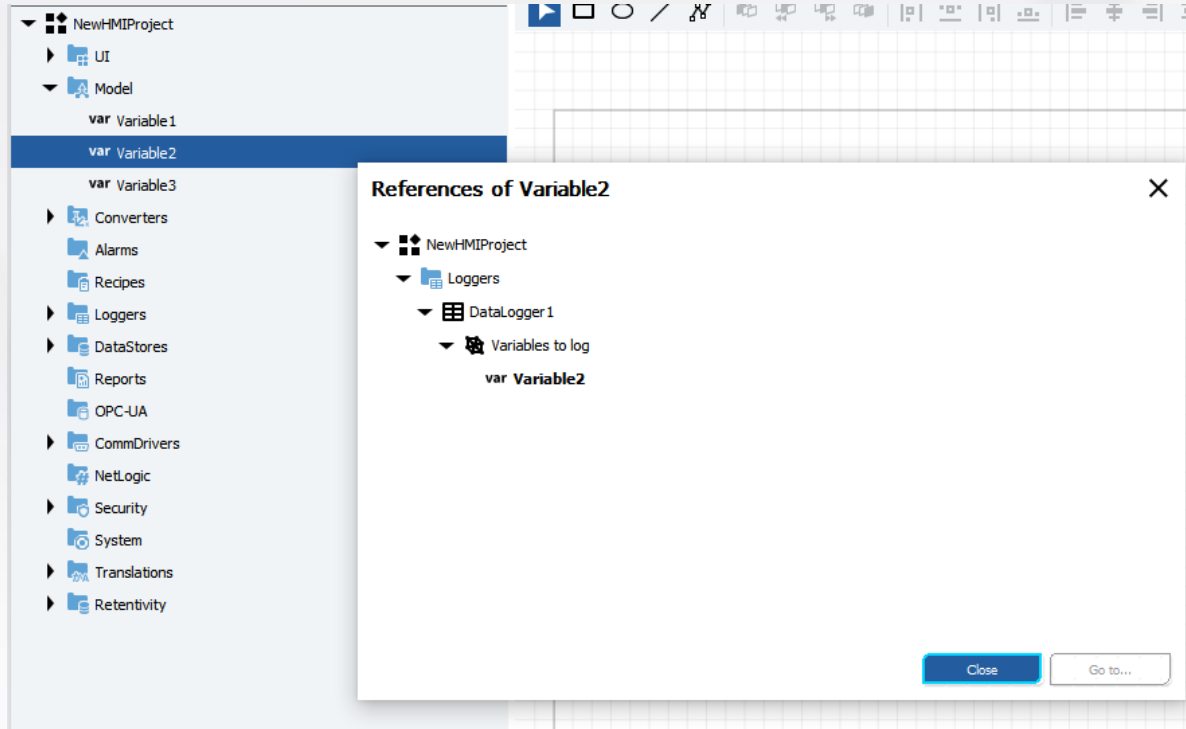
Show reference

- Answer the question:

"who refers to this object ?"

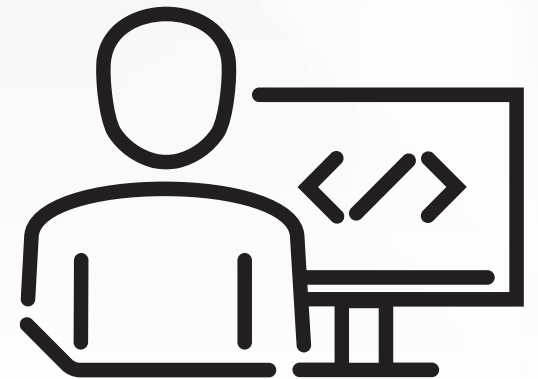
- Can be used mainly for Tags, but also, for other objects like

- Datastore,
- Scripts,
- ...



Hands-on session

- Define an "indexed dynamic link" to show the value of Variable1, Variable2, Variable3 using a SpinBox
- Use a Key-Value converter to change the background color of a rectangle based on the value of a variable
- Use a Key-Value converter to change the string of a label based on the value of a variable
- Use a Linear convert to show a scaled value



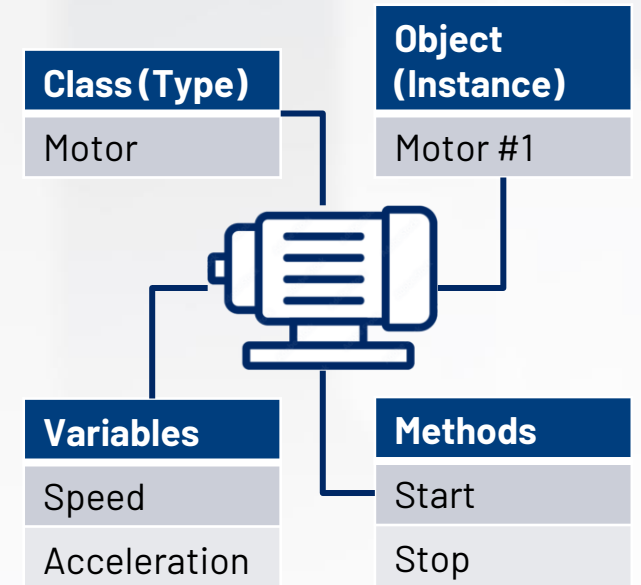
Work with reusable objects

Global objects, UDTs, Widgets



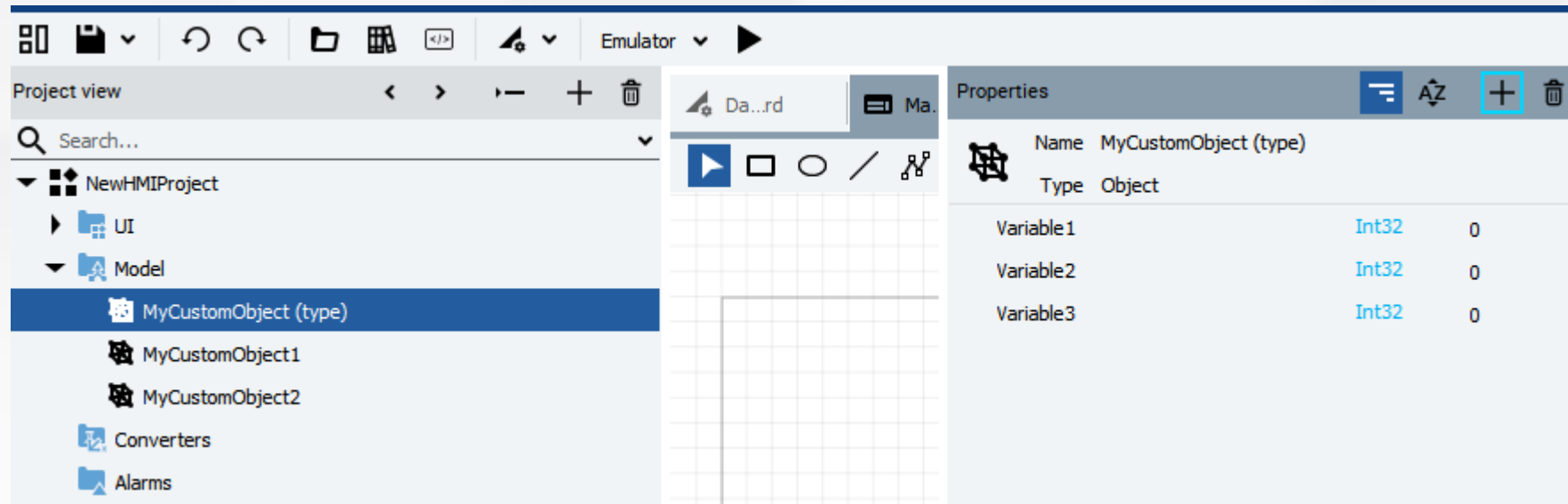
Object-oriented programming (OOP) paradigm

- **Elements of OOP:**
 - **Class:** as a core element, it's a collection of methods and variables and generally, it is regarded as a **type for instance**.
 - **Object:** as a basic element, it is regarded as an **instance** of a class
- **Capabilities of OOP:**
 - Encapsulation, Abstraction, Inheritance, Polymorphism
- **Object-Oriented Programming in Optix means**
 - **Native Types** already seen →
 - **Custom Types**



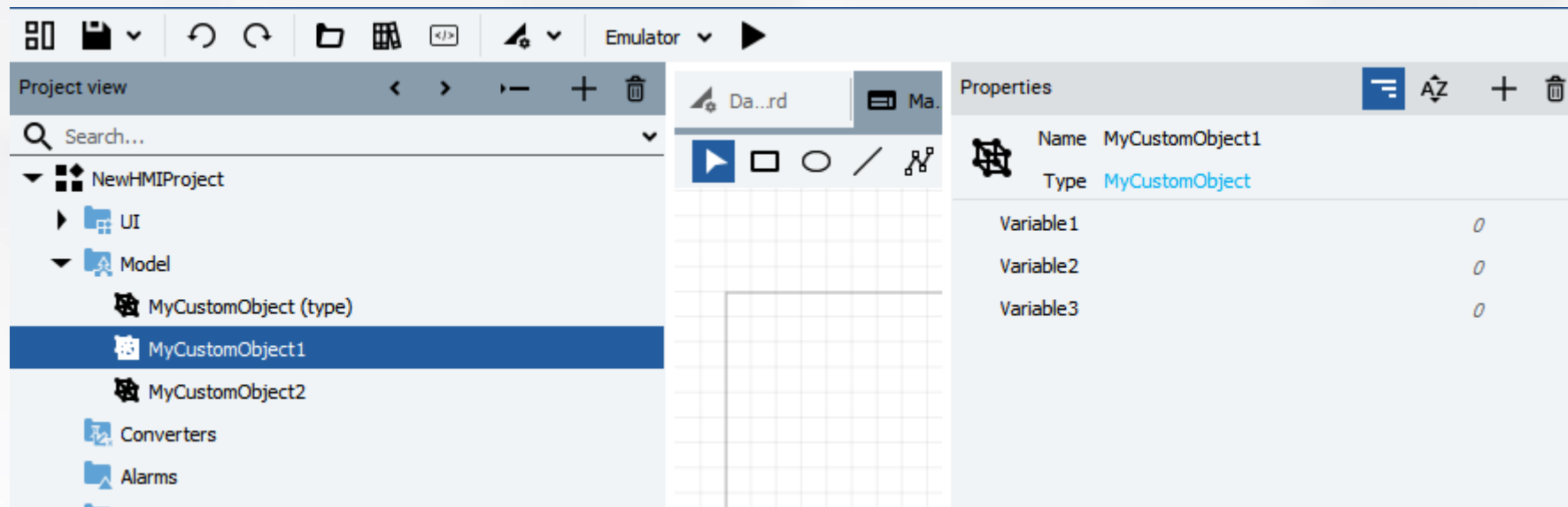
Custom types: reusable tags

- Object-Oriented Programming paradigm can be applied to Model Tags
 - "MyMotor(type)" = Class = Type
 - "MyMotor1", "MyMotor2" = Instances



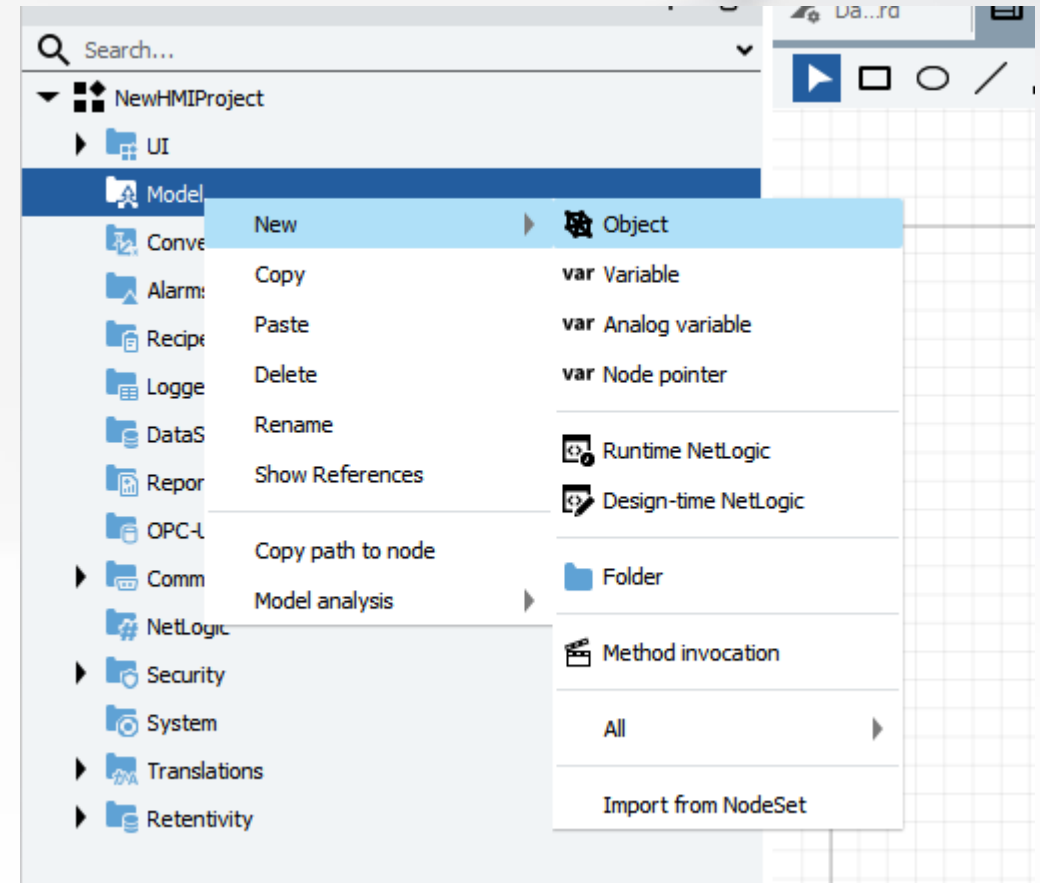
Custom types: reusable tags

- Object-Oriented Programming paradigm can be applied to Model Tags
 - "MyMotor(type)" = Class = Type
 - "MyMotor1", "MyMotor2" = Instances



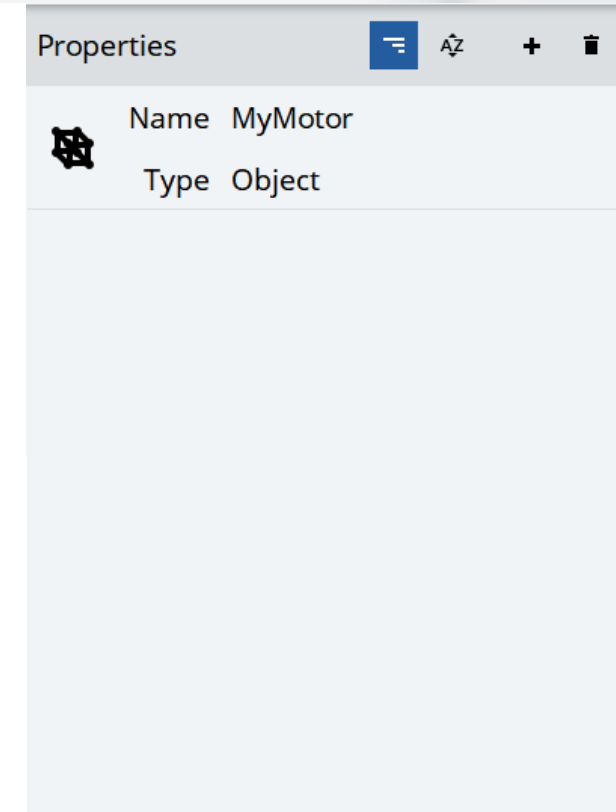
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



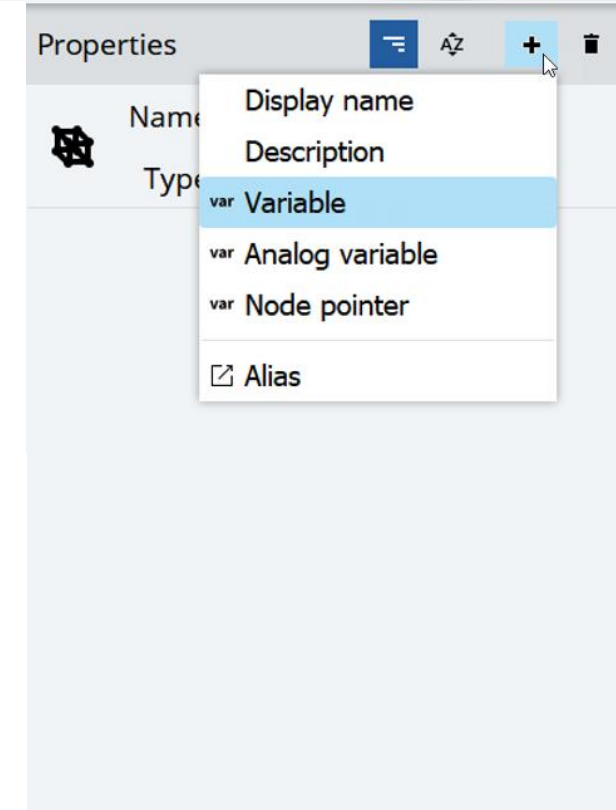
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



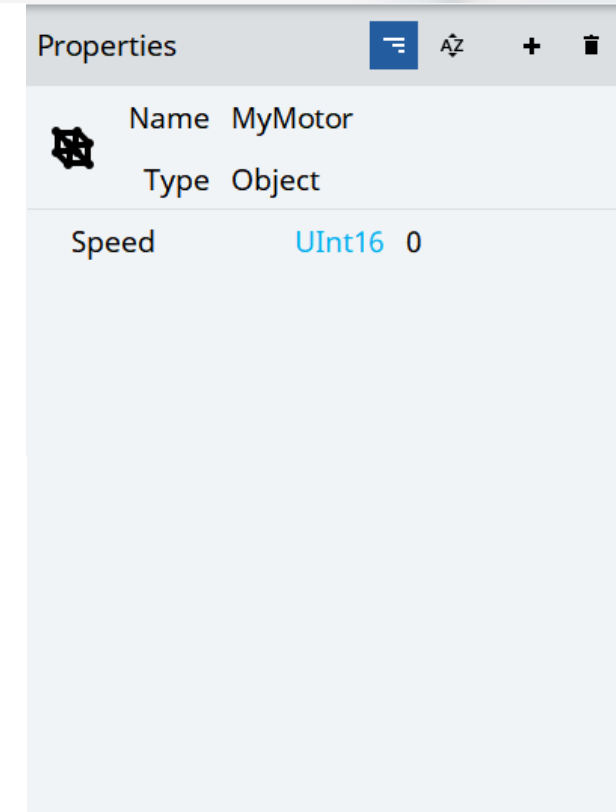
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



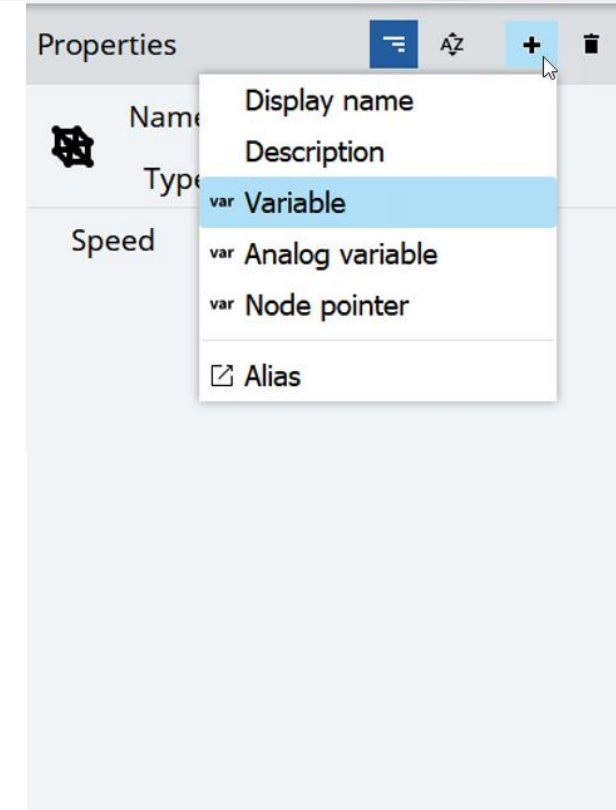
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



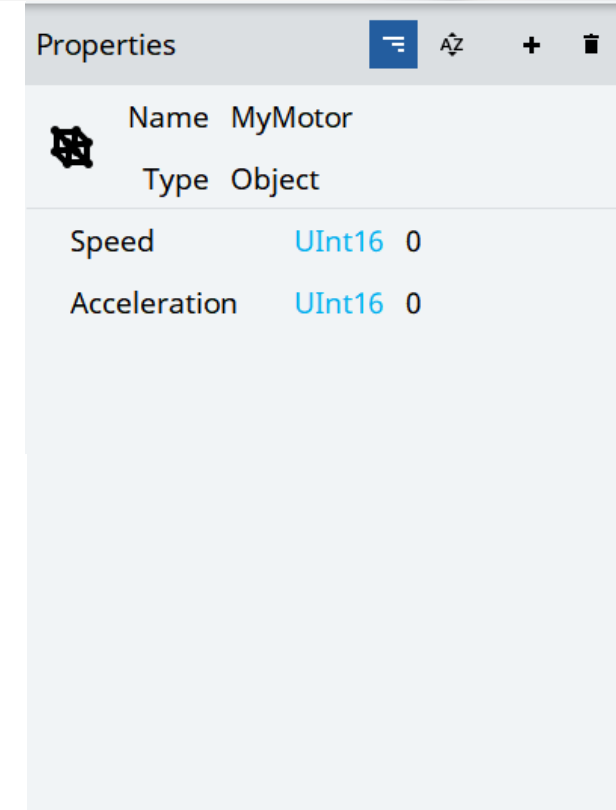
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



Custom types: reusable tags example

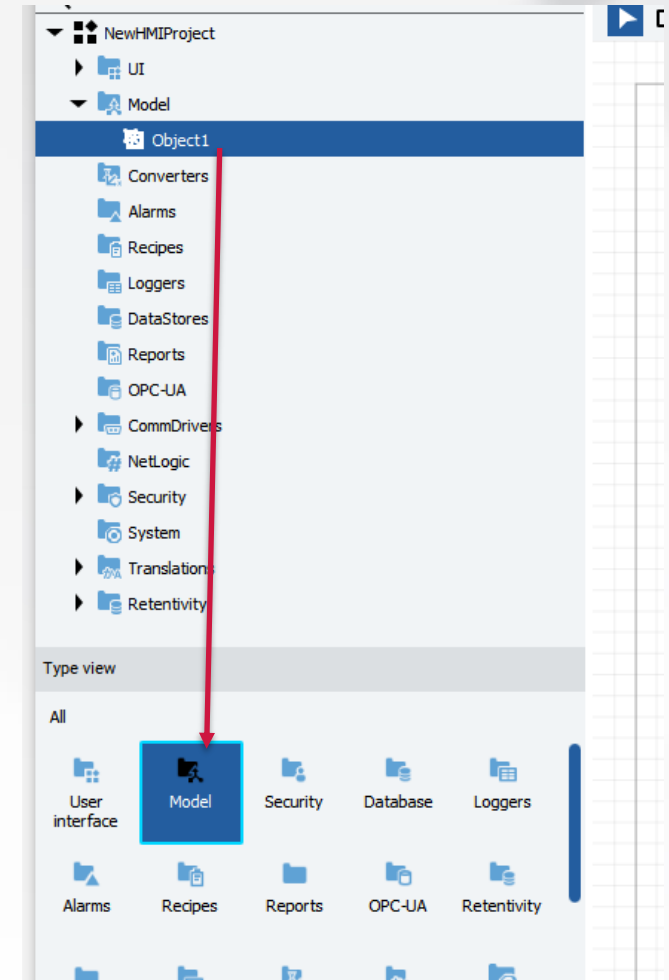
- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



Properties		
Name	MyMotor	
Type	Object	
Speed	UInt16	0
Acceleration	UInt16	0

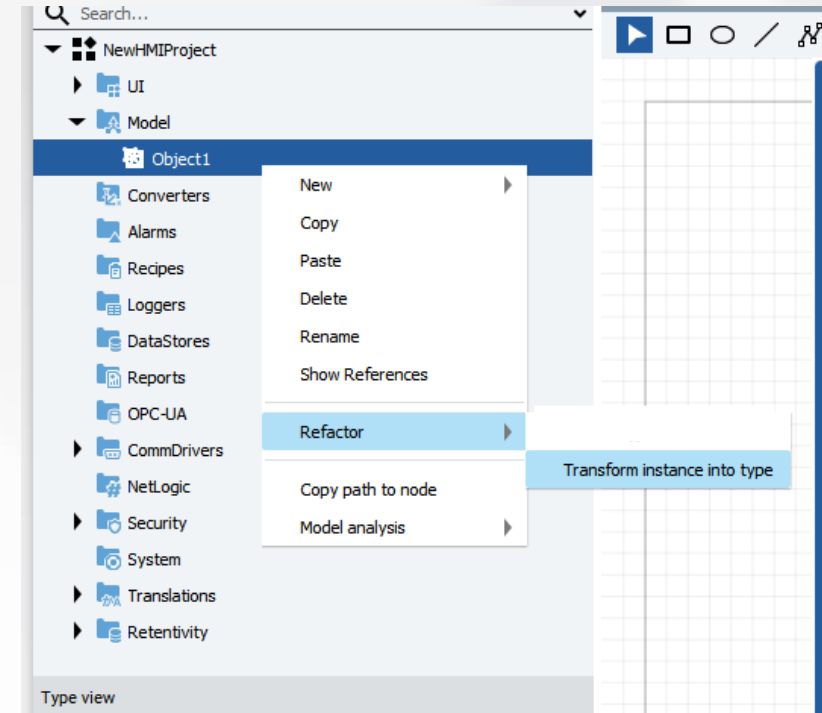
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



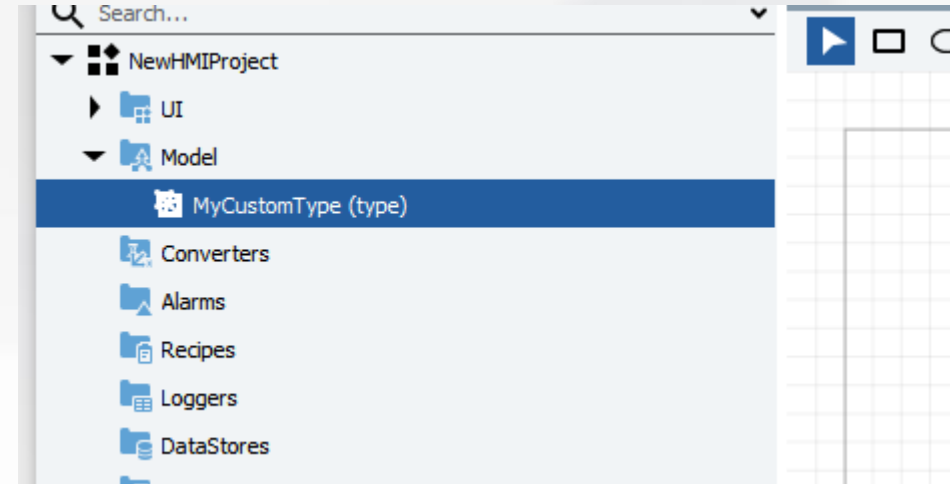
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



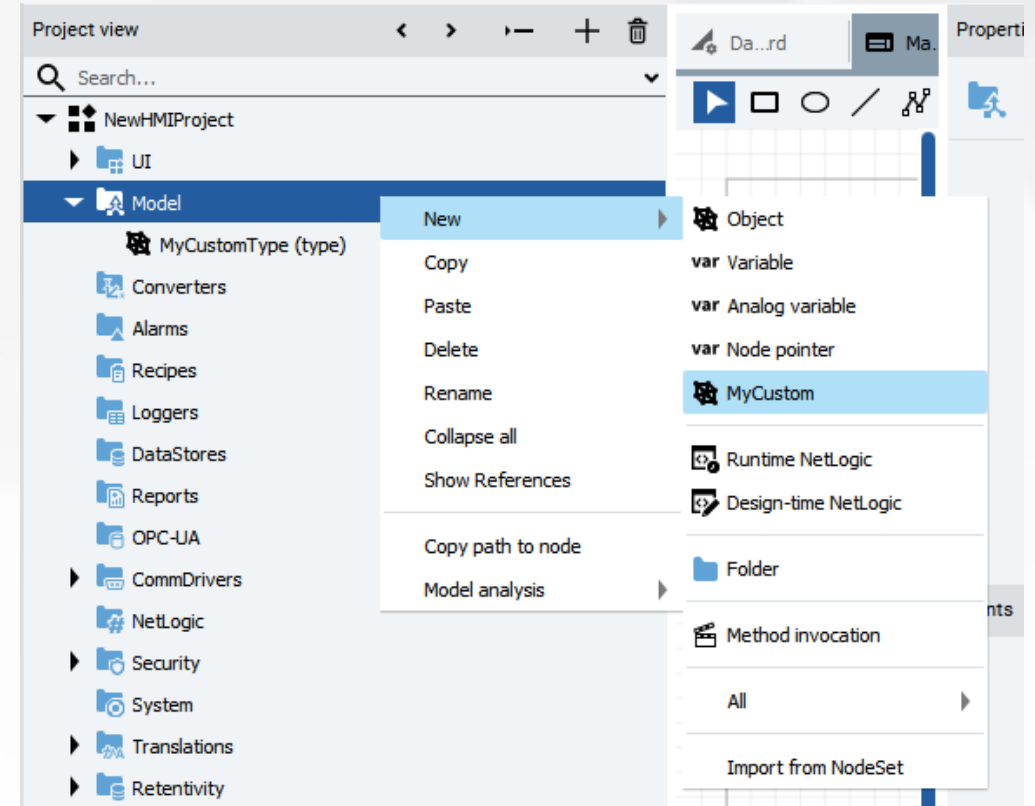
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



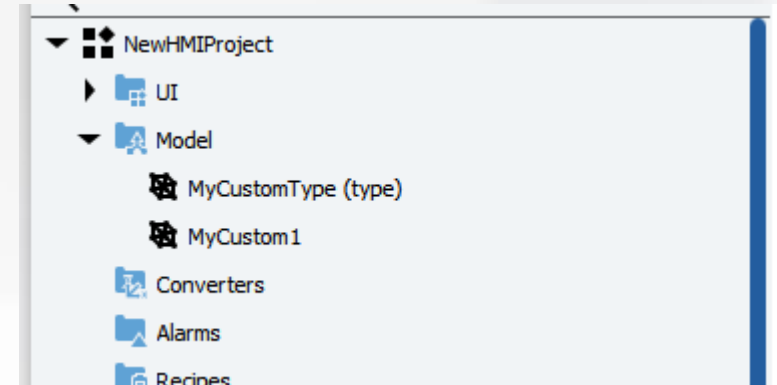
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



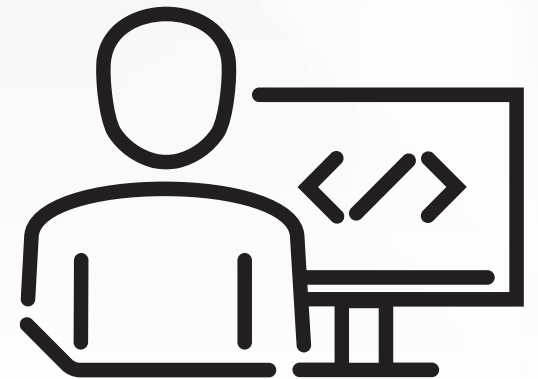
Custom types: reusable tags example

- Example: a Structured Tag into Model folder
- Created as a brand **new object** that becomes a **new type**
- How to make your object a Type:
 - Option 1: Drag&Drop into Types View panel
 - Option 2: in "Advanced Mode",
Right click > Refactor > Transform instance to type



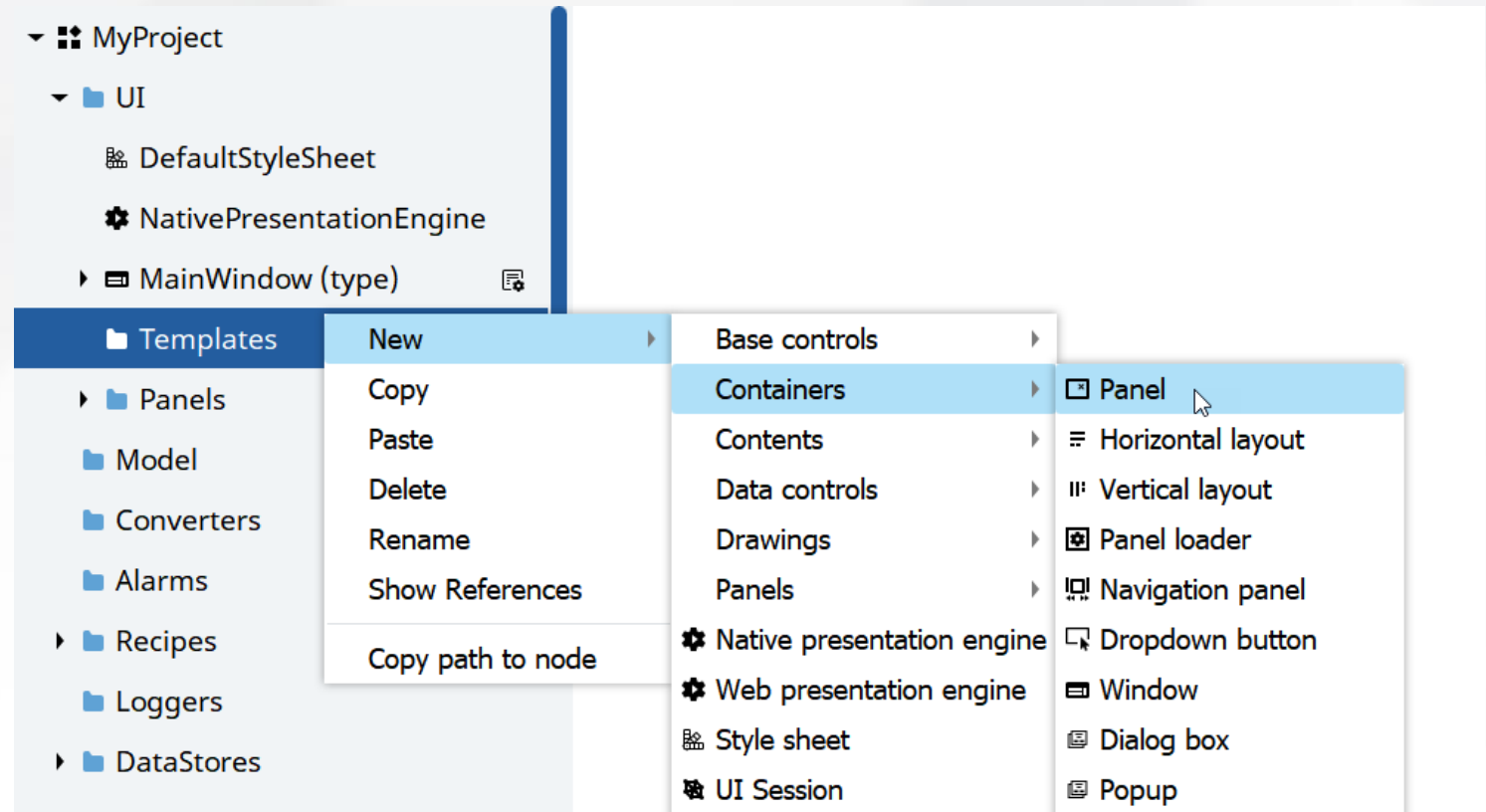
Hands-on session

- Create a Motor Type variable into the Model folder with these elements
 - Power(Float)
 - Speed(Int16)
- Create 3 instances of the Motor Type
 - Motor1, Motor2, Motor3...



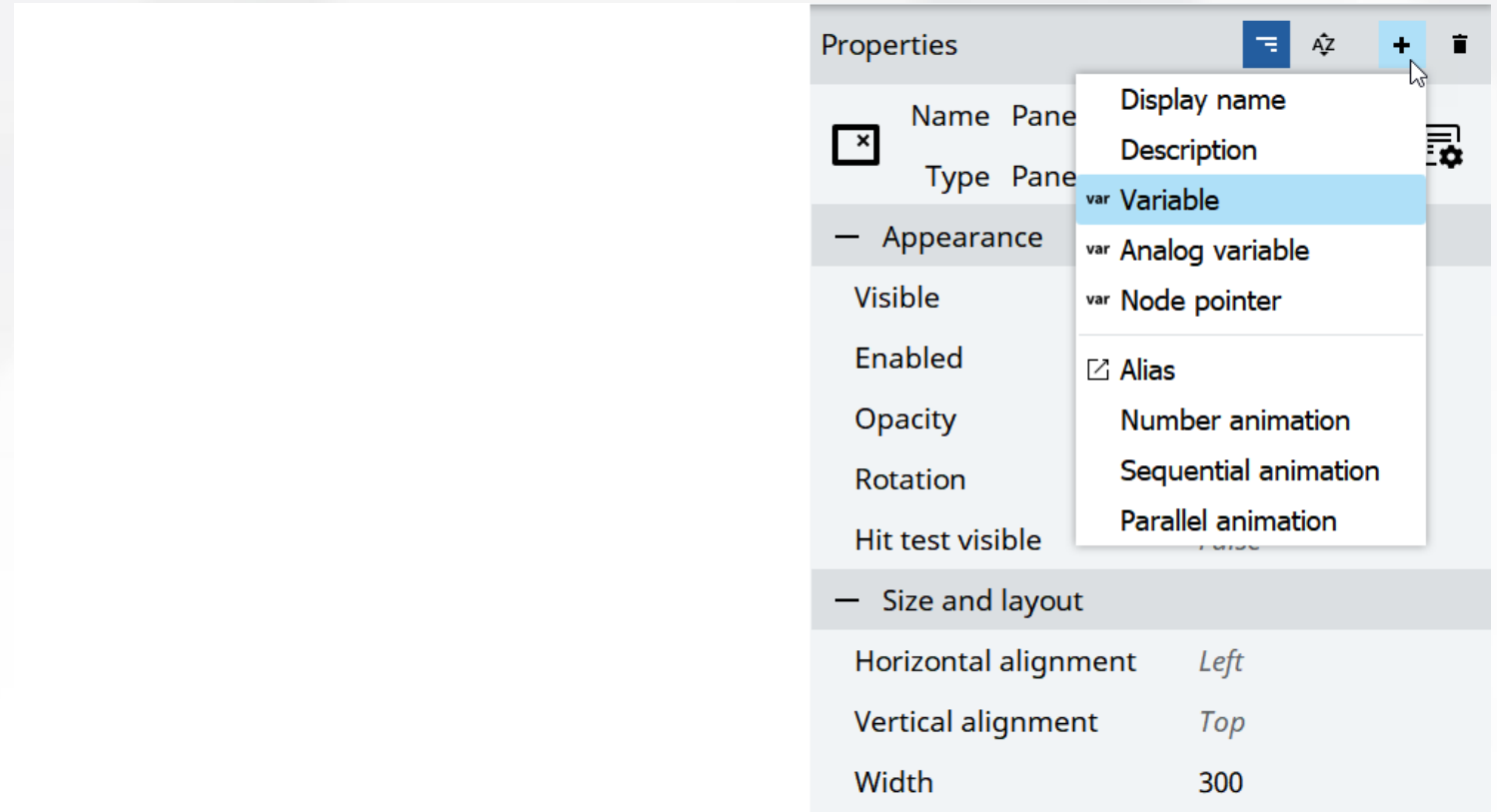
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



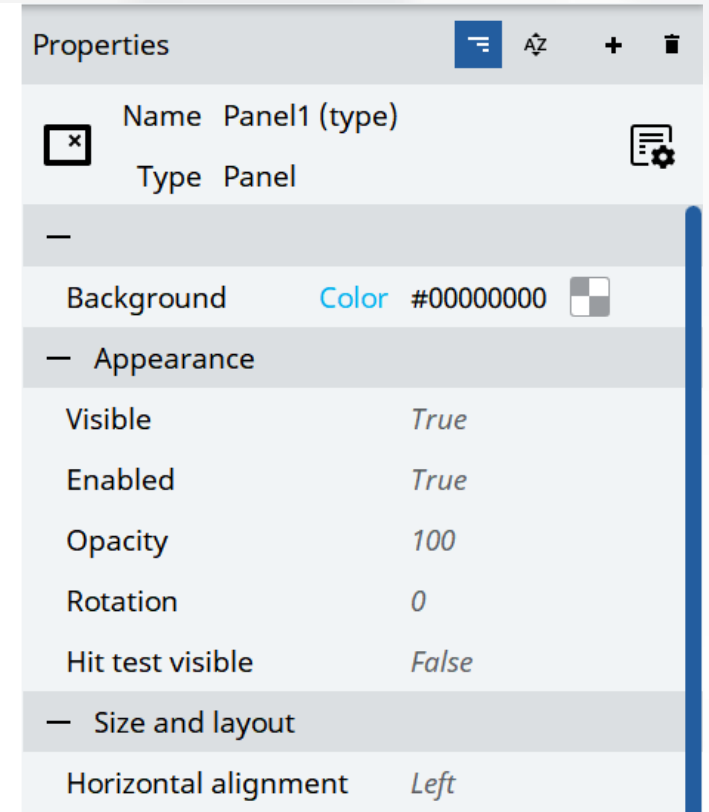
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



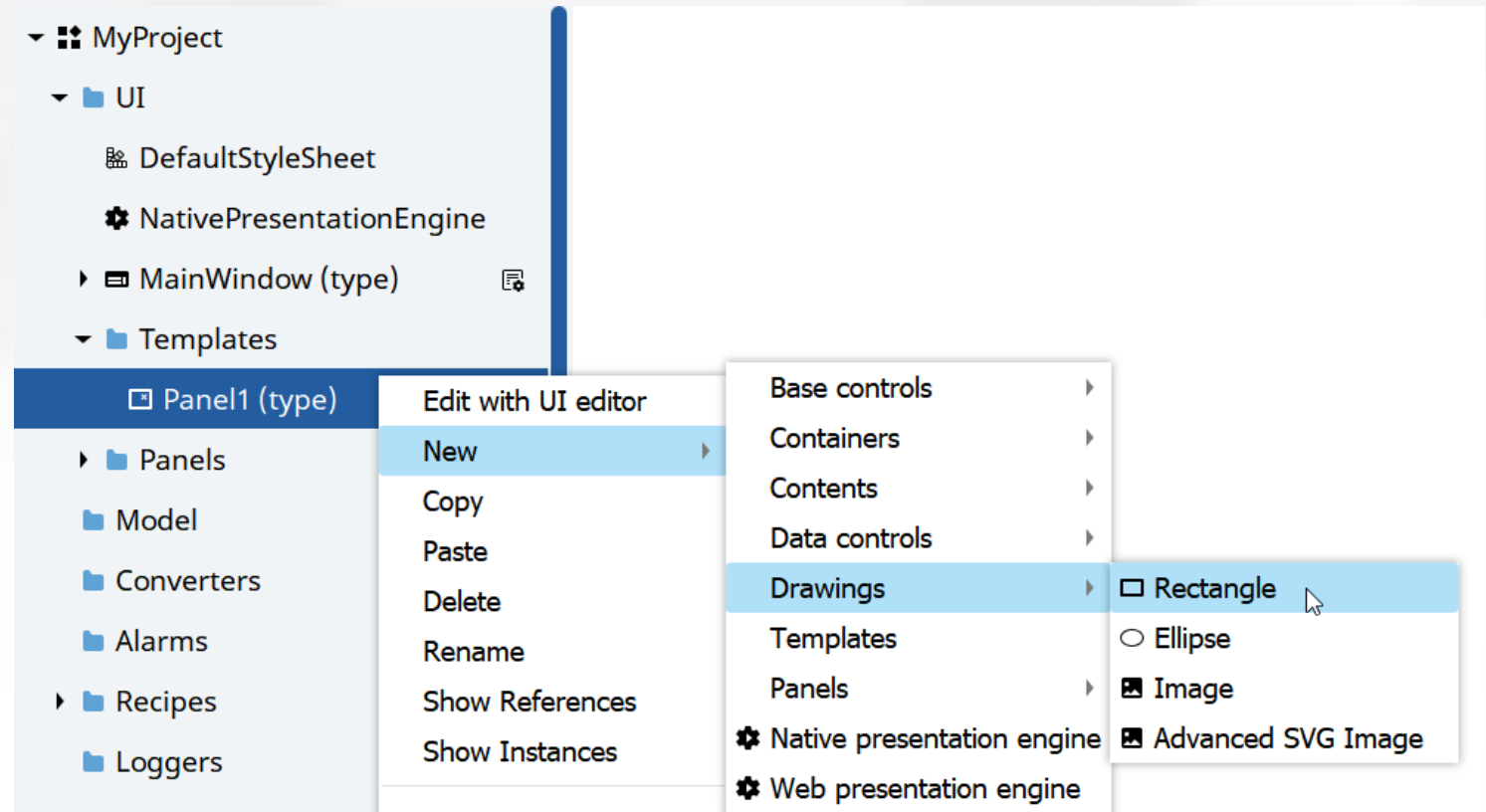
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



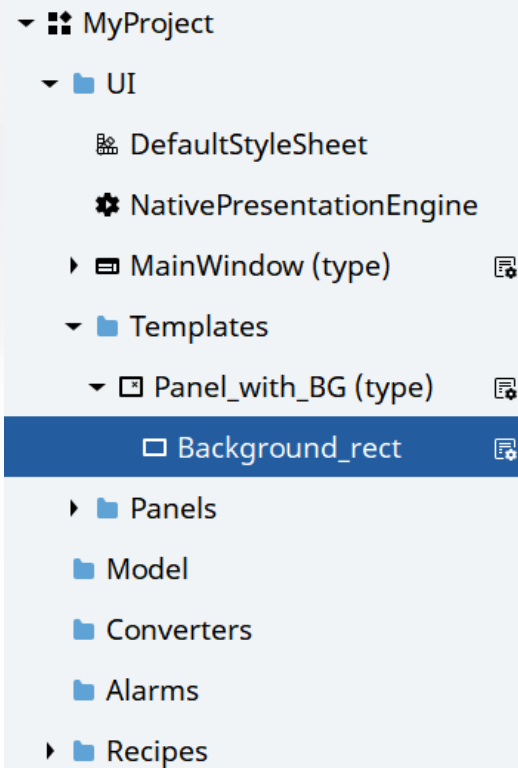
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



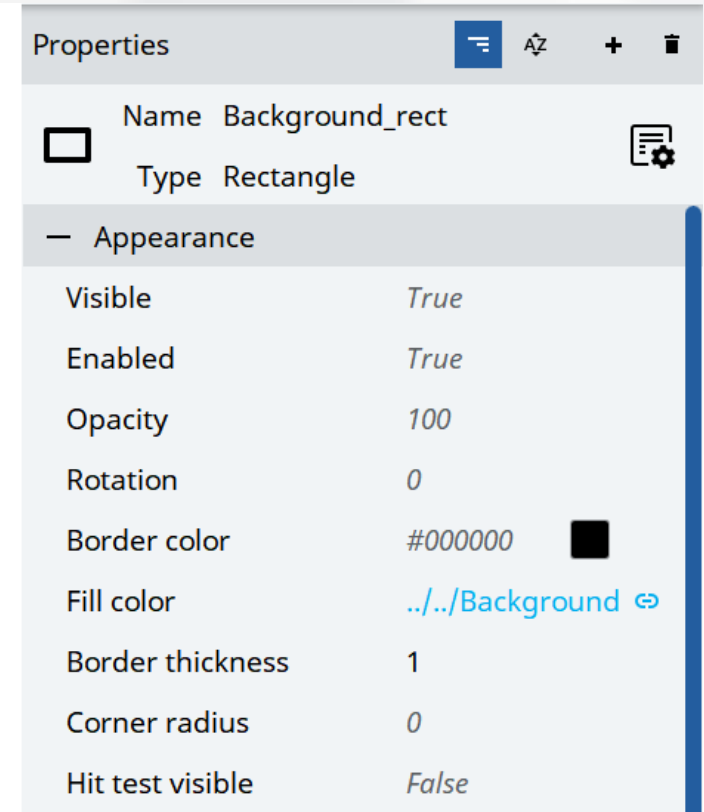
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



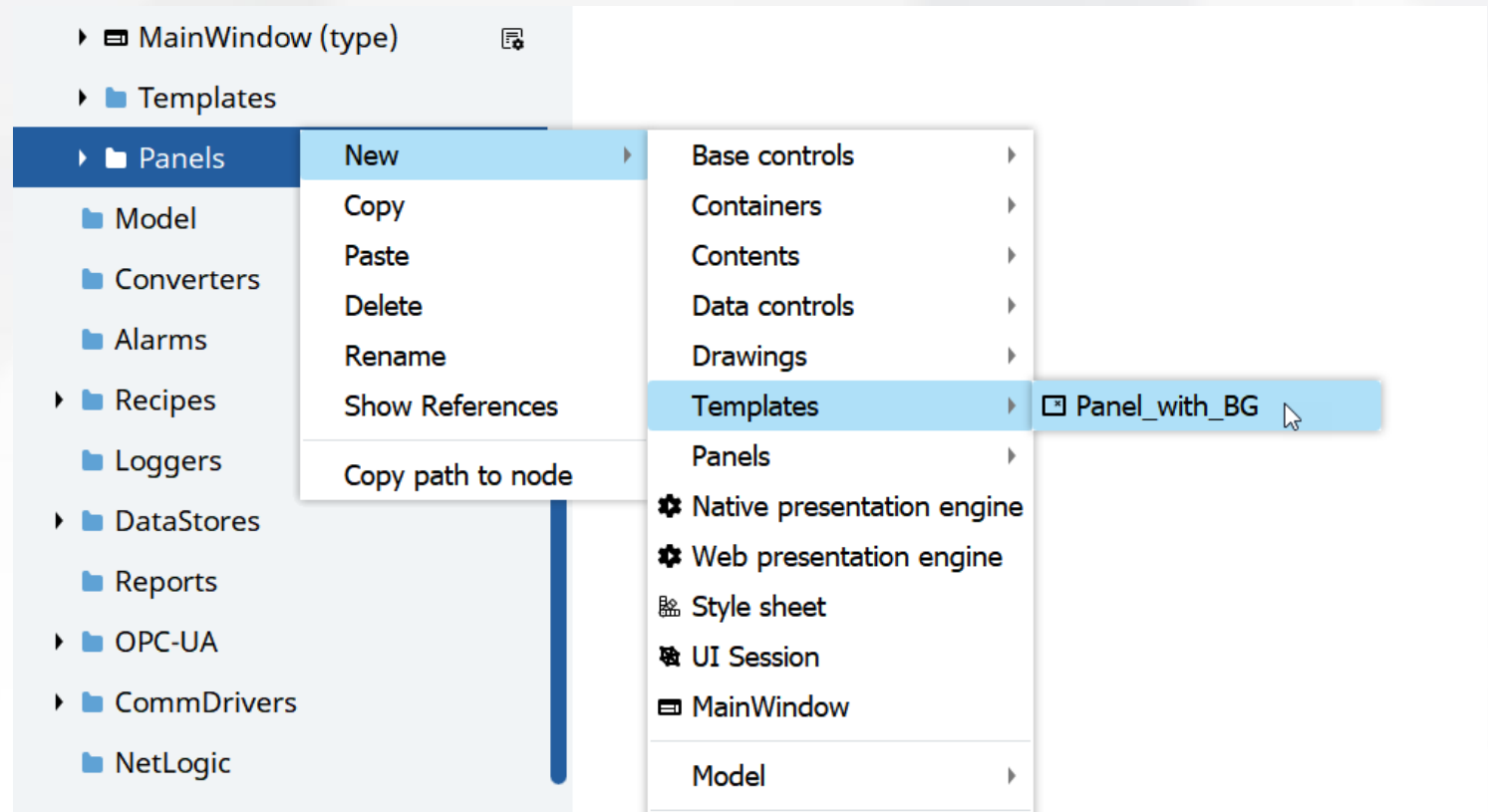
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



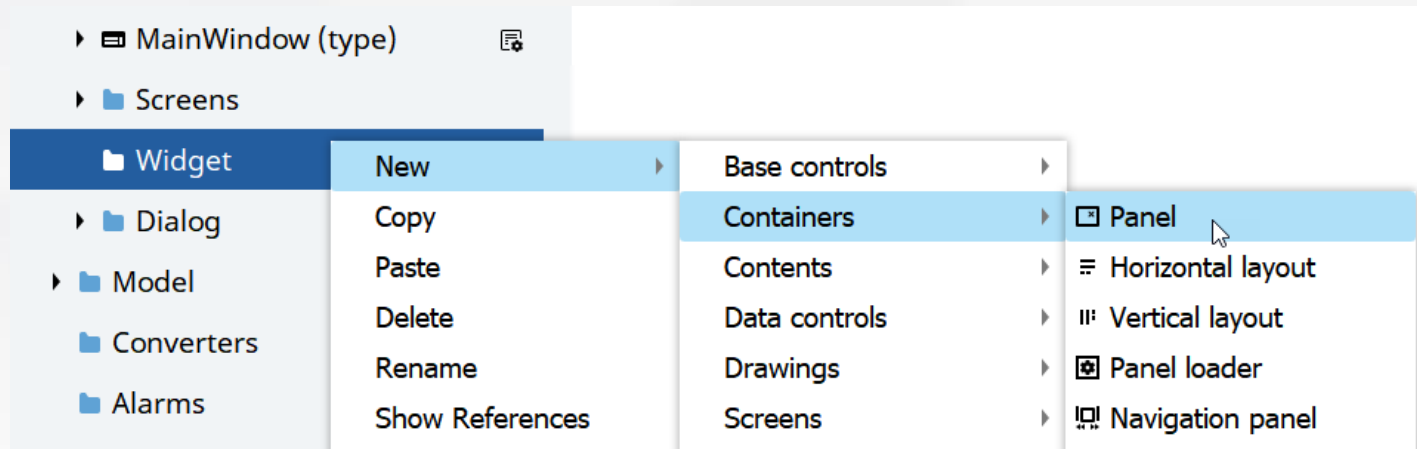
Custom types: reusable graphics with a "local variable"

- Object-Oriented Programming paradigm can be also applied to Graphics
 - "Panel_with_BG (type)" = Class = Type
 - "Panel1" = Instance
- Reusable Graphics can be
 - Widgets
 - Dialogs
 - Panels
- Example: a customized Panel with a rectangle used to set the background color



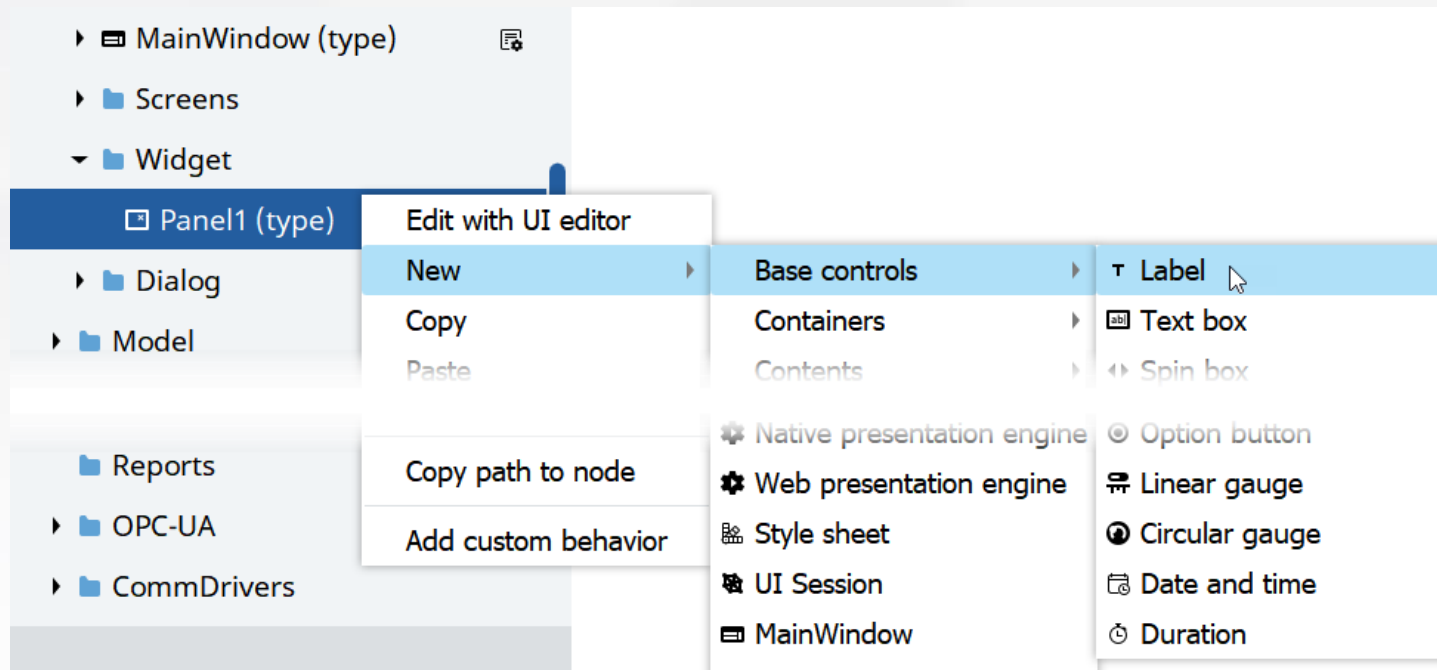
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



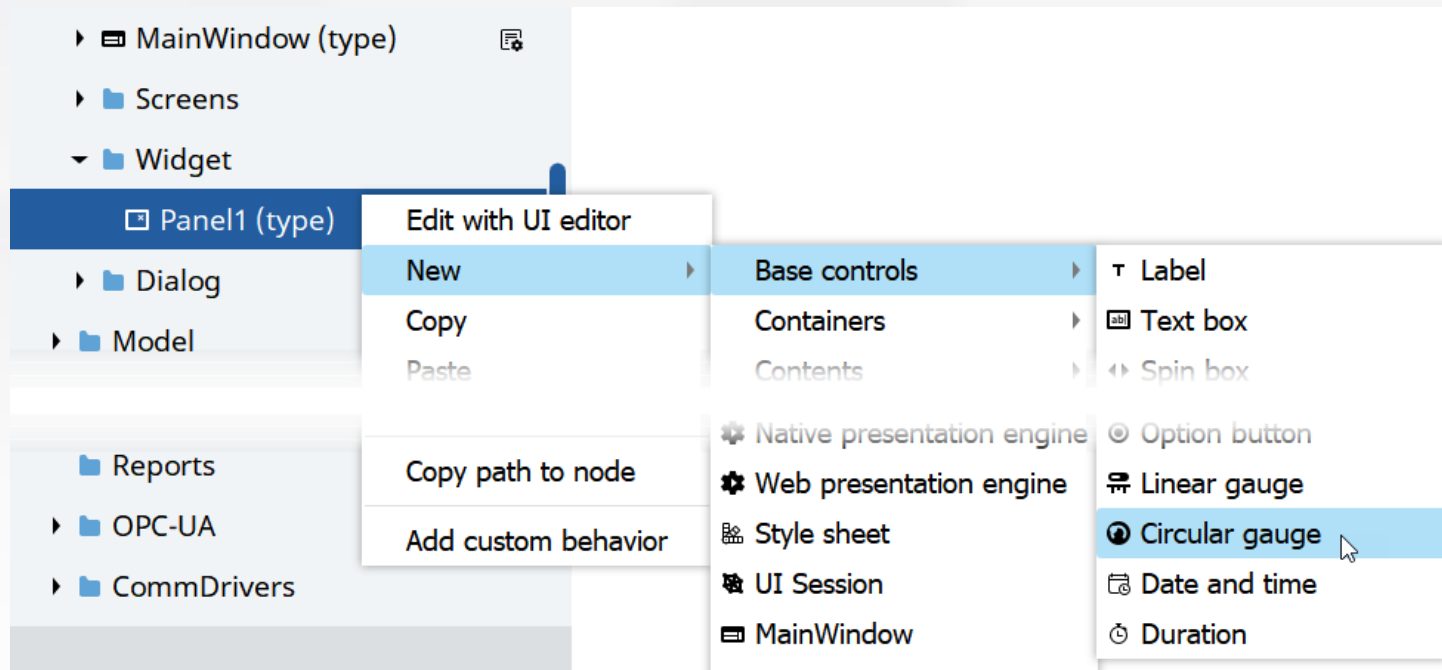
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



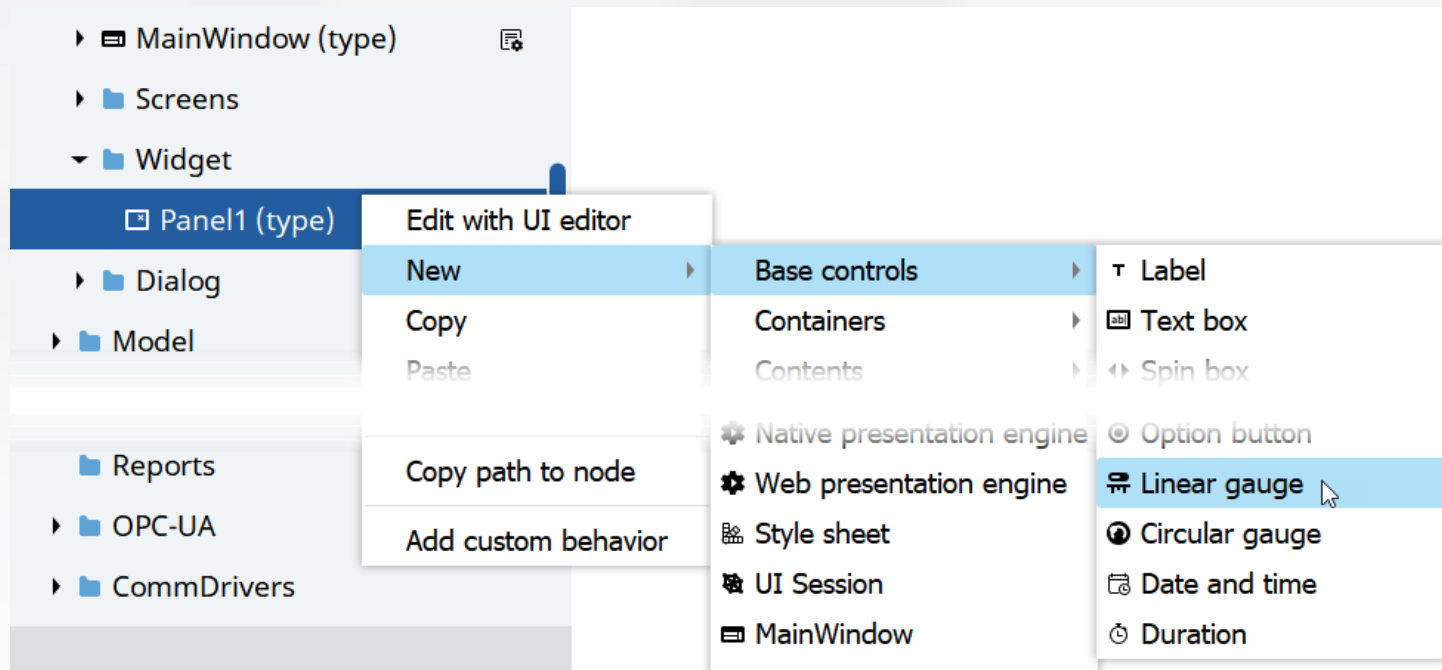
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



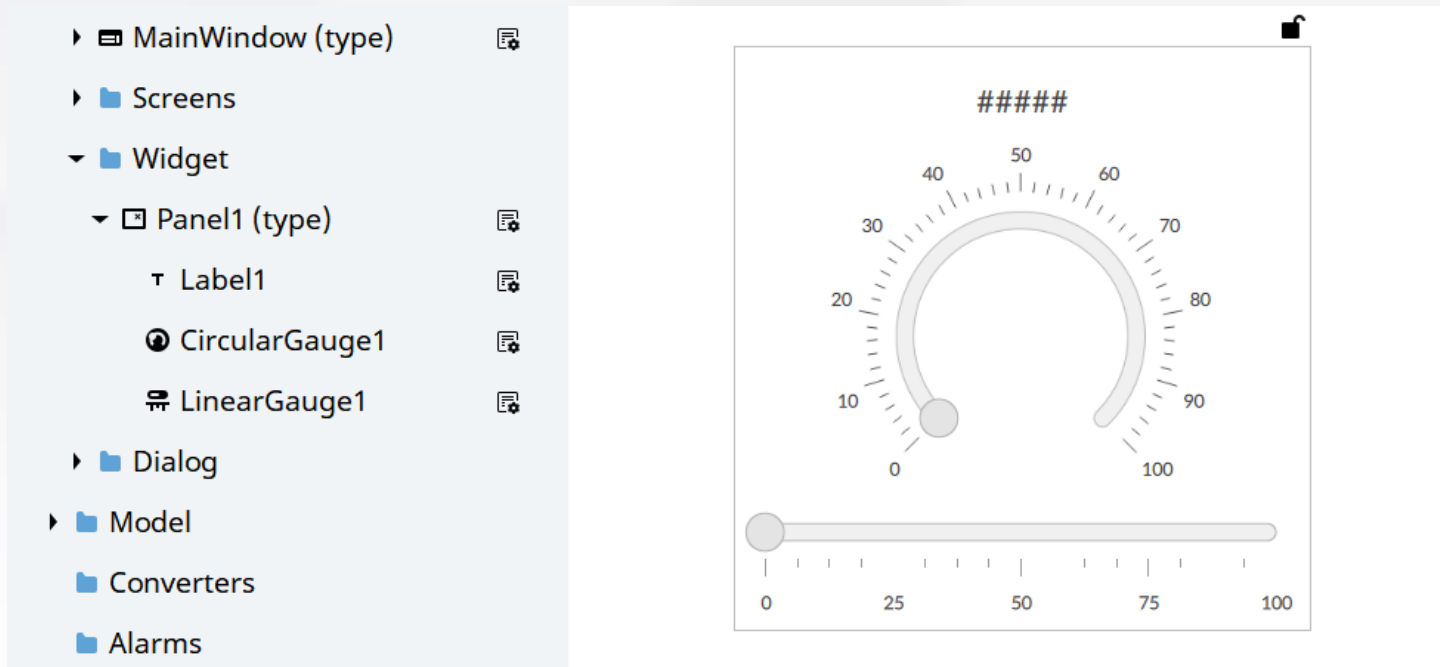
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



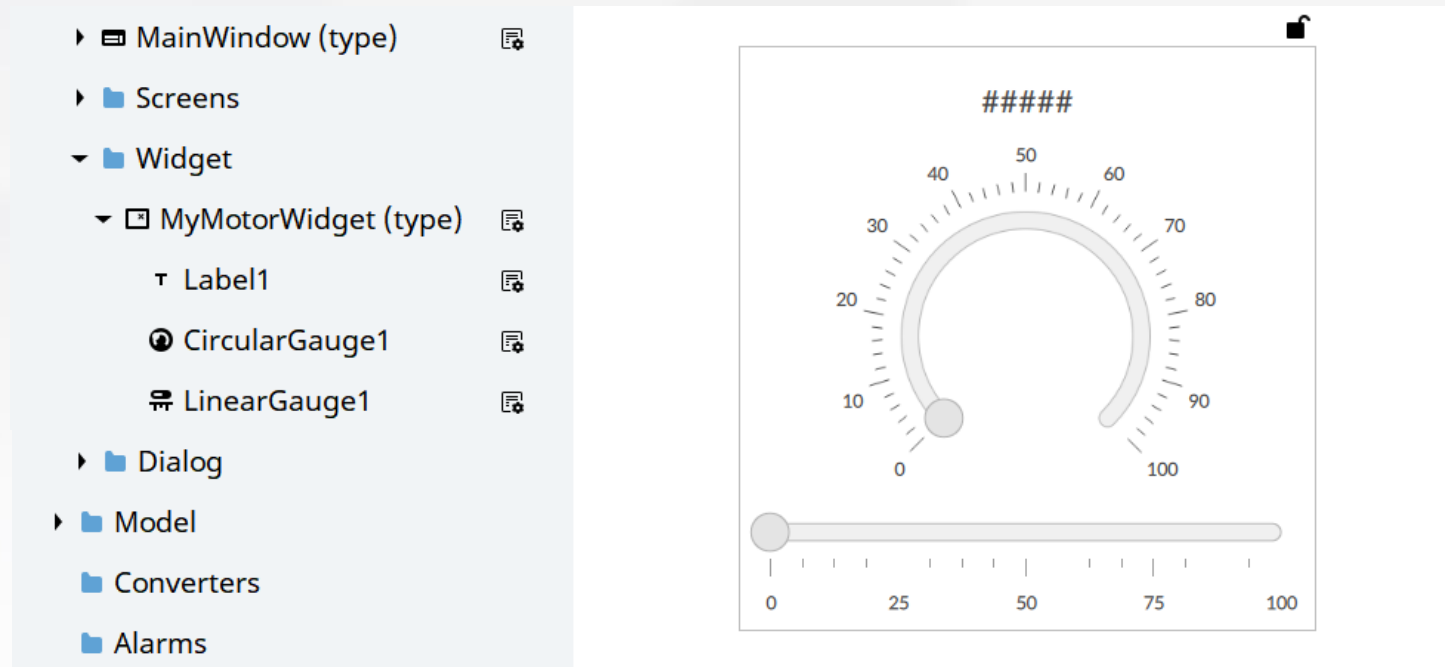
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



The image shows a software development environment with a project tree on the left and a motor widget visualization on the right.

Project Tree:

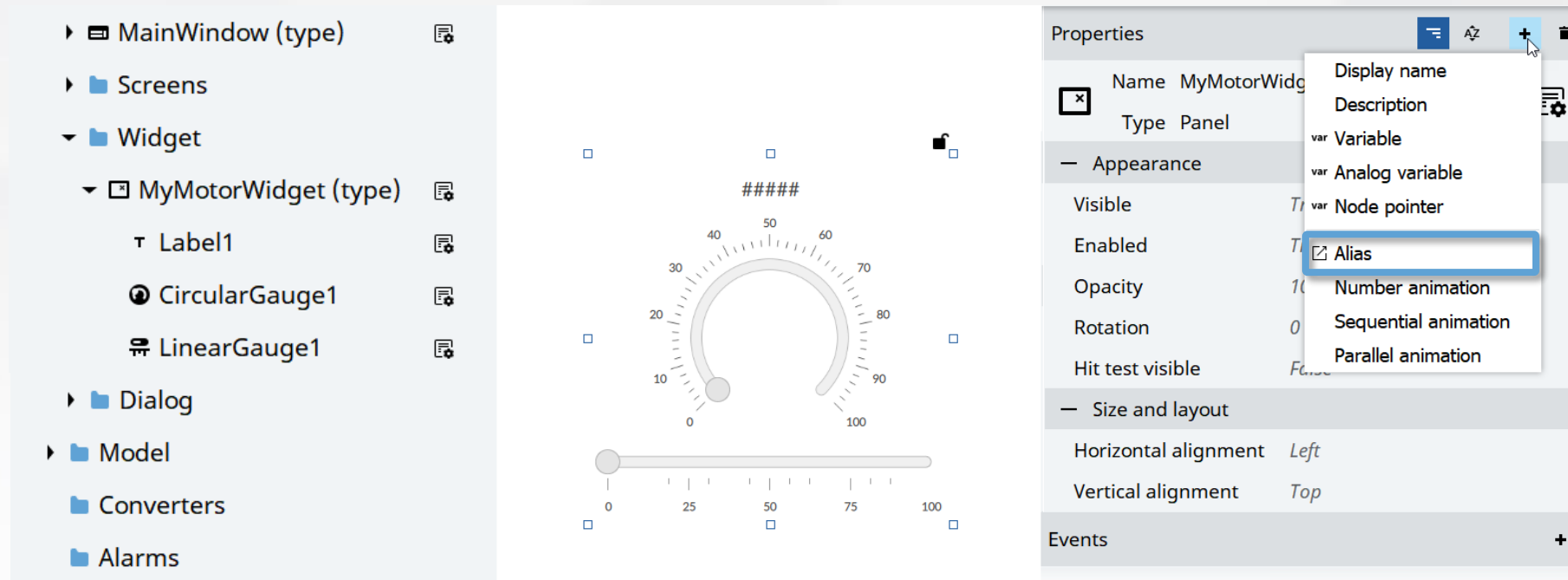
- ▶ **MainWindow (type)**
- ▶ **Screens**
- ▼ **Widget**
 - ▼ **MyMotorWidget (type)**
 - └ **Label1**
 - └ **CircularGauge1**
 - └ **LinearGauge1**
 - ▶ **Dialog**
 - ▶ **Model**
 - ▶ **Converters**
 - ▶ **Alarms**

Motor Widget Visualization:

The visualization displays a circular gauge and a linear gauge. The circular gauge has a scale from 0 to 100, with major ticks every 10 units and minor ticks every 2 units. The needle is positioned at approximately 35. The linear gauge is a horizontal bar with a scale from 0 to 100, with major ticks every 25 units and minor ticks every 5 units. The slider is positioned at approximately 35. Above the circular gauge, the text "#####" is displayed.

Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor

The screenshot displays the FactoryTalk Studio interface with three main components:

- Project Explorer (Left):** A tree view showing the project structure. It includes a 'Widget' folder containing 'MyMotorWidget (type)', which has sub-items 'Label1', 'CircularGauge1', and 'LinearGauge1'.
- Canvas (Center):** A workspace showing a preview of the 'MyMotorWidget'. It features a circular gauge with a needle pointing to approximately 35, a linear gauge below it, and a label '#####' at the top.
- Properties Panel (Right):** A panel showing the configuration for the selected 'MyMotorWidget (type)'. It includes fields for 'Name', 'Type', and a 'MotorAlias' section with a 'Kind' property set to 'MyMotor (type)'. Other properties like 'Visible', 'Enabled', 'Opacity', 'Rotation', and 'Hit test visible' are also listed.

Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor

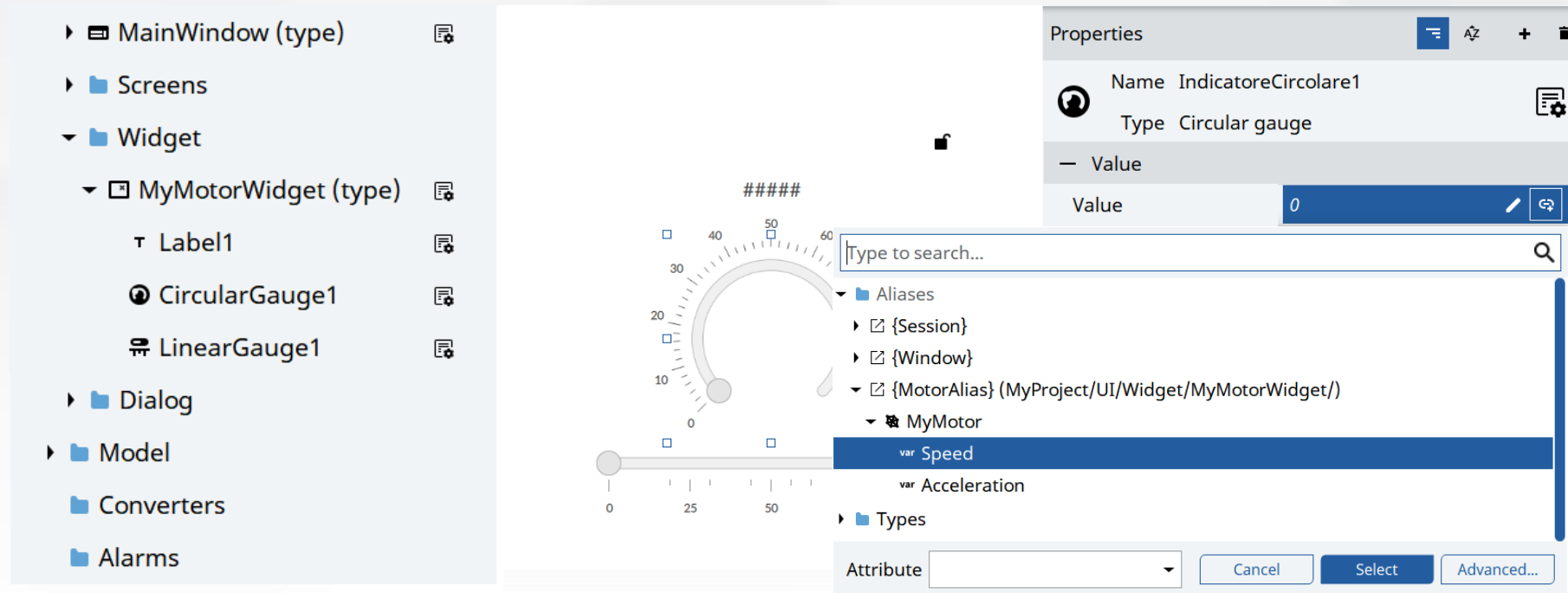
The screenshot displays the FactoryTalk Studio interface. On the left, the Project Explorer shows a hierarchy: **MainWindow (type)** > **Screens** > **Widget** > **MyMotorWidget (type)**. Under **MyMotorWidget (type)**, there are three sub-items: **Label1**, **CircularGauge1**, and **LinearGauge1**. Below this, there are folders for **Dialog**, **Model**, **Converters**, and **Alarms**.

In the center, a preview of the **MyMotorWidget** is shown. It features a circular gauge with a needle pointing to approximately 45, a linear gauge below it, and a label displaying "#####" above the gauges. The linear gauge has a scale from 0 to 100.

On the right, the **Properties** window is open for the selected **CircularGauge1** object. The **Name** is **IndicatoreCircolare1** and the **Type** is **Circular gauge**. The **Value** property is set to **0**, with **Minimum value** at **0** and **Maximum value** at **100**. The **Value change behavior** is set to **...ue change on edit finished**. The **Gauge** section shows **Start angle** at **-135**, **End angle** at **135**, and **Editable** set to **True**. The **Events** section is currently empty.

Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor

The screenshot displays the FactoryTalk Studio interface. On the left, a project tree shows the hierarchy: **MainWindow (type)** > **Screens** > **Widget** > **MyMotorWidget (type)**. Under **MyMotorWidget (type)**, the components **Label1**, **CircularGauge1**, and **LinearGauge1** are listed. The central workspace shows a preview of the **MyMotorWidget**, which includes a circular gauge with a needle pointing to approximately 45, a label **#####** above it, and a linear gauge below it. The right-hand **Properties** pane is open for the **CircularGauge1** widget. It shows the following configuration:

Properties	
Name	IndicatoreCircolare1
Type	Circular gauge
Value	
Value	{MotorAlias}/Speed
Minimum value	0
Maximum value	100
Value change behavior	...ue change on edit finished
Gauge	
Start angle	-135
End angle	135
Editable	True
Events	

Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor

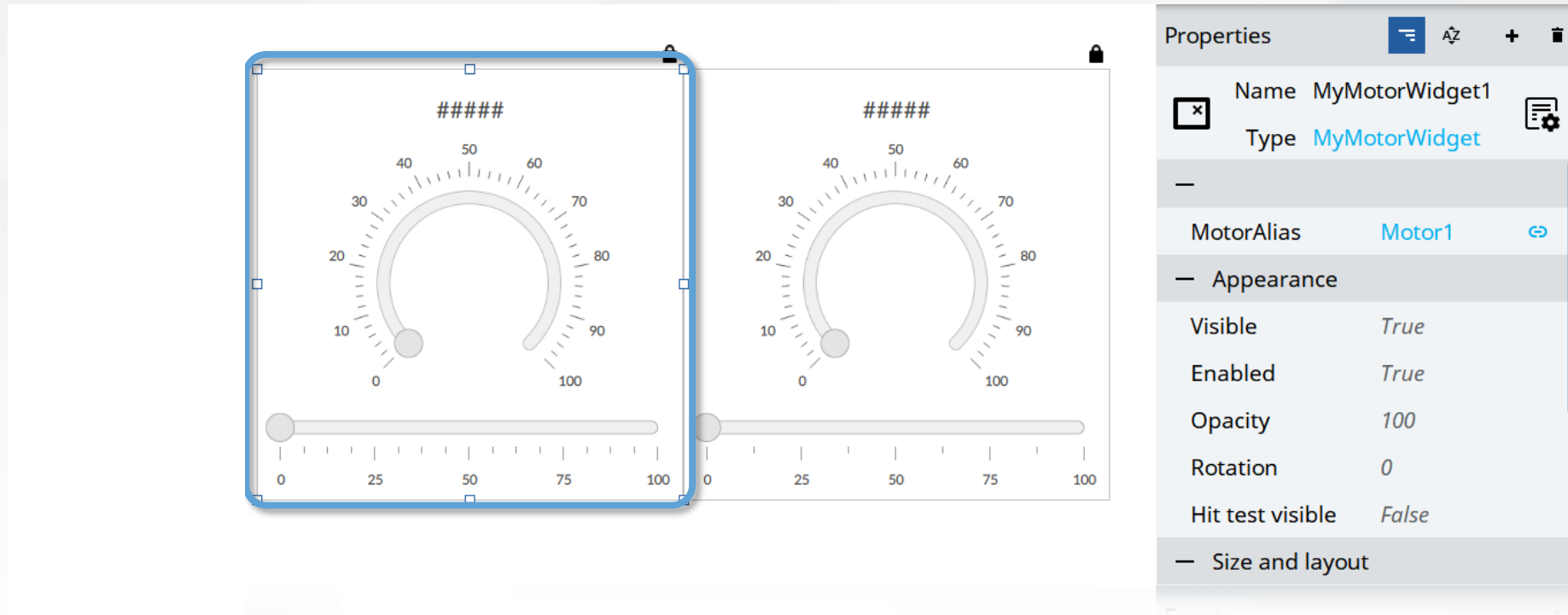
The screenshot displays the FactoryTalk Studio interface. On the left, the Project Explorer shows a hierarchy: **MainWindow (type)** > **Screens** > **Widget** > **MyMotorWidget (type)**. Under **MyMotorWidget (type)**, there are three sub-items: **Label1**, **CircularGauge1**, and **LinearGauge1**. Below this, there are folders for **Dialog**, **Model**, **Converters**, and **Alarms**.

The central canvas shows a preview of the **MyMotorWidget**. It features a circular gauge with a needle pointing to approximately 45, labeled with '#####' at the top. Below the circular gauge is a horizontal linear gauge with a slider bar and numerical markings from 0 to 100 in increments of 25.

On the right, the **Properties** pane is open for the selected **IndicatorLineare1** object. The **Type** is **Linear gauge**. The **Value** property is set to **{MotorAlias}/Acceleration**. The **Minimum value** is **0** and the **Maximum value** is **100**. The **Value change behavior** is set to **...ue change on edit finished**. The **Gauge** section shows **Orientation** as **Horizontal**, **Editable** as **True**, and **Major tick count** as **5**. The **Events** section is currently empty.

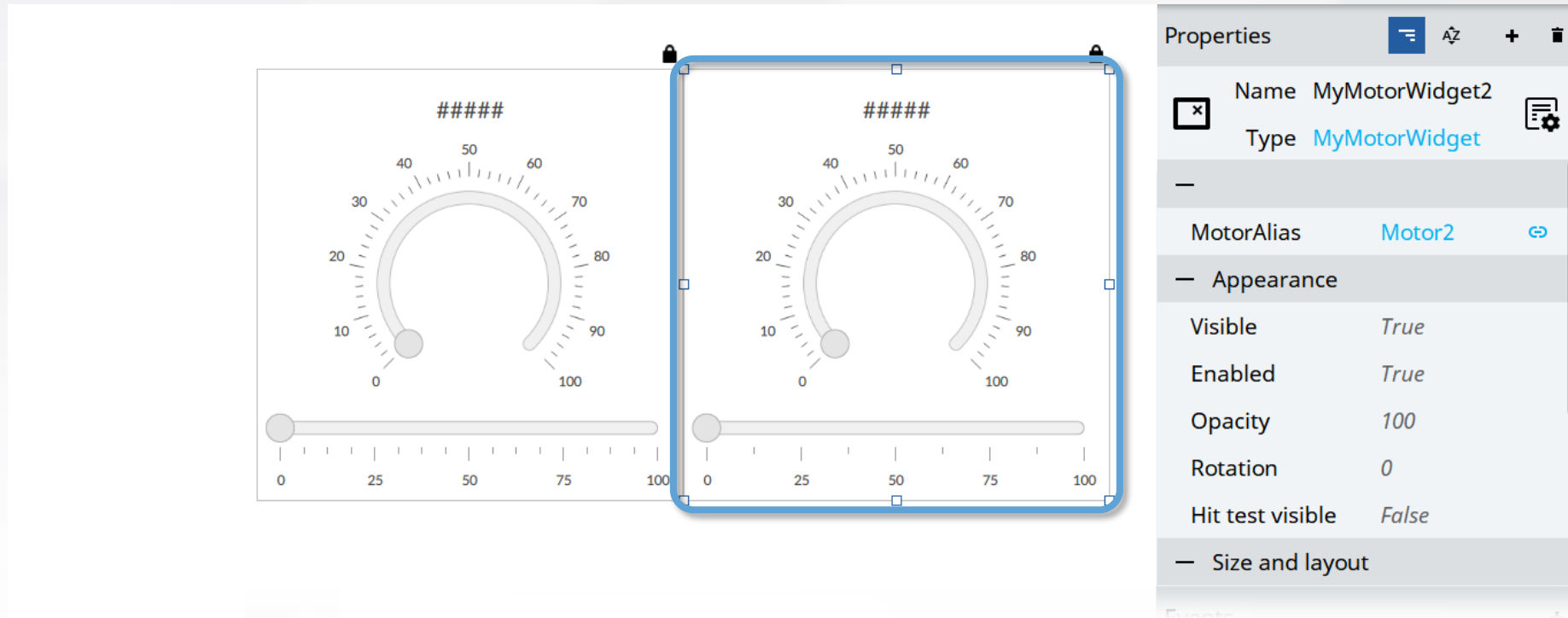
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



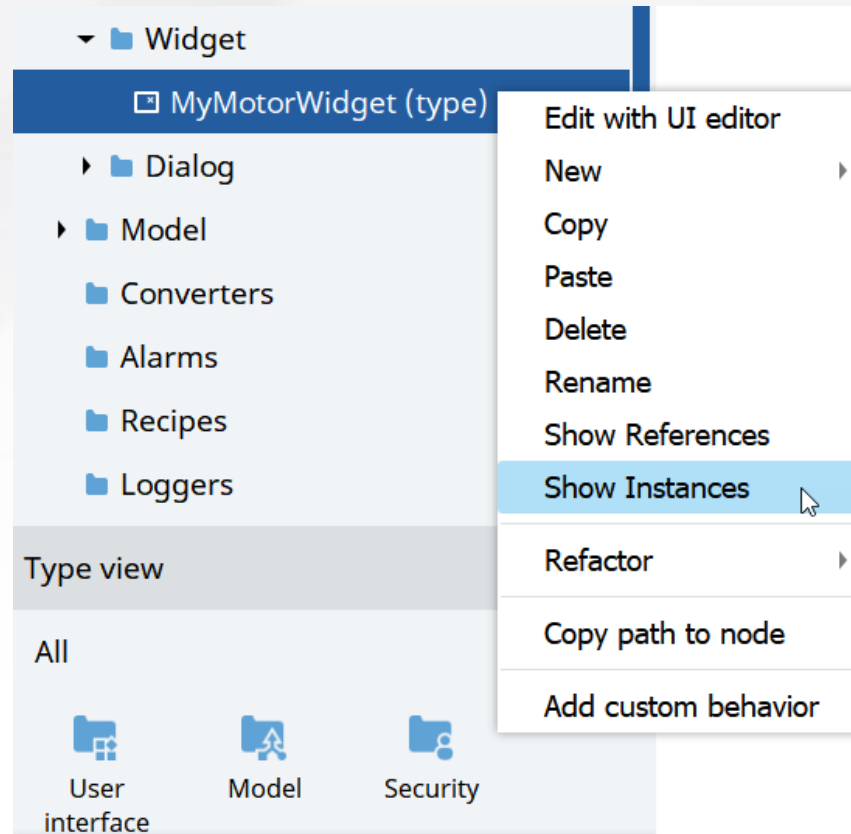
Custom types: reusable graphics with an alias

- Example: a motor widget that shows the values of different instances of the motor



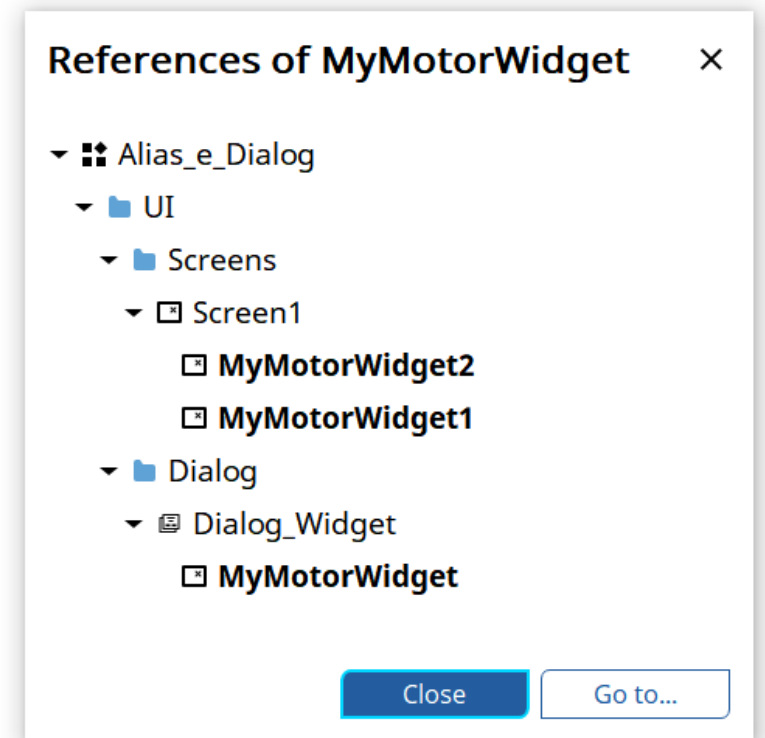
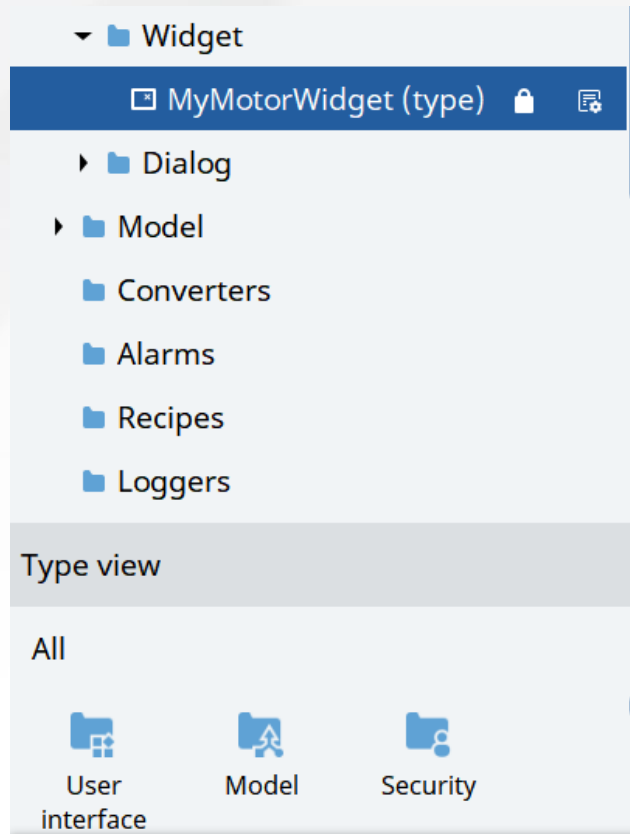
Custom types: show instances

- Answer the question: "**where are the instances of this type?**"
- Can be used to find instances of
 - Widgets,
 - Panels,
 - Tags (Structure)
 - "typed" Alarm
 - ...



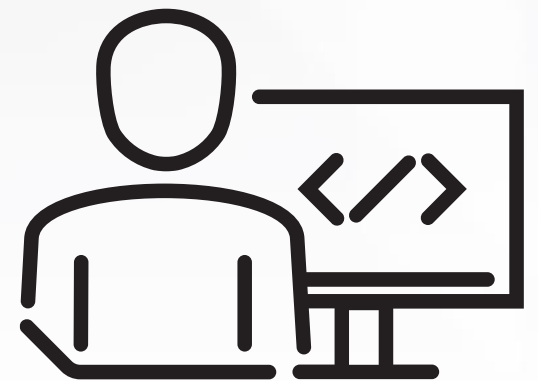
Custom types: show instances

- Answer the question: "**where are the instances of this type?**"
- Can be used to find instances of
 - Widgets,
 - Panels,
 - Tags (Structure)
 - "typed" Alarm
 - ...



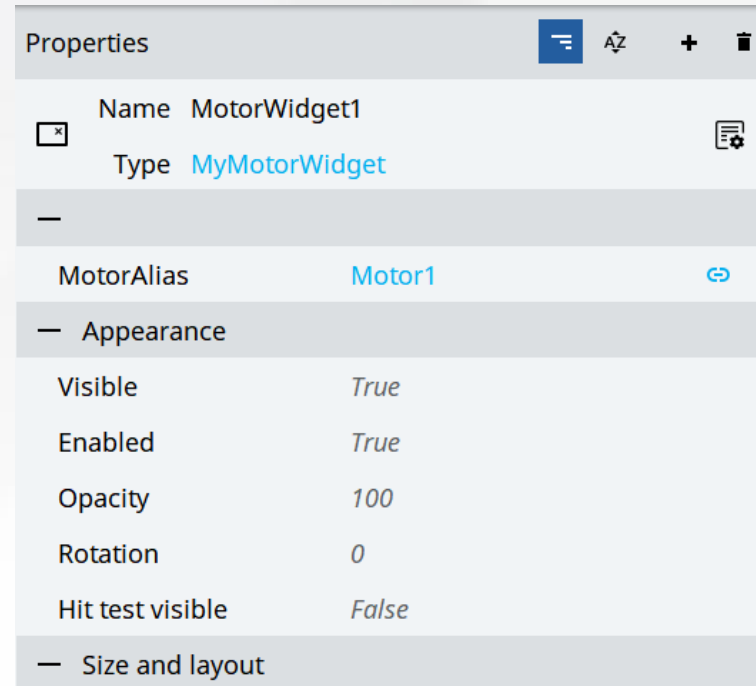
Hands-on session

- Create a widget called MotorWidget as a Type with an Alias to be used with the Motor variables
- Add 3 instances of the MotorWidget into a new Panel
- Specify, at design time, the alias on each MotorWidget instance associating to each Motor variable



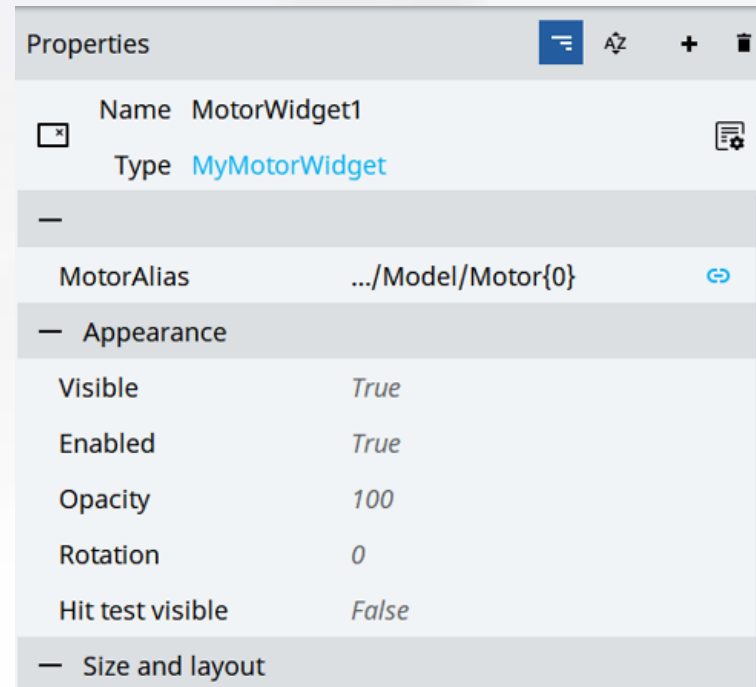
Reusable graphics: set aliases using indexing

- Alias variables can be indexed like simple variables
- Allow to easily change the alias of a widget



Reusable graphics: set aliases using indexing

- Alias variables can be indexed like simple variables
- Allow to easily change the alias of a widget



Reusable graphics: set aliases using indexing

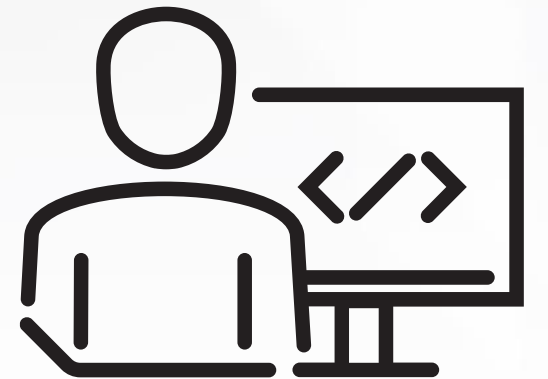
- Alias variables can be indexed like simple variables
- Allow to easily change the alias of a widget

The screenshot displays the FactoryTalk Graphics Designer interface. On the left, the **Properties** window for **MotorWidget1** (Type: **MyMotorWidget**) is shown. The **MotorAlias** property is set to `.../Model/Motor{0}`. Below this, the **Appearance** section lists properties: **Visible** (True), **Enabled** (True), **Opacity** (100), **Rotation** (0), and **Hit test visible** (False). The **Size and layout** section is partially visible.

On the right, the **MotorAlias** widget configuration window is shown. It contains a **Formatted dynamic link** set to `.../Model/Motor{0}`. Below this, a **{0} DynamicLink** is configured with the path `../..../SpinBox1/Value`. The window title bar indicates the hierarchy: **> MainWindow > MotorWidget1 > MotorAlias**.

Hands-on session

- Add just only one instance of the MotorWidget into a new Panel
- add also a Switch object in the Panel
- use the "indexing" method to define the Alias on the MotorWidget instance, associating the index to the Switch value



Reusable graphics: set aliases at runtime

- For Dialog and Panels, the Alias at Runtime can be set on:
 - Open Dialog command
 - Change Panel command
 - Navigation Panel properties
 - Panel Loader properties

Events		+
⚡	MouseClicked event	+
▼ ⚡	MouseDown event	+
▼	Method	.../UICommands/OpenDialog 🗑
▼	Input arguments	
	Dialog	Dialog_with_Alias (type) 🔗
	AliasNode	Motor1 🔗
⚡	MouseUp event	+

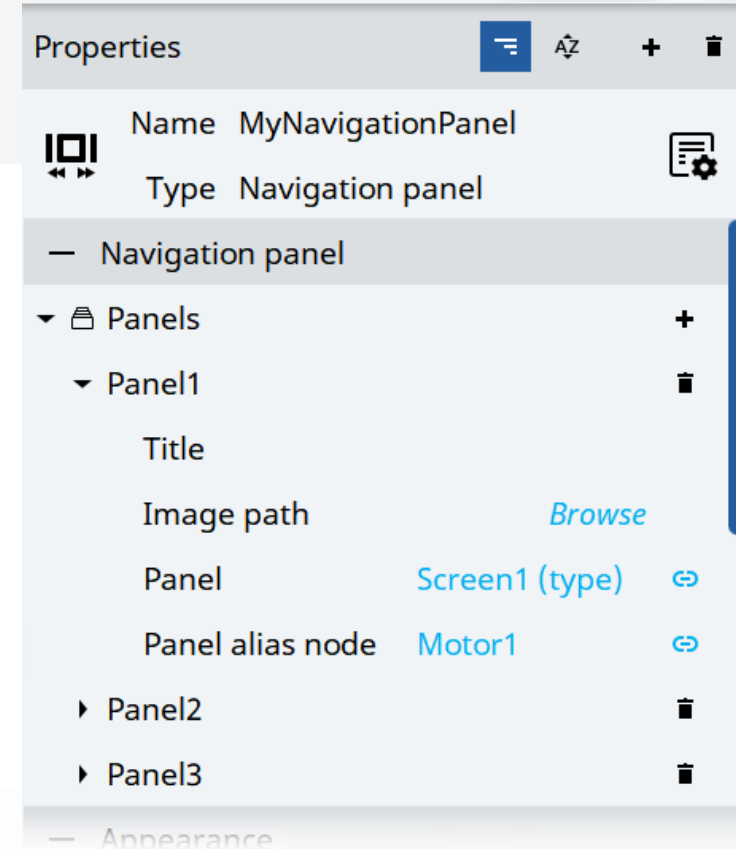
Reusable graphics: set aliases at runtime

- For Dialog and Panels, the Alias at Runtime can be set on:
 - Open Dialog command
 - Change Panel command
 - Navigation Panel properties
 - Panel Loader properties

Events			+
▼ ⚡	MouseClicked event		+
▼	Method	.../PanelLoader/ChangePanel	🗑
▼	Input arguments		
	NewPanel	Screen1 (type)	🔗
	AliasNode	Motor1	🔗
⚡	MouseDown event		+
⚡	MouseUp event		+

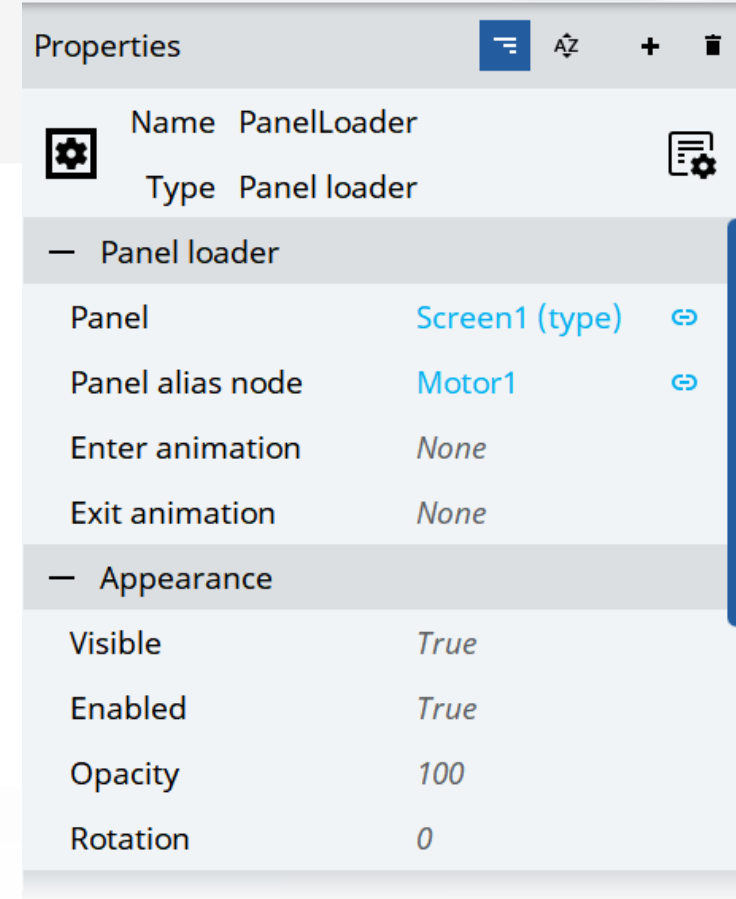
Reusable graphics: set aliases at runtime

- For Dialog and Panels, the Alias at Runtime can be set on:
 - Open Dialog command
 - Change Panel command
 - Navigation Panel properties
 - Panel Loader properties



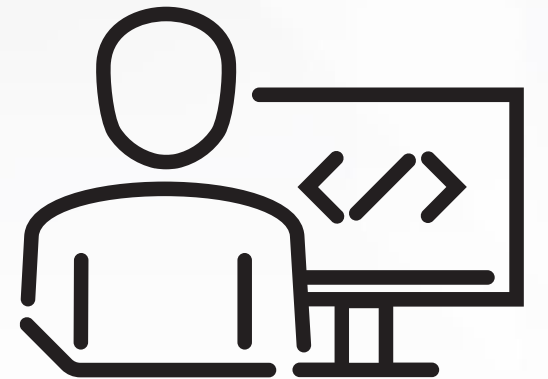
Reusable graphics: set aliases at runtime

- For Dialog and Panels, the Alias at Runtime can be set on:
 - Open Dialog command
 - Change Panel command
 - Navigation Panel properties
 - Panel Loader properties



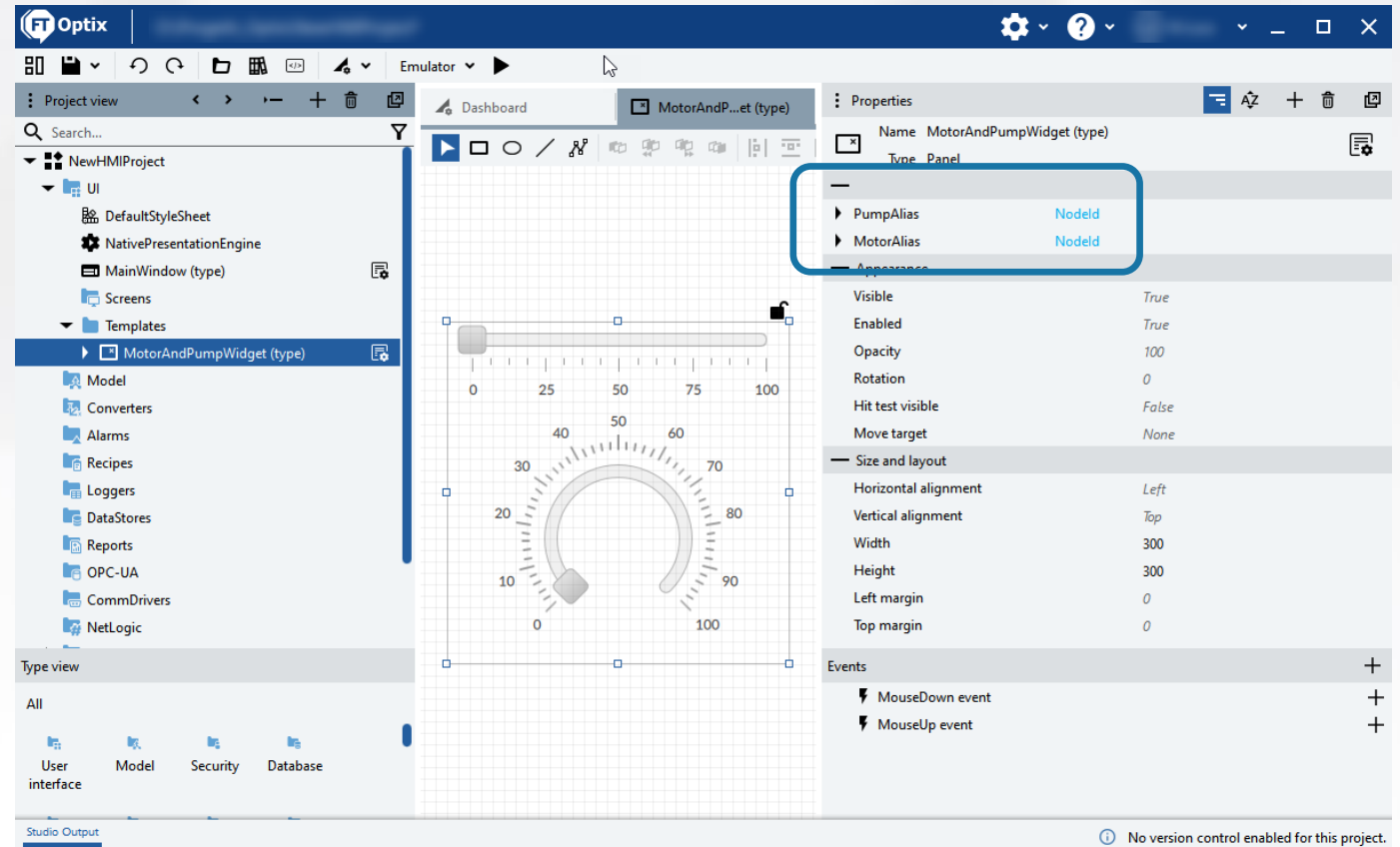
Hands-on session

- Add 3 Dropdown buttons (it's under the "container" group) into a new Panel
- Configure each button, with the following properties:
 - Panel = MotorWidget type
 - Panel alias node = the Alias of a Motor variable
- NOTE: you "probably" need to add a background to the MotorWidget type



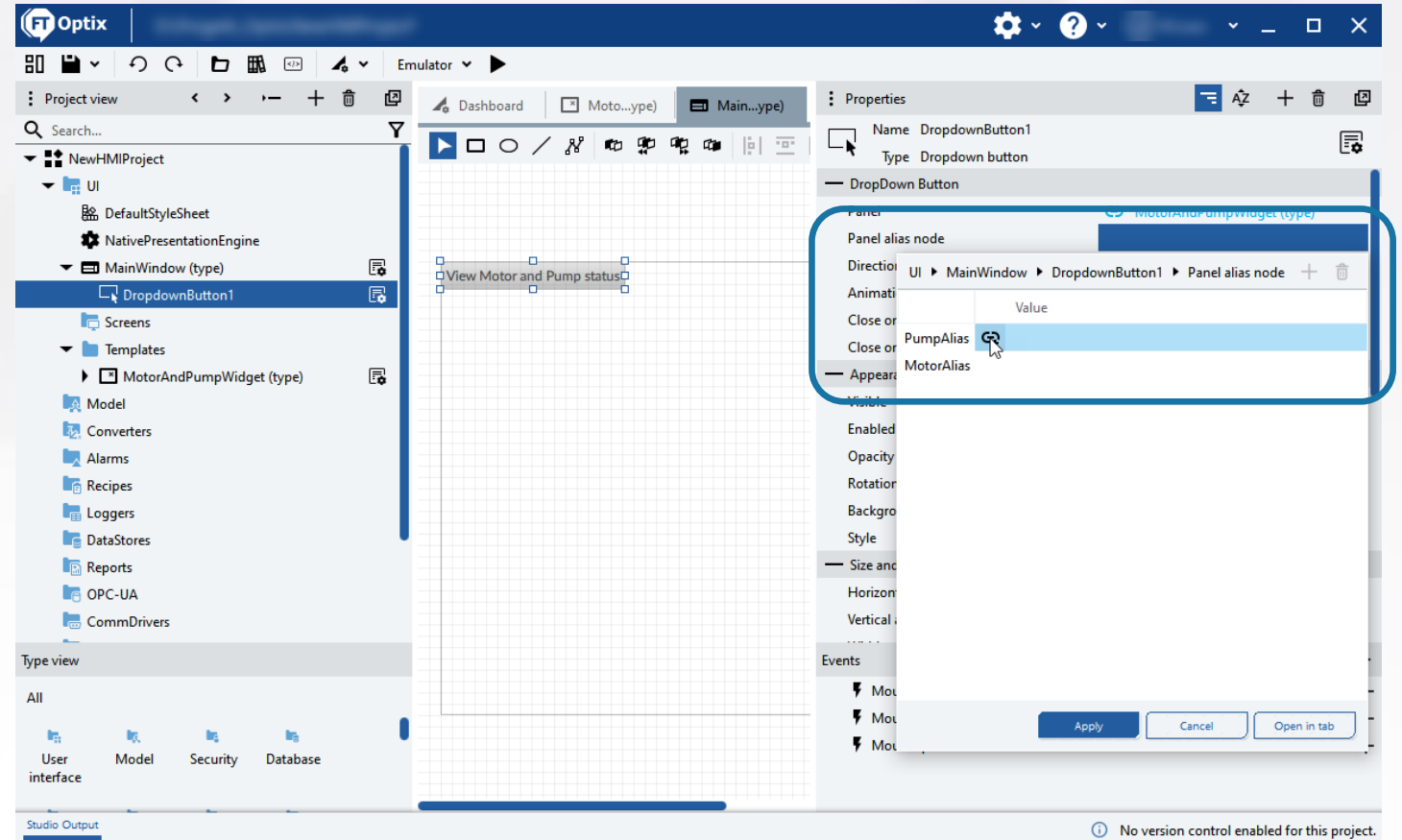
Multiple aliases

- If multiple aliases are used in a type, the Aliases editor panel will come out when configuring the Alias Node property of a container loader



Multiple aliases

- If multiple aliases are used in a type, the Aliases editor panel will come out when configuring the Alias Node property of a container loader

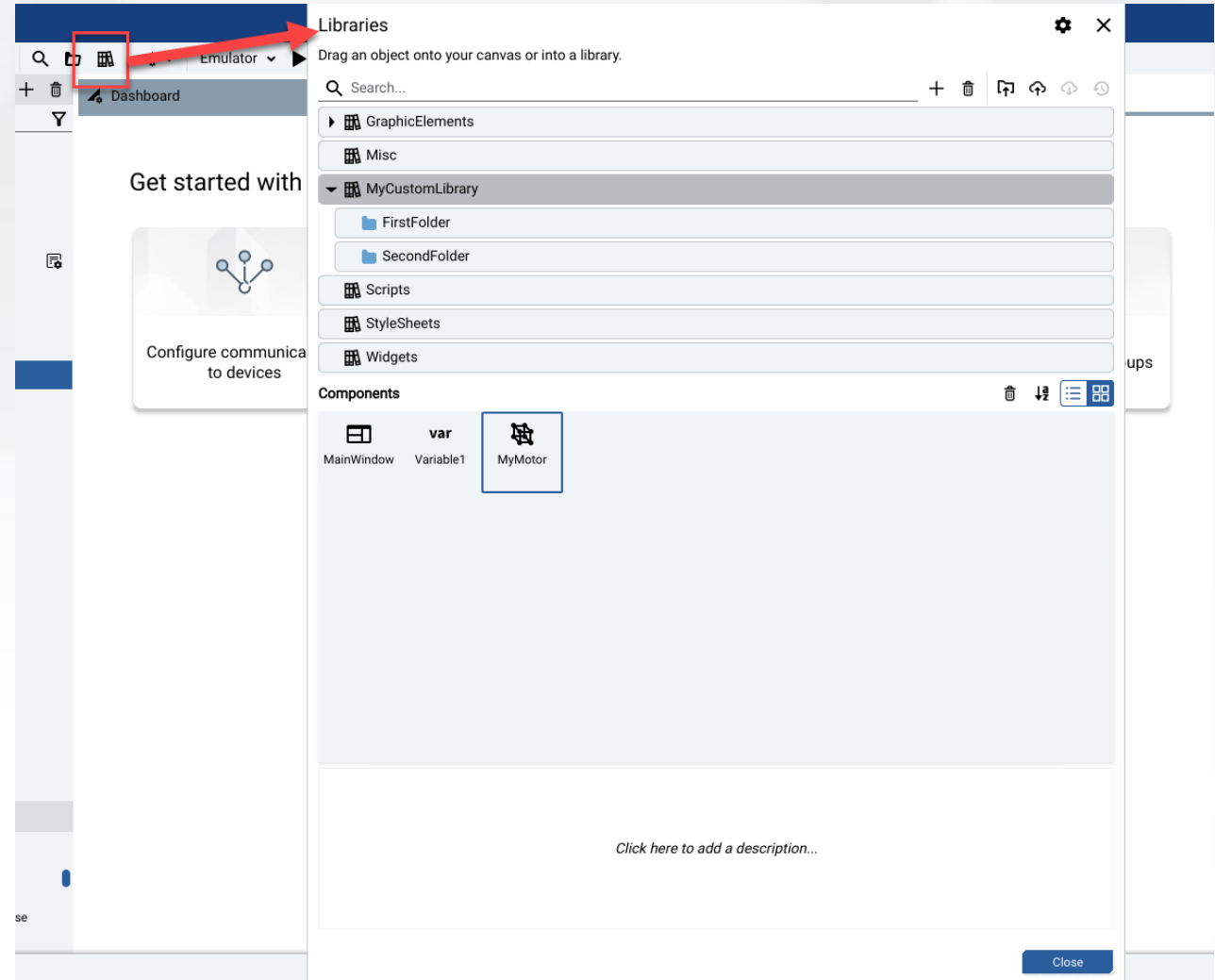


Template Library



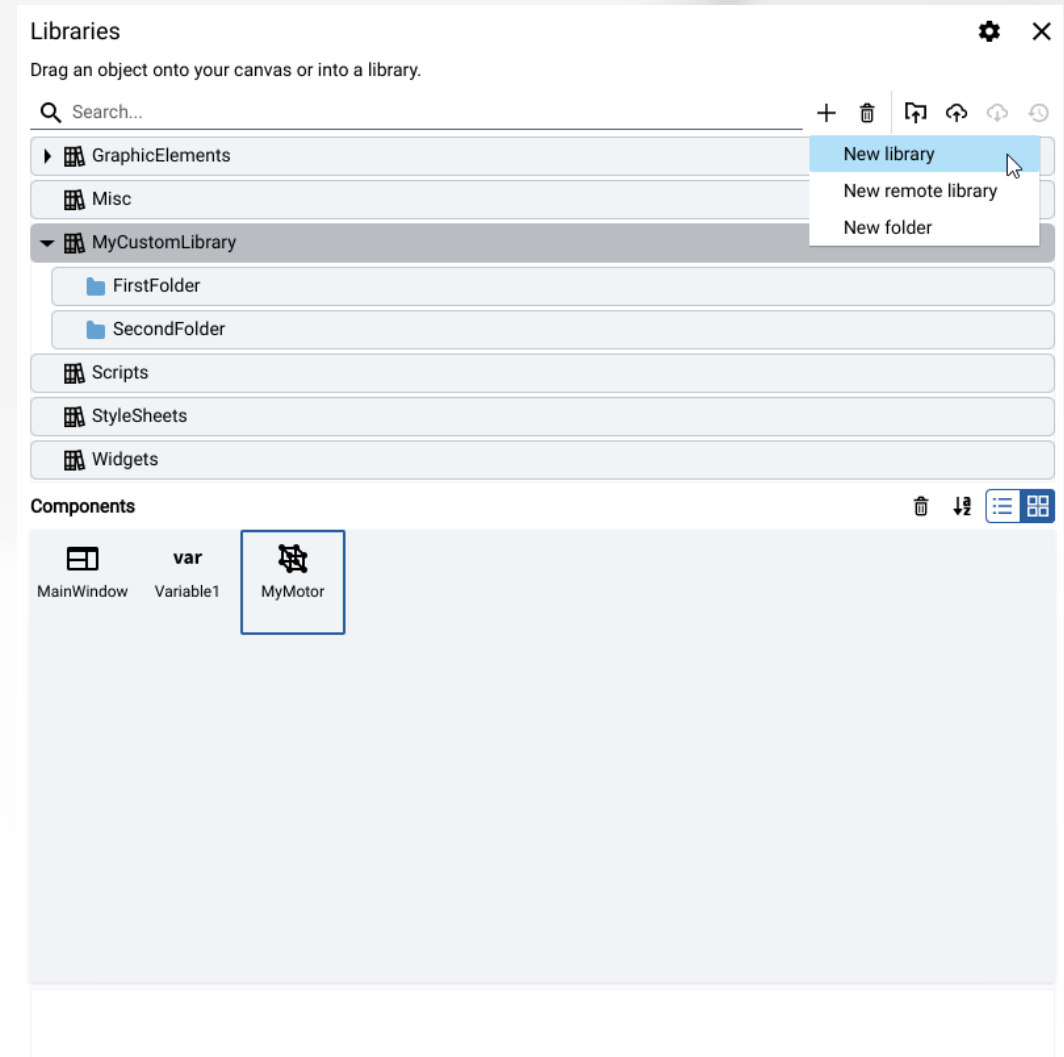
Template Library

- FactoryTalk Optix contains a good number of default libraries
- Default libraries cannot be modified (read-only)
- User can create unlimited libraries
 - Each library can have multiple subfolders
 - Libraries can contain any kind of element
- Libraries are shared across projects
- Libraries can be versioned with GIT
- Custom libraries are saved in the Documents folder
 - C:\Users\User\Documents\Rockwell Automation\FactoryTalk Optix\Libraries



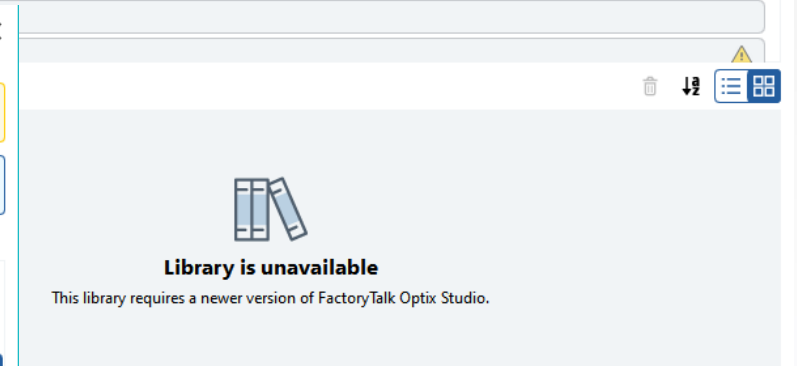
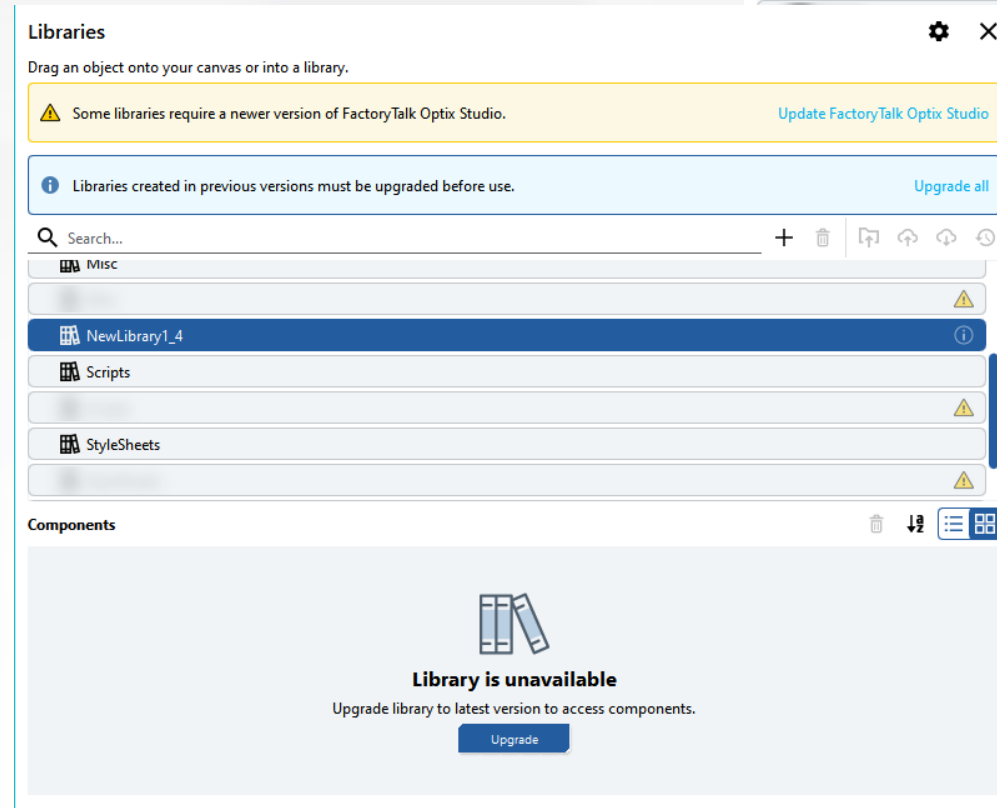
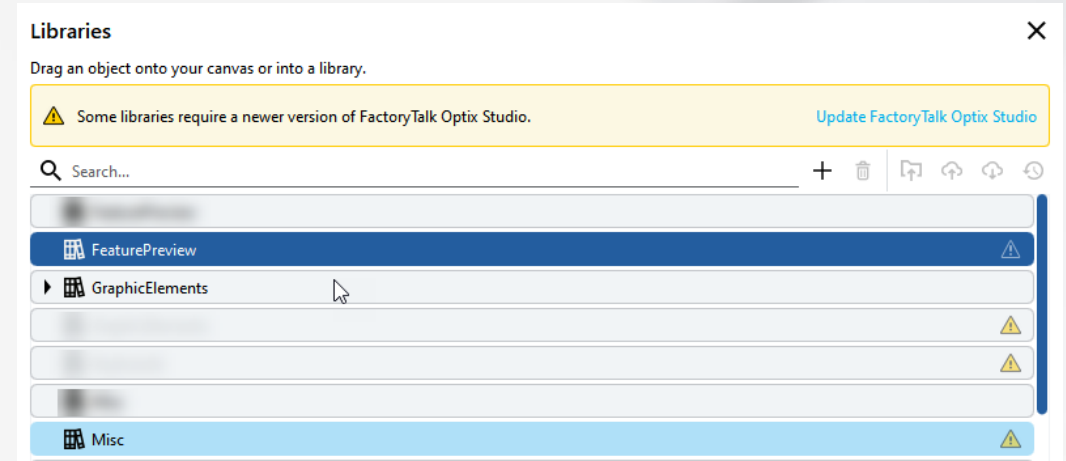
Template Library

- **New library:** create a new local library
 - Any new library can be versioned by using the dedicated buttons
- **New remote library:** pulls an existing library from a remote GIT provider
- **New folder:** create a subfolder in the selected library
- Version control history works for Template Library too (see the Version Control chapter)



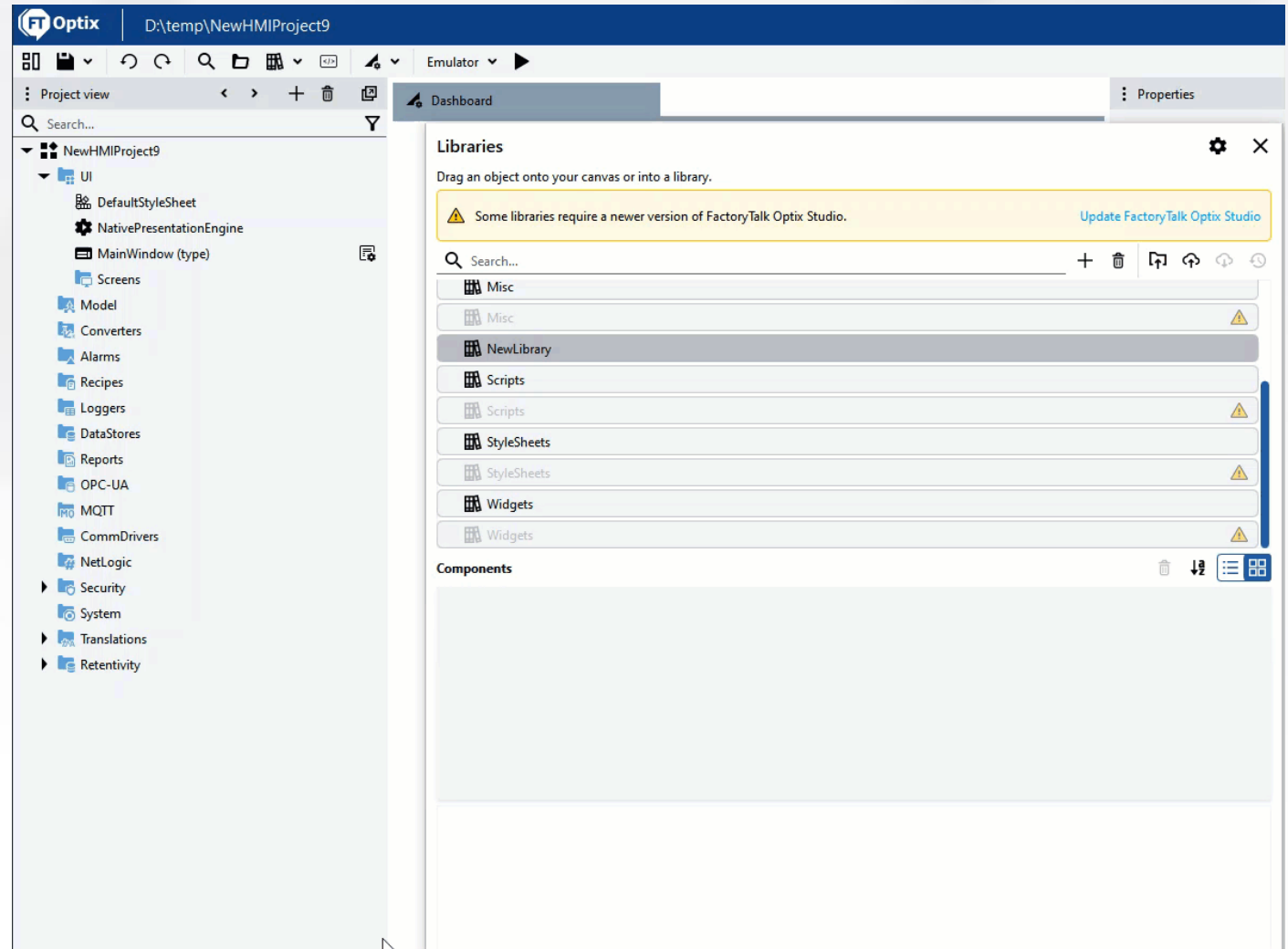
Template Library

- Libraries are only compatible with the same version of Optix that was used to create them
 - Libraries can be upgraded to a newer version if needed
 - Libraries cannot be downgraded



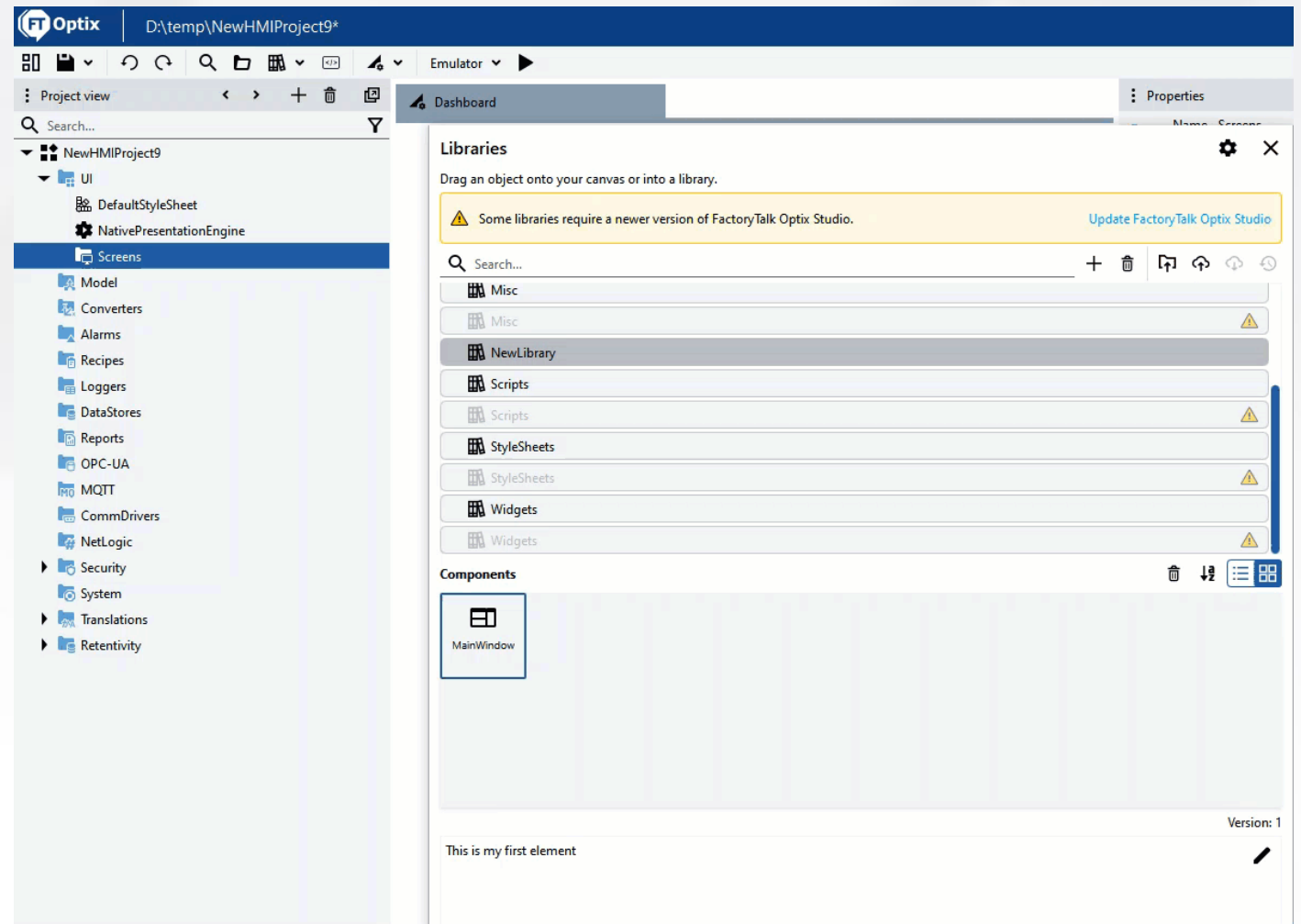
Template Library

- **Add an element to the library:** drag and drop the element from the project to the Template Library
 - Additional dependencies or types are imported automatically
 - Each element can have a description
- **Import an element from the library:** drag and drop the element from the library to the project
 - Additional dependencies or types are imported automatically
- **Resolving conflicts:** when importing or exporting any element, if the same type already exist, conflict resolution is prompt to the user



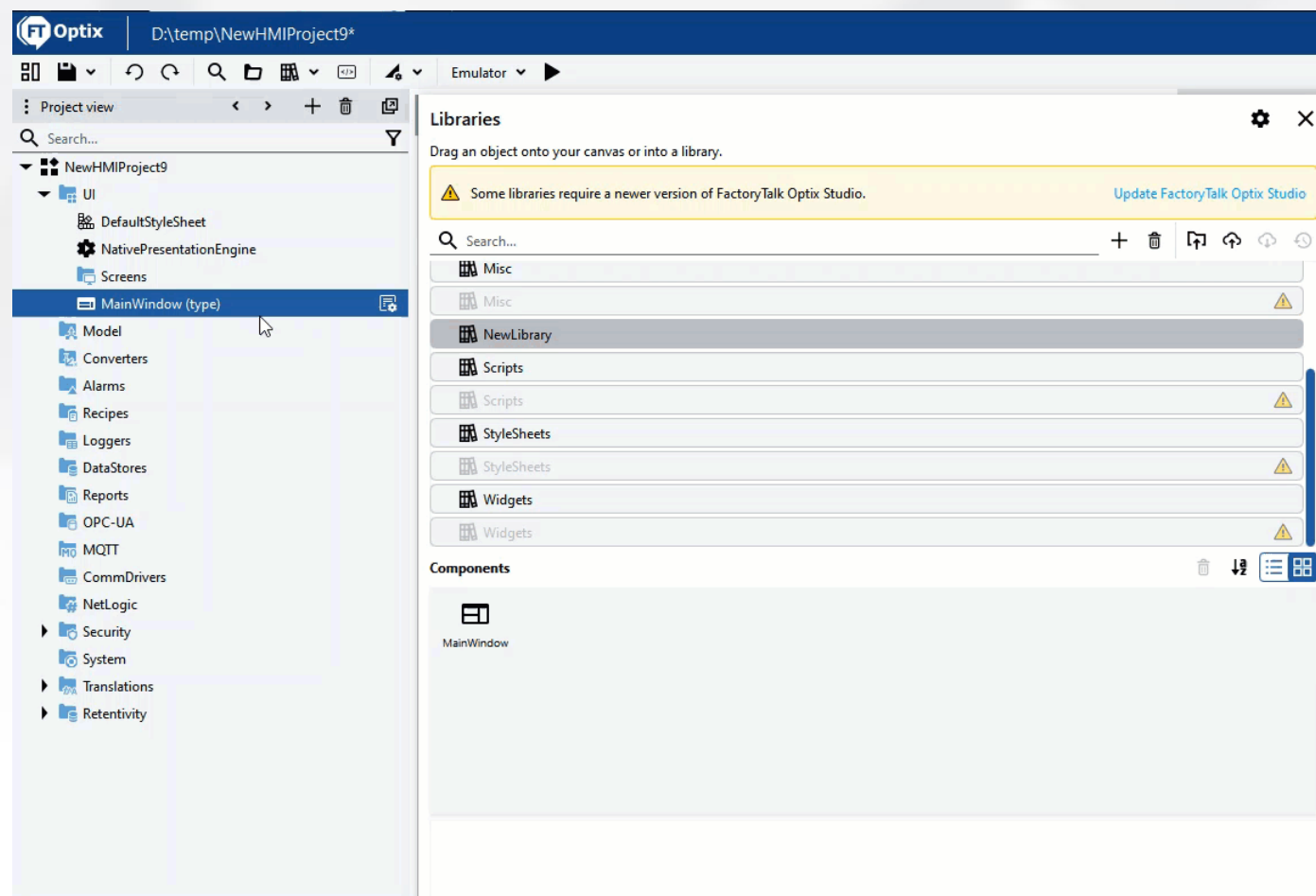
Template Library

- **Add an element to the library:** drag and drop the element from the project to the Template Library
 - Additional dependencies or types are imported automatically
 - Each element can have a description
- **Import an element from the library:** drag and drop the element from the library to the project
 - Additional dependencies or types are imported automatically
- **Resolving conflicts:** when importing or exporting any element, if the same type already exist, conflict resolution is prompt to the user



Template Library

- **Add an element to the library:** drag and drop the element from the project to the Template Library
 - Additional dependencies or types are imported automatically
 - Each element can have a description
- **Import an element from the library:** drag and drop the element from the library to the project
 - Additional dependencies or types are imported automatically
- **Resolving conflicts:** when importing or exporting any element, if the same type already exist, conflict resolution is prompt to the user

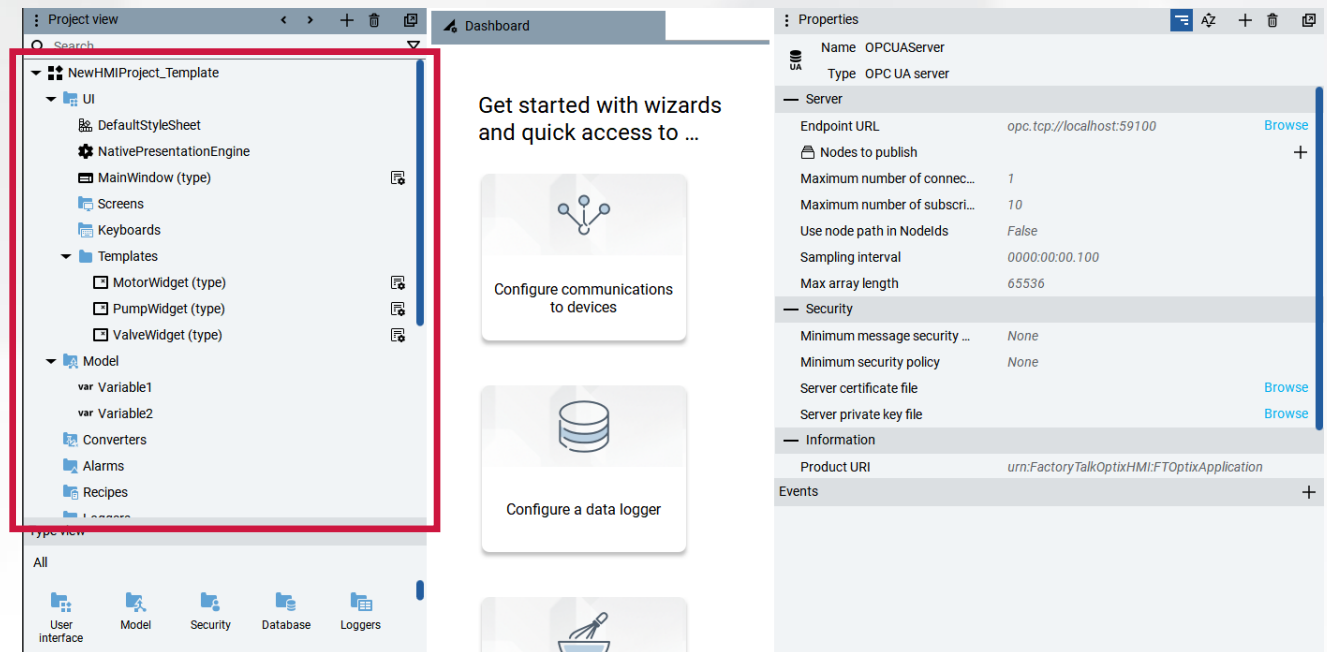


Project Templates



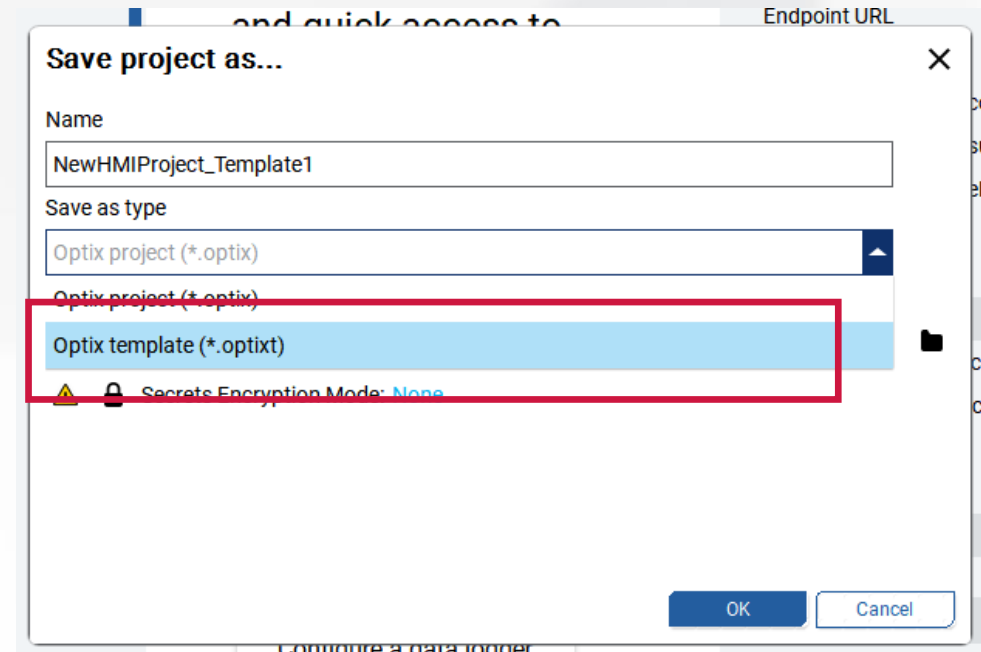
Project Templates

- **Reusable projects** that help speeding up development
- Any project can be saved as a template using the “Save As” dialog
 - Projects are saved at:
%UserProfile%\Documents\Rockwell Automation\FactoryTalk Optix\ProjectTemplates
- When creating a new project, the IDE will list all available templates
- Templates can be edited at any time



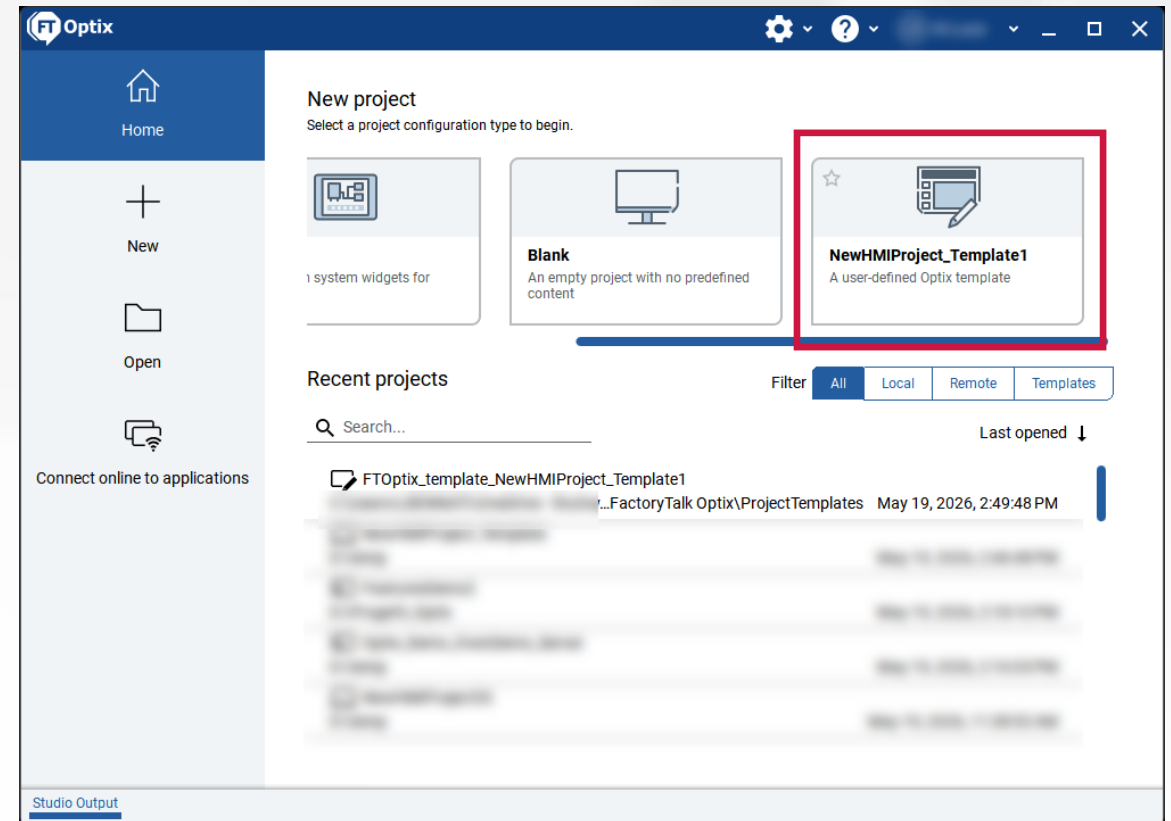
Project Templates

- **Reusable projects** that help speeding up development
- Any project can be saved as a template using the "Save As" dialog
 - Projects are saved at:
%UserProfile%\Documents\Rockwell Automation\FactoryTalk
Optix\ProjectTemplates
- When creating a new project, the IDE will list all available templates
- Templates can be edited at any time



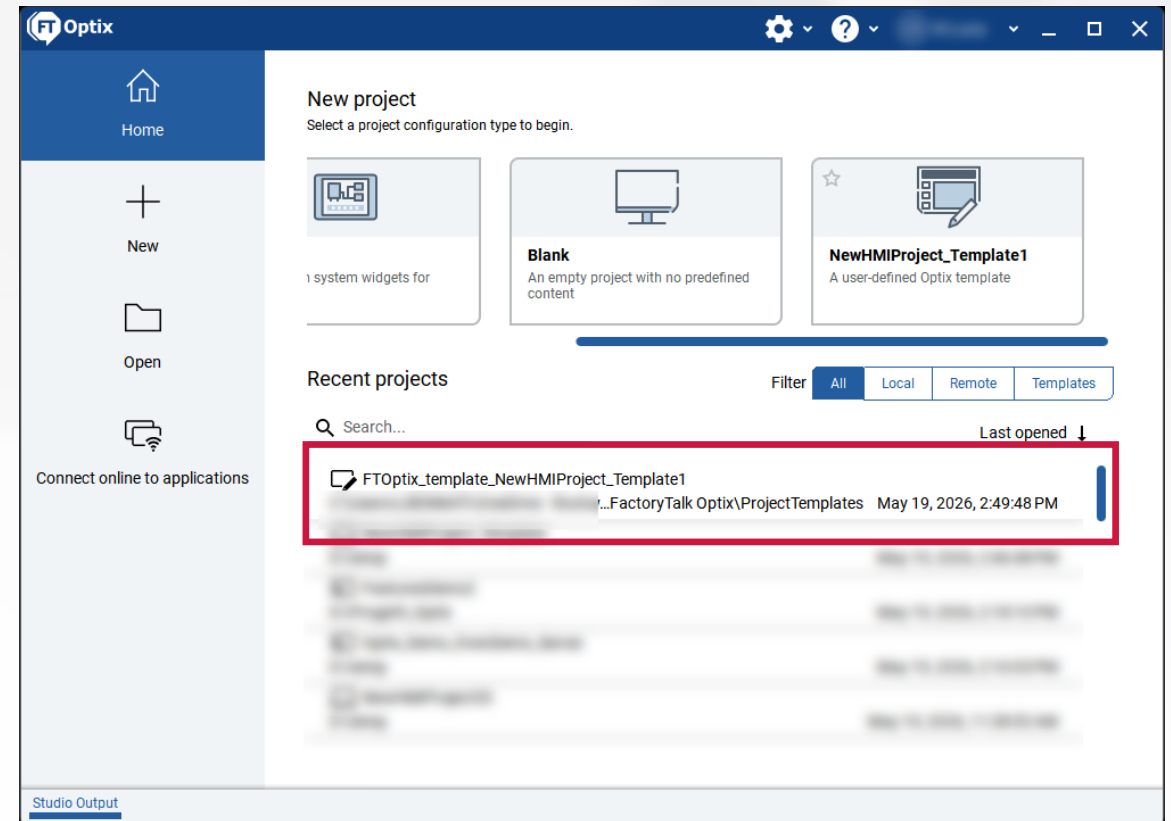
Project Templates

- **Reusable projects** that help speeding up development
- Any project can be saved as a template using the “Save As” dialog
 - Projects are saved at:
%UserProfile%\Documents\Rockwell Automation\FactoryTalk Optix\ProjectTemplates
- When creating a new project, the IDE will list all available templates
- Templates can be edited at any time



Project Templates

- **Reusable projects** that help speeding up development
- Any project can be saved as a template using the “Save As” dialog
 - Projects are saved at:
%UserProfile%\Documents\Rockwell Automation\FactoryTalk Optix\ProjectTemplates
- When creating a new project, the IDE will list all available templates
- Templates can be edited at any time

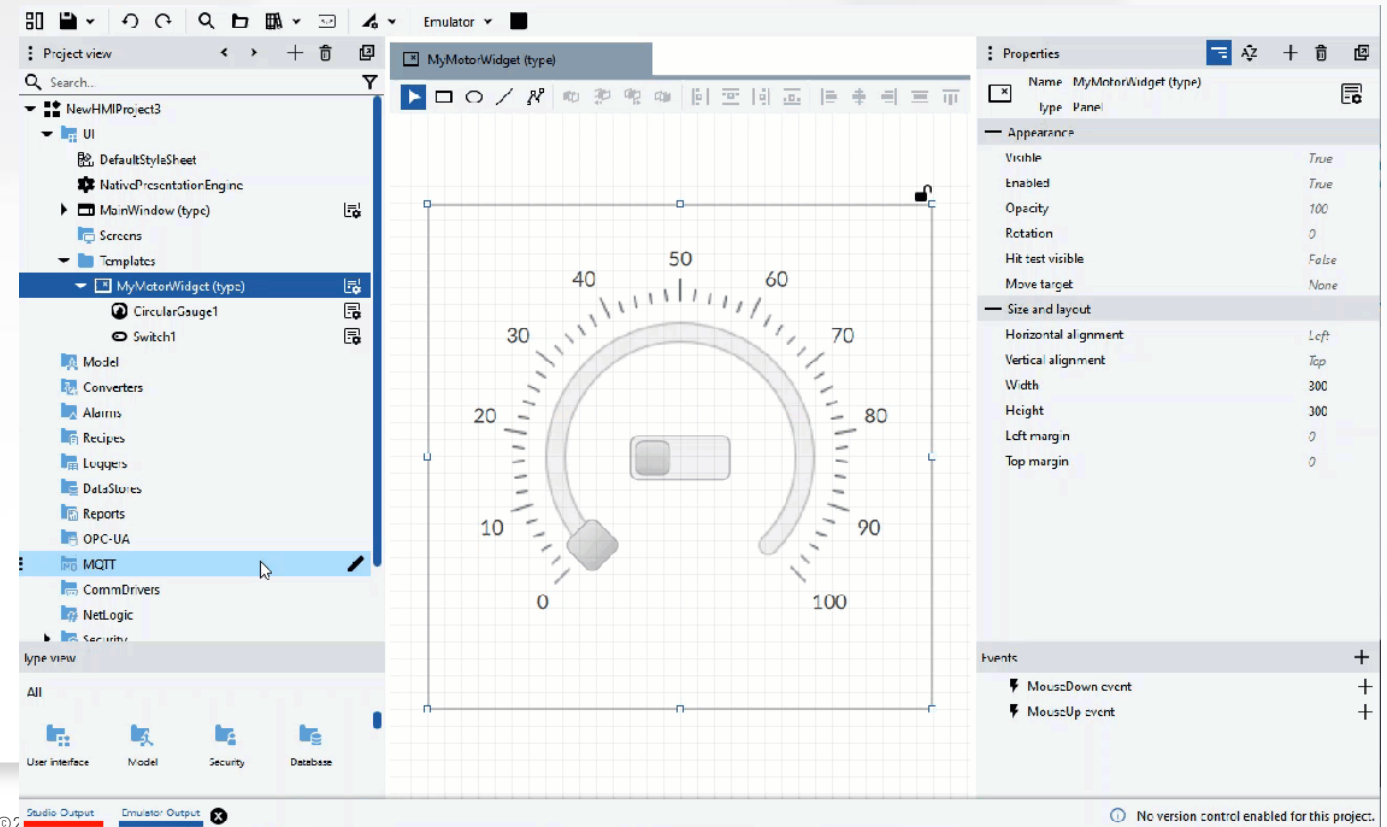


UI Objects Protection



UI Objects Protection

- UI Types can be protected to avoid accidental changes
 - Does not encrypt the content of the YAML file or prevents element manipulation from YAML source file
 - Does not encrypt the NetLogic in the UI Object (if any)
 - Used to avoid unauthorized changes of the children elements of an UI Type (e.g: sharing the object with colleagues)
 - Only works on UI Types

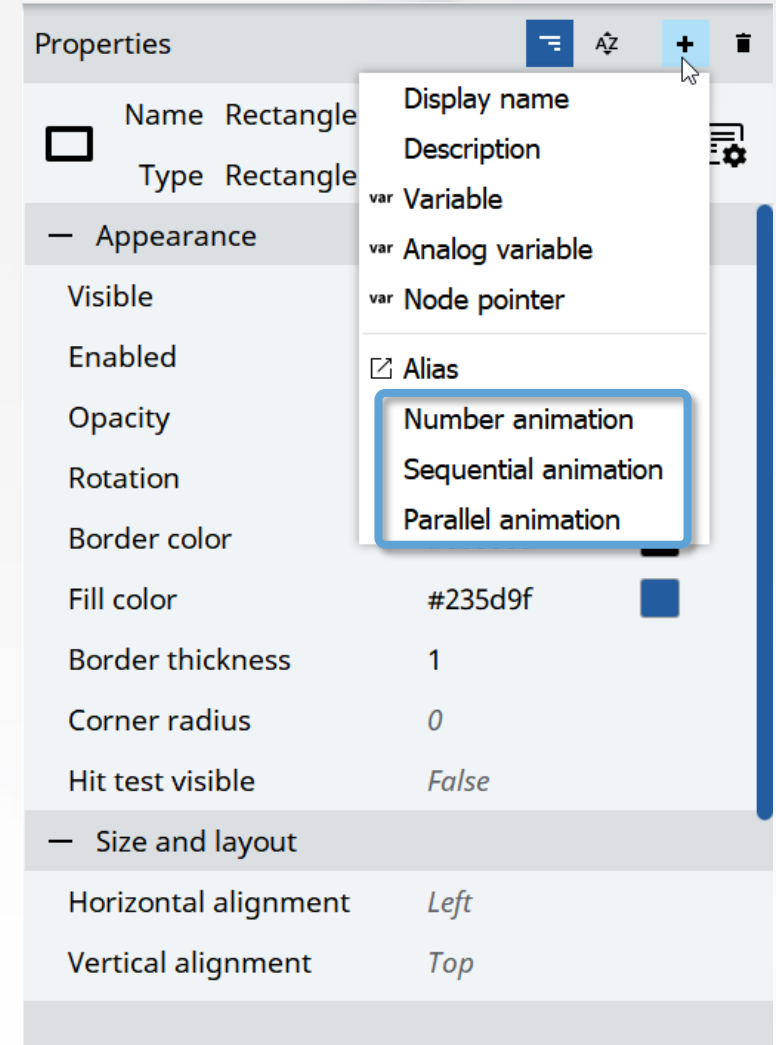


Animations



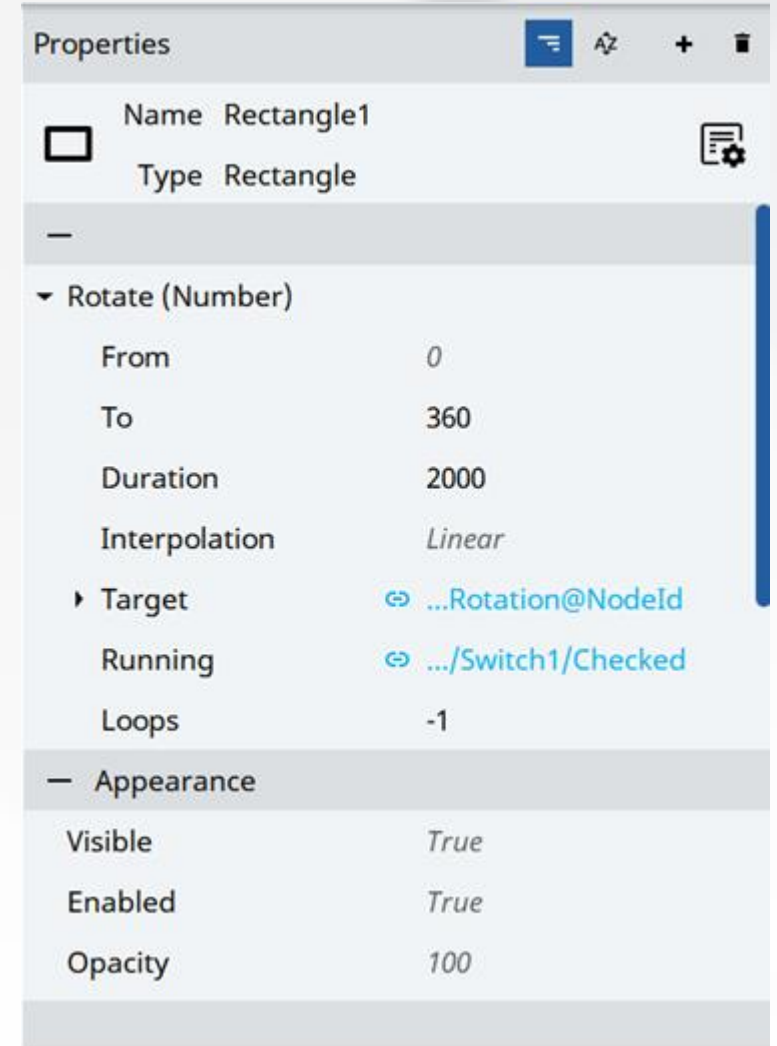
Animations

- Using Dynamic Links some basic animations can be defined as **just linking a Variable to a property**
 - Visibility, Enabling, Resizing, Change color...
- For "**eye-catching**" Animations:
 - Number animation: single animation
 - Sequential animation: container of animations executed in sequence
 - Parallel animation: container of animations executed in parallel (at the same time)
 - Animation of Behavior: allow to animate a property linked to a Variable



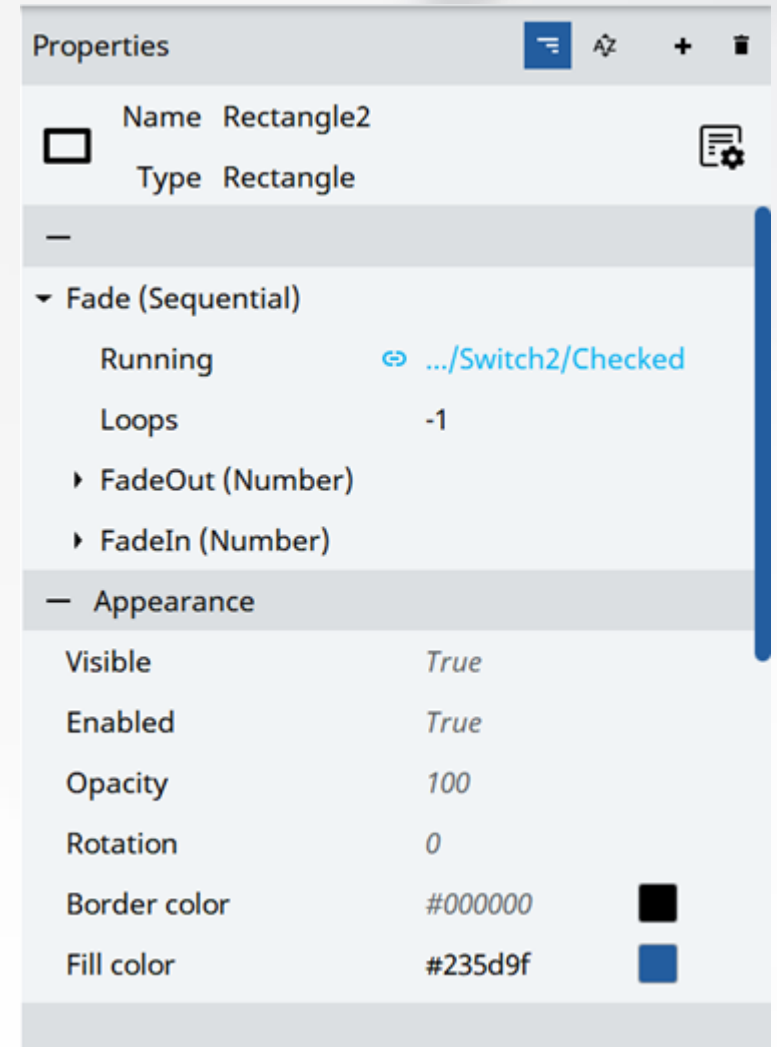
Animations

- Using Dynamic Links some basic animations can be defined as **just linking a Variable to a property**
 - Visibility, Enabling, Resizing, Change color...
- For **"eye-catching" Animations:**
 - Number animation: single animation
 - Sequential animation: container of animations executed in sequence
 - Parallel animation: container of animations executed in parallel (at the same time)
 - Animation of Behavior: allow to animate a property linked to a Variable



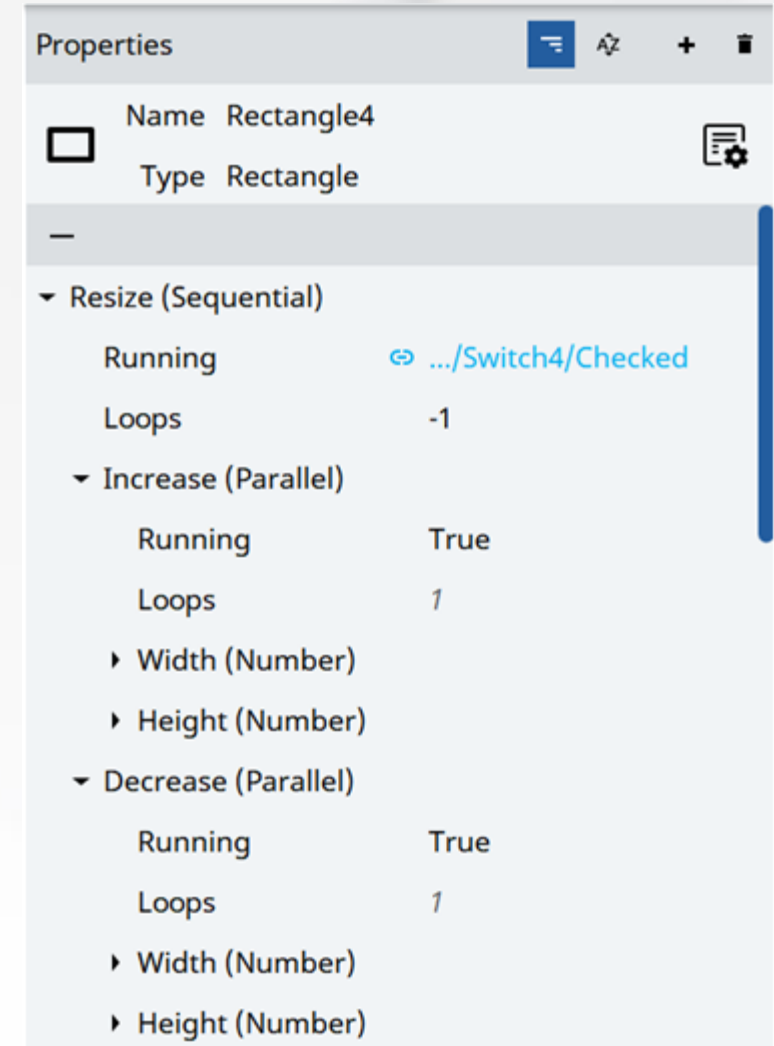
Animations

- Using Dynamic Links some basic animations can be defined as **just linking a Variable to a property**
 - Visibility, Enabling, Resizing, Change color...
- For **"eye-catching" Animations:**
 - Number animation: single animation
 - Sequential animation: container of animations executed in sequence
 - Parallel animation: container of animations executed in parallel (at the same time)
 - Animation of Behavior: allow to animate a property linked to a Variable



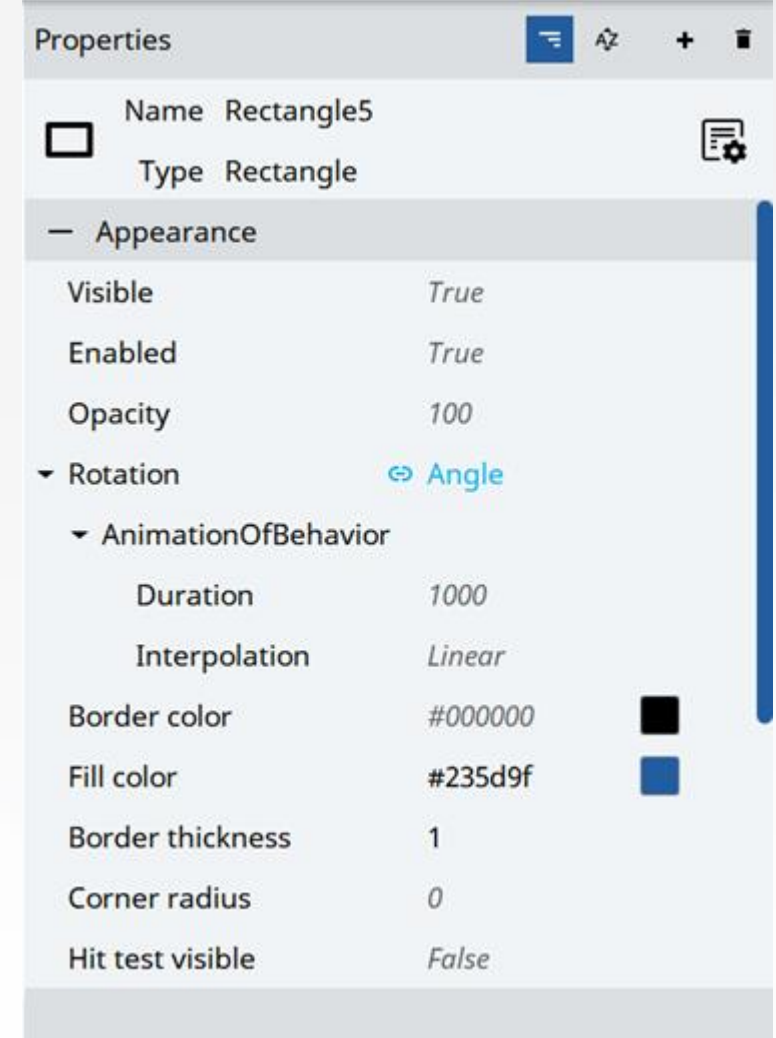
Animations

- Using Dynamic Links some basic animations can be defined as **just linking a Variable to a property**
 - Visibility, Enabling, Resizing, Change color...
- For **"eye-catching" Animations:**
 - Number animation: single animation
 - Sequential animation: container of animations executed in sequence
 - Parallel animation: container of animations executed in parallel (at the same time)
 - Animation of Behavior: allow to animate a property linked to a Variable



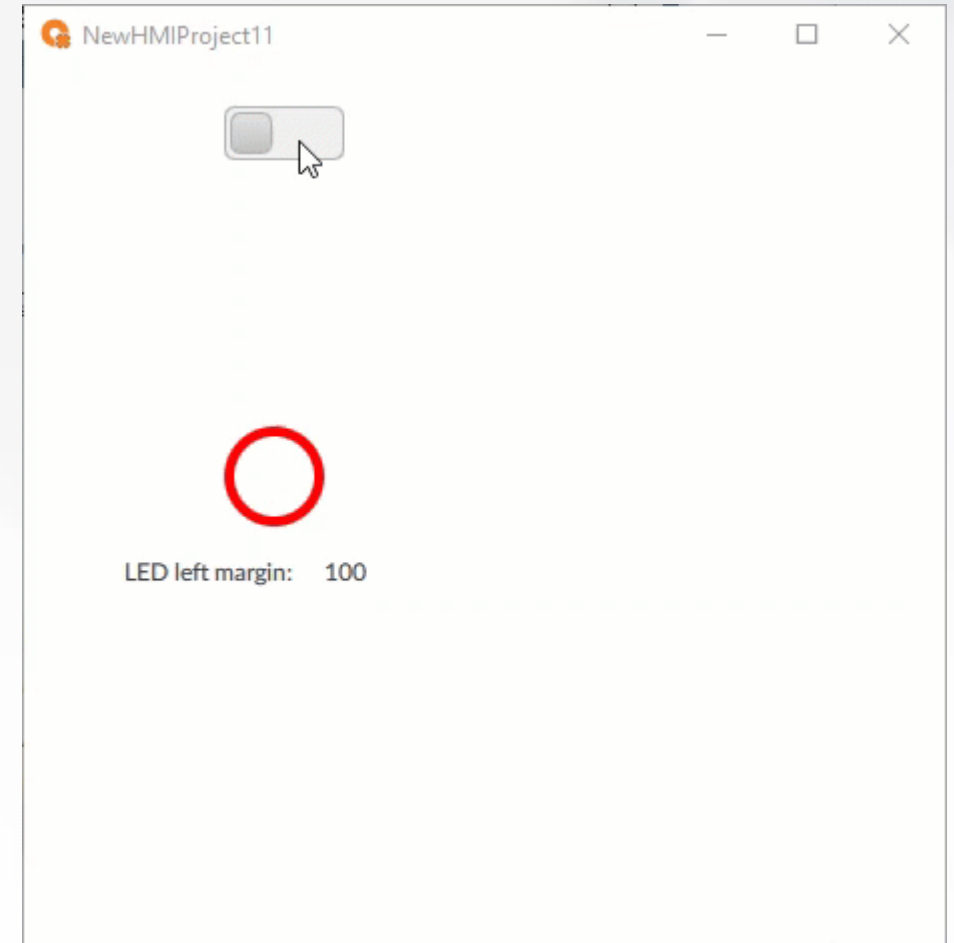
Animations

- Using Dynamic Links some basic animations can be defined as **just linking a Variable to a property**
 - Visibility, Enabling, Resizing, Change color...
- For **"eye-catching" Animations:**
 - Number animation: single animation
 - Sequential animation: container of animations executed in sequence
 - Parallel animation: container of animations executed in parallel (at the same time)
 - Animation of Behavior: allow to animate a property linked to a Variable



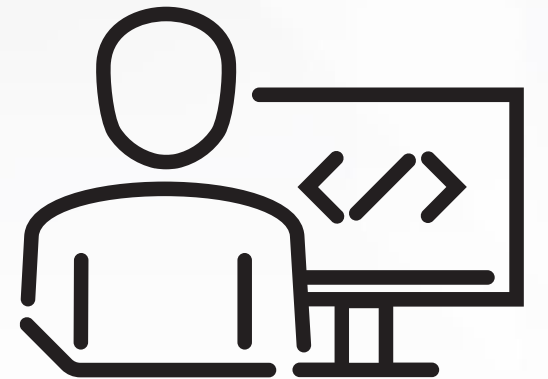
Things to remember

- Animations are just an “overlay” added before rendering to the variable
- The “real” value of the variable is left untouched



Hands-on session

- Add a Panel Loader object and add one or more buttons that call the method "Change Panel" to change the panel loaded
- Create a button object that calls the method "Open Dialog" to open a Dialog Box
- Define a Transitioned Event that executes a method like "Variable commands > Set Variable Value" or "Core commands > Close"



Events



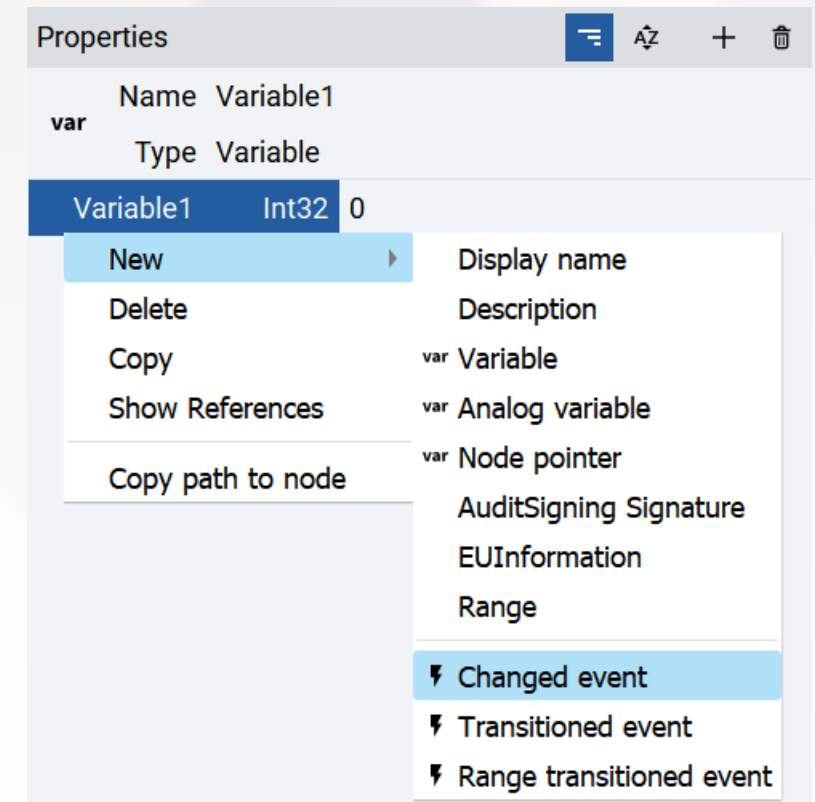
Commands, methods and events

- Commands/Methods can be executed on Events triggered by objects
- Event Types:
 - **Native Event** (Event View): like "MouseClicked", "MouseDown", "MouseUp" of a button
 - **Changed Event, Transitioned Event, Range Transitioned Event** (Property Panel): custom event
- Method Types:
 - Global Methods or **Commands**: like "Change User", or "Open Dialog"...
 - Local **Methods exposed by objects**: like "Log" of Datalogger, or "Refresh" of Datalogger viewer...

Events	+
⚡ MouseClick event	+
⚡ MouseDown event	+
⚡ MouseUp event	+

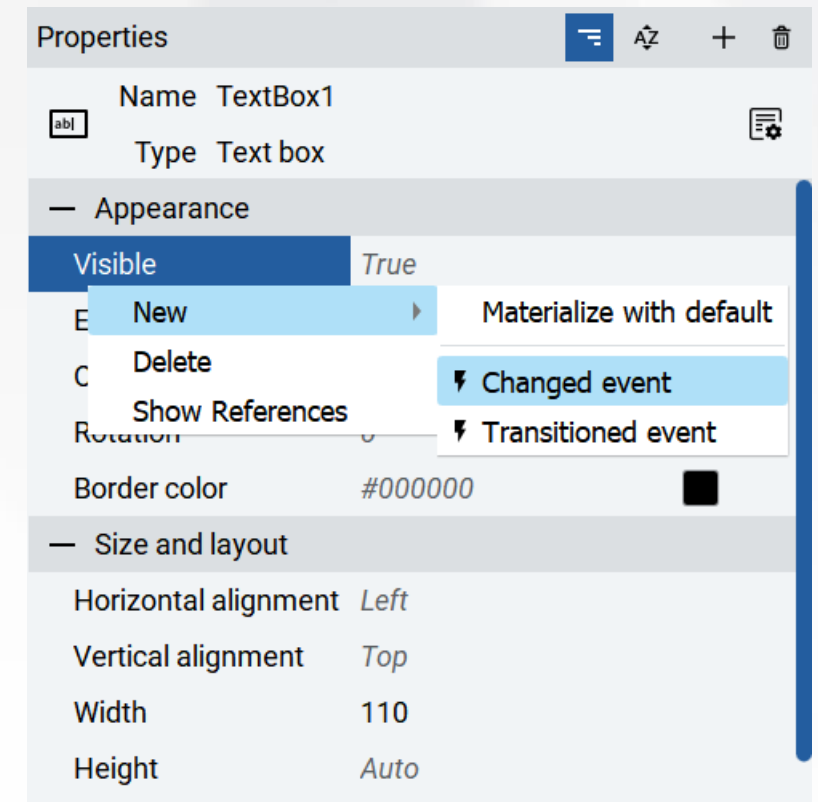
Commands, methods and events

- Commands/Methods can be executed on Events triggered by objects
- Event Types:
 - **Native Event** (Event View): like "MouseClicked", "MouseDown", "MouseUP" of a button
 - **Changed Event, Transitioned Event, Range Transitioned Event** (Property Panel): custom event
- Method Types:
 - Global Methods or **Commands**: like "Change User", or "Open Dialog"...
 - Local **Methods exposed by objects**: like "Log" of Datalogger, or "Refresh" of Datalogger viewer...



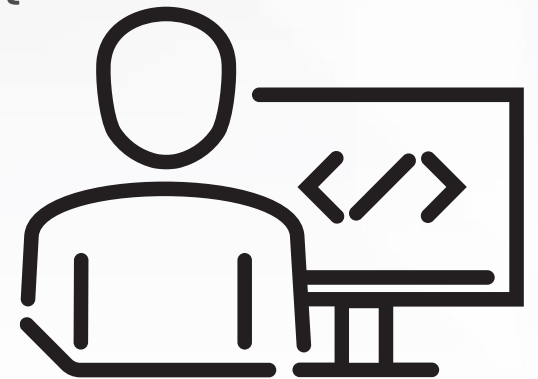
Commands, methods and events

- Commands/Methods can be executed on Events triggered by objects
- Event Types:
 - **Native Event** (Event View): like "MouseClicked", "MouseDown", "MouseUP" of a button
 - **Changed Event, Transitioned Event, Range Transitioned Event** (Property Panel): custom event
- Method Types:
 - Global Methods or **Commands**: like "Change User", or "Open Dialog"...
 - Local **Methods exposed by objects**: like "Log" of Datalogger, or "Refresh" of Datalogger viewer...



Hands-on session

- Create a button object that calls the method "Open Dialog" to open a Dialog Box
- Define a Transitioned Event that executes a method like "Variable commands > Set Variable Value" or "Core commands > Close"
- Define an event to display a ScreenSaver Screen after Idle Timeout
 - HINT: add the UISession object and assign Session in Presentation properties
 - Unzip Image file ra_logo.svg or use any image
 - C:\Program Files\Rockwell Automation\FactoryTalk Optix\Studio\Help\en\downloads

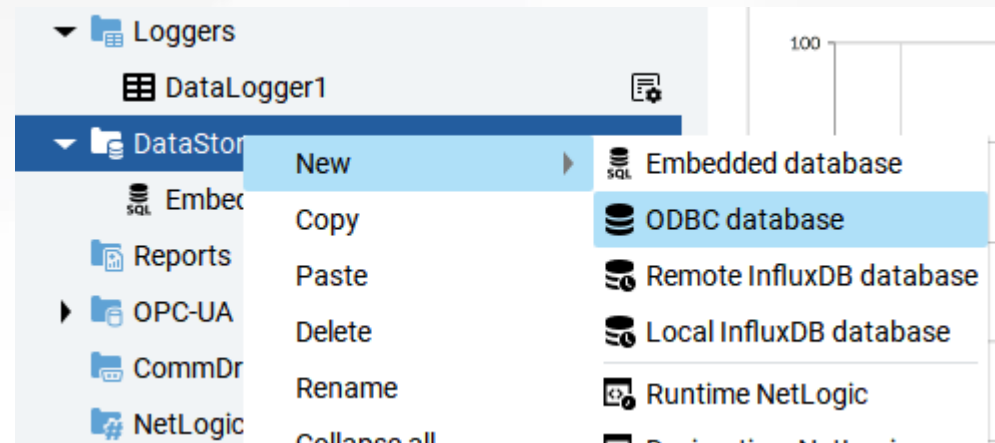


DataStores



DataStores

- A DataStore is a database connection
- Types of DataStores
 - **EmbeddedDatabase**: Records values of one or more variables at the same time
 - **ODBC database**: SQL based ODBC connection (SQL Server, MySQL or any derivative)
 - **Remote InfluxDB Database**: Remote PC Timeseries database connection (self-hosted)
 - **Local InfluxDB Connection**: Local PC Timeseries database connection (similar to an EmbeddedDatabase)



Working with Databases

Objects that can exchange data with a datastore:

- ***Data grid, List box, Combo box*** and *Trend* objects
query a table to display data at runtime
- *Data logger* and *Event logger* objects
store data in a table
- *Recipe* object
stores data in a table and execute a query to display data at runtime

Working with databases Design-Time

- Emulator will create a copy of the project at runtime
- C:\Users\<USER>\AppData\Local\Rockwell Automation\FactoryTalk Optix\Emulator\Projects
- At Runtime the Embedded database is Locked.
- Path: \ApplicationFiles\<DB Name>.sqlite
- Can be manipulated with SQLite editors like SQLite Browsers, Dbeaver and more.

PKI	11/14/2024 1:37 PM	File folder	
56db3d98fa3e3a8d3ae0e34c269688d6.sqlite	11/14/2024 1:39 PM	SQLITE File	8 KB
AlarmsRetentivityStorage	11/14/2024 1:48 PM	Data Base File	92 KB
DB_Example.sqlite	11/14/2024 1:48 PM	SQLITE File	8 KB
SecurityRetentivityStorage	11/14/2024 1:48 PM	Data Base File	92 KB

- Tables can be exported to CSV via script

	Name	GenericTableExporter		
	Type	NetLogic		
	CSVPath	AbsoluteResourceUri	...ktop\Recipe_Export.CSV	Browse
	FieldDelimiter	String	,	
▶	Table	NodeId	Mike_Recipe	
	Query	String	Complex Dynamic Link	
	WrapFields	Boolean	False	

Data Grid, List Box and Combo Box

Data Grid is a tabular view of data at Runtime

Connect to Database, Folder of objects, structure, Enumeration

- Model – DataStore or Folder or object
- Item kind (optional) – Design time structure
- Query – SQLite syntax
- Auto refresh time – how often data is queried – can be triggered on demand

Brands	
ID	Brands
1	Electra
2	Haro
3	Heller
4	Pure Cycles
5	Ritchey
6	Strider

Name dg_Brands	
Type Data grid	
Data grid	
Model	BikeStores
Item kind	DB_Brands
Kind	
Node class	Any
Query	SELECT * FROM brands
Auto refresh time	0000:00:30.000

Name DB_Brands		
Type Object		
brand_id	Int32	0
brand_name	String	0

Data Grid - Columns

Brands	
ID	Brands
1	Electra
2	Haro
3	Heller
4	Pure Cycles
5	Ritchey
6	Strider

Columns	
DataGridColumn1	
Data item template	
Text	{Item}/brand_id
Text color	#000000
Title	ID
Width	Auto
Visible	True
Header alignment	Left aligned
Content alignment	Left
Order by	{Item}/brand_id

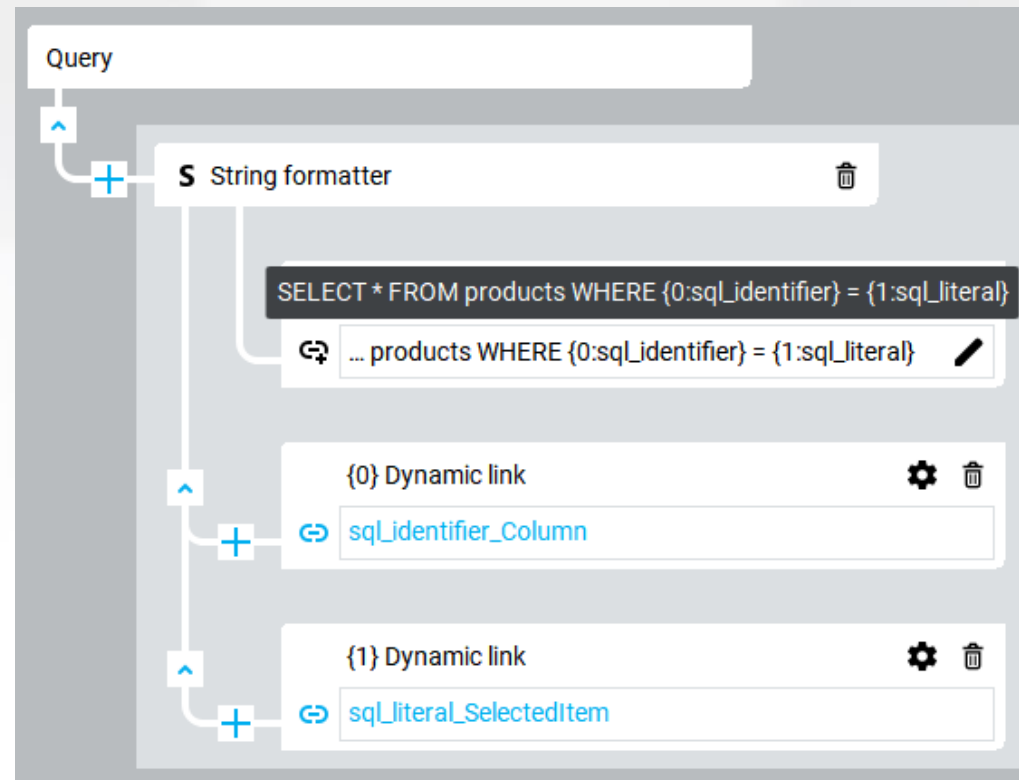
Aliases
{Session}
{Window}
{Item} (DB_BikeStores/UI/Screens/Select_Bil
DB_Brands
var brand_id
var brand_name

- Text – {Item}/ColumnName
 - For embedded Database
 - Browse into Aliases – {Item}/
 - Set Item kind to Datastore – Tables – columns
 - For ODBC Database
 - Browse into Aliases – {Item}/
 - Type in column name or create a model object for structure
- Title – Column label
- Query – SQLite syntax
- Order by – At least one column in DataGrid needs to be set

Database Queries

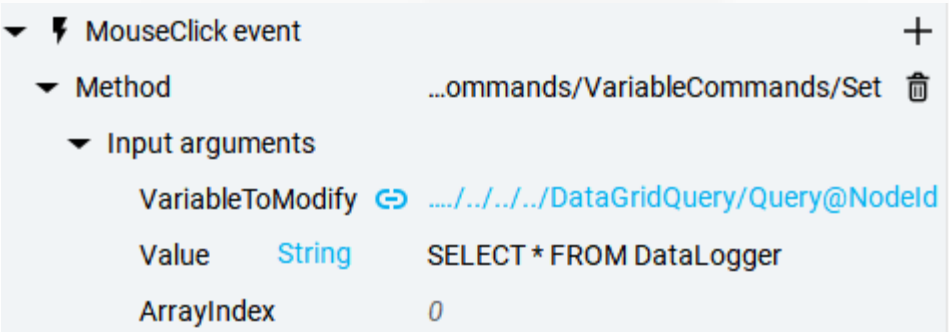
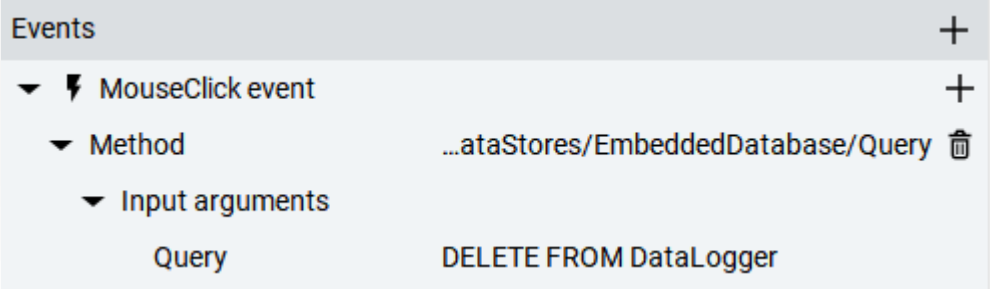
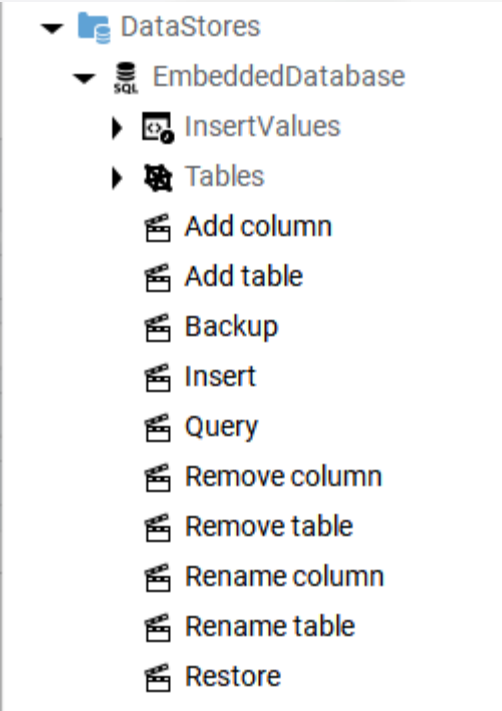
- Query – SQLite syntax supports a subset of statements such as:
- SELECT
- DELETE
- UPDATE
- CREATE and DROP
- JOIN
- Many more...

- When using the String formatter use placeholders
 - {0:sql_identifier} To define a column name for example
 - {1:sql_literal} To define a number for example



Database Queries

- There are 3 ways to Interact with a database
 1. Loggers
 - Data Logger or Event Logger can be set to “None” for OnDemand Writes or Periodic or OnChange
 2. SQL Methods
 - Use String Formatter to build query
 3. Scripts
 - See FeaturesDemo2 for examples

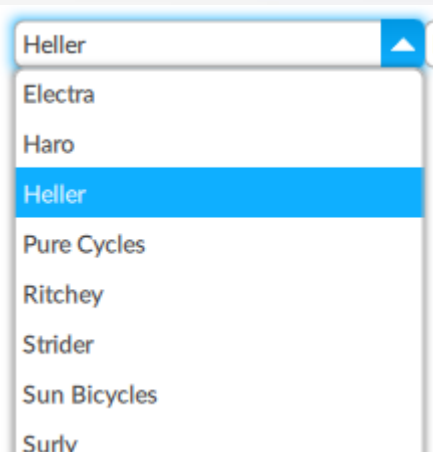


Combo box and List box

 Name ComboBox1
Type Combo box

— ComboBox

▶ Model	BikeStores
Query	SELECT * FROM brands
Mode	Normal
Display value path	{Item}/brand_name
Selected value path	BaseDataType {Item}/brand_id
▶ Selected item	
Selected value	BaseDataType brand_id
Allow deselection	False



The most flexible of objects

Similar to DataGrid but has additional functions

- Model – Datastore or Folder or Object
- Query – SQLite elect statement, use string formatter for advanced select
- Mode – Combo box – Normal, Search or Editable
- Display Value Path – Values to display
- Selected Value Path – Values to select
- Selected Item – Nodeld of selected (nodeld is the unique ID reference)
- Selected Value – Value of selected

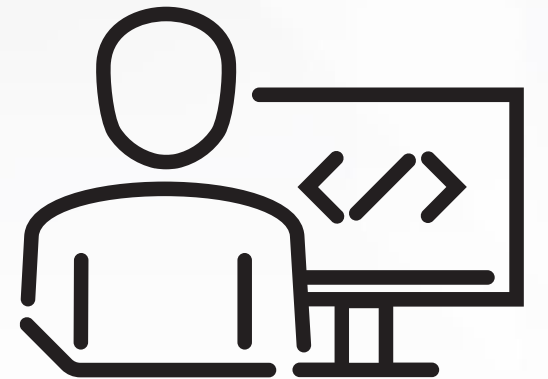
Queries limitations

- Support only for Standard SQL 2003 syntax
 - No support for specific dialects like SQL Server, MySQL or Flux
- List of supported clauses:
 - [SQL queries](#)
 - [Influx database limitations](#)

Operator	Example
IN	<pre>SELECT * FROM Table1 WHERE Column1 IN (10, 20, 30)</pre>
BETWEEN	<pre>SELECT * FROM Table1 WHERE Column1 BETWEEN 100 AND 200</pre>
LIKE	<pre>... WHERE column1 LIKE 'a'</pre> <pre>... WHERE column1 LIKE 'a%'</pre> <pre>... WHERE column1 LIKE '%bbp!%ppo%' ESCAPE '!'</pre>
EXISTS	<pre>... WHERE EXISTS (SELECT Table1.Column1 FROM Table2)</pre>
IS	<pre>... WHERE column1 IS NULL</pre>
NOT	<pre>... WHERE column1 IS NOT NULL</pre> <pre>... WHERE column1 NOT IS (10, 20)</pre>

Hands-on session

- Create a project with an EmbeddedDatabase and log some data
- Configure a DataGrid to show data from a DataBase
- Configure Combo box to filter by values
- Backup and Restore Embedded Database content

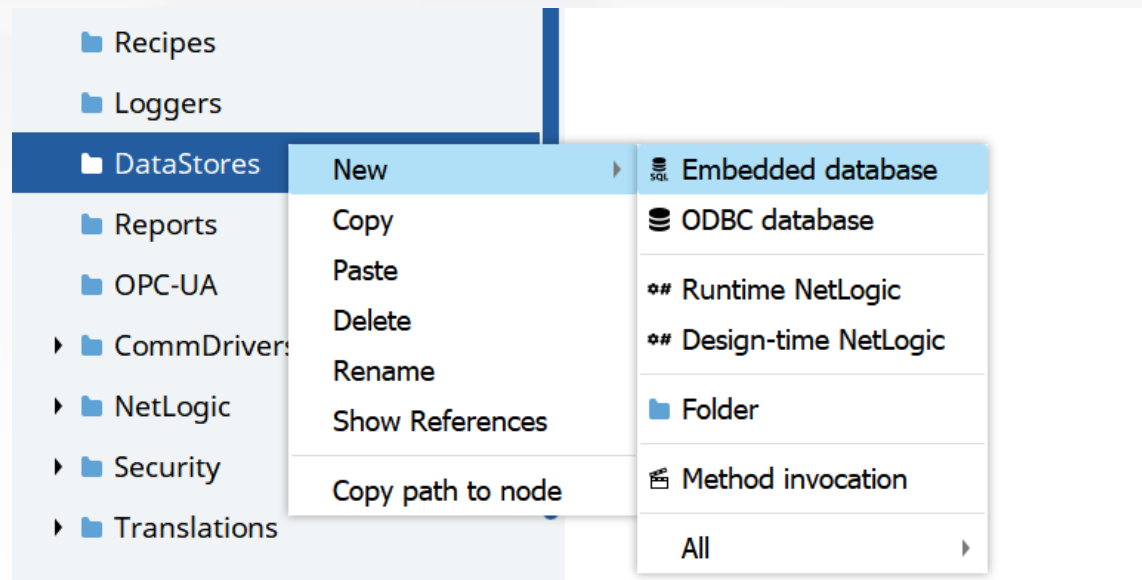


Save historical data with loggers



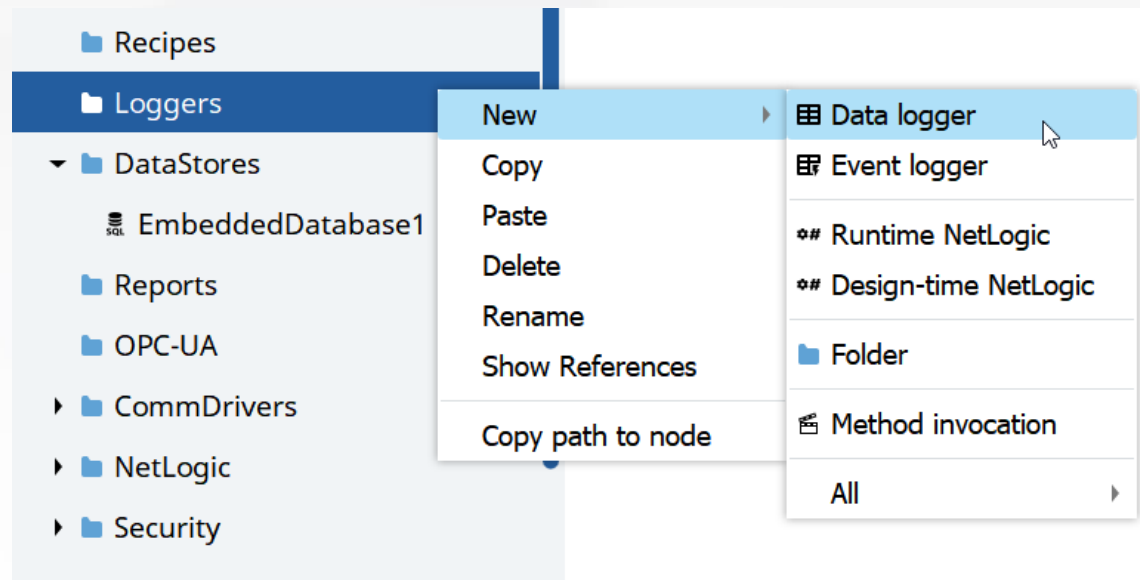
Loggers

- A logger is an object that records data and stores them in a database
- Types of logger
 - **Data logger**: Records values of one or more variables at the same time
 - **Event logger**: Records the properties of one or more events

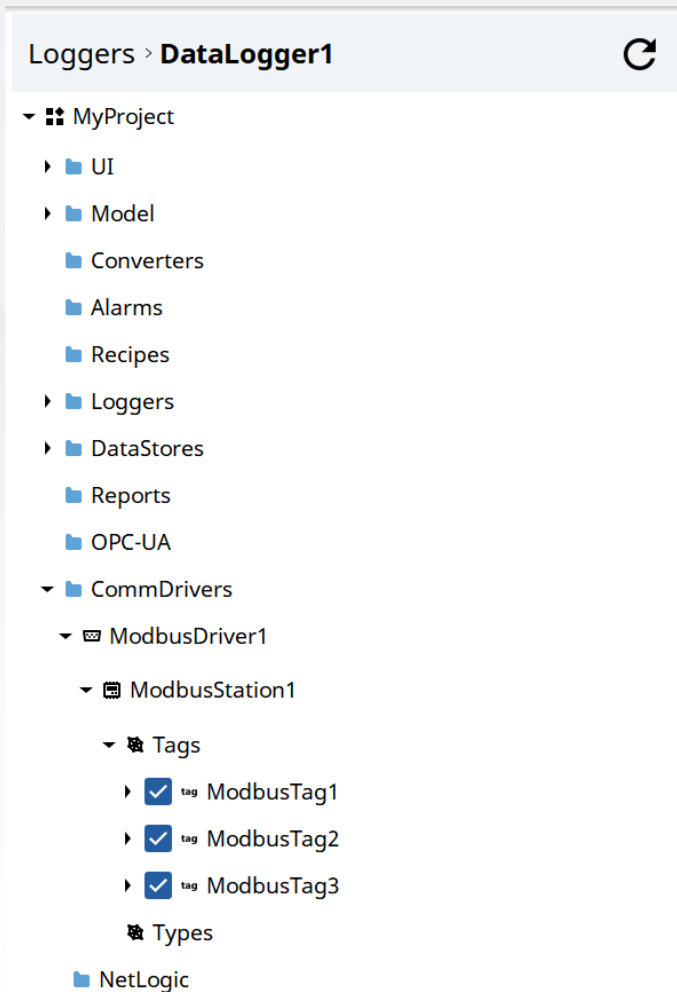


Loggers

- A logger is an object that records data and stores them in a database
- Types of logger
 - **Data logger**: Records values of one or more variables at the same time
 - **Event logger**: Records the properties of one or more events



Data logger



- At every sampling event the Datalogger create a new record into a table with the following structure:

Date/Time	Variable 1	Variable 2	...	Variable N
hh:mm:ss	value	value	...	value
hh:mm:ss	value	value	...	value

- Select Variables to record
 - Model tags, CommDrivers Tag, Objects properties...
- Configure Datalogger properties
 - Sampling Mode: Periodic/Change in Value
 - Sampling Period: dd:hh:mm:ss.fff
 - Table name (optional)
 - Store: DataStore used to save historical data

Data logger

Properties

Name DataLogger1

Type Data logger

Sampling mode

Periodic

Sampling period

0000:00:01.000

Log operation code...

False

Log timestamp...

False

Log local time

True

Table name

Variables to log

ModbusTag1

.../Tags/ModbusTag1

ModbusTag2

.../Tags/ModbusTag2

ModbusTag3

.../Tags/ModbusTag3

Store

EmbeddedDatabase1

- At every sampling event the Datalogger create a new record into a table with the following structure:

Date/Time	Variable 1	Variable 2	...	Variable N
hh:mm:ss	value	value	...	value
hh:mm:ss	value	value	...	value

- Select Variables to record
 - Model tags, CommDrivers Tag, Objects properties...
- Configure Datalogger properties
 - Sampling Mode: Periodic/Change in Value
 - Sampling Period: dd:hh:mm:ss.fff
 - Table name (optional)
 - Store: DataStore used to save historical data

Data logger "viewer"

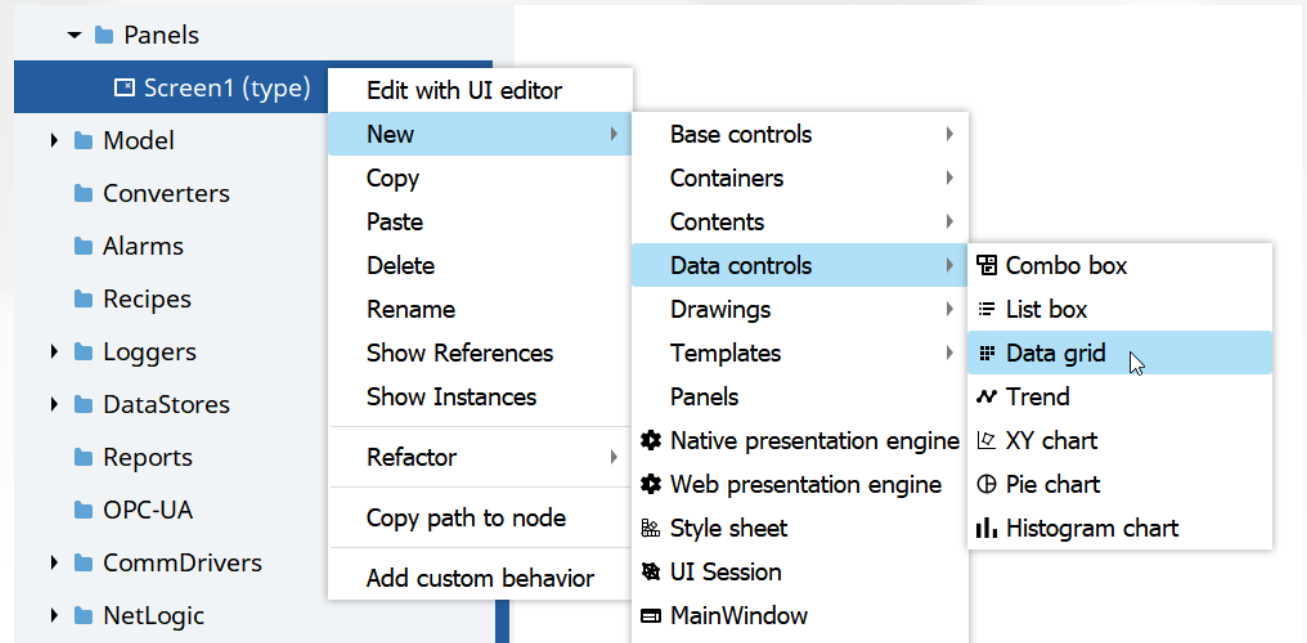
1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Datalogger to the Data grid object

- The object can be customized through Properties pane

- Query
- Columns format
- Sorting

- Export

- Data logger can be exported using the script "Data Logger exporter" available into Template Library. This script can be associated with a button event



Data logger "viewer"

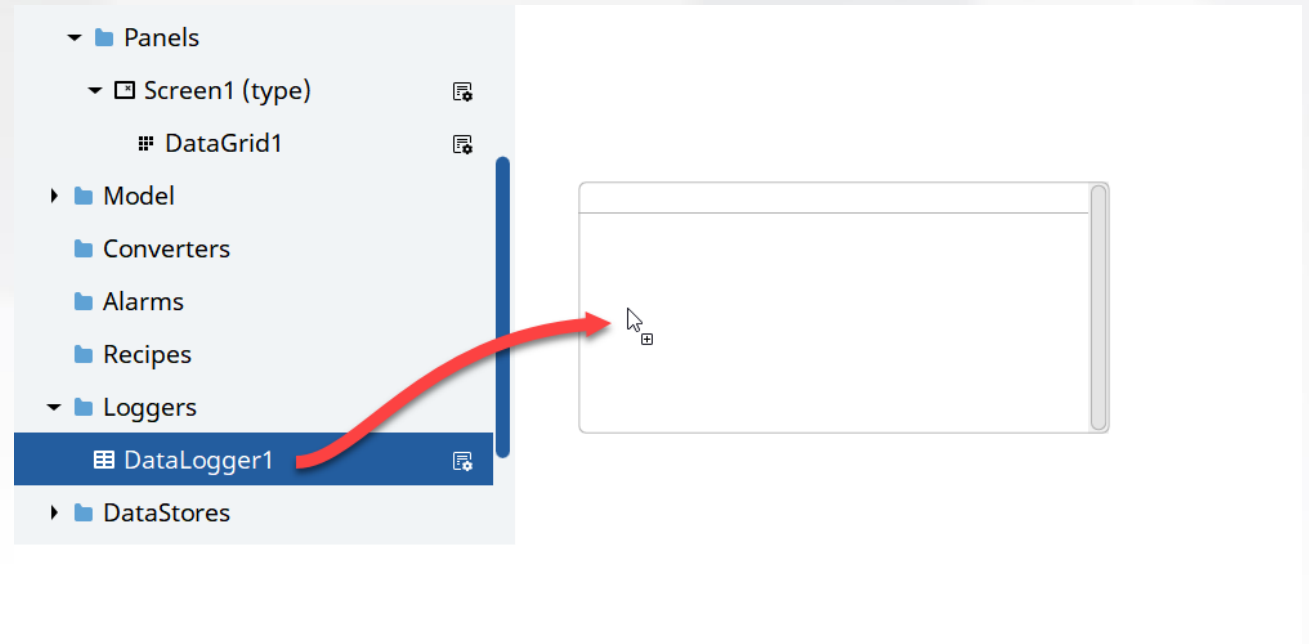
1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Datalogger to the Data grid object

- The object can be customized through Properties pane

- Query
- Columns format
- Sorting

- Export

- Data logger can be exported using the script "Data Logger exporter" available into Template Library. This script can be associated with a button event



Data logger "viewer"

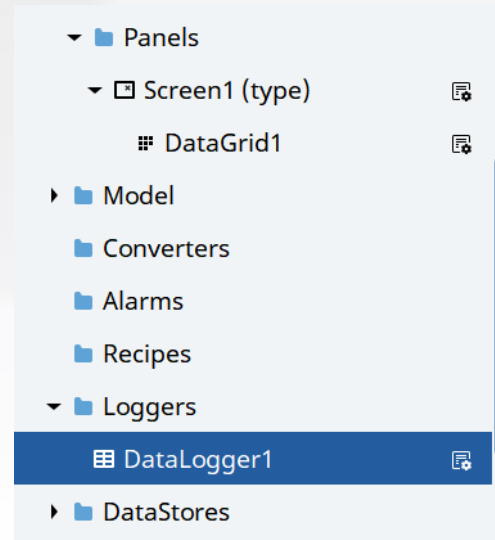
1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Datalogger to the Data grid object

- The object can be customized through Properties pane

- Query
- Columns format
- Sorting

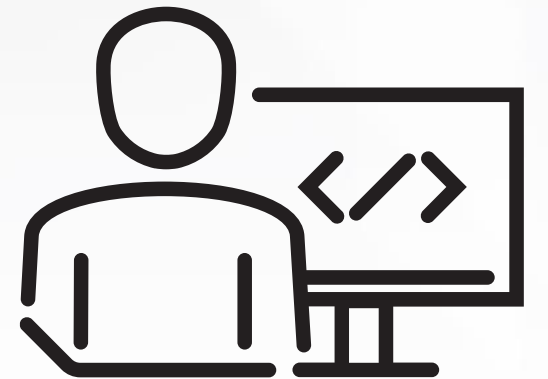
- Export

- Data logger can be exported using the script "Data Logger exporter" available into Template Library. This script can be associated with a button event

A screenshot of a data grid viewer. The grid has four columns: 'Timestamp', 'ModbusTag1', 'ModbusTag2', and 'ModbusTag3'. The grid is currently empty, showing only the header row.

Hands-on session

- Define a new Datalogger with a periodic sampling of 5 seconds
- Configure a Grid to show the datalogger table stored on the database
- Add a button to manually trigger a new record
- Add a button to refresh the grid
- Add a switch to start/stop the sampling



Event logger

Properties

Name

EventLogger1

Type

Event logger

Enabled

True

Locales

Event configurations

EventConfiguration1

Event source

Tags

Listen subtree

True

Event type

AuditWriteUpdateEventType

Non-interactive s...

False

Event fields to log

Database

EmbeddedDatabase1

- At every event the Event Logger creates a new record into the table with structure defined into "Event fields to log"

Time	Client User ID	Source Node	Old Value	New Value
hh:mm:ss	Guest	Variable1	0	5
hh:mm:ss	Admin	Variable2	2	8

- Configure the Source and Event to log
- Select Event fields to log
- Define the Datastore

Event logger

Loggers > **EventLogger1** 

▼ ⚡ BaseEventType

☐ var EventId

☐ var SourceName

☒ var SourceNode

☒ var Time

▼ ⚡ AuditEventType

☐ var ActionTimeStamp

☐ var ClientAuditEntryId

☒ var ClientUserId

☐ var ServerId

☐ var Status

▼ ⚡ AuditUpdateEventType

▼ ⚡ AuditWriteUpdateEventType

☐ var AttributeId

☐ var IndexRange

☒ var NewValue

☒ var OldValue

- At every event the Event Logger creates a new record into the table with structure defined into "Event fields to log"

Time	Client User ID	Source Node	Old Value	New Value
hh:mm:ss	Guest	Variable1	0	5
hh:mm:ss	Admin	Variable2	2	8

- Configure the Source and Event to log
- Select Event fields to log
- Define the Datastore

Event logger

Properties

Name

EventLogger1

Type

Event logger

Enabled

True

Locales

Event configurations

EventConfiguration1

Event source

Tags

Listen subtree

True

Event type

AuditWriteUpdateEventType

Non-interactive s...

False

Event fields to log

Time

UtcTime

../../EventArguments/Time

ClientUserId

String

../../EventArguments/ClientUserId

SourceNode

String

../../EventArguments/SourceNode@BrowseName

OldValue

Int32

../../EventArguments/OldValue@Value

NewValue

Int32

../../EventArguments/NewValue@Value

Database

EmbeddedDatabase1

- At every event the Event Logger creates a new record into the table with structure defined into "Event fields to log"

Time	Client User ID	Source Node	Old Value	New Value
hh:mm:ss	Guest	Variable1	0	5
hh:mm:ss	Admin	Variable2	2	8

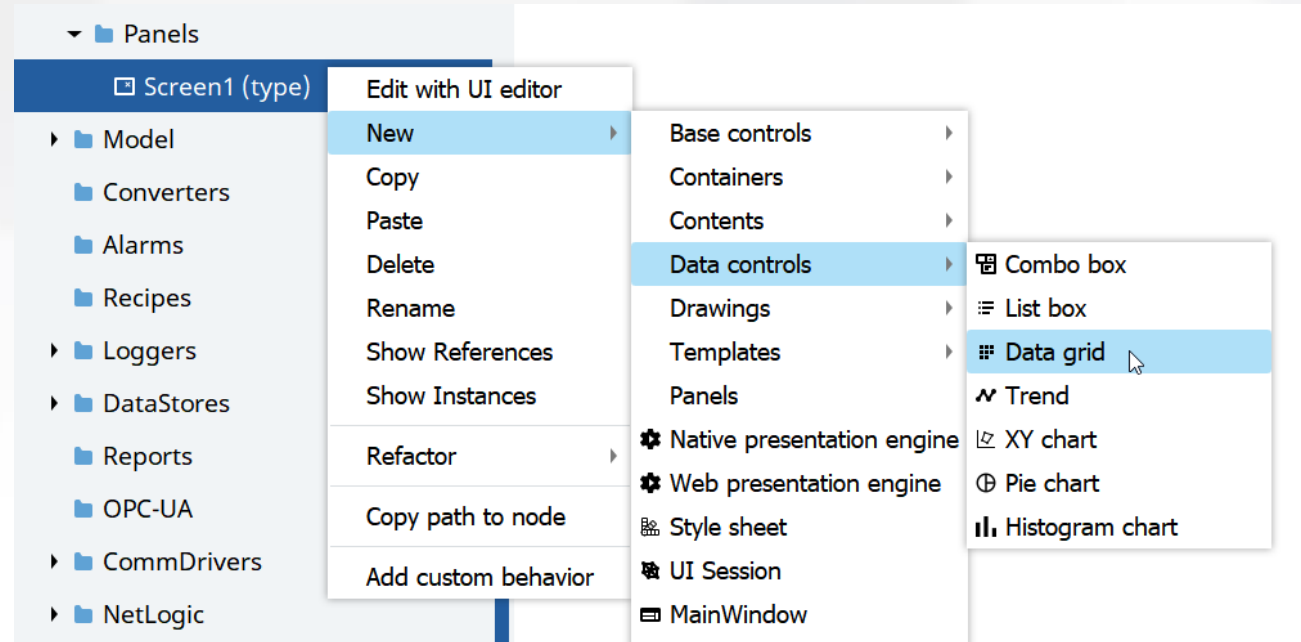
- Configure the Source and Event to log
- Select Event fields to log
- Define the Datastore

Event logger "viewer"

1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Event Logger to the Data grid object

- The object can be customized through Properties pane

- Query
- Columns format
- Sorting

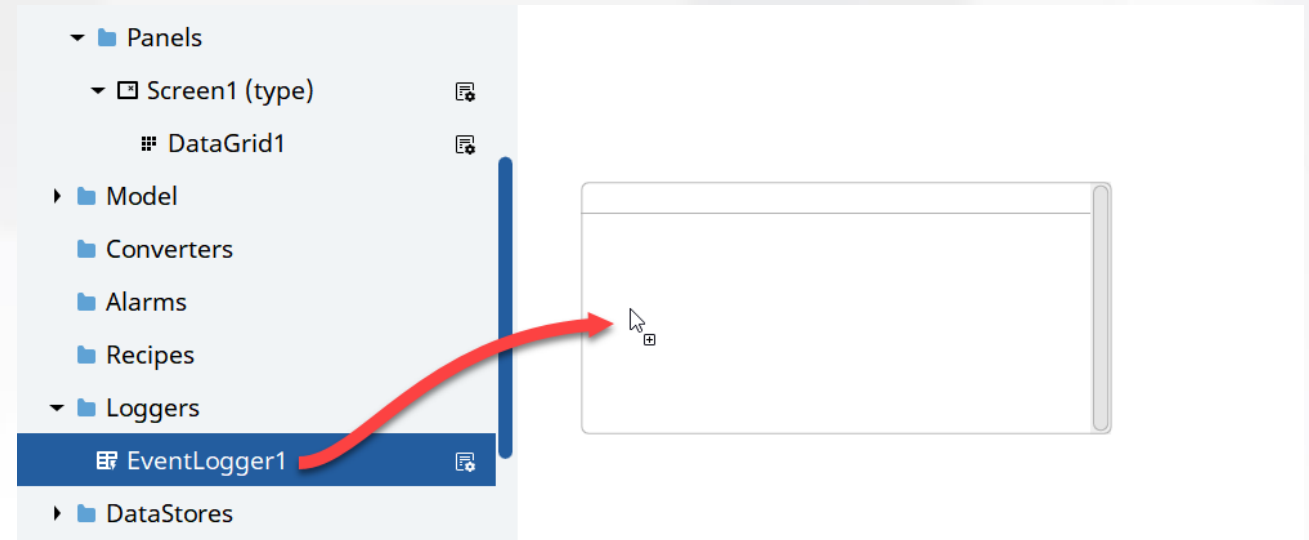


Event logger "viewer"

1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Event Logger to the Data grid object

- The object can be customized through Properties pane

- Query
- Columns format
- Sorting

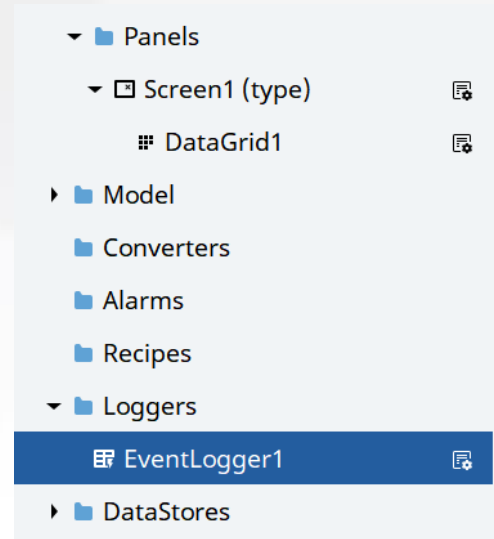


Event logger "viewer"

1. Add a "Data grid" object by New > Data controls > Data grid
2. Drag&Drop the Event Logger to the Data grid object

- The object can be customized through Properties pane

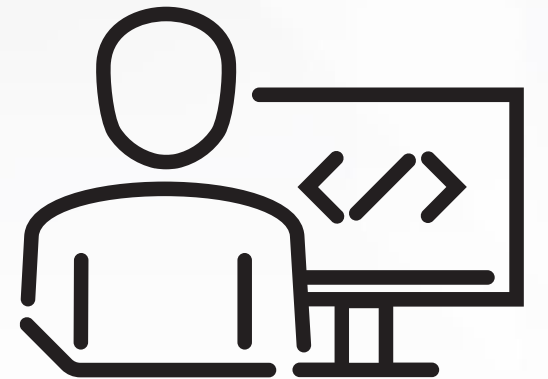
- Query
- Columns format
- Sorting



Time	ClientUserId	SourceNode	NewValue	OldValue

Hands-on session

- Configure an Event Logger to trace changes made to variables
- Add a Grid to show the Event Logger table
- Add a button to refresh the Grid

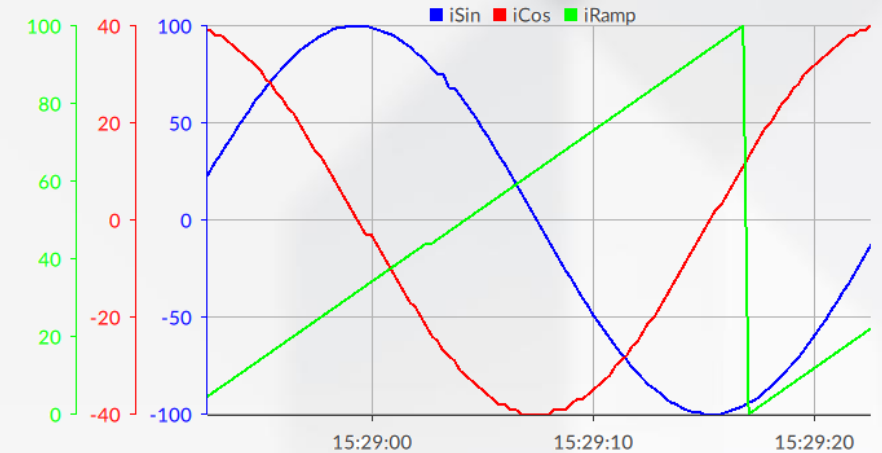


Represent data by Trends



Real-time Trends

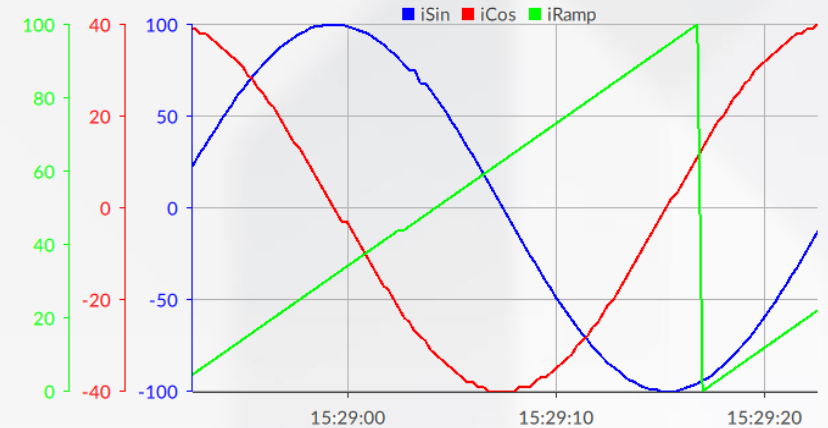
- Trend widget is available under the "Data Controls" group
- Real-time Trend = data showed in the Trend window, will be purged on the exit
- To configure a Trend as a Real-time one, just link a Pen with a Variable
- Optionally, a Y-Axis for each Pen can be configured



Properties	
Name	Trend1
Type	Trend
Pens	
iSin	
BaseDataType iSin	
Thickness	2
YAxis	
Automatic scaling	False
Interactive	True
Minimum value	-100
Maximum value	100
Color	#0000ff
Enabled	True

Historical Trends

- Historical Trends = when the Trend window is loaded, a query is executed to retrieve historical data
- To configure a Trend as an Historical one:
 - a DataLogger is needed to save historical data
 - Pens should be associated to DataLogger > Variables to Log > {name} > Last Value



Type to search...

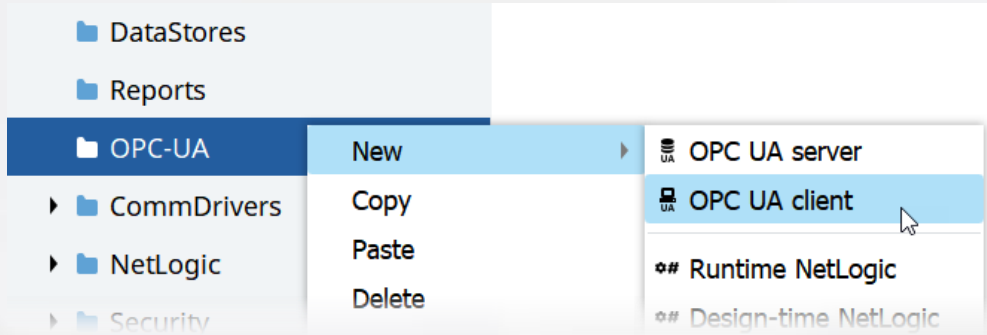
- ▼ Loggers
 - ▼ DataLogger1
 - var Sampling mode
 - var Table name
 - ▼ Variables to log
 - var iSin
 - var Last value**
 - var Timestamp

Attribute

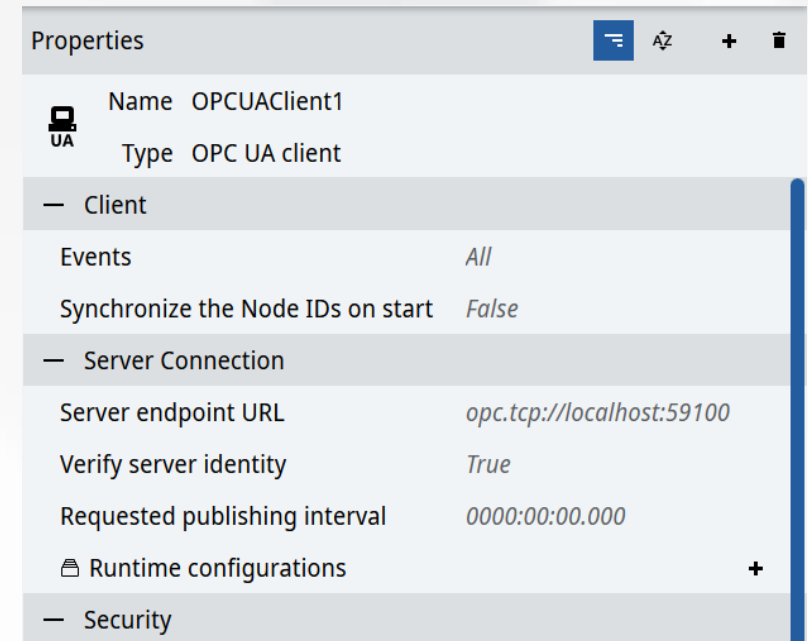
Exchange data via OPC-UA Client



OPC-UA Client: basic configuration

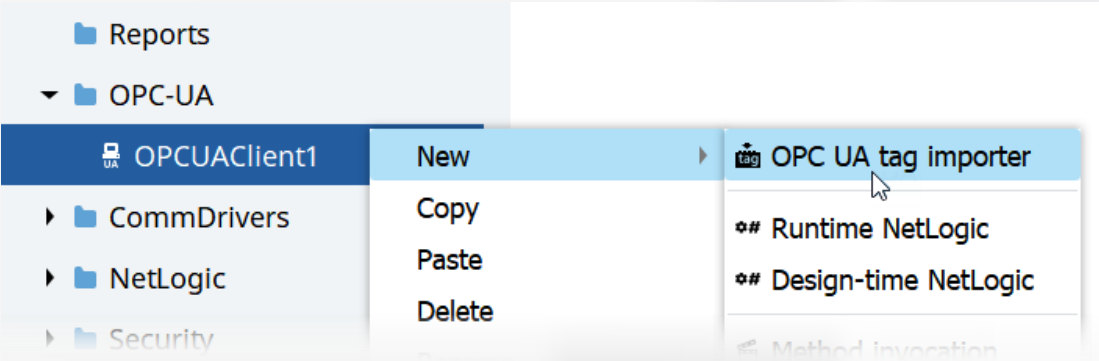


- *Events*: If not interested to read OPC-UA Server Events, set to None
- *Synchronize Node IDs on start*: In case node id of OPC-UA Server changes
- *Server Endpoint URL*: URL of OPC-UA Server to connect to
- *Requested publishing interval*: defines the rate that the OPC-UA Server returns data change notifications to the OPC UA client



OPC-UA Client: tag importer

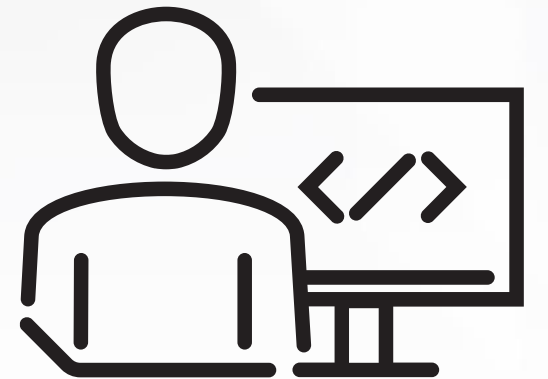
- Like any other Tag Importer
- Allow to import/synchronize tags exposed by an OPC-UA Server



OPC-UA > OPCUAClient1 > OPCUATagImporter					Type to search...	🔍	⚙️	↓	↑	↺
	PLC Item	HMI Item	DataType	Status						
☐	▶ Demo			New						
☐	▶ DeviceSet			New						
☐	▶ DeviceTopology			New						
☐	NetworkSet			New						
					Delete	Ignore	Synchronize			

Hands-on session

- Launch the OPC UA Server (UaAnsiCServer)
- Add the OPC UA Client node to the project and configure the server endpoint
- Open the OPC UA tag importer and import the folder "Demo > BoilerDemo"
- Into the UI:
 - attach the variable "FillLevel" to a Text box
 - attach the variable "Temperature" to a Text box
 - add a button that calls the method "Fill" setting the argument "SetPoint" as variable
 - add a button that calls the method "Heat" setting the argument "SetPoint" as variable



OPC-UA Client: security configuration

- *Verify server identity*
if True, accept only CA-issued certificates
if False, accept Self-signed certificates
- *Minimum message security mode*
Security Profile: none, Sign, Sign+Encrypt
- *Minimum security policy*
- *Client Certificate File*
"der" file stored in "...\ProjectFiles\PKI\Own\Client"
- *Client Private key file*
".pem" file stored in "...\ProjectFiles\PKI\Own\Client"
- Runtime, **will NOT automatically generate** a self-signed certificate for the Client so it's necessary to create one manually



— Client	
Events	All
Synchronize the Node IDs on start	False
— Server Connection	
Server endpoint URL	opc.tcp://localhost:59100
Verify server identity	True
Requested publishing interval	0000:00:00.000
⊞ Runtime configurations +	
— Security	
Minimum message security mode	None
Minimum security policy	None
Client certificate file	Browse
Client private key file	Browse

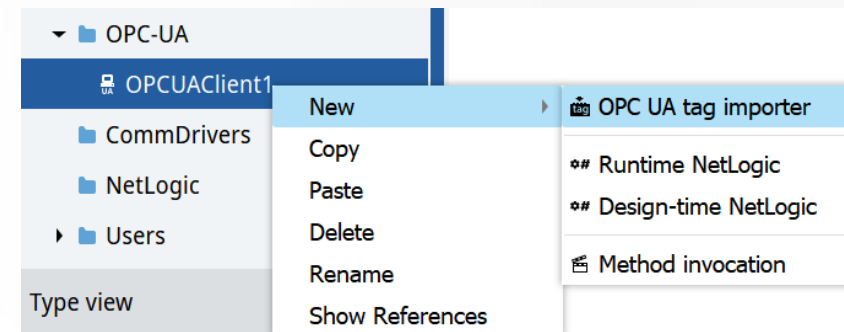
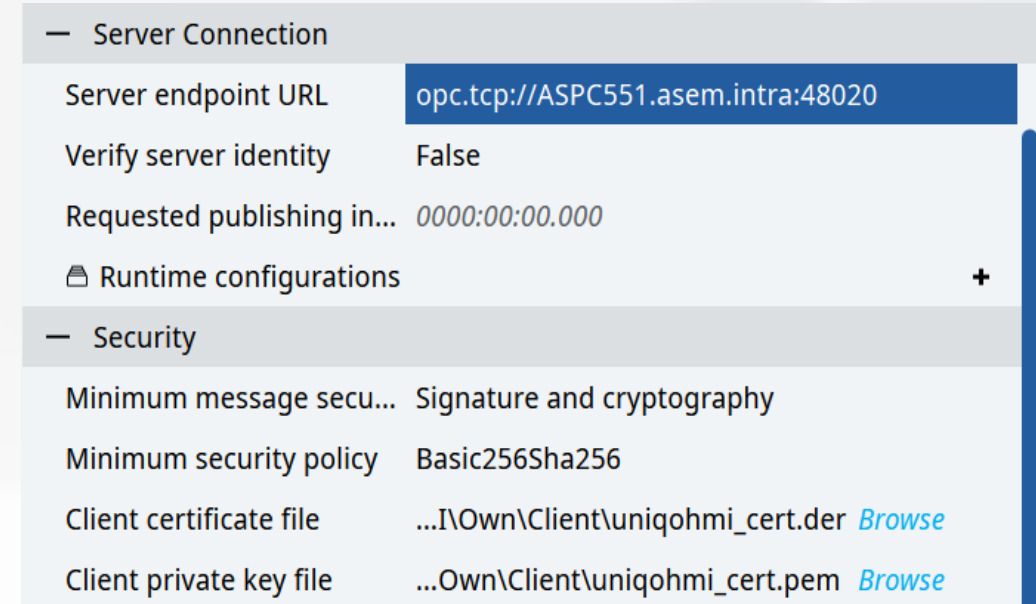
OPC-UA Client: connection using certificates

- Connection to a 3rd part OPC-UA Server

1. Configure OPC-UA Client Settings:

- Server endpoint URL
- Set "Verify server identity" to False
if OPC-UA Server uses a self-signed certificate
- Set Security Profile and Security Policy
- Import self-signed Client Certificate file
and Client Private Key file

2. Import tags adding the object OPC UA tag importer



OPC-UA Client: connection using certificates

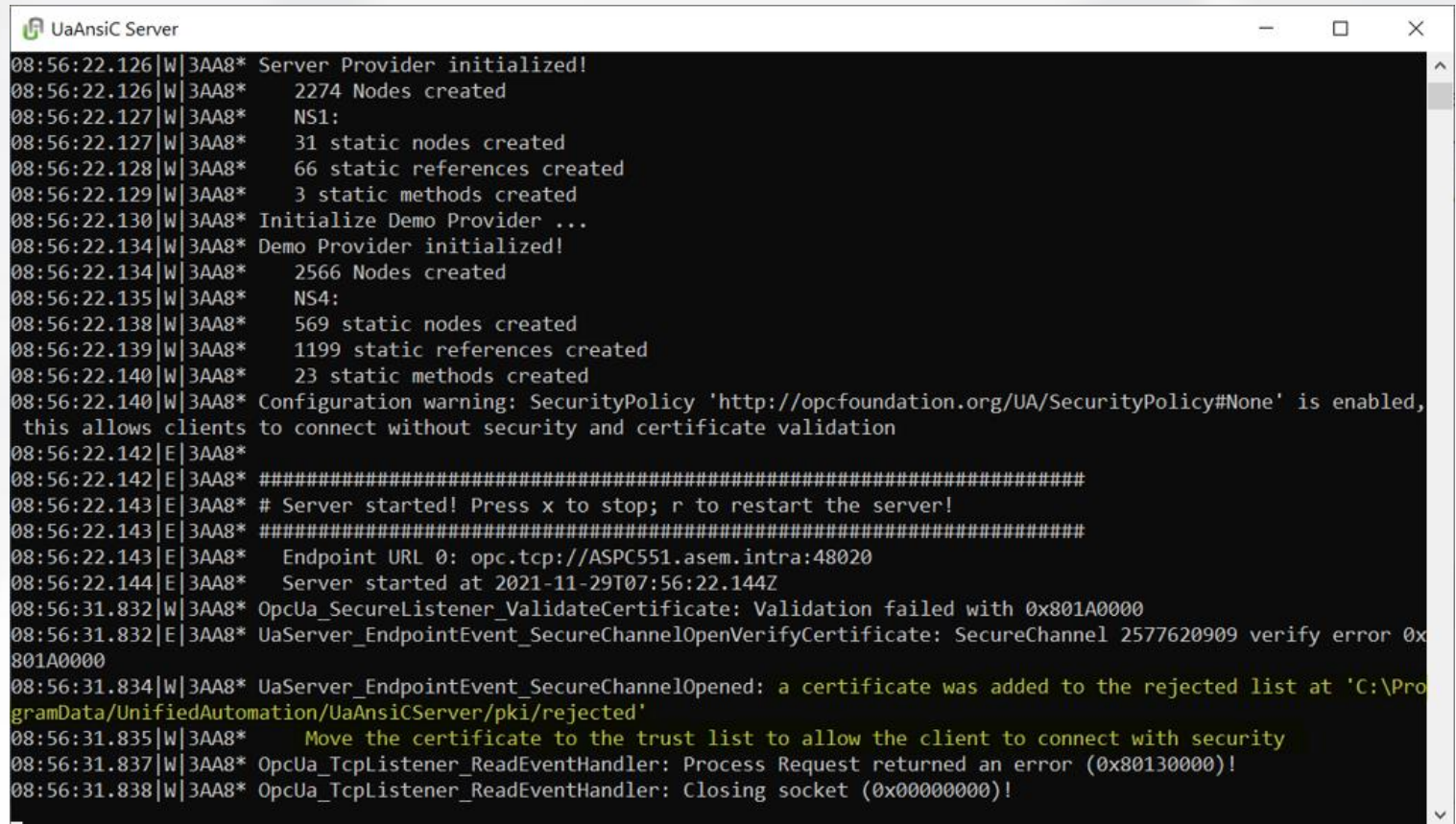
- OPC-UA Client sends his certificate to the Server, but in this case, the Server automatically set the certificate into the Rejected folder so "***BadSecurityChecksFailed***" is triggered

Code	Timestamp	Category	Object	Message
 43	2021-11-29 08:54:31.967	TaskConsumer		Exception caught: Connection failed with error BadSecurityChecksFailed

Q Studio Output

OPC-UA Client: connection using certificates

- In this case, we need to go to the OPC-UA Server and manually move the Client certificate from Rejected to Trusted folder
- After this operation, the Client will be able to connect to Server to Browse tag for import and exchange data.



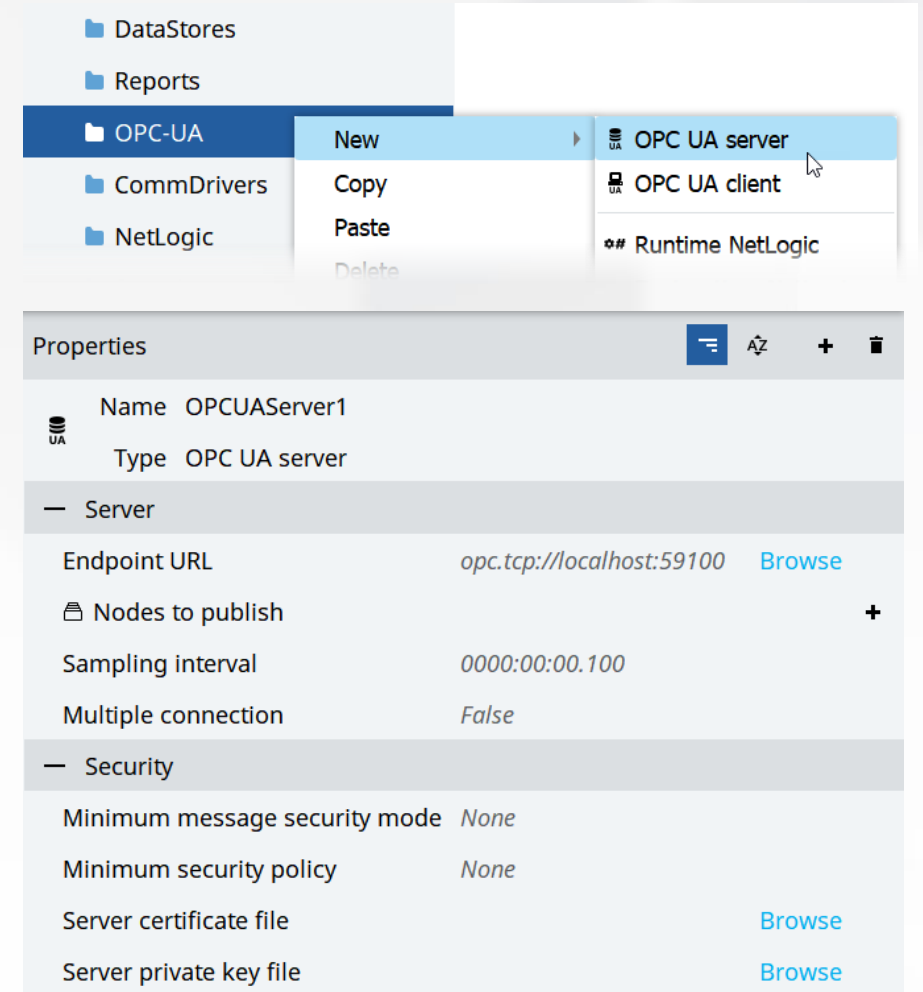
```
UaAnsiC Server
08:56:22.126|W|3AA8* Server Provider initialized!
08:56:22.126|W|3AA8* 2274 Nodes created
08:56:22.127|W|3AA8* NS1:
08:56:22.127|W|3AA8* 31 static nodes created
08:56:22.128|W|3AA8* 66 static references created
08:56:22.129|W|3AA8* 3 static methods created
08:56:22.130|W|3AA8* Initialize Demo Provider ...
08:56:22.134|W|3AA8* Demo Provider initialized!
08:56:22.134|W|3AA8* 2566 Nodes created
08:56:22.135|W|3AA8* NS4:
08:56:22.138|W|3AA8* 569 static nodes created
08:56:22.139|W|3AA8* 1199 static references created
08:56:22.140|W|3AA8* 23 static methods created
08:56:22.140|W|3AA8* Configuration warning: SecurityPolicy 'http://opcfoundation.org/UA/SecurityPolicy#None' is enabled,
this allows clients to connect without security and certificate validation
08:56:22.142|E|3AA8* #####
08:56:22.142|E|3AA8* # Server started! Press x to stop; r to restart the server!
08:56:22.143|E|3AA8* #####
08:56:22.143|E|3AA8* Endpoint URL 0: opc.tcp://ASPC551.asem.intra:48020
08:56:22.144|E|3AA8* Server started at 2021-11-29T07:56:22.144Z
08:56:31.832|W|3AA8* OpcUa_SecureListener_ValidateCertificate: Validation failed with 0x801A0000
08:56:31.832|E|3AA8* UaServer_EndpointEvent_SecureChannelOpenVerifyCertificate: SecureChannel 2577620909 verify error 0x
801A0000
08:56:31.834|W|3AA8* UaServer_EndpointEvent_SecureChannelOpened: a certificate was added to the rejected list at 'C:\Pro
gramData\UnifiedAutomation\UaAnsiCServer\pki\rejected'
08:56:31.835|W|3AA8* Move the certificate to the trust list to allow the client to connect with security
08:56:31.837|W|3AA8* OpcUa_TcpListener_ReadEventHandler: Process Request returned an error (0x80130000)!
08:56:31.838|W|3AA8* OpcUa_TcpListener_ReadEventHandler: Closing socket (0x00000000)!
```


Expose data as OPC-UA Server



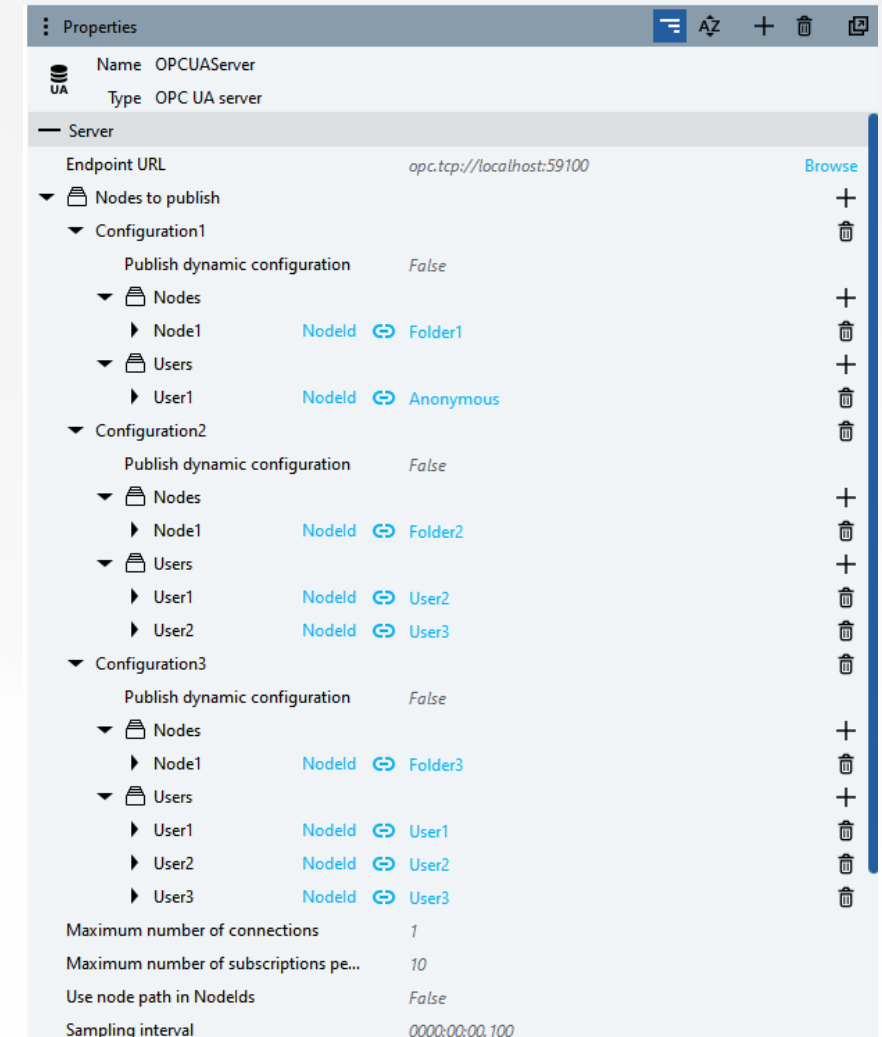
OPC-UA Server: basic configuration

- *Endpoint URL*
 - protocol, IP/hostname, port used by Server
- *Nodes to publish*
 - by default, the entire project will be published
- *Sampling Interval*
 - Defines a “best effort” cyclic rate that the Server uses to sample the item from its source.
i.e. the Communication Driver tag polling rate
- *Multiple connections*
 - Allow multiple OPC-UA Clients to connect
then OPC-UA Server cost 2 tokens instead of 1
- *Security settings*
 - See: [Appendix: See the light about OPC-UA Security](#)
 - See: [Appendix: Using OPC-UA certificates](#)
 - See: [Appendix: Generating OPC-UA certificates](#)



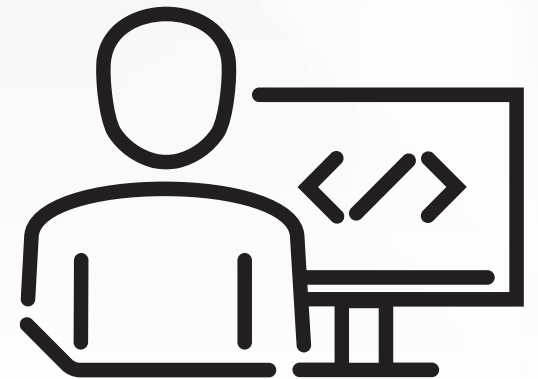
OPC-UA Server: Nodes to publish

- Specific users can be limited to specific nodes of the project
- When configuring a set of nodes to publish, a user entry must be provided (can be any project user or “Anonymous”)
- Multiple nodes can be assigned to multiple users
- Currently no way to set a node in “read-only” mode
- If no entry is created in the **Nodes to publish** collection, then the whole project is exposed via OPC-UA



Hands-on session

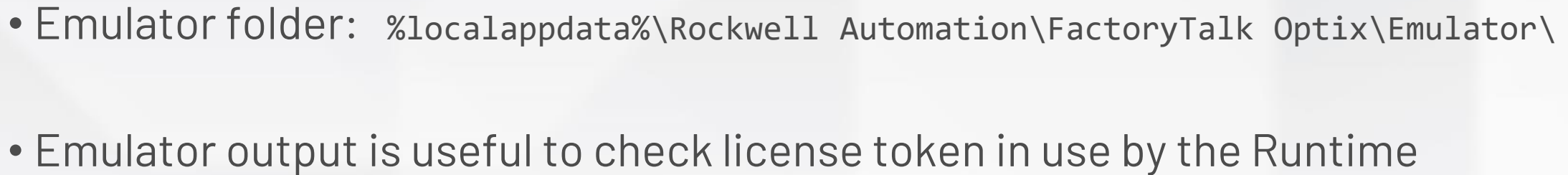
- Add the OPC UA Server node to the project
- Configure the OPC UA Server to expose the entire project
- Launch the Emulator
- Launch the OPC UA Client «UA Expert» and connect to the Emulator
- Browse some variables or call a method exposed like the "Log" method of a Datalogger



Project deployment

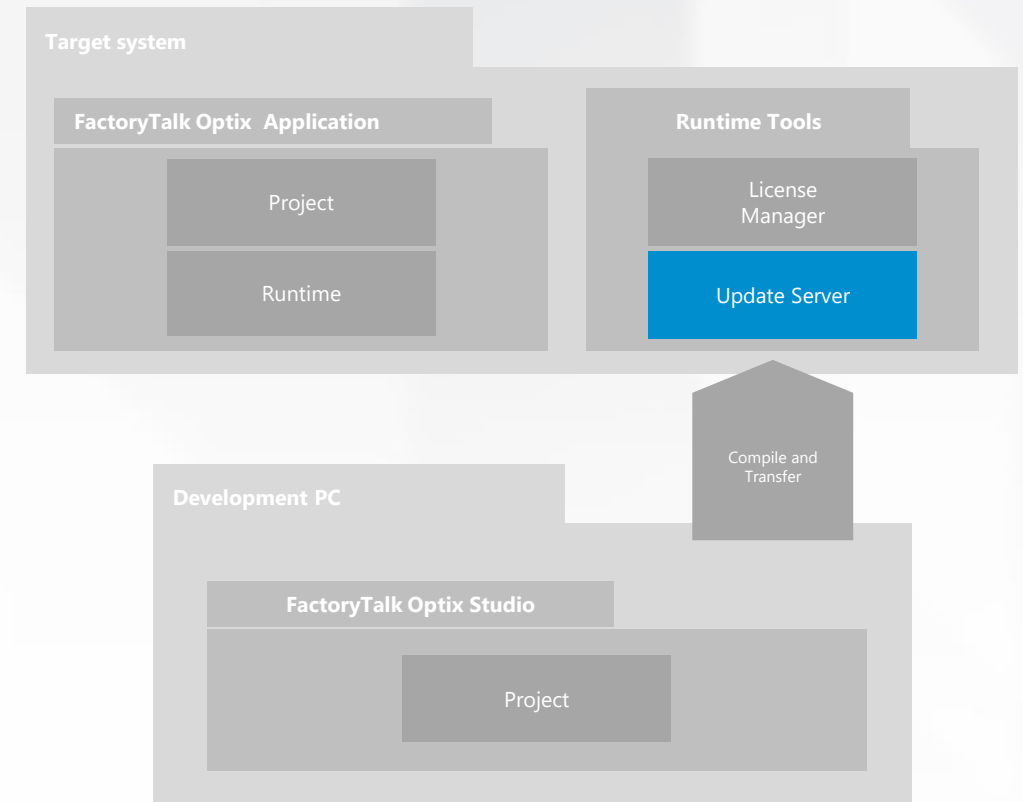


- Allow the execution of the project from Studio.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. **289**

Transfer the project to local target

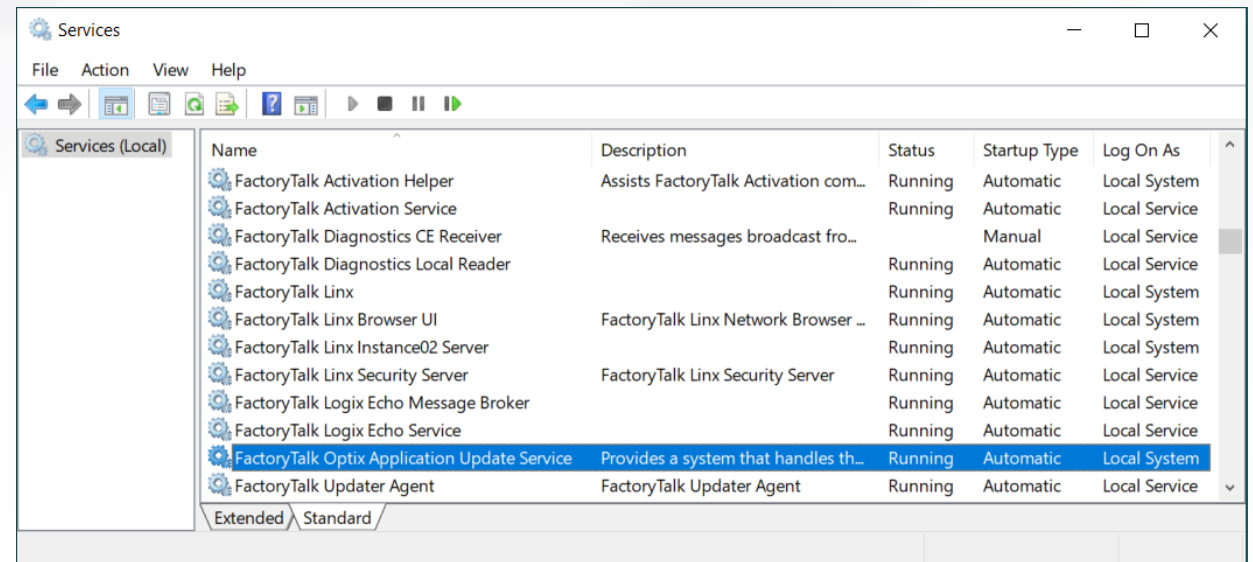
1. Target must have "Runtime Tools" installed and "Update Server" running
 - **Optix Panel:** Runtime Tools are already installed in production and Update the Server running as a service
 - **iPC:** Runtime Tools must be installed manually. Update Server will be installed as Service and set to automatically startup when Windows starts



Transfer the project to local target

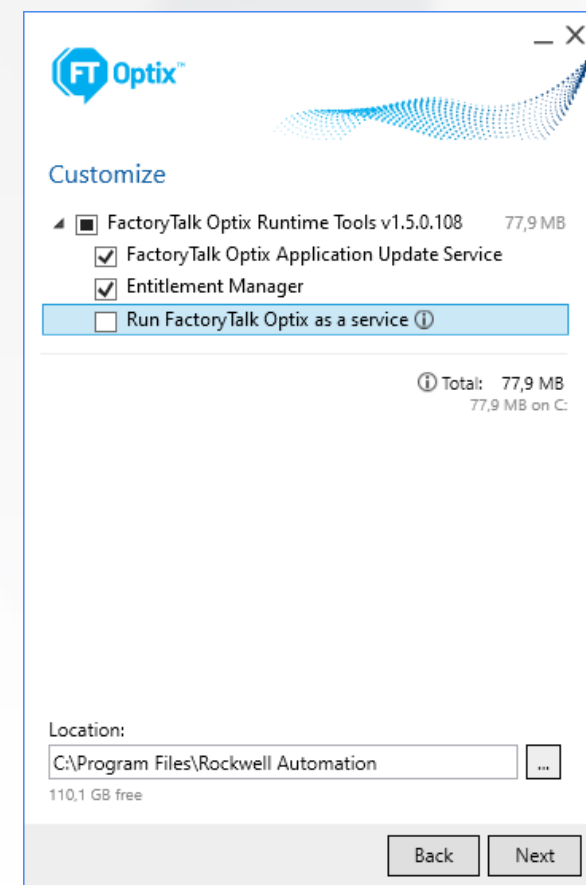
1. Target must have "Runtime Tools" installed and "Update Server" running

- **Optix Panel:** Runtime Tools are already installed in production and Update the Server running as a service
- **iPC:** Runtime Tools must be installed manually.
Update Server will be installed as Service and set to automatically startup when Windows starts
- On **OptixPanels**, the UpdateServer is automatically upgraded by FactoryTalk Optix Studio if needed (downgrade not supported)
- On **iPC**, the UpdateServer can be upgraded or downgraded by the user as needed
- FactoryTalk Optix version must be **same or newer** version than the UpdateServer version (no backward compatibility)



Run the project as Windows service

- Configure the RuntimeTools in Service mode during setup
 - Only available on Windows IPCs
 - Project cannot contain a NativePresentationEngine
 - WebPresentationEngine is allowed but not strictly needed
 - Useful when the application does not have any UI (data gateway) or when the user interface should only be accessed remotely



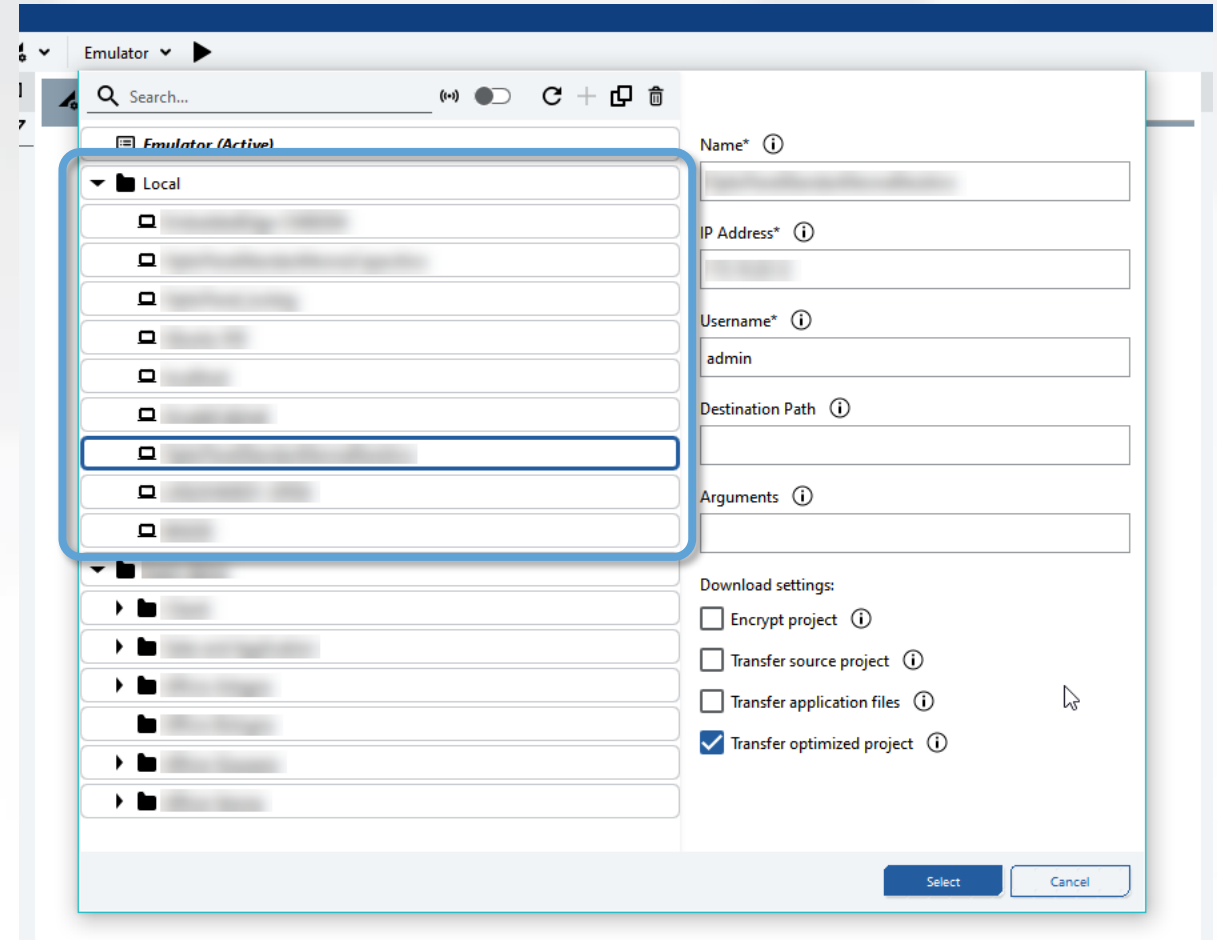
Transfer the project to local target

2. Configure Target device in Studio

- Name,
- IP Address,
- Username,
 - for Optix Panels, it's the same admin user that access the System Manager
 - for iPC, it's a Windows user with rights to access OS folders
- Destination Path: optional

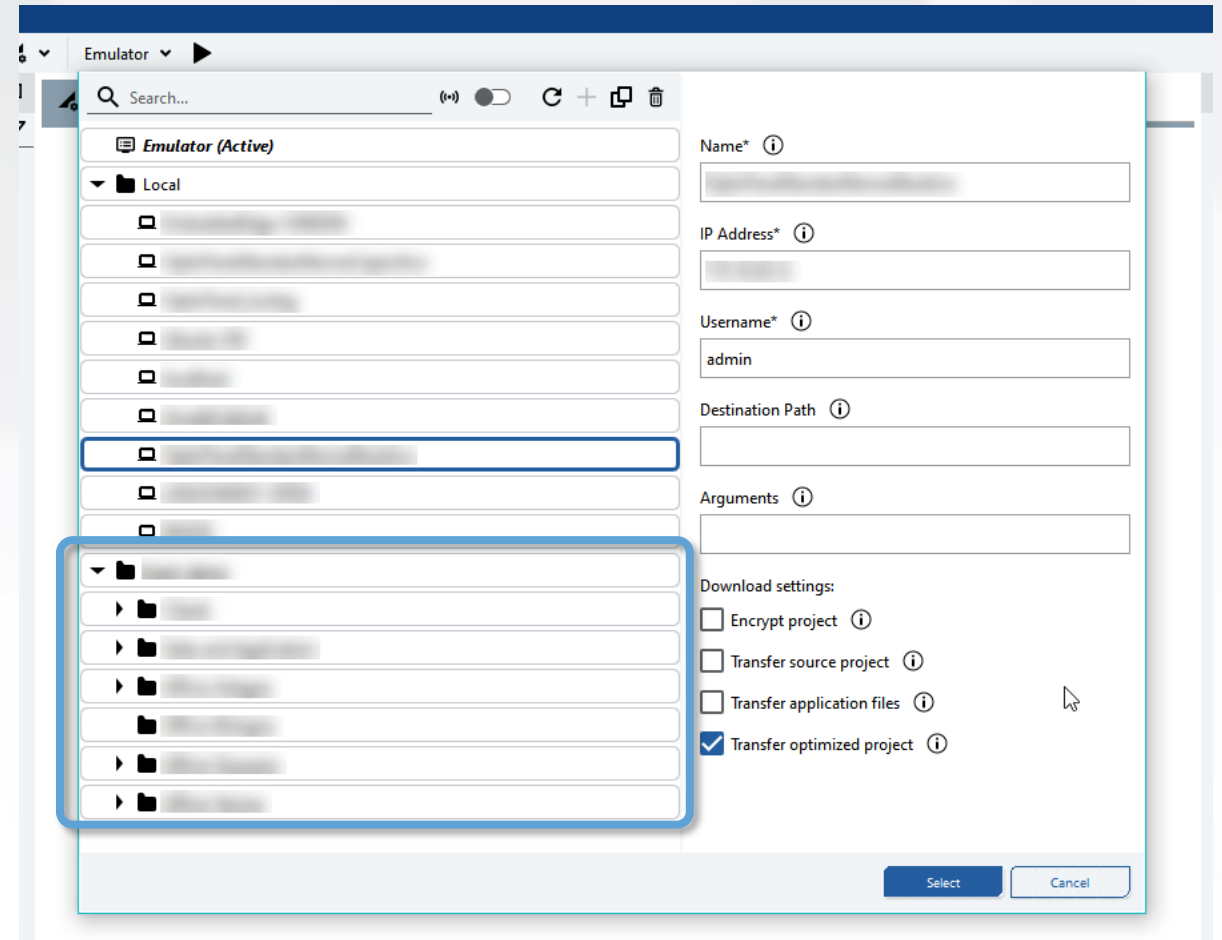
3. Transfer from Studio to Target

- set the Active path
- click the Play button



Transfer the project to remote target (FactoryTalk Remote Access)

- If you are signed in FactoryTalk Hub and you have also access to a FactoryTalk Remote Access domain
- FactoryTalk Optix Studio, lists the devices available through FactoryTalk Remote Access
- Then you will be able to:
 - activate a VPN connection to the device and transfer the project
 - open the FTRA screen mirroring feature (VNC-like)



Execute the project on target

4. Application is launched by Studio after download

- **HMI:** Application is also set to be executed automatically at HMI startup
- **IPC:** Application is also set to be automatically executed at Windows startup by using the shortcut of

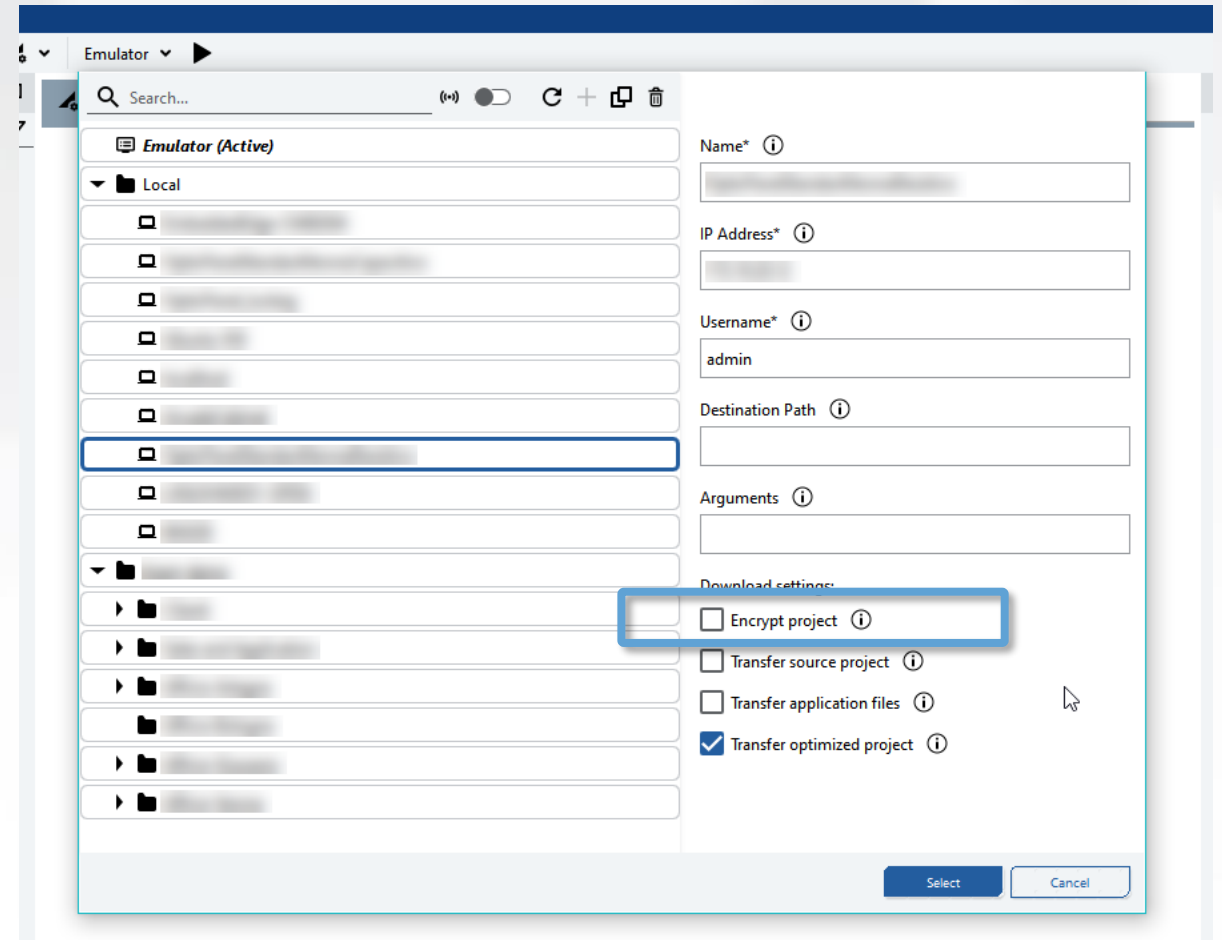
`%localappdata%\Rockwell Automation\FactoryTalk
Optix\Application\FTOptixRuntime.exe`

into:

`shell:startup`

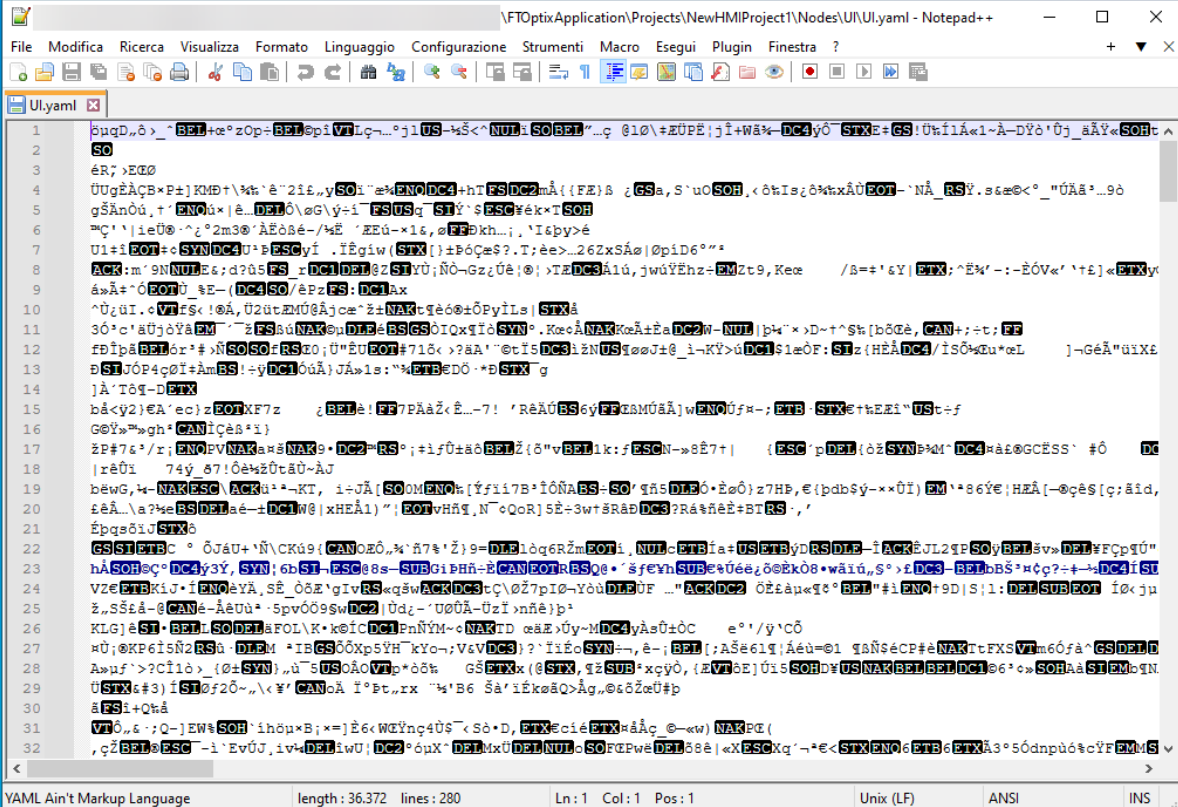
Deployment options

- Project can be encrypted with a hardware-based key
 - All YAML files will be fully encrypted using a unique and hardware-based encryption key
 - Once project is encrypted, it cannot be moved to a different PC or Panel (as the encryption is hardware based)
- Project is encrypted and decrypted using the UpdateServer
 - Encryption is only supported when deploying the project from FactoryTalk Optix Studio
 - Project can be decrypted and uploaded (if source files are transferred) only by using the Update Server



Deployment options

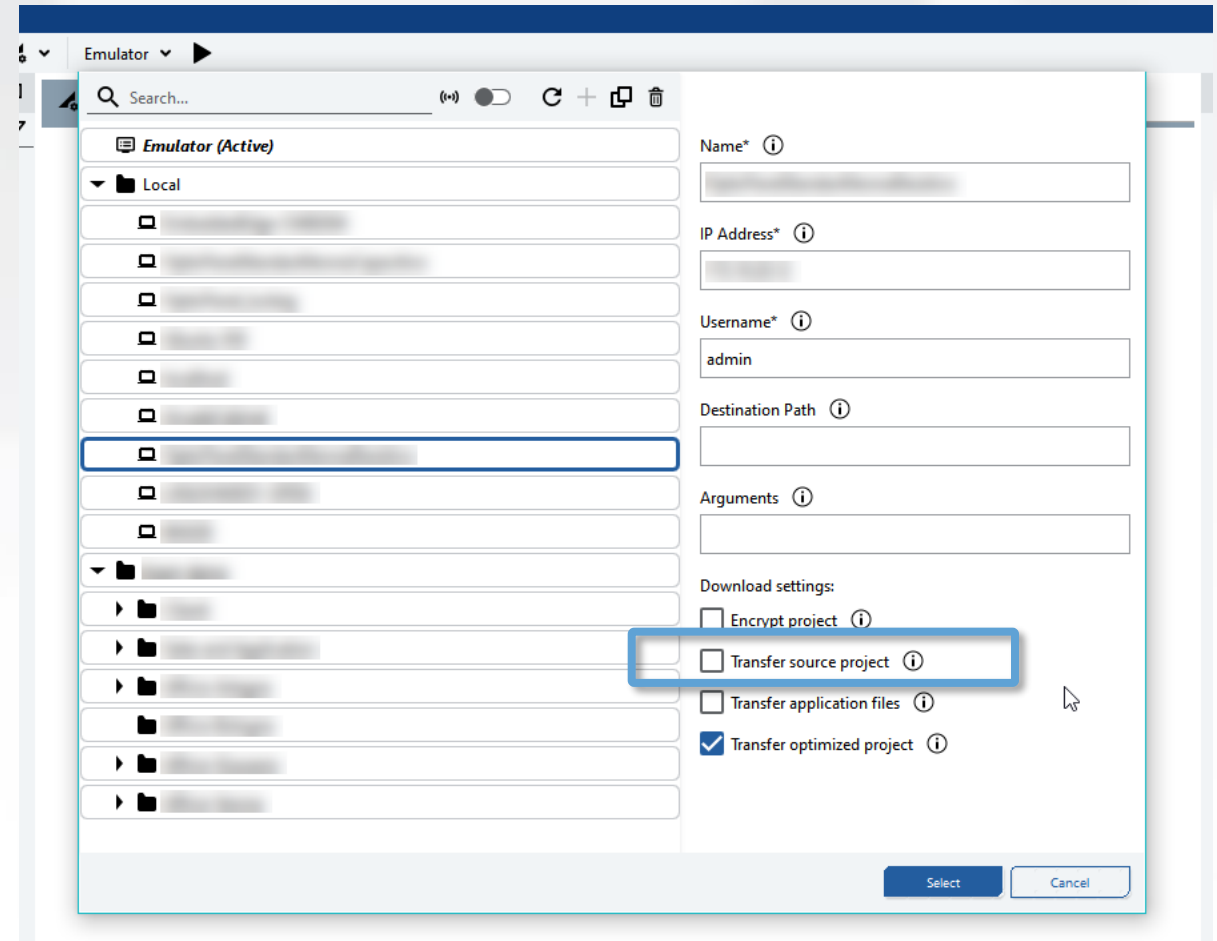
- Project can be encrypted with a hardware-based key
 - All YAML files will be fully encrypted using a unique and hardware-based encryption key
- Once project is encrypted, it cannot be moved to a different PC or Panel (as the encryption is hardware based)
- Project is encrypted and decrypted using the UpdateServer
 - Encryption is only supported when deploying the project from FactoryTalk Optix Studio
 - Project can be decrypted and uploaded (if source files are transferred) only by using the Update Server



```
1 1uqD,,ò>_ ^BETI+æ°zOp=BEI@piV7Lq~...°j1US-4S<^NUI:SOBEI"...q @10+æZUPE;jî+wâ%-DC4Y0°STXE=GS!Ükîlâ1~Â-DYô'Üj_âÄY«SOHt ^
2 SO
3 éR; >E00
4 ÜUgEÄÇB×P+J]KMB+\\%«'è~2iE,,yS0I'~æ%ENODC4+hT°FSDC2mÄ{({FE)S_çGSa,S'uoSOB<ð%Isçð%kxÄÜEOT~NÄ_RSÿ.sææ<°_ÜÄY«...9ð
5 gSÄn0ú,t'°ENQú×|è...DEIÖ\@G\ý=i°FSUSq°STY°°ESC#éK×T°SOH
6 °Ç'°|ieÜ0°^ç°2m30°ÄEðâé-/4E °ZEÜ-×16,øFEBkh...; °I6py>é
7 U1±1EOT+°SYND0C4U°°ESCyI °IÊgiw(STX){±bóÇæ$?.T;ée>...26ZxSÁø|@piD6°""
8 ACK:m'°9NNUT°E6;d?05ES_°DC1DEI°Ez°STYÜ;NÖ-GzzÜè;0; >EDEC3Ä1ú,jwüYËhz=EMZt9,Kee /B=±°sY°(STX;^E%'-:-EÖV«'°t±)«ETX°Y
9 á»Ä±°°EOTÜ°°E-(DC4SO/èPz°FS:DC1Ax
10 ^ÜçUI,°°V7fS<°!0Ä,Ü28t°EMÜ0Äjçæ°z°NARt°gè0±0PyiLs°(STXÄ
11 30°c°aÜjòYâEM°°z°FS8d°NAR°°DIB°BSGS0IQx°qI0°SYN°.Kæ°Ä°NAR°Kæ±Ea°DC2W-NUB|p4°×>D~+^°S%[bðÇE,CAN+;±t;°E
12 fDIPäBETI°ör'°#°N°SO°SO°°RS°E0;Ü°EUEOT°#710< >?ÄA'°°EtIS°DC3iZ°NUS°E0J±@_i-KY>ü°DC1S1æOF:STz(HEÄ°DC4/IS0°Eü°°æL }-GéÄ"Ü1XÉ
13 B°STJÓP4ç0I+ÄmBS!±y°DC1ÖüÄ)JÄ»1s:°°E°TB°EDÖ°°B°STX°g
14 jÄ'T0E-D°ETX
15 bÄ<y2)EÄ°ec}z°EOR°XF7z ç°BEB°E!°E37PÄäZ<È...-7! °ReÄÜ°BS6Y°E°E8MÜ0Äjw°ENQÜf~h-;°ETB°STX°e°tEzi°US°t~f
16 G0E°w°°gh°°CANIÇèÄ°i}
17 ZP#76°/r;°ENQ°PVNAR°a±°NAR°°DC2°°RS°°;±ifÜ±æ°BETI°Z°(°v°BETI1k:f°E°SCN-»8E7+| (°ESC°p°DEI°(òz°SYN°°M°DC4°HäE°GC°ESS°°#0°DC
18 |rèÜ1 74Y°87!0èkZÜtÄÜ-ÄJ
19 bèwG,°-NAR°ESC°ACR°u°°-KT, 1+JÄ[°SO°M°ENQ°[°f°i17B°°IÖN°BS°-SO°°E85°DIB°°E0°z7HP,e{pðb$ý-××ÜI)EM°°86Y°E;HEÄ[-0ç8$[ç;Äid,
20 eÄÄ...a?°æ°BS°DEI°æ±°DC1W(°jxHEÄ1)°°|°EOT°VH°E°N°°QoR]5E±3wt±SRÄD°DC3?Rä°E°E±E°BTR°S°,°
21 Èpqs0iJ°STX0
22 GS°SI°DIB°°°ÖJäU+°N°CKÜ9{°CAN°E0°,°Ä7°±°Z°}9=°DIB°10ç6RZm°EOT1.NUL°°ETB°ia°US°ETB°yD°RS°DLE-i°ACK°EJL2°EP°SOY°BETI°éV»°DEI°f°FC°p°qÜ"
23 hÄ°SOH°eç°DC4°y3Y,SYN;6b°ST°-ESC°e8s-SUB°G1P8H±E°CAN°EOT°RS°Qe°°°š°f°evh°SUB°e°üééç°è°E°K08°wäid,°S°>°I°DC3-BETI°B°S°°H°ç?°+±°DC4i°ST
24 VZe°ETB°KiJ°i°ENQ°EYÄ,SÈ°08E°gIv°RS°«qsw°ACR°DC3°Ç°0Z7pI0-Yòü°DIEÜF°...°ACK°DC2°ÖE±äü«°E°°BETI°°#1°ENQ°°9D;S;1:DEI°SUB°EOT°I0<jü
25 z,,SSÉÄ-0°CAN°E-ÄEÜà°°Spv0099sw°DC2|Üdç-'U0ÜÄ-ÜzI°nÄE)°p°
26 KLGjE°ST°BETI°L°SO°DEI°AFOL°K°k°E°IC°DC1°Pn°NÿM~°c°NAR°TD°æÄE°üY~M°DC4°yÄSÜ±0C e°°/°y°C0
27 °Ü;°0KP6iS°N2°RSä°DIE°M°°IB°GS°Ö0XpS°YH°°K°Yc~;V±V°DC3)?°I°i°E°c°SYN°-,-è;°BETI[;AS°E61q|Äü±01°°SÄ°S°E°CP°E°NAR°t°FXS°V7m6°fä°°GS°DEI°D
28 Äwuf°>°C110>_{0±°SYN}°ü°°5°US°ÖÄ°V7°p°°08% GS°ETX°K(°STX,°E±SUB°°xçy0,(E°V7°0E)Üis°SOH°D°Y°US°NAR°BETI°BETI°DC1°06°°«°SOH°Ää°ST°EM°°EIN
29 Ü°STX°#3)°S°I°0f20~°^°\°°CAN°Ä°I°°P°t°rx °°°B6°SÄ°°I°E°kæ°Q°Äg,°@s0Z°æÜ#p
30 ä°FS±°Q°Ä
31 V7°0°,°s°;Q-J°EW°°SOB°°ihöµ×B;°=j°E°6°WZ°Ync4Ü$°~°S0°D,°ETX°°c°i°E°ETX°°ÄÄÇ°E~°«w)°NAR°PE(
32 °,çZ°BETI°°ESC°-1°°EvÜJ,iV±°DEI°iwU;°DC2°°0µX°°DEI°MxÜ°DEI°NUL°°SO°F°EP°E±°DEI°08E|«°X°ESC°Kq°~°*°E<°STX°ENQ°°ETB°°ETX°Ä3°50dnpu0°k°c°Y°F°EM°°E
```

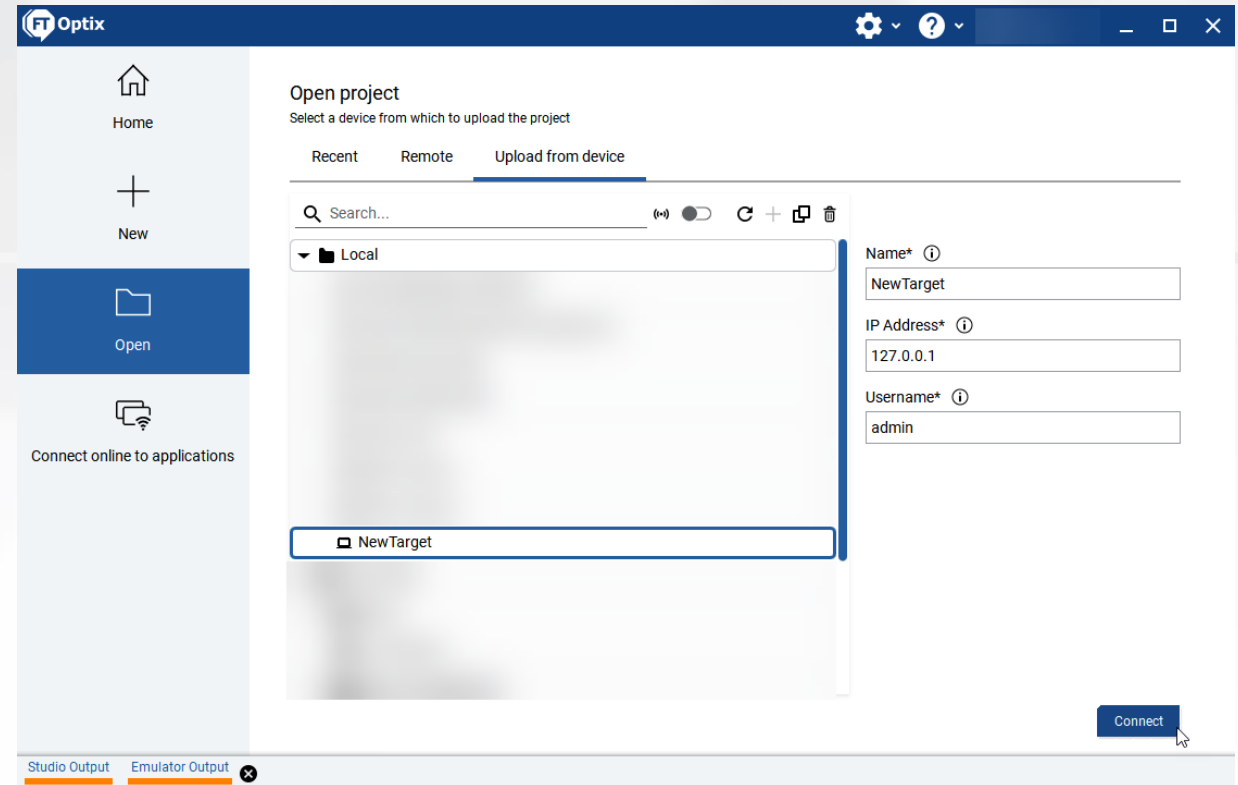
Deployment options

- Source code of the application can be deployed to the device
- This allows the project to be uploaded from the target device if needed



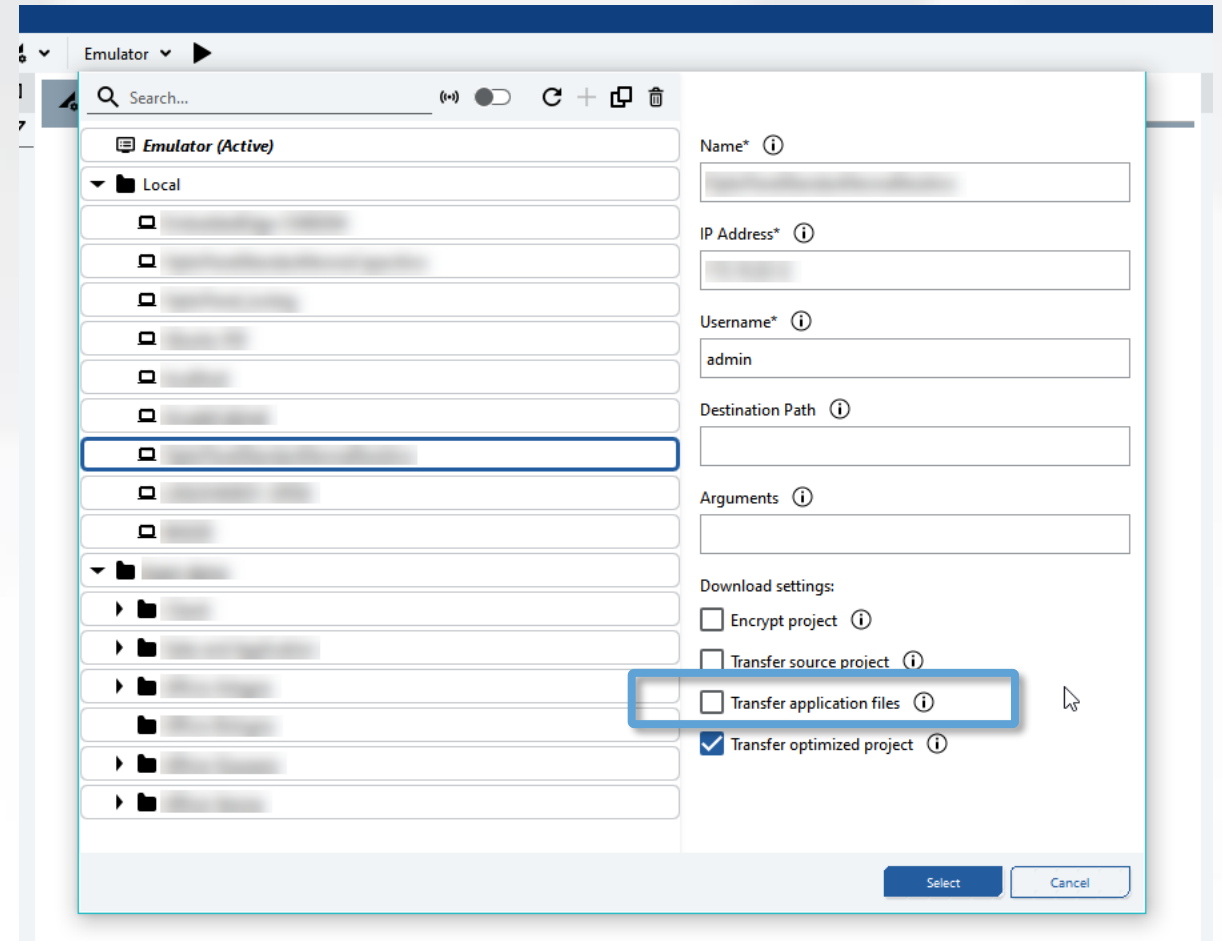
Deployment options

- Source code of the project can be transferred to the target device
- All YAML files, resources and NetSolution will be transferred to the target device
- If the project sources were transferred to the device, the project can be uploaded later from the device using the dedicated menu in the welcome screen of FactoryTalk Optix IDE



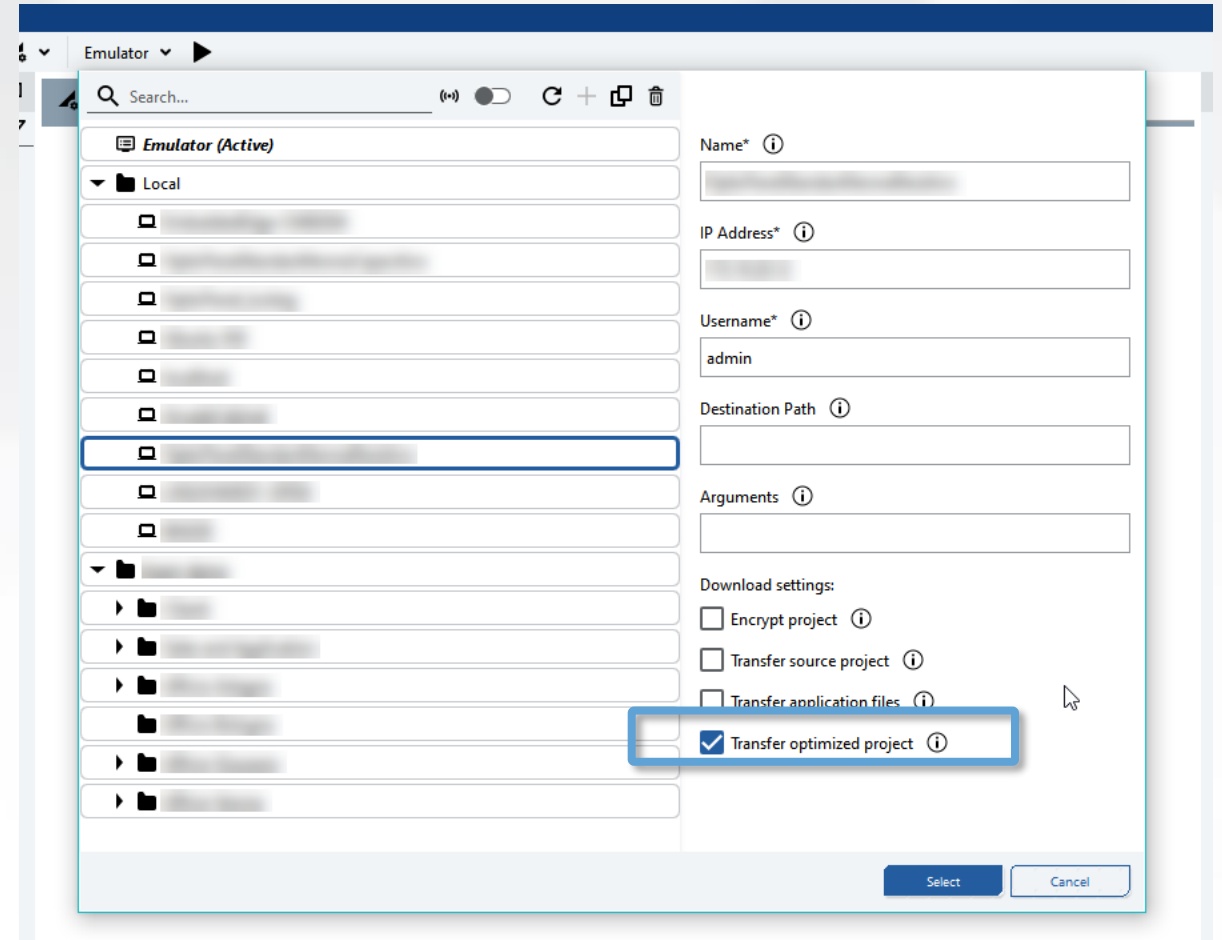
Deployment options

- Application files on the target device can be overwritten
- When a project is uploaded from a device, the files in the ApplicationDir can be uploaded too, this includes: retentivity, users, database, etc.
- Once the files were uploaded, they can be transferred to the same or a different device



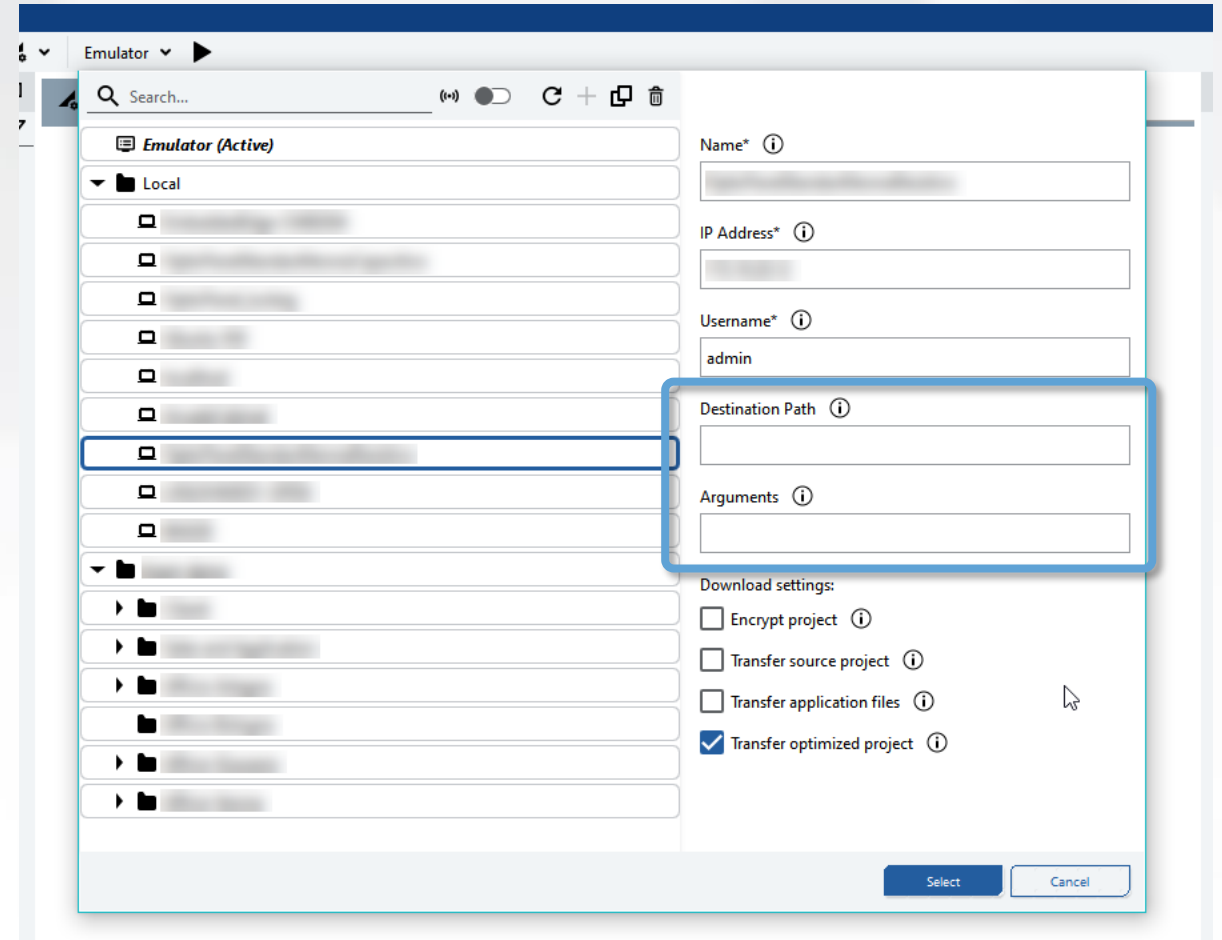
Deployment options

- Optimize the project before deployment
 - In some cases, a lot of tags were imported in the application even if not used in the application
 - The optimization process will move all imported tags from the YAML file to a dedicated binary repository where tags are retrieved only if needed.
 - This can significantly improve startup time and memory usage
 - Cannot be used if the OPC/UA server is exposing the PLC tags



Additional parameters

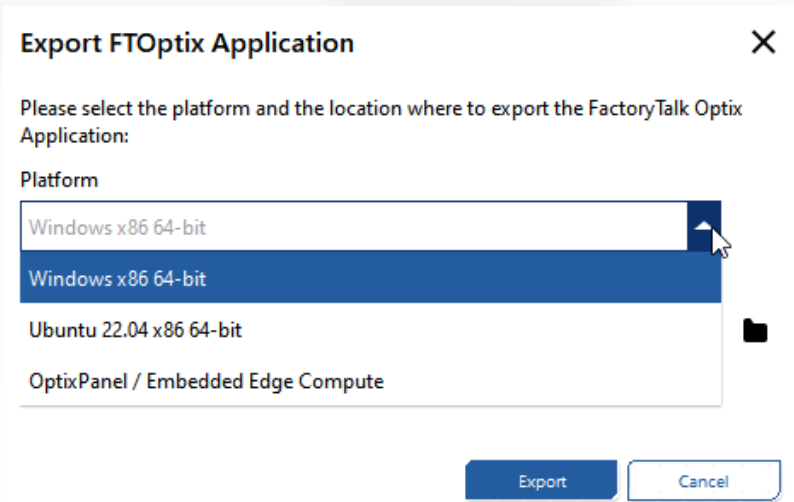
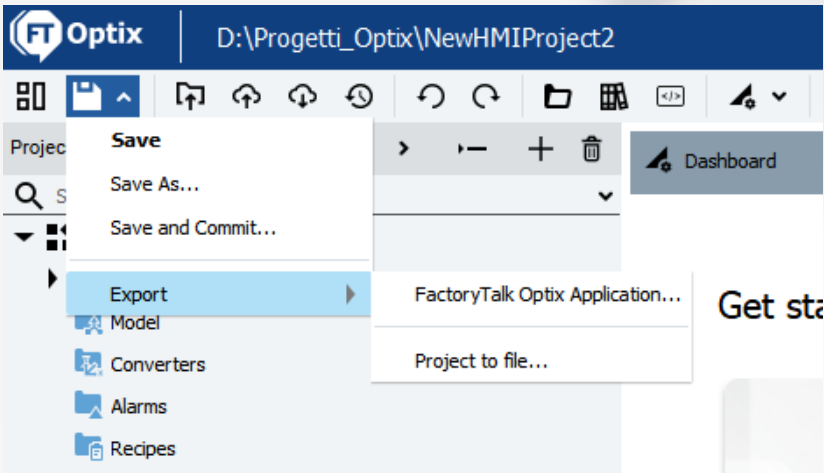
- A specific path on the remote device can be specified
 - Only for Windows-based devices
 - The user being selected for deployment must have write access to such path
- Additional startup arguments can be added
 - Only used for debugging if requested by SW support or R&D team, should be always blank



Transfer project to target manually

- Application can be transferred manually
 - **"Export FactoryTalk Optix Application"** : exports the (semi) compiled Application into a folder that needs to be manually transferred and executed on Target
 - No encryption can be enabled if exporting the application from this menu

Platform	HMI Model
Optix Panel	Optix Panel Standard or Compact
Windows (on x64 CPU)	iPC Windows
Ubuntu (on x64 CPU)	iPC Linux with Ubuntu 22

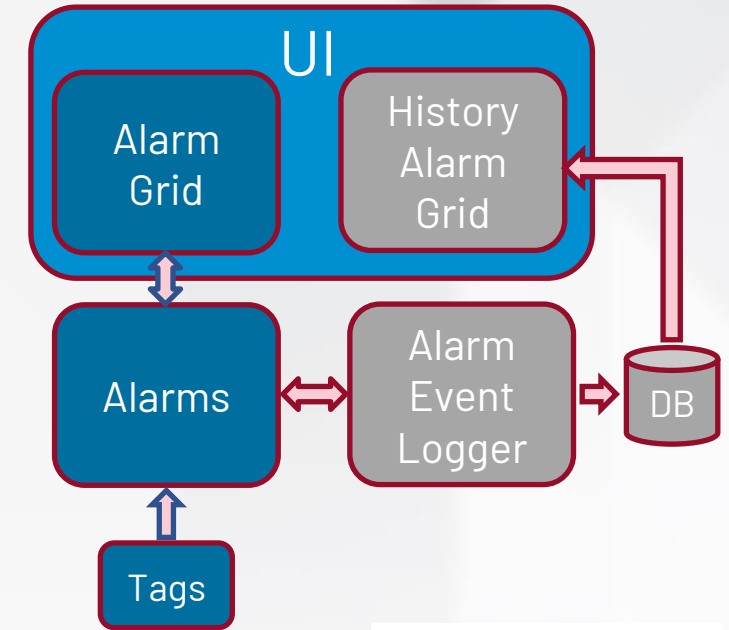


Manage alarms



Alarms

- Alarm functional module is used to define alarm events connected to Tags/Variables and manage them at runtime
- Active Alarms
 - just define alarms
 - drag&drop Alarm Grid widget from Template Library to UI
- History of Alarms
 - add a new datastore (Embedded database or ODBC)
 - drag&drop Alarm Event Logger object from Template Library, into Loggers node
 - drag&drop History Alarm Grid widget from Template Library to UI



OPC/UA Alarms
specifications

Alarm types

- Types of alarms:

- *Digital*: "two-state" alarm, typically used to monitor Boolean tags
- *Level*: an alarm that activates when the monitored variable exceed the range "Low – High"
- *Deviation*: alarms that activate when the monitored variable exceed the range defined by a Setpoint and Low/High deviation limit
- *Rate of Change*: like Level Alarm but evaluated with specific polling time (i.e. to filter spurious spikes)

- Exclusive/Non-Exclusive (when all four limits are used)

- Exclusive: exceeding High and High-high limits, only High-high alarms will be shown
- Non-Exclusive: exceeding High and High-high limits, both alarms will be shown

Properties		
	Name	DigitalAlarm1
	Type	Digital alarm
Input variable		0
Normal state value		0
Enabled		True
Auto acknowledge		False
Auto confirm		False
Message		
Severity		1

Alarm types

- Types of alarms:

- *Digital*: "two-state" alarm, typically used to monitor Boolean tags
- *Level*: an alarm that activates when the monitored variable exceed the range "Low – High"
- *Deviation*: alarms that activate when the monitored variable exceed the range defined by a Setpoint and Low/High deviation limit
- *Rate of Change*: like Level Alarm but evaluated with specific polling time (i.e. to filter spurious spikes)

- Exclusive/Non-Exclusive (when all four limits are used)

- Exclusive: exceeding High and High-high limits, only High-high level will be shown
- Non-Exclusive: exceeding High and High-high limits, both levels will be shown

Properties		
	Name	ExclusiveLevelAlarm1
	Type	Exclusive level alarm
Input variable		0
High-high limit		0
High limit		0
Low limit		0
Low-low limit		0
Enabled		True
Auto acknowledge		False
Auto confirm		False
Message		
Severity		1

Alarm types

- Types of alarms:
 - *Digital*: "two-state" alarm, typically used to monitor Boolean tags
 - *Level*: an alarm that activates when the monitored variable exceed the range "Low – High"
 - *Deviation*: alarms that activate when the monitored variable exceed the range defined by a Setpoint and Low/High deviation limit
 - *Rate of Change*: like Level Alarm but evaluated with specific polling time (i.e. to filter spurious spikes)
- Exclusive/Non-Exclusive (when all four limits are used)
 - Exclusive: exceeding High and High-high limits, only High-high level will be shown
 - Non-Exclusive: exceeding High and High-high limits, both levels will be shown

Properties	
	Name ExclusiveDeviationAlarm1
	Type Exclusive deviation alarm
Input variable	0
Set point value	0
High-high limit	0
High limit	0
Low limit	0
Low-low limit	0
Enabled	True
Auto acknowledge	False
Auto confirm	False
Message	
Severity	1

Alarm types

- Types of alarms:

- *Digital*: "two-state" alarm, typically used to monitor Boolean tags
- *Level*: an alarm that activates when the monitored variable exceed the range "Low – High"
- *Deviation*: alarms that activate when the monitored variable exceed the range defined by a Setpoint and Low/High deviation limit
- *Rate of Change*: like Level Alarm but evaluated with specific polling time (i.e. to filter spurious spikes)

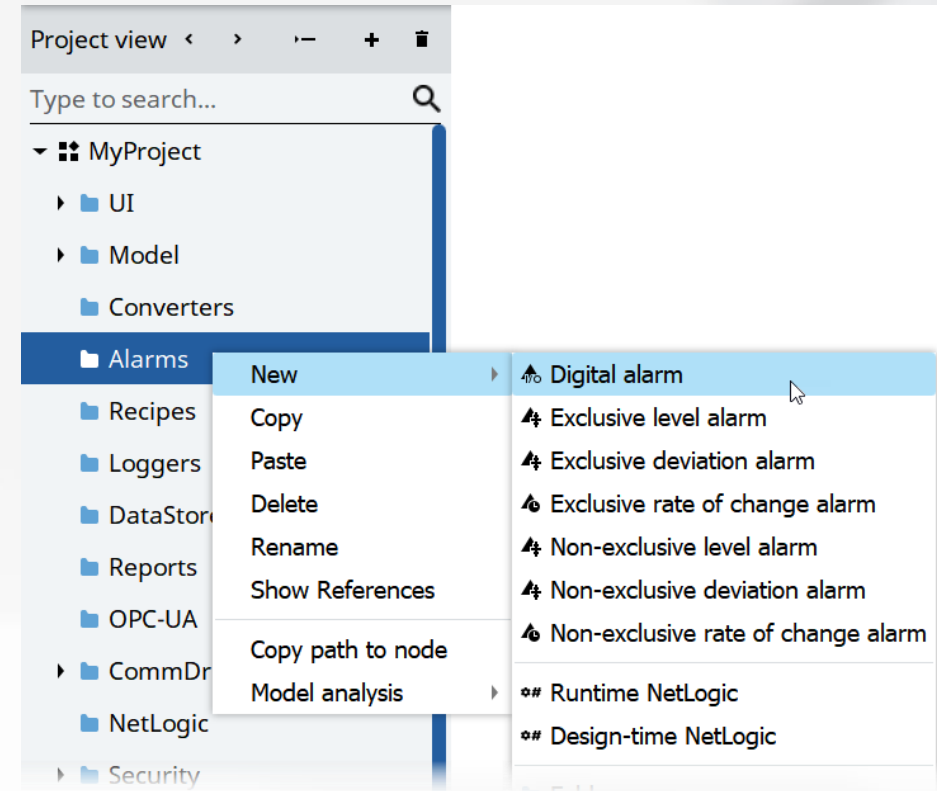
- Exclusive/Non-Exclusive (when all four limits are used)

- Exclusive: exceeding High and High-high limits, only High-high level will be shown
- Non-Exclusive: exceeding High and High-high limits, both levels will be shown

Properties		
	Name	ExclusiveRateOfChangeAlarm1
	Type	Exclusive rate of change alarm
	Input variable	0
	Polling time	0000:00:00.000
	High-high limit	0
	High limit	0
	Low limit	0
	Low-low limit	0
	Enabled	True
	Auto acknowledge	False
	Auto confirm	False
	Message	
	Severity	1

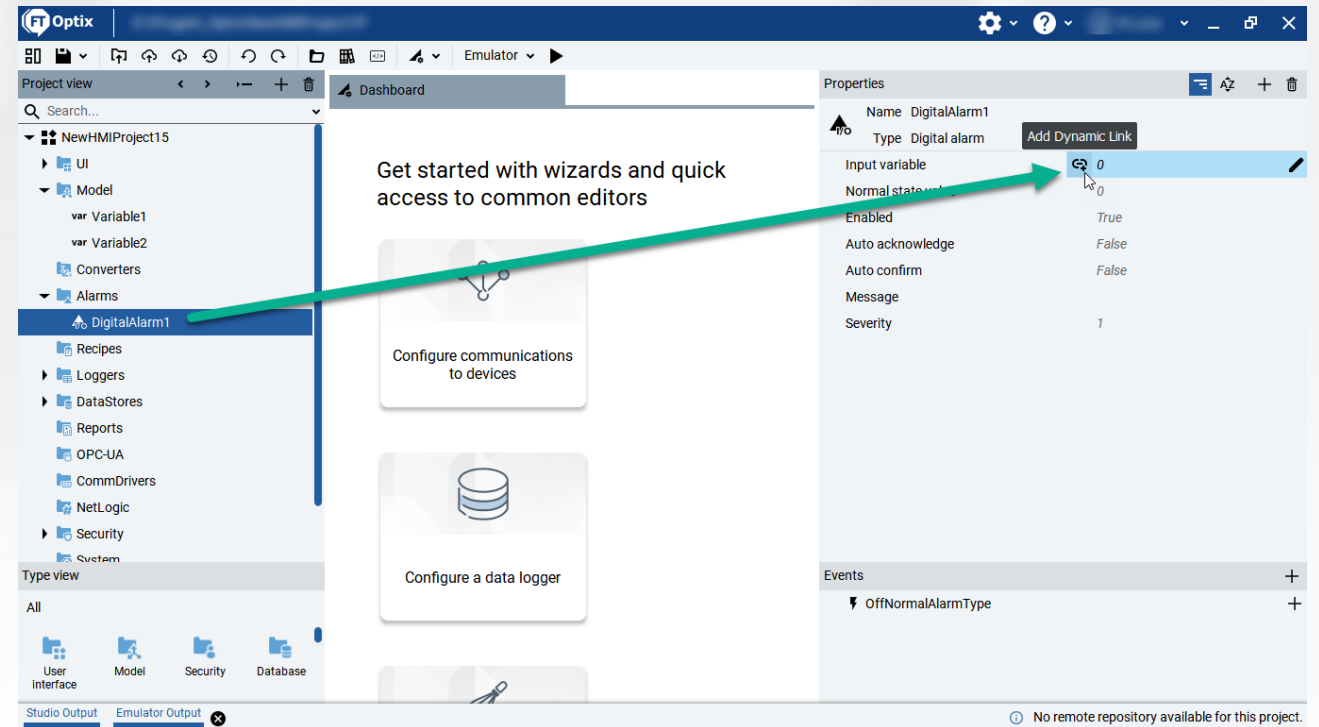
Alarm definition

- How to define alarms
 1. Choose the alarm type
 2. Configure Input Variable and properties
- For Tags where every bit bring alarm information
 1. Select the single tag
 2. Select the bit index
- Import/Export
 - Alarms Definition can be imported/exported to CSV file using a Template Library script (customizable as needed)
 - The path of the CSV file can be set to the current project fodler using the %PROJECTDIR% placeholder



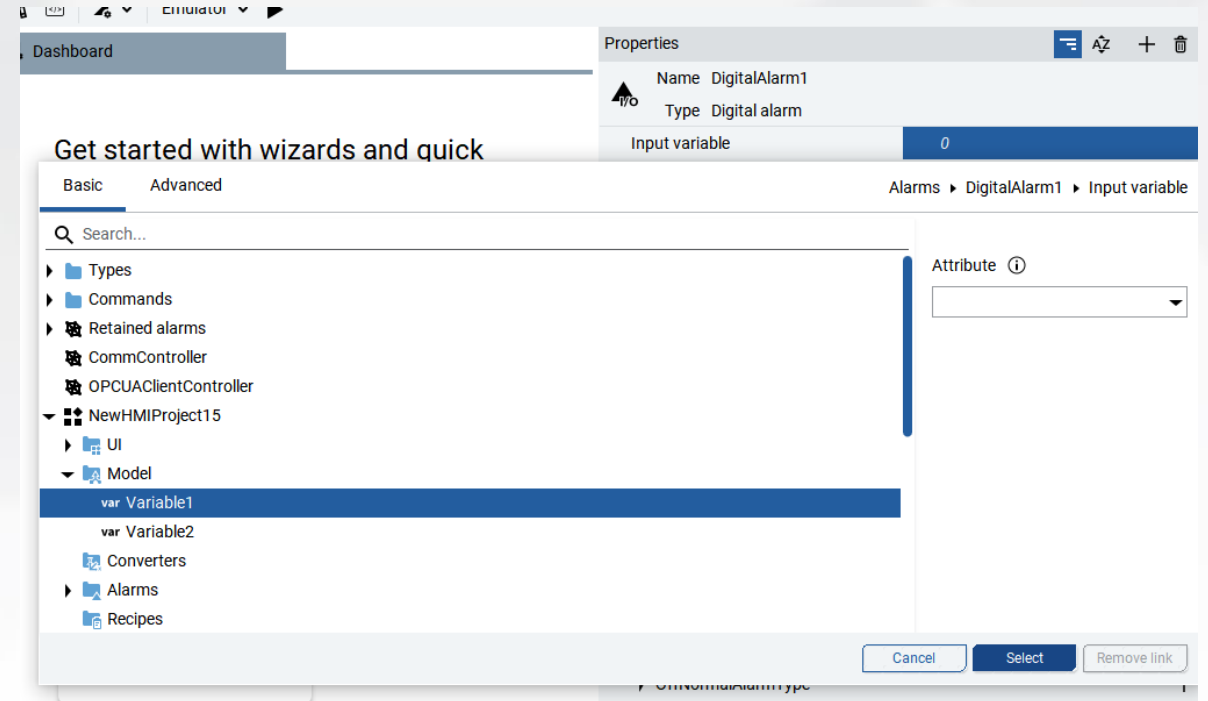
Alarm definition

- How to define alarms
 1. Choose the alarm type
 2. Configure Input Variable and properties
- For Tags where every bit bring alarm information
 1. Select the single tag
 2. Select the bit index
- Import/Export
 - Alarms Definition can be imported/exported to CSV file using a Template Library script (customizable as needed)
 - The path of the CSV file can be set to the current project fodler using the %PROJECTDIR% placeholder



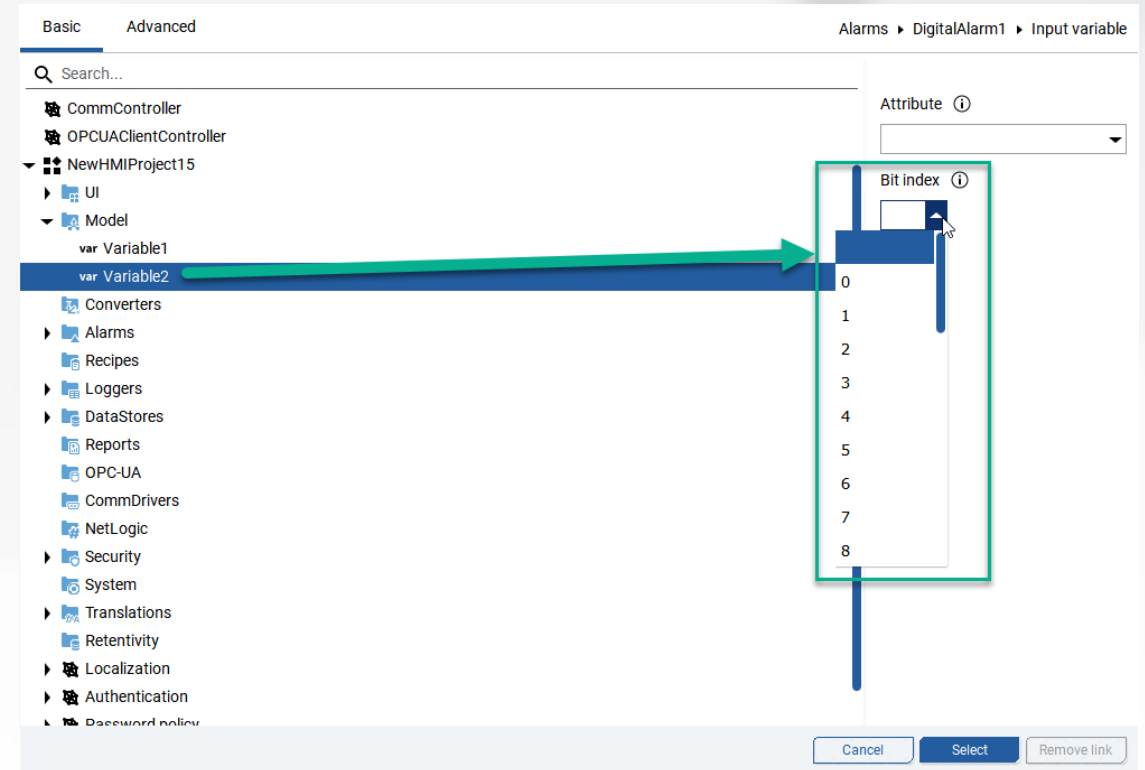
Alarm definition

- How to define alarms
 1. Choose the alarm type
 2. Configure Input Variable and properties
- For Tags where every bit bring alarm information
 1. Select the single tag
 2. Select the bit index
- Import/Export
 - Alarms Definition can be imported/exported to CSV file using a Template Library script (customizable as needed)
 - The path of the CSV file can be set to the current project fodler using the %PROJECTDIR% placeholder



Alarm definition

- How to define alarms
 1. Choose the alarm type
 2. Configure Input Variable and properties
- For Tags where every bit bring alarm information
 1. Select the single tag
 2. Select the bit index
- Import/Export
 - Alarms Definition can be imported/exported to CSV file using a Template Library script (customizable as needed)
 - The path of the CSV file can be set to the current project fodler using the %PROJECTDIR% placeholder



Show alarms

- Available Widgets

- into Template Library

- Alarm Grid,
- Alarm Banner,
- Alarm Icon

- customizable

- columns,
- buttons,
- the "Query" parameter allow to filter alarms of Alarm Grid

- Support Messages in different languages

- Supports different messages per each threshold of analog alarms

Timestamp	Name	Source Variable	Message	Active	Acked	Confirmed	Severity
Feb 10, 2022, 3:44:05 PM	DigitalAlarm1.1	ModbusTag1	Alarm on Tag1 bit 1				1
Feb 10, 2022, 3:44:04 PM	DigitalAlarm1.0	ModbusTag1	Alarm on Tag1 bit 0				1

AcknowledgeAcknowledge AllConfirmConfirm All

Properties

Name AlarmsDataGrid

Type Data grid

— Data grid

Model

Localized alarms

Query

Selected item

Allow deselection False

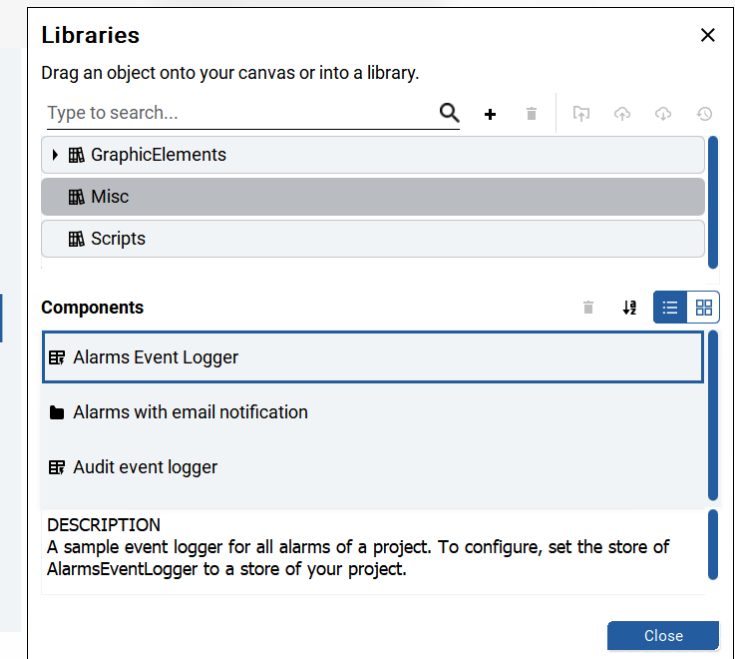
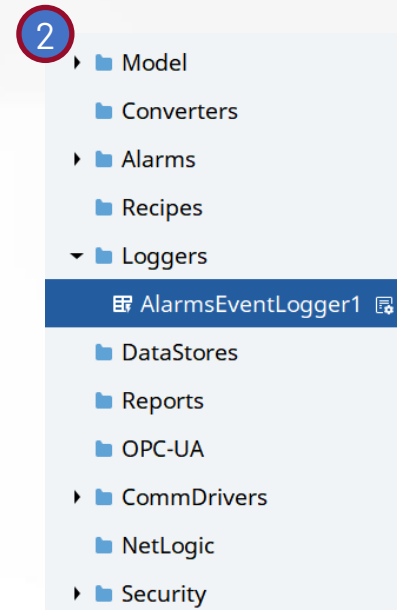
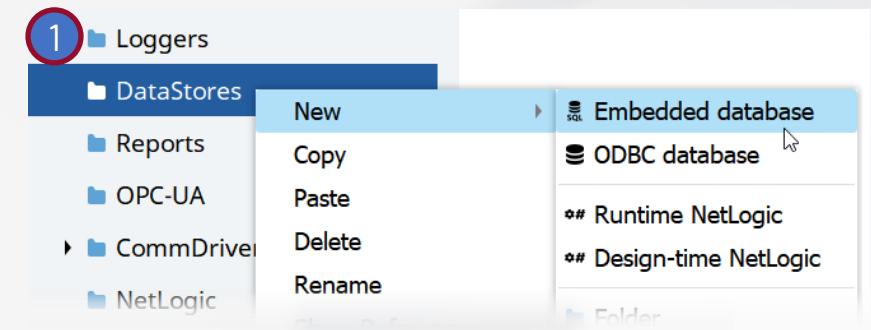
Columns

Timestamp

Name

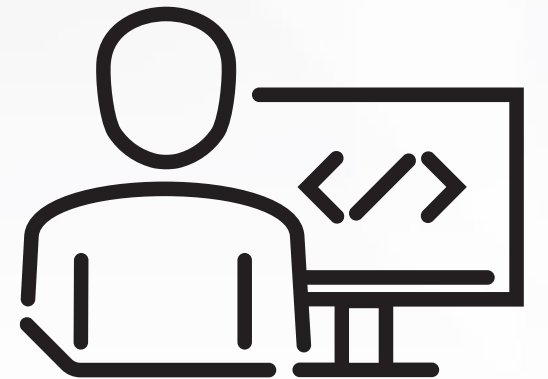
Show alarms history

- To record alarms history some additional resources are needed
 1. Datastore resource (if not already present)
 2. Alarm event logger (from Template Library) just drag&drop and configure datastore to be used
- Available Widgets
 - Alarm History grid
 - Alarm History grid with filter
- Export
 - Alarm History can be exported with script "Alarms History Exporter" available in Template Library



Hands-on session

- Define some digital alarms configuring the input variables
- Add the Alarm Grid widget to see alarms at runtime with the emulator
- Add the Alarm Event Logger to log historical alarm events
- Add the Alarm History Grid widget to see historical



Legacy Recipes



Why Legacy Recipes?

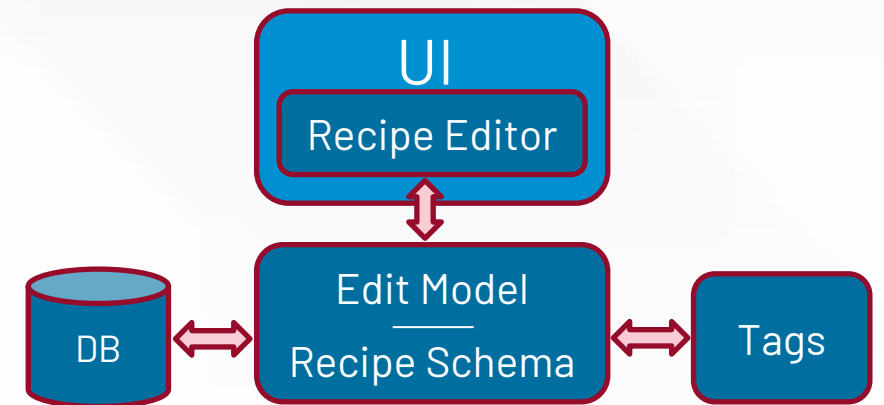
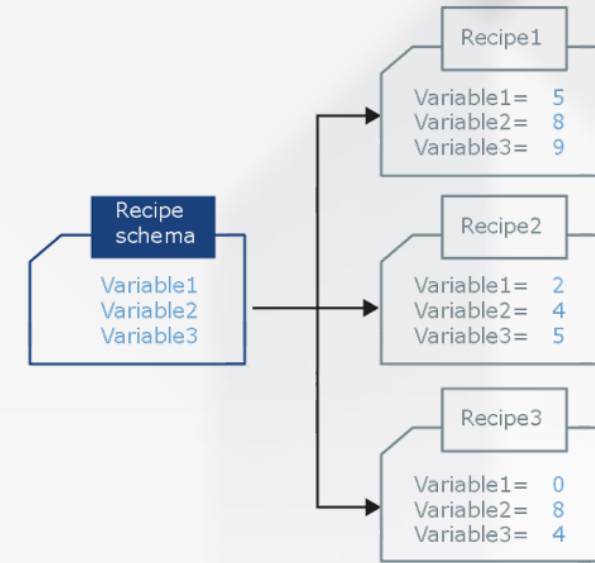
- The Legacy Recipe module has some limitations:
 - Maximum number of ingredients per recipe (2000 at most)
 - Relies on a single database table
 - No support for additional metadatas (difficult to track changes as per 21CFR compliance)
 - Requires manual UI setup (DesignTime NetLogic)
 - Single EditModel session (no concurrent update from multiple sessions)
 - Only supports simple (scalar) datatypes (no structures)

Legacy Recipes FAQ

- I have a project with the Legacy Recipe module, can I still use it?
 - Yes, the module is not going to be retired but no new features will be added
- I only use simple PLC tags, do I need to switch to the new Recipe module?
 - If the number of tags is below 2000, yes
 - If there are no complex datatypes, yes

Legacy Recipes

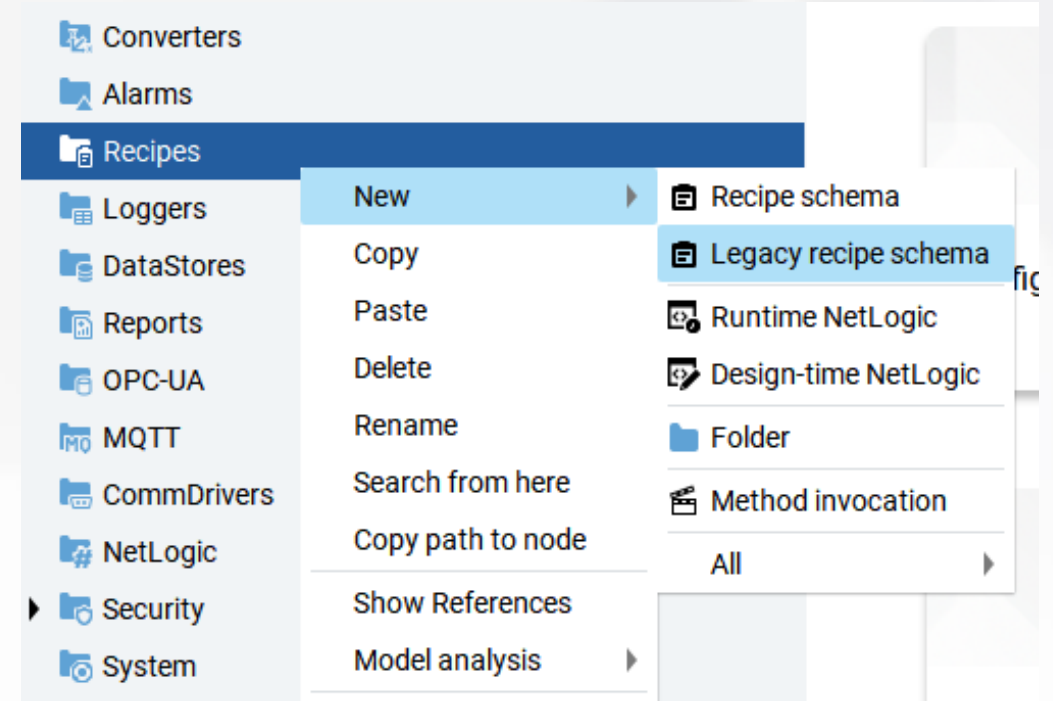
- Allows defining a group of values/set points stored in database archives (embedded or ODBC)
- in Studio
 - configure the Recipe Schema by defining datastore and selecting Tags
- in Runtime
 - User can create several recipes, by defining values manually or by reading values directly from PLC
 - User can store the recipe in the database for future use
 - User can retrieve the recipe from the database and transfer to the PLC



Configure a recipe schema

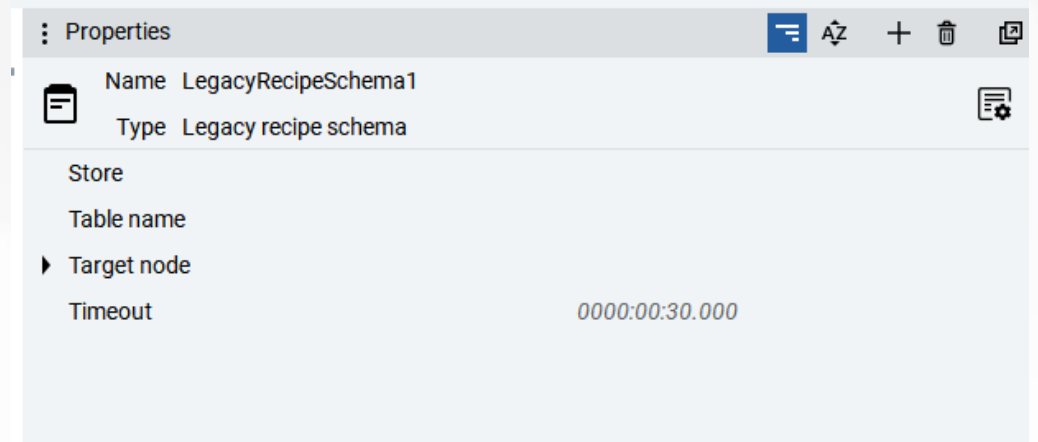
- Steps to configure Recipes

1. Add a Recipe Schema by
Recipes > New > Recipe Schema
2. Select the Datastore
(if not available add a new one)
3. Configure the Target Node
This is the project folder containing Variables
that will be used in the Recipe Schema
Example: Tags, Models...
4. Select which Variables add to the Schema
then confirm with "Apply"



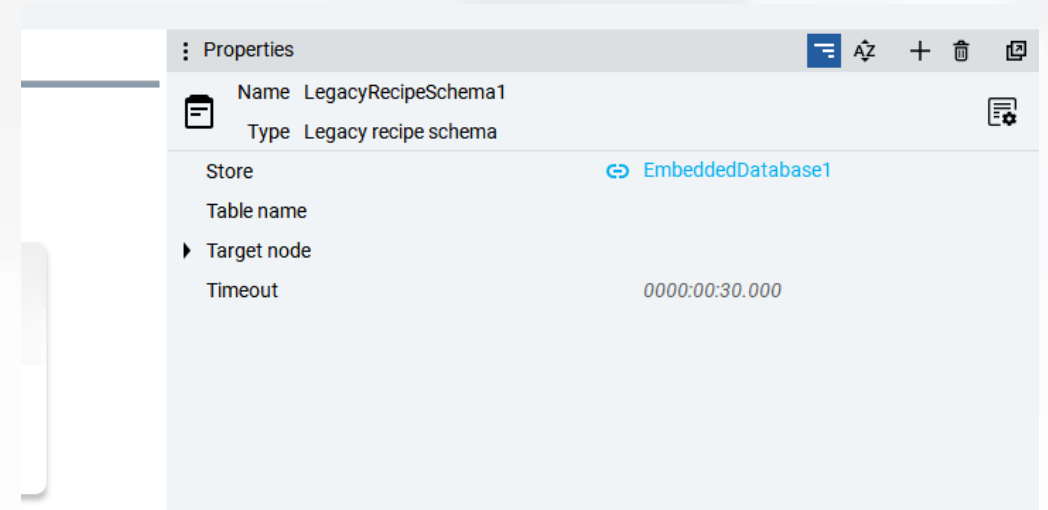
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"



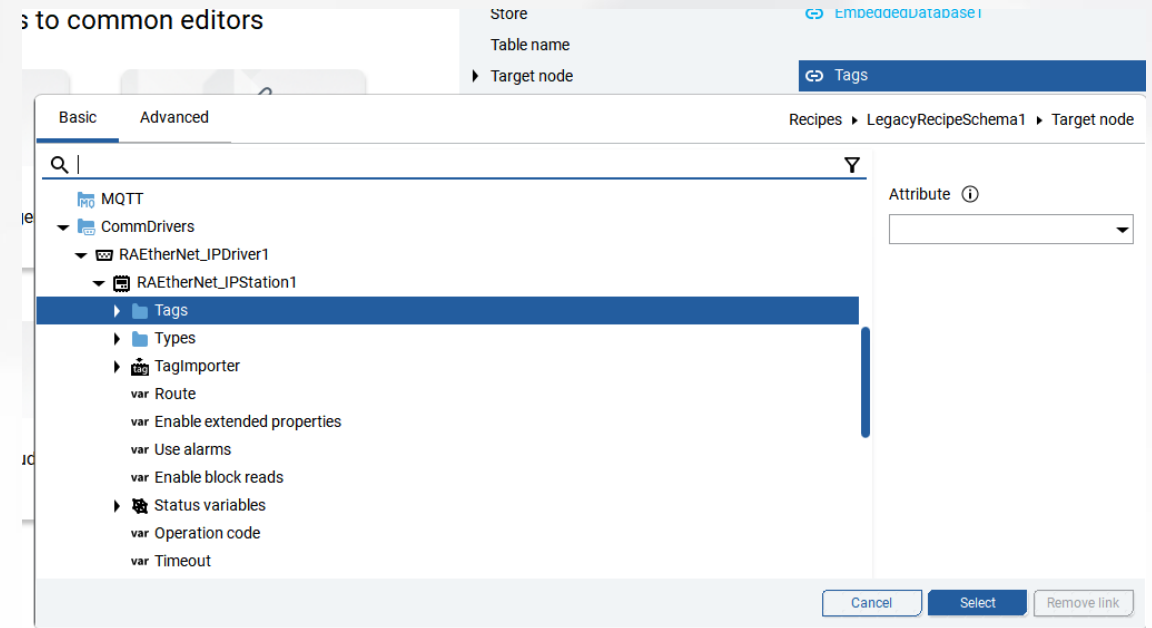
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"



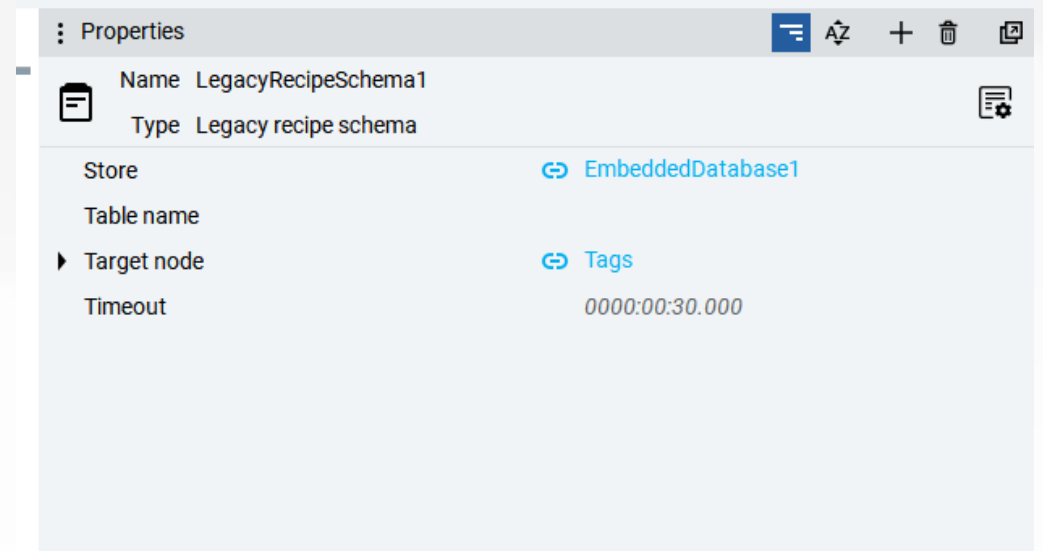
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"



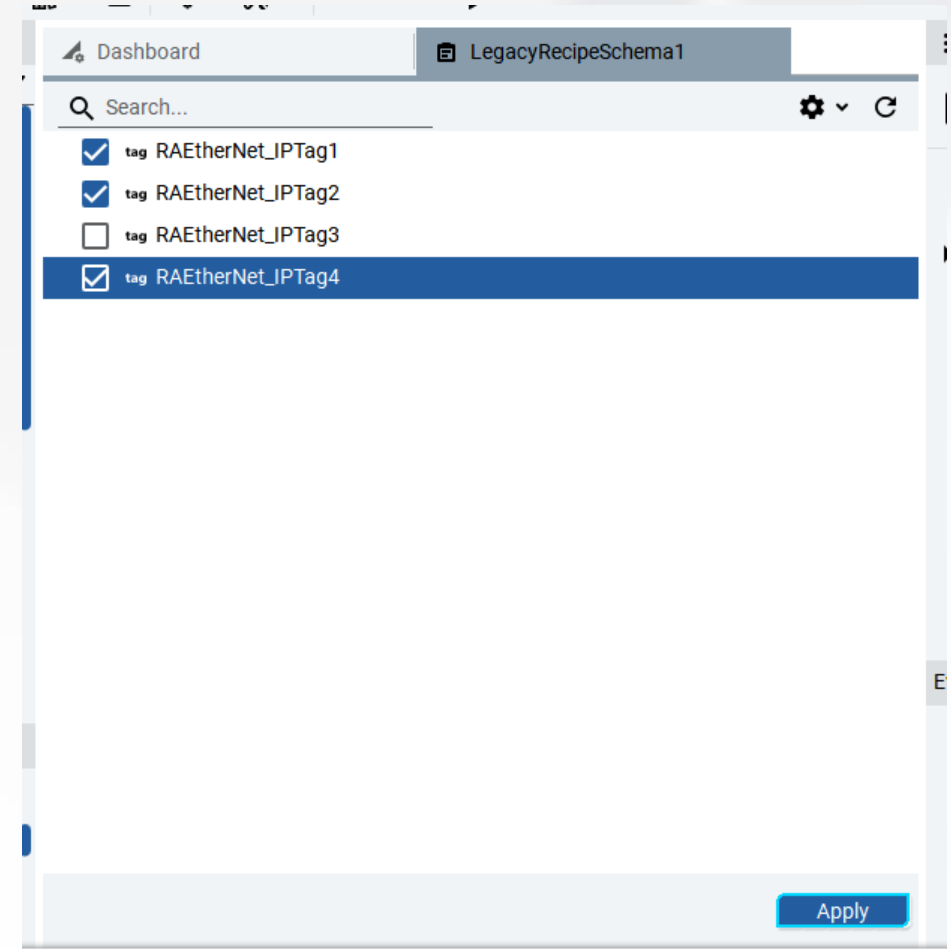
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"



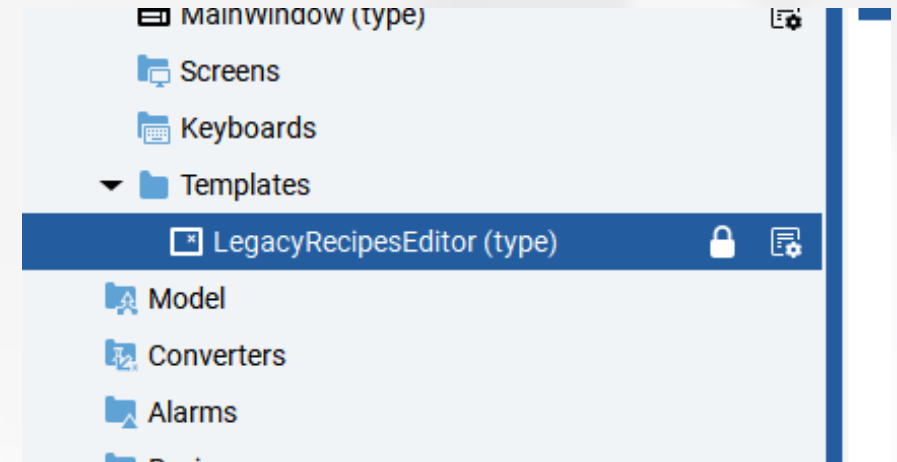
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"



Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
 3. Right-click on the "RecipeEditorUISetup" and select "Execute Setup"
- Widget:
 - Save: save/create the recipe into the database
 - Delete: delete the recipe from the database
 - Load: read Tag values and load them into recipe values
 - Apply: write recipe values into Tags
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget

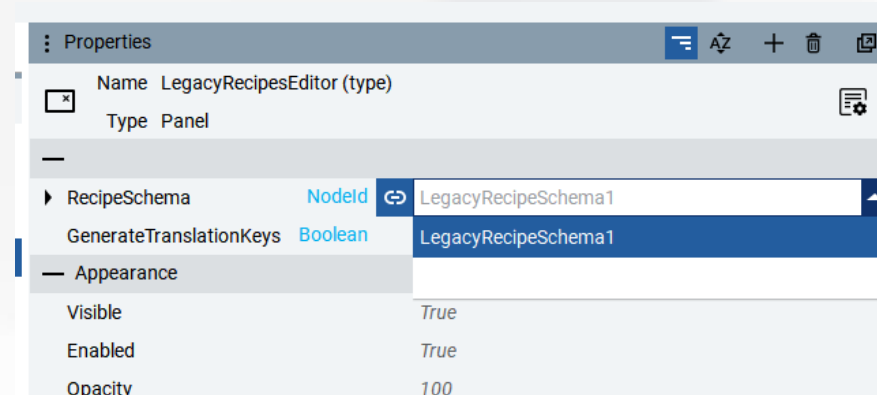
1. Drag&drop the widget from Template Library
2. Into Property pane,
select the recipe schema
3. Right-click on the "RecipeEditorUISetup"
and select "Execute Setup"

- Widget:

- Save: save/create the recipe into the database
- Delete: delete the recipe from the database
- Load: read Tag values and load them into recipe values
- Apply: write recipe values into Tags

- Import/Export

- Recipes can be exported/imported
through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget

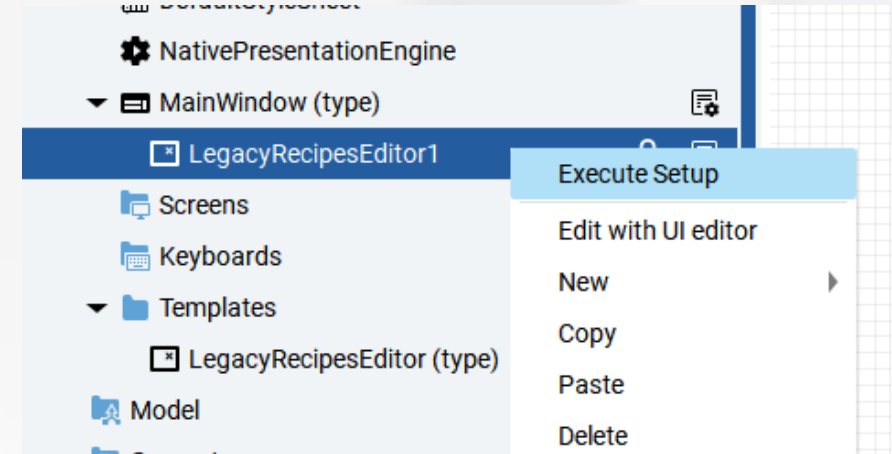
1. Drag&drop the widget from Template Library
2. Into Property pane,
select the recipe schema
3. Right-click on the "RecipeEditorUISetup"
and select "Execute Setup"

- Widget:

- Save: save/create the recipe into the database
- Delete: delete the recipe from the database
- Load: read Tag values and load them into recipe values
- Apply: write recipe values into Tags

- Import/Export

- Recipes can be exported/imported
through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget

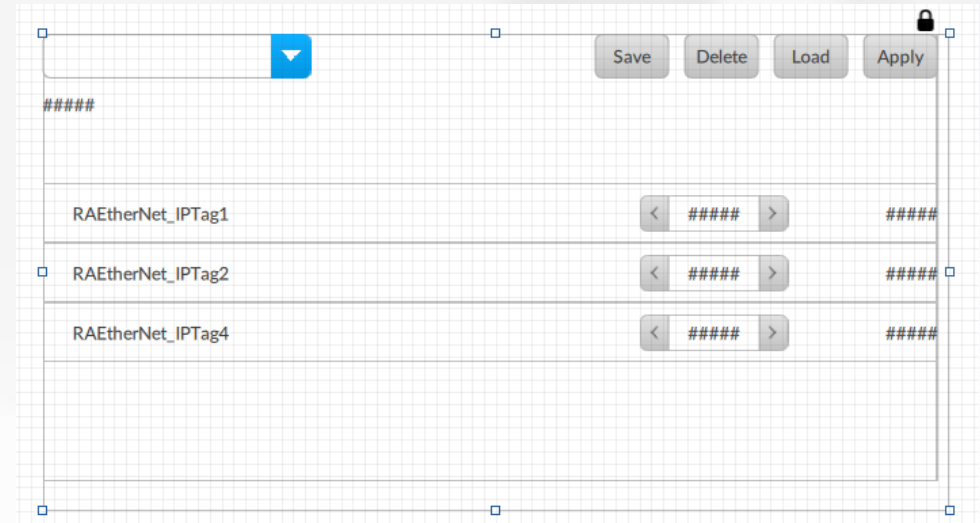
1. Drag&drop the widget from Template Library
2. Into Property pane, select the recipe schema
3. Right-click on the "RecipeEditorUISetup" and select "Execute Setup"

- Widget:

- Save: save/create the recipe into the database
- Delete: delete the recipe from the database
- Load: read Tag values and load them into recipe values
- Apply: write recipe values into Tags

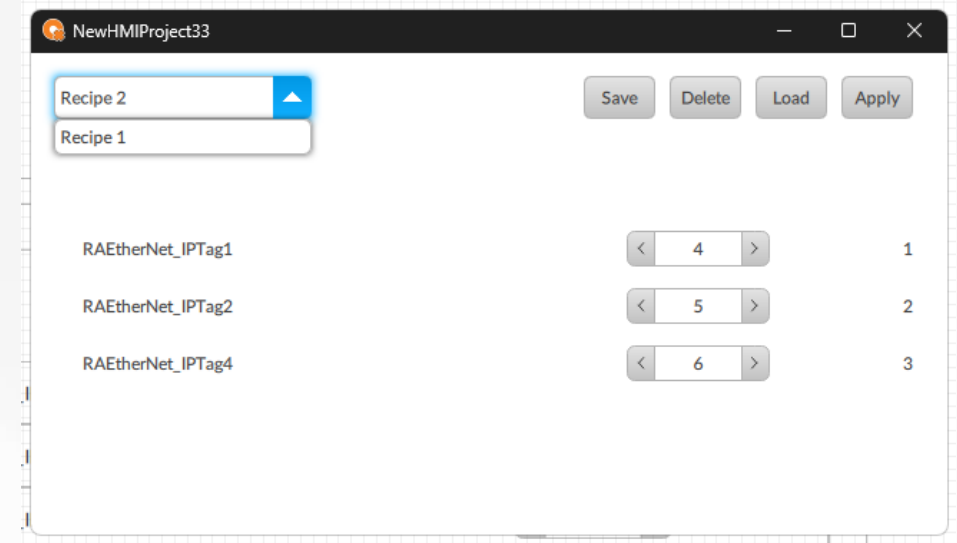
- Import/Export

- Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
 3. Right-click on the "RecipeEditorUISetup" and select "Execute Setup"
- Widget:
 - Save: save/create the recipe into the database
 - Delete: delete the recipe from the database
 - Load: read Tag values and load them into recipe values
 - Apply: write recipe values into Tags
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget

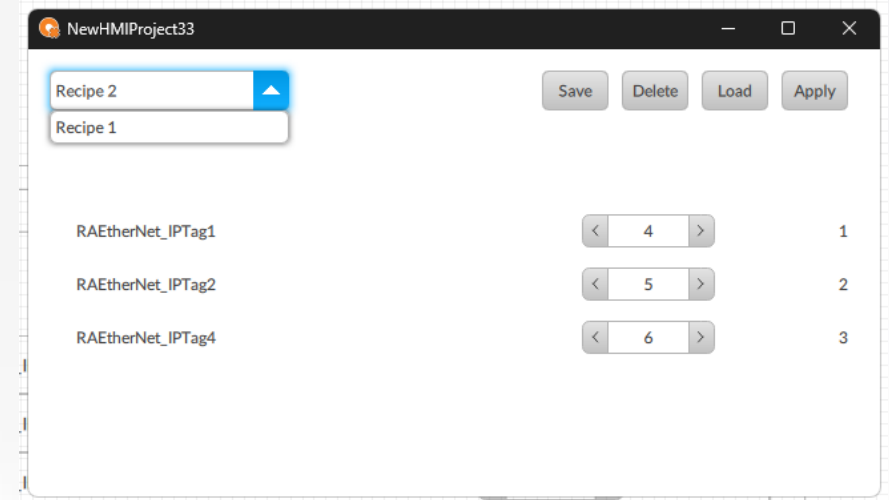
1. Drag&drop the widget from Template Library
2. Into Property pane, select the recipe schema
3. Right-click on the "RecipeEditorUISetup" and select "Execute Setup"

- Widget:

- Save: save/create the recipe into the database
- Delete: delete the recipe from the database
- Load: read Tag values and load them into recipe values
- Apply: write recipe values into Tags

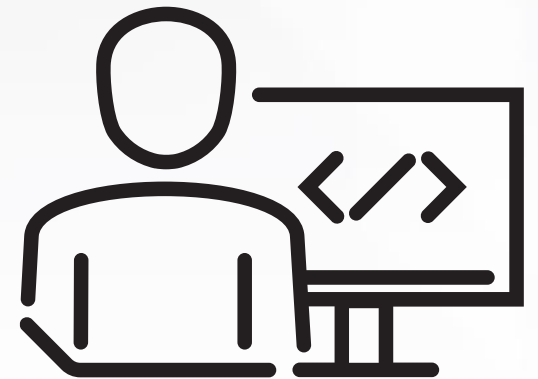
- Import/Export

- Recipes can be exported/imported through script available into Template Library



Hands-on session

- Create a Legacy Recipe schema using tags coming from the communication driver
- Add the Legacy Recipe Widget to the project
- Execute the Legacy Recipe Widget "setup"
- Play with the widget at runtime with Emulator



Recipes

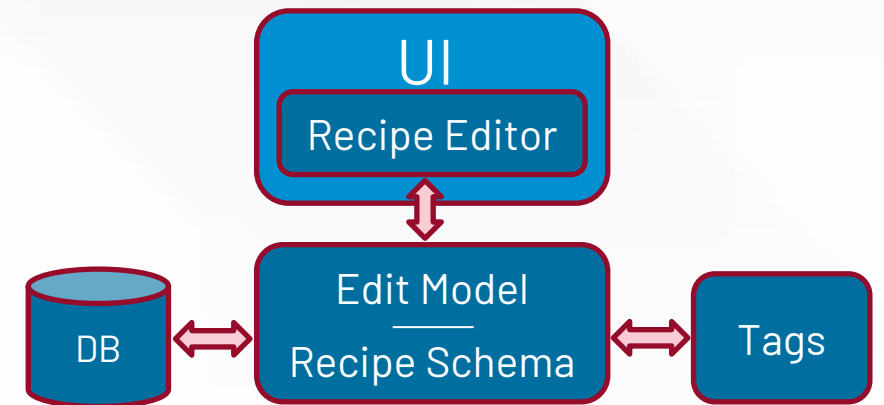
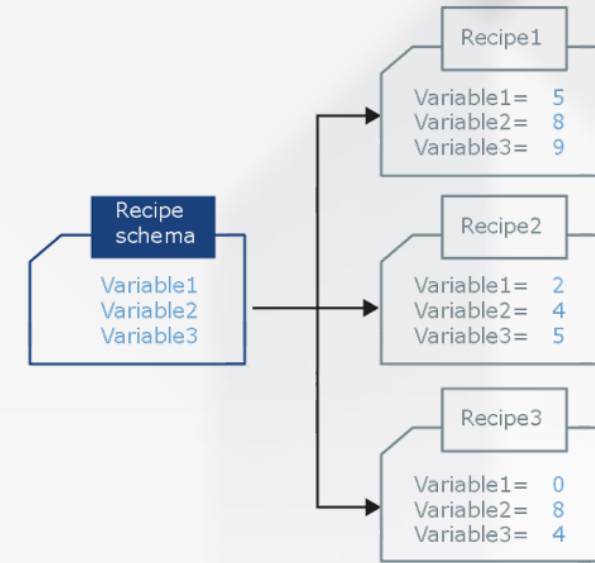


Why switching to the new Recipe module?

- No limit on ingredients size
- Support for recipe name and version
- Support for custom metadata
 - Additional user-defined attributes that can be added to each recipe
 - Useful to support additional information like the current status (approved, deprecated, etc)
- Support for complex datatypes (structures, UDTs, etc)

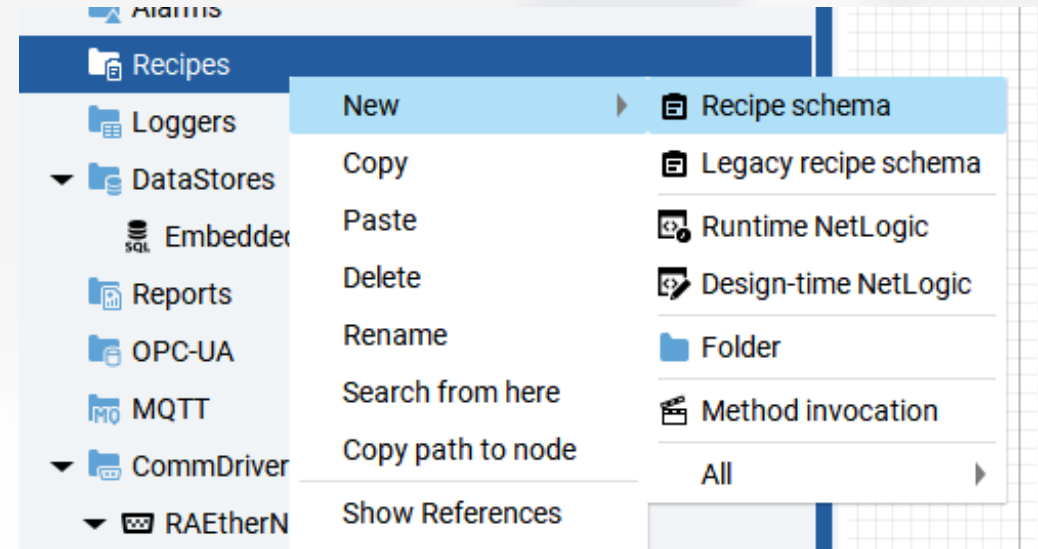
Recipes

- Allows defining a group of values/set points stored in database archives (embedded or ODBC)
- in Studio
 - Configure the Recipe Schema by defining datastore and selecting Tags
- in Runtime
 - User can create several recipes, by defining values manually or by reading values directly from PLC
 - User can store the recipe in the database for future use
 - User can retrieve the recipe from the database and transfer to the PLC
 - User can view and edit recipe metadatas



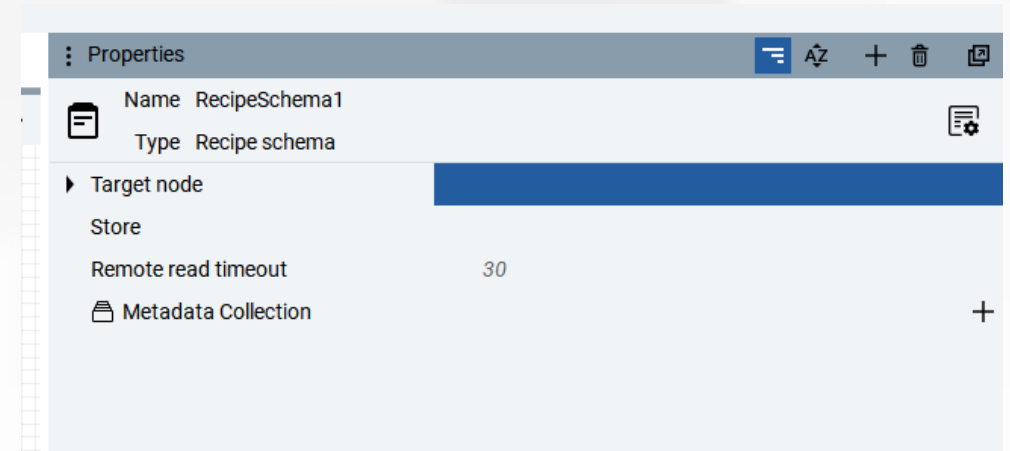
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



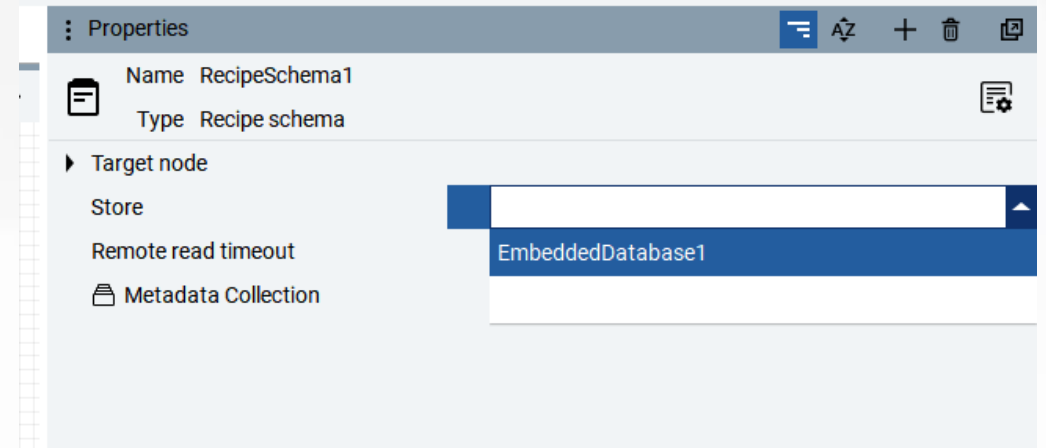
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



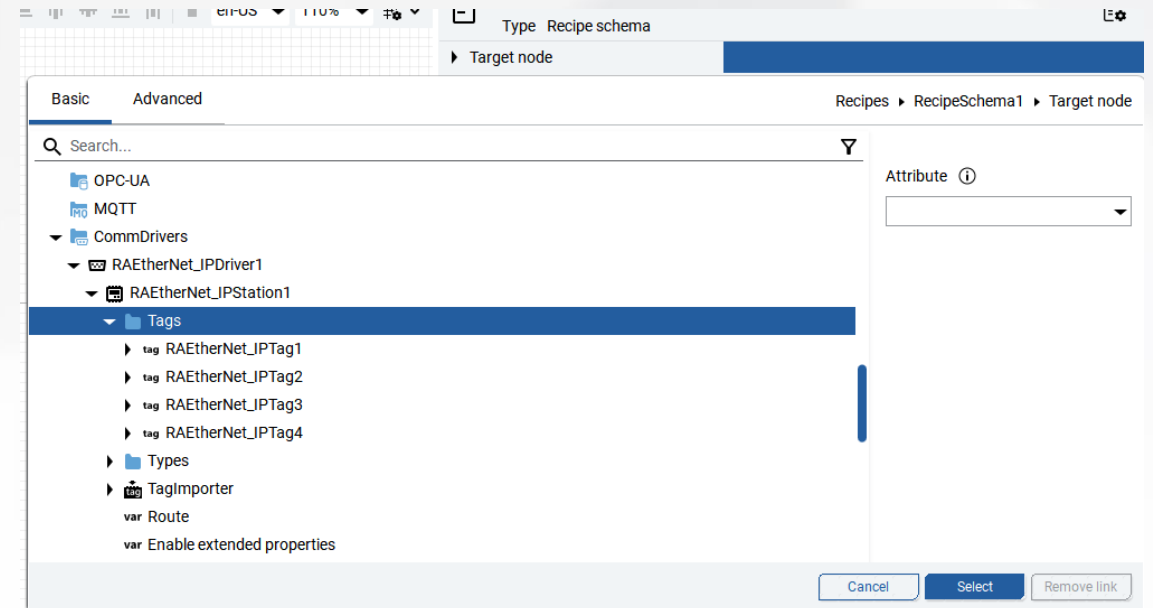
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



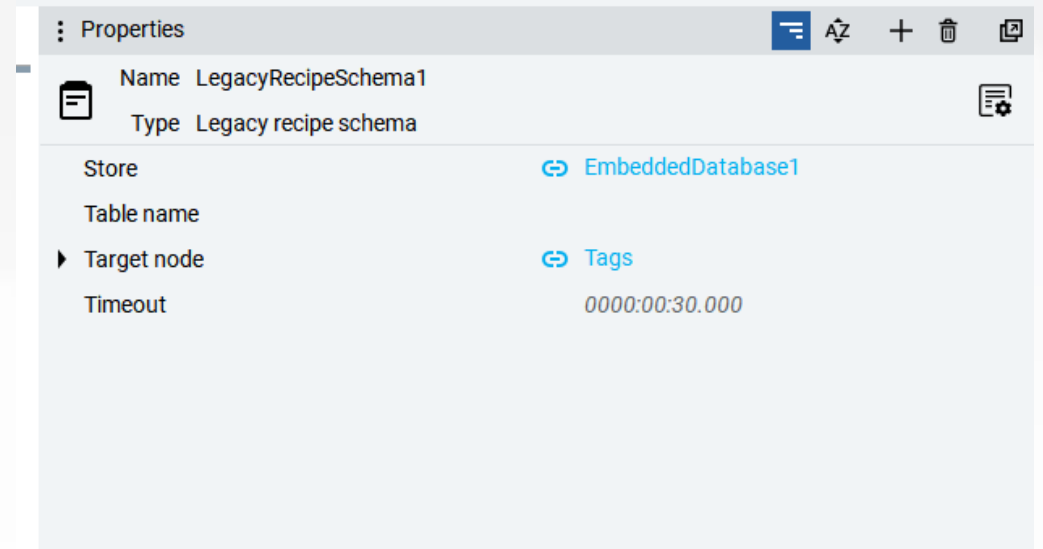
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



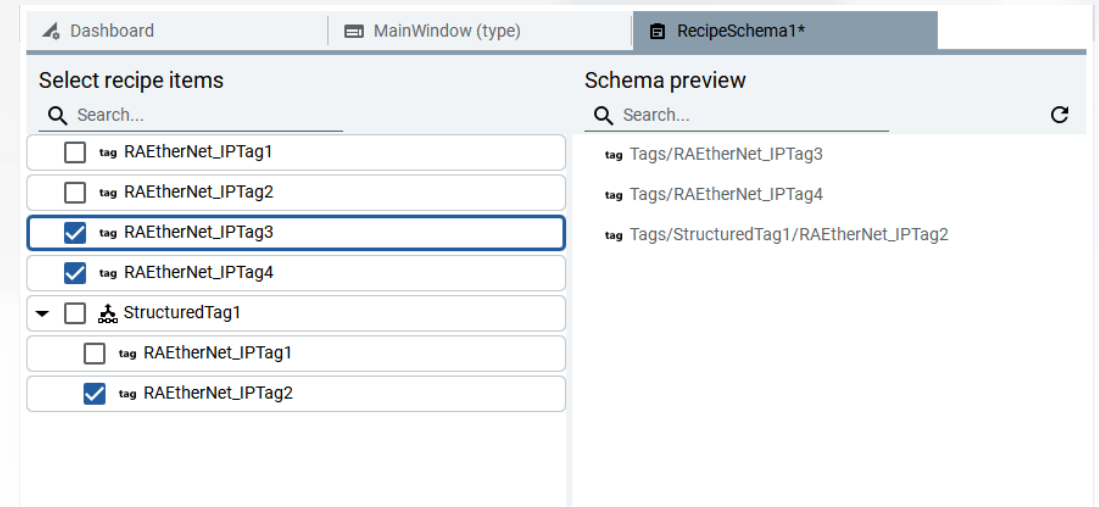
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



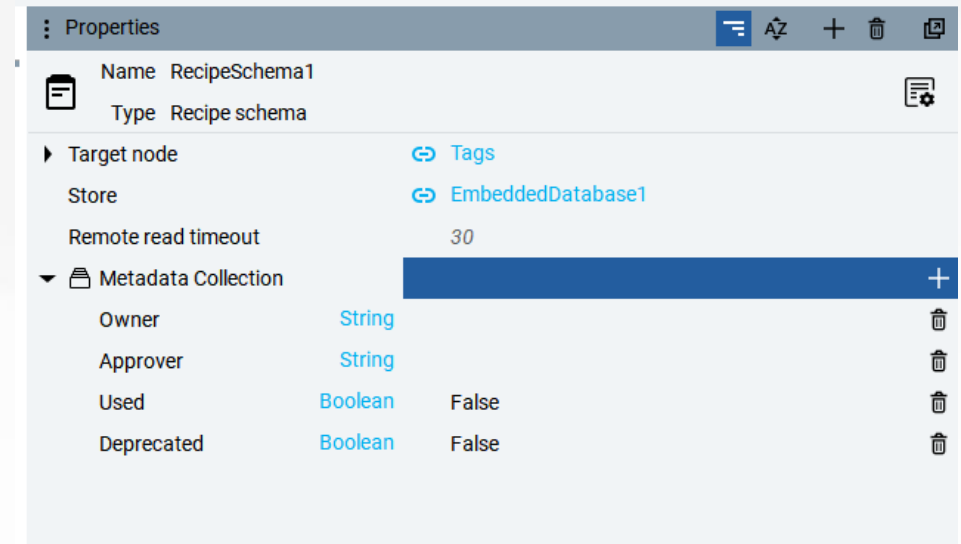
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



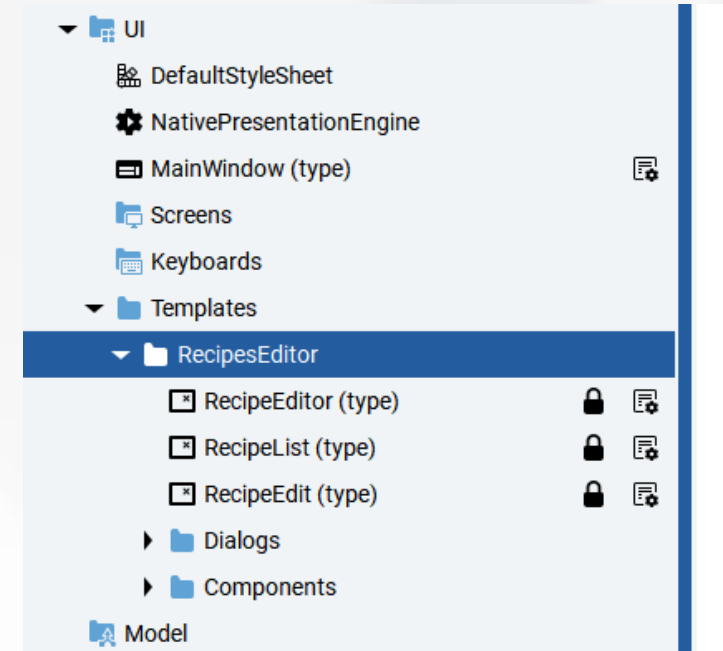
Configure a recipe schema

- Steps to configure Recipes
 1. Add a Recipe Schema by Recipes > New > Recipe Schema
 2. Select the Datastore (if not available add a new one)
 3. Configure the Target Node
This is the project folder containing Variables that will be used in the Recipe Schema
Example: Tags, Models...
 4. Select which Variables add to the Schema then confirm with "Apply"
 5. (optional) Configure metadata



Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget

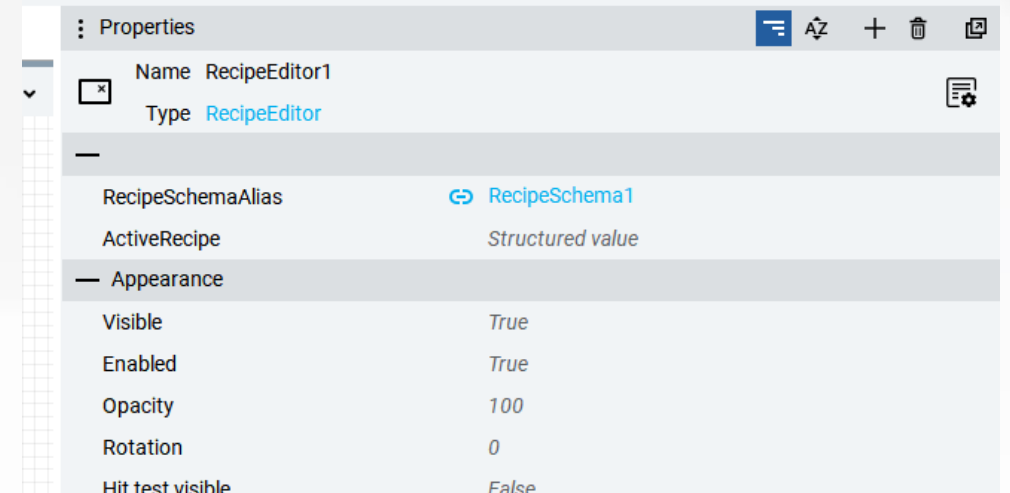
1. Drag&drop the widget from Template Library
2. Into Property pane,
select the recipe schema

- Widget:

- Create new recipe: new recipe with default values
- Create from target: new recipe starting from current tags value
- Refactor: updates the database when changes to the tags structure is performed and alignment is needed

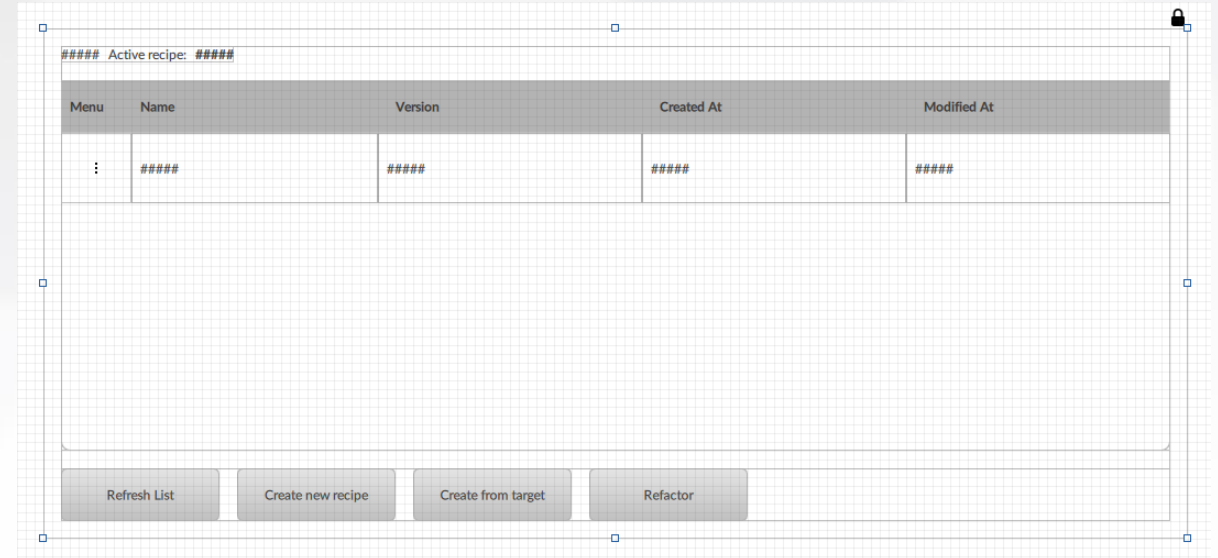
- Import/Export

- Recipes can be exported/imported through script available into Template Library



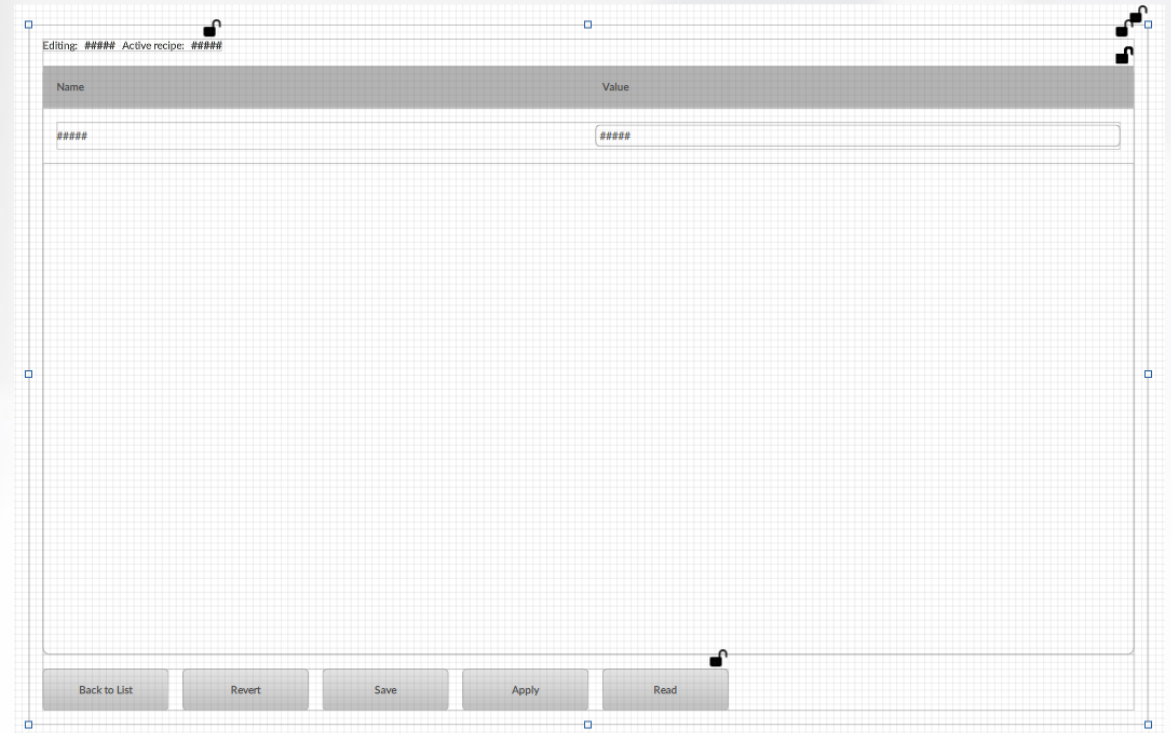
Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



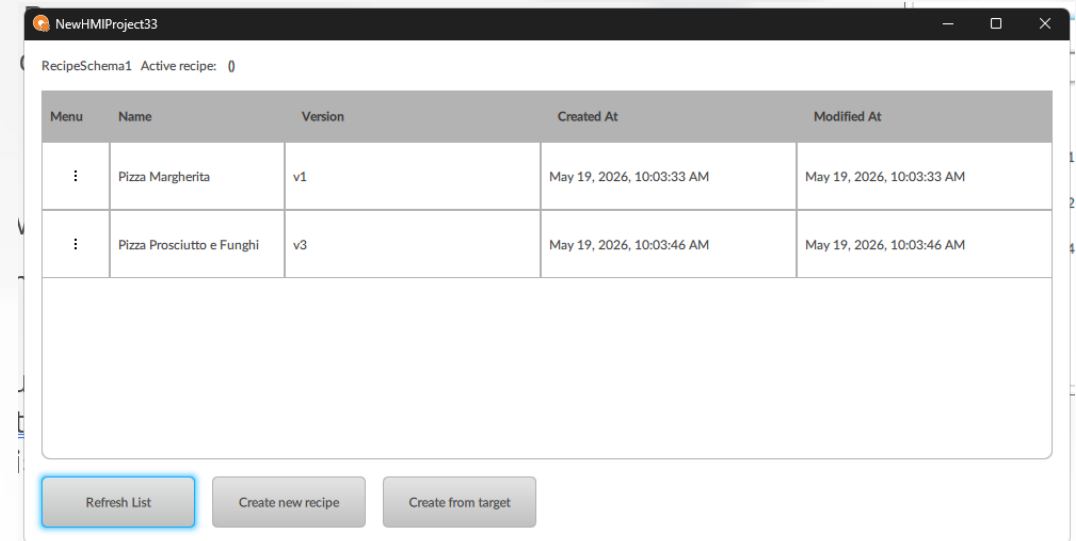
Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



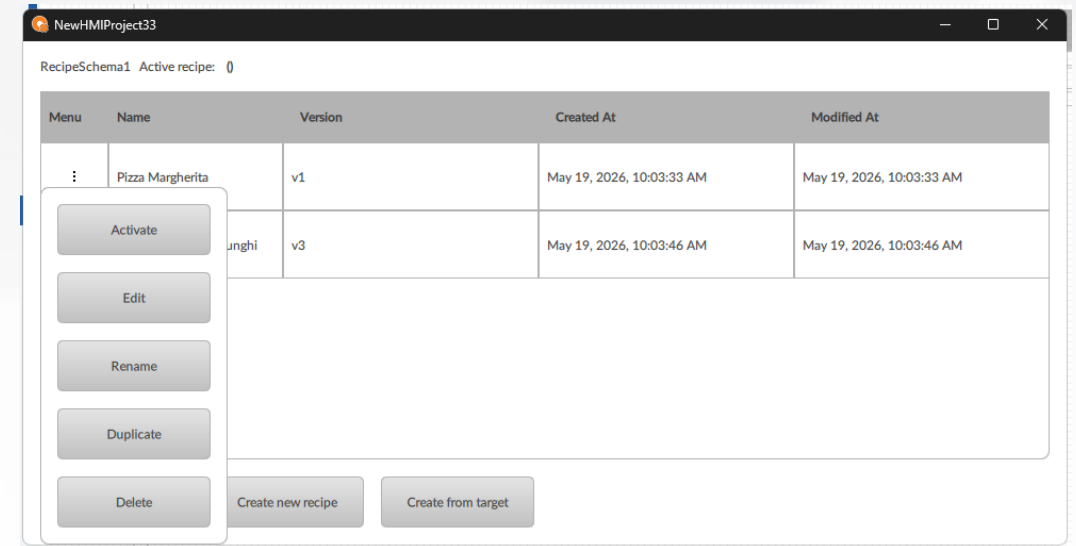
Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library

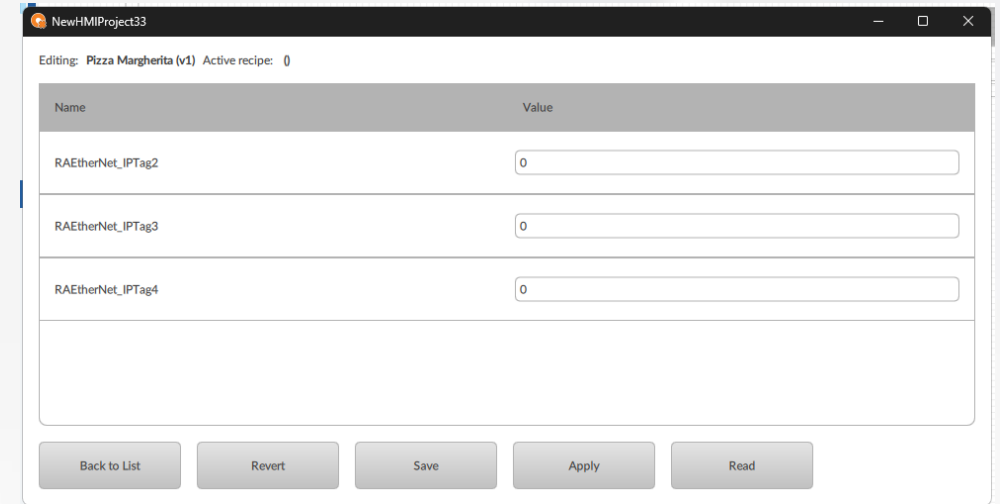
The screenshot shows a software window titled 'NewHMIPProject33'. Inside, the 'Editing: Pizza Margherita (v1)' section is active. Below this, there is a table with two columns: 'Name' and 'Value'. The table contains three rows of data:

Name	Value
RAEtherNet_IPTag2	0
RAEtherNet_IPTag3	0
RAEtherNet_IPTag4	0

Below the table is a large empty rectangular area. At the bottom of the window, there are five buttons: 'Back to List', 'Revert', 'Save', 'Apply', and 'Read'.

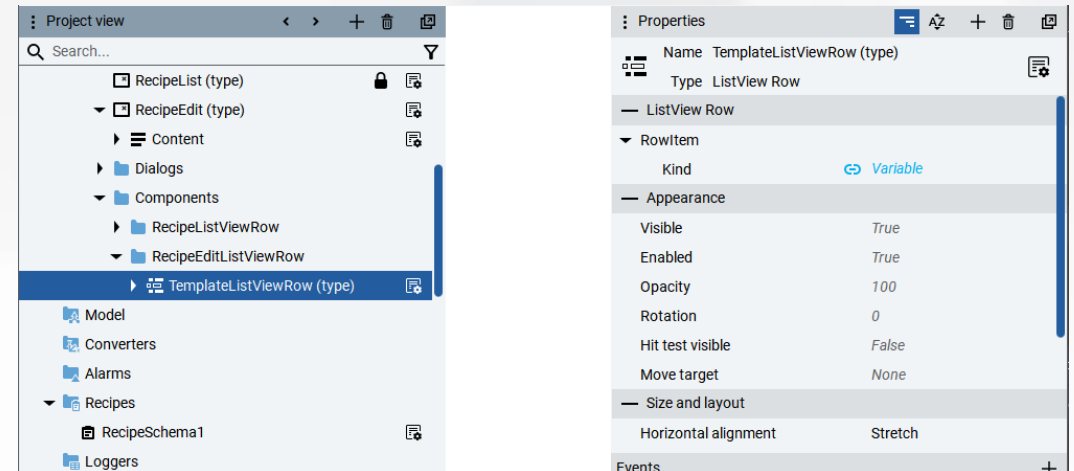
Recipe editor widget

- Steps to configure the Recipe Editor widget
 1. Drag&drop the widget from Template Library
 2. Into Property pane, select the recipe schema
- Widget:
 - Create new recipe: new recipe with default values
 - Create from target: new recipe starting from current tags value
 - Refactor: updates the database when changes to the tags structure is performed and alignment is needed
- Import/Export
 - Recipes can be exported/imported through script available into Template Library



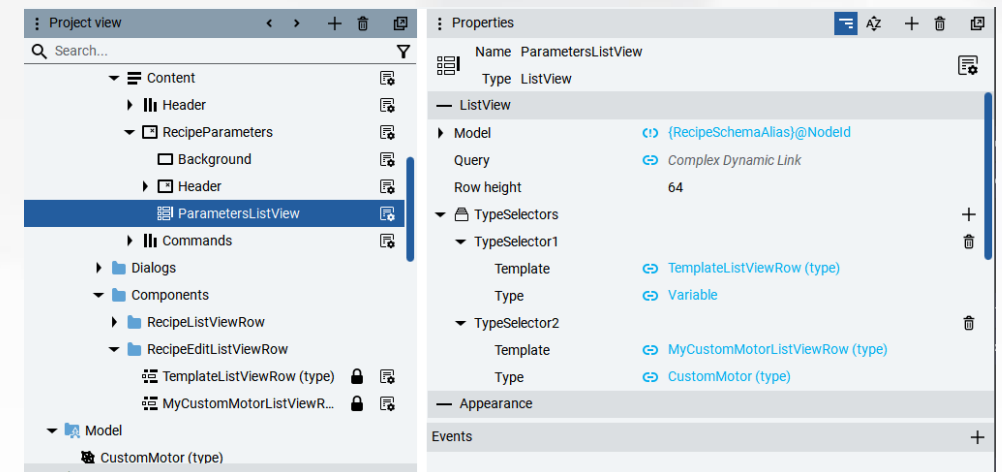
Things to know

- Recipes Editor from Template Library contains only basic datatypes support (Variable)
 - Custom datatypes require manual creation of custom ListViewRow items with the required datatype
 - Custom ListViewRows must be added to the ListView
- Recipes Metadata can be interacted as children of the recipe itself
 - Values can be manipulated with SetVariableValue, ToggleVariable, IncrementVariable, etc
 - Values can be displayed with standard UI elements



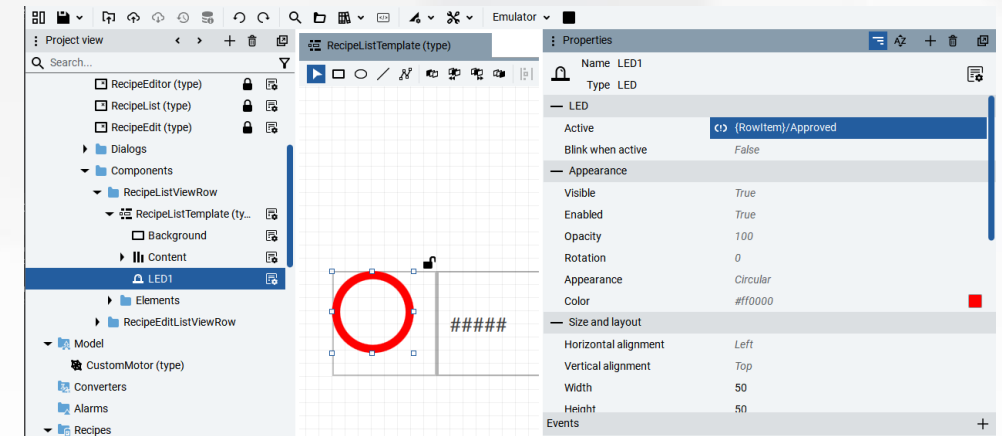
Things to know

- Recipes Editor from Template Library contains only basic datatypes support (Variable)
 - Custom datatypes require manual creation of custom ListViewRow items with the required datatype
 - Custom ListViewRows must be added to the ListView
- Recipes Metadata can be interacted as children of the recipe itself
 - Values can be manipulated with SetVariableValue, ToggleVariable, IncrementVariable, etc
 - Values can be displayed with standard UI elements



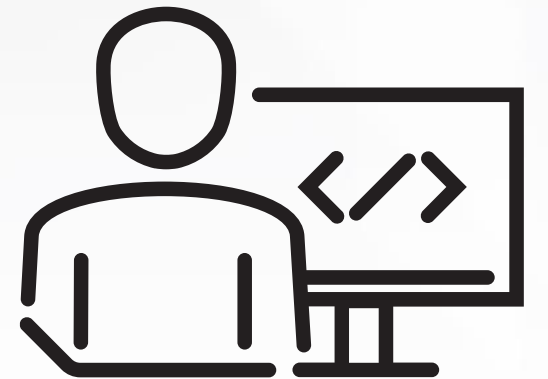
Things to know

- Recipes Editor from Template Library contains only basic datatypes support (Variable)
 - Custom datatypes require manual creation of custom ListViewRow items with the required datatype
 - Custom ListViewRows must be added to the ListView
- Recipes Metadata can be interacted as children of the recipe itself
 - Values can be manipulated with SetVariableValue, ToggleVariable, IncrementVariable, etc
 - Values can be displayed with standard UI elements



Hands-on session

- Create a Recipe schema using tags coming from the communication driver (possibly with custom UDTs)
- Add the Recipe Widget to the project
- Configure a new ListViewRow for the custom UDTs
- Play with the widget at runtime with Emulator

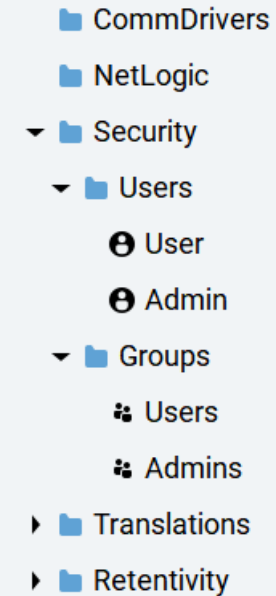


Configure users and groups



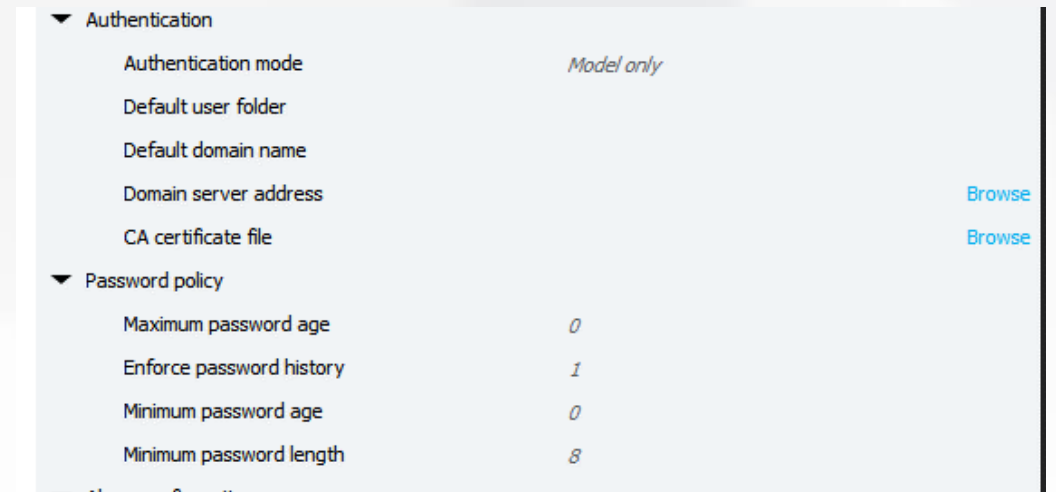
Users and groups

- Allow defining a level of Security for commands/UI objects
- Project Properties
 - Authentication Mode: Model = Project's users, Local = iPC Local users, Domain = Active Directory users
 - Default User Folder
 - Password Policy
- Presentation Engine properties
 - Login Window: on Web Presentation Engine only, allow to specify a Window that should contain a Login Form to be prompted at startup.
 - Starting user: "Anonymous" it's the user logged in at the startup (it's not visible in the Users folder)



Users and groups

- Allow defining a level of Security for commands/UI objects
- Project Properties
 - Authentication Mode: Model = Project's users, Local = iPC Local users, Domain = Active Directory users
 - Default User Folder
 - Password Policy
- Presentation Engine properties
 - Login Window: on Web Presentation Engine only, allow to specify a Window that should contain a Login Form to be prompted at startup.
 - Starting user: "Anonymous" it's the user logged in at the startup (it's not visible in the Users folder)

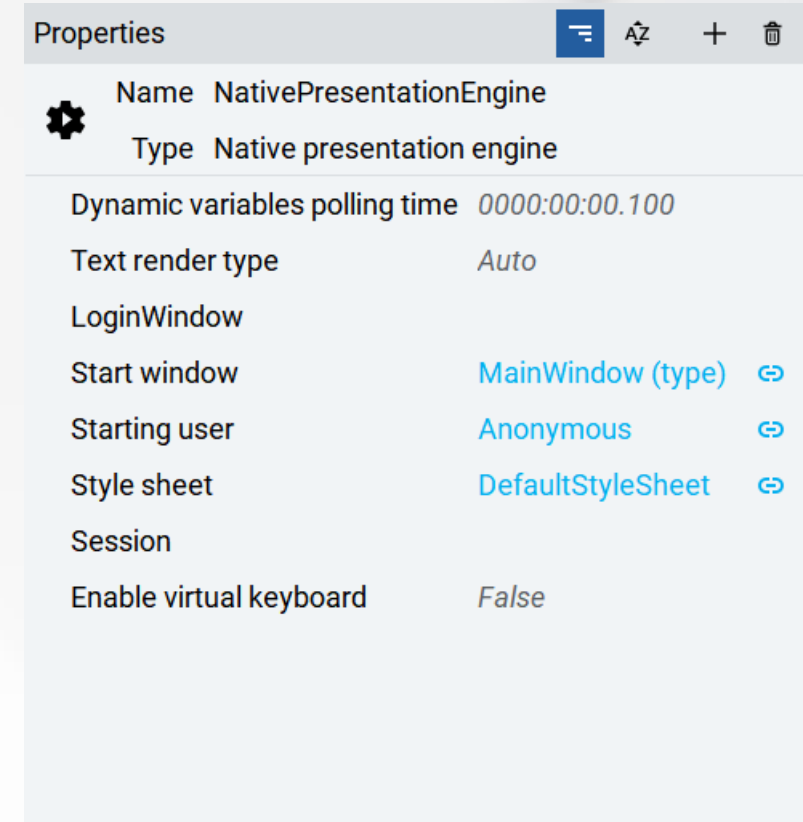


The screenshot shows the 'Project Properties' dialog box with the 'Authentication' and 'Password Policy' sections expanded. The 'Authentication' section includes fields for 'Authentication mode' (set to 'Model only'), 'Default user folder', 'Default domain name', 'Domain server address' (with a 'Browse' button), and 'CA certificate file' (with a 'Browse' button). The 'Password Policy' section includes fields for 'Maximum password age' (0), 'Enforce password history' (1), 'Minimum password age' (0), and 'Minimum password length' (8).

Section	Property	Value	Action
Authentication	Authentication mode	Model only	
	Default user folder		
	Default domain name		
	Domain server address		Browse
	CA certificate file		Browse
Password policy	Maximum password age	0	
	Enforce password history	1	
	Minimum password age	0	
	Minimum password length	8	

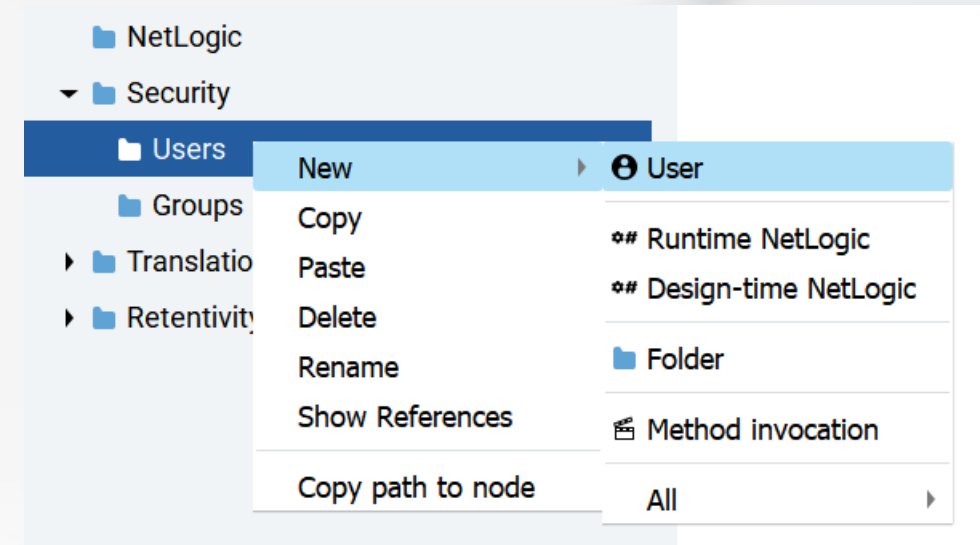
Users and groups

- Allow defining a level of Security for commands/UI objects
- Project Properties
 - Authentication Mode: Model = Project's users, Local = iPC Local users, Domain = Active Directory users
 - Default User Folder
 - Password Policy
- Presentation Engine properties
 - Login Window: on Web Presentation Engine only, allow to specify a Window that should contain a Login Form to be prompted at startup.
 - Starting user: "Anonymous" it's the user logged in at the startup (it's not visible in the Users folder)



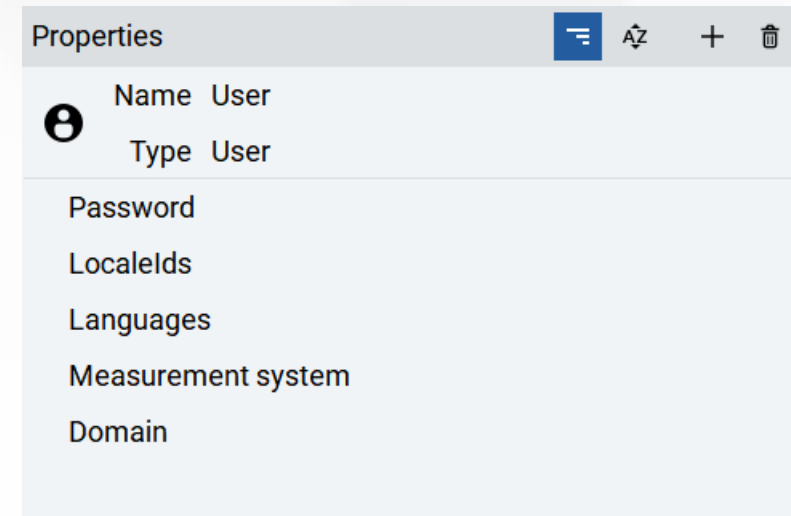
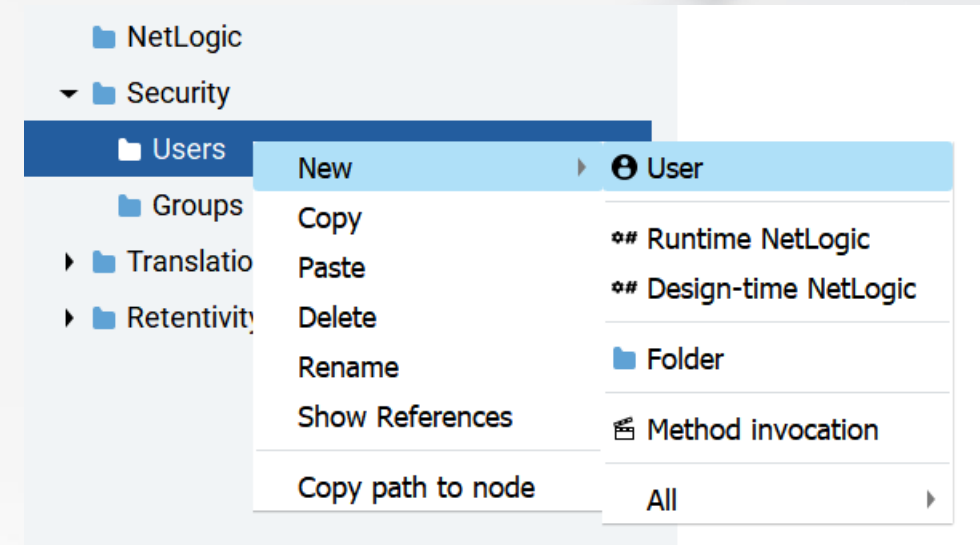
Create users and groups

- User/Group Creation:
 - Users and Groups can be created manually, by the right click on Users or Groups folders (except for Local/Domain users)
- User properties:
 - Password
 - LocaleIds: a combination of Regional Settings, Language and Measurement System
 - Languages: Translation table column to be used (overrides the default language of the LocaleId)
 - Measurement system: International System, British imperial or US Customary (overrides the default Measurement system of the LocaleId)
 - Domain: Domain name used in case of authentication mode: Domain only, Domain and Local, or Any (it is automatically populated after a successful login) – blank by default



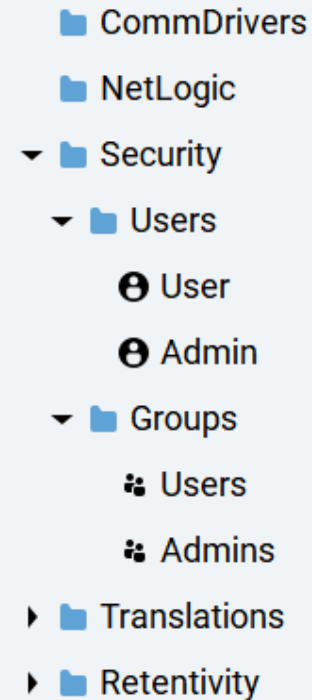
Create users and groups

- User/Group Creation:
 - Users and Groups can be created manually, by the right click on Users or Groups folders (except for Local/Domain users)
- User properties:
 - Password
 - Localelds: a combination of Regional Settings, Language and Measurement System
 - Languages: Translation table column to be used (overrides the default language of the Localeld)
 - Measurement system: International System, British imperial or US Customary (overrides the default Measurement system of the Localeld)
 - Domain: Domain name used in case of authentication mode: Domain only, Domain and Local, or Any (it is automatically populated after a successful login) – blank by default



Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users



Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users



Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users

Manage Users and Groups

Users can belong to multiple, single or no groups.

Users

Assign users to groups

☐ User
☐ Admin

Groups

Assign groups to users

☐ Admins
☐ Users

Exit

Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users

Manage Users and Groups

Users can belong to multiple, single or no groups.

Users

Assign users to groups

☒ User

☐ Admin

Groups

Assign groups to users

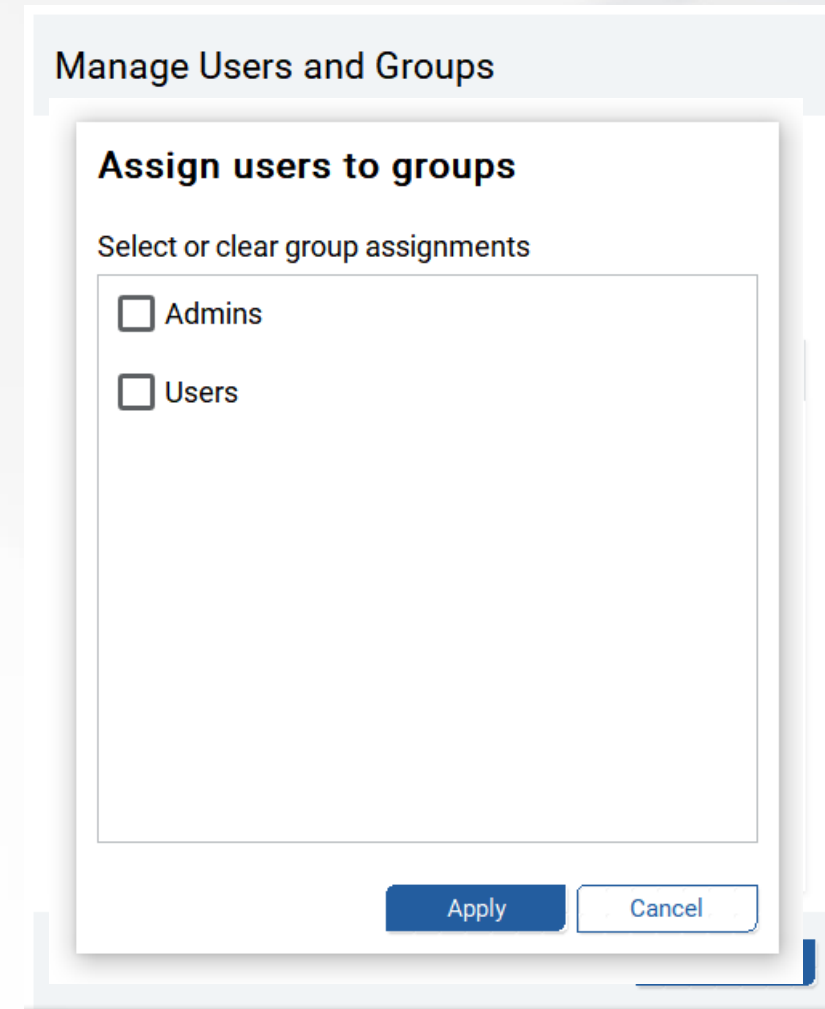
☐ Admins

☐ Users

Exit

Assign users to groups (and viceversa)

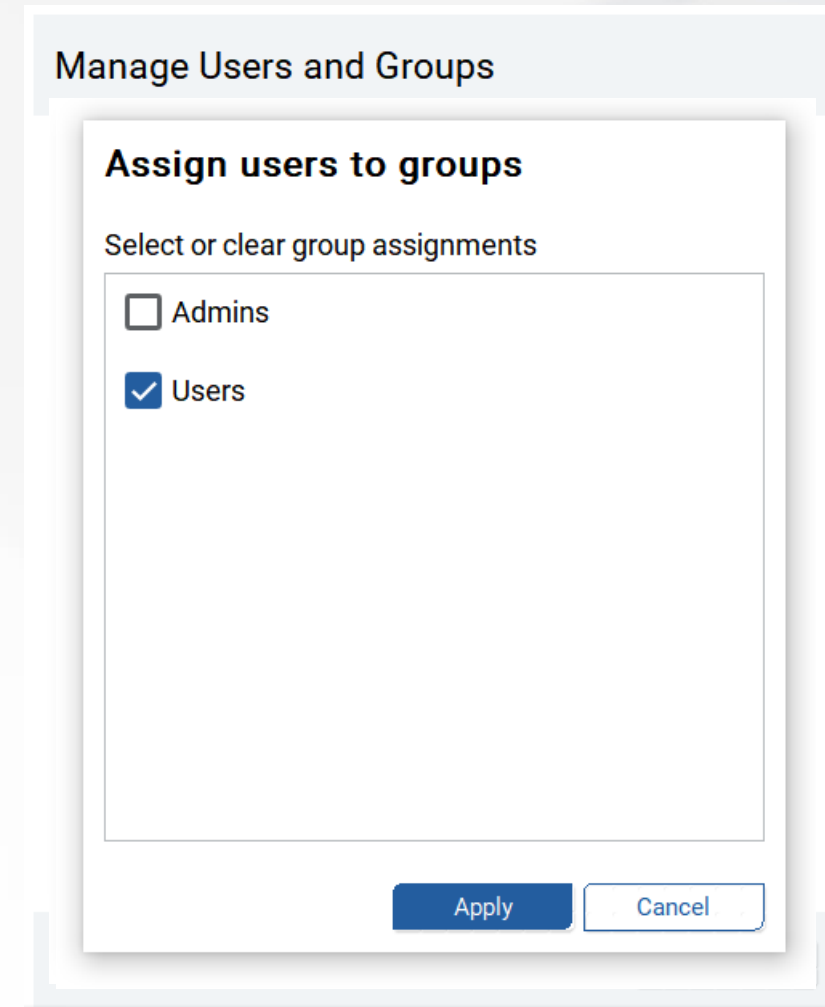
- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users



The screenshot shows a software interface for managing users and groups. At the top, a light blue header bar contains the text "Manage Users and Groups". Below this, a white dialog box titled "Assign users to groups" is displayed. Inside the dialog, the instruction "Select or clear group assignments" is followed by two unchecked checkboxes: "Admins" and "Users". At the bottom right of the dialog, there are two buttons: a blue "Apply" button and a white "Cancel" button with a blue border.

Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users



The screenshot displays a software interface for managing users and groups. At the top, a light blue header bar contains the text "Manage Users and Groups". Below this, a white dialog box titled "Assign users to groups" is shown. Inside the dialog, the instruction "Select or clear group assignments" is present. There are two checkboxes: "Admins" which is unchecked, and "Users" which is checked with a blue checkmark. At the bottom right of the dialog, there are two buttons: a blue "Apply" button and a white "Cancel" button with a blue border.

Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users

Manage Users and Groups

Users can belong to multiple, single or no groups.

Users

Assign users to groups

☒ User

☐ Users

☐ Admin

Groups

Assign groups to users

☐ Admins

☒ Users

☐ User

Exit

Assign users to groups (and viceversa)

- Assignment must be done through the wizard "manage users and groups"
- The wizard allows assigning users to groups but also groups to users

Manage Users and Groups

Users can belong to multiple, single or no groups.

Users

Assign users to groups

☐ User

- Users

☐ Admin

- Admins
- Users

Groups

Assign groups to users

☐ Admins

- Admin

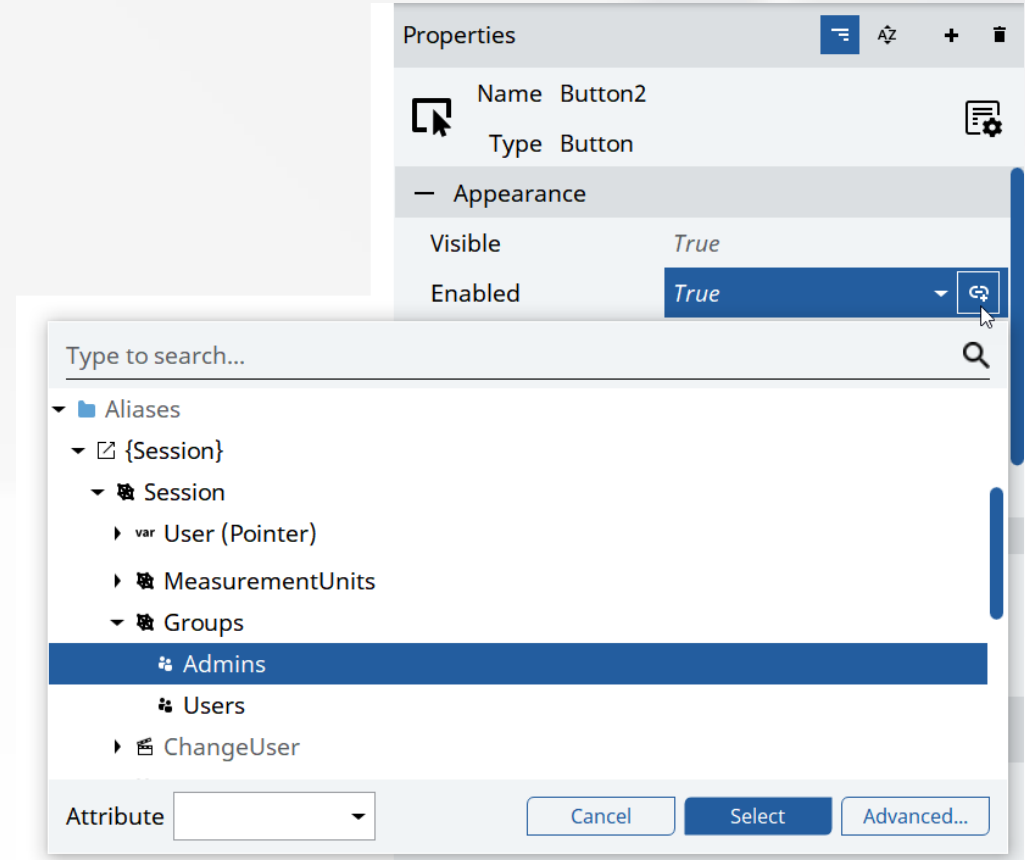
☐ Users

- User
- Admin

Exit

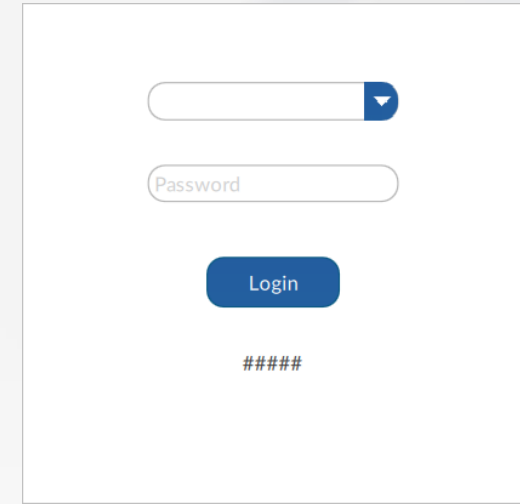
Define security into UI

- It's possible to make an object
 - "Visible/Invisible"
 - "Enabled/Disabled"
- just linking a Group to the needed property
{Session} > Session > Groups > ...
- In this example "Button2" will be "Enabled" only for users of group "Admins"

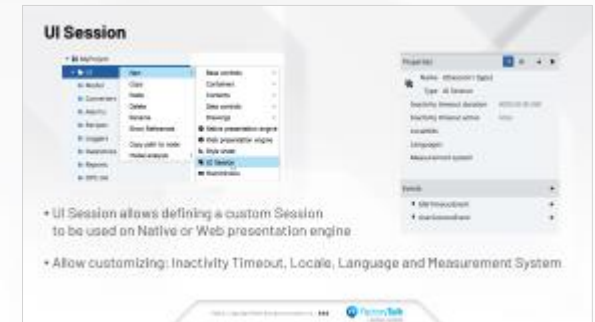


Manage user access at runtime

- Login options
 - using the **widget "Login Form"** of Template Library.
Widget comes as a folder that contains the "LoginForm" to be placed in a Panel or called by a Dropdown button
 - using the **method** "Aliases > {Session} > Session > Login"
or "Aliases > {Session} > Session > Change User"
 - using the **method** "Commands > Core Commands > Change User"



- Logout options
 - using the **method** "Aliases > {Session} > Session > Logout"
 - using an **UISession object** and use the IdleTimeoutEvent to call the method "UI Session > Logout" or "UI Session > Change User"
 - using the **script "Idle Timeout"** available in Template Library, place into the MainWindow and associate a "Logout" command



Manage user access at runtime

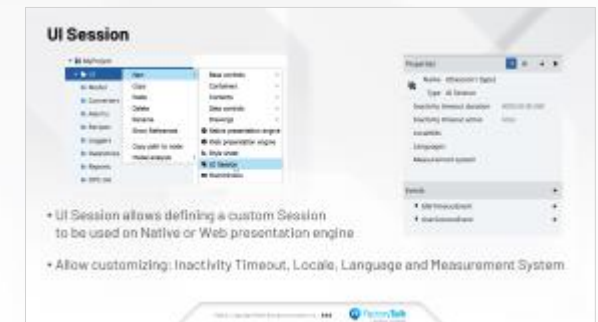
- Login options

- using the **widget "Login Form"** of Template Library.
Widget comes as a folder that contains the "LoginForm" to be placed in a Panel or called by a Dropdown button
- using the **method** "Aliases > {Session} > Session > Login" or "Aliases > {Session} > Session > Change User"
- using the **method** "Commands > Core Commands > Change User"

- Logout options

- using the **method** "Aliases > {Session} > Session > Logout"
- using an **UISession object** and use the IdleTimeoutEvent to call the method "UI Session > Logout" or "UI Session > Change User"
- using the **script "Idle Timeout"** available in Template Library, place into the MainWindow and associate a "Logout" command

Events		+
▼ ⚡	MouseClicked event	+
▼	Method	{Session}/Login
▼	Input arguments	
	Username	admin
	Password	admin
⚡	MouseDown event	+
⚡	MouseUp event	+



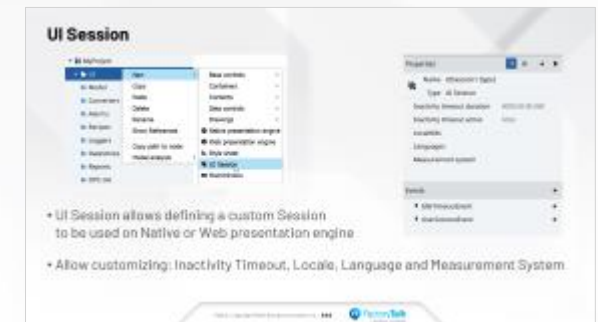
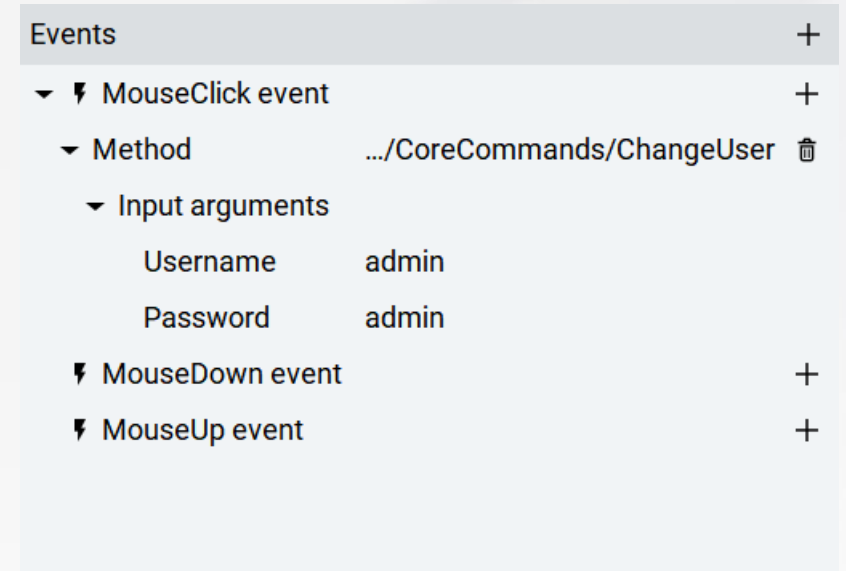
Manage user access at runtime

- Login options

- using the **widget "Login Form"** of Template Library.
Widget comes as a folder that contains the "LoginForm" to be placed in a Panel or called by a Dropdown button
- using the **method** "Aliases > {Session} > Session > Login" or "Aliases > {Session} > Session > Change User"
- using the **method** "Commands > Core Commands > Change User"

- Logout options

- using the **method** "Aliases > {Session} > Session > Logout"
- using an **UISession object** and use the IdleTimeoutEvent to call the method "UI Session > Logout" or "UI Session > Change User"
- using the **script "Idle Timeout"** available in Template Library, place into the MainWindow and associate a "Logout" command



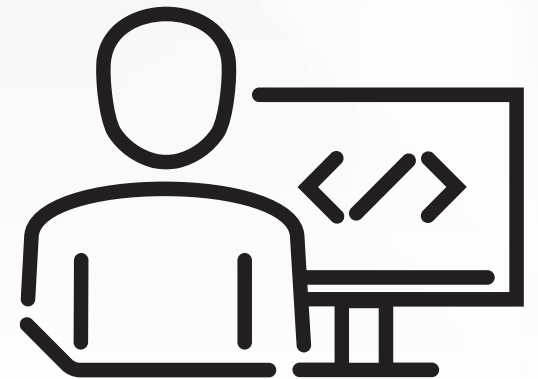
Edit users at runtime

- using the **widget "User Editor"** of Template Library
- Widget, comes as a folder that contains the "UserEditorOverview" to be placed in a Panel
- It is required to link Users and Groups folder to the widget

The screenshot displays the 'User Editor' widget interface. On the left, a list of users is shown with 'user' as the header and 'admin' as the only entry. Below this list are 'Create' and 'Delete' buttons. The central area contains form fields for 'Name' (pre-filled with 'user'), 'Password' (an empty text box), and 'Locale' (a dropdown menu). To the right, under the 'Groups' header, there are two options: 'users' with a checked checkbox and 'admins' with an unchecked checkbox. An 'Apply' button is located at the bottom center of the form area.

Hands-on session

- Create a couple of different users, and groups and make "user – group" association
- Define security at some UI objects (Visibility/Enabling)
- Add the Login form to a Panel
- Optional: call the Login form from a dropdown button paying attention to the background...
- Try at runtime with different users



Configure Reports

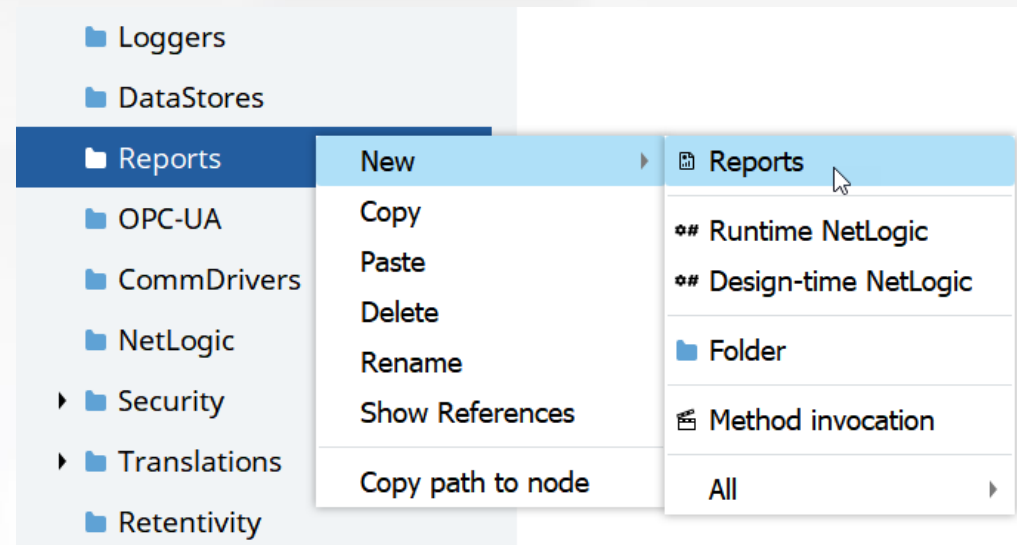


Reports

- Design and generate reports in PDF format
- Data coming from FTOptixApplication and/or from a database

- Components:

- StyleSheet
- Header
- Sections
- Footer

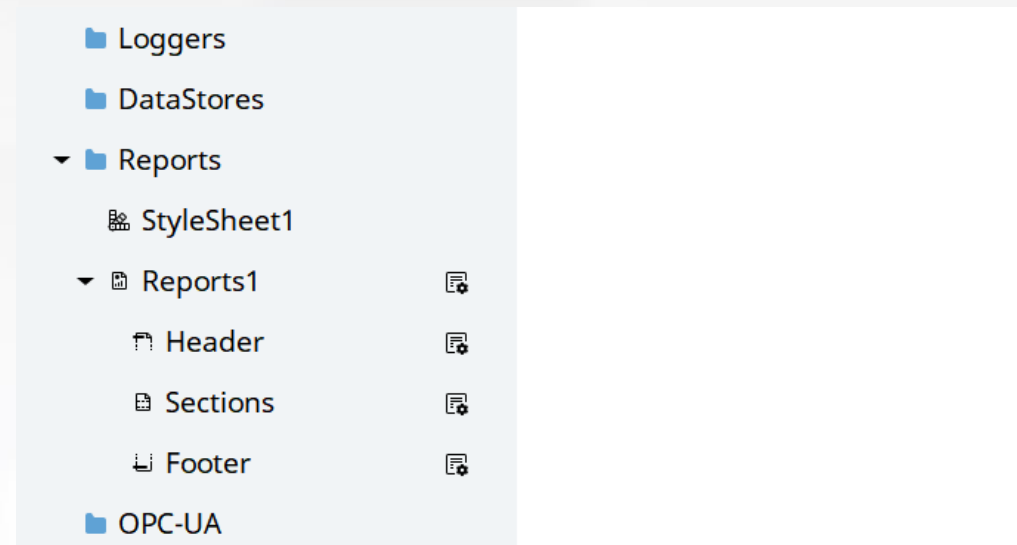


Reports

- Design and generate reports in PDF format
- Data coming from FTOptixApplication and/or from a database

- Components:

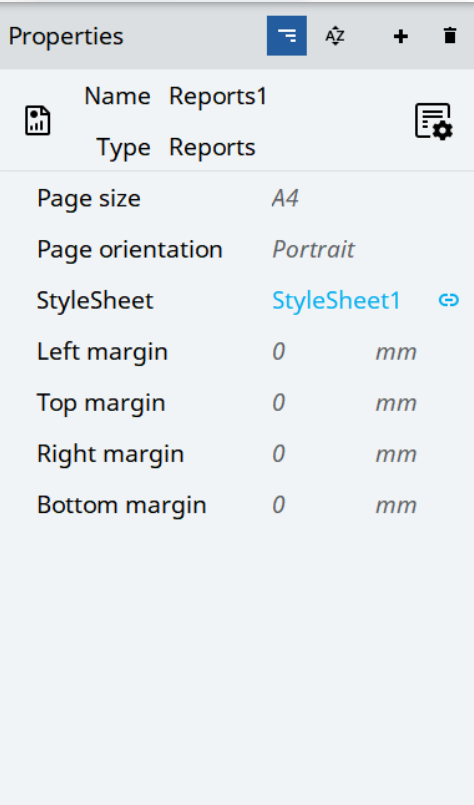
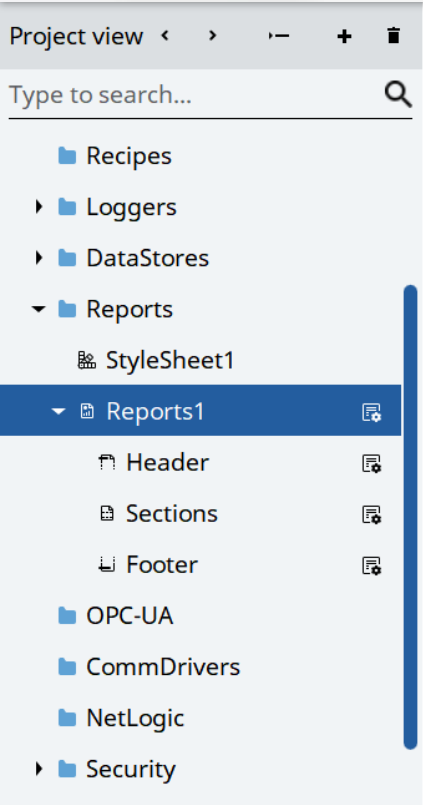
- StyleSheet
- Header
- Sections
- Footer



Report properties

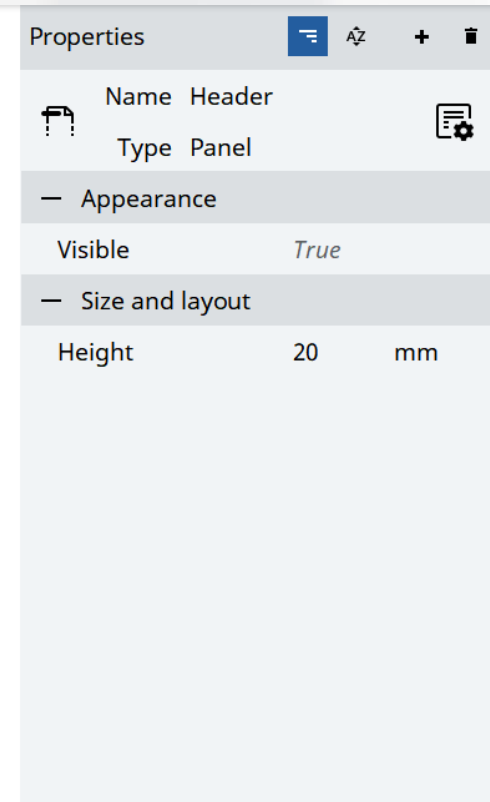
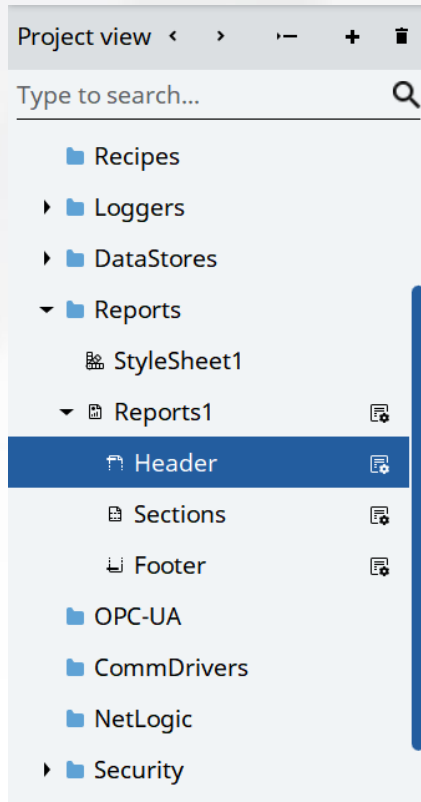
- Allow setting the report's global layout properties, which apply to all pages of the final PDF document

- Page size,
- Page orientation,
- Style sheet used,
- Page margins

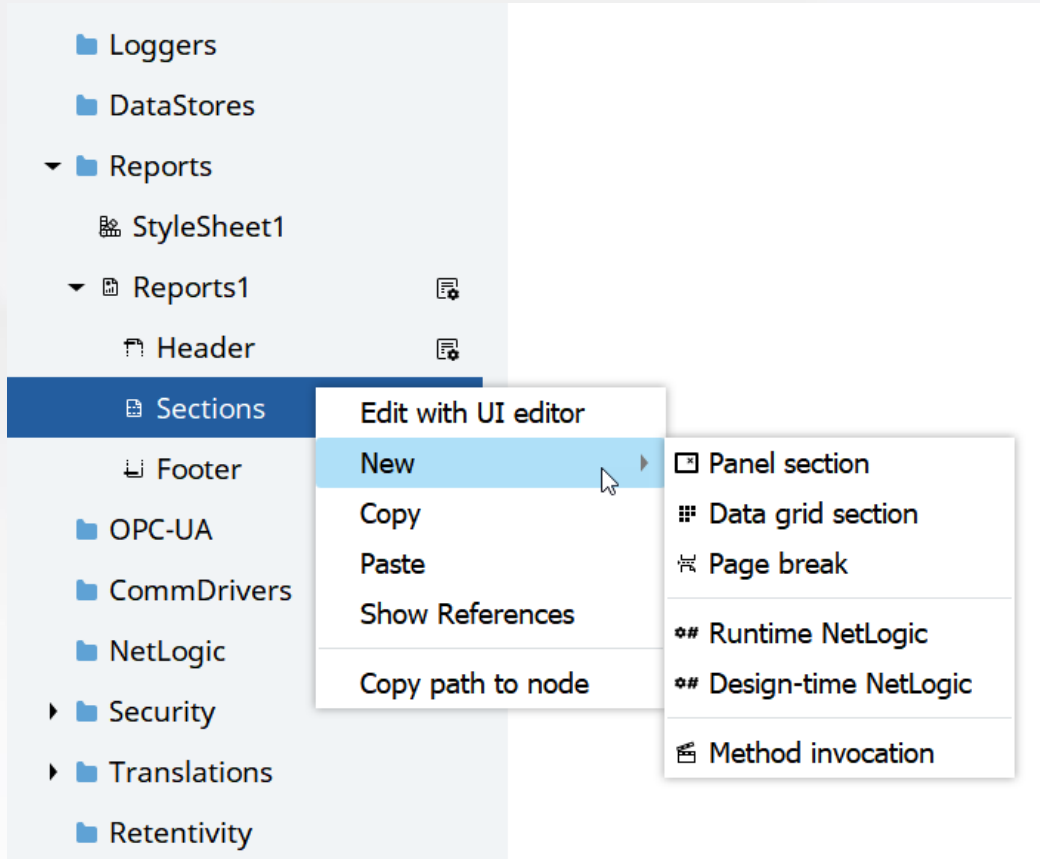


Report header and footer

- Header/Footer is content that appears at the top/bottom of a report page
- Properties:
 - Visible
 - Height

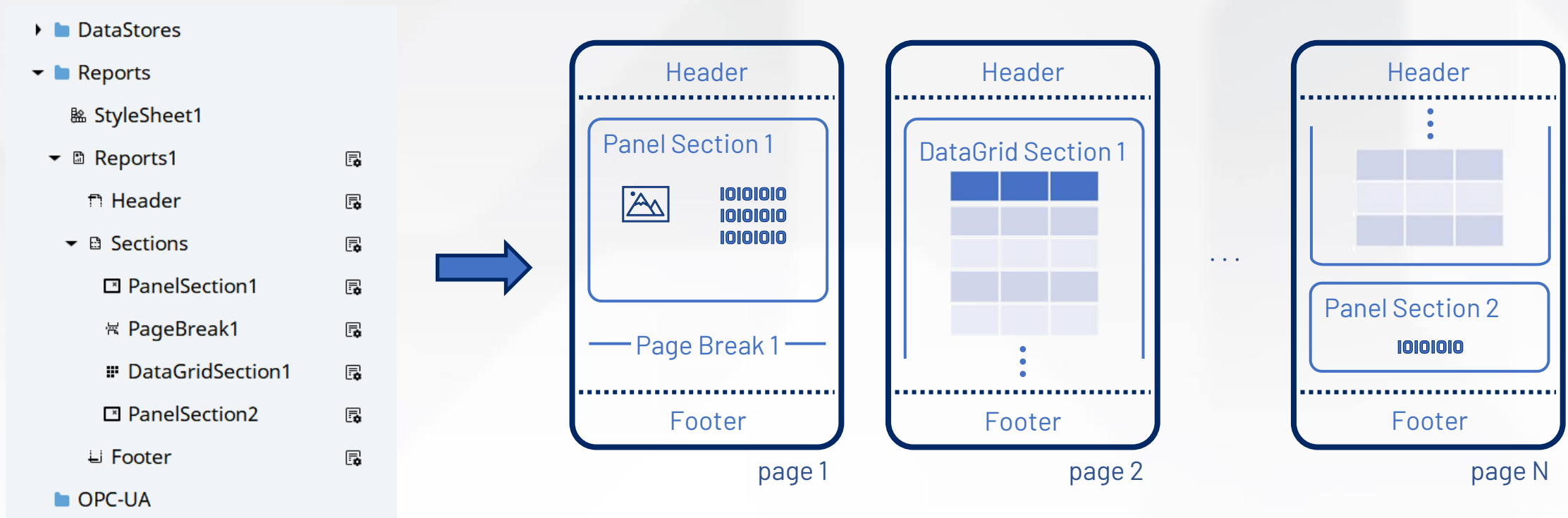


Report sections



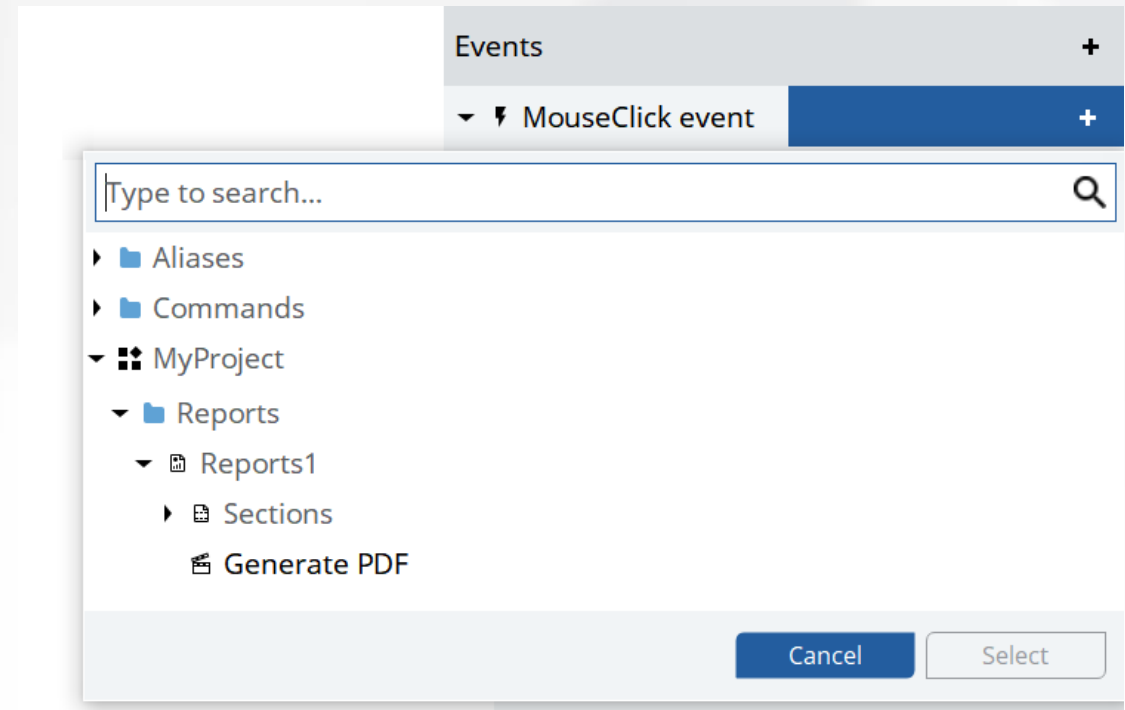
- **Panel:** container of graphical objects as Label, Rectangle, Panel, Image.
Can be used to show variable values.
- **DataGrid:** container of data read from a database in table format.
Can be used to show historical data collected by a Datalogger or Alarms.
- **Page break:** placeholder that forces the report to continue to another page.

Report sections example



Generate report at runtime

- Report resource expose method "Generate PDF"
- Can be called for example on Mouse Click event
- Properties:
 - Output Path: the path of the PDF file
 - Locale Id (optional): language to be used inside report



Generate report at runtime

- Report resource expose method "Generate PDF"
- Can be called for example on Mouse Click event
- Properties:
 - Output Path: the path of the PDF file
 - Locale Id (optional): language to be used inside report

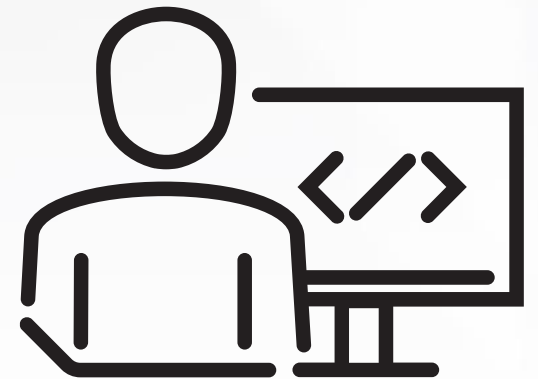
Events		+
▼ ⚡	MouseClicked event	+
▼	Method	.../GeneratePdf
▼	Input arguments	
	OutputPath	...\test.pdf Browse
	LocaleId	
⚡	MouseDown event	+
⚡	MouseUp event	+

Output Path

Shortcut	Description
<code>%PROJECTDIR%</code>	A directory that contains files transferred at runtime (images, documents, and so on). Example: <code>C:\Users\UserName\Documents\Rockwell Automation\FactoryTalk Optix \ProjectName</code>
<code>%APPLICATIONDIR%</code>	A directory that contains objects used at runtime, including these objects: • Embedded database • Retentivity storage • Exported files
<code>%USB<i>N</i>%</code> , where <i>N</i> is a number associated with the USB device.	A path to a specific USB device connected to the target device.

Hands-on session

- Define a report that includes a Datagrid to show values from a datalogger table
- Add a button that calls the "Generate PDF" method



Set Locale, Languages and Measurement System



Localization

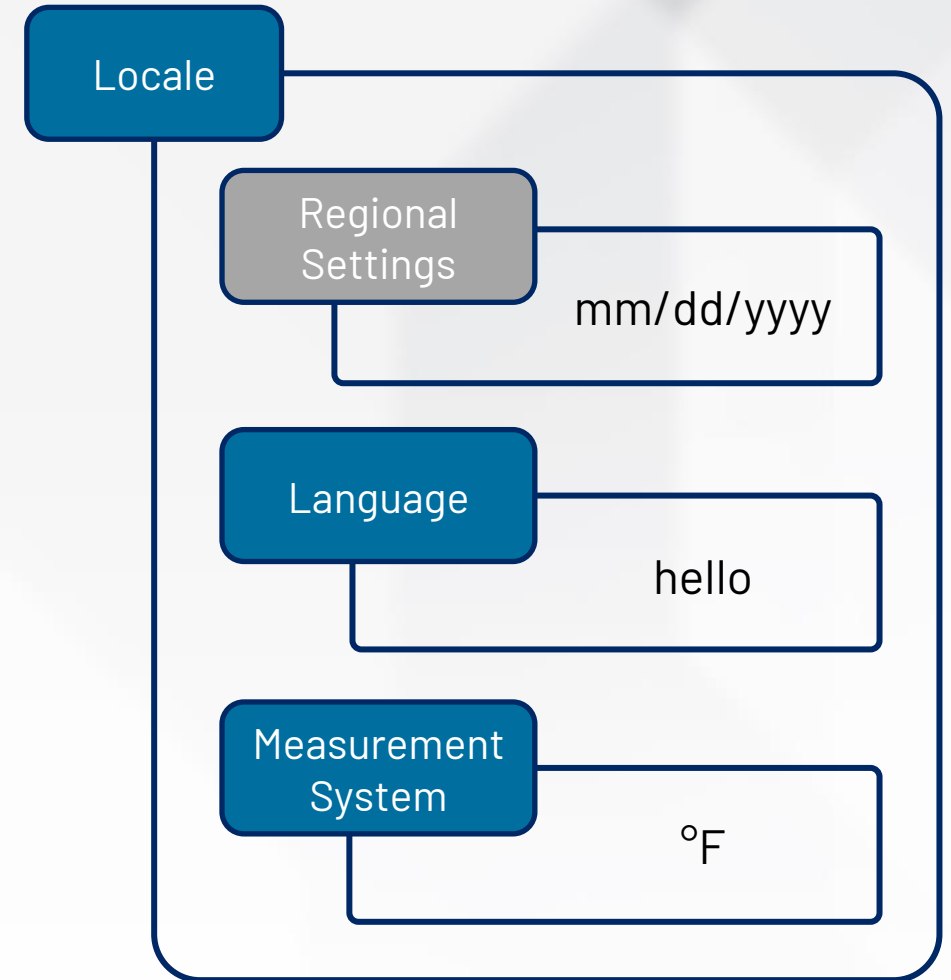
- a Locale is **a set of User Interface settings** based on country

Locale	en-US	it-IT
Regional Settings	mm/dd/yyyy ... 10.5	dd/mm/yyyy ... 10,5
Language/Translations	Hello	Ciao
Measurement System	°F	°C

- Locale, Language and Measurement System are **related to the Session**
 - A session is a Runtime context in which a user executes operations
 - Examples: a session of Native Presentation Engine can look different from a session of Web Presentation Engine.
A session of User1 can look different from a session of User2

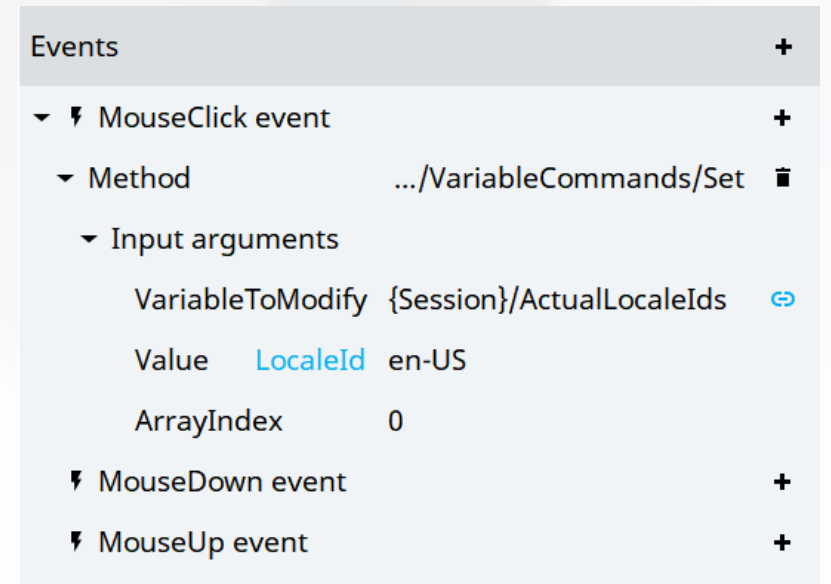
Locale

- Changing Locale will set altogether:
 - Regional Settings
 - Language
 - Measurement System
- How to change the Locale at Runtime
 - Method
 - Command > Variable commands > Set variable value
 - Variable to Modify
 - Aliases > {Session} > Session > ActualLocaleIds
 - Value
 - Select the Locale ISO code
- Language and Measurement System can be set independently from Locale



Locale

- Changing Locale will set altogether:
 - Regional Settings
 - Language
 - Measurement System
- How to change the Locale at Runtime
 - Method
 - Command > Variable commands > Set variable value
 - Variable to Modify
 - Aliases > {Session} > Session > ActualLocaleIds
 - Value
 - Select the Locale ISO code



- Language and Measurement System can be set independently from Locale

Languages - Translations

- The translation editor makes it possible to translate strings used in multilingual projects

- **Translation Table**

- shows the translation of strings (Keys) in different languages (Locales)

- **Translation Key References**

- shows all strings of the project that can be synched with the Translation Table (to become a key)

Translation Table		Remove translations	Add locale	View Translation References
Key	en-US			
Acked	Acked			
Acknowledge	Acknowledge			
Acknowledge All	Acknowledge All			
Active	Active			
Apply	Apply			
Confirm	Confirm			
Confirm All	Confirm All			
Confirmed	Confirmed			
Delete	Delete			
Failed	Failed			
Load	Load			
Message	Message			
Name	Name			
RecipeApply	Recipe applied			

Languages - Translations

- The translation editor makes it possible to translate strings used in multilingual projects

- **Translation Table**

- shows the translation of strings (Keys) in different languages (Locales)

- **Translation Key References**

- shows all strings of the project that can be synched with the Translation Table (to become a key)

Project view

Type to search...

Model

Converters

Alarms

▶ Recipes

Loggers

▶ DataStores

Reports

OPC-UA

▶ CommDrivers

NetLogic

▶ Security

▼ Translations

⌕ LocalizationDictionary1

Retentivity

Translation Key References				View Translation Table
String	↑ Key	Path	Synchronized	
Acked	Acked	MyProject/UI/Panels/Screen2/...	☒	
Acknowledge	Acknowledge	MyProject/UI/Panels/Screen2/...	☒	
Acknowledge All	Acknowledge All	MyProject/UI/Panels/Screen2/...	☒	
Active	Active	MyProject/UI/Panels/Screen2/...	☒	
Apply	Apply	MyProject/UI/Panels/Screen1/...	☒	
Confirm	Confirm	MyProject/UI/Panels/Screen2/...	☒	
Confirm All	Confirm All	MyProject/UI/Panels/Screen2/...	☒	
Confirmed	Confirmed	MyProject/UI/Panels/Screen2/...	☒	
Delete	Delete	MyProject/UI/Panels/Screen1/...	☒	
Label1		MyProject/UI/Panels/Screen3/...	☐	
Label2		MyProject/UI/Panels/Screen3/...	☐	
Label3		MyProject/UI/Panels/Screen3/...	☐	
Load	Load	MyProject/UI/Panels/Screen1/...	☒	
Message	Message	MyProject/UI/Panels/Screen2/...	☒	

Translation table

- Remove Translations
 - to remove one or more keys from the table
- Add locale
 1. Click to add a new language
 2. Select the language in a standard Language-Country ISO code
 3. Confirm adding the new language to the Project properties
- View Translation Reference
 - switch to Key Reference table

Translation Table		Remove translations	Add locale	View Translation References
Key	↑ en-US ✕			
Acked	Acked			
Acknowledge	Acknowledge			
Acknowledge All	Acknowledge All			
Active	Active			
Apply	Apply			
Confirm	Confirm			
Confirm All	Confirm All			
Confirmed	Confirmed			
Delete	Delete			

- Import/Export
 - Translation Table can be exported and imported back using the "Translation Importer Exporter" script available in Template Library

Translation table

- Remove Translations

- to remove one or more keys from the table

- Add locale

1. Click to add a new language
2. Select the language in a standard Language-Country ISO code
3. Confirm adding the new language to the Project properties

- View Translation Reference

- switch to Key Reference table

Translation Table

Remove translations

Add locale

View Translation References

Key	en-US
Acked	Acked
Acknowledge	Acknowledge
Acknowledge All	Acknowledge All
Active	Active
Apply	Apply
Confirm	Confirm
Confirm All	Confirm All
Confirmed	Confirmed
Delete	Delete

Select the new locale...

it-IT

it-CH

it-IT

iu-CA

✓

- Import/Export

- Translation Table can be exported and imported back using the "Translation Importer Exporter" script available in Template Library

Translation table

- Remove Translations

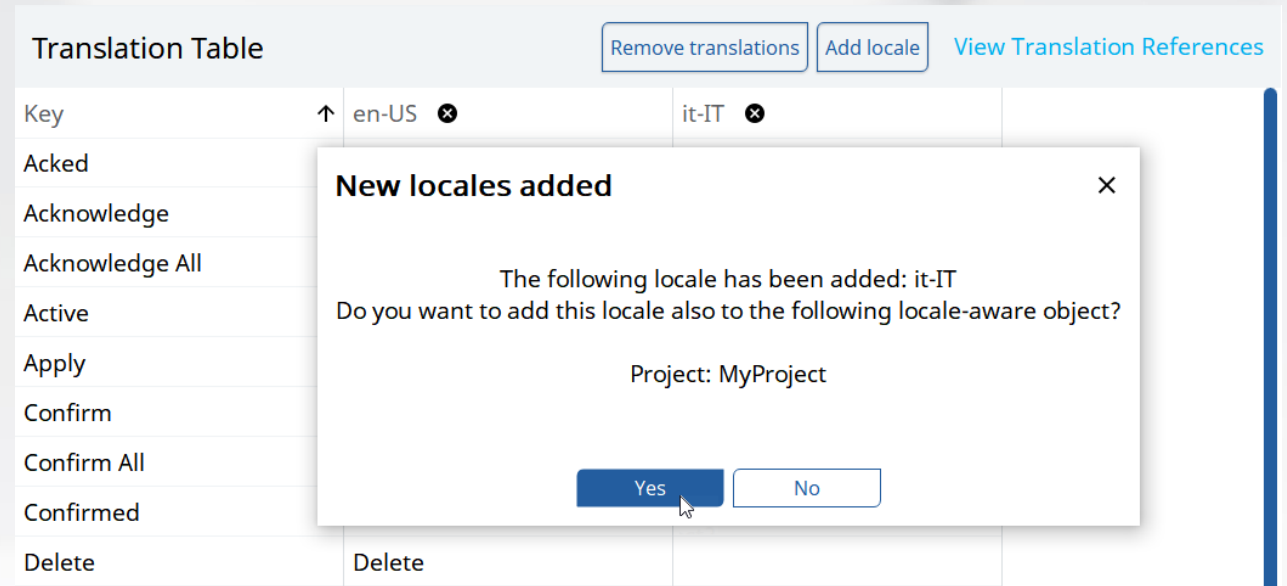
- to remove one or more keys from the table

- Add locale

1. Click to add a new language
2. Select the language in a standard Language-Country ISO code
3. Confirm adding the new language to the Project properties

- View Translation Reference

- switch to Key Reference table



- Import/Export

- Translation Table can be exported and imported back using the "Translation Importer Exporter" script available in Template Library

Translation key references

- Key Reference table
 - shows all strings of the project
- Setting "Synchronized" to true
 - creates a key into Translation Table with the same text
 - copy the key text into the default language (en-US)

Translation Key References

 [View Translation Table](#)

String	↑ Key	Path	Synchronized
Acked	Acked	MyProject/UI/Panels/Screen2/...	☒
Acknowledge	Acknowledge	MyProject/UI/Panels/Screen2/...	☒
Acknowledge All	Acknowledge All	MyProject/UI/Panels/Screen2/...	☒
Active	Active	MyProject/UI/Panels/Screen2/...	☒
Apply	Apply	MyProject/UI/Panels/Screen1/...	☒
Confirm	Confirm	MyProject/UI/Panels/Screen2/...	☒
Confirm All	Confirm All	MyProject/UI/Panels/Screen2/...	☒
Confirmed	Confirmed	MyProject/UI/Panels/Screen2/...	☒
Delete	Delete	MyProject/UI/Panels/Screen1/...	☒
Label1		MyProject/UI/Panels/Screen3/...	☐
Label2		MyProject/UI/Panels/Screen3/...	☐
Label3		MyProject/UI/Panels/Screen3/...	☐
Load	Load	MyProject/UI/Panels/Screen1/...	☒
Message	Message	MyProject/UI/Panels/Screen2/...	☒

Translation key references

- Key Reference table
 - shows all strings of the project
- Setting "Synchronized" to true
 - creates a key into Translation Table with the same text
 - copy the key text into the default language (en-US)

Translation Key References			 View Translation Table
String	↑ Key	Path	Synchronized
Acked	Acked	MyProject/UI/Panels/Screen2/...	☒
Acknowledge	Acknowledge	MyProject/UI/Panels/Screen2/...	☒
Acknowledge All	Acknowledge All	MyProject/UI/Panels/Screen2/...	☒
Active	Active	MyProject/UI/Panels/Screen2/...	☒
Apply	Apply	MyProject/UI/Panels/Screen1/...	☒
Confirm	Confirm	MyProject/UI/Panels/Screen2/...	☒
Confirm All	Confirm All	MyProject/UI/Panels/Screen2/...	☒
Confirmed	Confirmed	MyProject/UI/Panels/Screen2/...	☒
Delete	Delete	MyProject/UI/Panels/Screen1/...	☒
Label1	Label1	MyProject/UI/Panels/Screen3/...	☒
Label2		MyProject/UI/Panels/Screen3/...	☐
Label3		MyProject/UI/Panels/Screen3/...	☐
Load	Load	MyProject/UI/Panels/Screen1/...	☒
Message	Message	MyProject/UI/Panels/Screen2/...	☒

Translation usage

- Two ways to set the "starting" language:

1. Language or Locale Id of the "Starting user"
2. First language listed in the "Locales" project property

- How to change the language at Runtime

- Method

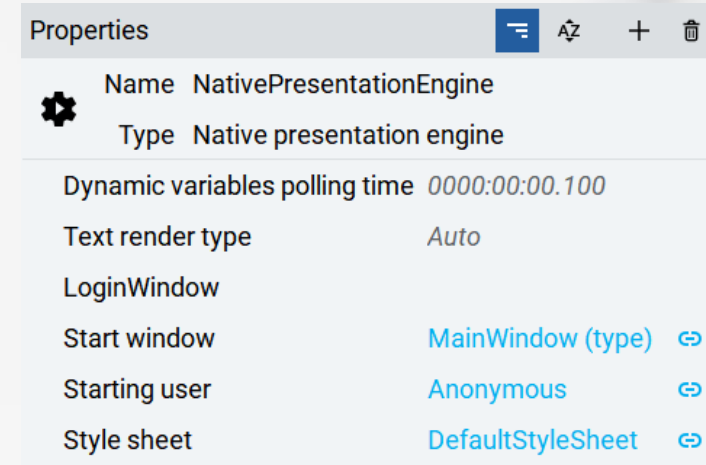
- Command > Variable commands > Set variable value

- Variable to Modify

- Aliases > {Session} > Session > ActualLanguages

- Value

- Select the language ISO code



Translation usage

- Two ways to set the "starting" language:

1. Language or Locale Id of the "Starting user"
2. First language listed in the "Locales" project property

- How to change the language at Runtime

- Method

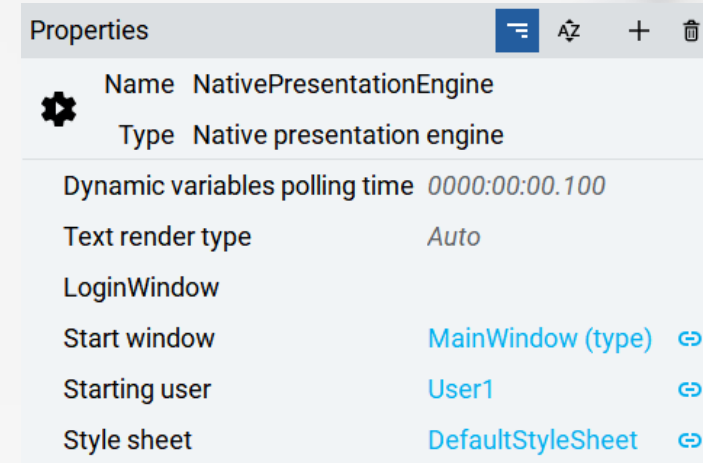
- Command > Variable commands > Set variable value

- Variable to Modify

- Aliases > {Session} > Session > ActualLanguages

- Value

- Select the language ISO code



Translation usage

- Two ways to set the "starting" language:

1. Language or Locale Id of the "Starting user"
2. First language listed in the "Locales" project property

- How to change the language at Runtime

- Method

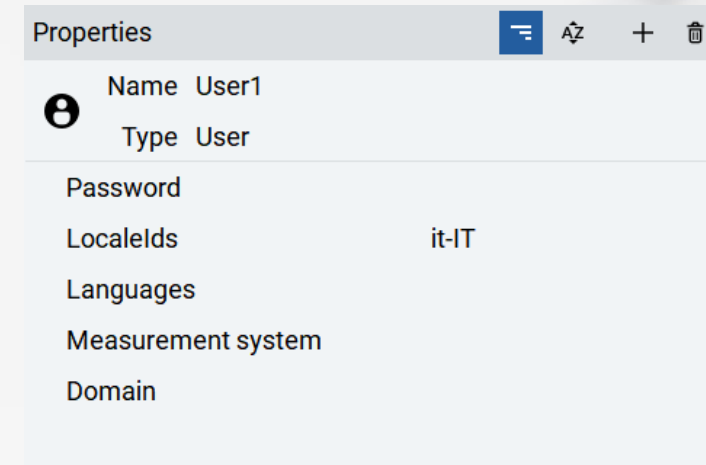
- Command > Variable commands > Set variable value

- Variable to Modify

- Aliases > {Session} > Session > ActualLanguages

- Value

- Select the language ISO code



Translation usage

- Two ways to set the "starting" language:

1. Language or Locale Id of the "Starting user"
2. First language listed in the "Locales" project property

- How to change the language at Runtime

- Method

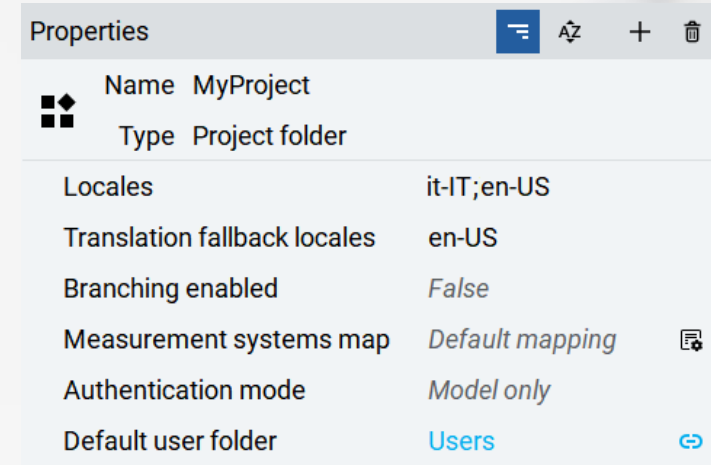
- Command > Variable commands > Set variable value

- Variable to Modify

- Aliases > {Session} > Session > ActualLanguages

- Value

- Select the language ISO code



Properties		
Name	MyProject	
Type	Project folder	
Locales	it-IT;en-US	
Translation fallback locales	en-US	
Branching enabled	False	
Measurement systems map	Default mapping	
Authentication mode	Model only	
Default user folder	Users	

Translation usage

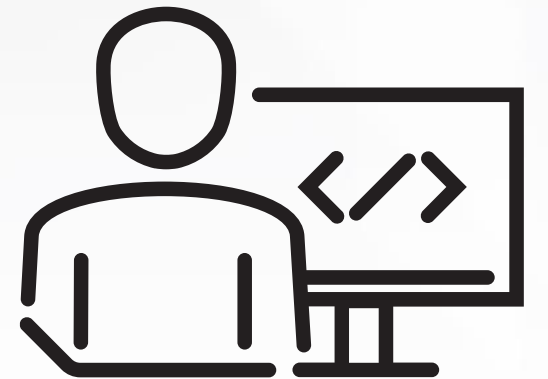
- Two ways to set the "starting" language:
 1. Language or Locale Id of the "Starting user"
 2. First language listed in the "Locales" project property
- How to change the language at Runtime
 - Method
 - Command > Variable commands > Set variable value
 - Variable to Modify
 - Aliases > {Session} > Session > ActualLanguages
 - Value
 - Select the language ISO code

Properties		
Name	MyProject	
Type	Project folder	
Locales	it-IT;en-US	
Translation fallback locales	en-US	
Branching enabled	False	
Measurement systems map	Default mapping	
Authentication mode	Model only	
Default user folder	Users	

Events		
▼  MouseClick event		
▼ Method	.../VariableCommands/Set	
▼ Input arguments		
VariableToModify	{Session}/ActualLanguages	
Value	 en-US	
ArrayIndex	0	
 MouseDown event		
 MouseUp event		

Hands-on session

- Use some strings to label objects around the project
- Import strings into the Translation Table using the Key Reference Table then edit the translation manually
- Change the language at Runtime with the Emulator
- Optionally
 - try to export the Translation table,
 - modify with an external editor,
 - import back in Optix Studio

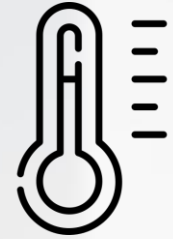


Measurement system

- Choose between:
 - International System of Units (SI)
 - US Customary measurement system
 - British Imperial units
- is enough to set the Unit of Measure on the Variable, then at Runtime, the conversion will be automatically applied depending on the Measurement System chosen



liter / quart



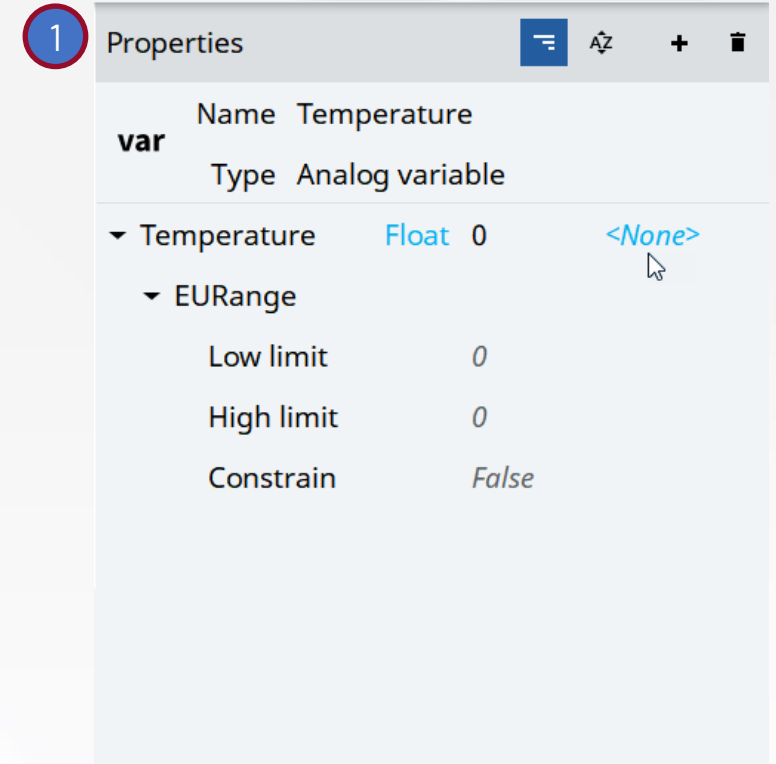
°C / °F



meter / yard

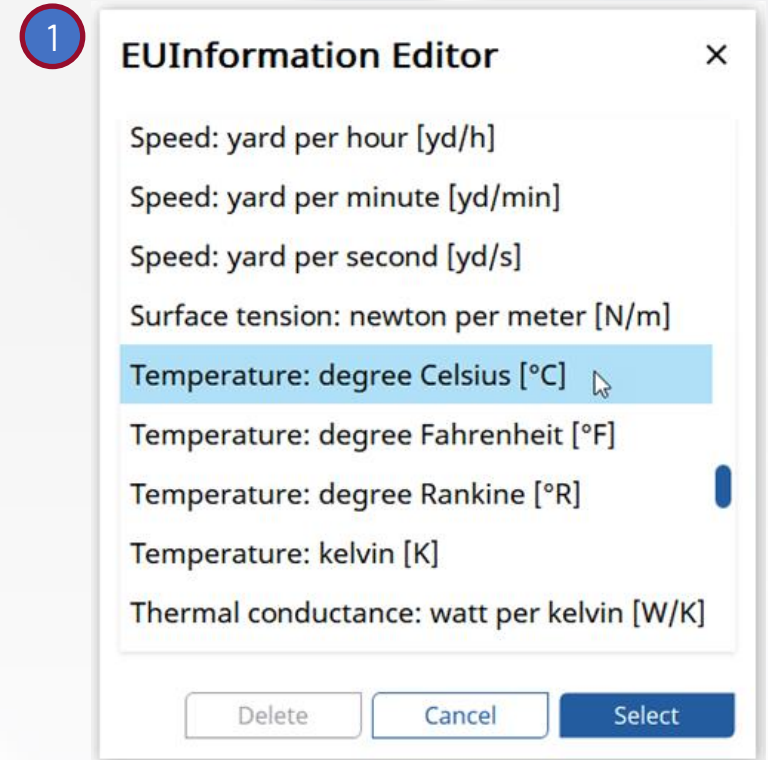
Localize an Analog Variable

1. Define a new Analog Variable with Engineering Unit into the Model folder
 2. Add an Analog Variable to the object
 3. Into the object's Analog Variable set an Advanced Dynamic Link using "Binding mode between Source and Parent engineering unit"
 4. Link the object's value to the object's Analog Variable
- Unit of measure can be defined directly within a Comm.Driver Tag as well



Localize an Analog Variable

1. Define a new Analog Variable with Engineering Unit into the Model folder
 2. Add an Analog Variable to the object
 3. Into the object's Analog Variable set an Advanced Dynamic Link using "Binding mode between Source and Parent engineering unit"
 4. Link the object's value to the object's Analog Variable
- Unit of measure can be defined directly within a Comm.Driver Tag as well



Localize an Analog Variable

1. Define a new Analog Variable with Engineering Unit into the Model folder
 2. Add an Analog Variable to the object
 3. Into the object's Analog Variable set an Advanced Dynamic Link using "Binding mode between Source and Parent engineering unit"
 4. Link the object's value to the object's Analog Variable
- Unit of measure can be defined directly within a Comm.Driver Tag as well

1

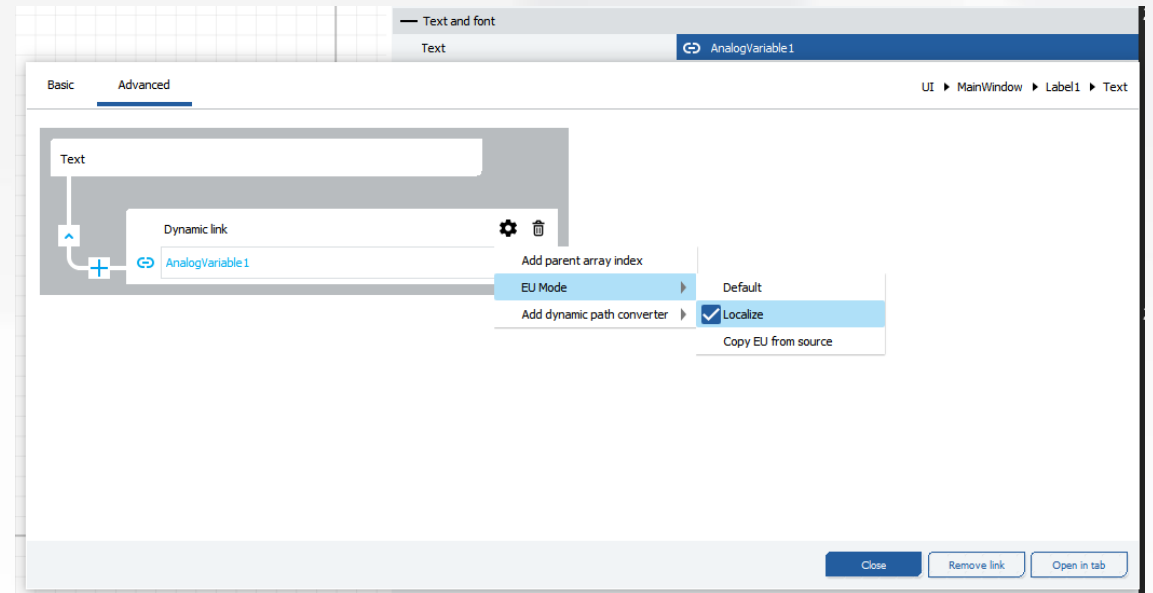
Properties

	Name	Temperature
var	Type	Analog variable
▼ Temperature	Float	0 °C
▼ EURange		
	Low limit	0
	High limit	0
	Constrain	False

Localize an Analog Variable

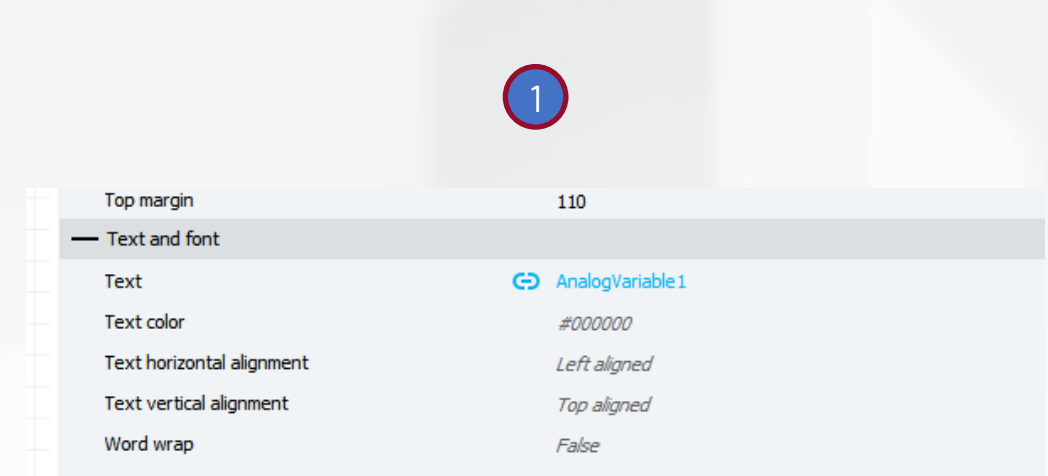
1. Define a new Analog Variable with Engineering Unit into the Model folder
 2. Add an Analog Variable to the object
 3. Into the object's Analog Variable set an Advanced Dynamic Link using "Binding mode between Source and Parent engineering unit"
 4. Link the object's value to the object's Analog Variable
- Unit of measure can be defined directly within a Comm.Driver Tag as well

2



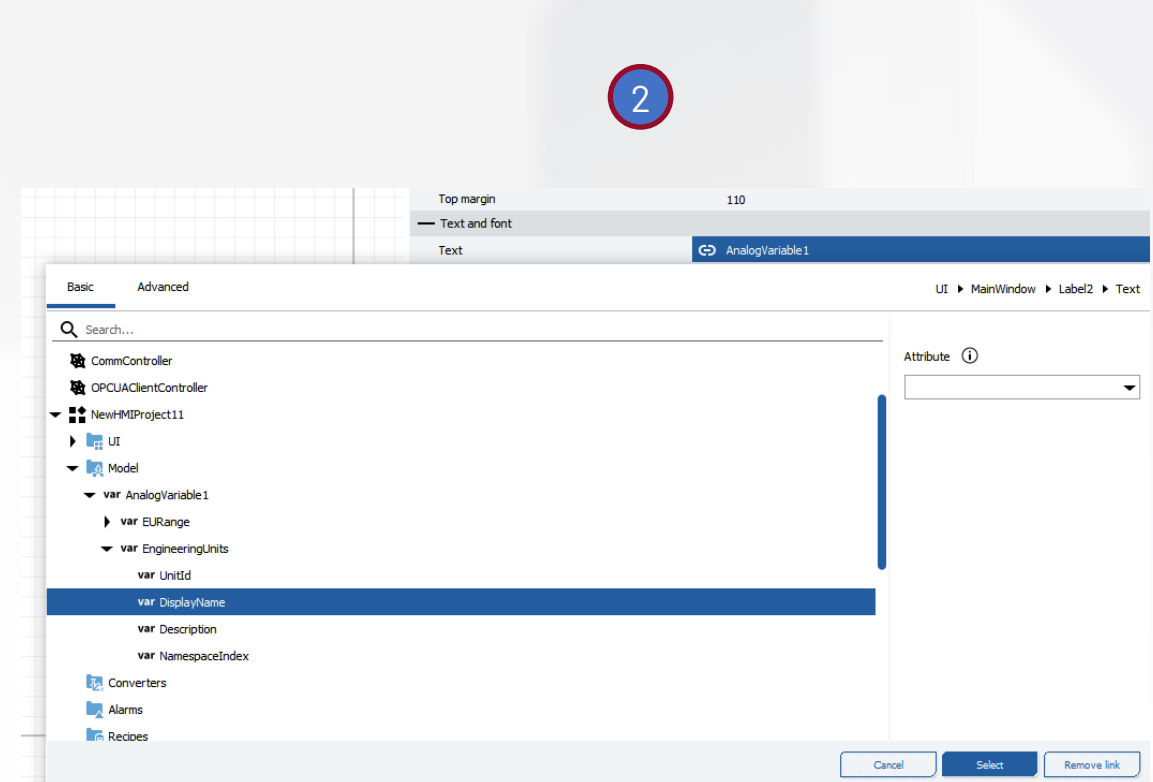
Display and localize the Engineering Unit

1. Add a label to the screen and create a DynamicLink from the Analog Variable to the Text property of the label
2. Open the DynamicLink popup, expand the Analog Variable and select Engineering Unit > Display Name
3. Reopen the DynamicLink popup, move to the Advanced tab and localize the Engineering Unit



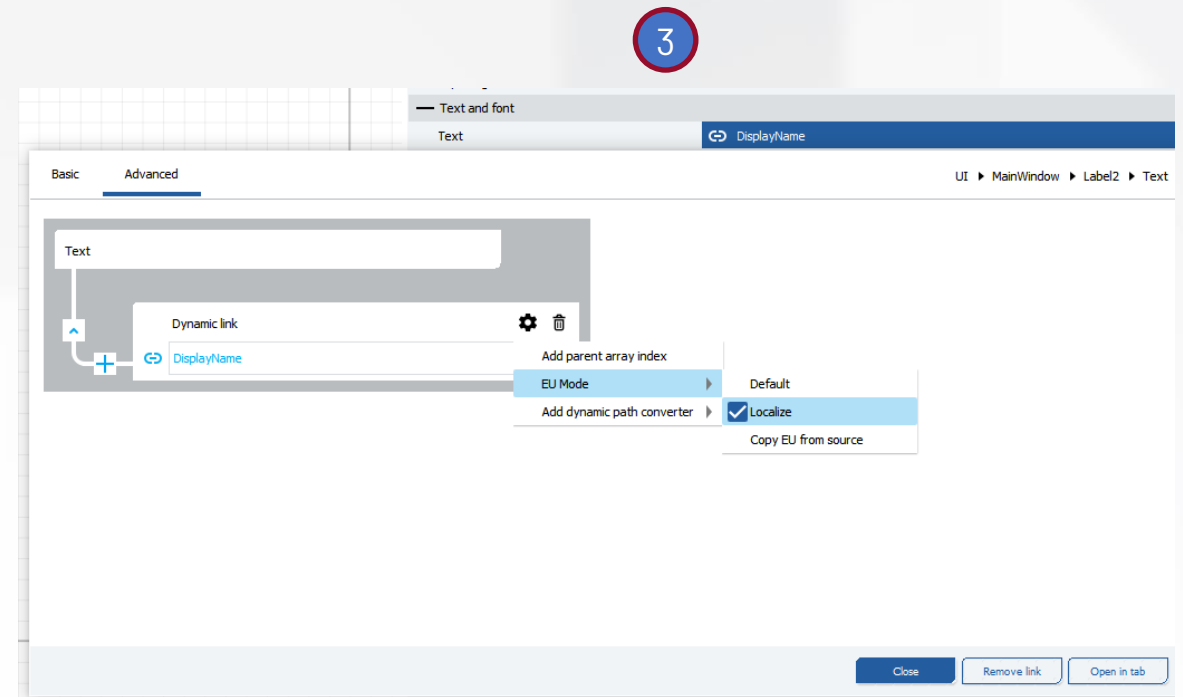
Display and localize the Engineering Unit

1. Add a label to the screen and create a DynamicLink from the Analog Variable to the Text property of the label
2. Open the DynamicLink popup, expand the Analog Variable and select Engineering Unit > Display Name
3. Reopen the DynamicLink popup, move to the Advanced tab and localize the Engineering Unit



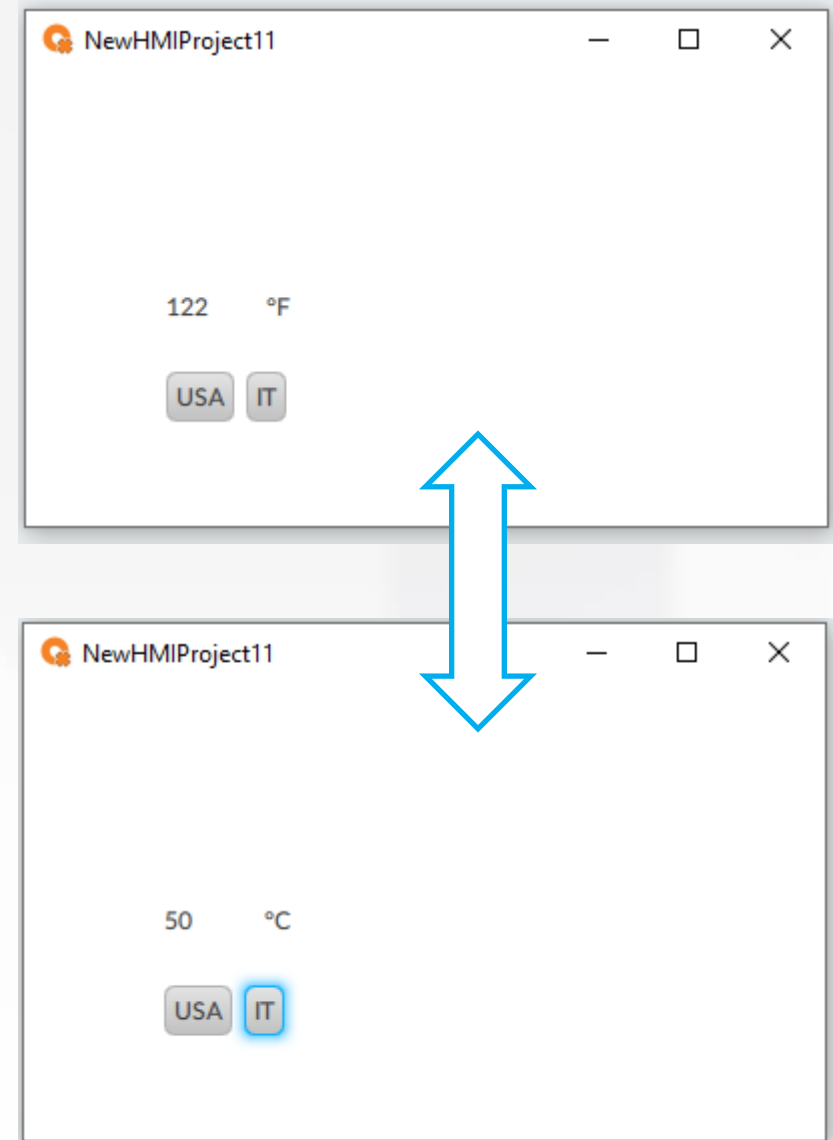
Display and localize the Engineering Unit

1. Add a label to the screen and create a DynamicLink from the Analog Variable to the Text property of the label
2. Open the DynamicLink popup, expand the Analog Variable and select Engineering Unit > Display Name
3. Reopen the DynamicLink popup, move to the Advanced tab and localize the Engineering Unit



Test the conversion

1. Add two buttons to change the Measurement System
2. Execute the runtime and check the conversion



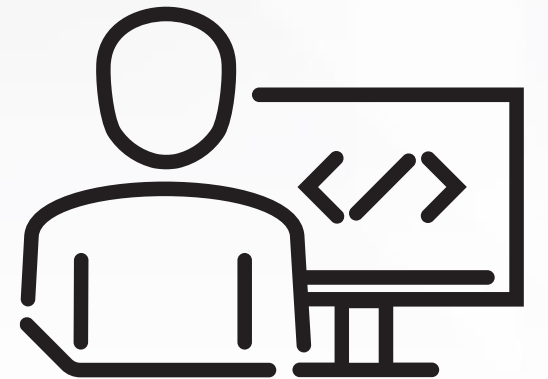
Set a measurement system

- How to change the Measurement System at Runtime
 - Method
 - Command > Variable commands > Set variable value
 - Variable To Modify
 - Aliases > {Session} > Session > ActualMeasurementSystem
 - Value
 - Select one of the available Measurement Systems

Events		+
▼ ⚡	MouseClicked event	+
▼	Method	.../Commands/VariableCommands/Set
▼	Input arguments	
	VariableToModify	{Session}/ActualMeasurementSystem
	MeasurementSystem	International system of units
	ArrayIndex	0
⚡	MouseDown event	+
⚡	MouseUp event	+

Hands-on session

- Set one or more Units of Measure to one or more variables
- Add a button to change the Measurement System (or the Locale)
- Check the conversion is applied





Develop using version control and collaboration



Version control and Collaboration

- **Version Control**

- used for keeping track of incrementally-different versions
- work also locally (offline)

- **Collaboration**

- distributed version control
- form of version control in which the full history, is mirrored on every developer's computer
- enables automatic management of merges,
- does not rely on a single location for backups
- Git is a distributed version control system
- GitHub is an Internet hosting service using Git

The screenshot shows the 'FactoryTalk Optix Studio Options' dialog box with the 'Version control' tab selected. The 'Version control User Configuration' section is highlighted with a blue border and contains two input fields: 'User Name' and 'User Email'. Below this, the 'Version control providers and IDs' section is visible, featuring a 'Select a provider' list with 'GitHub' and 'GitLab (on-premise)' options, and a 'Provider identity login' area showing a 'GitHub' entry with a green checkmark. At the bottom right, there are 'Save' and 'Cancel' buttons.

Version control and Collaboration

- **Version Control**

- used for keeping track of incrementally-different versions
- work also locally (offline)

- **Collaboration**

- distributed version control
- form of version control in which the full history, is mirrored on every developer's computer
- enables automatic management of merges,
- does not rely on a single location for backups
- Git is a distributed version control system
- GitHub is an Internet hosting service using Git

FactoryTalk Optix Studio Options

Optix Studio Version control

Version control User Configuration

User Name

User Email

Version control providers and IDs

Select a provider Provider identity login

GitHub

GitLab (on-premise)

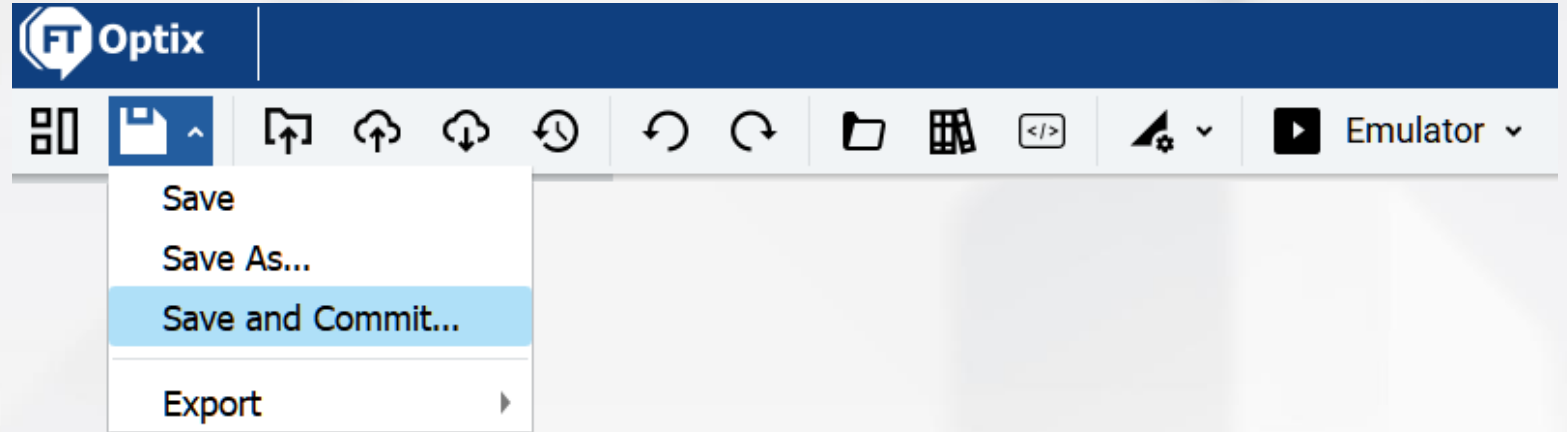
GitHub

Add identity

Save Cancel

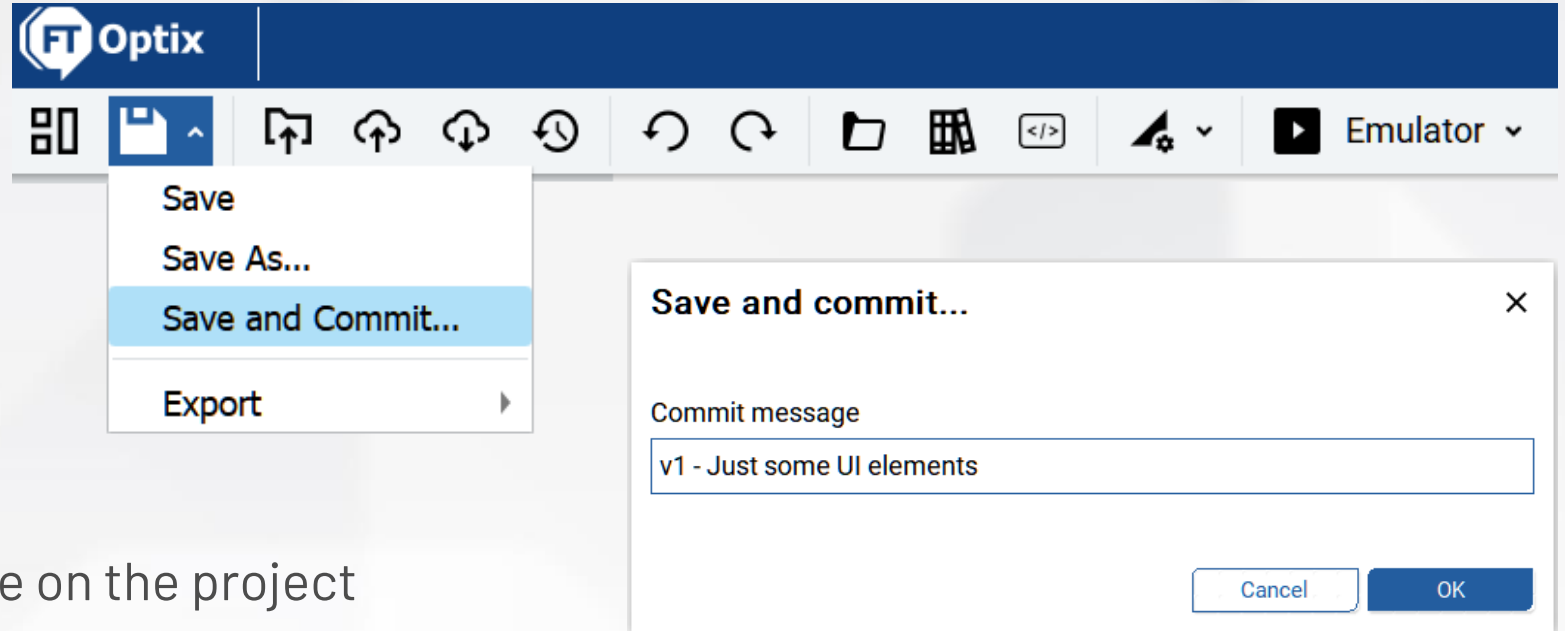
Version control: save and commit locally

- Save and Commit, allow saving the project keeping track of changes
- Works "Local"
- Ideal for those who work alone with no colleagues that collaborate on the project
- No more needs to make zip files named v1, v2...
- The history button show differences, and allow restoring a previous version



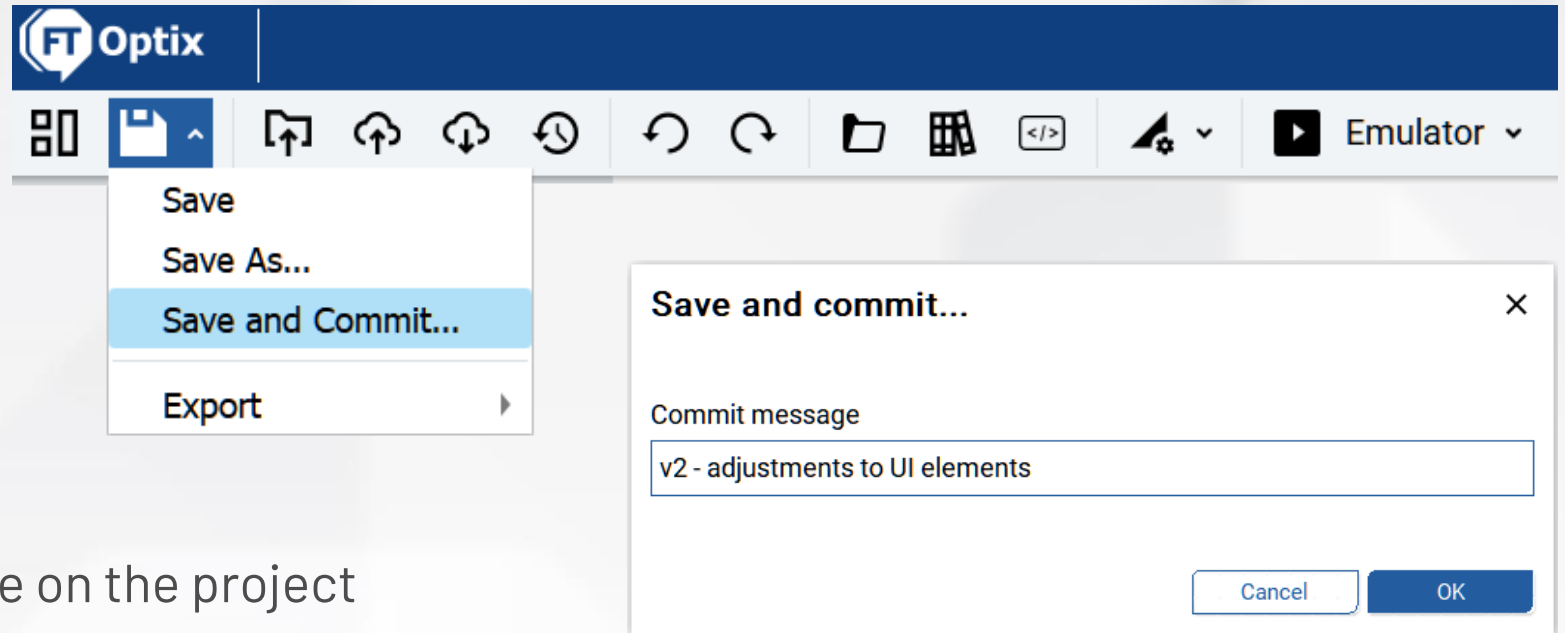
Version control: save and commit locally

- Save and Commit, allow saving the project keeping track of changes
- Works "Local"
- Ideal for those who work alone with no colleagues that collaborate on the project
- No more needs to make zip files named v1, v2...
- The history button show differences, and allow restoring a previous version



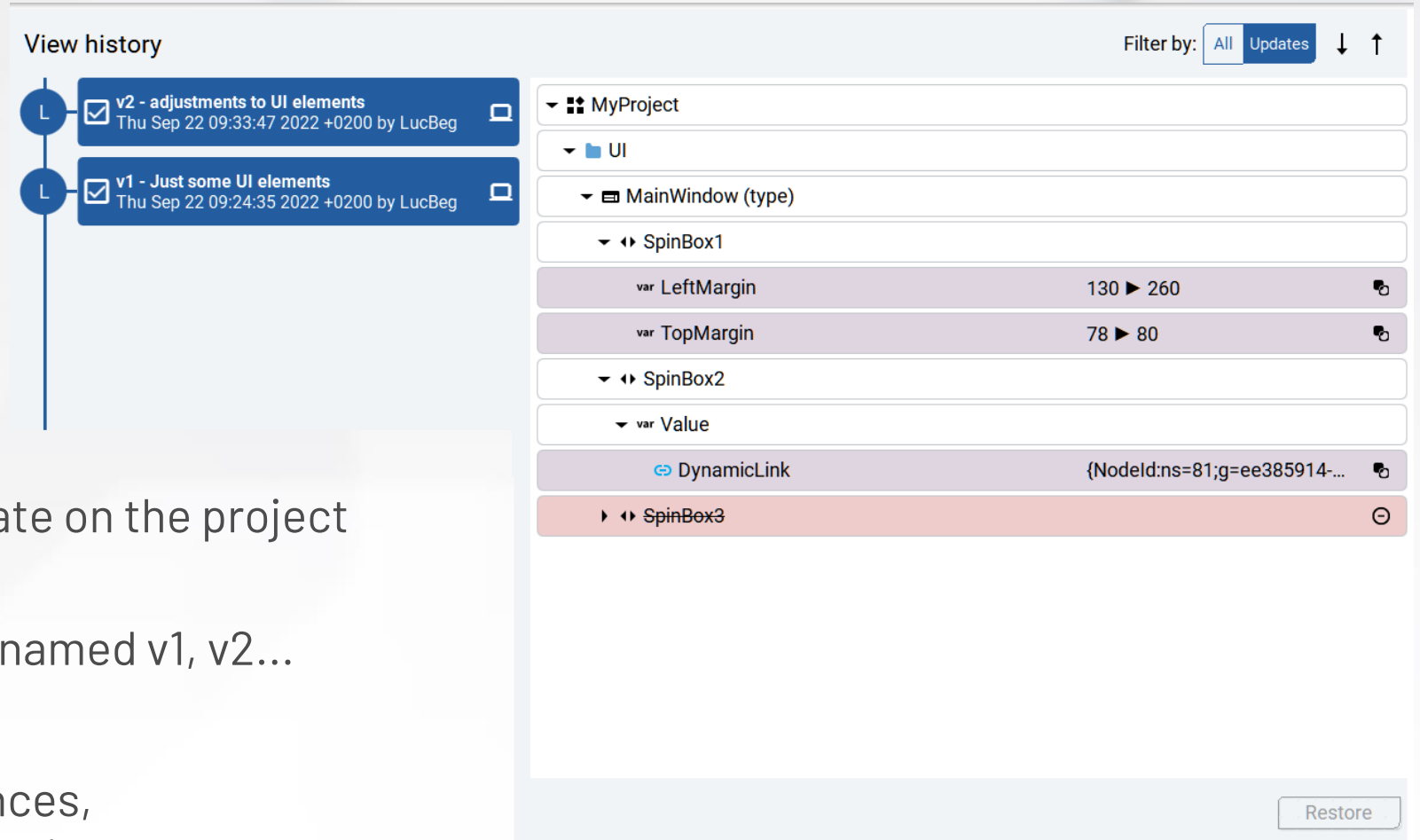
Version control: save and commit locally

- Save and Commit, allow saving the project keeping track of changes
- Works "Local"
- Ideal for those who work alone with no colleagues that collaborate on the project
- No more needs to make zip files named v1, v2...
- The history button show differences, and allow restoring a previous version



Version control: save and commit locally

- Save and Commit, allow saving the project keeping track of changes
- Works "Local"
- Ideal for those who work alone with no colleagues that collaborate on the project
- No more needs to make zip files named v1, v2...
- The history button show differences, and allow restoring a previous version



Version control: save and commit locally

- Save and Commit, allow saving the project keeping track of changes
- Works "Local"
- Ideal for those who work alone with no colleagues that collaborate on the project
- No more needs to make zip files named v1, v2...
- The history button show differences, and allow restoring a previous version

View history

Filter by: All Updates ↓ ↑

v2 - adjustments to UI elements
Thu Sep 22 09:33:47 2022 +0200 by LucBeg

v1 - Just some UI elements
Thu Sep 22 09:24:35 2022 +0200 by LucBeg

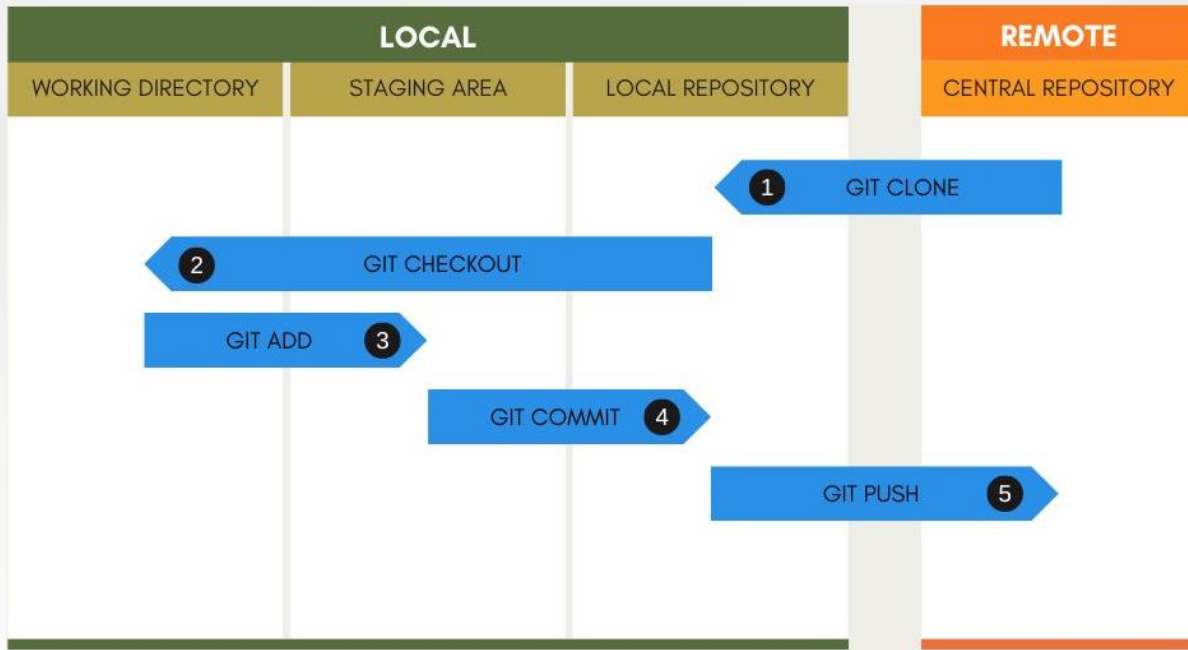
MyProject

- var BranchingEnabled 0
- PasswordPolicy
- var Locales {en-US}
- var LocaleFallbackList {en-US}
- UI
- Model
- Converters
- Alarms
- Recipes
- Loggers
- DataStores
- Reports
- OPC-UA
- CommDrivers

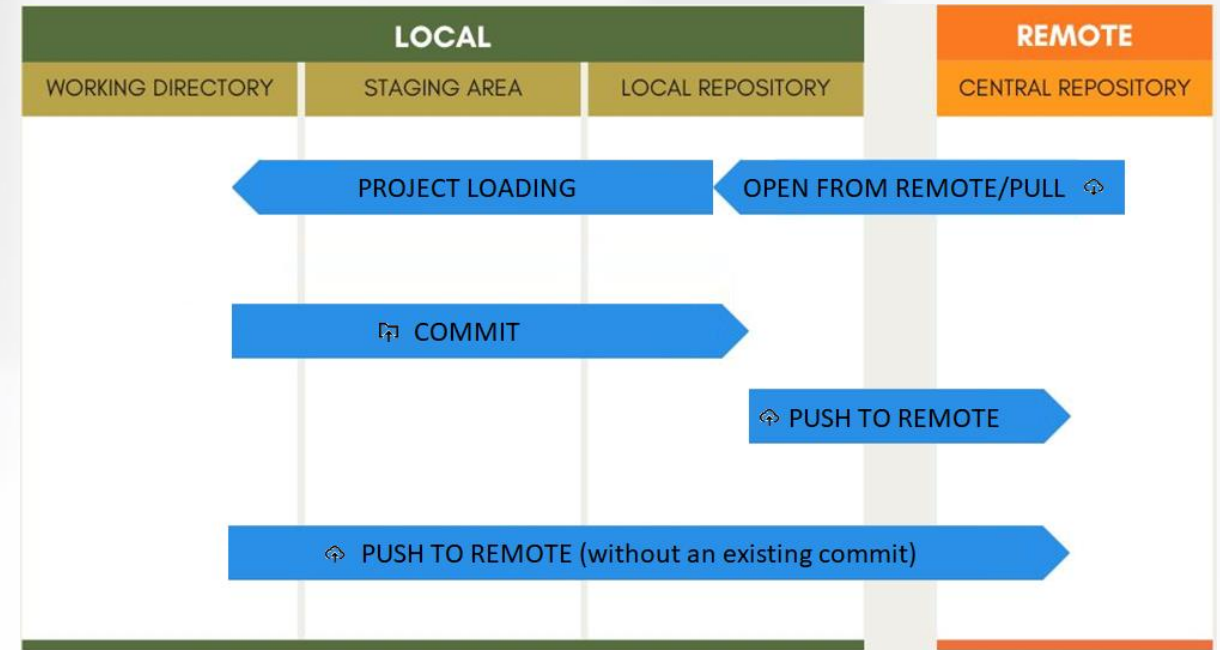
⚠ Commits that occurred after a restored version will be removed permanently. [Restore](#)

GIT workflow vs. Optix workflow

GIT Workflow



Optix Workflow



- Optix «hides» the GIT commands to improve the user experience
- User does not need to know anything about GIT commands and workflow

Collaboration tips

- Always configure/verify the GitHub access from Studio options
- General Git-related tips
 - Repository name **MUST** have the same name of Optix project
 - Some git providers does not allow non-alphanumerical characters
 - You can set the repository to be Private, then share with colleagues

The screenshot shows the 'FactoryTalk Optix Studio Options' dialog box with the 'Version control' tab selected. The 'Version control User Configuration' section has fields for 'User Name' and 'User Email'. The 'Version control providers and IDs' section has a list of providers on the left and a 'Provider identity login' area on the right. The providers listed are Azure DevOps (on-premise), Bitbucket (on-premise), GitHub, GitLab (cloud), and GitLab (on-premise). The 'Add identity' button is at the bottom of the list. The 'Provider identity login' area shows 'GitHub' selected with a green checkmark. At the bottom right are 'Save' and 'Cancel' buttons.

FactoryTalk Optix Studio Options

Optix Studio Version control

Version control User Configuration

User Name

User Email

Version control providers and IDs

Select a provider

- Azure DevOps (on-premise)
- Bitbucket (on-premise)
- GitHub
- GitLab (cloud)
- GitLab (on-premise)

Add identity

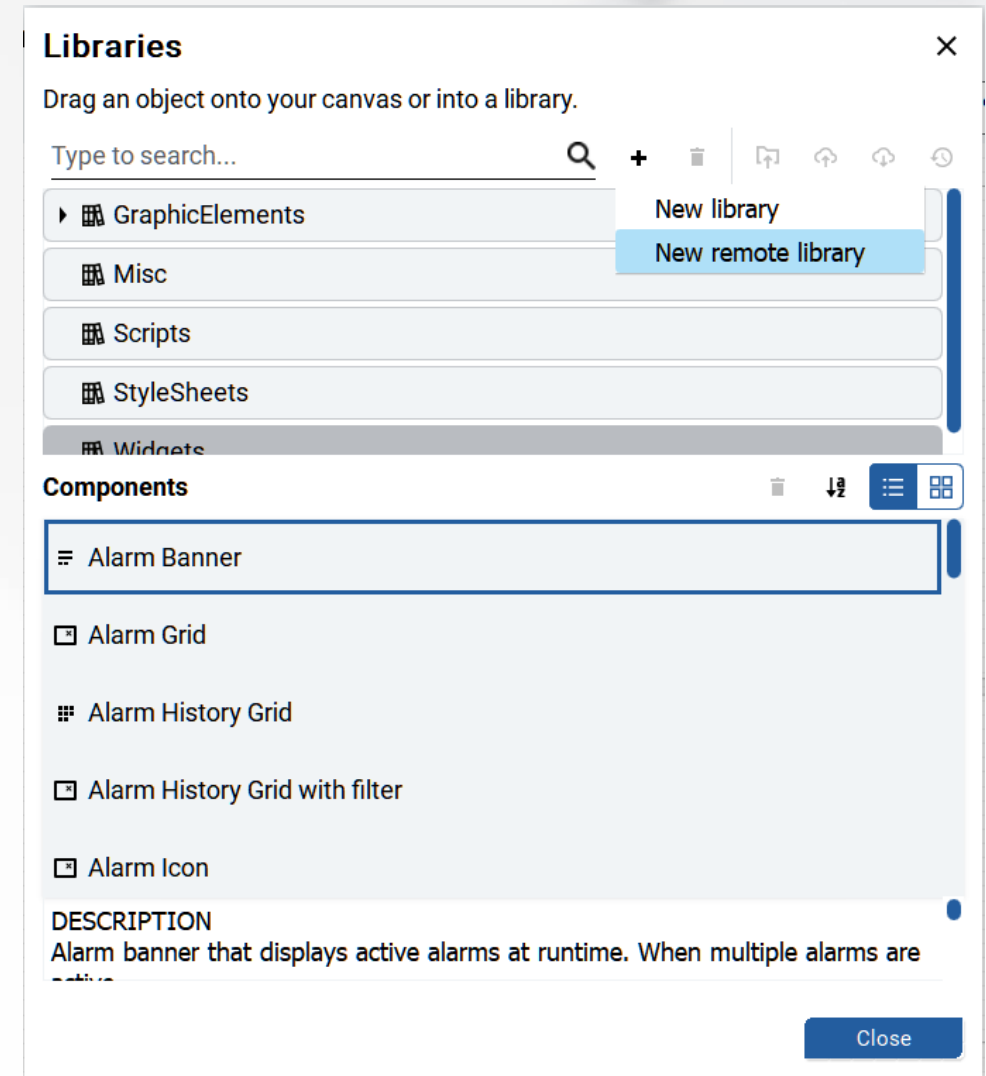
Provider identity login

GitHub ☒

Save Cancel

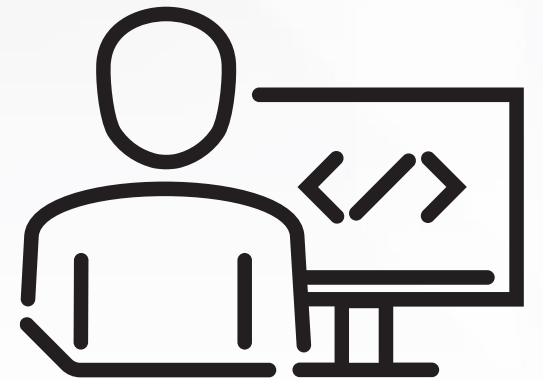
Sharing libraries using Version Control

- a Custom Library can be shared exactly the same as the projects
- a Shared Library is useful when the company has built his own graphical or widget library in the way all projects reuse the same basic components
- Libraries does not offer back compatibility, once somebody from the collaborators upgrades a library, everybody must upgrade the IDE to use it (same as projects)



Hands-on session

- Execute some Save and commit
- Explore the version history
- Try to restore to a previous state

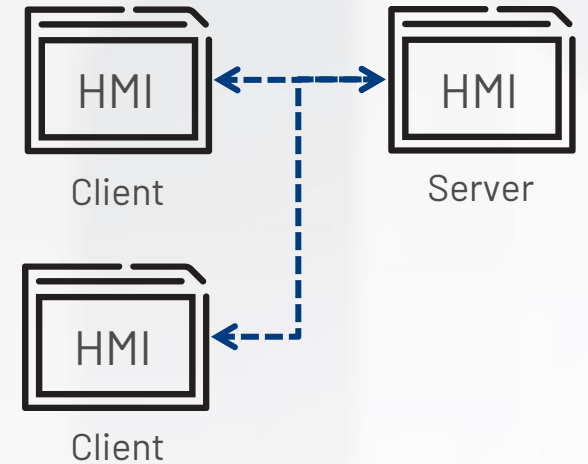


Define client-server architectures via OPC-UA



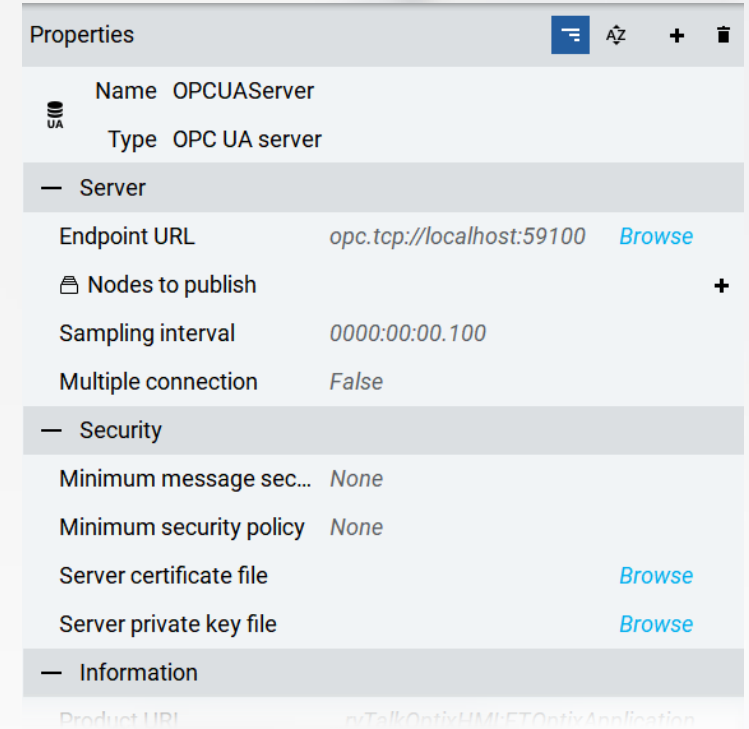
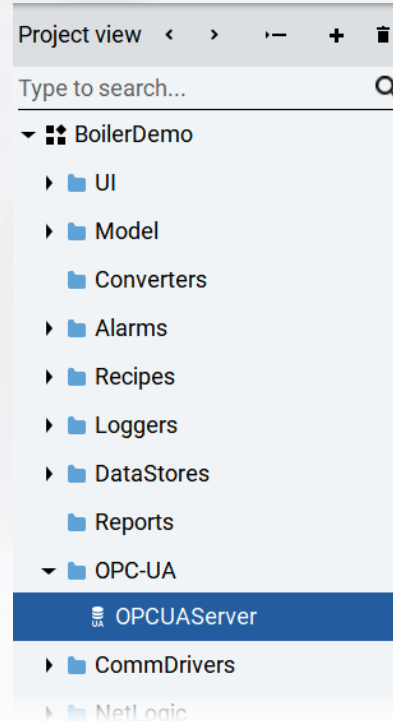
Define client-server architectures via OPC-UA

- Thanks to the Full-featured OPC-UA architecture of FactoryTalk Optix It's possible to define client-server architectures
- Clients will have access to:
 - Variables
 - Methods
 - Alarms and Historical Alarms
 - Datalogger and Eventlogger
 - Recipes
- Client: HMI project with OPC-UA Client (Full OPC-UA Client = 3 tokens)
- Server: HMI project with OPC-UA Server (1 client = 1 token, 2 or more clients = 3 tokens)



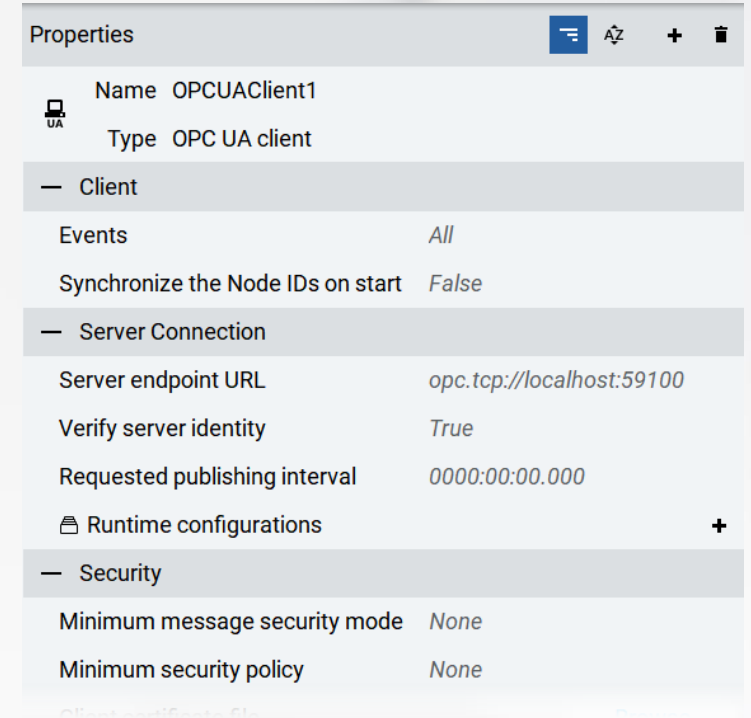
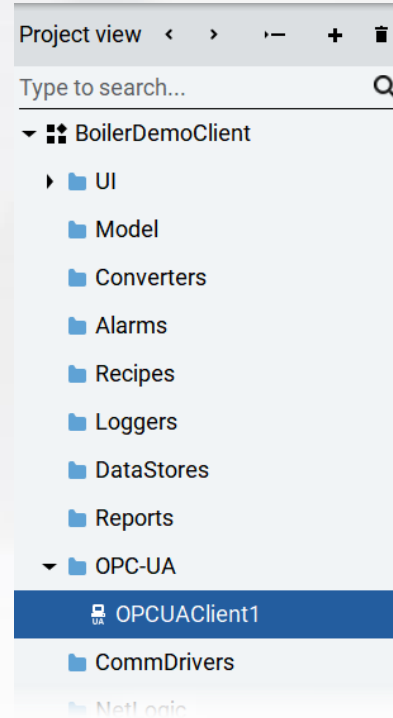
Configure the server

- On the HMI project, that is going to act as a Server, add the OPCUAServer node
- Endpoint can be manually changed
- Also Nodes to publish can be manually defined



Configure the client

- On the HMI project, that is going to act as a Client, add the OPCUAClient node
- Server endpoint must be the one defined into the Server project
- Events property should be set to "All" to be able to attach to Methods, Alarms, Loggers...



Configure the client

- Using the OPCUA Tag importer, it's possible to import not only variables but also objects!
- For example, select the entire folders Model, Alarms, Loggers and DataStores
- Elements like Alarms, Datalogger and AlarmsEventLogger can be used as they were part of the Client project

OPC-UA > OPCUAClient1 > **OPCUATagImporter**

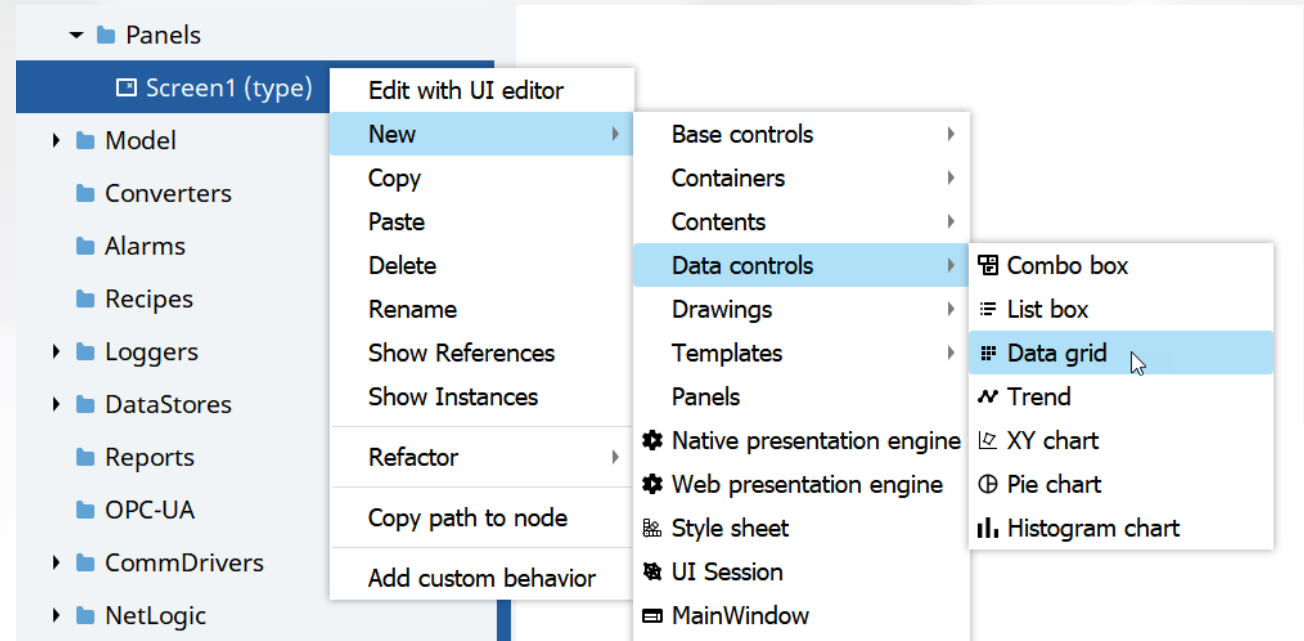
☐ Online Type to search... 🔍 ↺

▼ ☒ BoilerDemo	Project folder
☐ var BranchingEnabled	Variable Boolean
☐ var Locales	Variable String[0]
☐ var LocaleFallbackList	Variable String[0]
▶ ☐ UI	Folder
▶ ☒ Model	Folder
☐ Converters	Folder
▶ ☒ Alarms	Folder
▶ ☐ Recipes	Folder
▶ ☒ Loggers	Folder
▶ ☒ DataStores	Folder
☐ Reports	Folder
▶ ☐ OPC-UA	Folder
☐ var MeasurementSystemMap	Variable MeasurementSystem...
▶ ☐ PasswordPolicy	Password policy

Apply

Configure the client

- Using the OPCUA Tag importer, it's possible to import not only variables but also objects!
- For example, select the entire folders Model, Alarms, Loggers and DataStores
- Elements like Alarms, Datalogger and AlarmsEventLogger can be used as they were part of the Client project



Configure the client

- Using the OPCUA Tag importer, it's possible to import not only variables but also objects!
- For example, select the entire folders Model, Alarms, Loggers and DataStores
- Elements like Alarms, Datalogger and AlarmsEventLogger can be used as they were part of the Client project



Configure the client

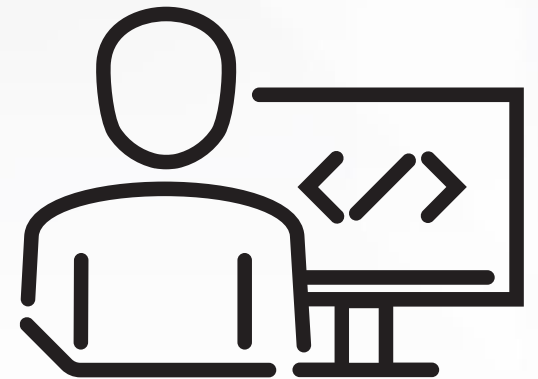
- Using the OPCUA Tag importer, it's possible to import not only variables but also objects!
- For example, select the entire folders Model, Alarms, Loggers and DataStores
- Elements like Alarms, Datalogger and AlarmsEventLogger can be used as they were part of the Client project

The screenshot displays the FactoryTalk configuration interface. On the left, the 'Project view' pane shows a hierarchical tree structure. Under the 'OPC-UA' folder, 'OPCUAClient1' is expanded, showing 'OPCUATagImporter' and 'Objects'. Under 'Objects', 'BoilerDemo' is expanded, showing 'Model', 'Alarms', 'Loggers', and 'DataStores'. The 'Loggers' folder is selected, showing 'DataLogger' and 'AlarmsEventLogger'. A red arrow points from 'AlarmsEventLogger' in the Project view to the 'AlarmsEventLogger' entry in the Properties panel on the right.

The 'Properties' panel on the right shows the configuration for 'FilteredAlarmHistoryGrid1'. The 'Name' is 'FilteredAlarmHistoryGrid1' and the 'Type' is 'Panel'. The 'AlarmsEventLogger' section is expanded, showing the 'Kind' as 'Event logger'. The 'From' and 'To' properties are both set to 'DateTime' and 'Not set'. The 'Appearance' section shows 'Visible' as 'True', 'Enabled' as 'True', 'Opacity' as '100', 'Rotation' as '0', and 'Hit test visible' as 'False'. The 'Size and layout' section shows 'Horizontal alignment' as 'Left', 'Vertical alignment' as 'Top', and 'Width' as '760'.

Hands-on session

- Use the BoilerDemo project as Server
- Create a new project that acts as a Client
- Try to visualize and manage the Server's Alarms, Alarm History and Datalogger from the Client project

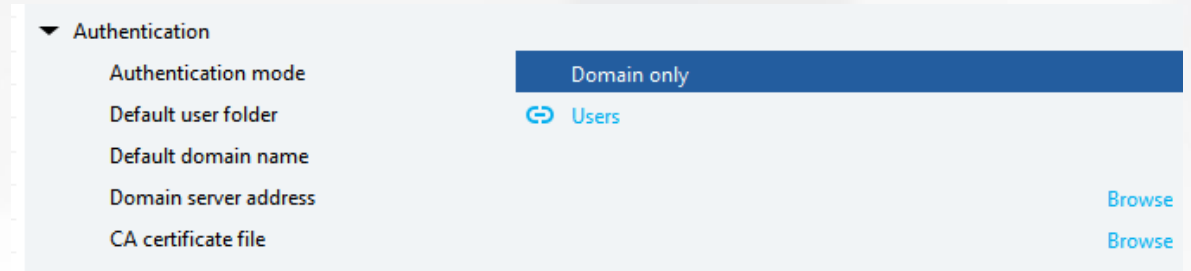


Domain authentication



Domain authentication

- Login can be forwarded to external services
 - Active Directory
 - LDAP with SSL (no LDAPS)
 - Oauth 2.0 (only with PKCE and JWKD endpoint)
- Need to configure the authentication server properties
 - Type
 - Address
 - Certificates
 - Tokens (for Oauth 2.0)

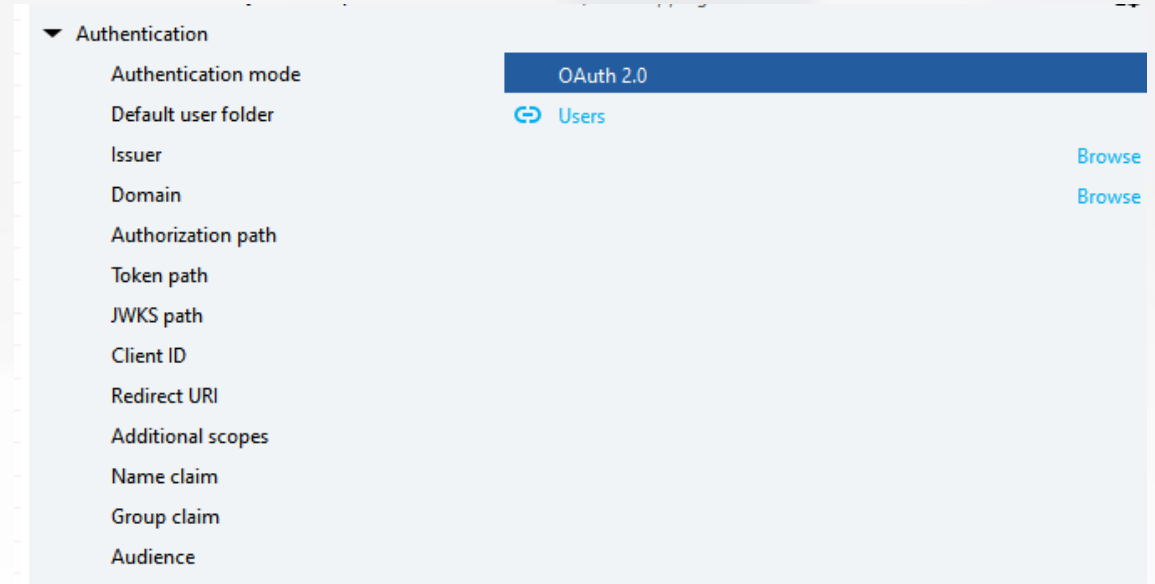


The screenshot shows a configuration window for authentication. On the left, a sidebar lists the following settings: Authentication mode, Default user folder, Default domain name, Domain server address, and CA certificate file. On the right, the 'Authentication mode' is set to 'Domain only' (highlighted in blue). Below it, 'Default user folder' is set to 'Users' with a blue link icon. The 'Domain server address' and 'CA certificate file' fields have 'Browse' buttons next to them.

Setting	Value
Authentication mode	Domain only
Default user folder	Users
Default domain name	
Domain server address	Browse
CA certificate file	Browse

Domain authentication

- Login can be forwarded to external services
 - Active Directory
 - LDAP with SSL (no LDAPS)
 - OAuth 2.0 (only with PKCE and JWKS endpoint)
- Need to configure the authentication server properties
 - Type
 - Address
 - Certificates
 - Tokens (for OAuth 2.0)

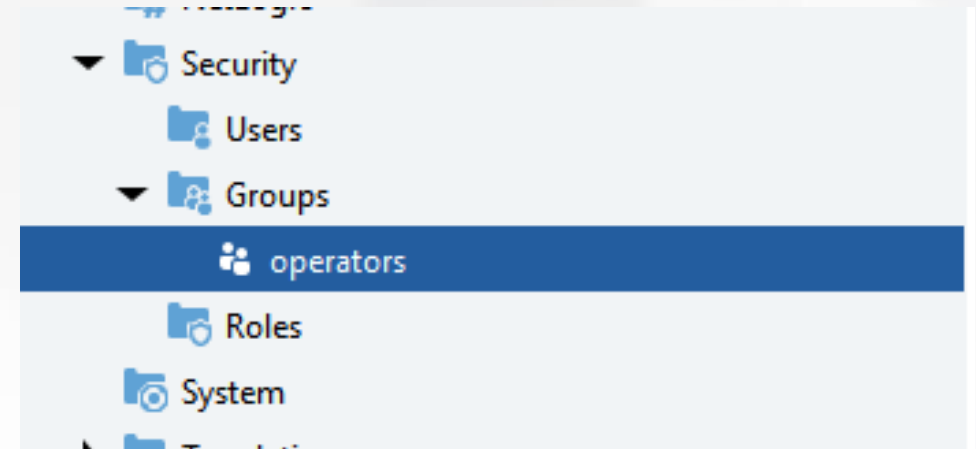


The screenshot shows the 'Authentication' configuration page in the FactoryTalk interface. The left sidebar lists various settings: Authentication mode, Default user folder, Issuer, Domain, Authorization path, Token path, JWKS path, Client ID, Redirect URI, Additional scopes, Name claim, Group claim, and Audience. The 'Authentication mode' is currently set to 'OAuth 2.0'. The 'Default user folder' is set to 'Users', with a 'Browse' button next to it. The 'Domain' field also has a 'Browse' button. The other fields are currently empty.

Property	Value
Authentication mode	OAuth 2.0
Default user folder	Users
Issuer	
Domain	
Authorization path	
Token path	
JWKS path	
Client ID	
Redirect URI	
Additional scopes	
Name claim	
Group claim	
Audience	

Domain authentication

- No need to create local users
 - If the authentication is validated by the server, a local user is automatically added to the project
- Login is always validated by the authentication server
 - No password caching can be done
 - Cannot login if the authentication server is unreachable
- FactoryTalk Optix groups are automatically mapped to the authentication server groups
 - Designer must have the list of groups in advance
 - When logging to the authentication server, the list of groups where the users belongs is returned, a mapping to the Optix user with the same name is performed automatically
 - For example, if "user1" is part of the "operators" group on the authentication server, all controls linked to the local "operators" group will be accessible



Domain authentication

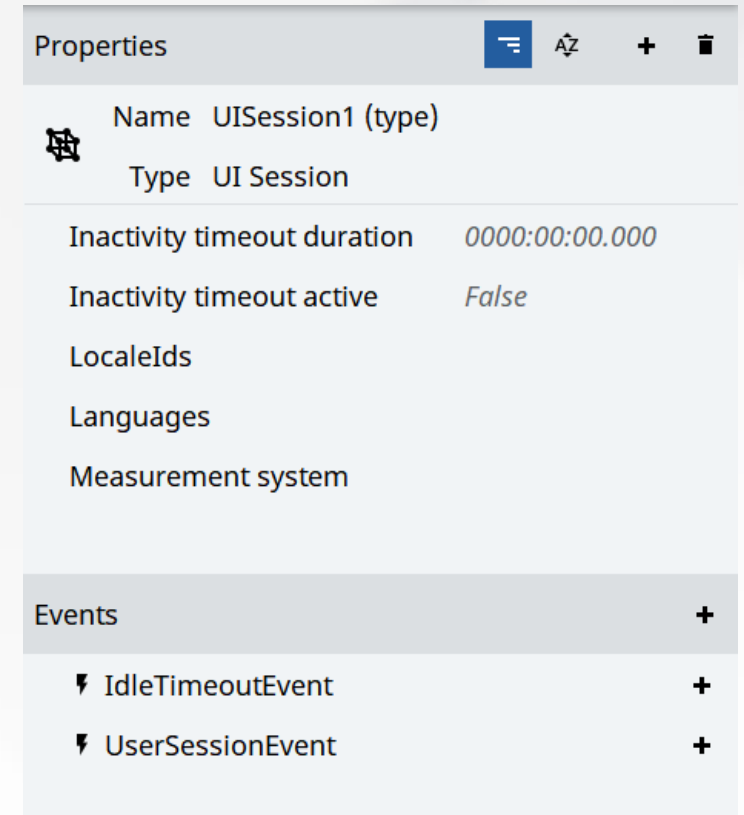
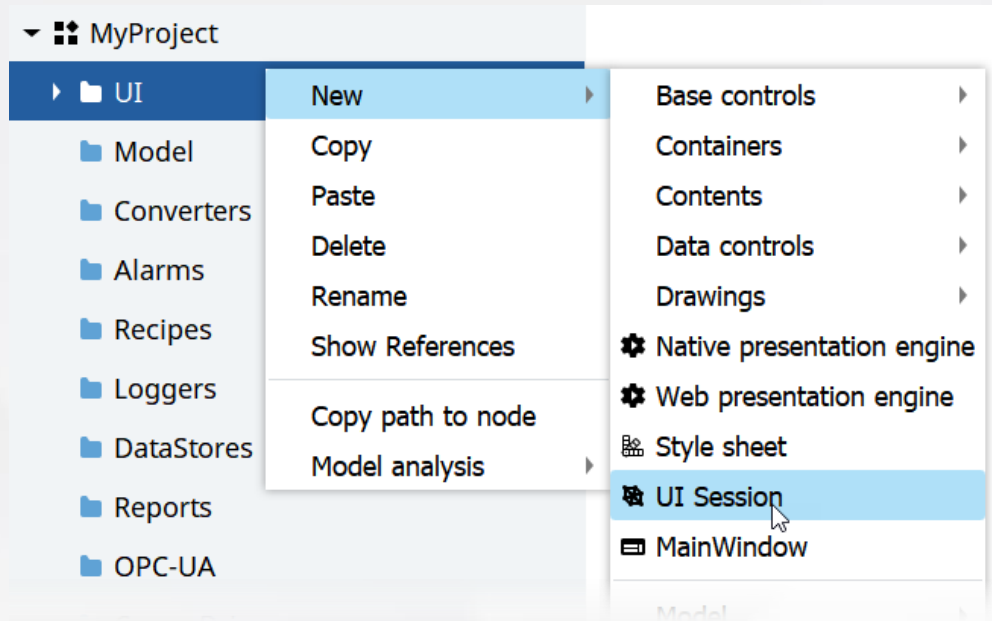
- If the designer does not have a list of the groups from the authentication server, roles can be used
 - All UI controls are tied to the Roles defined in the FactoryTalk Optix application
 - Groups can be created at runtime, then each group can be assigned to one or more roles during after deploying the project
- Full example is available in the GitHub organization
 - [Optix_Sample_AssignRolesToGroupsAtRuntime: How to associate Active Directory groups to FactoryTalk Optix roles at runtime](#)

The screenshot shows the 'RolesToGroups' configuration window in the FactoryTalk Optix designer. The window has two tabs: 'Permissions' and 'RolesToGroups', with 'RolesToGroups' being the active tab. On the left, there is a list of groups: 'operators' (highlighted in blue), 'users', and 'administrators'. Below this list is a 'Remove group' button. On the right, there is a section for 'Add new group' with a text input field containing 'administrators' and an 'Add' button. Below that is a section for 'Associated roles' with three checkboxes: 'Role1' (unchecked), 'Role2' (checked with a blue checkmark), and 'Role3' (unchecked). At the bottom right, there is an 'Apply' button.

Define UI Session and Session variables



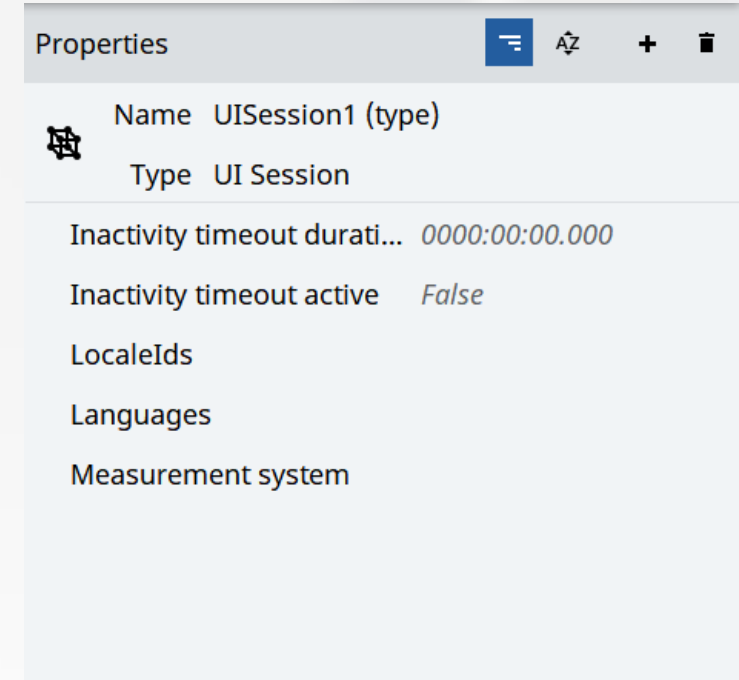
UI Session



- UI Session allows defining a custom Session to be used on Native or Web presentation engine
- Allow customizing: Inactivity Timeout, Locale, Language and Measurement System

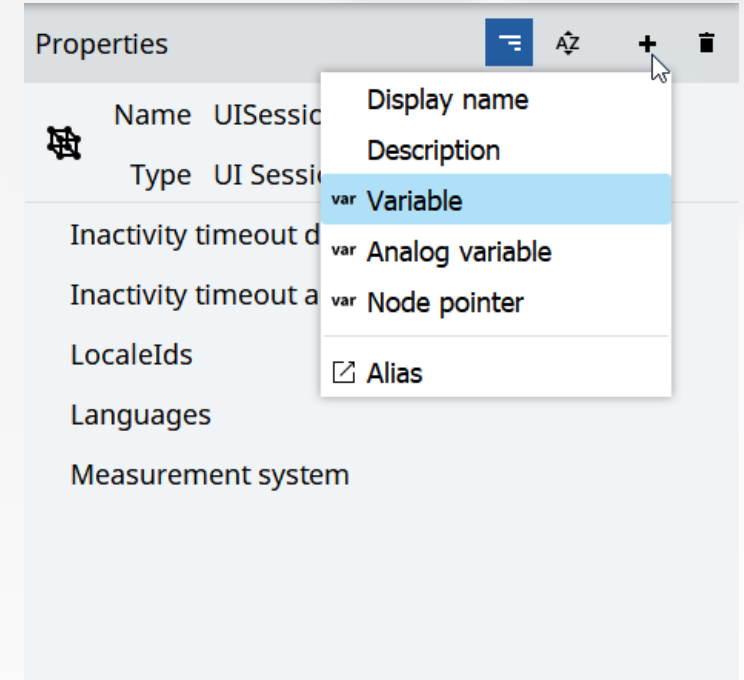
Session variables vs others

- Model/Comm. Driver variables
 - belongs to the Project context
 - display the same value regardless of sessions or panels
- Panel variables
 - belongs to the Panel context
 - different sessions show different values, but when you change the panel the value is lost
- Session variables
 - belongs to the Session context
 - different sessions show different values, but when you change the panel the value is retained



Session variables vs others

- Model/Comm. Driver variables
 - belongs to the Project context
 - display the same value regardless of sessions or panels
- Panel variables
 - belongs to the Panel context
 - different sessions show different values, but when you change the panel the value is lost
- Session variables
 - belongs to the Session context
 - different sessions show different values, but when you change the panel the value is retained

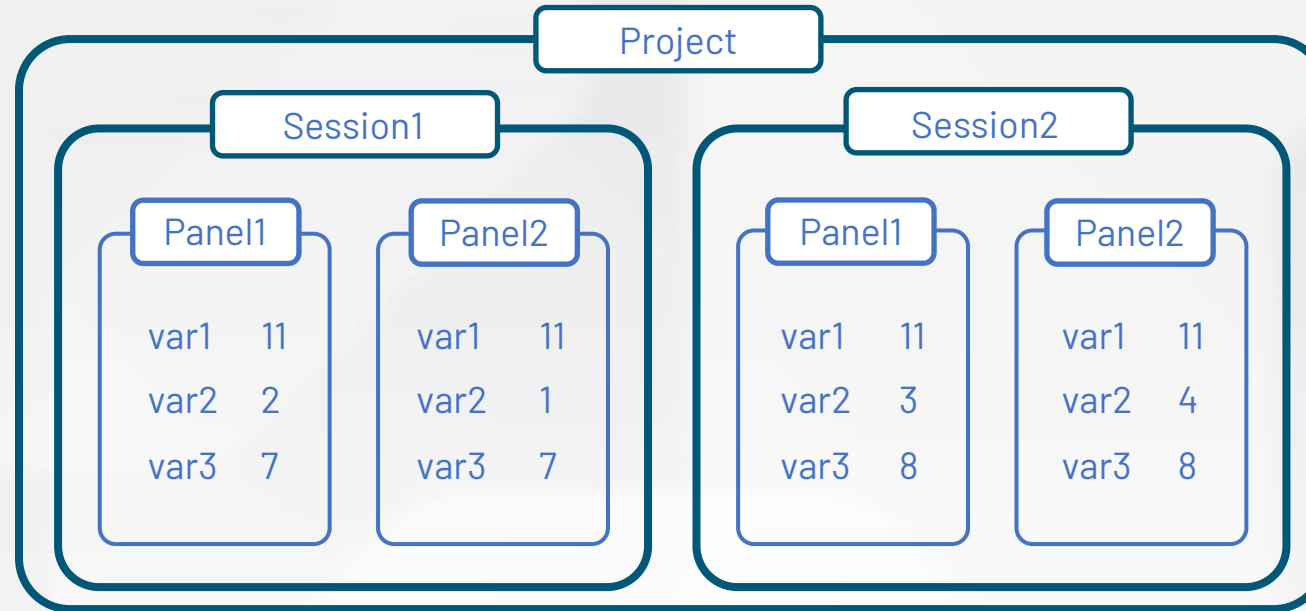


Session variables vs others

- Model/Comm. Driver variables
 - belongs to the Project context
 - display the same value regardless of sessions or panels
- Panel variables
 - belongs to the Panel context
 - different sessions show different values, but when you change the panel the value is lost
- Session variables
 - belongs to the Session context
 - different sessions show different values, but when you change the panel the value is retained

Properties		
	Name	UISession1 (type)
	Type	UI Session
Inactivity timeout durati...		0000:00:00.000
Inactivity timeout active		False
LocaleIds		
Languages		
Measurement system		
Session_Variable1		Int32 0

Session variables example



var1 = Model variable
var2 = Panel variable
var3 = Session variable

- Examples of different UI Sessions
 - Example1: one Session active on Native Presentation Engine, and another Session on the Web Presentation Engine
 - Example2: two browsers connected to the same Web Presentation Engine

Work with NetLogic scripts



.NET

- **What is .NET?**

- .NET is a free, open-source development platform

- **What is .NET Standard?**

- .NET Standard is an API specification that lets you develop class libraries for multiple implementations of .NET

- **What are .NET Framework, Mono and .NET Core?**

- They are formally known as *implementations* of .NET Standard
- .NET Framework is the original implementation of .NET (runs only on Windows)
- Mono is used when a small runtime is required (formerly used by UNIQO)
- .NET Core is the latest implementation and runs on any platform (used by OPTIX)

- **What is C#?**

- it's one of the programming languages supported by .NET

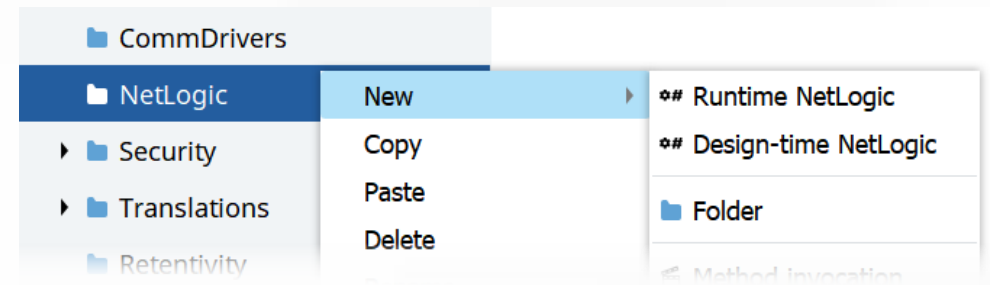
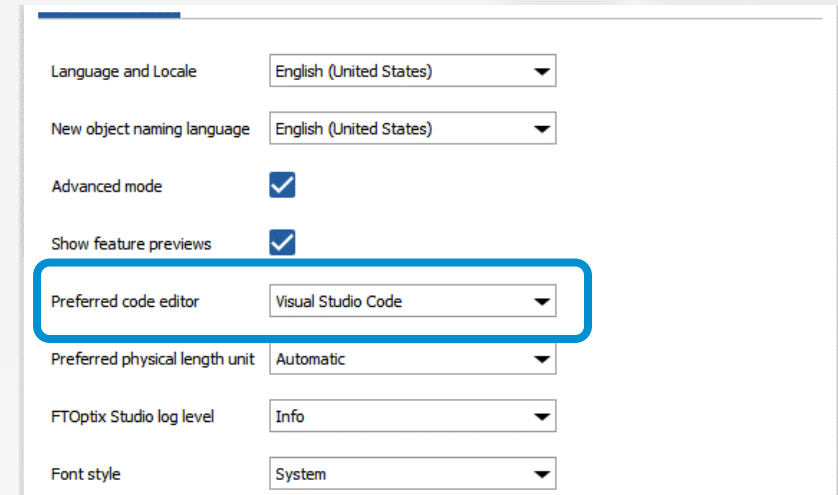
- **What .NET version does Optix support?**

- Depends on the FT Optix version, check requirements in the Install Guide
- Additional packages or libraries must be compatible with the specific .NET version (NuGet packages for example)

	Version	Start Date	End Date
Optix 1.5+	.NET 8 LTS	Nov 14, 2023	2026
	.NET 7	Nov 8, 2022	May 14, 2024
Optix 1.4	.NET 6.0 LTS	Nov 8, 2021	Nov 12, 2024
	.NET 5.0	Nov 10, 2020	May 10, 2022
	.NET 3.1 LTS	Dec 3, 2019	Dec 13, 2022
	.NET 3.0	Sep 23, 2019	Mar 3, 2020
	.NET 2.2	Dec 4, 2018	Dec 23, 2019
	.NET 2.1 LTS	May 30, 2018	Aug 21, 2021
	.NET 2.0	Aug 14, 2017	Oct 1, 2018
	.NET 1.1	Nov 16, 2016	Jun 27, 2019
	.NET 1.0	Jun 27, 2016	Jun 27, 2019

NetLogic

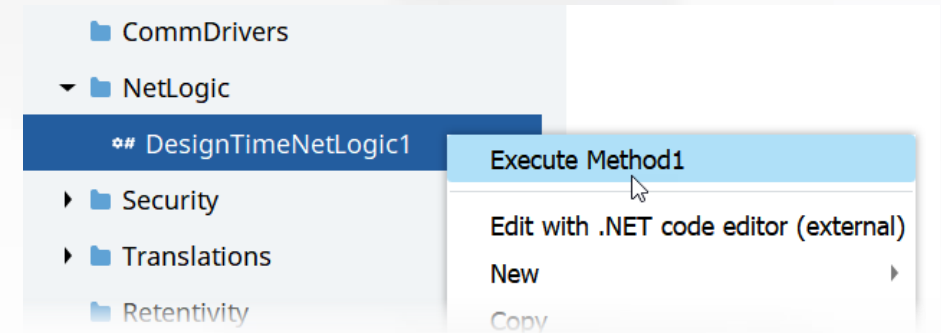
- It's an object that contains **C#** code compatible with the .NET version used by the current release of FactoryTalk Optix
- Supported NetLogic Editors:
 - Microsoft Visual Studio Code ([Download](#))
 - C# Extension needs to be installed
 - Microsoft Visual Studio 2022 (no previous versions)
- Types of NetLogic:
 - **Runtime**: to be executed at Runtime
 - **Design-time**: to be executed in IDE



Design-time NetLogic

- **Where** can be placed?
 - Anywhere!
But usually stay in NetLogic folder
- **When** is the method executed?
 - When the "exported method" is executed/called from Studio
- Notes on [ExportMethod]:
 - Allow exposing the method outside the script
 - Must be used before every method that need to be exposed
 - Exported methods, must be «public»
 - Methods must be void, output are passed as arguments using the «out» modifier

```
public class DesignTimeNetLogic1 : BaseNetLogic
{
    [ExportMethod]
    public void Method1()
    {
        // Insert code to be executed by the method
    }
}
```



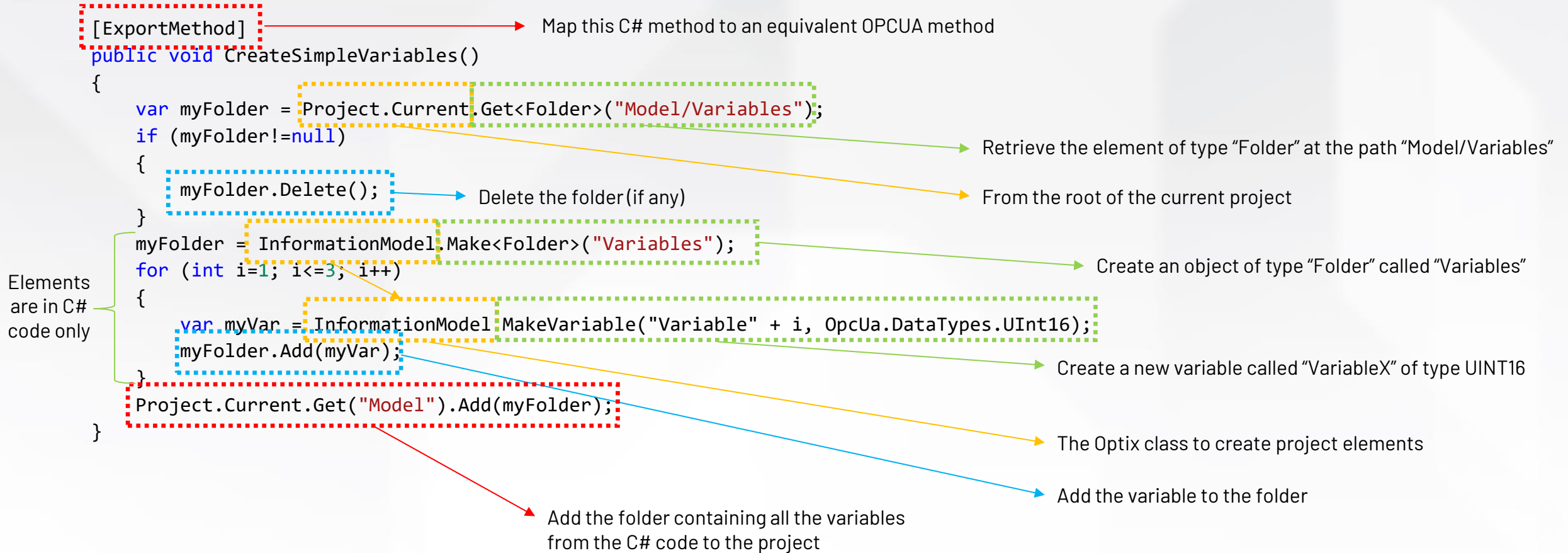
Design-time NetLogic: create simple variables

```
[ExportMethod]
public void CreateSimpleVariables()
{
    var myFolder = Project.Current.Get<Folder>("Model/Variables");
    if (myFolder!=null)
    {
        myFolder.Delete();  Delete the folder(if any)
    }
    myFolder = InformationModel.Make<Folder>("Variables");
    for (int i=1; i<=3; i++)
    {
        var myVar = InformationModel.MakeVariable("Variable" + i, OpcUa.DataTypes.UInt16);
        myFolder.Add(myVar);
    }
    Project.Current.Get("Model").Add(myFolder);
}
```

TIP: Optix uses some custom APIs to create and manage project elements of any kind.
Do not use the `var myVar = new IUAVariable();` syntax from plain C# as it will not work, it will only create an useless blank class

TIP: When creating multiple elements under the same parent folder, a best practice is to create the container, add the element and finally add the container to the project. This will avoid executing multiple calls to the FactoryTalk Optix information model

Design-time NetLogic: create simple variables APIs overview



Design-time NetLogic: create Motor Type and instances

```
[ExportMethod]
public void CreateMotorType()
{
    // Create an UAObject and call it MotorType
    var motorType = InformationModel.MakeObjectType("MotorType");
    // Create some properties for the motor object
    var speed = InformationModel.MakeVariable("Speed", OpcUa.DataTypes.UInt16);
    var running = InformationModel.MakeVariable("Running", OpcUa.DataTypes.Boolean);
    // Attach the properties of the motor to the motor object
    motorType.Add(speed);
    motorType.Add(running);
    // Attach the motor type (with the child properties) to the project
    Project.Current.Get("Model").Add(motorType);
}

[ExportMethod]
public void CreateMotorInstances()
{
    // Create a new folder
    var motorInstancesFolder = InformationModel.Make<Folder>("MotorInstances");
    // Create some motors and add them to the new folder
    for (int i = 1; i <= 3; i++)
    {
        var motorType = Project.Current.Get("Model/MotorType");
        var motorInstance = InformationModel.MakeObject("Motor" + i, motorType.NodeId);
        motorInstancesFolder.Add(motorInstance);
    }
    // Attach the folder (with the motors) to the current project
    Project.Current.Get("Model").Add(motorInstancesFolder);
}
```

Design-time NetLogic: create widget instances

[ExportMethod]

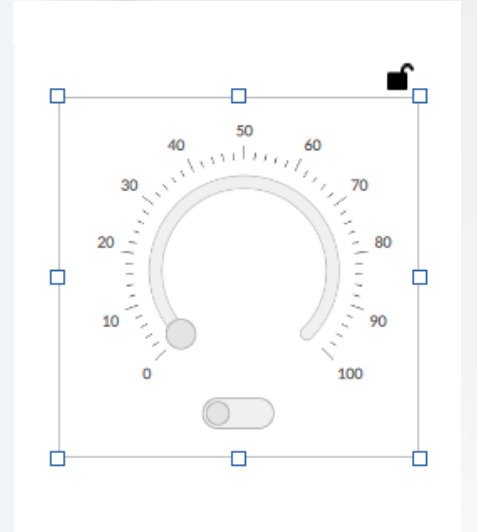
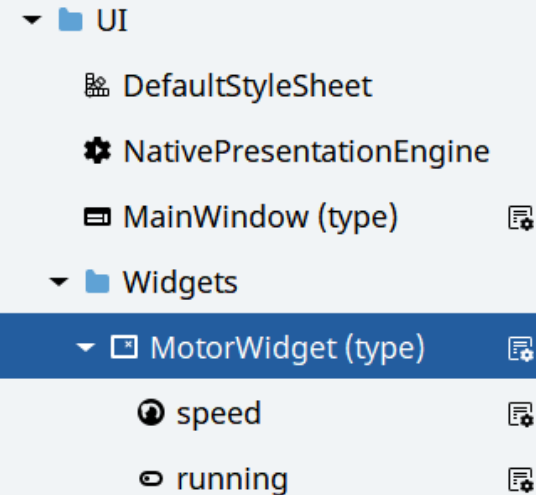
```
public void CreateMotorWidgetInstances()
{
```

```
    // NOTE: the "MotorWidget" type with "MotorAlias" alias,
    // must be already available at UI/Widgets
```

```
    for (int i = 1; i <= 3; i++)
    {
```

```
        var motorWidgetType = Project.Current.Get("UI/Widgets/MotorWidget");
        var motorWidgetInstance = InformationModel.MakeObject("MotorWidget" + i, motorWidgetType.NodeId);
        motorWidgetInstance.SetAlias("MotorAlias", Project.Current.Get("Model/MotorInstances/Motor" + i));
        ((Panel)motorWidgetInstance).LeftMargin = ((Panel)motorWidgetInstance).Width + 50 * (i - 1);
        Project.Current.Get("UI/MainWindow").Add(motorWidgetInstance);
    }
```

```
}
```



Cast to Panel is needed as the MakeObject would return a generic OPC-UA Object which does not contain the margin properties

Design-time NetLogic: script parameters

- Variables added to the NetLogic object behaves like script parameters
- Value of these variables can be read using `LogicObject.GetVariable("{variable}").Value`

```
[ExportMethod]
public void ReadParameters()
{
    string varName = LogicObject.GetVariable("Variable_Name").Value;
    string folderName = LogicObject.GetVariable("Folder_Name").Value;
    Log.Info(LogicObject.BrowseName, "Variable_Name = " + varName);
    Log.Info(LogicObject.BrowseName, "Folder_Name = " + folderName);
}
```

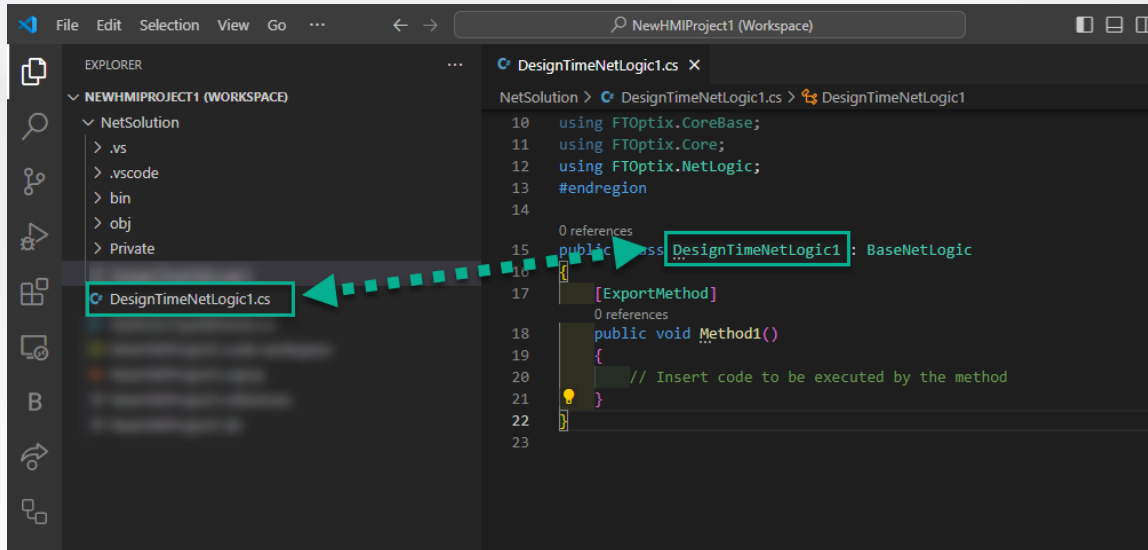
Properties			
Name DesignTimeNetLogic1			
Type NetLogic			
Variable_Name	String	MyVariable	
Folder_Name	String	MyFolder	

Access node path

- Using `Log.Node(IUANode)`; a string is returned containing the project path to such node

```
var myNode = Project.Current.Get("Model/Variable1");  
var nodePath = Log.Node(myNode); // returns /Objects/Model/Variable1
```

NetLogic don'ts !



1. NetLogic class name must be the same of the NetLogic object name

- **Don't change the name from the source code!**
- Rename it from Studio instead!

2. NetLogic objects are also browsable from the NetSolution

- **Don't delete a NetLogic from the solution!**
- Remove it from Studio instead!

Cast vs Type parameter

- Use of a **"type parameter"**

- Type parameter can be used with "generic methods" that allows using different types
- A generic method can be called either without an argument or by specifying the type of argument within angle brackets
- Returns null if MyFolder is not an instance of Folder

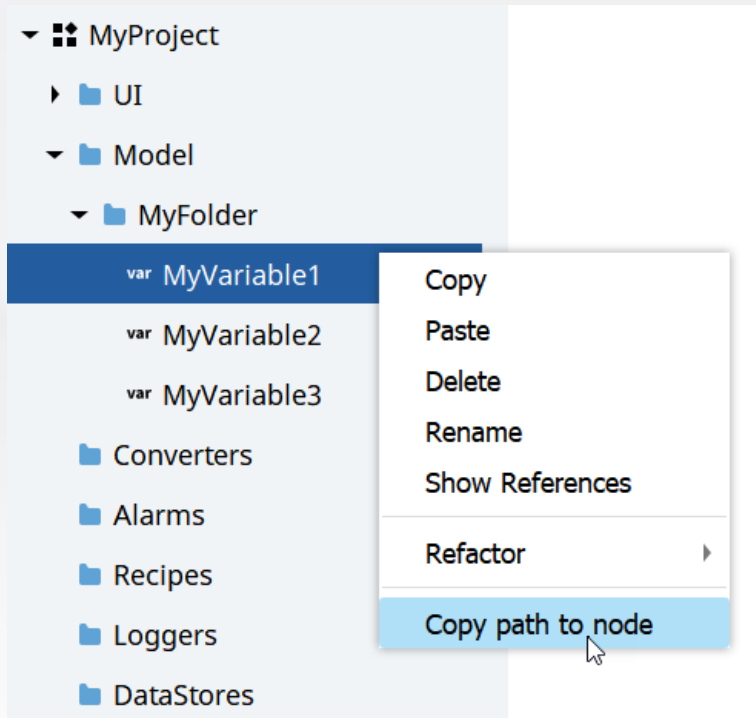
```
var myfolder = Project.Current.Get<Folder>("Model\MyFolder");
```

- Use of a **"cast"**

- a cast is an explicit conversion
- To perform a cast, specify the type that you are casting to in parentheses in front of the value or variable to be converted.
- Throws InvalidCastException if MyFolder cannot be assigned to Folder type

```
var myfolder = (Folder)Project.Current.Get("Model\MyFolder");
```

Copy path to node



- Can be useful when using absolute addressing with `Project.Current.Get`
- Provides the absolute path of the node to clipboard

`"MyProject/Model/MyFolder/MyVariable1"`

```
Project.Current.GetVariable("Model/MyFolder/MyVariable1");
```

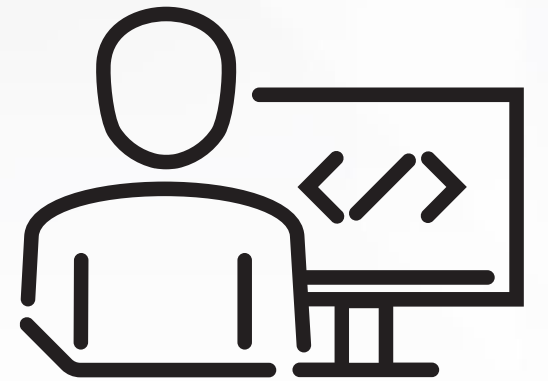
- NOTE: remember to remove the project name

NetLogic(s) FAQ

- Can I add ActiveX controls to my FactoryTalk Optix application?
 - No, you can't add ActiveX to FactoryTalk Optix, this is due to safety concerns (deprecated technology)
- I have my custom WPF application, can I embed it into FactoryTalk Optix?
 - No, graphics can only come from FactoryTalk Optix UI, there's no chance to embed other applications
 - If the custom application exposes a HTML interface, the WebBrowser object can be used
- Can I use any NuGet package in my FactoryTalk Optix application?
 - Yes, if they are compatible with the .NET version used in FactoryTalk Optix
 - Yes, if they are cross-platform capable (compiler target must be AnyCPU)
 - Exceptions are:
 - Windows Forms (WPF and derivatives are not cross-platform and cannot be used)
 - Microsoft.SQL (customers should use ODBC connector instead) → Limitation was removed in Optix 1.6.X
- Any limitations in a DesignTime script?
 - There is no way to access value of PLC tags (CommDriver only exist at Runtime)
 - There is no way to interact with ODBC connectors, EmbeddedDatabase or InfluxDB using Optix modules

Hands-on session

- Write one or more Design-Time script
- See what happens if multiple variables are created with the same name



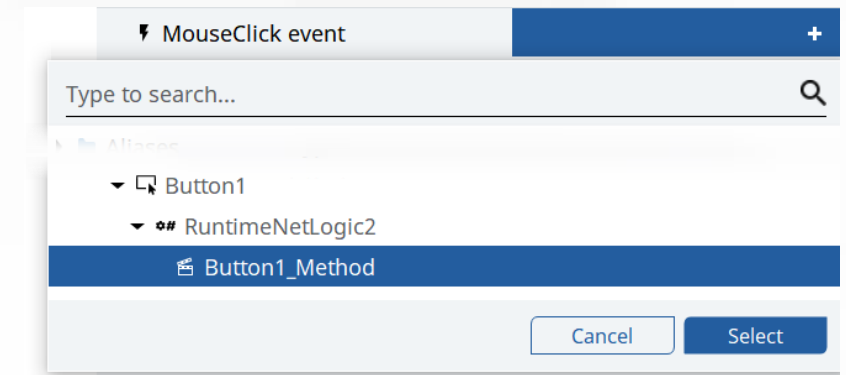
Runtime Netlogic

- **Where** can it be placed?
 - into the NetLogic folder
 - into the Main Window
 - into a Panel
 - into a graphical object (like a Button)
- **When** are the methods executed?
 - *Start*: automatically executed when the node that contain the NetLogic is created/opened
 - *Stop*: automatically executed when the node that contain the NetLogic is destroyed/closed
 - *Exported Method*: executed when called by an Event

```
public class RuntimeNetLogic1 : BaseNetLogic
{
    public override void Start()
    {
        // code executed when the logic is started
    }

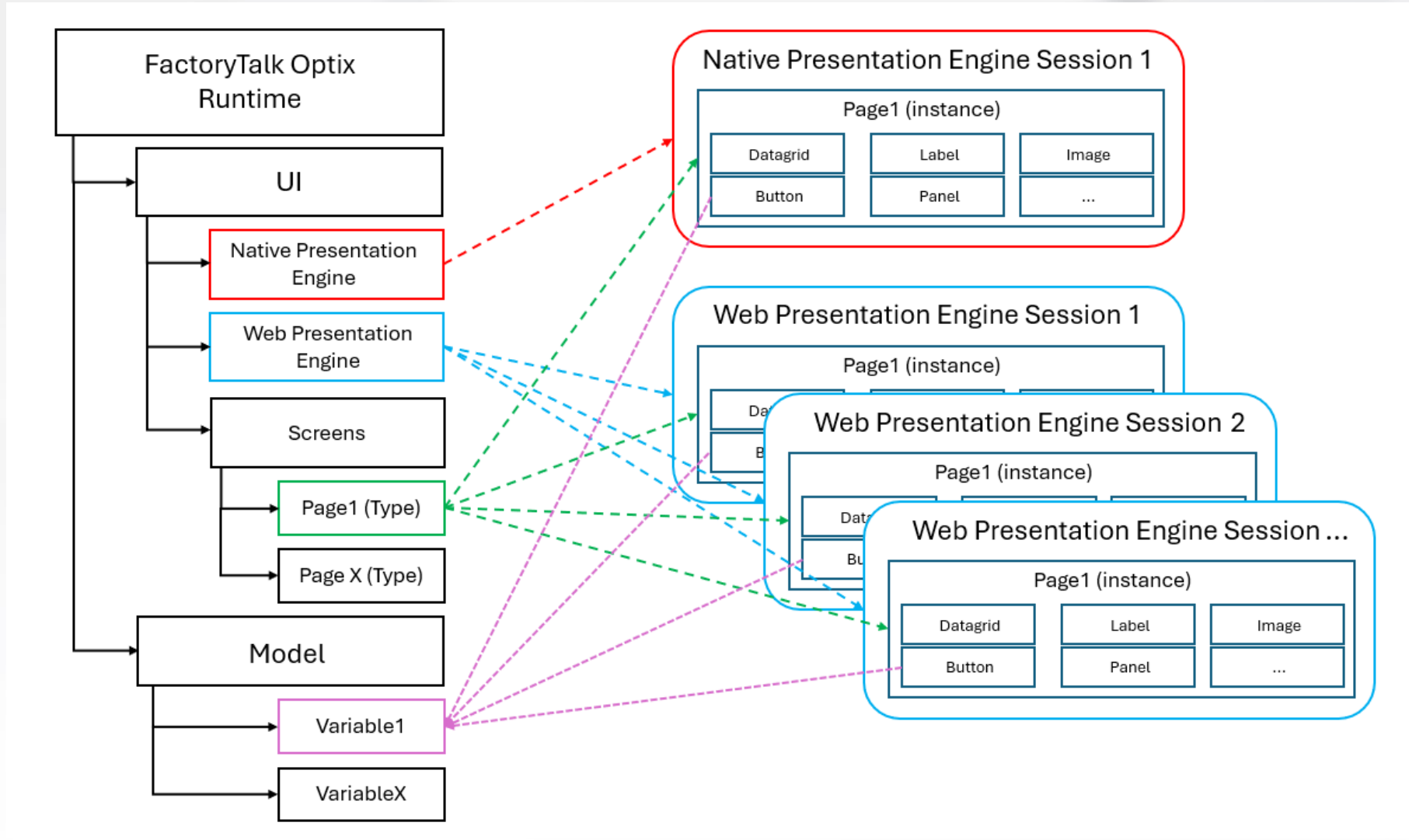
    public override void Stop()
    {
        // code executed when the logic is stopped
    }

    [ExportMethod]
    public void Button1_Method()
    {
        // code executed by the method
    }
}
```



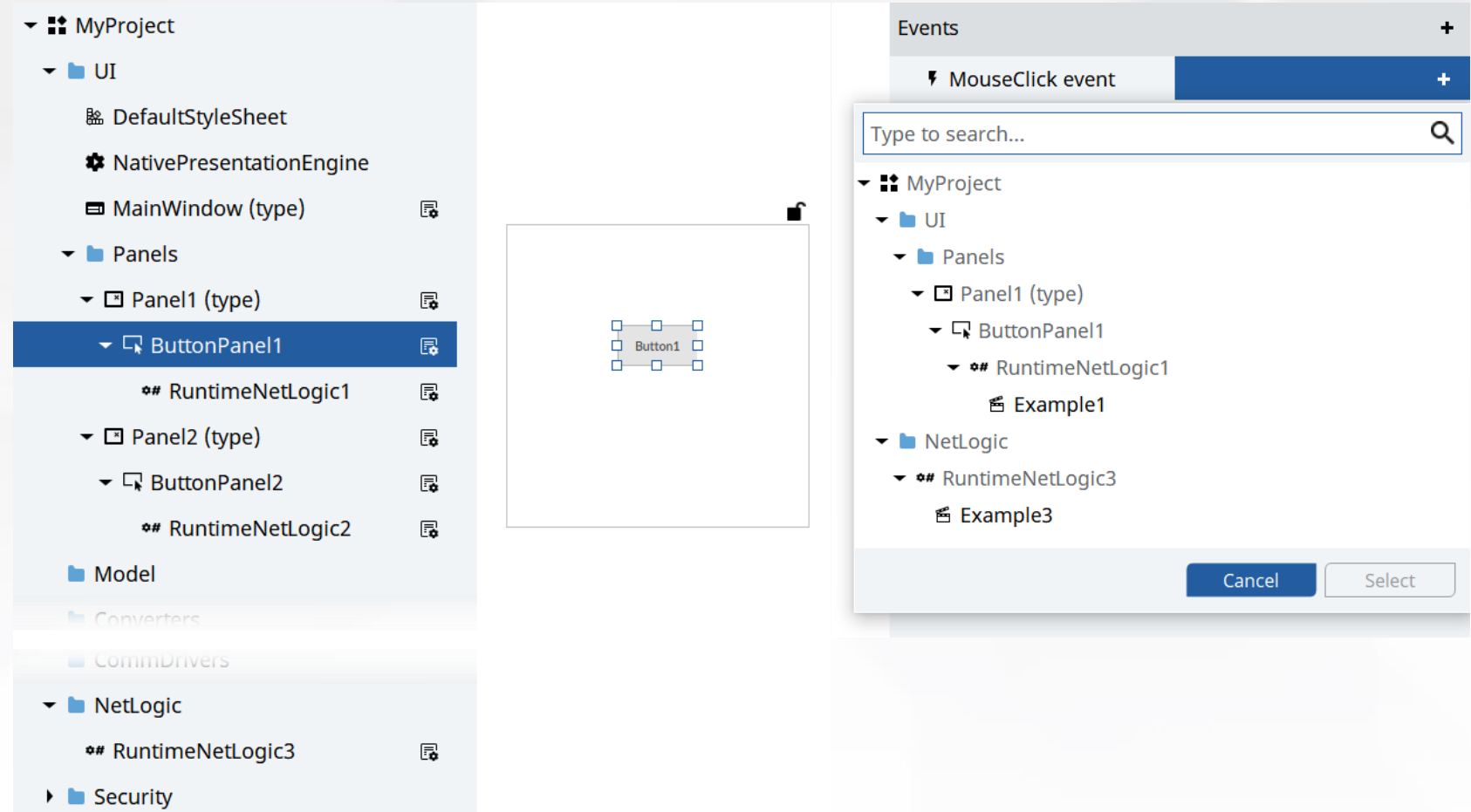
Runtime NetLogic: scope

- Every session of the Runtime will create an instance of the UI type that was designed (session)
- Elements of a session can only interact with elements from the same session or global scope
- Elements from a global scope cannot interact with elements of a specific session



Runtime NetLogic: scope

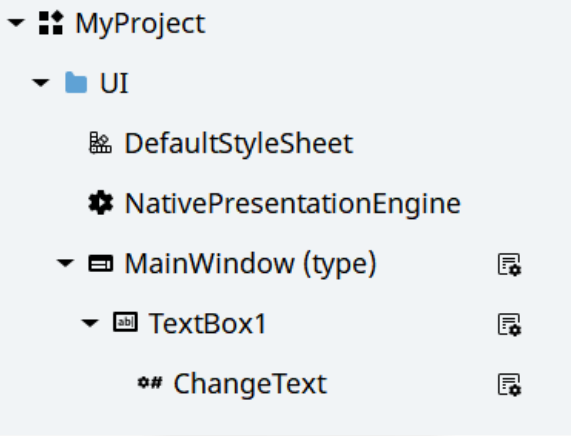
- Button1 of Panel1 cannot access method exposed by RuntimeNetLogic2 because it is related to Panel2 (different type and different scope).
- Panel2 and therefore RuntimeNetLogic2 do not exist when Button1 is loaded
- Dynamic links from a global scope (model variables or PLC tags) cannot point to objects from a session scope



Runtime NetLogic: usage of "owner"

```
public class ChangeText : BaseNetLogic
{
    [ExportMethod]
    public void ChangeTextRelativePath()
    {
        // WORKING AT RUNTIME!
        Owner.GetVariable("Text").Value = "Text changed using Owner";
    }

    [ExportMethod]
    public void ChangeTextAbsolutePath()
    {
        // NOT WORKING AT RUNTIME!
        Project.Current.Get<TextBox>("UI/MainWindow/TextBox1").Text = "Text changed using Project.Current";
    }
}
```



NetLogic is a child of TextBox1, so Owner of the NetLogic is the TextBox

	Project.Current...	Owner...
Type of path	Absolute	Relative
Runtime usage	can be used to refer nodes <u>outside the UI</u> *	must be used to refer nodes <u>inside the UI</u> *

* This is due to the UI Sessions, dynamically created at Runtime

Runtime NetLogic: read/write variables value

The screenshot displays the Rockwell Automation Studio interface for a NetLogic project. On the left, the project tree shows 'Panel1 (type)' with controls like Button1, SpinBox1, Label1, SpinBox2, Label2, SpinBox3, and a Calculate button. Below this is a 'Model' folder containing three variables: Variable1, Variable2, and Variable3. The center pane shows a visual representation of a calculator interface with two input fields, an equals button, and a 'sum' button. The right pane shows the properties for 'Button1', including appearance and events.

Name	Type
Button1	Button

Appearance	
Visible	True
Enabled	True
Opacity	100

Events	
MouseDown event	+
Method	../..../Calculate/Sum
MouseDown event	+
MouseUp event	+

```
[ExportMethod]
public void Sum()
{
    var addend1 = (Int32)Project.Current.GetVariable("Model/Variable1").Value;
    var addend2 = (Int32)Project.Current.GetVariable("Model/Variable2").Value;
    Project.Current.GetVariable("Model/Variable3").Value = addend1 + addend2;
}
```

Runtime NetLogic: subscribe to VariableChange event

```
private IUAVariable addend1;
private IUAVariable addend2;

public override void Start()
{
    addend1 = Project.Current.GetVariable("Model/Variable1");
    addend2 = Project.Current.GetVariable("Model/Variable2");
    addend1.VariableChange += addend1_VariableChange;
    addend2.VariableChange += addend2_VariableChange;
}

private void addend1_VariableChange(object sender, VariableChangeEventArgs e)
{
    Project.Current.GetVariable("Model/Variable3").Value = (Int32)e.NewValue + (Int32)addend2.Value;
}

private void addend2_VariableChange(object sender, VariableChangeEventArgs e)
{
    Project.Current.GetVariable("Model/Variable3").Value = (Int32)addend1.Value + (Int32)e.NewValue;
}

public override void Stop()
{
    addend1.VariableChange -= addend1_VariableChange;
    addend2.VariableChange -= addend2_VariableChange;
}
```

This example executes the Sum when an addend changes, without the need to click the Button

Runtime NetLogic: set a variable always in use

```
private IUAVariable addend1;
private IUAVariable addend2;
private RemoteVariableSynchronizer variableSynchronizer;

public override void Start()
{
    addend1 = Project.Current.GetVariable("Model/Variable1");
    addend2 = Project.Current.GetVariable("Model/Variable2");
    addend1.VariableChange += addend1_VariableChange;
    addend2.VariableChange += addend2_VariableChange;

    variableSynchronizer = new RemoteVariableSynchronizer();
    variableSynchronizer.Add(addend1);
    variableSynchronizer.Add(addend2);
}

private void addend1_VariableChange(object sender, VariableChangeEventArgs e)
{
    Project.Current.GetVariable("Model/Variable3").Value = (Int16)e.NewValue + (Int16)addend2.Value;
}

private void addend2_VariableChange(object sender, VariableChangeEventArgs e)
{
    Project.Current.GetVariable("Model/Variable3").Value = (Int16)addend1.Value + (Int16)e.NewValue;
}
```

Starting from the previous example, here Variable1 and Variable2 are associated with Modbus registers, but they are not in use (because not linked to any object on page) so they need to be added to a RemoteVariableSynchronizer

Runtime NetLogic: execute code in asynchronous mode *

```
public override void Start()
{
    variable1 = Project.Current.GetVariable("Model/Variable1");
    myPeriodicTask = new PeriodicTask(IncrementVariable, 250, LogicObject);
    myPeriodicTask.Start();
    dCounter = 0;
}

private void IncrementVariable(PeriodicTask task)
{
    dCounter = dCounter + 0.05;
    variable1.Value = 50 + Math.Sin(dCounter) * 50;
}

public override void Stop()
{
    myPeriodicTask.Dispose();
}

private PeriodicTask myPeriodicTask;
private IUAVariable variable1;
private double dCounter;
```

* The asynchronous classes available are:

PeriodicTask to create a task that executes code at regular time intervals.

DelayedTask to create a task that executes code after a time delay.

LongRunningTask to create tasks that require significant time and/or CPU resources.

The default "async" methods in C# cannot be used to access project nodes (to avoid concurrency issues), use the FactoryTalk Optix classes from above instead

Runtime NetLogic: execute DB query

```
private Store myStore;  
private Table myTable;  
private object[,] resultSet;  
private string[] header;  
  
public override void Start()  
{  
    myStore = Project.Current.Get<Store>("DataStores/EmbeddedDatabase1");  
    myTable = myStore.Tables.Get<Table>("Table1");  
}  
  
[ExportMethod]  
public void Insert(string Lot, string Prod, uint Quantity)  
{  
    object[,] rawValues = new object [1,3];    // insert 1 row with 3 columns  
    rawValues[0,0] = Lot;  
    rawValues[0,1] = Prod;  
    rawValues[0,2] = Quantity;  
  
    string[] columns = new string[3] {"Lot","Product","Quantity"};  
    myTable.Insert(columns,rawValues);  
}
```

Properties

Name	EmbeddedDatabase1
Type	Embedded database

— Configuration

Filename	
In memory	False

— Structure

▼ Tables

▼ Table1

Record limit	0
▼ Columns	
Lot	String
Product	String
Quantity	UInt32 0

Runtime NetLogic: execute DB query

```
[ExportMethod]
public void Delete(string Lot)
{
    myStore.Query($"DELETE FROM Table1 WHERE Lot='{Lot}'", out header, out resultSet);
}
```

```
[ExportMethod]
public void Update(string Lot, uint Quantity)
{
    myStore.Query($"UPDATE Table1 SET Quantity='{Quantity}' WHERE Lot='{Lot}'", out header, out resultSet);
}
```

```
[ExportMethod]
public void Select(string Lot)
{
    myStore.Query($"SELECT Product, QUantity FROM Table1 WHERE Lot='{Lot}'", out header, out resultSet);
    for (int i = 0; i < resultSet.GetLength(0); i++)
    {
        Log.Info(LogicObject.BrowseName, $"Product = '{resultSet[i,0]}' - Quantity = '{resultSet[i,1]}'");
    }
}
```

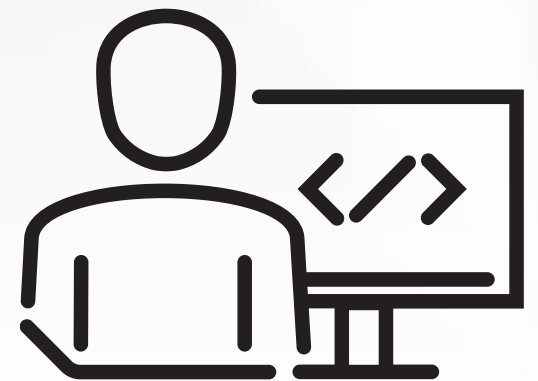
NetLogic: look at some library scripts

- MQTT push-agent (Runtime)
- FTP Client/Server (Runtime)
- ClockLogic (Runtime)
- Import/Export Alarm (Design-time)
- Import/Export Translations (Design-time)

Most Design-time scripts from TemplateLibrary can be modified to be used at Runtime too!

Hands-on session

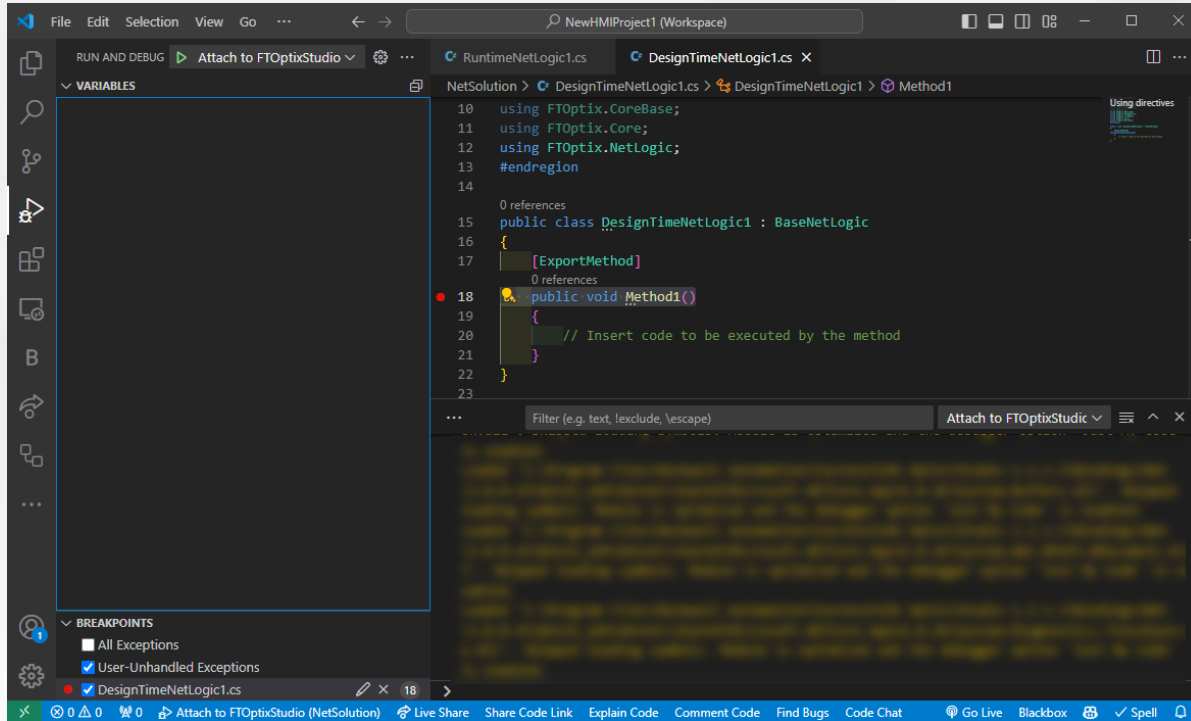
- Write one or more Runtime script
- See what happens when adding an element to a page using `Project.Current()` vs. `Owner.Get()`



Debugging NetLogic scripts



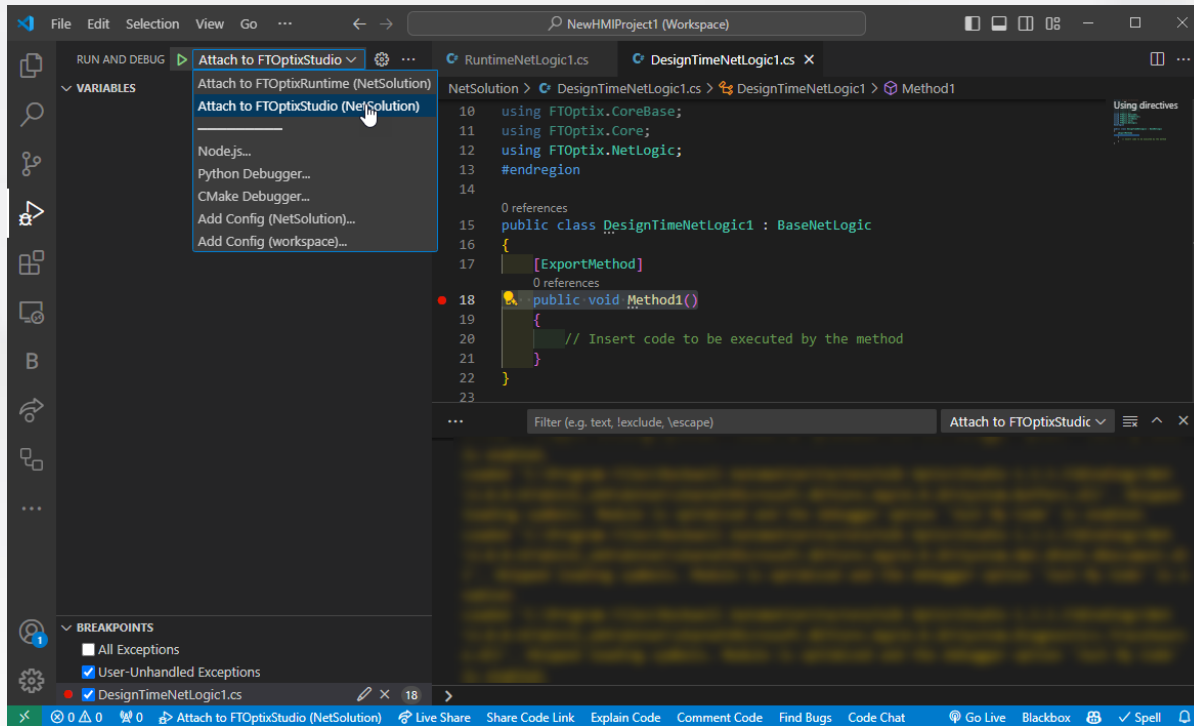
Debug with Visual Studio Code



Due to a bug of the C# Extension of VSCode, versions of the plugin between 2.4.4 and 2.34.10 may not work when debugging Designtime NetLogic, please upgrade or downgrade as needed.

- Set a breakpoint by clicking to the left of the row number (a red dot will appear)
- Select
 - Attach to FTOptixStudio for Design time NetLogic
 - Attach to FTOptixRuntime for Runtime NetLogic
- Click "Run and Debug" on the left toolbar or hit F5 on the keyboard
- Execute the script from FactoryTalk Optix Studio or FactoryTalk Optix Runtime
- Execute step-by-step
 - Step Over (F10)
 - Step Into (F11)

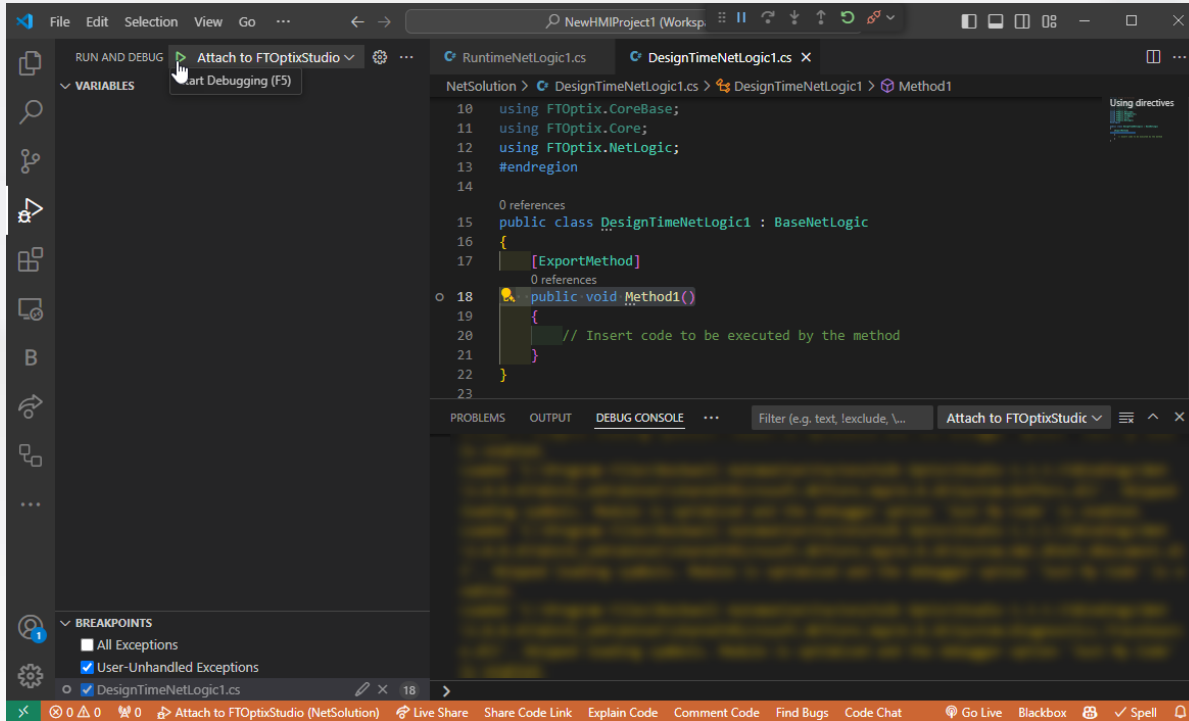
Debug with Visual Studio Code



- Set a breakpoint by clicking to the left of the row number (a red dot will appear)
- Select
 - Attach to FTOptixStudio for Design time NetLogic
 - Attach to FTOptixRuntime for Runtime NetLogic
- Click "Run and Debug" on the left toolbar or hit F5 on the keyboard
- Execute the script from FactoryTalk Optix Studio or FactoryTalk Optix Runtime
- Execute step-by-step
 - Step Over (F10)
 - Step Into (F11)

Due to a bug of the C# Extension of VSCode, versions of the plugin between 2.4.4 and 2.34.10 may not work when debugging Designtime NetLogic, please upgrade or downgrade as needed.

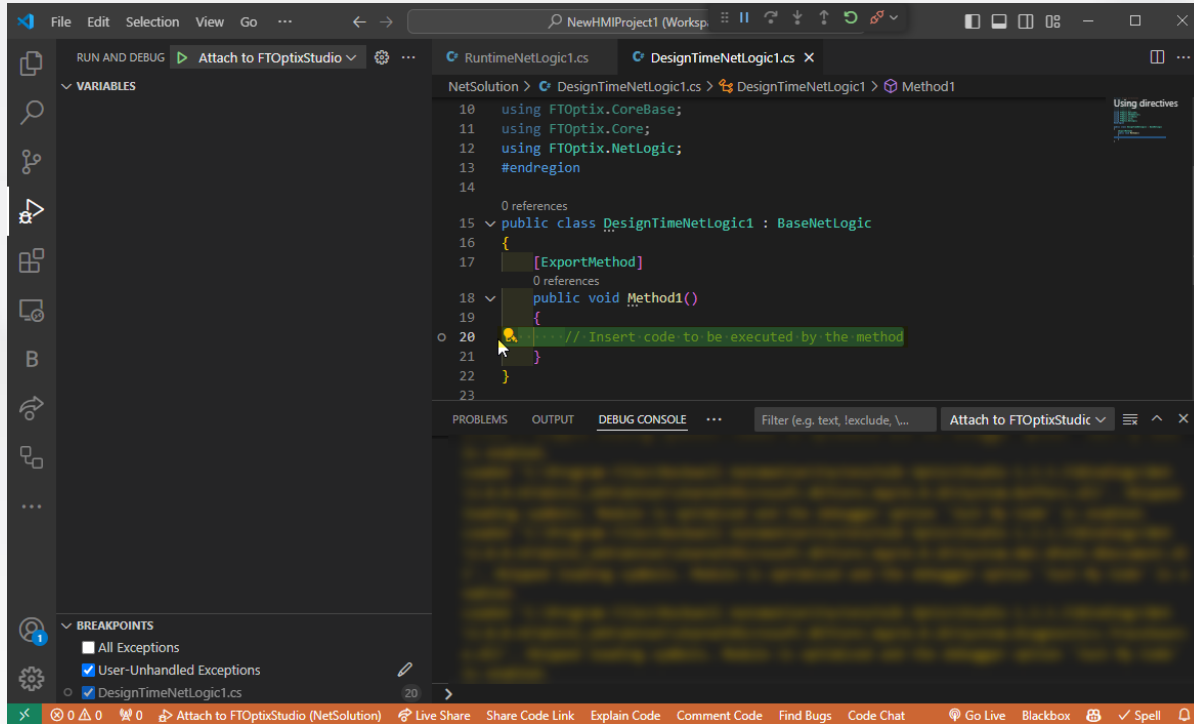
Debug with Visual Studio Code



- Set a breakpoint by clicking to the left of the row number (a red dot will appear)
- Select
 - Attach to FTOptixStudio for Design time NetLogic
 - Attach to FTOptixRuntime for Runtime NetLogic
- Click "Run and Debug" on the left toolbar or hit F5 on the keyboard
- Execute the script from FactoryTalk Optix Studio or FactoryTalk Optix Runtime
- Execute step-by-step
 - Step Over (F10)
 - Step Into (F11)

Due to a bug of the C# Extension of VSCode, versions of the plugin between 2.4.4 and 2.34.10 may not work when debugging Designtime NetLogic, please upgrade or downgrade as needed.

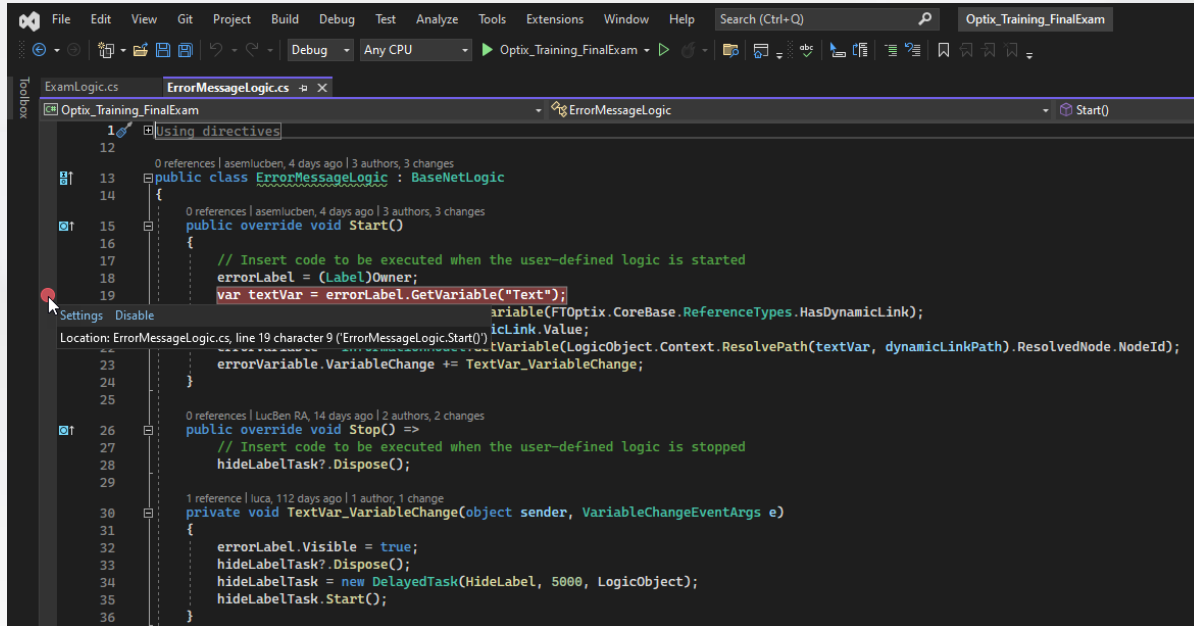
Debug with Visual Studio Code



- Set a breakpoint by clicking to the left of the row number (a red dot will appear)
- Select
 - Attach to FTOptixStudio for Design time NetLogic
 - Attach to FTOptixRuntime for Runtime NetLogic
- Click "Run and Debug" on the left toolbar or hit F5 on the keyboard
- Execute the script from FactoryTalk Optix Studio or FactoryTalk Optix Runtime
- Execute step-by-step
 - Step Over (F10)
 - Step Into (F11)

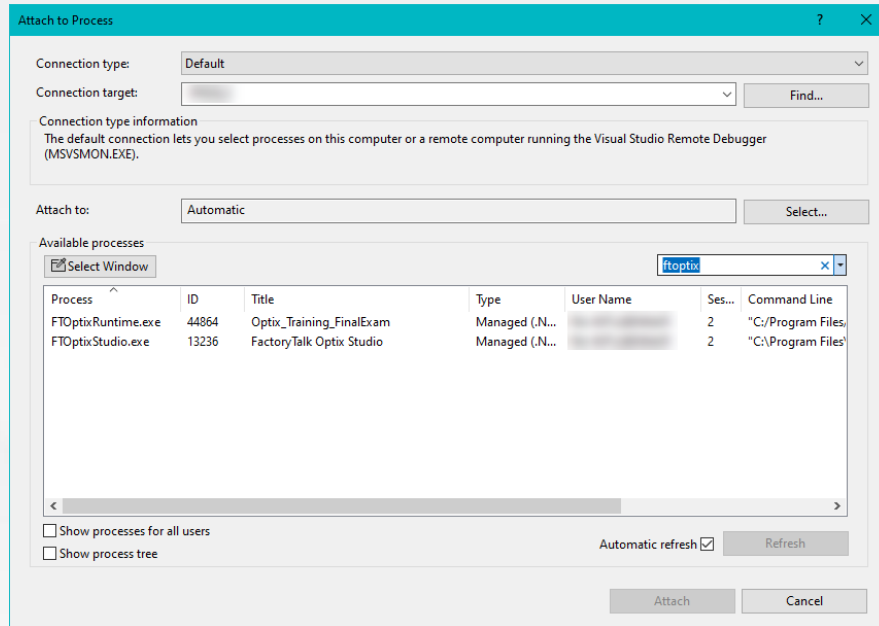
Due to a bug of the C# Extension of VSCode, versions of the plugin between 2.4.4 and 2.34.10 may not work when debugging Designtime NetLogic, please upgrade or downgrade as needed.

Debug with Visual Studio 2022



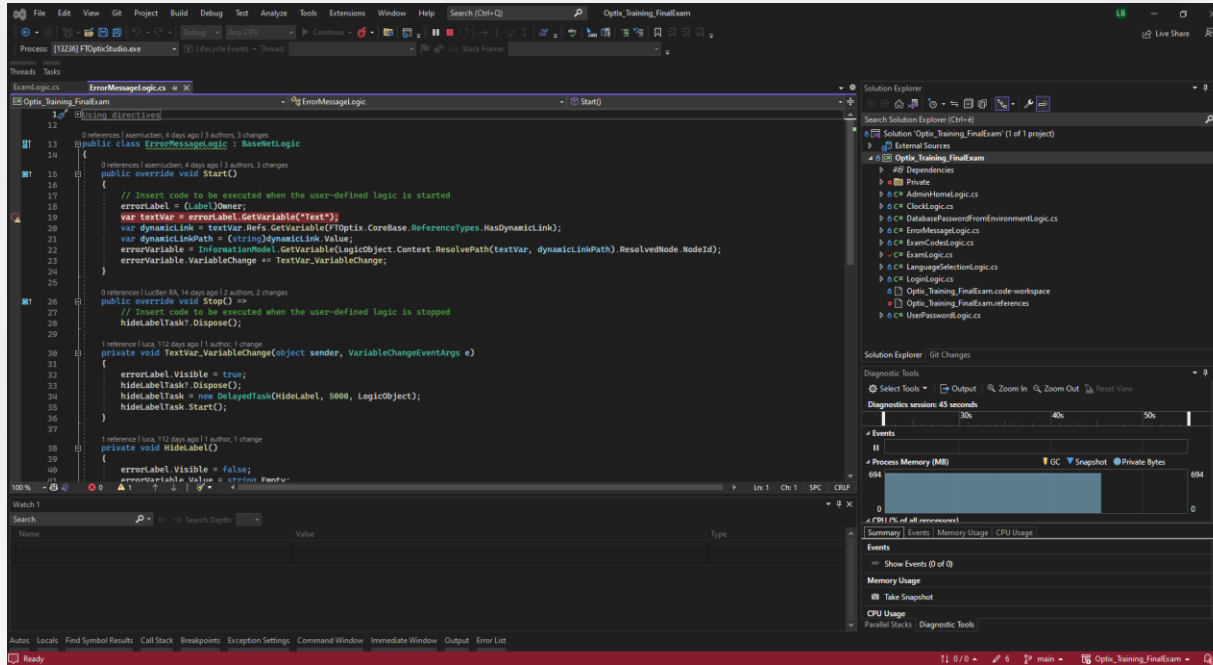
- Open the .NET Solution with VS
- Select the .cs file to debug
- Set a breakpoint by clicking on the left column of the code editor
- DesignTime NetLogic:
 - From the Debug menu, select «Attach to process»
 - Select FTOptixStudio from the menu
 - Click «Attach»
 - Execute the DesignTime method
- RunTime NetLogic
 - From the Debug menu select «Attach to process»
 - Select FactoryTalkOptixRuntime from the menu
 - Click «Attach»
 - Execute the RunTime method

Debug with Visual Studio 2022



- Open the .NET Solution with VS
- Select the .cs file to debug
- Set a breakpoint by clicking on the left column of the code editor
- DesignTime NetLogic:
 - From the Debug menu, select «Attach to process»
 - Select FTOptixStudio from the menu
 - Click «Attach»
 - Execute the DesignTime method
- RunTime NetLogic
 - From the Debug menu select «Attach to process»
 - Select FactoryTalkOptixRuntime from the menu
 - Click «Attach»
 - Execute the RunTime method

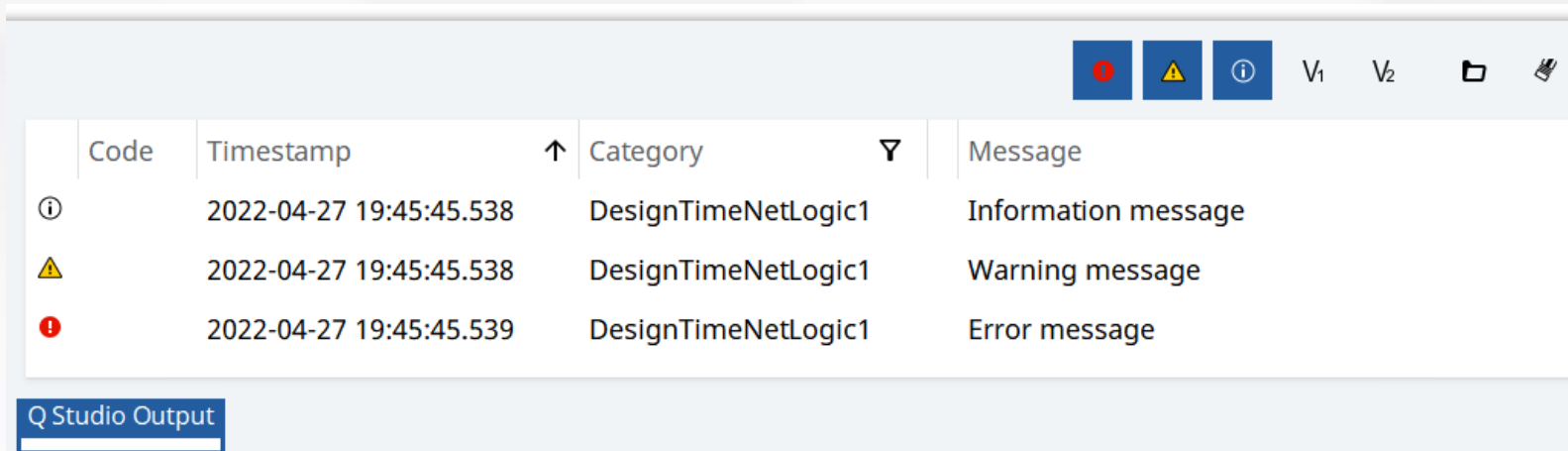
Debug with Visual Studio 2022



- Open the .NET Solution with VS
- Select the .cs file to debug
- Set a breakpoint by clicking on the left column of the code editor
- DesignTime NetLogic:
 - From the Debug menu, select «Attach to process»
 - Select FTOptixStudio from the menu
 - Click «Attach»
 - Execute the DesignTime method
- RunTime NetLogic
 - From the Debug menu select «Attach to process»
 - Select FactoryTalkOptixRuntime from the menu
 - Click «Attach»
 - Execute the RunTime method

Debug by logging to output

- `Log.Info(LogicObject.BrowseName, "Information message");`
- `Log.Warning(LogicObject.BrowseName, "Warning message");`
- `Log.Error(LogicObject.BrowseName, "Error message");`



The screenshot shows the 'Q Studio Output' window. At the top right, there are three filter buttons (red circle, yellow triangle, blue circle with 'i'), and two view buttons labeled 'V1' and 'V2'. Below these is a table with the following data:

	Code	Timestamp	Category	Message
ⓘ		2022-04-27 19:45:45.538	DesignTimeNetLogic1	Information message
⚠		2022-04-27 19:45:45.538	DesignTimeNetLogic1	Warning message
❗		2022-04-27 19:45:45.539	DesignTimeNetLogic1	Error message

At the bottom left of the window, the text 'Q Studio Output' is displayed in a blue box.

Why to debug?

- Debugger is a powerful tool to understand what's happening to the project
 - Helps to understand why some «weird» errors appear, like «Object not set to an instance of an object» (when trying to access a variable that does not exist) or «Unhandled exception»
- When debugging some logic in the Start method of a RunTime NetLogic multiple options are available
 - Add the «ExportMethod» modifier to the Start method and use a button to call it
 - Use a NavigationPanel or PanelLoader to first connect to the debugger and then load the page containing the script
- If you get «The break point will not currently be hit» please check for:
 - Make sure you attached to the right process
 - Make sure the .NET solution compiles
 - Make sure you saved the project both on VS/VSCode and FactoryTalk Optix
 - Make sure the code in the FactoryTalk Optix Runtime is the latest (restart the runtime if needed)
- Remote debugging is also available via SSH on iPC
 - Cannot be used on closed systems (like OptixPanel) where SSH is not available

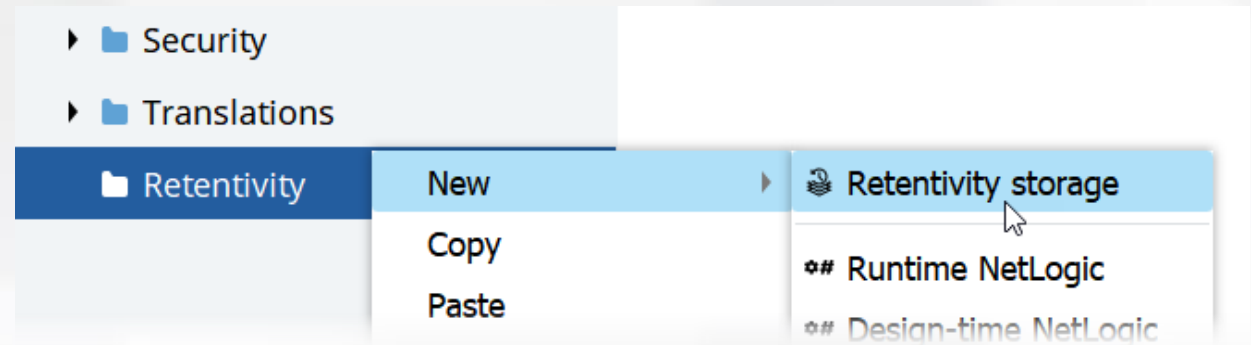


Use Retentivity



Retentivity usage

- Provide storage of changes made at Runtime
- Retentivity applies to:
 - **Variables,**
 - **Users,**
 - **Dictionary of Translations,**
 - or every node of the project
- Every Retentivity storage object is related to an embedded database



Retentivity properties and location

- Properties

- **Nodes:** Folder/node of the project to be retained.
It's possible to define several nodes on the same Retentivity Storage, or create separated storage
- **Write delay:** define how often to save into the database
Zero means values are saved as soon as a change is detected
- **Delta observer enabled:** allow to enable/disable the Retentivity

Properties



Name

RetentivityStorage1

Type

Retentivity storage

▼

 Nodes

+

▶ Node1

NodeId

Model



▶ Node2

NodeId

Security



▶ Node3

NodeId

LocalizationDictionary1



Write delay

0000:00:00.000

Delta observer enabled

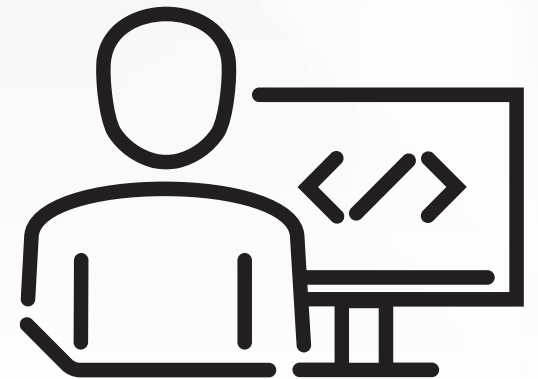
True

- Location:

- Database file can be found in the ApplicationFiles folder of FTOptixApplication
- Example: %localappdata%\Rockwell Automation\FactoryTalk Optix\Emulator\Projects\<ProjectName>\ApplicationFiles

Hands-on session

- Add retentivity storage to the project
- Add a project's node to the retentivity (like the Model folder)
- Verify that changes made on Model's variable are retained

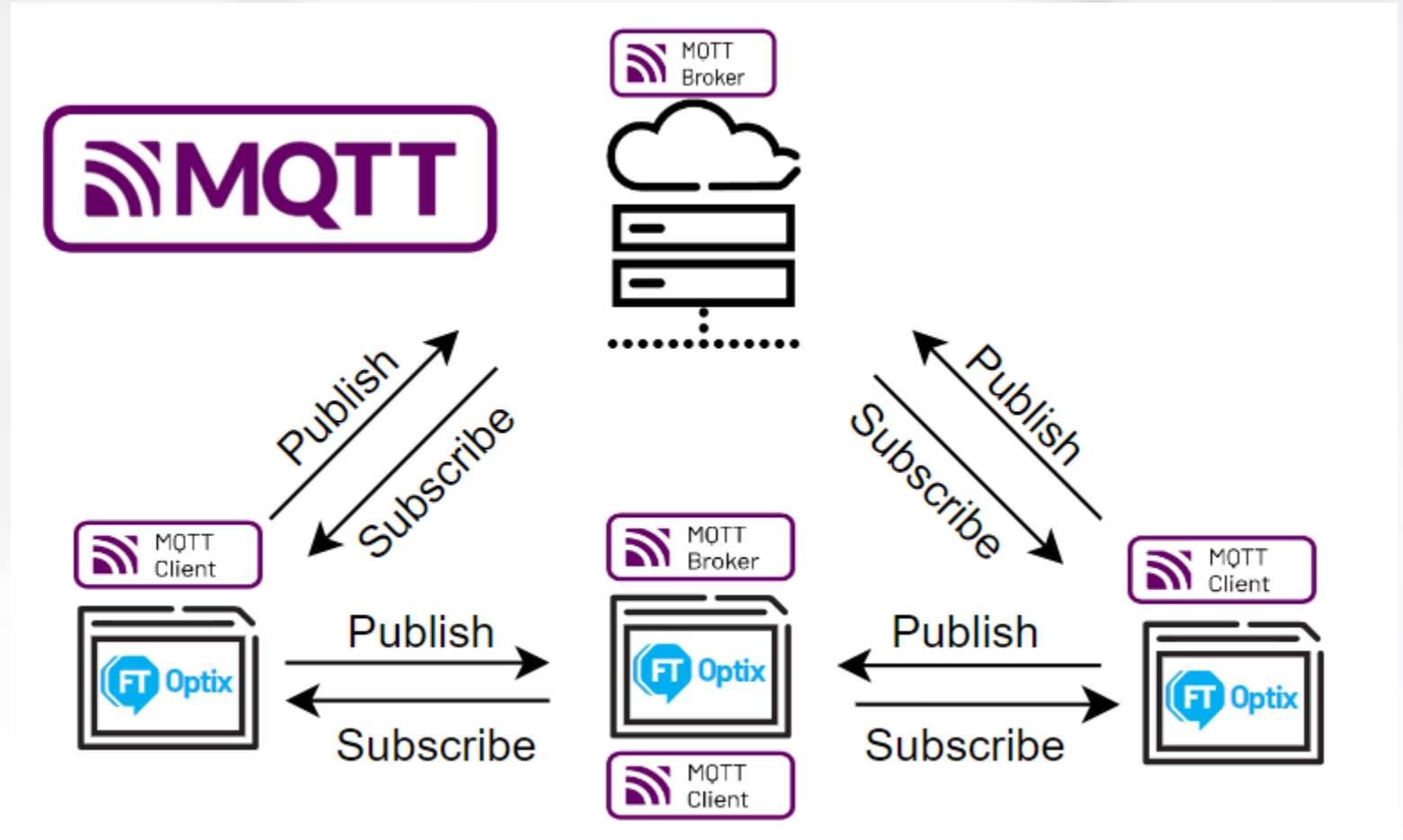


MQTT



Basic configuration

- **MQTT Broker:** The central server that routes messages between clients in the publish/subscribe model.
- **MQTT Client:** Interacts with the broker to send or receive messages.
 - **Publisher:** A client that sends messages to a specific topic on the broker.
 - **Subscriber:** A client that listens to topics and receives messages from the broker.



MQTT broker

General Configuration

Listener port: The port on which the MQTT broker listens for incoming client connections (default: 8883 for secure communication using SSL/TLS).

Max connections: The maximum number of simultaneous client connections that the broker can handle (e.g., 25).

Max QoS: The highest Quality of Service level supported by the broker:

- 0: At most once (fire-and-forget).
- 1: At least once (acknowledged delivery).
- 2: Exactly once (guaranteed delivery).

Max in-flight messages: The maximum number of unacknowledged messages per client that the broker allows to remain "in flight."

Max queued messages: The maximum number of messages that can be queued for offline clients.

Message size limit: The maximum size (in bytes) of a single MQTT message. A value of 0 indicates no limit.

Retain available: Indicates whether the broker supports retained messages. Retained messages store the last published message for a topic and send it to new subscribers.

: Properties		
MQ	Name	MQTTBroker
	Type	MQTT Broker
— General configuration		
	Listener port	8883
	Max connections	25
	Max QoS	2 - Exactly once
	Max in-flight messages	20
	Max queued messages	1000
	Message size limit	0
	Retain available	True
— Security		
	Require client certificate	False
	CA certificate file	Browse
	Broker certificate file	Browse
	Broker private key file	Browse

MQTT broker

Security

- Require client certificate:** Determines whether clients must provide a certificate for mutual TLS authentication (True or False).
 - CA certificate file:** The path to the Certificate Authority (CA) certificate file, used to verify the authenticity of client certificates.
 - Broker certificate file:** The path to the broker's SSL/TLS certificate file, used to encrypt communication.
 - Broker private key file:** The path to the private key file corresponding to the broker's SSL/TLS certificate, used for secure communication.
- This setup is typical for a secure and configurable MQTT broker deployment, ensuring flexibility and security.

Properties

Name MQTTBroker

Type MQTT Broker

General configuration

Listener port

8883

Max connections

25

Max QoS

2 - Exactly once

Max in-flight messages

20

Max queued messages

1000

Message size limit

0

Retain available

True

Security

Require client certificate

False

CA certificate file

[Browse](#)

Broker certificate file

[Browse](#)

Broker private key file

[Browse](#)

MQTT Client

Client

Broker address: The address of the broker to connect to.

Port: Communication port number.

ClientID: The Client identifier that is unique on the server.

Properties

MQ

Name

MQTTClient1

Type

MQTT Client

Client

Broker address

localhost

Port

8883

Client Id

FTOptix-1

Security

SSL/TLS enabled

True

Validate broker certificate

True

CA certificate file

Browse

Client certificate file

Browse

Client private key file

Browse

User identity

User identity type

Anonymous

MQTT Client

Security

SSL/TLS Enabled: Enabling secure connection between FT Optix MQTT Client and MQTT Broker. Once activated 'mqqt's' protocol is being used.

Validate broker certificate: Activates/deactivates validation process of MQTT broker certificate. During the SSL certificate verification process, the FT Optix MQTT client checks the digital signature of the broker certificate to ensure that it has been issued by a trusted certificate authority (CA). The client also verifies that the certificate has not expired.

CA certificate file: A Certificate Authority certificate that has signed the server certificate on the MQTT Broker. It is required once SSL/TLS is activated to establish secure connection with the broker. It must be found in \FactoryTalk Optix\Projects\ProjectName\ProjectFiles\PKI\Own\Certs folder.

Client certificate file: FT Optix MQTT Client certificate. A client certificate identifies the client just like the server certificate identifies the server. If this property stays empty only broker certificate is user for authentication. It must be found in \FactoryTalk Optix\Projects\ProjectName\ProjectFiles\PKI\Own\Certs folder.

Client private key file: FT Optix MQTT Client private key. It must be found in \FactoryTalk Optix\Projects\ProjectName\ProjectFiles\PKI\Own\Certs folder.

Properties

MQ

NameMQTTClient1

TypeMQTT Client

Client

Broker addresslocalhost

Port8883

Client IdFTOptix-1

Security

SSL/TLS enabledTrue

Validate broker certificateTrue

CA certificate fileBrowse

Client certificate fileBrowse

Client private key fileBrowse

User identity

User identity typeAnonymous

MQTT Client

User identity

User Identity type: Specify user when accessing the MQTT Broker:

- Anonymous (default) - No login is performed, access the server as Anonymous user
- Username/Password - Specify a username and password combination for logon.

Properties

MQ

Name

MQTTClient1

Type

MQTT Client

Client

Broker address

localhost

Port

8883

Client Id

FTOptix-1

Security

SSL/TLS enabled

True

Validate broker certificate

True

CA certificate file

Browse

Client certificate file

Browse

Client private key file

Browse

User identity

User identity type

Anonymous

MQTT Client/Publisher

Sampling mode:

- **None.** Automatic recording is disabled, can record by invoking the Sample method;
- **Periodic.** At regular intervals records the values of all the selected variables;
- **Change in value.** At regular intervals records only the values of the selected variables that have changed with respect to the previous sampling.

Sampling period: Interval in hours, with millisecond precision, between one sampling and the next, in the Periodic mode.

Pooling time: Interval in hours, with millisecond precision, between one sampling and the next, in the Value change mode.

Folder: This folder contains variables whose values will be sent to the broker. The Folder node must be a global object and cannot be a session-based object.

Topic: Defines MQTT topic on which we are sending/publishing data. (Topic is required. Once topic is empty Publisher will generate warning message during first publish. RT will not stop and other mqtt functions will keep working, only this specific Publisher will be stopped.)

: Properties		
	Name	MQTTPublisher1
	Type	MQTT Publisher
Sampling mode		Periodic
Sampling period		0000:00:01.000
Folder		
Topic		
QoS		0 - At most once
Retain		False

MQTT Client/Publisher

QoS: MQTT Quality of Service (QoS), defines the guarantee of delivery for a specific message (0, 1, 2).

- 0 - At most once** (May lose messages).
- 1 - At least once** (Guarantees the message delivery but potentially duplicate messages).
- 2 - Exactly once** (Ensures messages are delivered exactly once without duplication).

Retain: Retain message on the topic even after read

- True - Enables Retain functionality
- False - Disables Retain functionality

⋮ Properties

MQ

Name

MQTTPublisher1

Type

MQTT Publisher

Sampling mode

Periodic

Sampling period

0000:00:01.000

Folder

Topic

QoS

0 - At most once

Retain

False

MQTT Client/Subscriber

Folder: This folder node contains variables on which the Subscriber will copy the values read from the broker.

Topic: A single topic on a Broker to subscribe. Topic is required.

QoS: MQTT Quality of Service (QoS), defines the guarantee of delivery for a specific message (0, 1, 2).

- 0 - At most once** (May lose messages).
- 1 - At least once** (Guarantees the message delivery but potentially duplicate messages).
- 2 - Exactly once** (Ensures messages are delivered exactly once without duplication).

: Properties		
	Name	MQTTPublisher1
	Type	MQTT Publisher
Sampling mode		Periodic
Sampling period		0000:00:01.000
Folder		
Topic		
QoS		0 - At most once
Retain		False

Current Limitations

Supported Data Types:

- The publish/subscribe mechanism currently supports **only scalar variables**.
- Structures (UDTs), folders, or other complex data types are **not supported**.

Sparkplug B:

- Coming soon.

Advanced topic configurations:

- Wildcards are not supported

MQTT Token usage

Basic Concept of the License

1. MQTT Clients without a Broker (pointing to localhost):

- Each client that connects to its local broker(localhost) consumes 1 token each.
- If you have n clients, the total cost will be n tokens.

2. MQTT Clients with a Central Broker:

- If you configure a central broker and all clients connect to it:
 - The clients do not consume tokens.
 - Only the broker consumes 1 token, regardless of the number of connected clients.

3. MQTT Clients with Different Brokers:

- If the clients connect to different brokers (e.g., Broker A, Broker B, etc.):
 - Each broker consumes 1 token.
 - If there are n different brokers, the total cost will be n tokens, regardless of the number of clients connected to each broker.

MQTT Utilities

MQTT Official website: [MQTT - The Standard for IoT Messaging](#)

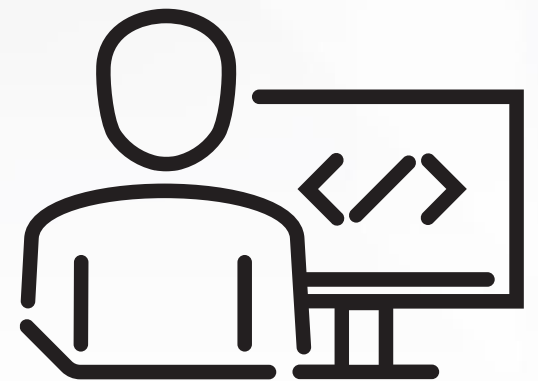
MQTT 5 Standard: [MQTT Version 5.0](#)

This tool helps us to connect to a broker and to subscribe to a specific topic, checking all the packets sent. It also allows us to publish to a topic.

[MQTTX: Your All-in-one MQTT Client Toolbox](#)

Hands-on session

- Define an MQTT Broker: Set up the MQTT broker to act as the central hub for communication.
 - Key configurations to cover:
 - Listener port (e.g., 1883 for standard or 8883 for SSL/TLS).
 - Retained messages (optional, if applicable to your example).
 - Max connections (adjust as needed based on class size).
- Define an MQTT Client: introduce the concept of an MQTT client, explaining its dual roles as a Publisher and Subscriber.
 - Publisher: configure the client to publish updates to a specific topic (e.g., sensor/temperature).
 - Create a variable that triggers a message when its value changes.
 - Subscriber: Configure the client to subscribe to the same topic (sensor/temperature) and react to updates published by the Publisher.
- Key learning points: Ensure the topic names match exactly (case-sensitive).
- Create Variables that Synchronize via MQTT: Demonstrate how to bind variables to MQTT topics in the software or platform being used.
- Example: create a "source" variable (e.g., temperature) that updates periodically or based on variable change. Create a "target" variable (e.g., displayTemperature) that listens for updates via MQTT and synchronizes automatically. Ensure the variable changes are observable in real-time.
- Test the configuration: Publish a value change (e.g., simulate a sensor reading update). Verify that the subscriber's variable updates automatically when the publisher sends data.



REST APIs



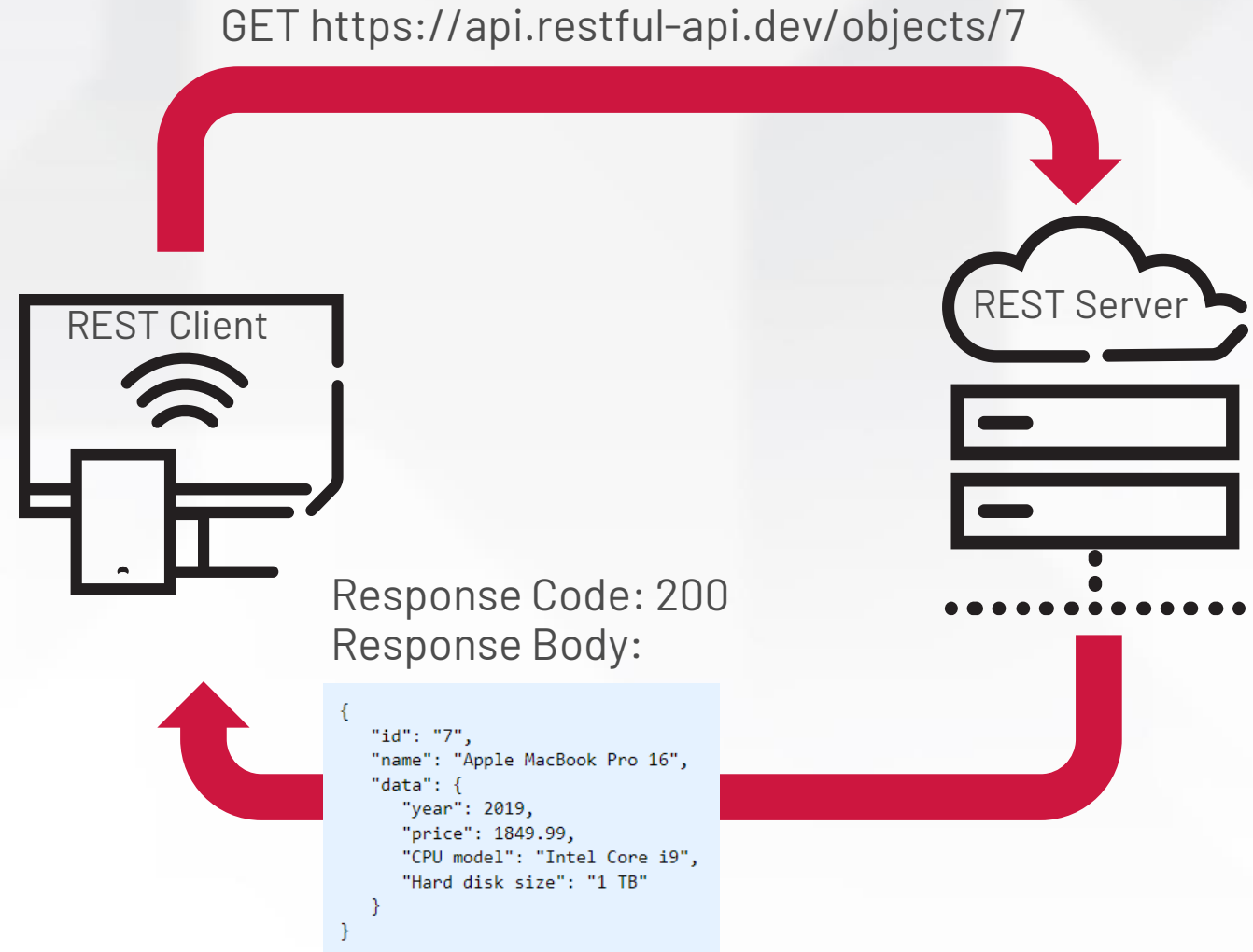
REST APIs – Intro to REST

- REST: Representational State Transfer
- Most commonly used protocol for interacting with Web Services
- Protocol uses HTTP / HTTPS methods as the delivery mechanism
- Example Web Services:
 - Google Search (Search the web)
 - Google Maps (Get directions)
 - National Weather Service (Get weather conditions)
 - Twilio (Send Email / Text)
 - Yahoo Finance (Get stock information)
 - SAP (Get business data)
 - ...



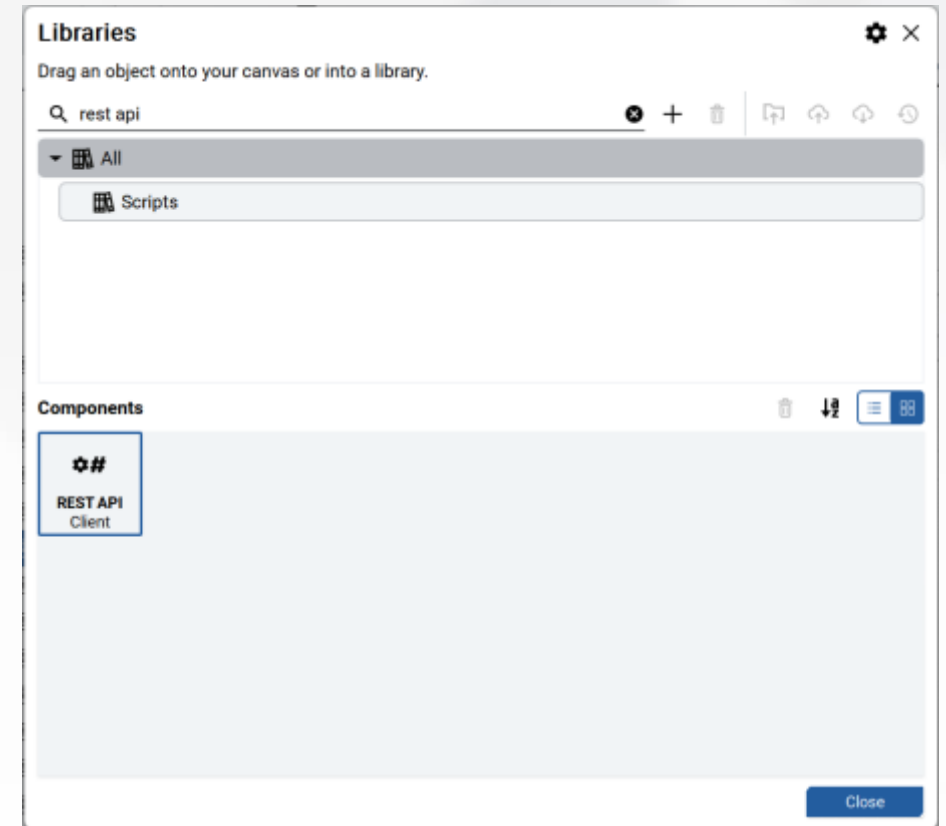
REST APIs – HTTP Methods / Response

- CRUD: CREATE, READ, UPDATE, DELETE
- HTTP CRUD Methods
 - CREATE: **POST**
 - READ: **GET**
 - UPDATE: **PUT**
 - DELETE: **DELETE**
- HTTP Response:
 - Status Code Integer
 - 200: OK
 - 404: Not Found
 - ...
 - Response Body
 - Typically JSON format



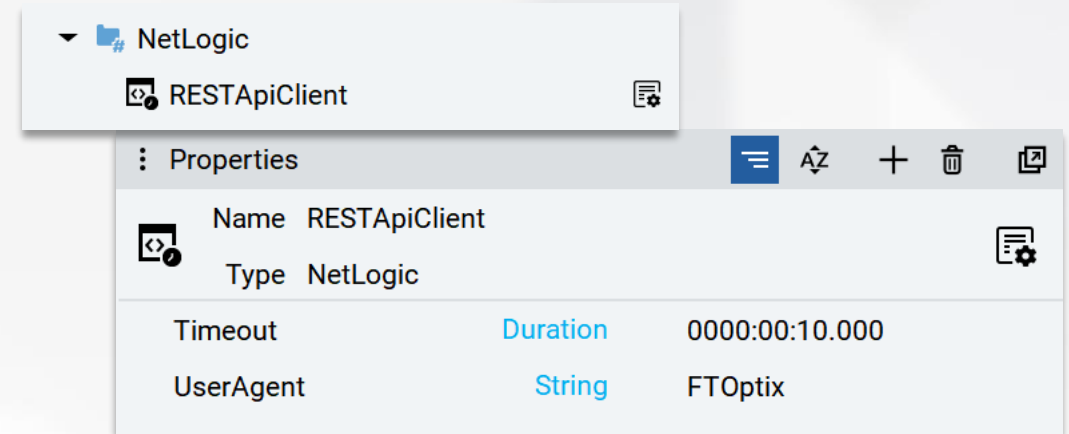
REST APIs – Implementing in Optix

- Prebuilt NetLogic script in Optix Scripts Library
- 'REST API Client' Script supported HTTP methods:
 - **GET**
 - **POST**
 - **PUT**



REST APIs – Import / Configure RESTApiClient

- Add RESTApiClient to Project
- Configure:
 - Timeout: Defaults to 10s
 - UserAgent: User defined string for client identification. Most REST web services consider this optional. Default: "FTOptix"



REST APIs – RESTApiClient NetLogic Exploration

- Three Methods available: Get, Post, Put
- Get contains 3 input parameters:
 - apiURL
 - queryString
 - bearerToken
- Post and Put contain 4 input parameters:
 - apiURL
 - requestBody
 - bearerToken
 - contentType
- Each Method contains 'out' parameters with the Response Code (code) and Response Body (response)

```
[ExportMethod]
public void Get(string apiUrl, string queryString, string bearerToken, out string response, out int code)
{
    TimeSpan timeout = TimeSpan.FromMilliseconds(GetTimeout());
    UriBuilder uriBuilder = new UriBuilder(apiUrl);
    uriBuilder.Query = queryString;

    var requestMessage = BuildMessage(HttpMethod.Get, uriBuilder.Uri, "", bearerToken, "");
    var requestTask = PerformRequest(requestMessage, timeout);
    var httpResponse = requestTask.Result;

    (response, code) = (httpResponse.Payload, httpResponse.Code);
}

[ExportMethod]
public void Post(string apiUrl, string requestBody, string bearerToken, string contentType, out string response, out int code)
{
    TimeSpan timeout = TimeSpan.FromMilliseconds(GetTimeout());
    UriBuilder uriBuilder = new UriBuilder(apiUrl);

    var requestMessage = BuildMessage(HttpMethod.Post, uriBuilder.Uri, requestBody, bearerToken, contentType);
    var requestTask = PerformRequest(requestMessage, timeout);
    var httpResponse = requestTask.Result;

    (response, code) = (httpResponse.Payload, httpResponse.Code);
}

[ExportMethod]
public void Put(string apiUrl, string requestBody, string bearerToken, string contentType, out string response, out int code)
{
    TimeSpan timeout = TimeSpan.FromMilliseconds(GetTimeout());
    UriBuilder uriBuilder = new UriBuilder(apiUrl);

    var requestMessage = BuildMessage(HttpMethod.Put, uriBuilder.Uri, requestBody, bearerToken, contentType);
    var requestTask = PerformRequest(requestMessage, timeout);
    var httpResponse = requestTask.Result;

    (response, code) = (httpResponse.Payload, httpResponse.Code);
}
```


REST APIs – GET Example (Get a list of all objects)

- Get contains 3 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - queryString: NULL (No Parameters)
 - bearerToken: NULL (No Security)

Method	Action
GET	LIST OF ALL OBJECTS
GET	LIST OF OBJECTS BY IDS
GET	SINGLE OBJECT
POST	ADD OBJECT
PUT	UPDATE OBJECT
PATCH	PARTIALLY UPDATE OBJECT
DELETE	DELETE OBJECT

Request
`https://api.restful-api.dev/objects`

Response
200

```
[
  {
    "id": "1",
    "name": "Google Pixel 6 Pro",
    "data": {
      "color": "Cloudy White",
      "capacity": "128 GB"
    }
  },
  {
    "id": "2",
    "name": "Apple iPhone 12 Mini, 256GB, Blue",
    "data": null
  },
  {
    "id": "3",
    "name": "Apple iPhone 12 Pro Max",
    "data": {
      "color": "Cloudy White",
      "capacity GB": 512
    }
  }
]
```

*<https://restful-api.dev>

REST APIs – GET Example (Get a list of all objects)

- Get contains 3 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - queryString: NULL (No Parameters)
 - bearerToken: NULL (No Security)

1. Call the RESTApiClient/Get method with 'apiURL' set.
2. Use VariableCommands/Set to gather the RESApiClient/Get Output Parameters



The screenshot displays the 'Events' panel in the FactoryTalk interface. It shows the configuration for the 'REST/NetLogic/RESTApiClient/Get' method. The 'Input arguments' section lists 'apiUrl', 'queryString', and 'bearerToken', with 'apiUrl' linked to the variable 'urlGetListOfAllObjects'. Below this, the 'VariableCommands/Set' method is configured to capture the output parameters. The first 'Set' command is linked to 'statusCode' (Int32) and the variable '.../.../MethodContainer1/OutputArguments/code'. The second 'Set' command is linked to 'jsonResponse' (String) and the variable '.../.../MethodContainer1/OutputArguments/response'. Red arrows from the text on the left point to these specific configurations in the interface.

REST APIs – GET Example (Get a list of specific objects)

- Get contains 3 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - queryString: "id=3&id=5&id=10" (everything after "?" in request URL)
 - bearerToken: NULL (No Security)

		Request	Response
GET	LIST OF ALL OBJECTS	https://api.restful-api.dev/objects?id=3&id=5&id=10	200
GET	LIST OF OBJECTS BY IDS		
GET	SINGLE OBJECT		
POST	ADD OBJECT		
PUT	UPDATE OBJECT		
PATCH	PARTIALLY UPDATE OBJECT		
DELETE	DELETE OBJECT		

```
[
  {
    "id": "3",
    "name": "Apple iPhone 12 Pro Max",
    "data": {
      "color": "Cloudy White",
      "capacity GB": 512
    }
  },
  {
    "id": "5",
    "name": "Samsung Galaxy Z Fold2",
    "data": {
      "price": 689.99,
      "color": "Brown"
    }
  },
  {
    "id": "10",
    "name": "Apple iPad Mini 5th Gen",
    "data": {
```

*https://restful-api.dev

REST APIs – GET Example (Get a list of specific objects)

- Get contains 3 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - queryString: "id=3&id=5&id=10" (everything after "?" in request URL)
 - bearerToken: NULL (No Security)

1. Call the RESTApiClient/Get method with 'apiURL' and 'queryString' set.
2. Use VariableCommands/Set to gather the RESApiClient/Get Output Parameters



The screenshot shows the 'Events' panel in the FactoryTalk interface. It lists several events, including 'MouseClicked event', 'MouseDown event', and 'MouseUp event'. The 'MouseClicked event' is expanded, showing its 'Method' as 'REST/NetLogic/RESTApiClient/Get'. This method has three 'Input arguments': 'apiUrl' (linked to 'urlGetListOfObjectsByIds'), 'queryString' (linked to 'queryGetListOfObjectsByIds'), and 'bearerToken'. Below this, there are two 'VariableCommands/Set' entries. The first 'Set' command has 'VariableToModify' as 'statusCode' (linked to 'statusCode'), 'Value' as 'Int32' (linked to 'Int32'), and 'ArrayIndex' as '0'. The second 'Set' command has 'VariableToModify' as 'jsonResponse' (linked to 'jsonResponse'), 'Value' as 'String' (linked to 'String'), and 'ArrayIndex' as '0'. Red arrows from the text instructions point to these elements: one to the 'apiUrl' input argument, one to the first 'Set' command, and one to the 'jsonResponse' input argument.

REST APIs – POST Example (Add new object)

- POST contains 4 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - requestBody: "{ \"name\": \"Apple MacBook Pro 16\", \"data\": { \"year\": 2019, \"price\": 1849.99, \"CPU model\": \"Intel Core i9\", \"Hard disk size\": \"1 TB\" } }"
 - bearerToken: NULL (No Security)
 - contentType: "application/json" (See [Media type – Wikipedia](#) for other content-type examples)

The screenshot displays a REST client interface with a list of HTTP methods on the left and a detailed view of a POST request and response on the right.

Method	Description
GET	LIST OF ALL OBJECTS
GET	LIST OF OBJECTS BY IDS
GET	SINGLE OBJECT
POST	ADD OBJECT
PUT	UPDATE OBJECT
PATCH	PARTIALLY UPDATE OBJECT
DELETE	DELETE OBJECT

Request
https://api.restful-api.dev/objects

```
{
  "name": "Apple MacBook Pro 16",
  "data": {
    "year": 2019,
    "price": 1849.99,
    "CPU model": "Intel Core i9",
    "Hard disk size": "1 TB"
  }
}
```

Response
200

```
{
  "id": "7",
  "name": "Apple MacBook Pro 16",
  "data": {
    "year": 2019,
    "price": 1849.99,
    "CPU model": "Intel Core i9",
    "Hard disk size": "1 TB"
  },
  "createdAt": "2022-11-21T20:06:23.986Z"
}
```

*https://restful-api.dev

REST APIs – POST Example (Add new object)

- POST contains 4 input parameters:
 - apiURL: "https://api.restful-api.dev/objects"
 - requestBody: "{ \"name\": \"Apple MacBook Pro 16\", \"data\": { \"year\": 2019, \"price\": 1849.99, \"CPU model\": \"Intel Core i9\", \"Hard disk size\": \"1 TB\" } }"
 - bearerToken: NULL (No Security)
 - contentType: "application/json" (See [Media type – Wikipedia](#) for other content-type examples)

1. Call the RESTApiClient/Post method with 'apiURL', 'requestBody', and contentType set.
2. Use VariableCommands/Set to gather the RESApiClient/Post Output Parameters



REST APIs – PUT Example (Update object)

- PUT contains 4 input parameters:
 - apiURL: "https://api.restful-api.dev/objects/7"
 - requestBody: "{ \"name\": \"Apple MacBook Pro 16\", \"data\": { \"year\": 2019, \"price\": 2049.99, \"CPU model\": \"Intel Core i9\", \"Hard disk size\": \"1 TB\", \"color\": \"silver\" } }"
 - bearerToken: NULL (No Security)
 - contentType: "application/json" (See [Media type – Wikipedia](#) for other content-type examples)

The screenshot displays the RESTful API interface with a list of methods on the left and a detailed view of a PUT request and response on the right.

Method	Description
GET	LIST OF ALL OBJECTS
GET	LIST OF OBJECTS BY IDS
GET	SINGLE OBJECT
POST	ADD OBJECT
PUT	UPDATE OBJECT
PATCH	PARTIALLY UPDATE OBJECT
DELETE	DELETE OBJECT

Request
https://api.restful-api.dev/objects/7

```
{
  "name": "Apple MacBook Pro 16",
  "data": {
    "year": 2019,
    "price": 2049.99,
    "CPU model": "Intel Core i9",
    "Hard disk size": "1 TB",
    "color": "silver"
  }
}
```

Response
200

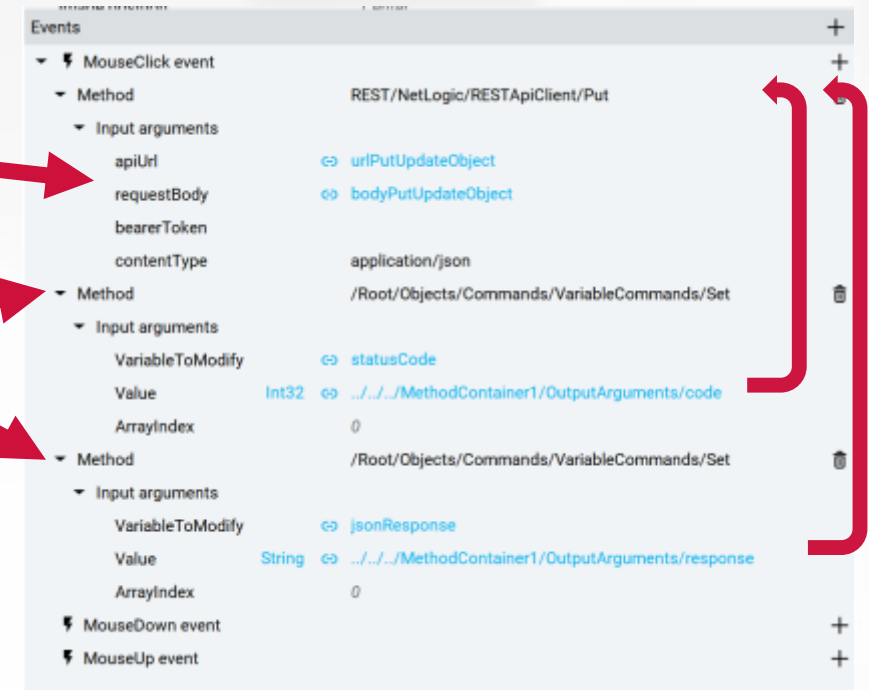
```
{
  "id": "7",
  "name": "Apple MacBook Pro 16",
  "data": {
    "year": 2019,
    "price": 2049.99,
    "CPU model": "Intel Core i9",
    "Hard disk size": "1 TB",
    "color": "silver"
  },
  "updatedAt": "2022-12-25T21:08:41.986Z"
}
```

*https://restful-api.dev

REST APIs – PUT Example (Update object)

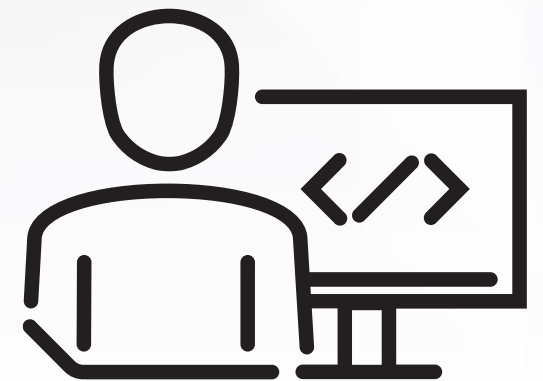
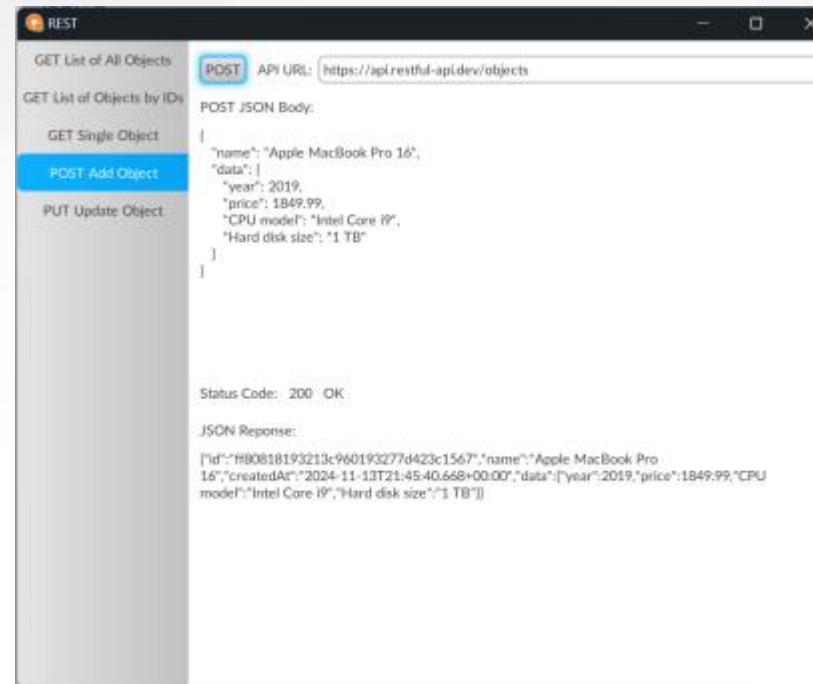
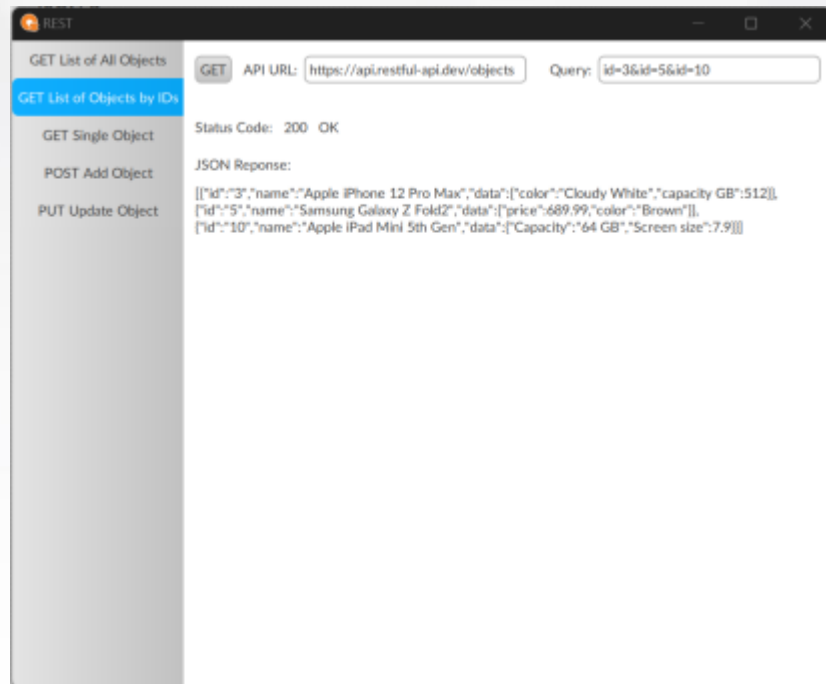
- PUT contains 4 input parameters:
 - apiURL: "https://api.restful-api.dev/objects/7"
 - requestBody: "{ \"name\": \"Apple MacBook Pro 16\", \"data\": { \"year\": 2019, \"price\": 2049.99, \"CPU model\": \"Intel Core i9\", \"Hard disk size\": \"1 TB\", \"color\": \"silver\" } }"
 - bearerToken: NULL (No Security)
 - contentType: "application/json" (See [Media type – Wikipedia](#) for other content-type examples)

1. Call the RESTApiClient/Put method with 'apiURL', 'requestBody', and contentType set.
2. Use VariableCommands/Set to gather the RESApiClient/Put Output Parameters



Hands-on session

- Open D01_01 REST_Start
- Add RESTAPIClient NetLogic from Library
- Create method calls on premade GET, POST, and PUT push buttons to complete their associated tasks against the RESTAPIClient NetLogic



Appendix – API Key vs Bearer Token

- Many Public REST APIs require an API Key and or Bearer Token for access
- API Keys are used for Application Authorization
- Bearer Tokens are used for User Authorization
- More Detail here: [Why and when to use API keys | Cloud Endpoints with OpenAPI | Google Cloud](#)

Appendix – Optix as REST Web Service

- [FactoryTalk-Optix/REST_WEB_WS_Server: Sample of hosting a REST, WEB and Web Socket Server in Optix](#)

Features

- All-in-one object to host WEB Server, REST Server, and Web Sockets in FTOptix
- Currently, no TLS is supported, but maybe in the future
- Provided as sample code
- When you use a REST Server, you will need to adjust the C# code to deal with all the different resources (Endpoints) of the server.
- If used as a Web Socket, there is a variable “Message” which is updated automatically.

Optix Runtime on Docker containers



Optix Runtime on Docker containers

- Optix supports running on x86_64 Ubuntu 22.04 devices
- Creating a container with Ubuntu 22.04 LTS allows the Update Server to be installed in the container
- Two ways to create containers:
 - Container with FactoryTalk Optix Update Server: the Runtime can be downloaded from the IDE to the container
 - Container with the FactoryTalk Optix Runtime: the exported Linux binaries are packed to a Docker container



Optix Runtime on Docker containers

- Procedure overview:
 1. Create base container with Ubuntu 22.04
 2. Install required libraries and dependencies
 3. Copy the FactoryTalk Optix Application Update Service installer
 4. Install the Update Service process
 5. Set the Update Service as entrypoint for the container
 6. Deploy the container



Full procedure with links
and images

Optix Runtime on Docker containers

- Limitations:
 - Cannot use Native Presentation Engine
 - Containers can be licensed only if permanently connected to the internet
 - No offline licensing mechanism (as per V1.6)
 - License is verified to the Rockwell infrastructure every 30 mins
 - Entitlement number is passed as Environment variable to the container
 - No support for arm64 devices (Raspberry Pi, etc)
 - Update Server cannot be upgraded from the IDE
 - A new container must be rebuilt

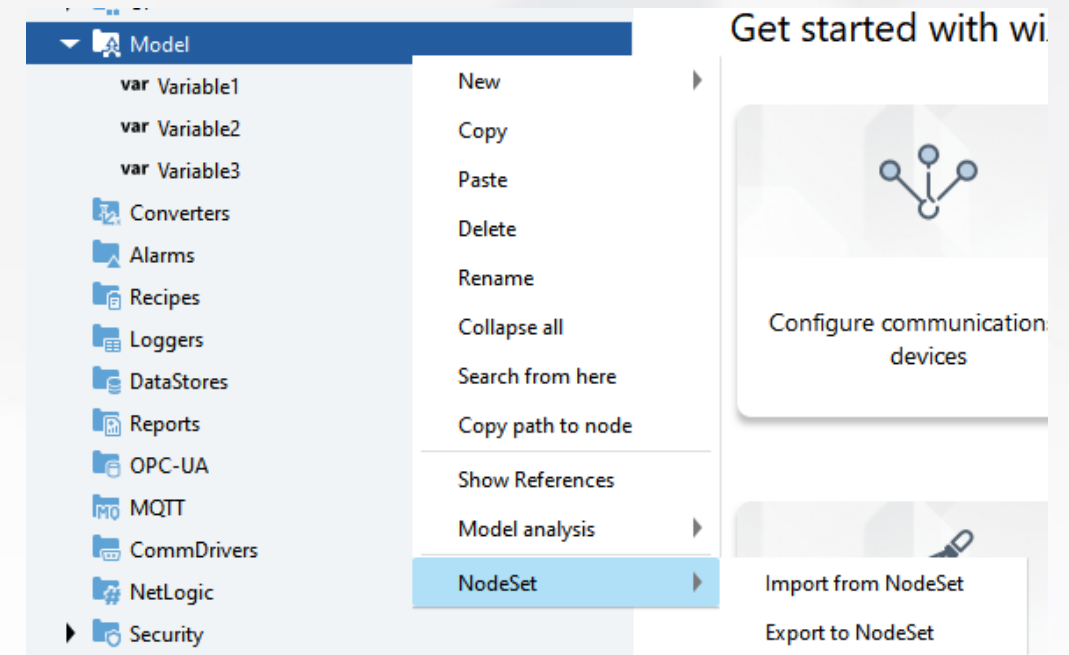


OPC-UA NodeSet



OPC-UA NodeSet

- A NodeSet is an XML file which describes a set of OPC-UA elements
- Can be used to:
 - Replicate a portion of the project tree
 - Import a set of tags from an OPC-UA server (offline import)
 - Perform bulk changes on project elements



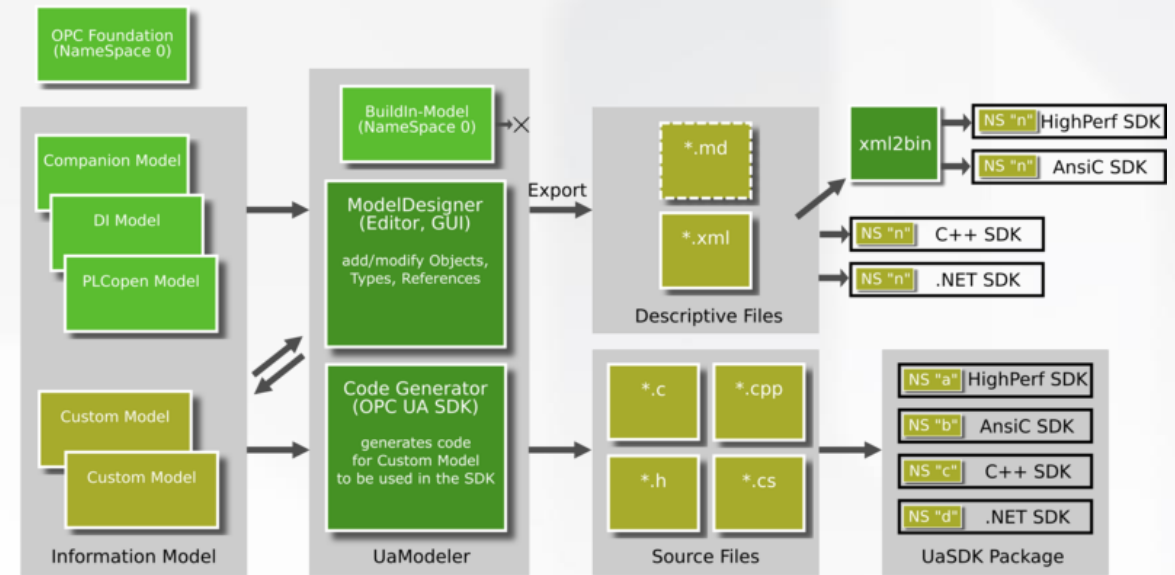
OPC-UA NodeSet

- A NodeSet is an XML file which describes a set of OPC-UA elements
- Can be used to:
 - Replicate a portion of the project tree
 - Import a set of tags from an OPC-UA server (offline import)
 - Perform bulk changes on project elements

```
<?xml version="1.0" encoding="UTF-8"?>
<UANodeSet xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:uax="http://opcfoundation.org/UA/2008/02/Types.xsd" xmlns:
  <NamespaceUris>
    <Uri>NewHMIPProject35</Uri>
    <Uri>urn:FTOptix:HMIPProject</Uri>
  </NamespaceUris>
  <Models>
    <Model ModelUri="NewHMIPProject35">
  </Models>
  <Aliases/>
  <UAVariable NodeId="ns=1;g=daa41e1b-1136-bbe5-f7f1-970d2b2fdde7" ParentNodeId="ns=1;g=8664f4b1-8e8c-4962-2e12-97fe51b45d41" Br
    <References>
      <Value>
        <uax:Int32>0</uax:Int32>
      </Value>
    </UAVariable>
    <UAVariable NodeId="ns=1;g=168c3a30-d679-7d23-42d1-8a8bc38802ee" ParentNodeId="ns=1;g=8664f4b1-8e8c-4962-2e12-97fe51b45d41" Br
      <References>
        <Value>
          <uax:Int32>0</uax:Int32>
        </Value>
      </UAVariable>
      <UAVariable NodeId="ns=1;g=112d5751-d851-9962-ba4d-8e5611dcb970" ParentNodeId="ns=1;g=8664f4b1-8e8c-4962-2e12-97fe51b45d41" Br
        <References>
          <Value>
            <uax:Int32>0</uax:Int32>
          </Value>
        </UAVariable>
        <UAObject NodeId="ns=1;g=8664f4b1-8e8c-4962-2e12-97fe51b45d41" BrowseName="1:Model" WriteMask="69876">
          <References>
        </UAObject>
  </UANodeSet>
```

OPC-UA NodeSet

- NodeSet can be created with a variety of tools like UaModeler from the OPC Foundation
- Most third-parties software can generate NodeSet files

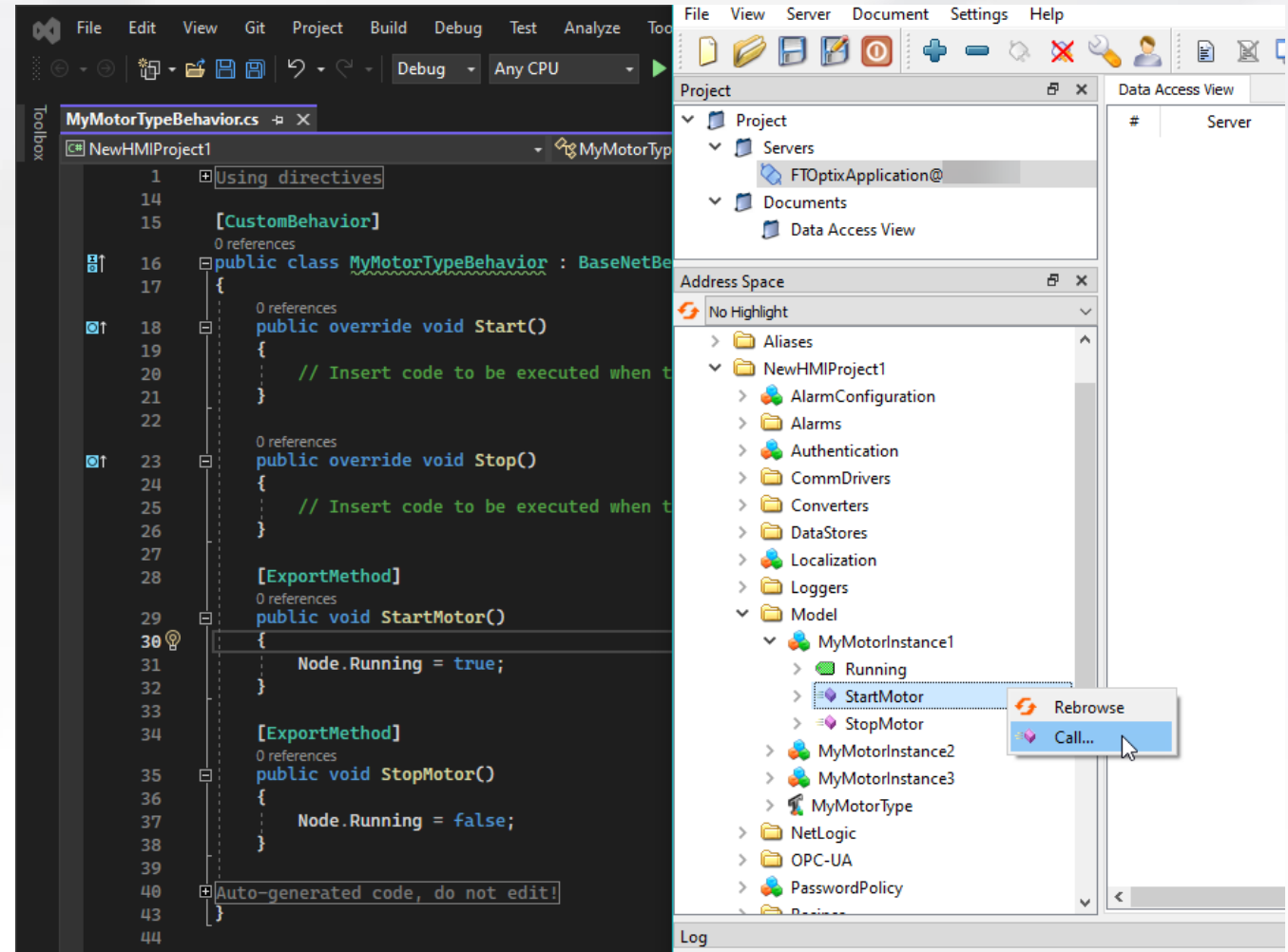


OPC-UA Custom Behavior



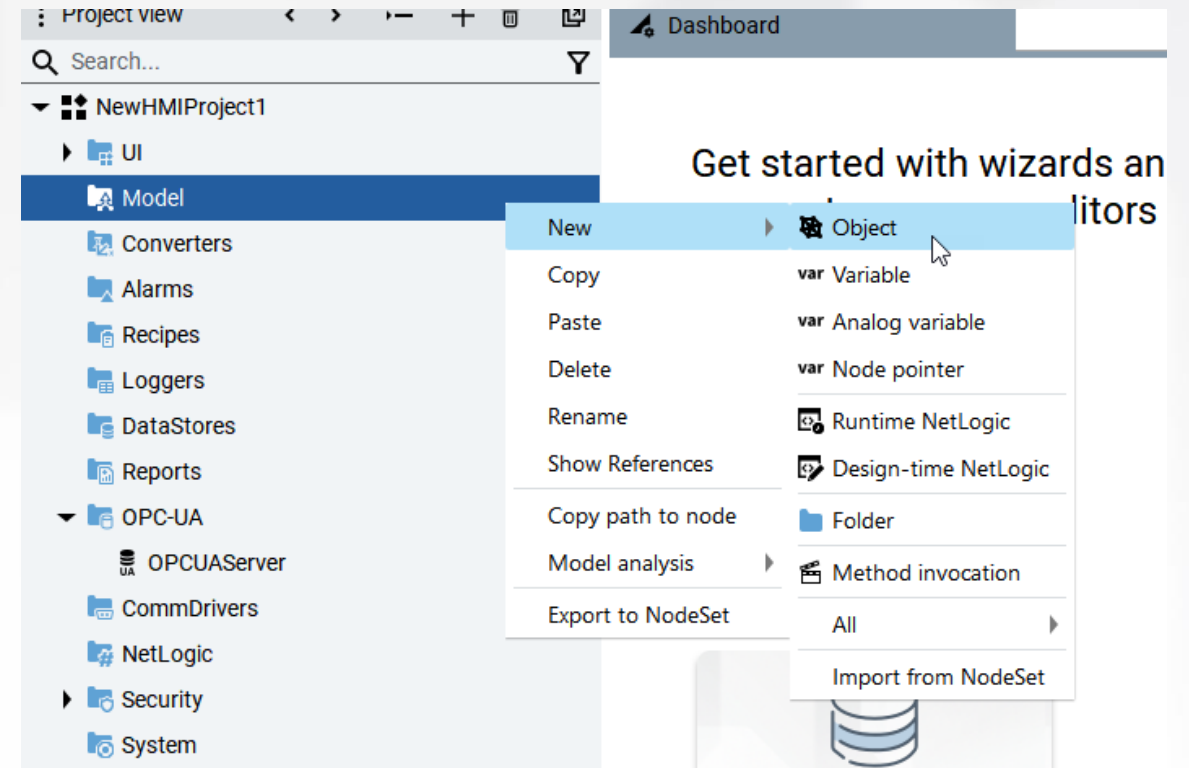
OPC-UA Custom Behavior

- When creating custom OPC-UA Object Types in FactoryTalk Optix, some methods can be exposed via OPC-UA to be called from a client
- Custom behaviors can only be added to IUAObject types
- Instances will inherit all the exposed methods
- Methods must return void, output shall be passed via the out modifier of method's arguments
- Properties of the IUAObject can be accessed using the «node» syntax



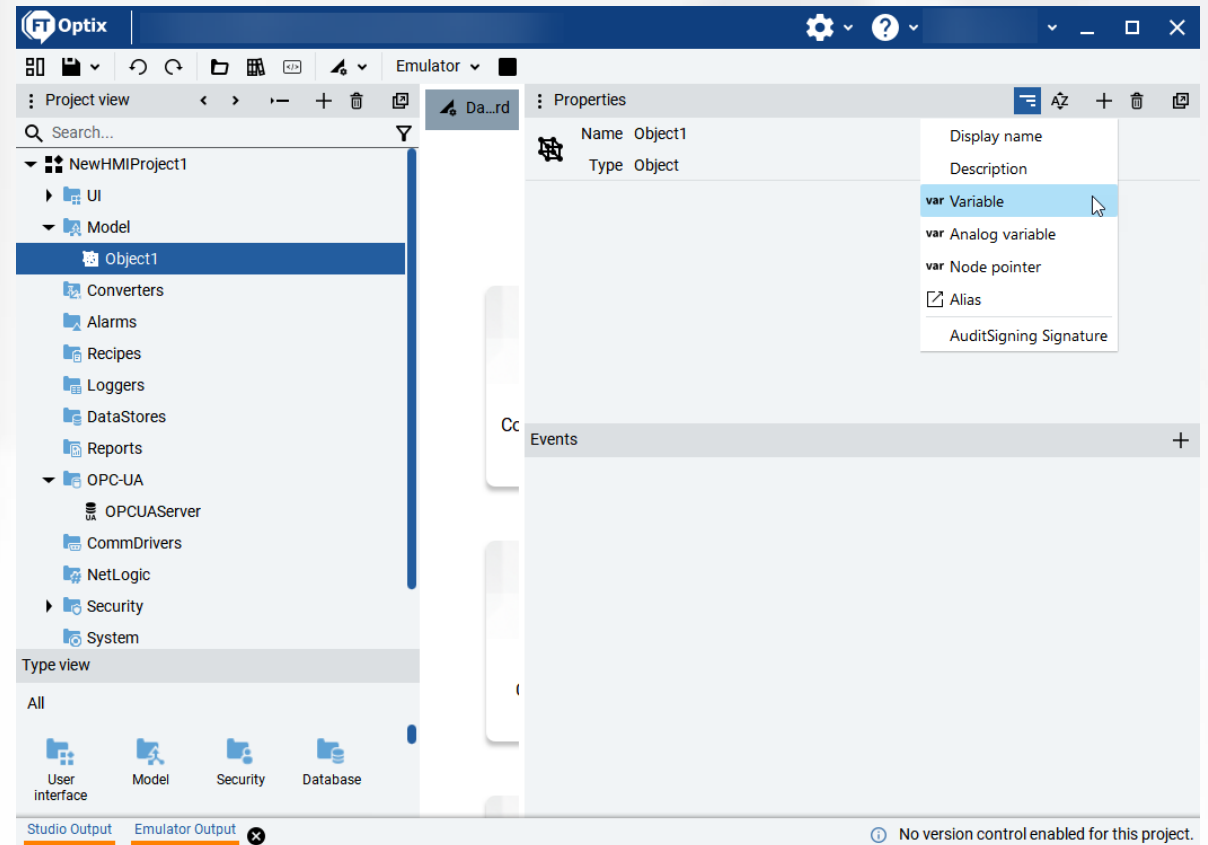
Creating an IUObject with a custom behavior

- Create new IUObject
- Add properties to the object
- Convert object to object type
- Right click the object type to add the custom behavior
- Double click the object type to open the code editor and configure the methods



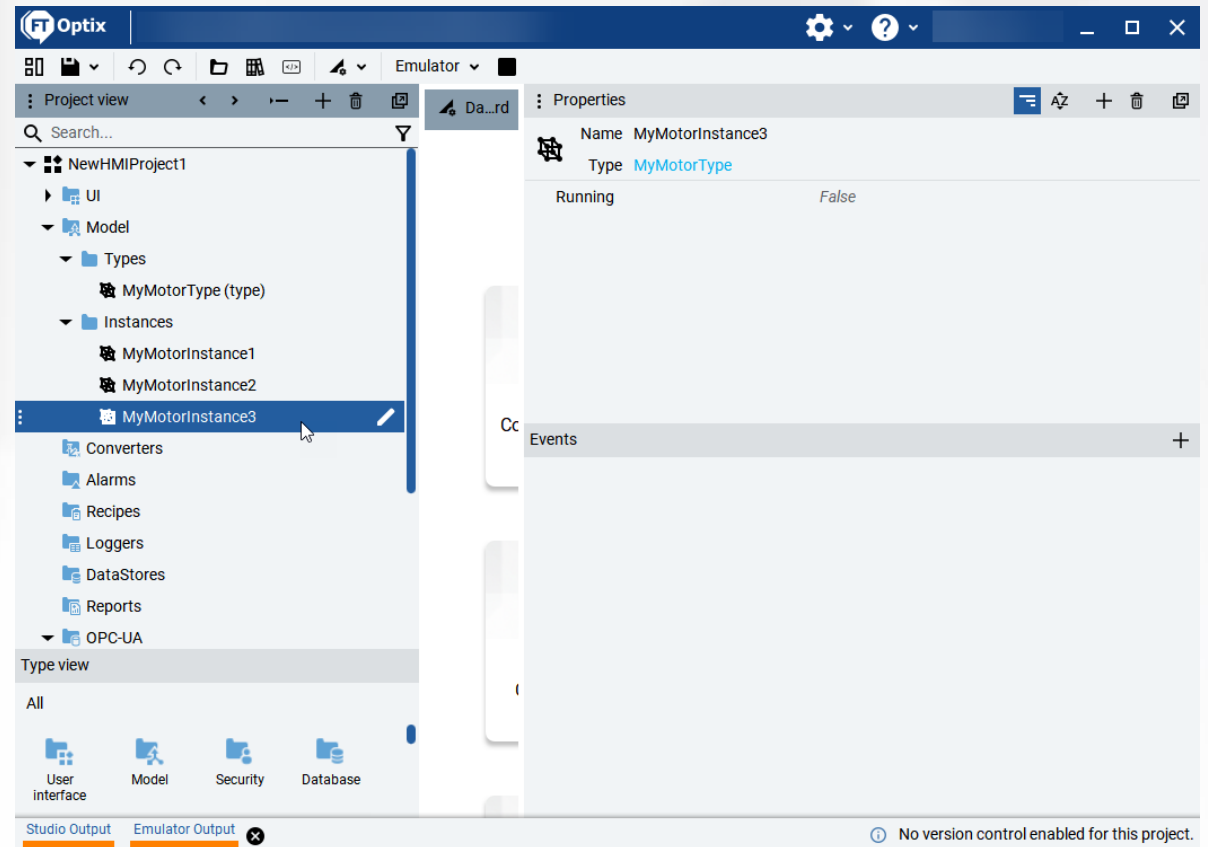
Creating an IUObject with a custom behavior

- Create new IUObject
- Add properties to the object
- Convert object to object type
- Right click the object type to add the custom behavior
- Double click the object type to open the code editor and configure the methods



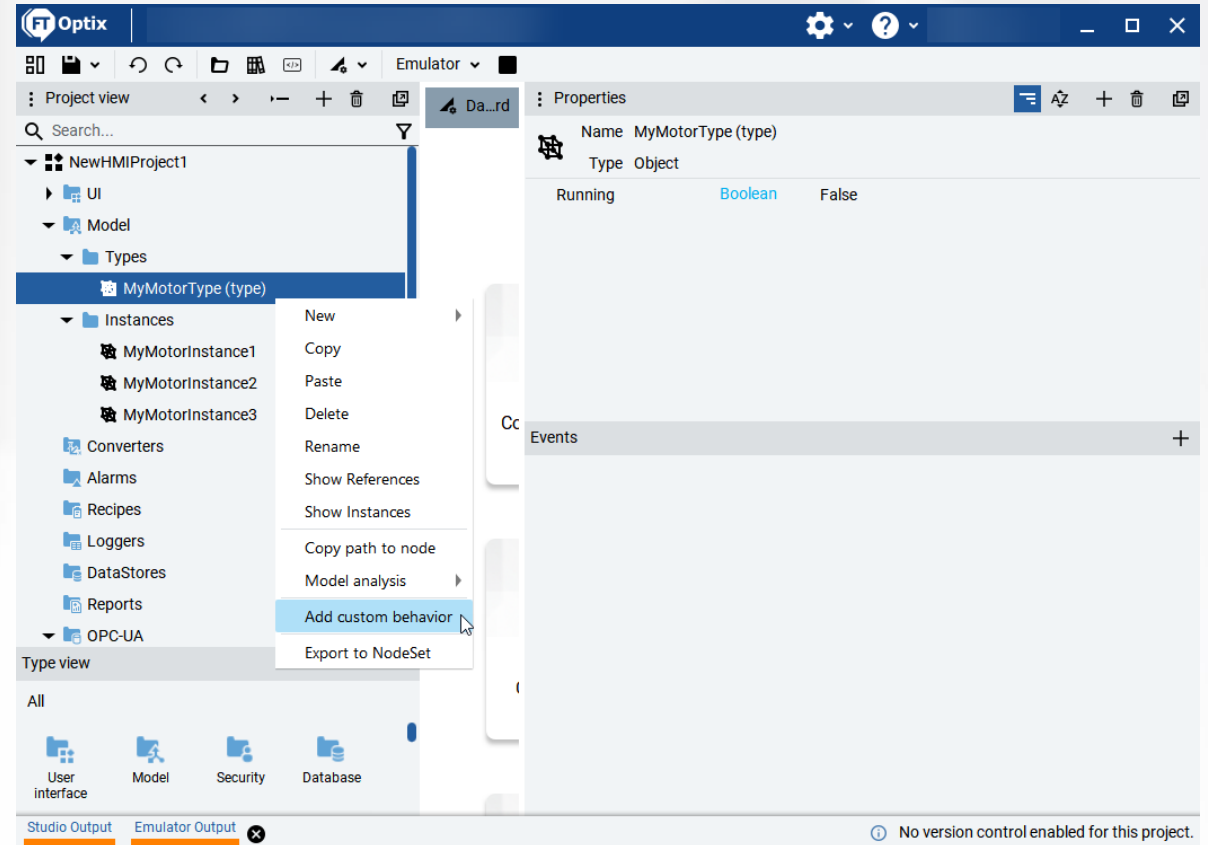
Creating an IUAObject with a custom behavior

- Create new IUAObject
- Add properties to the object
- Convert object to object type
- Right click the object type to add the custom behavior
- Double click the object type to open the code editor and configure the methods



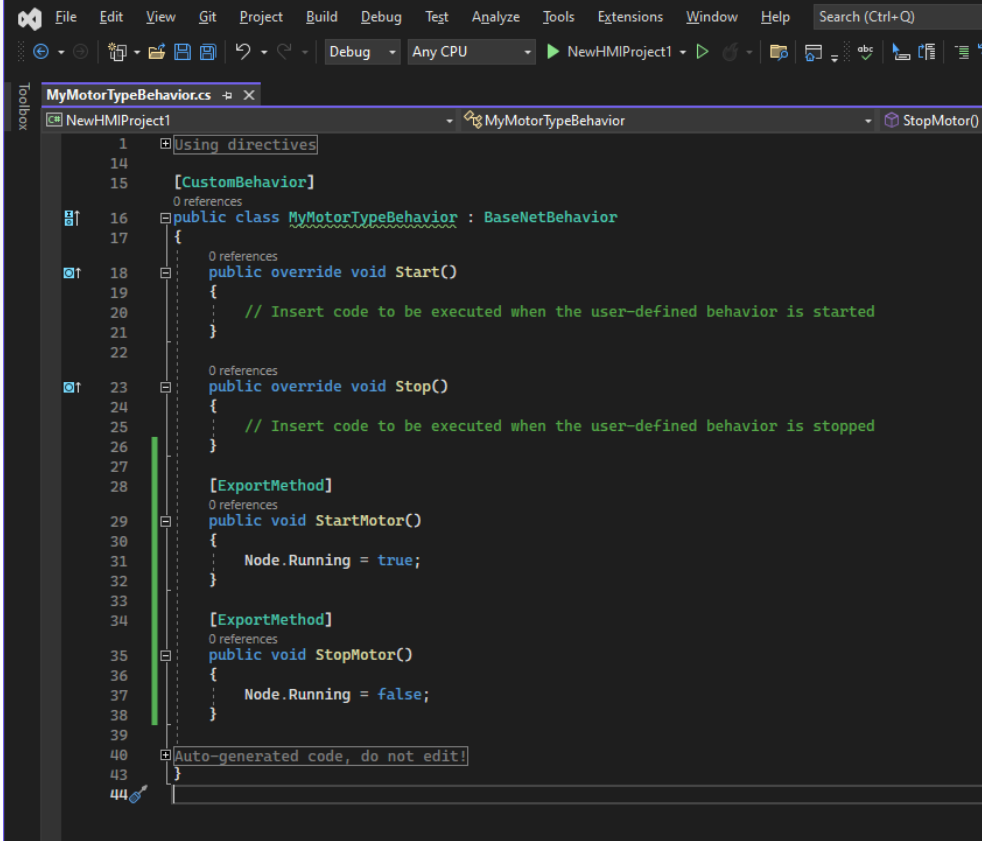
Creating an IUAObject with a custom behavior

- Create new IUAObject
- Add properties to the object
- Convert object to object type
- Right click the object type to add the custom behavior
- Double click the object type to open the code editor and configure the methods



Creating an IUObject with a custom behavior

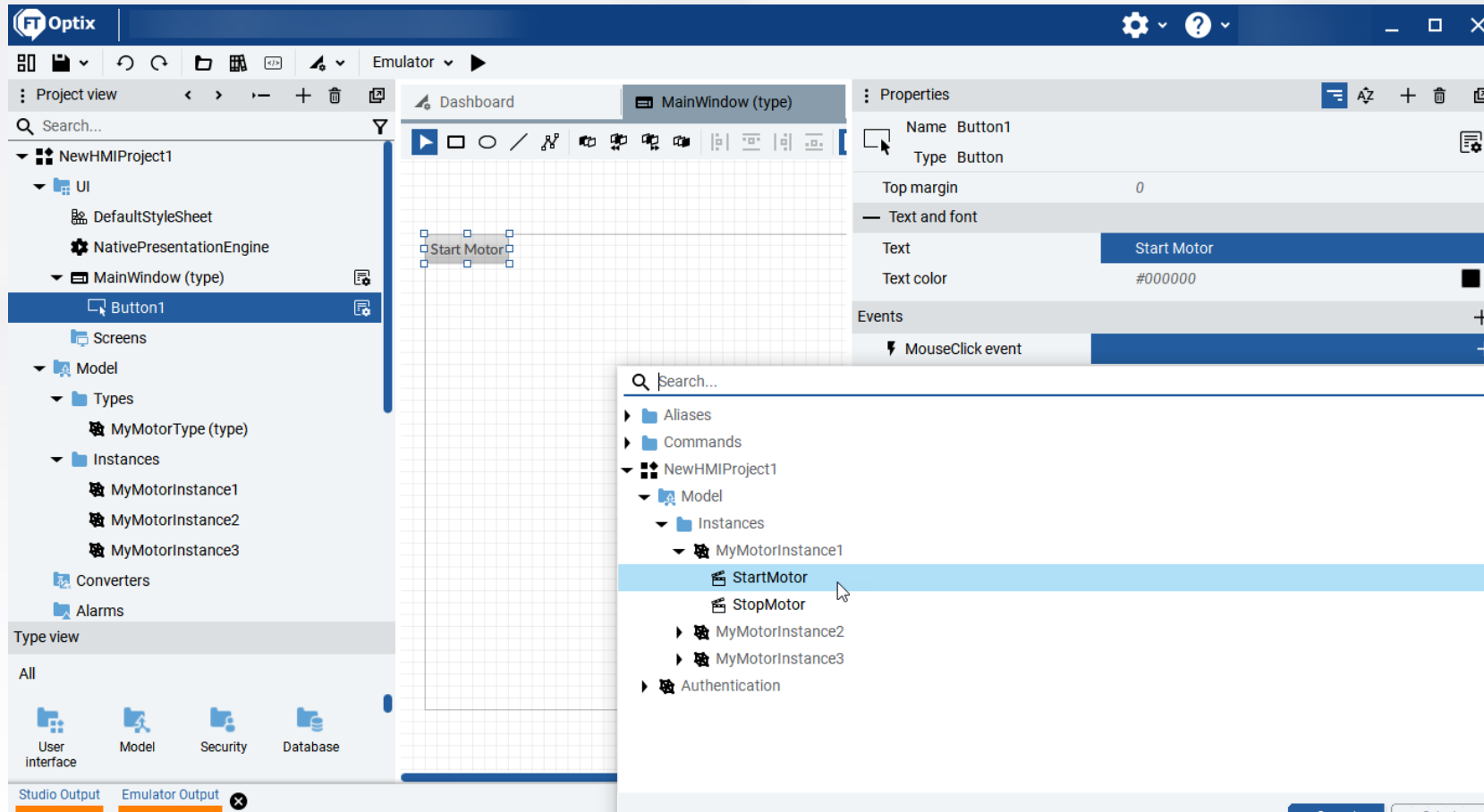
- Create new IUObject
- Add properties to the object
- Convert object to object type
- Right click the object type to add the custom behavior
- Double click the object type to open the code editor and configure the methods



```
1  [Using directives]
14
15  [CustomBehavior]
16  0 references
17  public class MyMotorTypeBehavior : BaseNetBehavior
18  {
19      0 references
20      public override void Start()
21      {
22          // Insert code to be executed when the user-defined behavior is started
23      }
24
25      0 references
26      public override void Stop()
27      {
28          // Insert code to be executed when the user-defined behavior is stopped
29      }
30
31      [ExportMethod]
32      0 references
33      public void StartMotor()
34      {
35          Node.Running = true;
36      }
37
38      [ExportMethod]
39      0 references
40      public void StopMotor()
41      {
42          Node.Running = false;
43      }
44  }
```

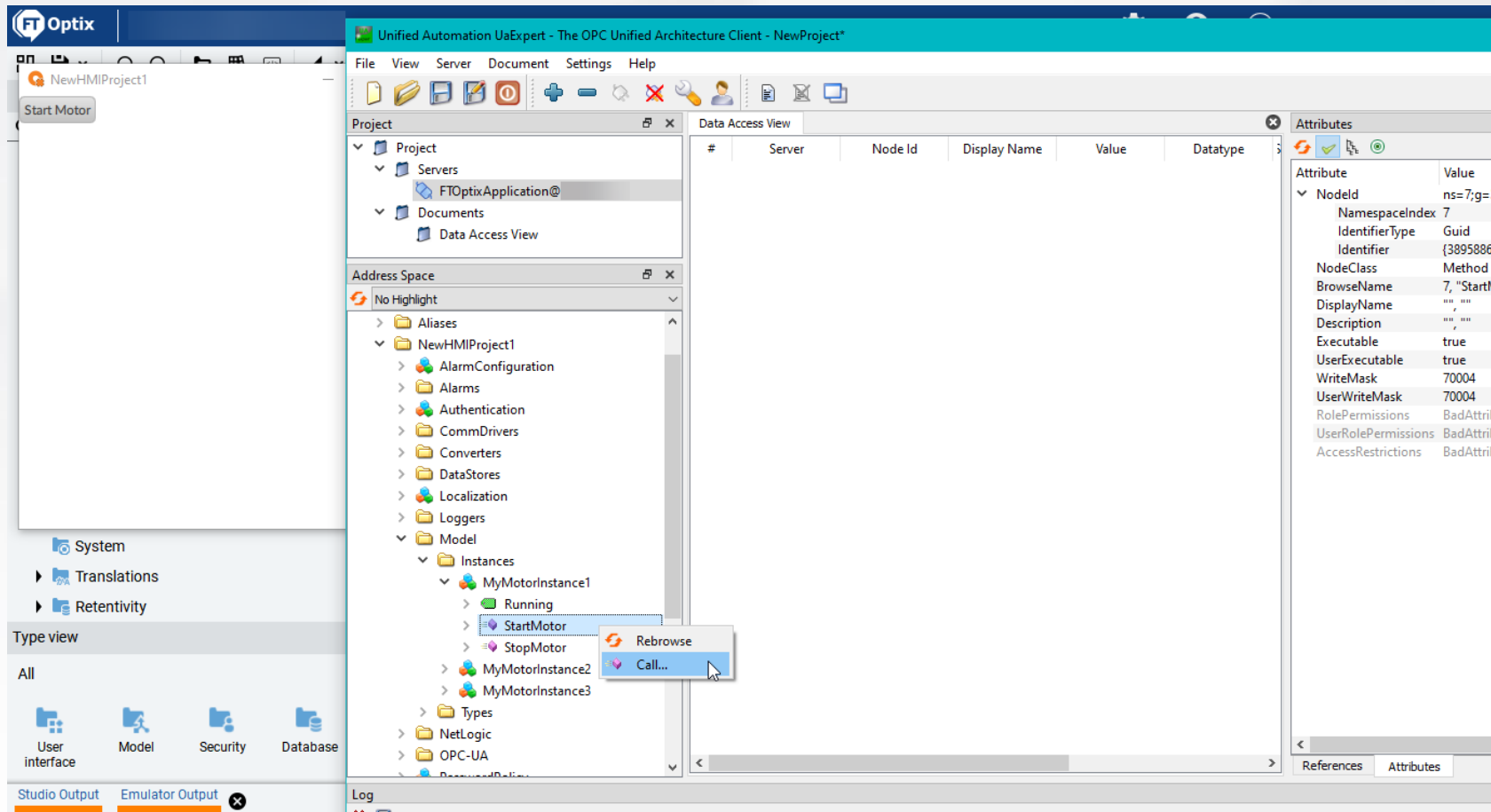
Usage of a custom behavior

- Custom behaviors can be used to call methods via OPC-UA
- Custom behaviors can be used to call methods from FactoryTalk Optix UI Controls or events



Usage of a custom behavior

- Custom behaviors can be used to call methods via OPC-UA
- Custom behaviors can be used to call methods from FactoryTalk Optix UI Controls or events

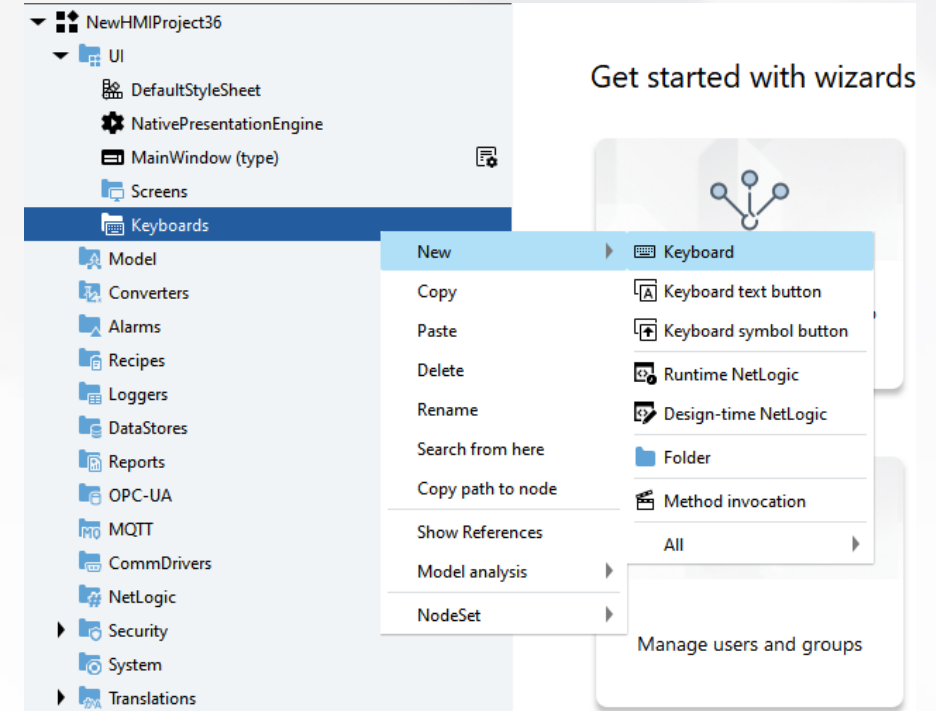


Custom keyboards



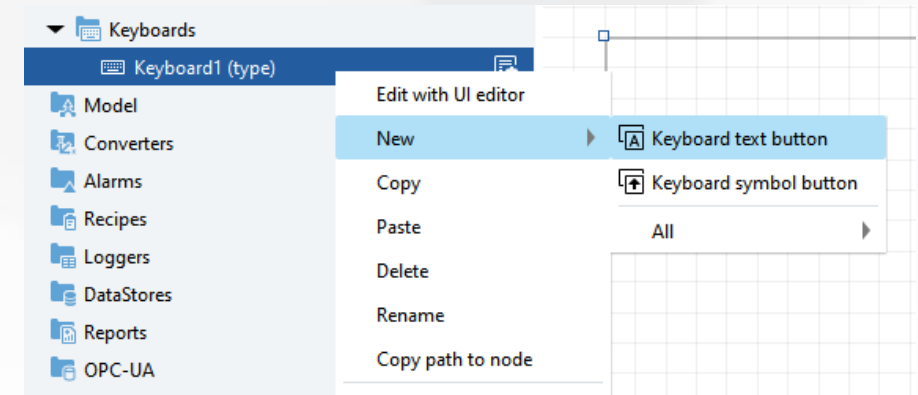
Custom keyboards

- Allows overriding the default Qt keyboard
- Allows creating custom logics before data is passed to the field
- Works both on Native UI and Web UI
- Can be fully customized
 - Custom layouts
 - Custom keys for characters
 - Custom keys for actions or symbols



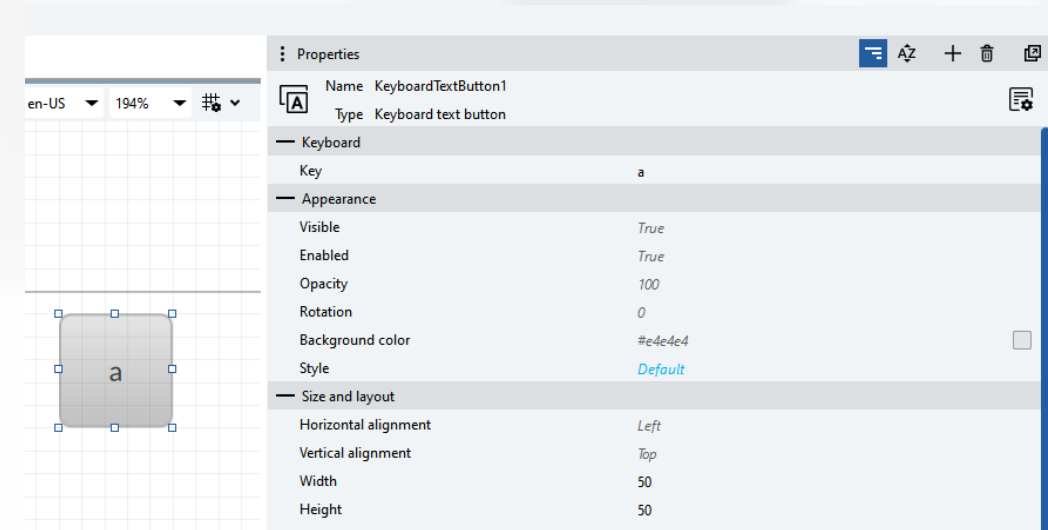
Custom keyboards

- Allows overriding the default Qt keyboard
- Allows creating custom logics before data is passed to the field
- Works both on Native UI and Web UI
- Can be fully customized
 - Custom layouts
 - Custom keys for characters
 - Custom keys for actions or symbols



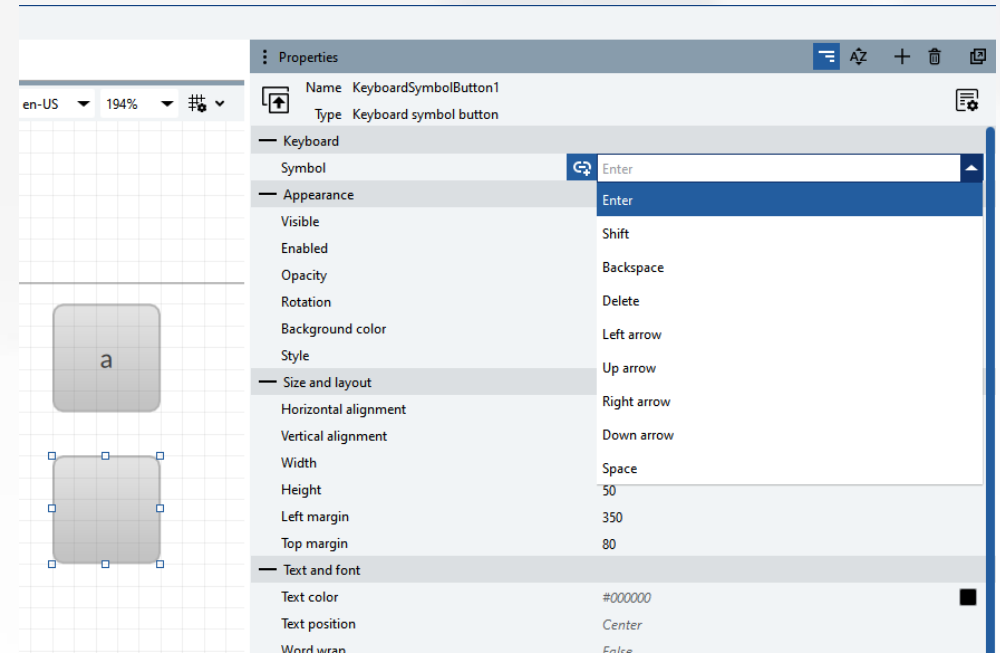
Custom keyboards

- Allows overriding the default Qt keyboard
- Allows creating custom logics before data is passed to the field
- Works both on Native UI and Web UI
- Can be fully customized
 - Custom layouts
 - Custom keys for characters
 - Custom keys for actions or symbols



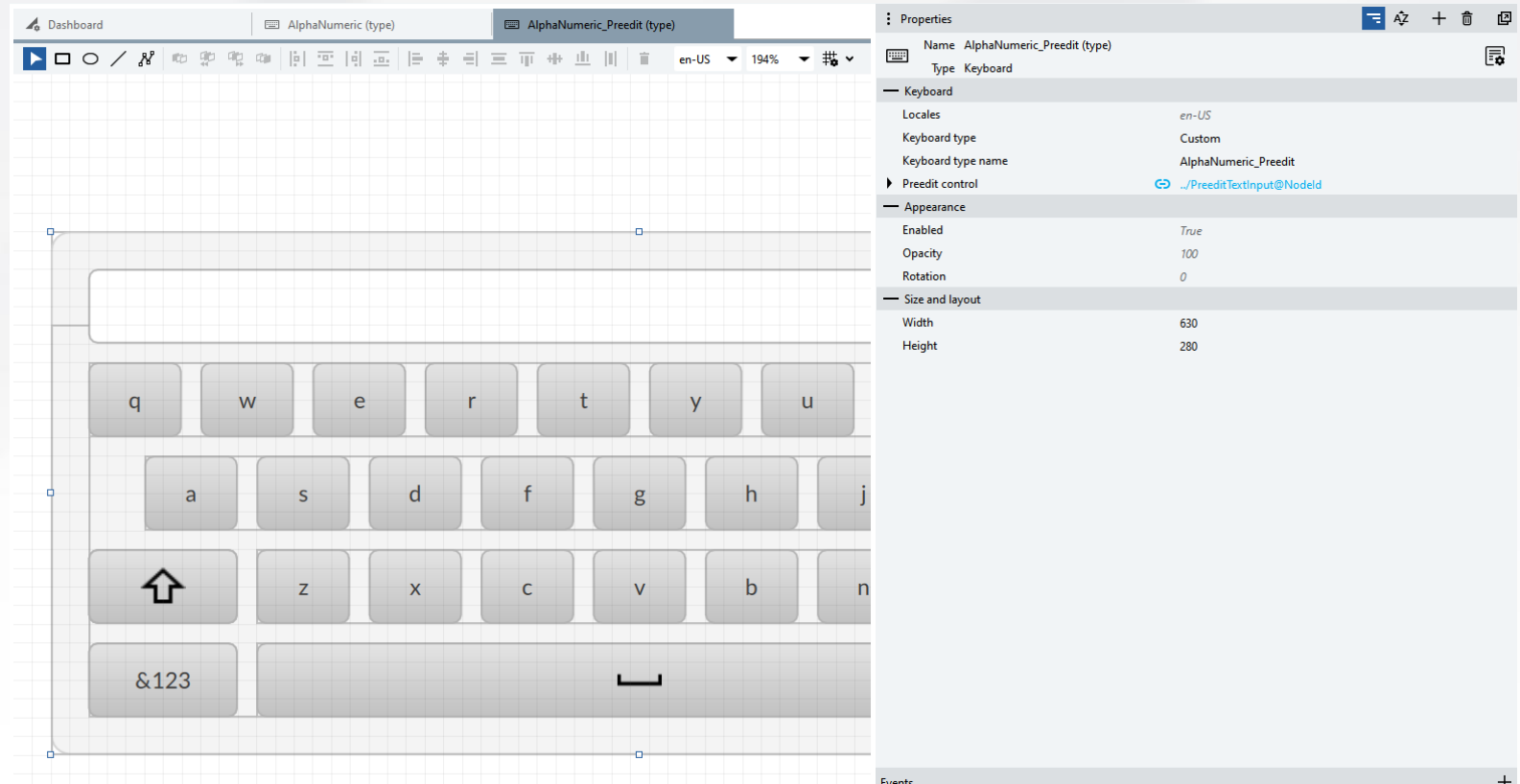
Custom keyboards

- Allows overriding the default Qt keyboard
- Allows creating custom logics before data is passed to the field
- Works both on Native UI and Web UI
- Can be fully customized
 - Custom layouts
 - Custom keys for characters
 - Custom keys for actions or symbols



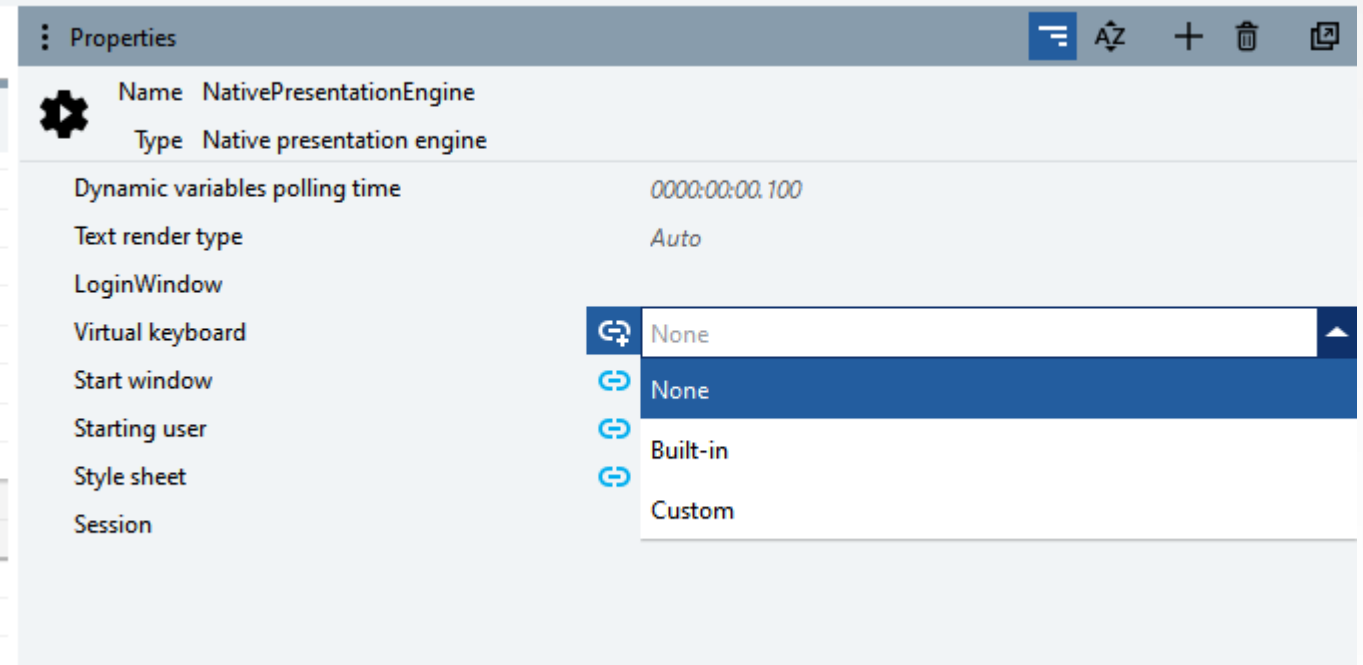
Custom keyboards

- A control (TextBox) can be added as preedit interface
 - Useful to double check the value before sending it to the field
 - Useful to create custom logic to the value (length check, RegEx, etc)



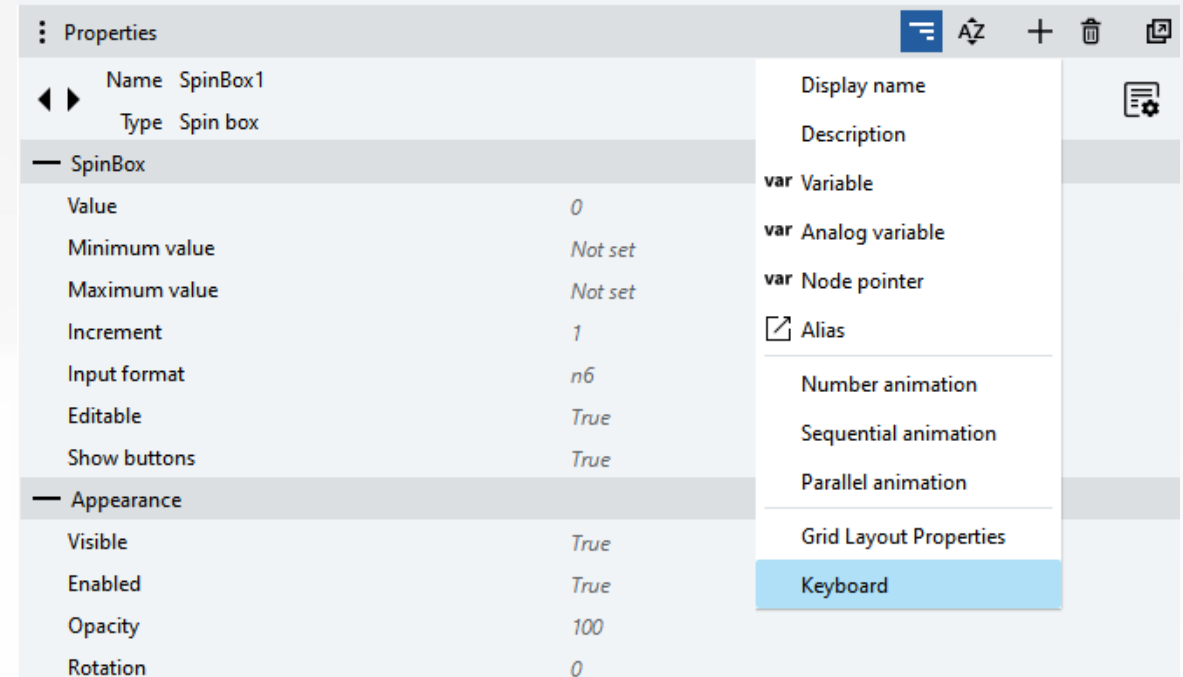
Custom keyboards

- Keyboard selection is done at the Presentation Engine level
 - Built-in: default Qt keyboard
 - Custom: the custom keyboard for that control type



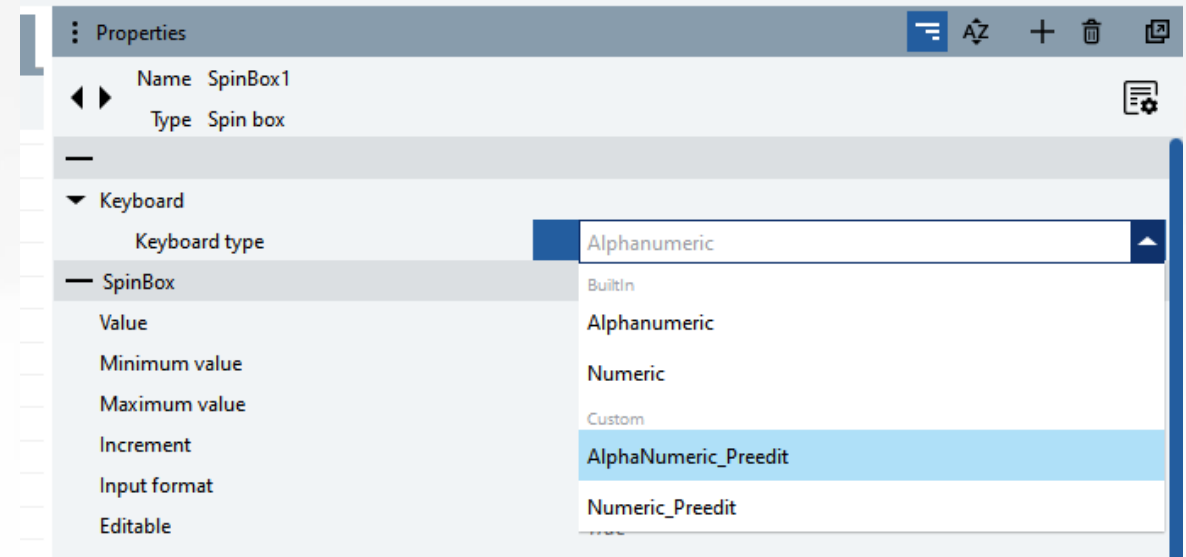
Custom keyboards

- Different UI controls can trigger different keyboards
 - Need to add the attached property "Keyboard" to the control
 - Configure the specific keyboard to be invoked
 - If no custom attached property is added, the default custom keyboard for that kind of control (numeric or alphanumeric) is triggered



Custom keyboards

- Different UI controls can trigger different keyboards
 - Need to add the attached property "Keyboard" to the control
 - Configure the specific keyboard to be invoked
 - If no custom attached property is added, the default custom keyboard for that kind of control (numeric or alphanumeric) is triggered

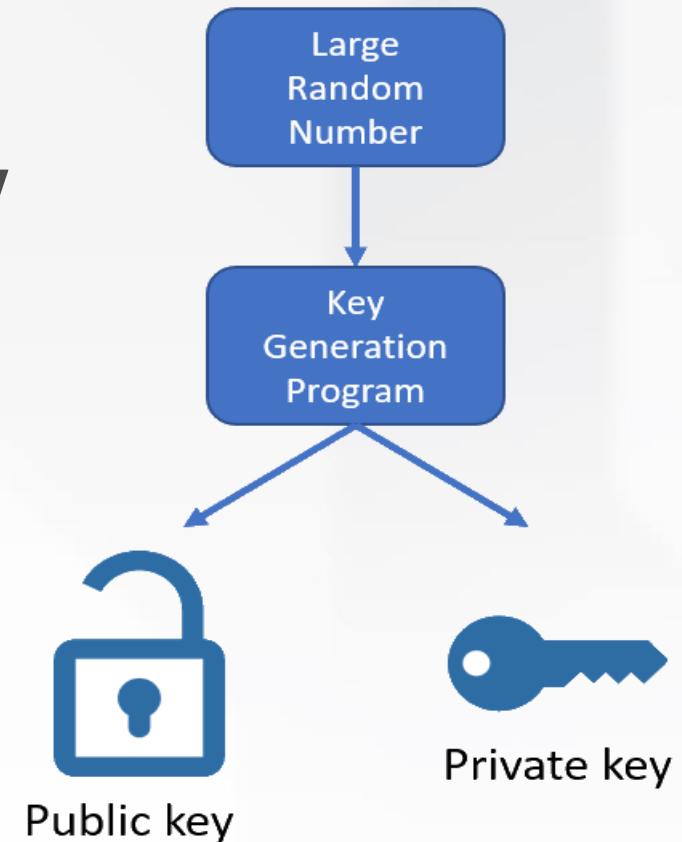


Appendix: see the light about OPC-UA Security



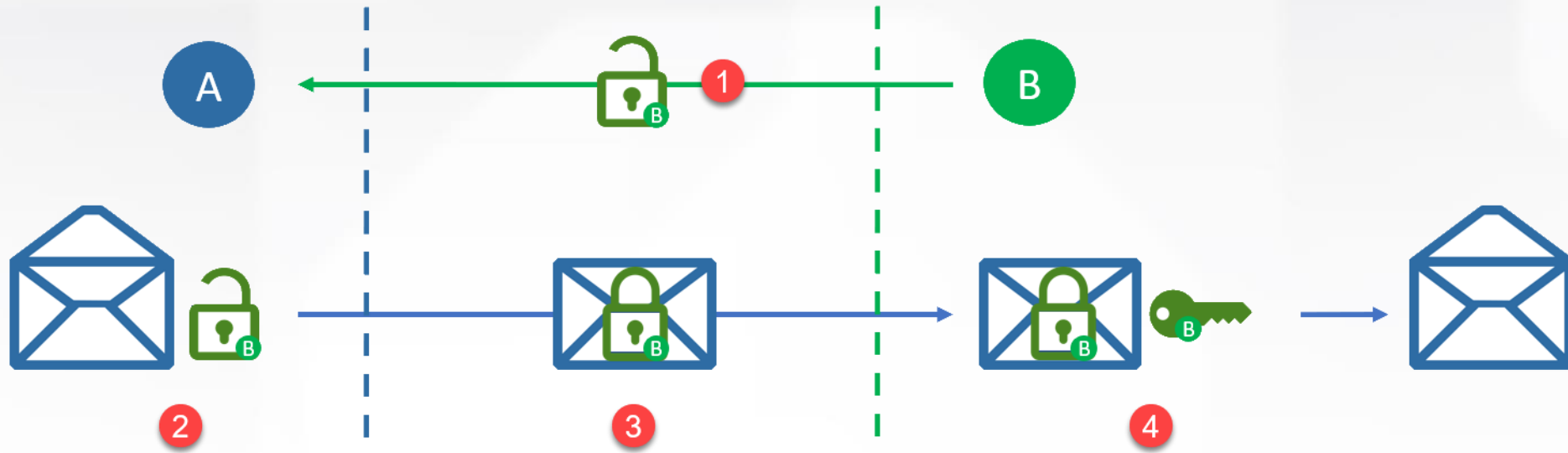
Asymmetric cryptography

- is a cryptographic system that **uses pairs of keys**
- each pair consists of a **public key** and a **private key**
- the generation of such key pairs depends on cryptographic algorithms



How does encryption work ?

1. Alice asks and receives Bob's padlock (Public Key exchange)
2. Alice uses the padlock to close the letter
3. Alice sends the locked letter to Bob
4. When Bob receives the letter, he can open it with his key (Private Key) of which he is the only owner.



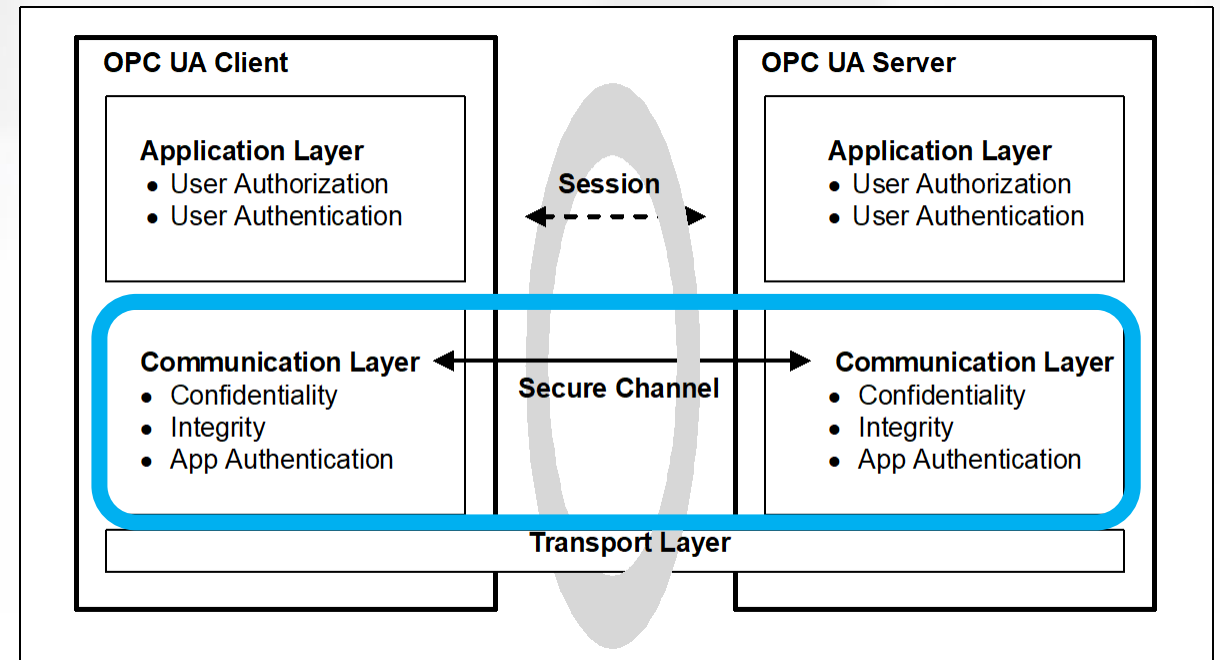
How does encryption apply to OPC-UA?

- When the OPC-UA Client connects to the OPC-UA Server:

1. X.509 Certificates are exchanged so a **Secure Channel** is established

Certificates used here are called
"Application Instance Certificates"

These are the certificates
exchanged when establishing
a secure communication
between OPC-UA Client and
Server



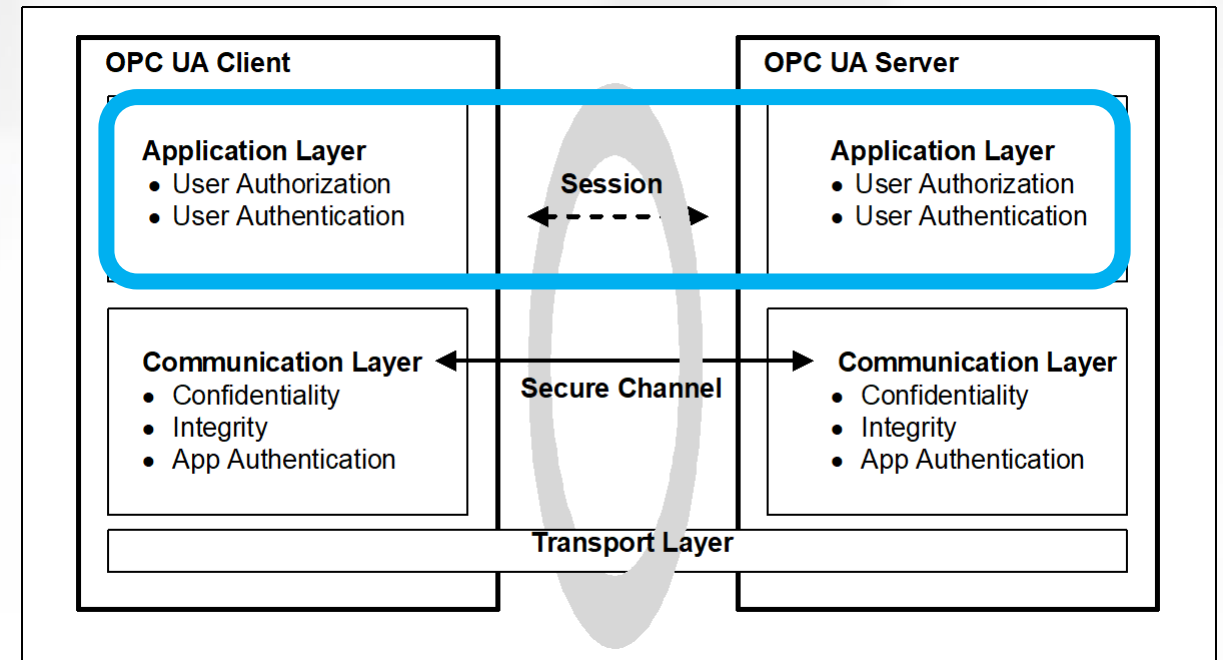
How does encryption apply to OPC UA?

2. Over the Secure Channel, a "**Session**" at "Application Layer" is defined.

Here is where the OPC-UA Client application and OPC-UA Server application transmit information, settings, and commands.

at Application Layer
there is the User Authentication
and User Authorization

- *anonymous,*
- *username/password*
- *X.509 Certificate*
(*User Instance Certificate*)



Secure channel security

- Secure Channel security = Security Profile + Security Policy

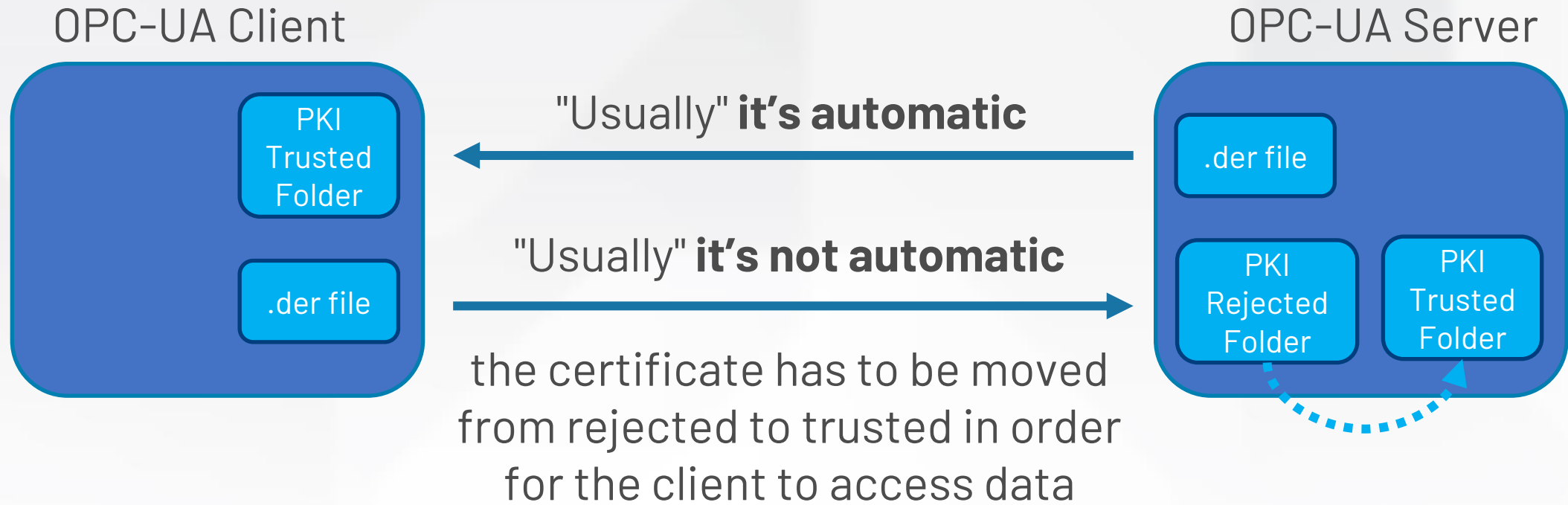
Security Profiles	Description
None	No security
Sign	The sender uses <u>his private key</u> to digitally sign the data. The receiver verifies whether the data comes from the expected sender using the <u>sender's public key</u>
Sign+Encrypt	In addition to Sign, the sender encrypt data using <u>receiver's public key</u>

Security Policies (length of the key and the algo)
None
Basic256Sha256
Aes128-Sha256-RsaOaep
Aes256-Sha256-RsaPss

PKI and X.509 Certificates

- Public Key Infrastructure (PKI)
 - is a system for the creation, storage, and distribution of digital certificates
- X.509
 - is a standard defining the format of public key certificates
- In OPC-UA:
 - **".der"** is the certificate file containing **public key**,
 - **".pem"** is the **private key** file

OPC-UA certificate exchange



Generating OPC-UA certificates

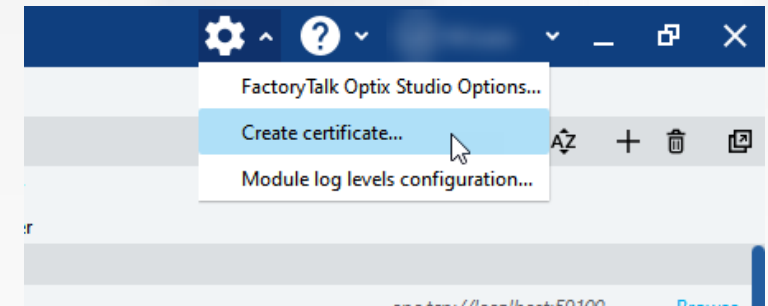


Generating OPC-UA certificates

- Certificates are generated by authorities
 - Authorities are special entities, trusted by a number of users/clients that can **validate a certificate** and inform all users/clients that a certificate is valid and can be trusted
- Authorities (CA) can be public or local
 - Public authorities are well-known and trusted organizations that can sign and validate certificates (almost) worldwide. Example include **DigiCert, Let's Encrypt, Comodo** and many more.
 - Private authorities are trusted services that can sign and validate certificates in a local (confined) environment. An example of a private authority can be the IT department of a company
- All certificates must be signed by an authority
 - After a certificate is generated, **an authority must sign it**
 - The certificate authority provides to all users/clients a notice with the list of valid and signed certificates
 - After a certificate is signed by an authority, all users/clients performing the handshake with the public key and the thumbprint of a signed certificate can be sure that nothing was tampered, and the connection can be established.
 - If the certificate is signed by an untrusted CA, it is rejected by default and the user is warned about potential risks.

Creating a self-signed certificate from FactoryTalk Optix Studio

- From the dropdown of the **Settings** menu, select **Create certificate**
- Fill-in all the details in the certificate creation form
- Select the certificate name
- Select the output folder
- Click **Create** to generate the certificates
- **Certificates created with FactoryTalk Optix Studio are self-signed and must be trusted manually by the clients**



Creating a self-signed certificate from FactoryTalk Optix Studio

- From the dropdown of the **Settings** menu, select **Create certificate**
- Fill-in all the details in the certificate creation form
- Select the certificate name
- Select the output folder
- Click **Create** to generate the certificates

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ + -
ftoptixruntime.local
localhost

IP addresses ⓘ + -

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

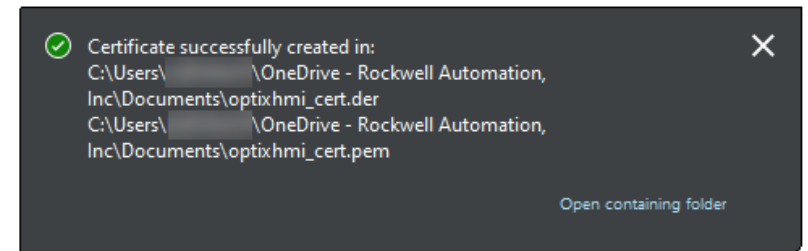
Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

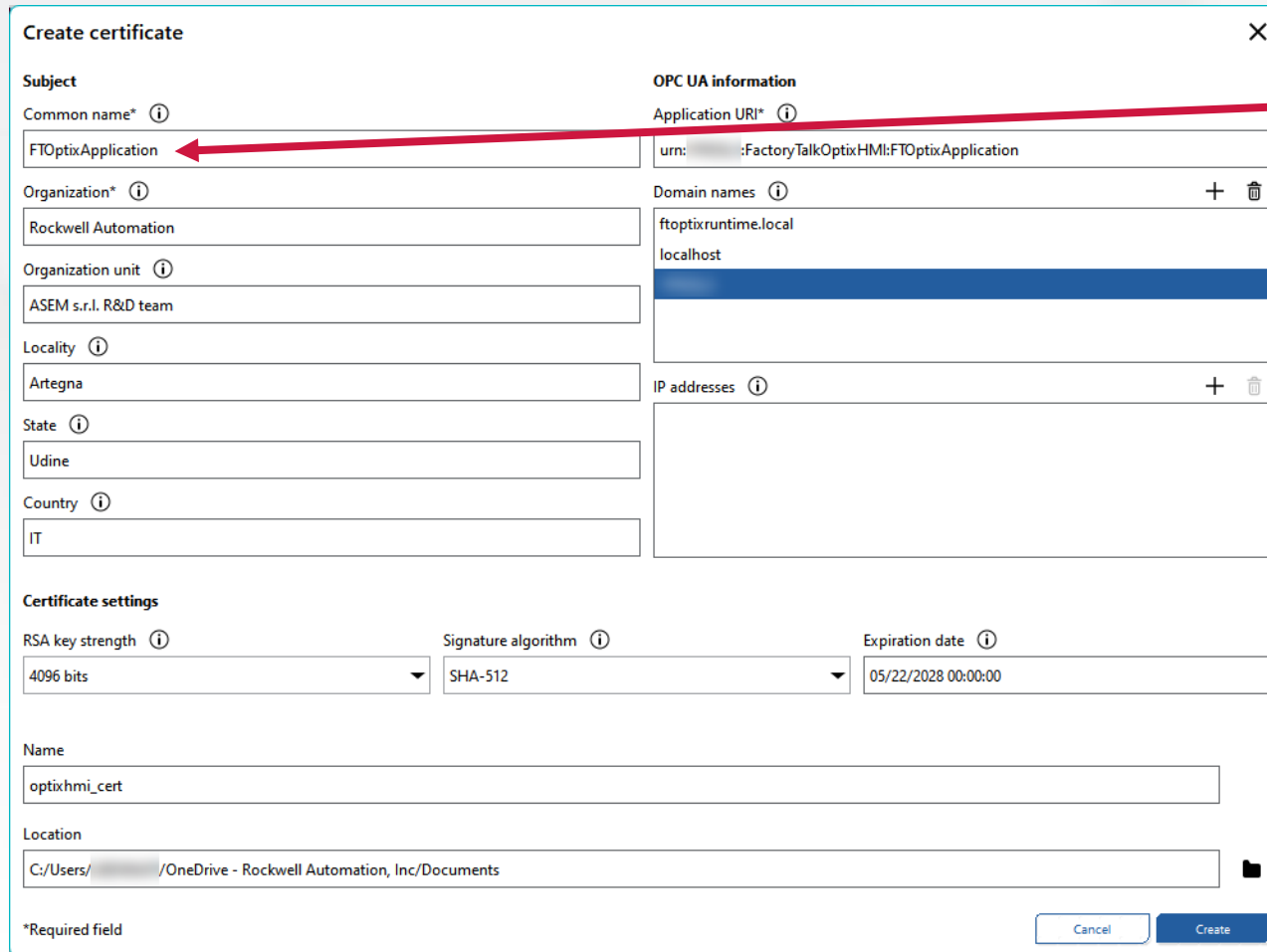
Cancel Create

Creating a self-signed certificate from FactoryTalk Optix Studio

- From the dropdown of the **Settings** menu, select **Create certificate**
- Fill-in all the details in the certificate creation form
- Select the certificate name
- Select the output folder
- Click **Create** to generate the certificates



How to fill the self-signed certificate generation fields



The screenshot shows a 'Create certificate' dialog box with the following fields and values:

Subject		OPC UA information	
Common name*	FTOptixApplication	Application URI*	urn: :FactoryTalkOptixHMI:FTOptixApplication
Organization*	Rockwell Automation	Domain names	ftoptixruntime.local localhost
Organization unit	ASEM s.r.l. R&D team	IP addresses	
Locality	Artegna		
State	Udine		
Country	IT		

Certificate settings

RSA key strength	Signature algorithm	Expiration date
4096 bits	SHA-512	05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Buttons: Cancel, Create

Common name: name of the OPC-UA application. By default, in FactoryTalk Optix, this is always “FTOptixApplication”

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI570plus-application

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

Organization: registration name of the company responsible for the certificate. Do not use abbreviations or non-alphanumeric characters.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 567

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegia

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
+ -
roptixruntime.local
localhost

IP addresses ⓘ
+ -

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel Create

Organization unit: local unit, office or group responsible for the certificate inside the Organization

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
192.168.1.100

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

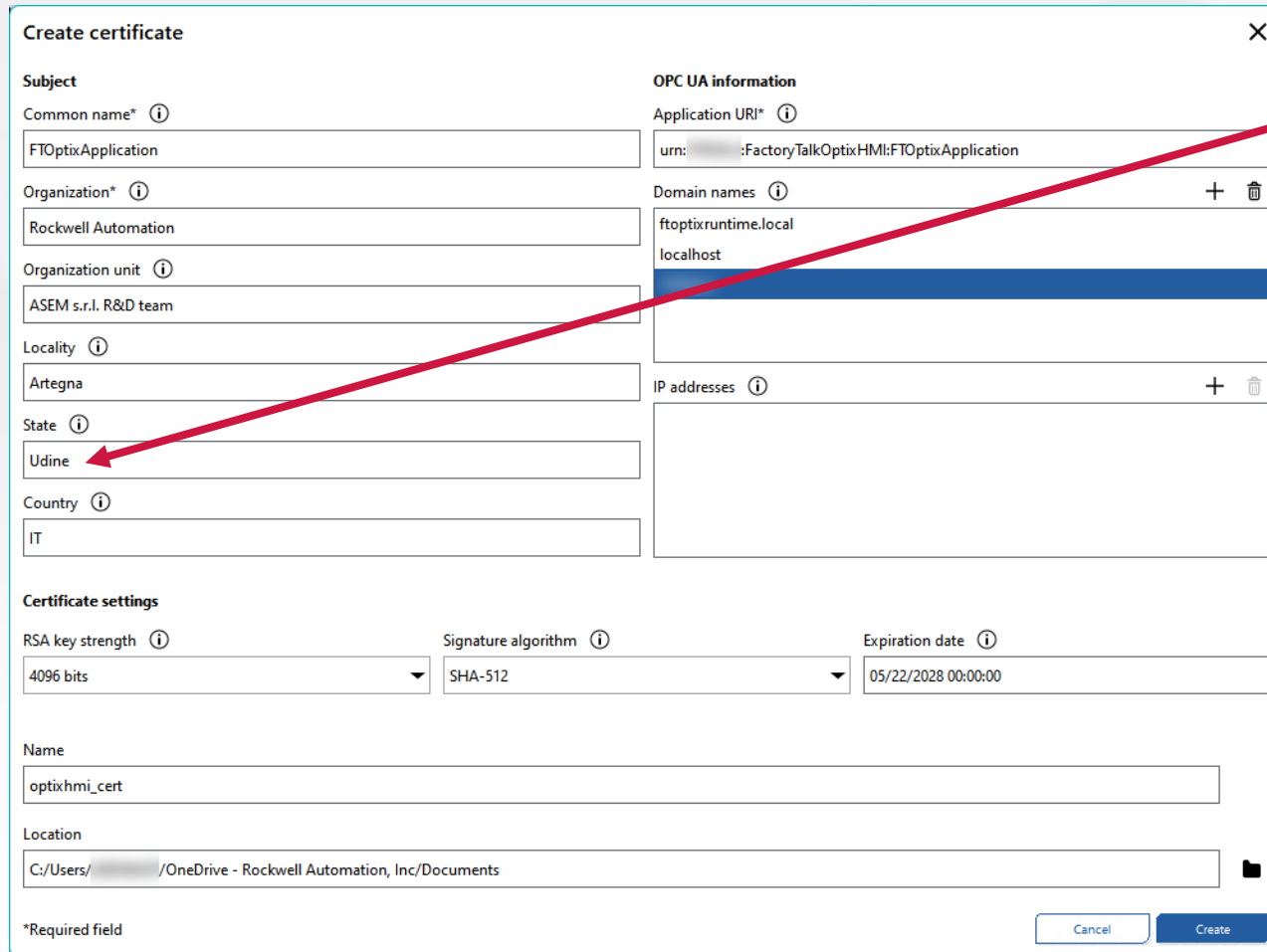
Locality: hometown of the Organization (or the Organization unit).

Must be the proper town name as specified in the postal address. Do not use abbreviations.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 569

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields



The screenshot shows a 'Create certificate' dialog box with the following fields and values:

- Subject**
 - Common name*: FTOptixApplication
 - Organization*: Rockwell Automation
 - Organization unit: ASEM s.r.l. R&D team
 - Locality: Artegna
 - State: Udine (highlighted with a red arrow)
 - Country: IT
- OPC UA information**
 - Application URI*: urn: :FactoryTalkOptixHMI:FTOptixApplication
 - Domain names: ftoptixruntime.local, localhost
 - IP addresses: (empty)
- Certificate settings**
 - RSA key strength: 4096 bits
 - Signature algorithm: SHA-512
 - Expiration date: 05/22/2028 00:00:00
- Name**: optixhmi_cert
- Location**: C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Buttons: Cancel, Create

State: geographical area where the Organization (or Organization unit) is located.

This can be an US state or a province.

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegia

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

Country: Two-letters ISO-3166 country code where the Organization (or Organization unit) is located.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc.

571

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

Application URI: the unique identifier for the project on a specific machine.

In FactoryTalk Optix, this must be the same as the Product URI in the server configuration

Properties

OPCUAServer

OPC UA server

Server

Endpoint URL
opc.tcp://localhost:59100

Nodes to publish

Maximum number of connections
1

Maximum number of subscriptions pe...
10

Use node path in NodeIds
False

Sampling interval
0000:00:00.100

Message array length
65536

Security

Minimum message security mode
Signature and cryptography

Minimum security policy
Basic256Sha256

Server certificate file
%PROJECTDIR%\PKI\Own\Certs\optixhmi_cert.der

Server private key file
%PROJECTDIR%\PKI\Own\Certs\optixhmi_cert.pem

Information

Product URI
urn:FactoryTalkOptixHMI:FTOptixApplication

Product name
FactoryTalkOptix HMI FTOptixApplication

Manufacturer name
Rockwell Automation, Inc.

Events

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 572

 FactoryTalk
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ + -
ftoptixruntime.local
localhost

IP addresses ⓘ + -

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel Create

Domain names: a list of domain names that can be used to access the machine.

This field contains a list of URLs (like myapplication.example.com) or Active Directory name for the host machine.

This field should contain all the available domains used to access the host machine either locally or remotely.

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

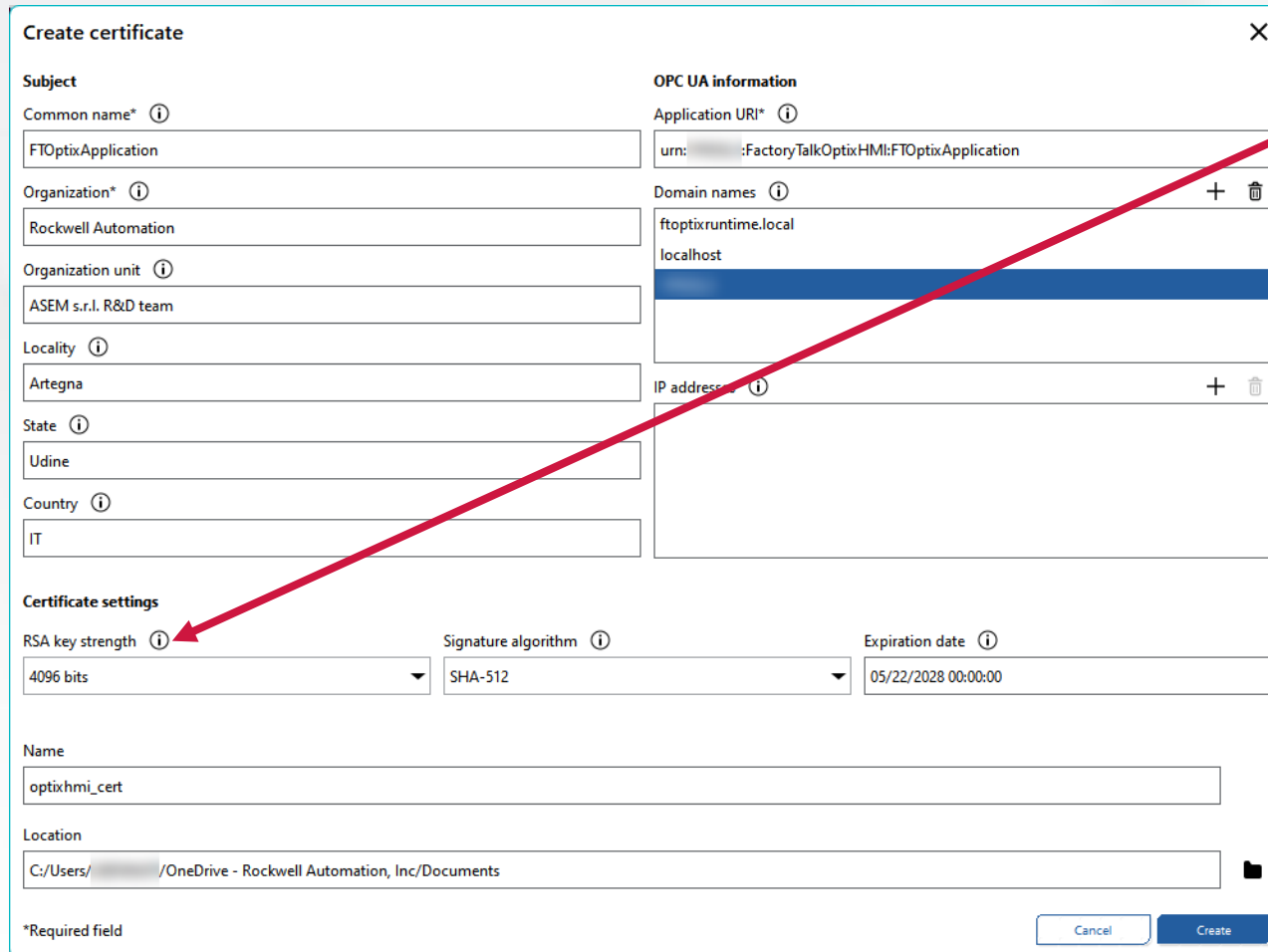
IP address: a list of IP addresses that can be used to access the machine.

This field is optional and should not be used in normal condition as the certificate would be invalidated if the IP address changes.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 574

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields



The screenshot shows a 'Create certificate' dialog box with the following fields and values:

- Subject**
 - Common name*: FTOptixApplication
 - Organization*: Rockwell Automation
 - Organization unit: ASEM s.r.l. R&D team
 - Locality: Artegna
 - State: Udine
 - Country: IT
- OPC UA information**
 - Application URI*: urn: :FactoryTalkOptixHMI:FTOptixApplication
 - Domain names: ftoptixruntime.local, localhost
 - IP address: (empty)
- Certificate settings**
 - RSA key strength: 4096 bits (highlighted by a red arrow)
 - Signature algorithm: SHA-512
 - Expiration date: 05/22/2028 00:00:00
- Name**: optixhmi_cert
- Location**: C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Buttons: Cancel, Create

RSA key strength: length in bits of the RSA key.

Longer keys are more secure, but need more computing resources to be processed.

It is preferable to choose the highest value that does not impact on the host hardware.

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegnà

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

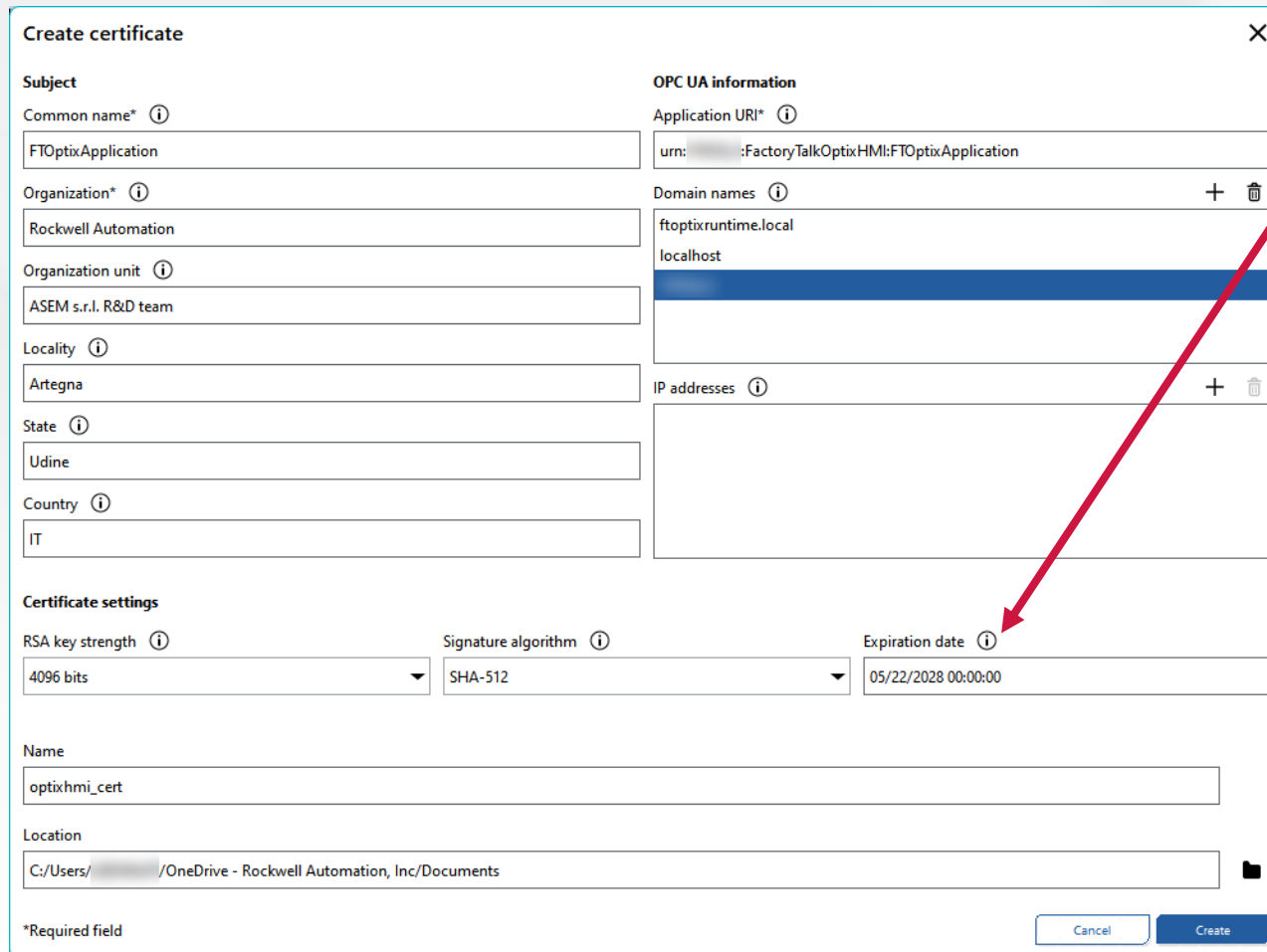
Create

Signature algorithm: algorithm used by the devices to sign the data using these certificates.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 576

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields



The screenshot shows a 'Create certificate' dialog box with the following sections:

- Subject**
 - Common name*: FTOptixApplication
 - Organization*: Rockwell Automation
 - Organization unit: ASEM s.r.l. R&D team
 - Locality: Artegna
 - State: Udine
 - Country: IT
- OPC UA information**
 - Application URI*: urn: :FactoryTalkOptixHMI:FTOptixApplication
 - Domain names: ftoptixruntime.local, localhost
 - IP addresses: (empty)
- Certificate settings**
 - RSA key strength: 4096 bits
 - Signature algorithm: SHA-512
 - Expiration date: 05/22/2028 00:00:00
- Name**: optixhmi_cert
- Location**: C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

Buttons: Cancel, Create

*Required field

Expiration date: how long the certificate will be valid before a new certificate should be requested/generated.

Longer times means less maintenance but are more prone to tampering, hijacking or identity thefts (as the certificate remains the same for longer time, increasing attack surface).

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

Name: name of the output certificate.
Can be any valid file name.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 578

 **FactoryTalk**
by ROCKWELL AUTOMATION

How to fill the self-signed certificate generation fields

Create certificate

Subject

Common name* ⓘ
FTOptixApplication

Organization* ⓘ
Rockwell Automation

Organization unit ⓘ
ASEM s.r.l. R&D team

Locality ⓘ
Artegna

State ⓘ
Udine

Country ⓘ
IT

OPC UA information

Application URI* ⓘ
urn: :FactoryTalkOptixHMI:FTOptixApplication

Domain names ⓘ
ftoptixruntime.local
localhost

IP addresses ⓘ

Certificate settings

RSA key strength ⓘ
4096 bits

Signature algorithm ⓘ
SHA-512

Expiration date ⓘ
05/22/2028 00:00:00

Name
optixhmi_cert

Location
C:/Users/ /OneDrive - Rockwell Automation, Inc/Documents

*Required field

Cancel

Create

Location: where to store the generated certificate.

Can be any valid path.

PUBLIC • Copyright ©2024 Rockwell Automation, Inc. 579

 **FactoryTalk**
by ROCKWELL AUTOMATION

Creating a self-signed certificate with OpenSSL

- Generate a Private Key
- Create a Certificate Signing Request (CSR)
 - Fill in all the details in the command line (refer to the procedure using Factory Optix to fill the fields)
- Generate the self-signed certificate
- Convert private key to a “pem” file
- Extract the public key and convert it to a “der” file

```
# Generate the private key
openssl genrsa -out private.key 2048
# Create the certificate sign request
openssl req -new -key private.key -out request.csr
# Generate the self-signed certificate
openssl x509 -req -days 365 -in request.csr -signkey private.key -out certificate.crt
# Convert the private key to a .pem file
openssl rsa -in private.key -out private.pem
# Extract the public key from the generated key
openssl rsa -in private.key -pubout -out public.pem
# Convert the public key to a .der file
openssl rsa -pubin -in public.pem -outform der -out public.der
```

Creating a CA-trusted certificate with OpenSSL

- Choose a Certificate Authority
 - Many online providers are available, some for free, some other require a fee
- Generate a Private Key
- Create a Certificate Signing Request (CSR)
 - Fill in all the details in the command line (refer to the procedure using Factory Optix to fill the fields)
- Send the CSR file to the Certificate Authority
 - Depending on the Certificate Authority, additional validations or checks might be needed
 - Depending on the Certificate Authority, the returned file might need to be converted to the “pem” and “der” files

```
# Generate the private key
openssl genrsa -out private.key 2048
# Create the certificate sign request
openssl req -new -key private.key -out request.csr
# Now send the generated certificate signing request file to the CA
# to proceed with the signing process.
```

Using OPC-UA certificates

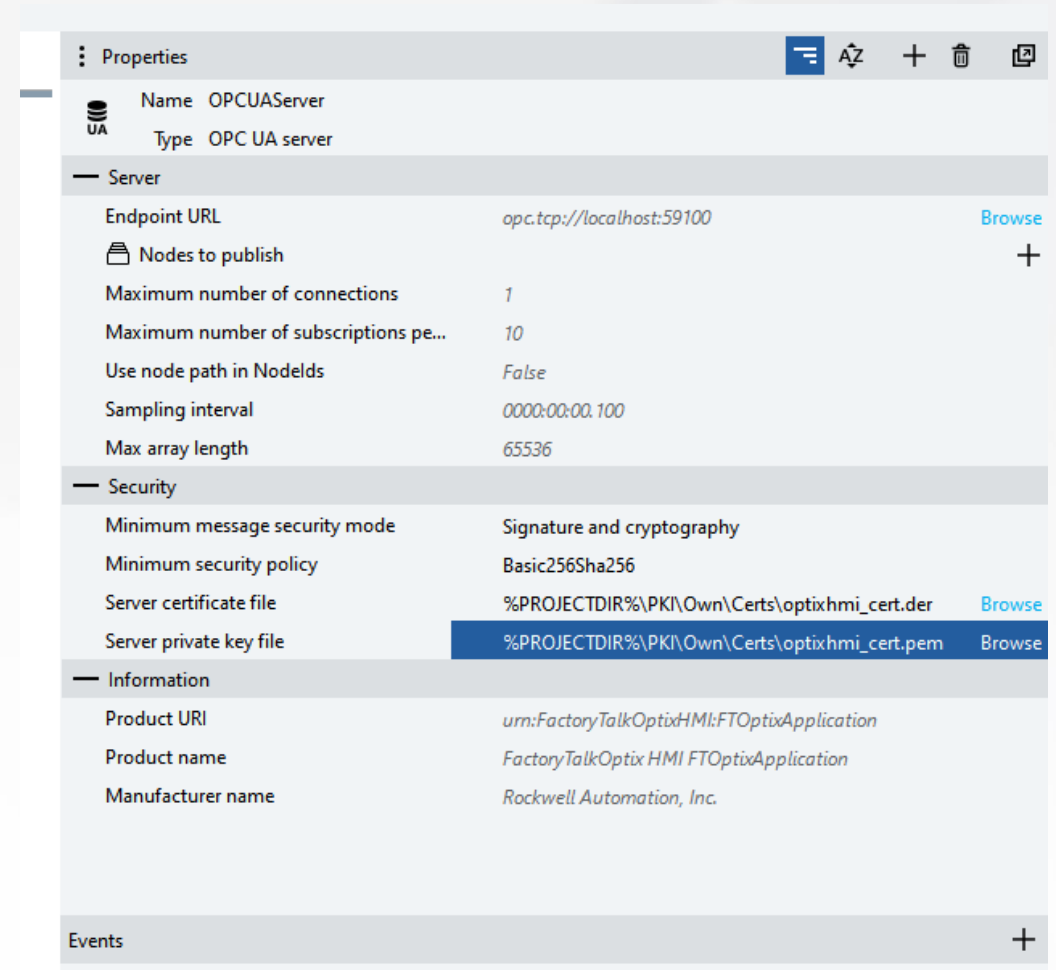


Things to know about OPC-UA certificates

- A FactoryTalk Optix OPC-UA server always uses certificates to perform some actions, this depends on the server configuration on the “Minimum message security mode”:
 - None → Certificates are only used by the client to login to the server
 - Signature only → Certificates are used to sign messages, but content of messages is not encrypted
 - Signature and cryptography → Certificates are used to sign messages and to encrypt the content of the message
- Even if the “Minimum message security mode” is set to “None”, certificates are still required to complete the login process
- In addition to the security mode, the server can also allow only specific encryption algorithms to be used by the connected clients
- FactoryTalk Optix automatically creates a self-signed certificate for testing purposes, **the automatically created self-signed certificate should never be used in a production environment**
 - Certificates should be issued by a trusted authority
 - FactoryTalk Optix Studio includes a dedicated tool to generate custom self-signed certificates if needed

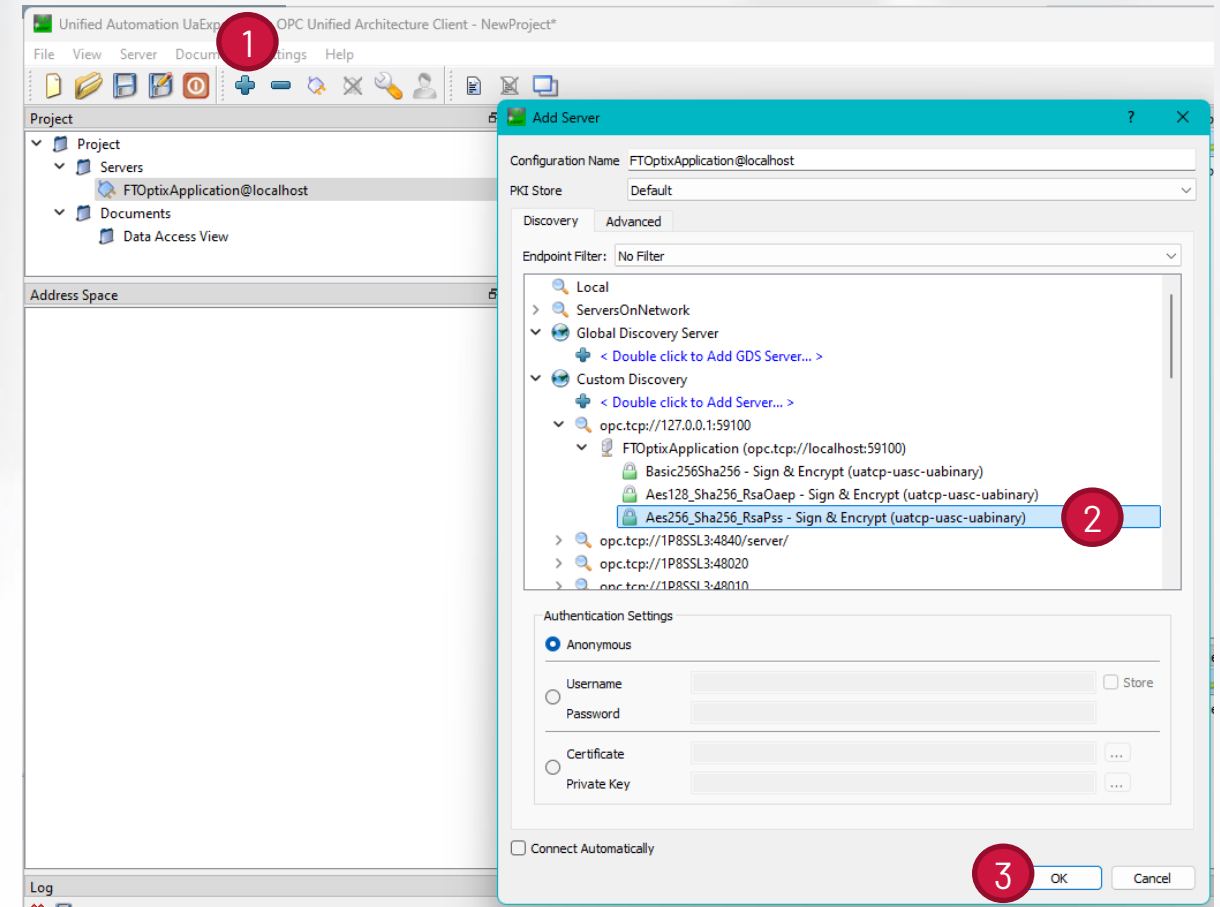
Using a custom certificate in the OPC-UA server

- To apply custom certificates to an OPC-UA server, simply import them in the project and set the relevant properties in the server configuration:
 - Server certificate file: public key of the certificate (“der” format)
 - Server private key file: private key of the certificate (“pem” format)
- Additional settings can be used to enforce a minimum-security policy for connected clients
 - This allows forcing stronger encryption and signing algorithm for all connected clients



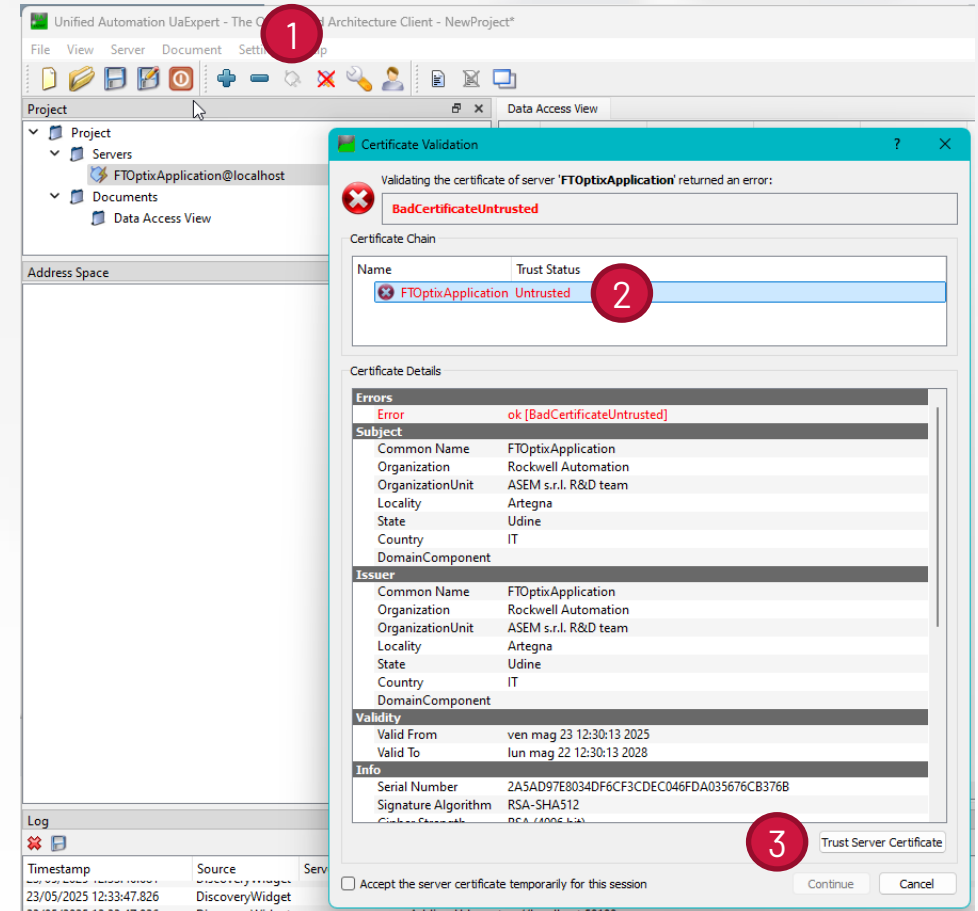
Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder



Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder



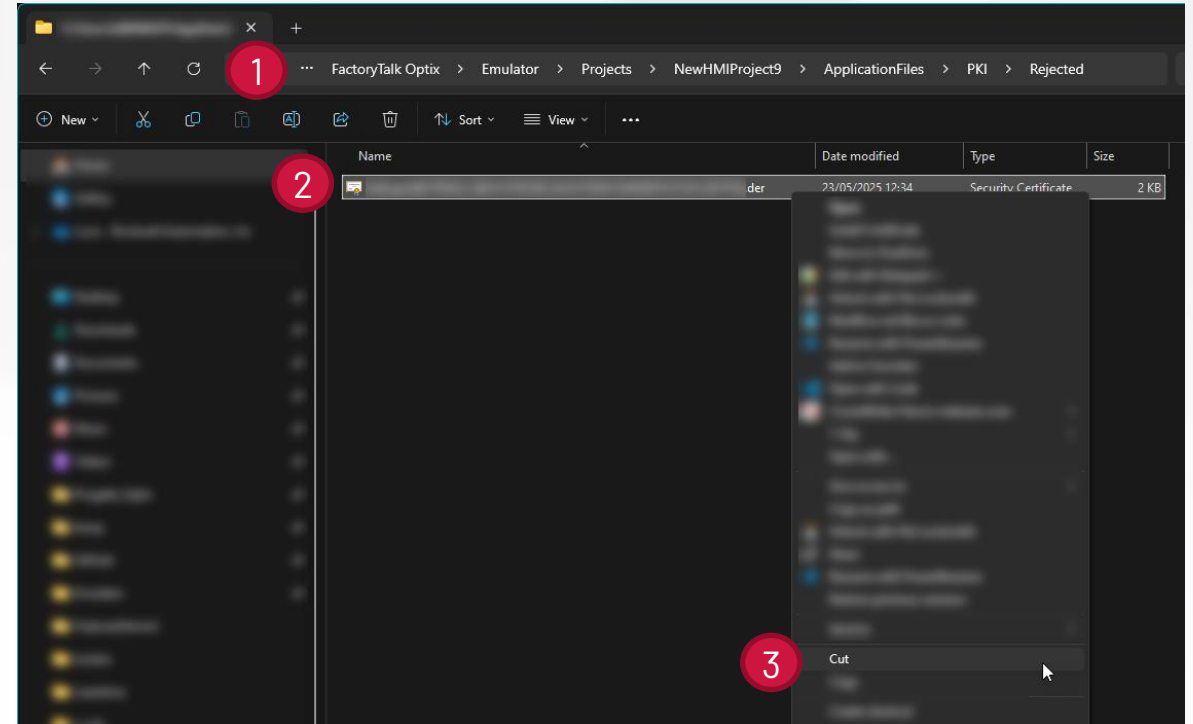
Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder

Timestamp	Source	Server	Message
23/05/2025 12:33:47.826	DiscoveryWidget		Adding Url opc.tcp://localhost:59100
23/05/2025 12:34:09.580	Server Node	FTOptixApplicat...	Endpoint: 'opc.tcp://localhost:59100'
23/05/2025 12:34:09.580	Server Node	FTOptixApplicat...	Security policy: 'http://opcfoundation.org/UA/SecurityPolicy#Aes256_Sha256_RsaPss'
23/05/2025 12:34:09.580	Server Node	FTOptixApplicat...	ApplicationUri: 'urn: :FactoryTalkOptixHMI:FTOptixApplication'
23/05/2025 12:34:09.580	Server Node	FTOptixApplicat...	Used UserTokenType: Anonymous
23/05/2025 12:34:27.949	Server Node	FTOptixApplicat...	Error 'BadSecurityChecksFailed' was returned during OpenSecureChannel
23/05/2025 12:34:27.949	Server Node	FTOptixApplicat...	Connection status of server 'FTOptixApplication@localhost' changed to 'Disconnected'.

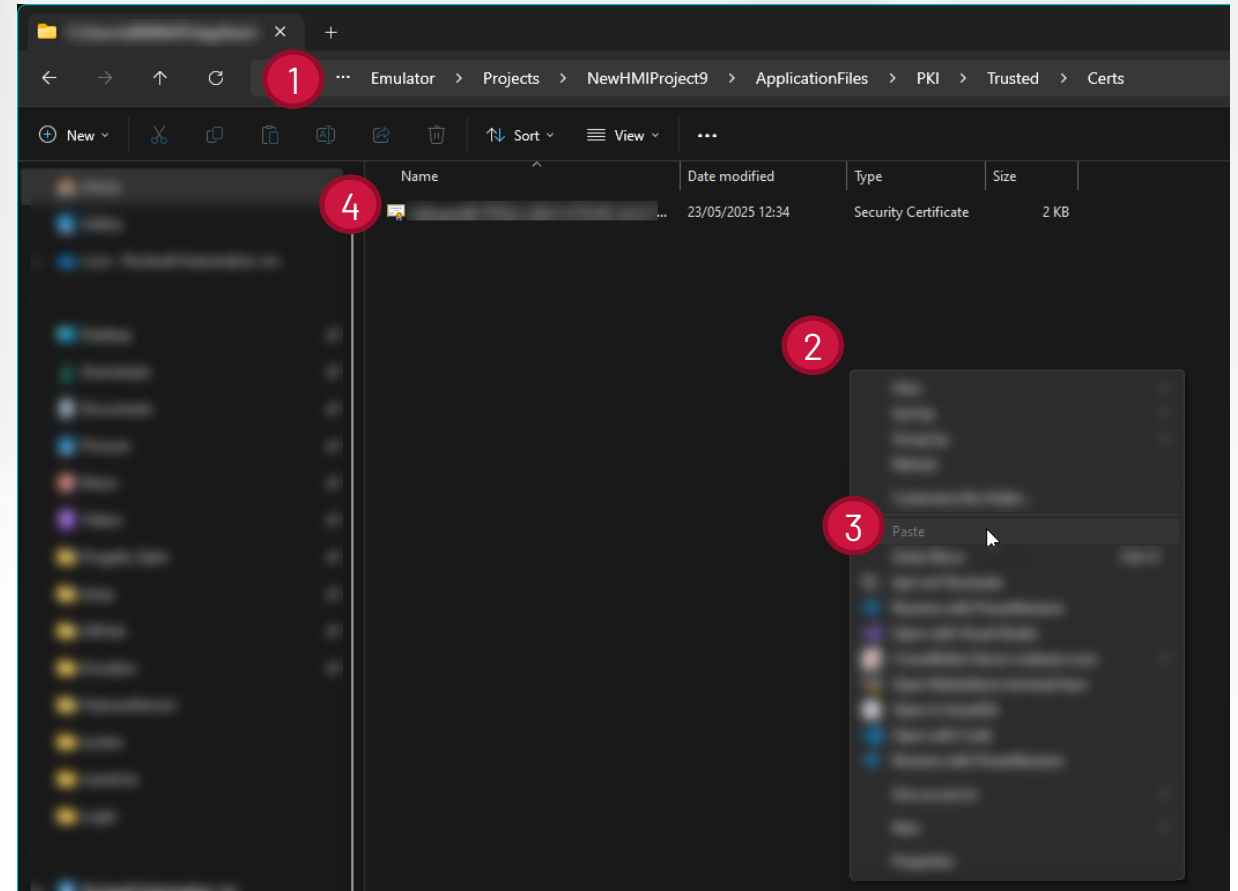
Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder



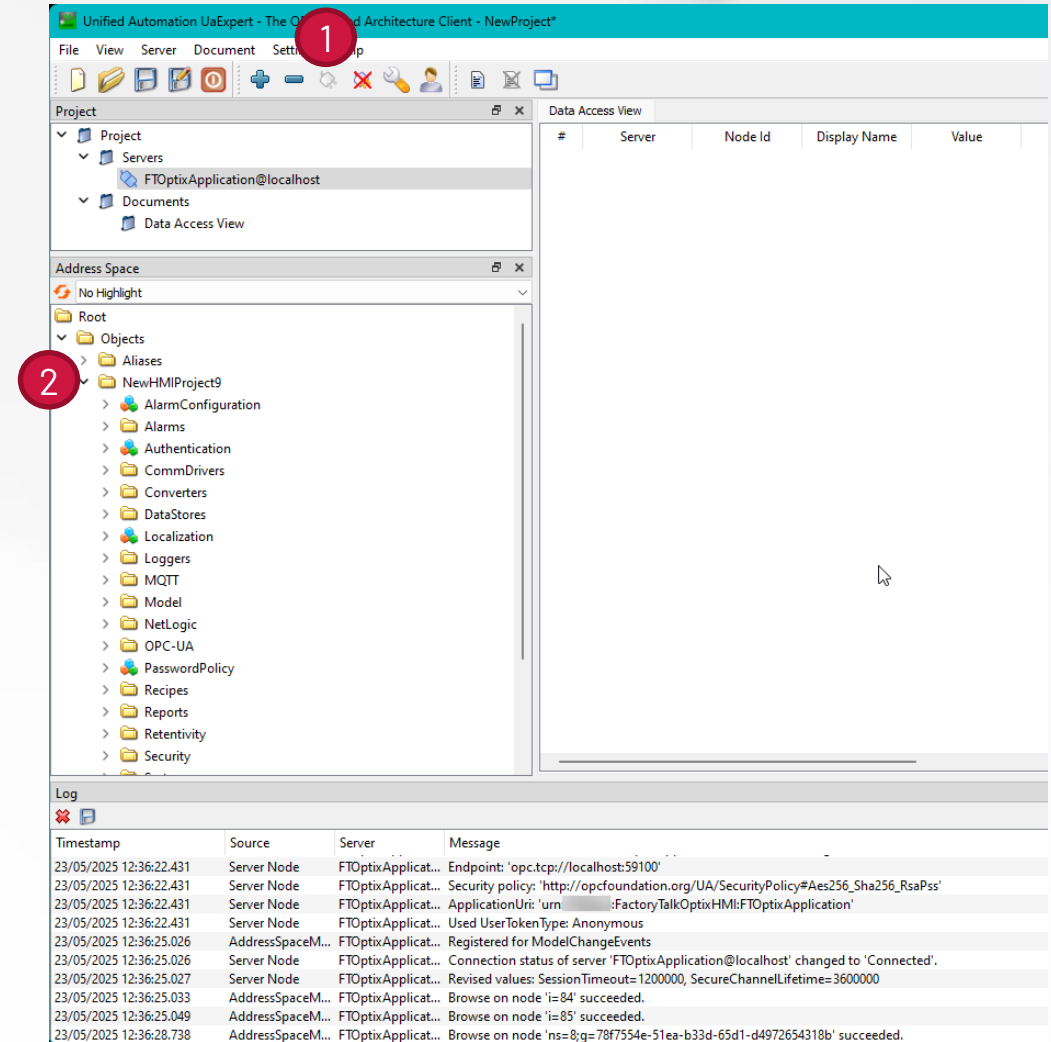
Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder



Connecting to an OPC-UA server using custom certificates

- If the server is forcing clients to use certificates, when browsing the available servers, no unsecure connections are now listed
- Select the required sign and encryption protocol to be used
- If a self-signed certificate was used, the client will prompt the user to accept or reject the server certificate
- After trusting the certificate, the client can now attempt a connection
- FactoryTalk Optix will reject the client certificate by default, which needs to be trusted manually by moving it to the Trusted folder



Return parameters from NetLogic methods



Return parameters from NetLogic methods

- All methods to be used in FactoryTalk Optix must be void (returning nothing)
- Parameters can be returned by using the «out» modifier in the arguments list

```
// This one will not work, returned value is discarded
```

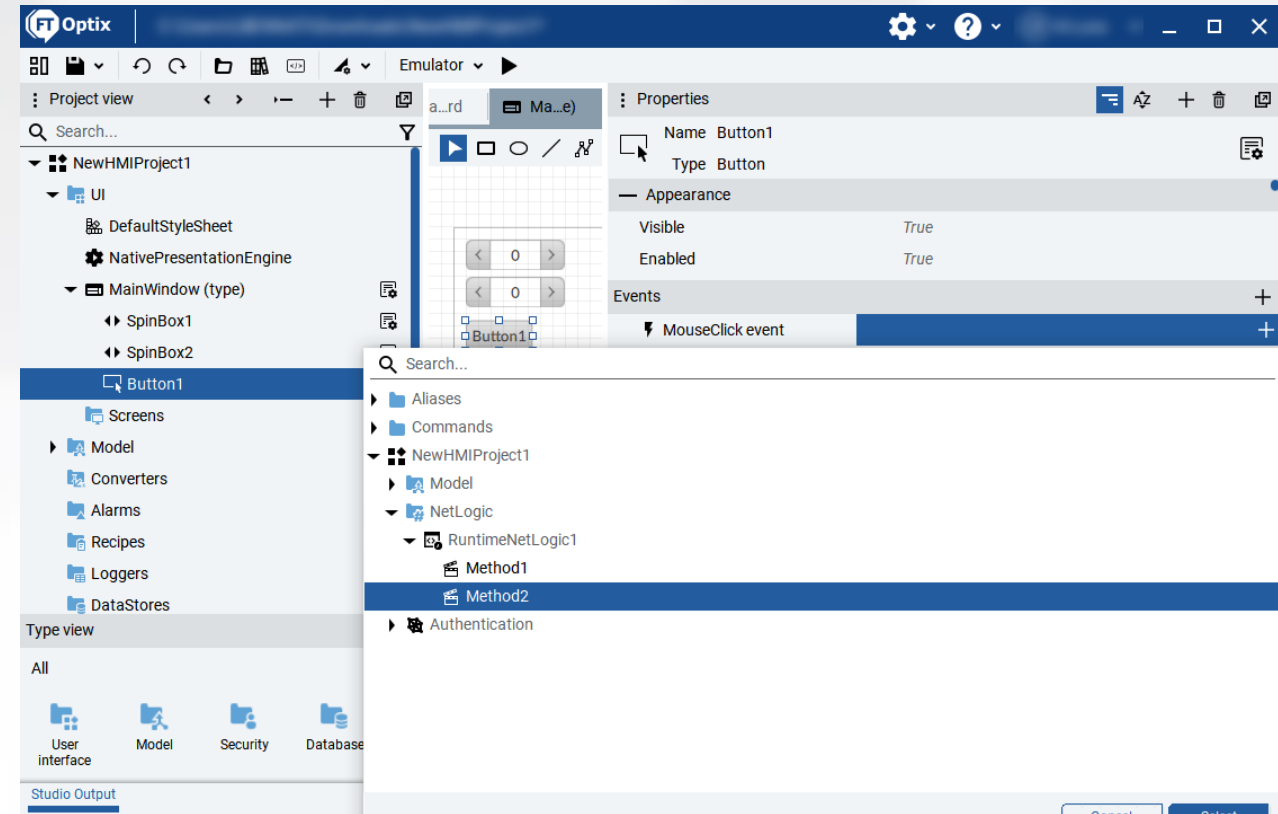
```
[ExportMethod]  
public int Method1(int a, int b) {  
    return a + b;  
}
```

```
// This one will work, returned value can be used
```

```
[ExportMethod]  
public void Method2(int a, int b, out int c) {  
    c = a + b;  
}
```

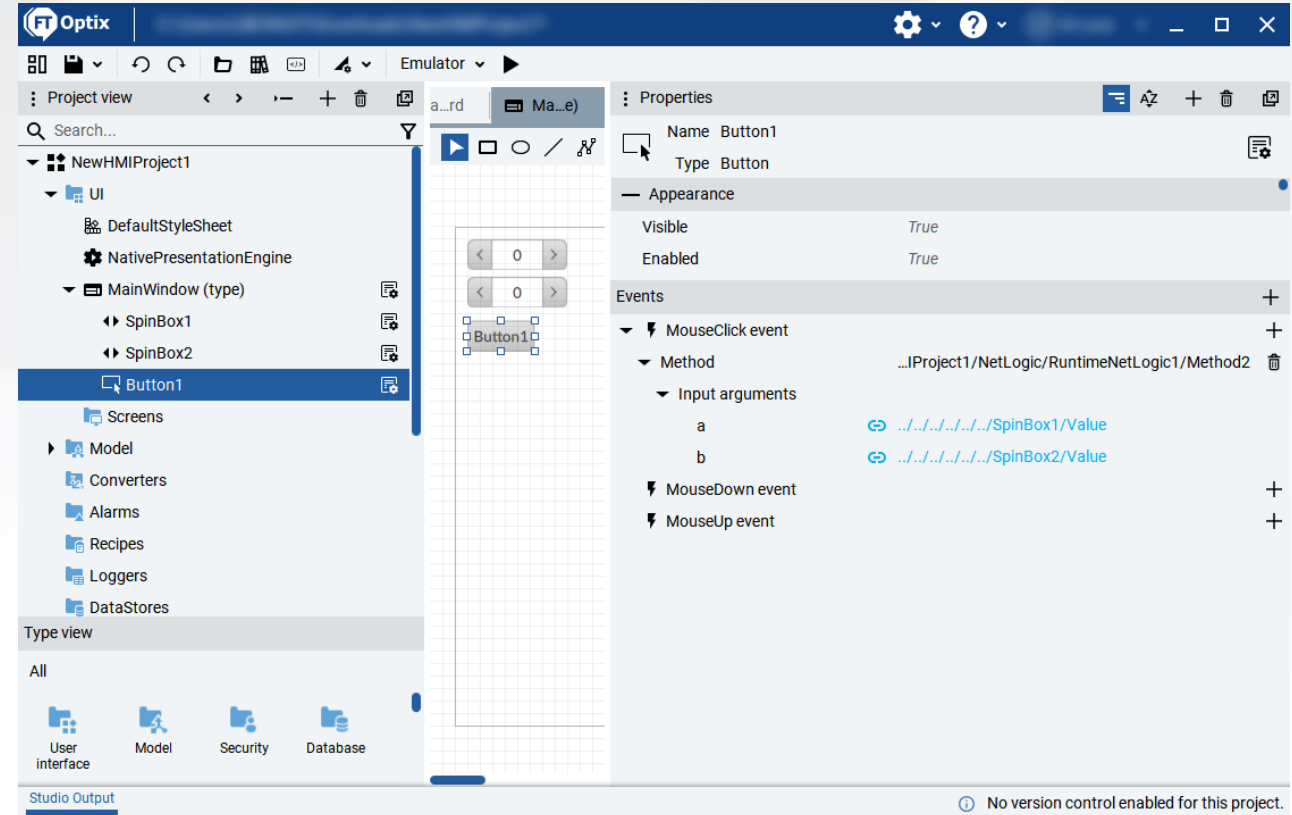
Return parameters from NetLogic methods

- On any method that can trigger an event, call the method exposed by the NetLogic
- Configure the input parameters (if any)
- Add a second method invocation
- Use a SetVariableValue to get the output value from the MethodContainer1 and set it to a different variable (or property)



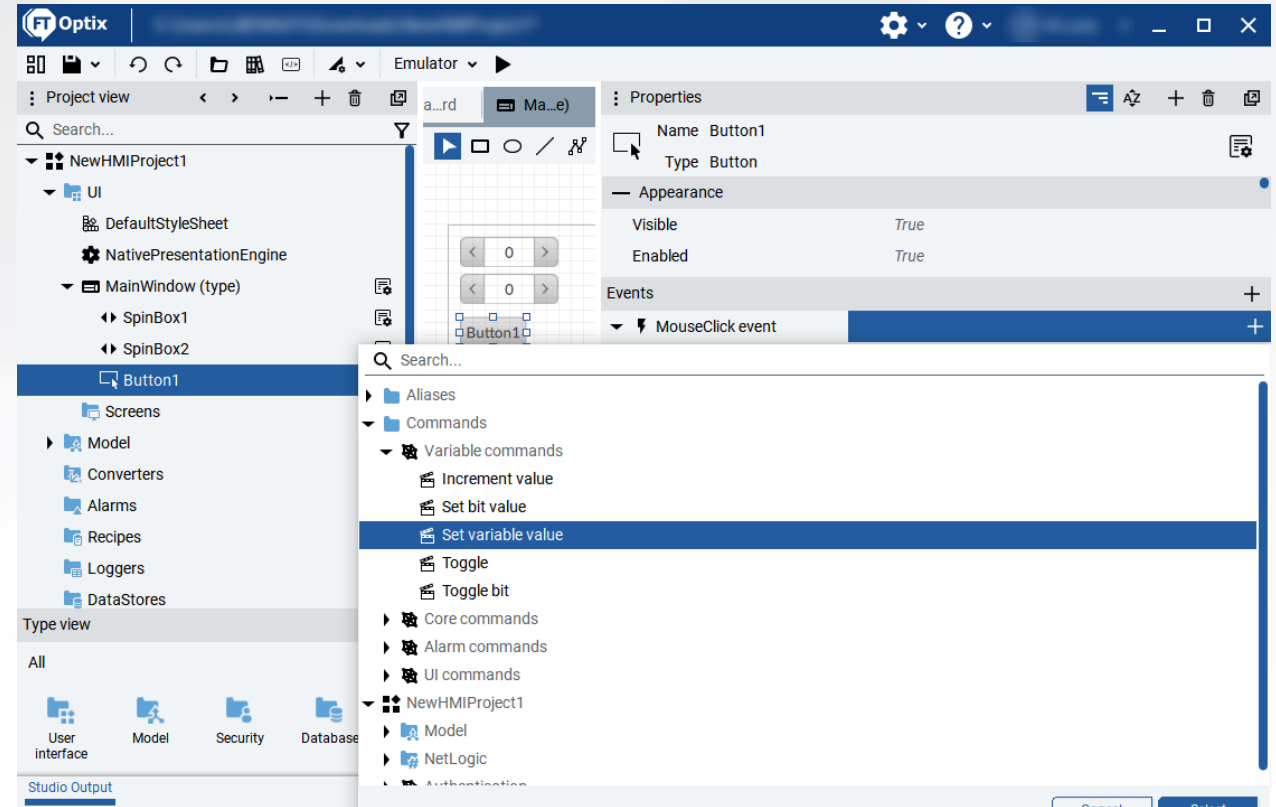
Return parameters from NetLogic methods

- On any method that can trigger an event, call the method exposed by the NetLogic
- Configure the input parameters (if any)
- Add a second method invocation
- Use a SetVariableValue to get the output value from the MethodContainer1 and set it to a different variable (or property)



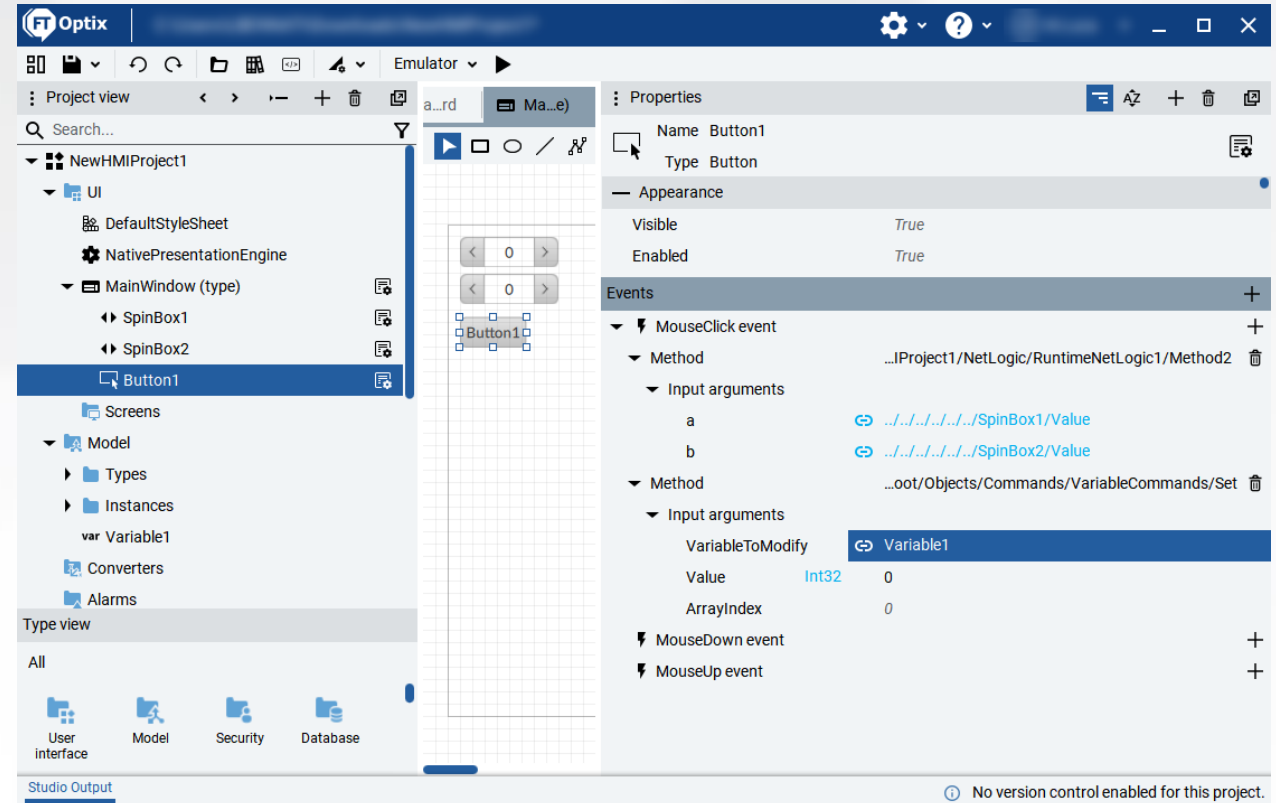
Return parameters from NetLogic methods

- On any method that can trigger an event, call the method exposed by the NetLogic
- Configure the input parameters (if any)
- Add a second method invocation
- Use a SetVariableValue to get the output value from the MethodContainer1 and set it to a different variable (or property)



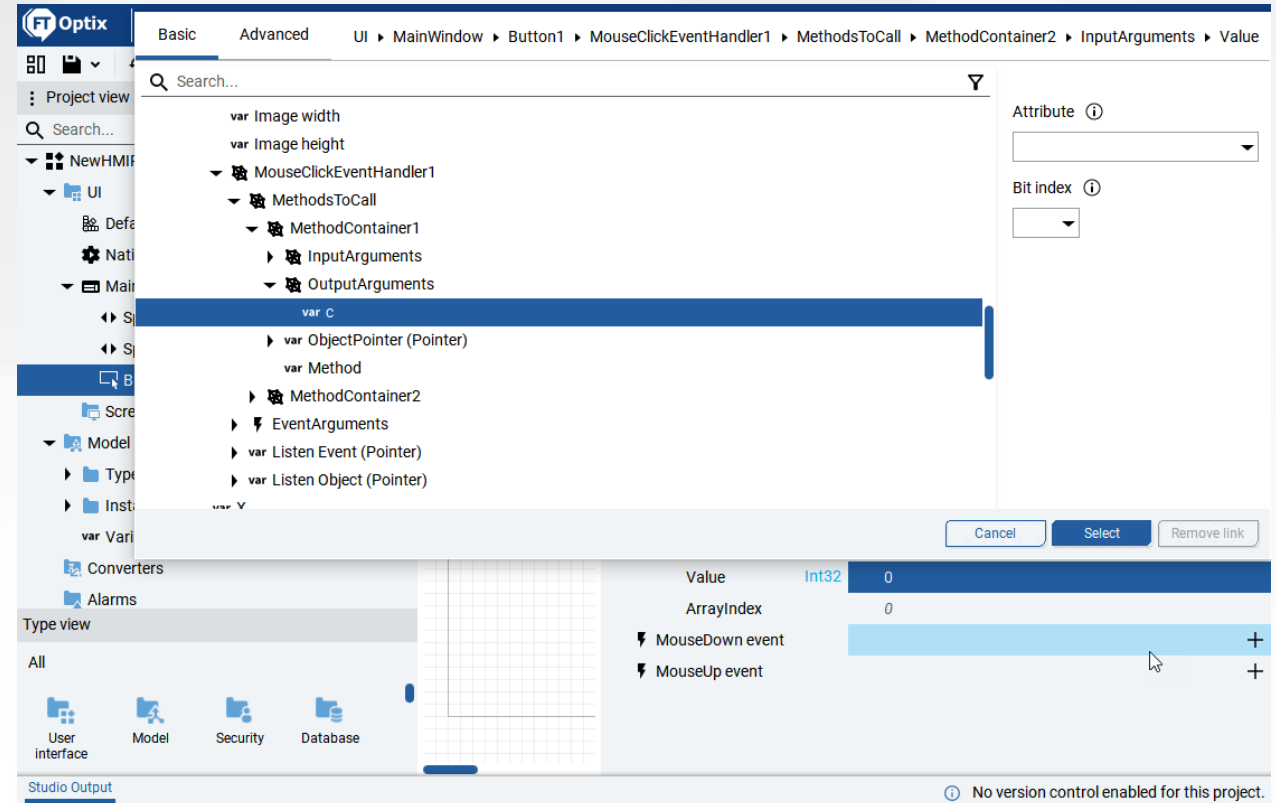
Return parameters from NetLogic methods

- On any method that can trigger an event, call the method exposed by the NetLogic
- Configure the input parameters (if any)
- Add a second method invocation
- Use a SetVariableValue to get the output value from the MethodContainer1 and set it to a different variable (or property)



Return parameters from NetLogic methods

- On any method that can trigger an event, call the method exposed by the NetLogic
- Configure the input parameters (if any)
- Add a second method invocation
- Use a SetVariableValue to get the output value from the MethodContainer1 and set it to a different variable (or property)



Connect to an External Database

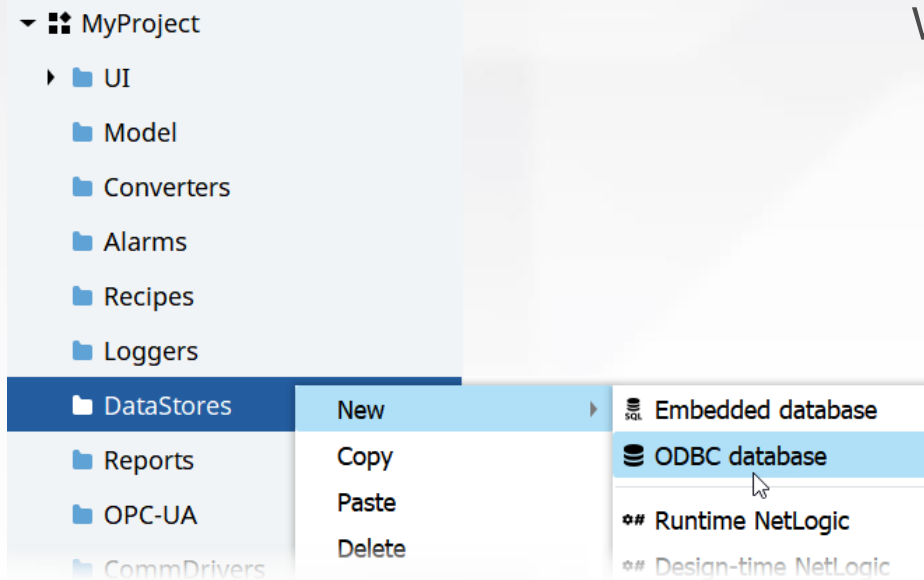


ODBC Database

- It's used to query/populate a relational database external to the FTOptixApplication
- Supported DBMS:
 - SQL Server or SQL Server Express 2012 (or higher)
 - MySQL Server

- Objects that can exchange data with an ODBC database:

- *Data grid, List box, Combo box and Trend* objects query a table to display data at runtime
- *Data logger and Event logger* objects store data in a table
- *Recipe* object stores data in a table and execute a query to display data at runtime

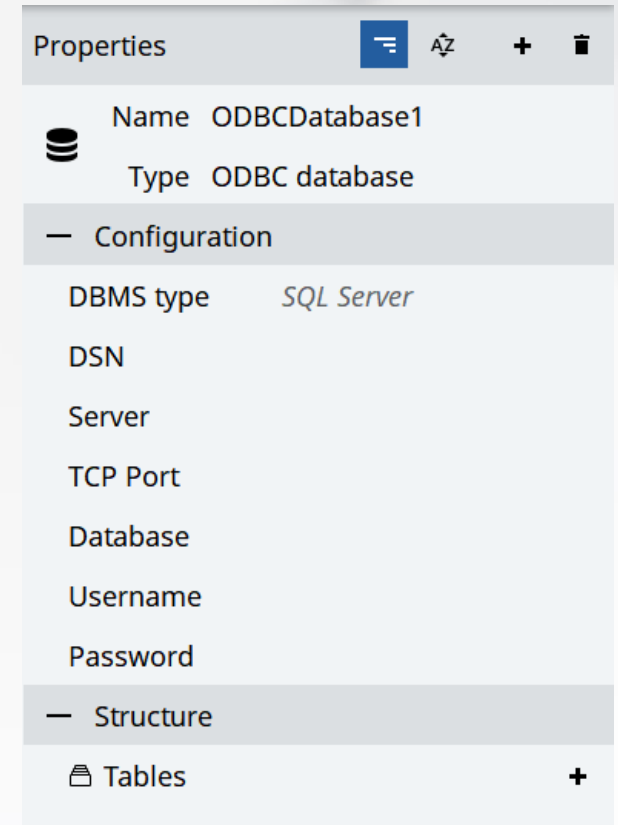


ODBC Database

- *DBMS type*: SQL Express or MySQL
- *DSN (optional)*: only in case ODBC is used
- *Server*: can be IP Address, Hostname or "Hostname\Instance Name"
- *TCP Port (optional)*: to be used if the port is different from the default
- *Database*: Name of the database

NOTE: database must be created manually

- *Username/Password*: credential of the user, used to read/write into database tables

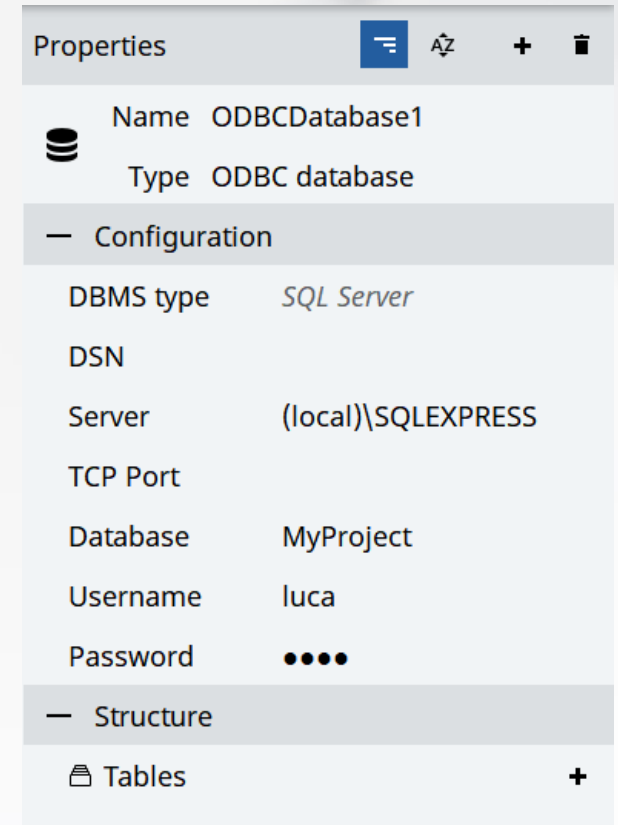


ODBC Database

- *DBMS type*: SQL Express or MySQL
- *DSN (optional)*: only in case ODBC is used
- *Server*: can be IP Address, Hostname or "Hostname\Instance Name"
- *TCP Port (optional)*: to be used if the port is different from the default
- *Database*: Name of the database

NOTE: database must be created manually

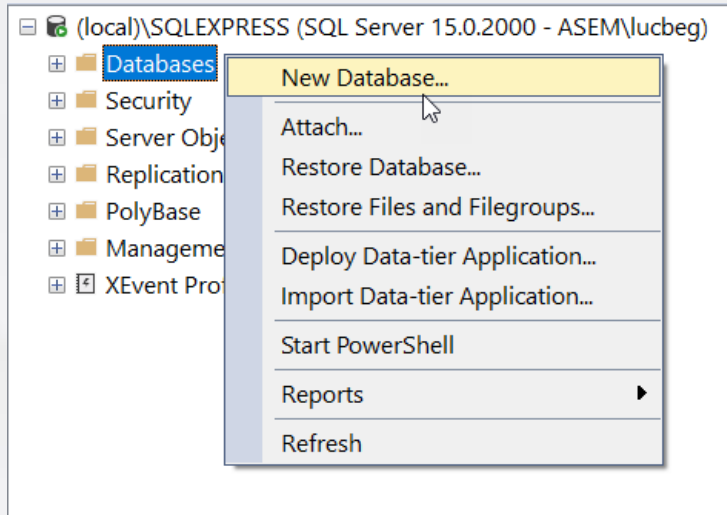
- *Username/Password*: credential of the user, used to read/write into database tables



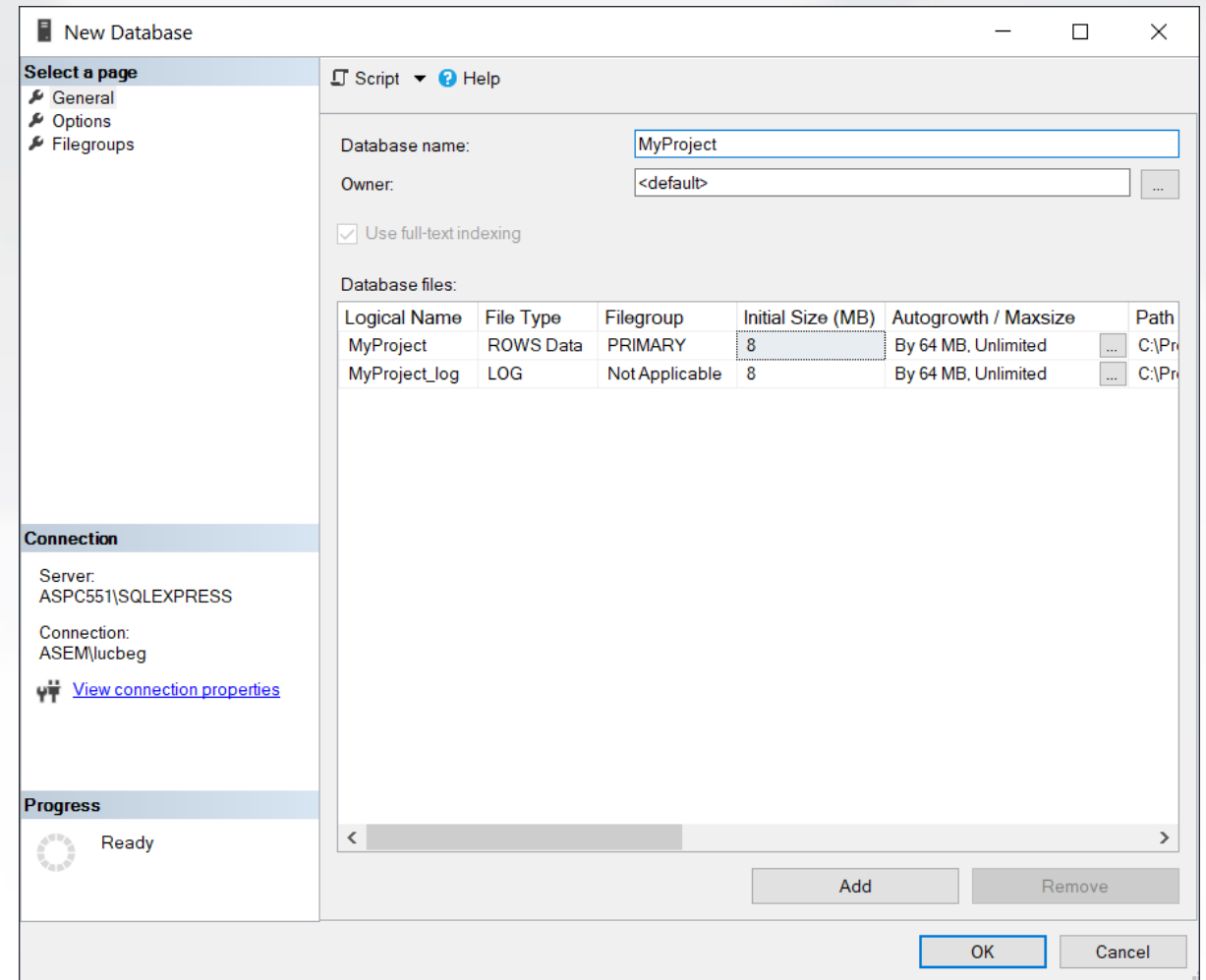
The screenshot shows the 'Properties' dialog box for an ODBC database. It has a title bar with a menu icon, a refresh icon, and add/delete icons. The main content is divided into sections: 'Name' (ODBCDatabase1), 'Type' (ODBC database), 'Configuration', and 'Structure'. The 'Configuration' section is expanded, showing fields for DBMS type (SQL Server), DSN, Server ((local)\SQLEXPRESS), TCP Port, Database (MyProject), Username (luca), and Password (masked with dots). The 'Structure' section is collapsed, showing a 'Tables' entry with a plus icon.

Properties	
Name	ODBCDatabase1
Type	ODBC database
— Configuration	
DBMS type	SQL Server
DSN	
Server	(local)\SQLEXPRESS
TCP Port	
Database	MyProject
Username	luca
Password	•••••
— Structure	
Tables	+

SQL Server: create database via SQL Server Management Studio

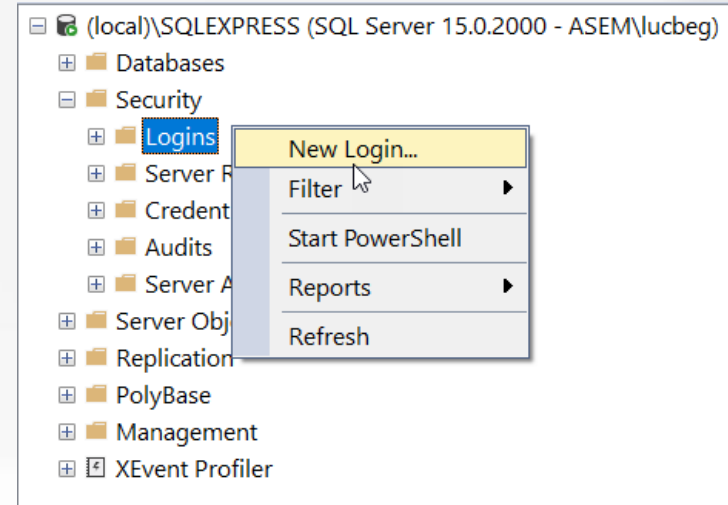


- Is suggested to create the database before the creation of the user
- To create a database
Databases > New Database...
and is enough to provide the name



SQL Server: create user via SQL Server Management Studio

1. Security > Logins > New Login...
2. Once created a User, it's necessary to assign a role for the needed database
3. User Mapping > Database > db_owner



SQL Server: create user via SQL Server Management Studio

1. Security > Logins > New Login...
2. Once created a User, it's necessary to assign a role for the needed database
3. User Mapping > Database > db_owner

Login - New

Select a page

- General
- Server Roles
- User Mapping
- Securables
- Status

Script Help

Login name: QApplication Search...

☐ Windows authentication

☒ SQL Server authentication

Password:

Confirm password:

☐ Specify old password

Old password:

☒ Enforce password policy

☐ Enforce password expiration

☐ User must change password at next login

☐ Mapped to certificate

☐ Mapped to asymmetric key

☐ Map to Credential

Mapped Credentials

Credential	Provider
------------	----------

Add Remove

Default database: master

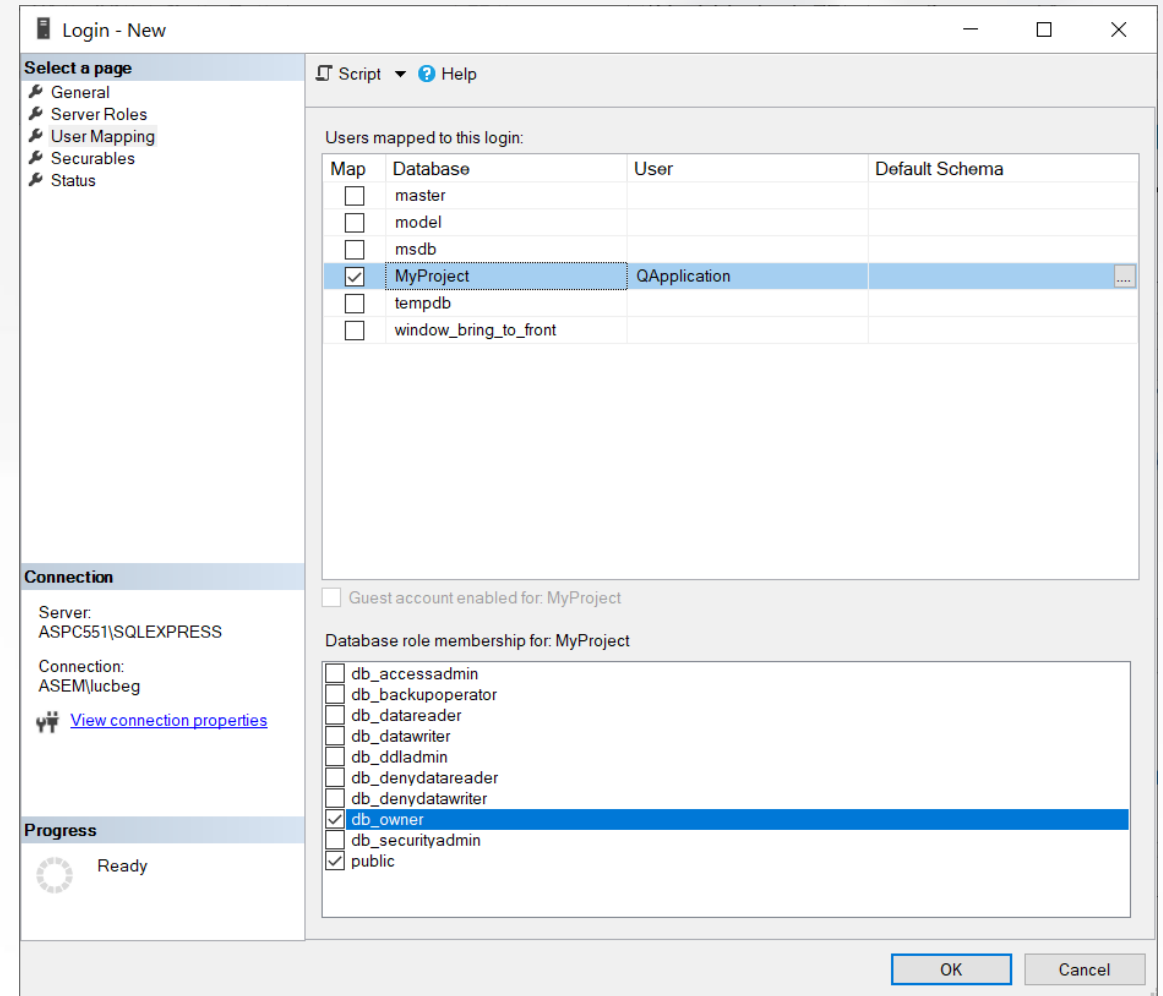
Default language: <default>

Progress: Ready

OK Cancel

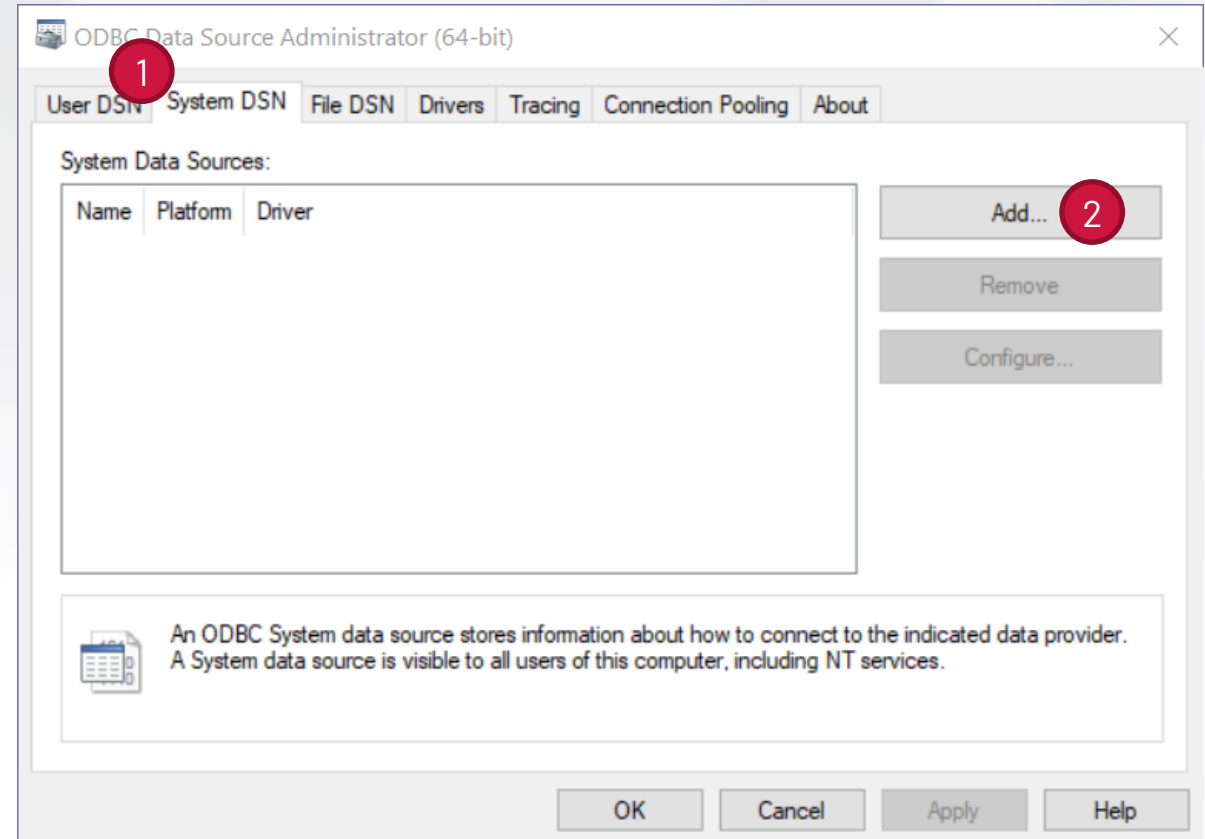
SQL Server: create user via SQL Server Management Studio

1. Security > Logins > New Login...
2. Once created a User, it's necessary to assign a role for the needed database
3. User Mapping > Database > db_owner



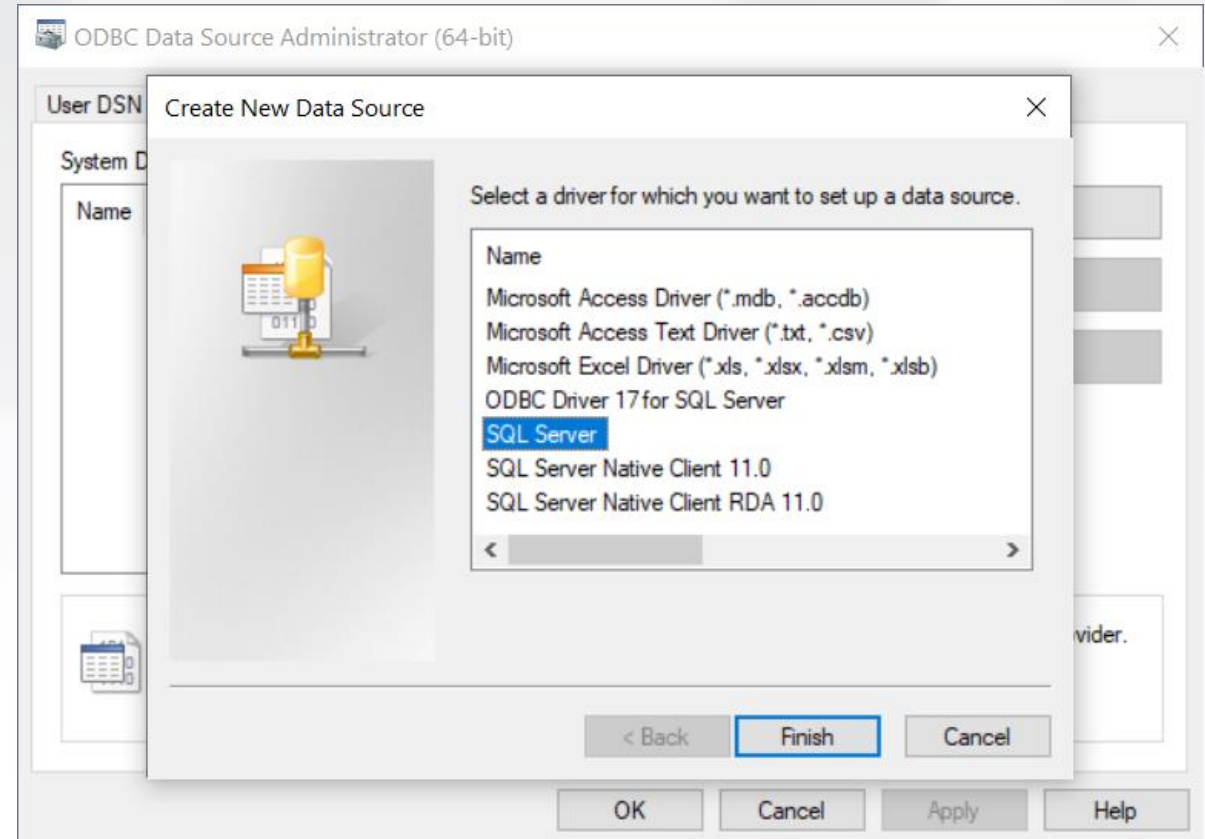
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



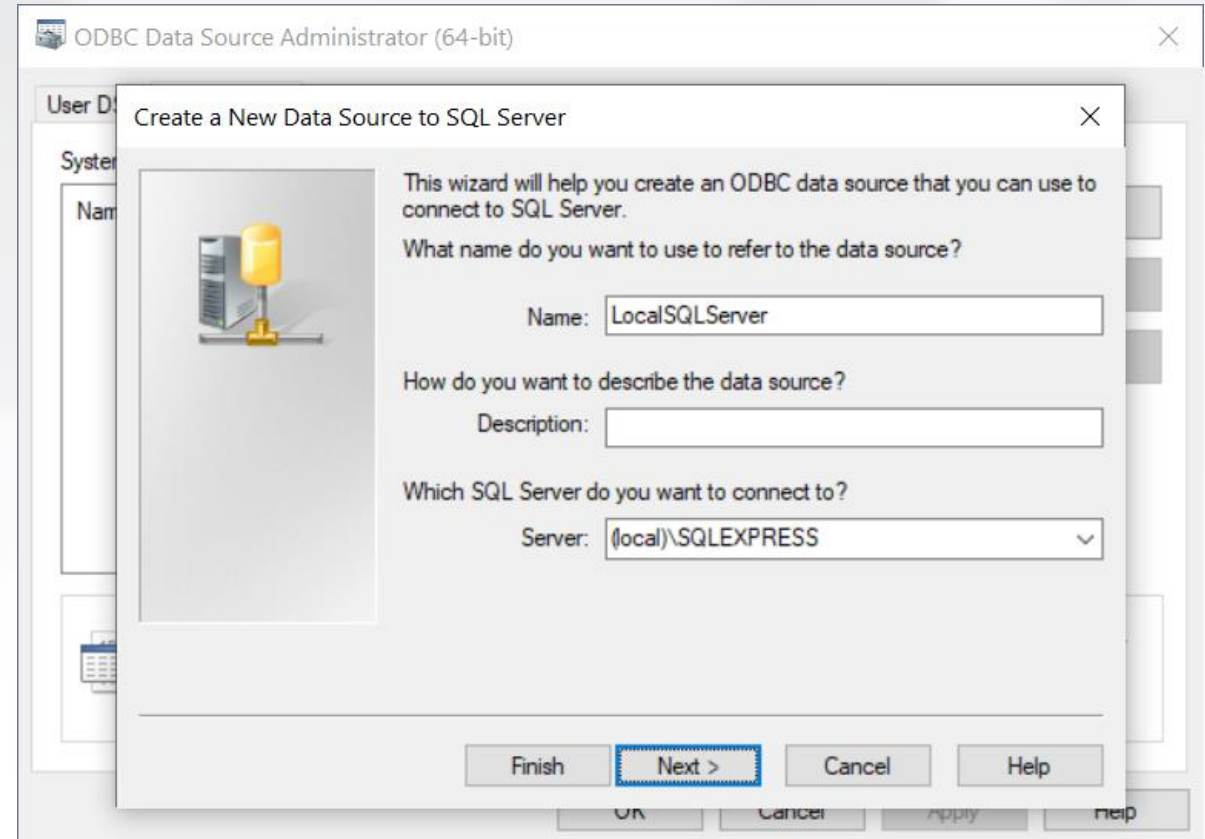
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



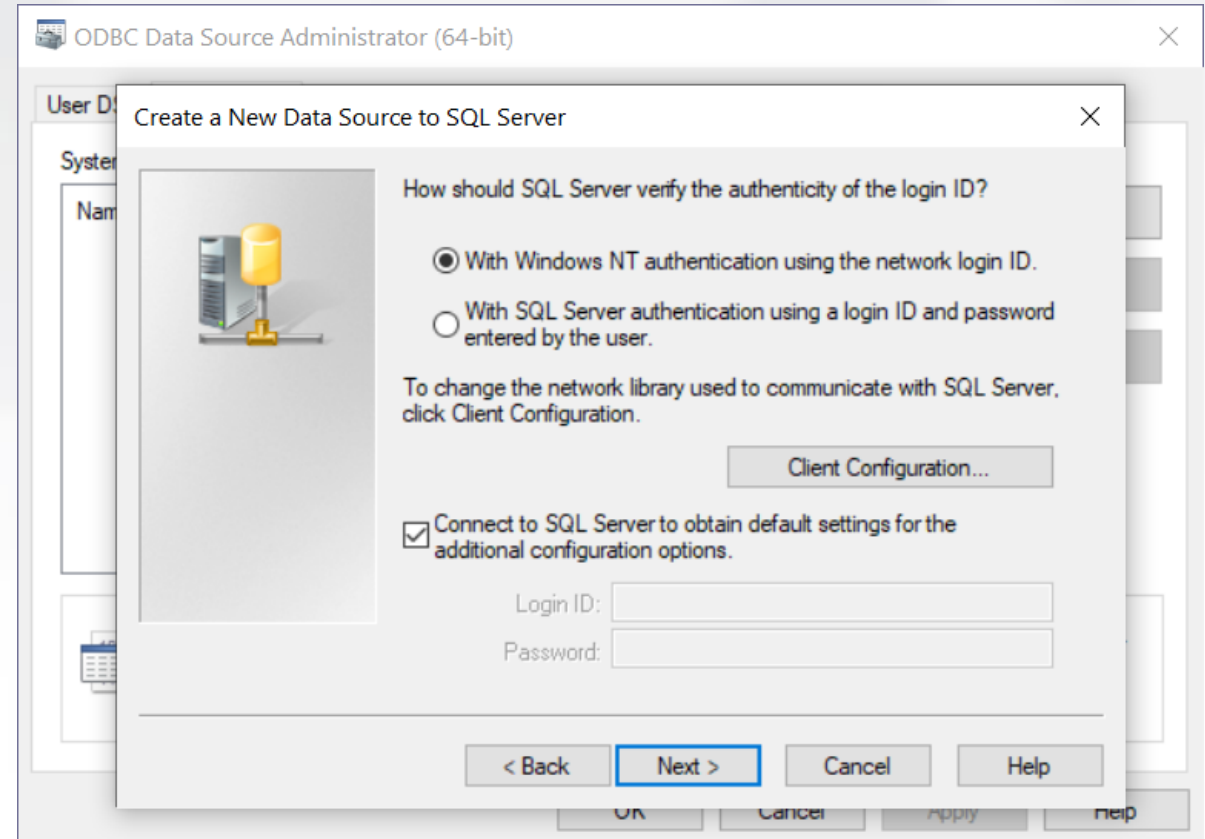
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



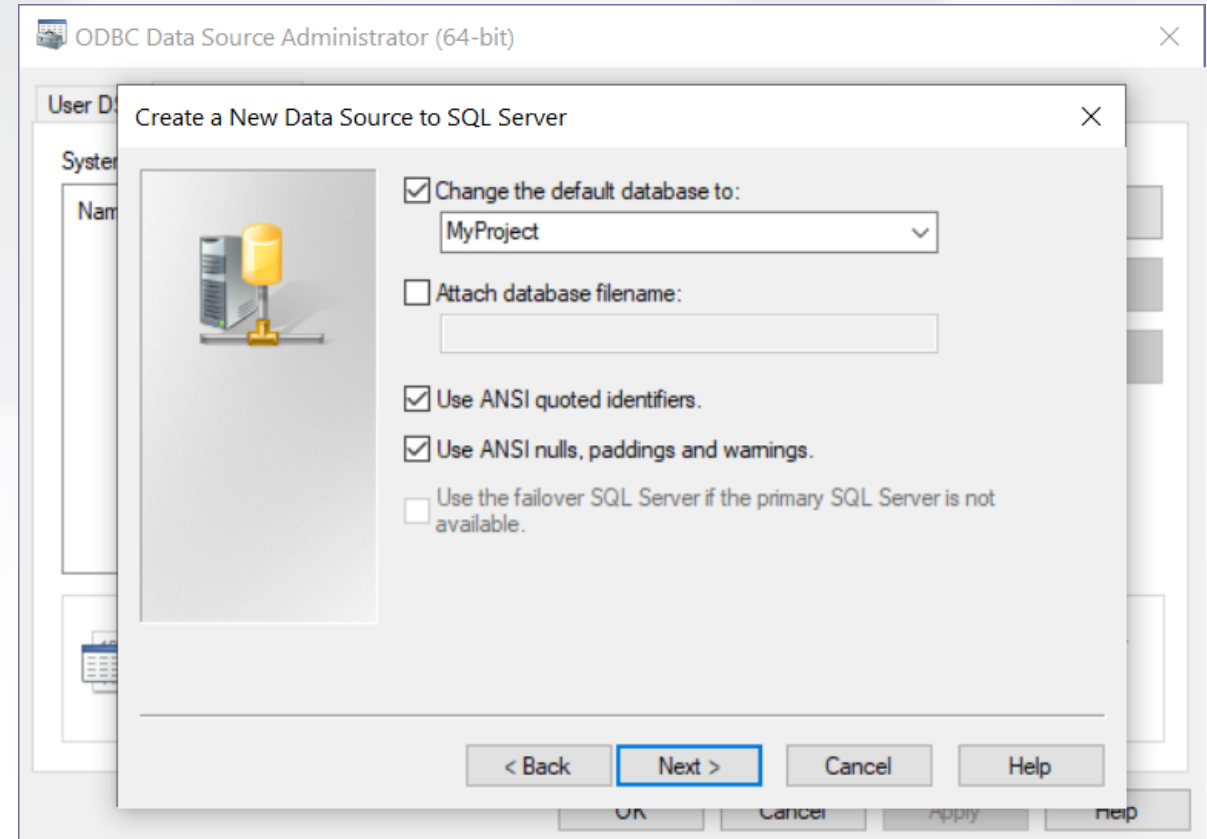
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



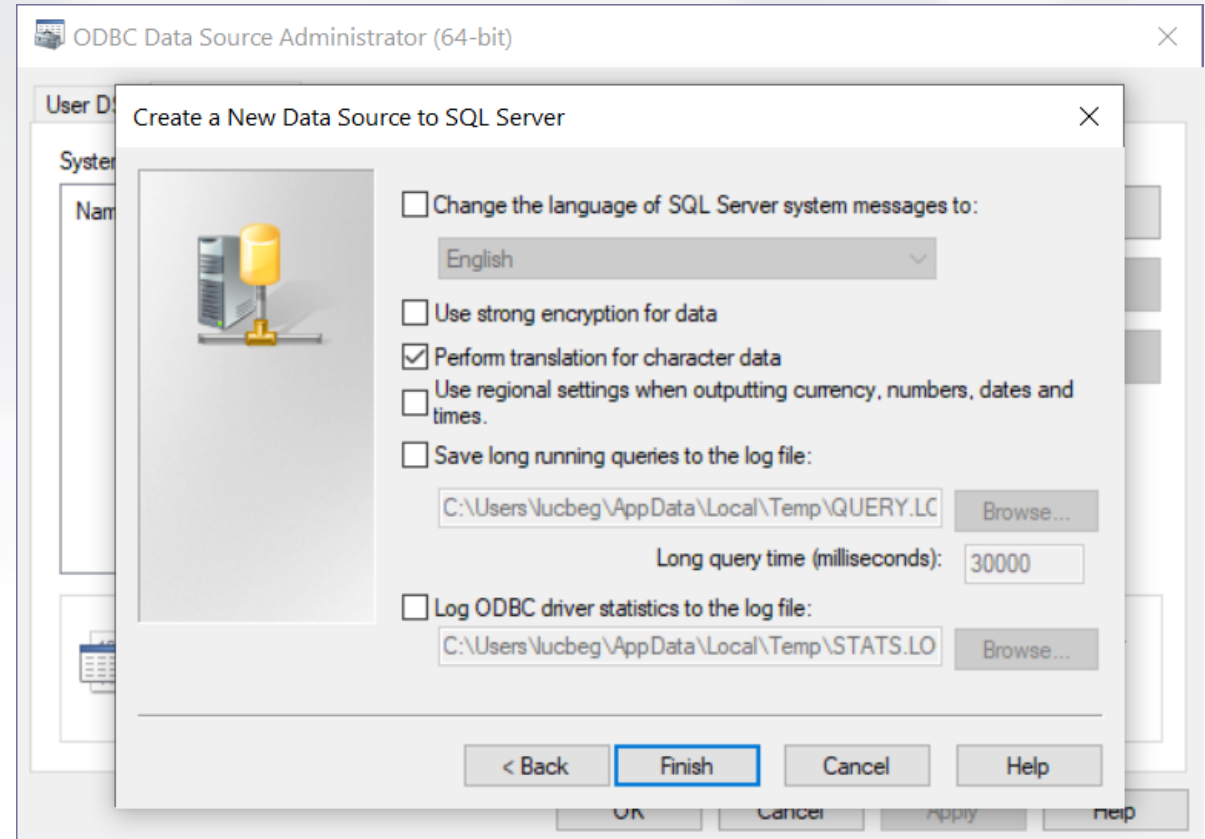
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



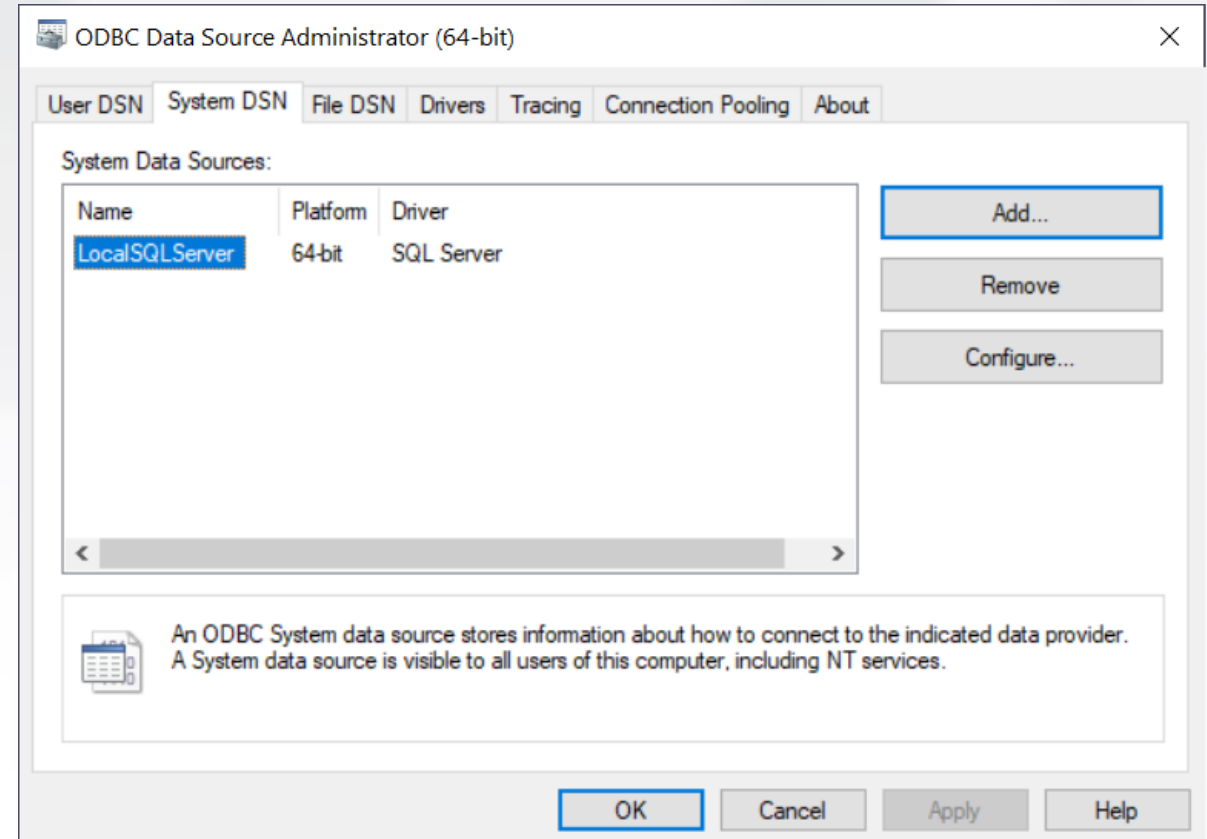
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



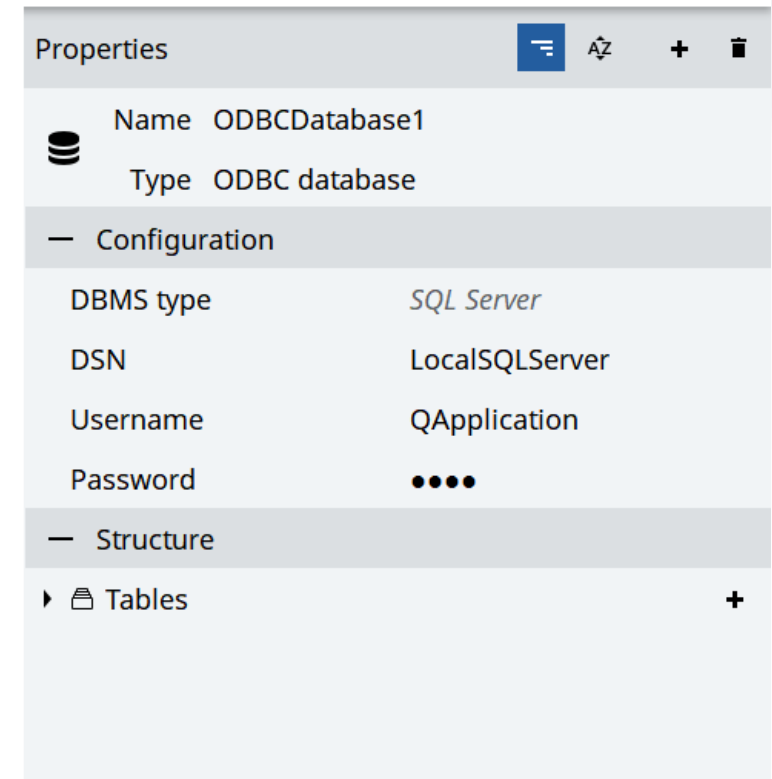
Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



Usage of ODBC Data Source (optional)

- To configure a DSN use "ODBC Data Sources (64-bit)"
- ODBC Driver supported:
 - SQL Server
 - MySQL ANSI
 - MySQL Unicode
- Key settings:
 - DSN Name: defined into ODBC DSN, must be the same in FactoryTalk Optix Studio
 - Default Database: is the Database created for the FTOptixApplication
 - Username/Password: valid credential to access the Database



Connect to an External InfluxDB instance



Sample InfluxDB server using Docker

services:

influxdb:

image: influxdb:latest

container_name: influxdb

ports:

- '8086:8086' # Port to set in the Optix project

volumes:

- influxdb-storage:/var/lib/influxdb2

restart: unless-stopped

environment:

- DOCKER_INFLUXDB_INIT_MODE=setup
- DOCKER_INFLUXDB_INIT_USERNAME=admin
- DOCKER_INFLUXDB_INIT_PASSWORD=admin_pass
- DOCKER_INFLUXDB_INIT_ORG=organization # Name of the organization to set in the Optix project
- DOCKER_INFLUXDB_INIT_BUCKET=bucket # Name of the bucket to set in the Optix project
- DOCKER_INFLUXDB_INIT_RETENTION=4w # How long to retain data
- DOCKER_INFLUXDB_INIT_ADMIN_TOKEN=MyVeryLongAndS3cr3tTok3nToWriteD4t4 # Access token to set in the Optix project

volumes:

influxdb-storage:

Remote InfluxDB configuration

- Configure the connection parameters to reach the InfluxDB instance
- Usage of SSL is recommended
 - Certificate can be exported from the InfluxDB server and loaded in the Optix application
- Tables and columns are only required if inserting new data (same as ODBC)

The screenshot shows the 'Properties' window for a remote InfluxDB database. The 'Name' is 'RemoteInfluxDBDatabase1' and the 'Type' is 'Remote InfluxDB database'. The 'Configuration' section lists: Server (172.19.17.72), TCP Port (8086), Organization (organization), Bucket (bucket), Token (masked with dots), Use SSL (False), and Server Certificate (with a 'Browse' link). The 'Store and Forward' section shows SF Enabled (False) and SF Buffer max size (1000). The 'Structure' section shows a table named 'DataLogger2' with a record limit of 0. It lists columns: Timestamp (UtcTime, Not set), LocalTimestamp (DateTime, Not set), Sine (Int32, 0), Cosine (Int32, 0), and Ramp (Int32, 0). Each column has a trash icon for deletion.

Properties		
Name	RemoteInfluxDBDatabase1	
Type	Remote InfluxDB database	
Configuration		
Server	172.19.17.72	
TCP Port	8086	
Organization	organization	
Bucket	bucket	
Token	••••	
Use SSL	False	
Server Certificate	Browse	
Store and Forward		
SF Enabled	False	
SF Buffer max size	1000	
Structure		
Tables	+	
DataLogger2	-	
Record limit	0	
Tags	+	
Columns	+	
Timestamp	UtcTime	Not set
LocalTimestamp	DateTime	Not set
Sine	Int32	0
Cosine	Int32	0
Ramp	Int32	0

Configuration example

services:

influxdb:

image: influxdb:latest

container_name: influxdb

ports:

- '8086:8086'

volumes:

- influxdb-storage:/var/lib/influxdb2

restart: unless-stopped

environment:

- DOCKER_INFLUXDB_INIT_MODE=setup
- DOCKER_INFLUXDB_INIT_USERNAME=admin
- DOCKER_INFLUXDB_INIT_PASSWORD=admin_pass
- DOCKER_INFLUXDB_INIT_ORG=organization
- DOCKER_INFLUXDB_INIT_BUCKET=bucket
- DOCKER_INFLUXDB_INIT_RETENTION=4w
- DOCKER_INFLUXDB_INIT_ADMIN_TOKEN=MyVeryLongAndS3cr3tTok3nToWriteD4t4

volumes:

influxdb-storage:

Properties

Name RemoteInfluxDBDatabase1
Type Remote InfluxDB database

Configuration

Server	172.19.17.72
TCP Port	8086
Organization	organization
Bucket	bucket
Token	••••
Use SSL	False
Server Certificate	Browse

Store and Forward

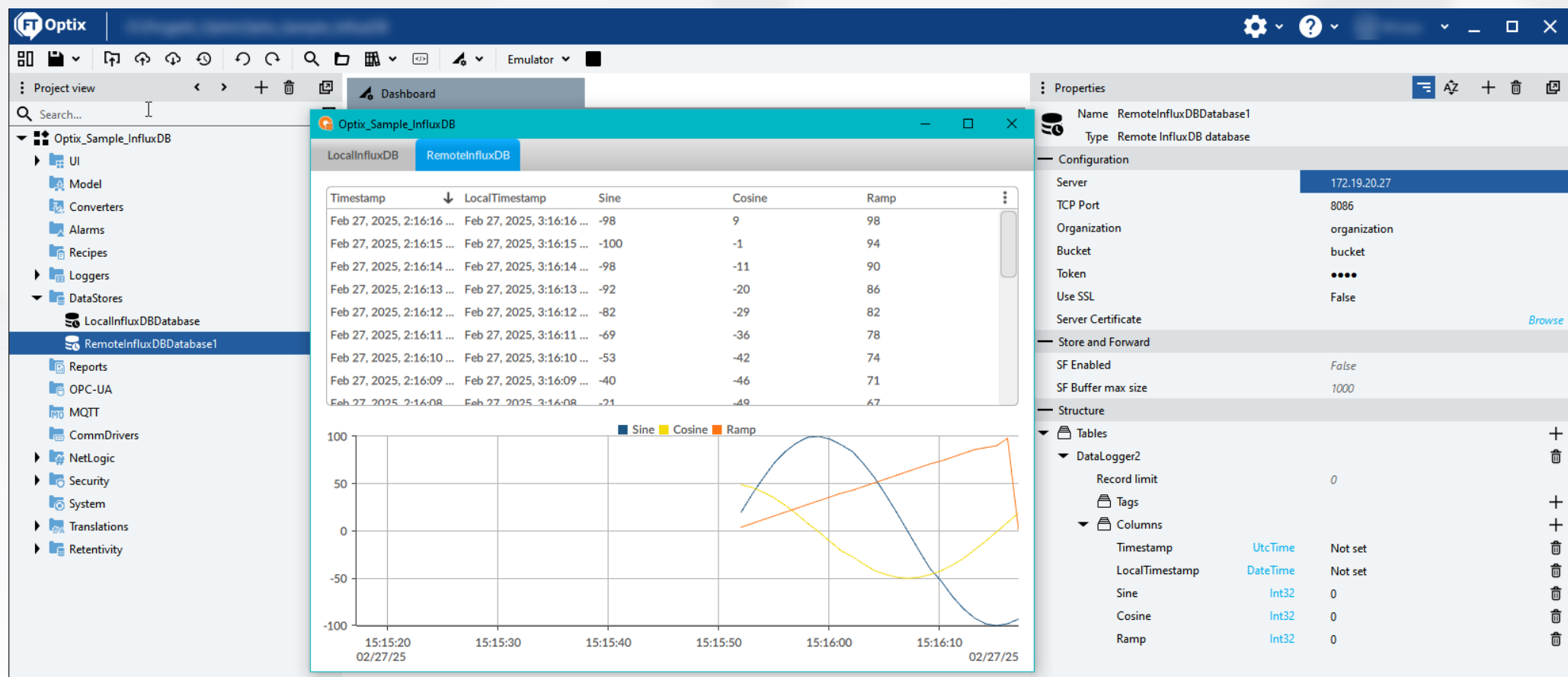
SF Enabled	False
SF Buffer max size	1000

Structure

Tables

- DataLogger2
 - Record limit: 0
 - Tags
 - Columns
 - Timestamp: [UtcTime](#) Not set
 - LocalTimestamp: [DateTime](#) Not set
 - Sine: [Int32](#) 0
 - Cosine: [Int32](#) 0
 - Ramp: [Int32](#) 0

Configuration example

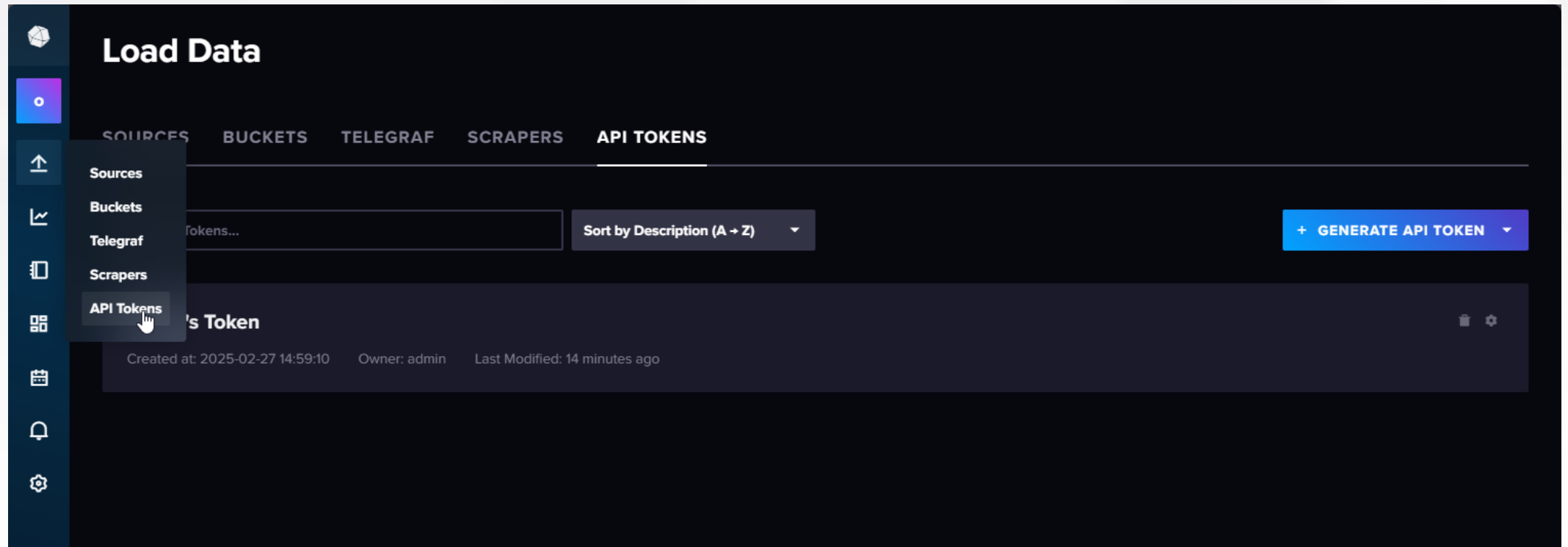


InfluxDB limitations

- No support for Flux query language
- Limited set of queries (see InfluxDB documentation in Optix help)
- Recipes cannot be used with an InfluxDB store

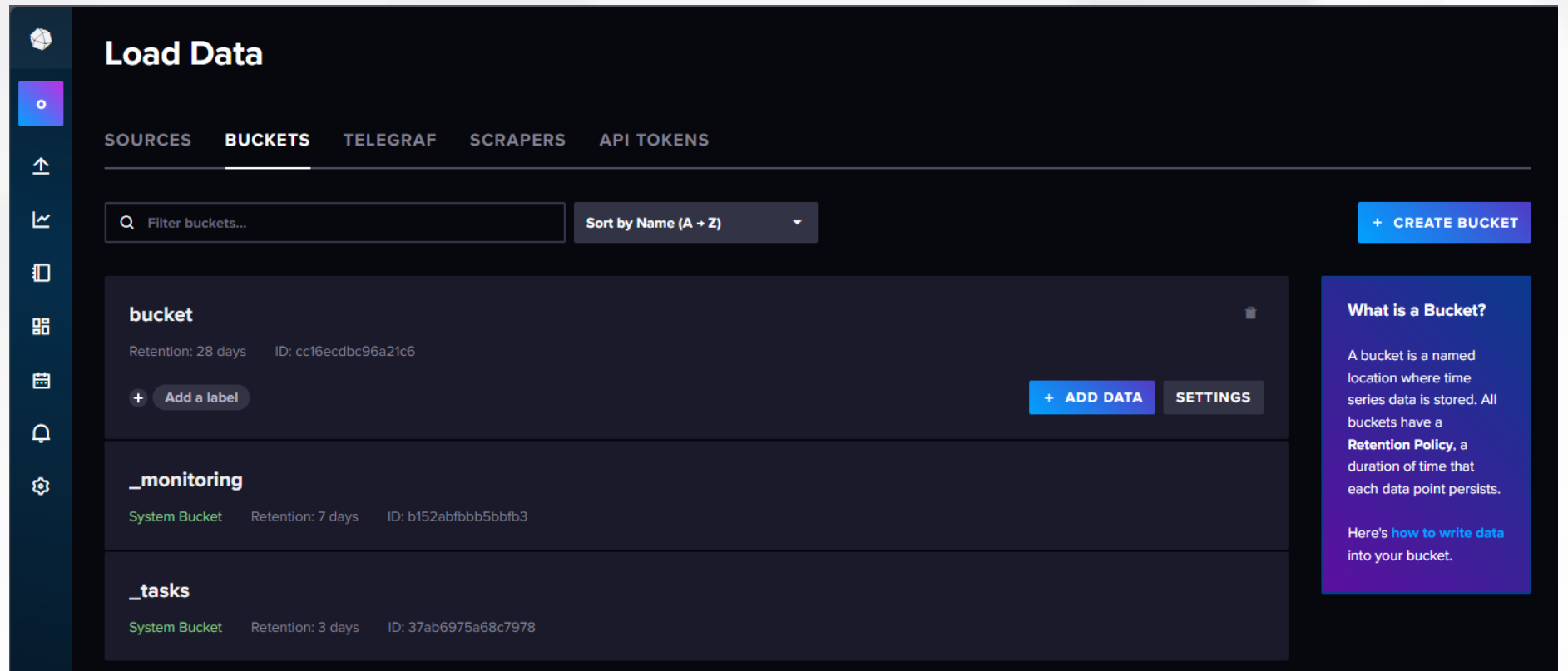
Creating a new token

- Access tokens are created from the InfluxDB web interface



Creating or configuring a bucket

- Buckets, retention and user access are configured using the InfluxDB web interface



Additional resources



FactoryTalk Optix GitHub organization

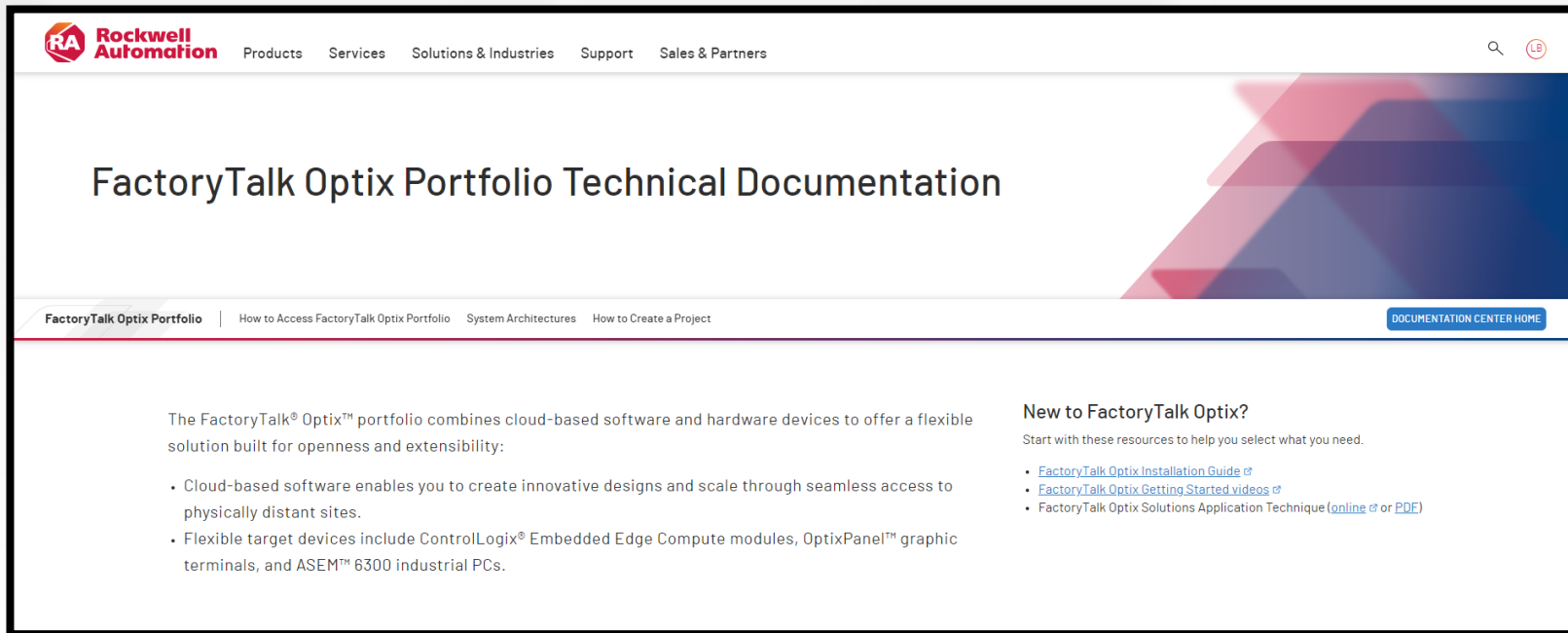
- Contains a big list of projects with a lot of features that can be studied and reused in custom projects
- Contains a lot of sample projects
- Contains a NetLogic CheatSheet with a lot of useful snippets



[FactoryTalk-Optix\(github.com\)](https://github.com/FactoryTalk-Optix)

FactoryTalk Optix landing page

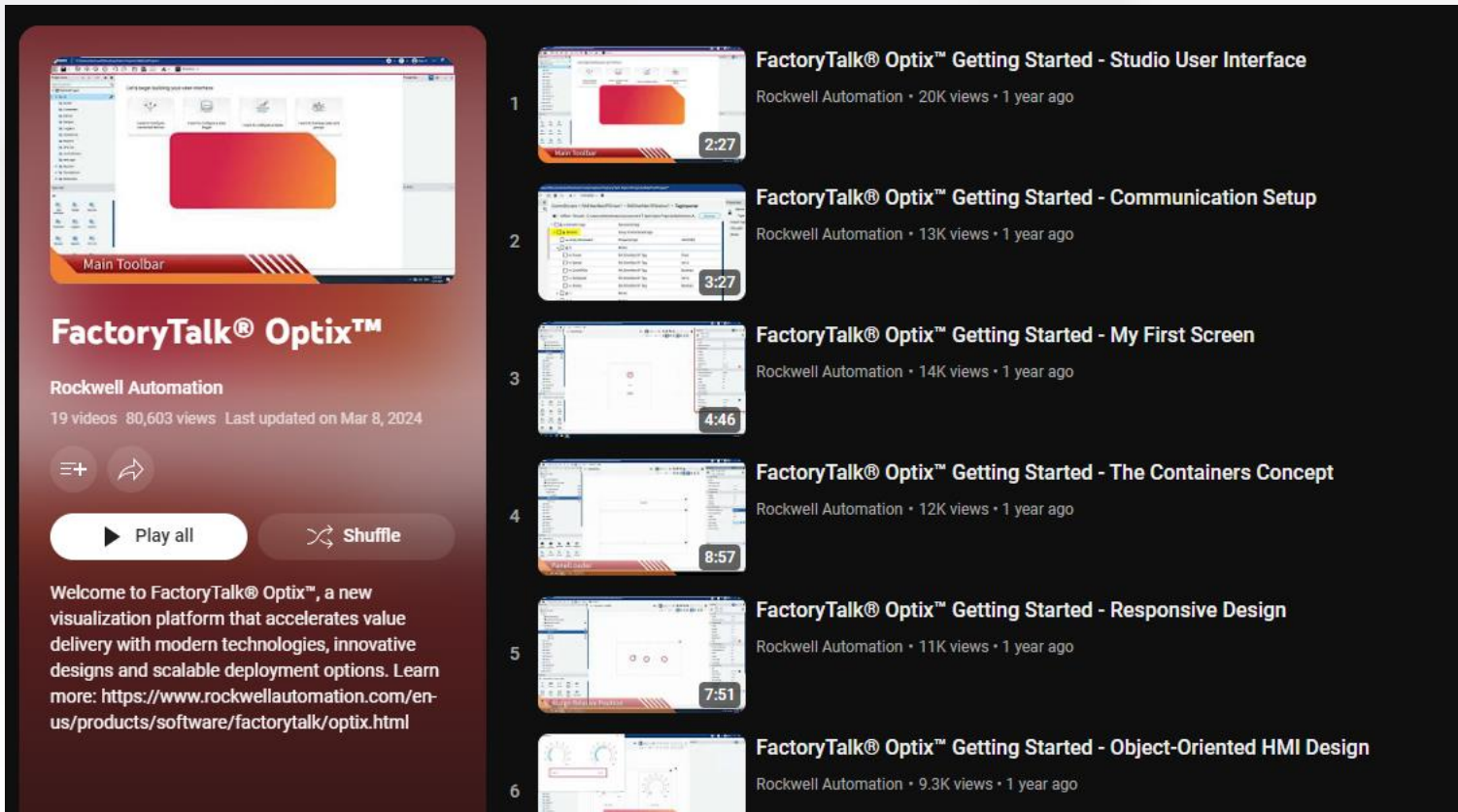
- Contains all the FactoryTalk Optix related resources (help, install guide, release notes, etc)
- Constantly updated



[FactoryTalk Optix Portfolio Technical Documentation | Rockwell Automation](#)

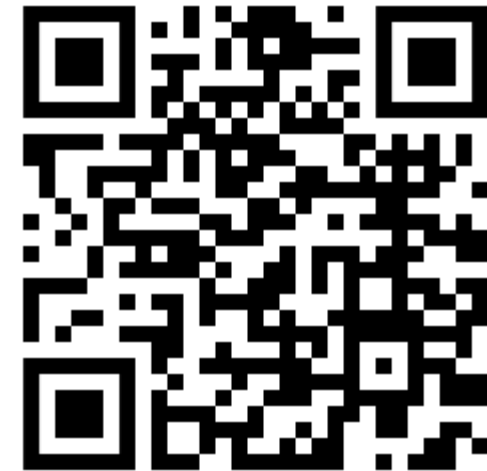
FactoryTalk Optix YouTube channel

- Contains videos on how to build your first project (check the playlists tab)



The screenshot shows the FactoryTalk Optix YouTube channel page. On the left, there's a video player showing the 'Main Toolbar' of the software. Below the player, the channel name 'FactoryTalk® Optix™' is displayed, followed by 'Rockwell Automation' and statistics: '19 videos · 80,603 views · Last updated on Mar 8, 2024'. There are buttons for 'Play all' and 'Shuffle'. A welcome message states: 'Welcome to FactoryTalk® Optix™, a new visualization platform that accelerates value delivery with modern technologies, innovative designs and scalable deployment options. Learn more: <https://www.rockwellautomation.com/en-us/products/software/factorytalk/optix.html>'. On the right, a list of six videos is shown, each with a thumbnail, title, and view count.

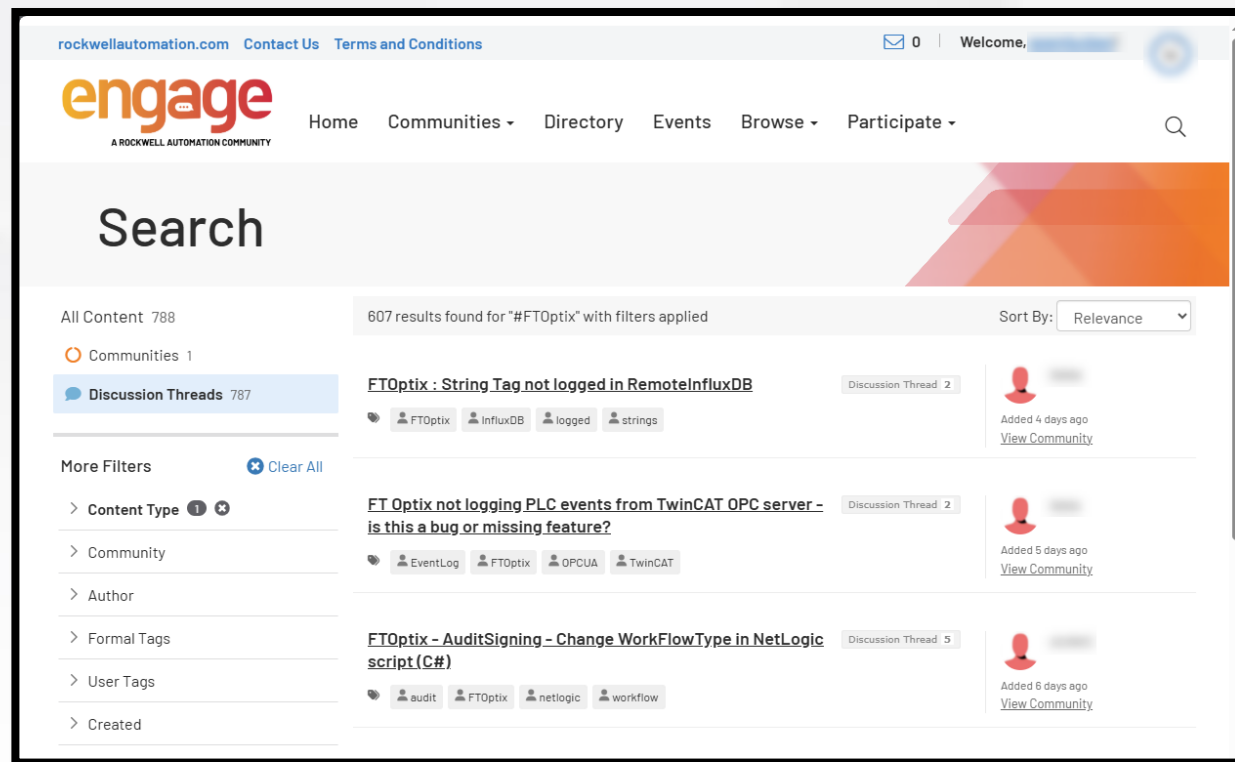
Video Number	Video Title	Views	Age
1	FactoryTalk® Optix™ Getting Started - Studio User Interface	20K views	1 year ago
2	FactoryTalk® Optix™ Getting Started - Communication Setup	13K views	1 year ago
3	FactoryTalk® Optix™ Getting Started - My First Screen	14K views	1 year ago
4	FactoryTalk® Optix™ Getting Started - The Containers Concept	12K views	1 year ago
5	FactoryTalk® Optix™ Getting Started - Responsive Design	11K views	1 year ago
6	FactoryTalk® Optix™ Getting Started - Object-Oriented HMI Design	9.3K views	1 year ago



[Rockwell Automation - YouTube](#)

FactoryTalk Optix forum

- Used by customers and colleagues to exchange informations



[Search - Engage, A Rockwell Automation Community](#)

FactoryTalk Hub

- Contains all the releases of FactoryTalk Optix
 - Contains additional resources like RuntimeTools
 - Used to access the WebIDE
- Contains some interactive demos

The screenshot displays the FactoryTalk Hub interface. On the left, the 'Releases' section shows a table of FactoryTalk Optix releases. On the right, the 'Resources' section lists various interactive demos.

FactoryTalk Optix Release

Download or open Release build of FactoryTalk Optix.

[Release Notes](#) [Changelog](#)

Search...

Date created	Version	
6/13/2024	1.4.0.938	Open in Browser
4/15/2024	1.3.3.3	Open in Browser
12/1/2023	1.2.1.4	Open in Browser
12/1/2023	1.1.1.3	Open in Browser Download
12/1/2023	1.0.3.363	Open in Browser Download
12/1/2023	1.0.2.801	Open in Browser Download

Resources

Here you can find some resources representing how FactoryTalk Optix features work and how to use them.

Search...

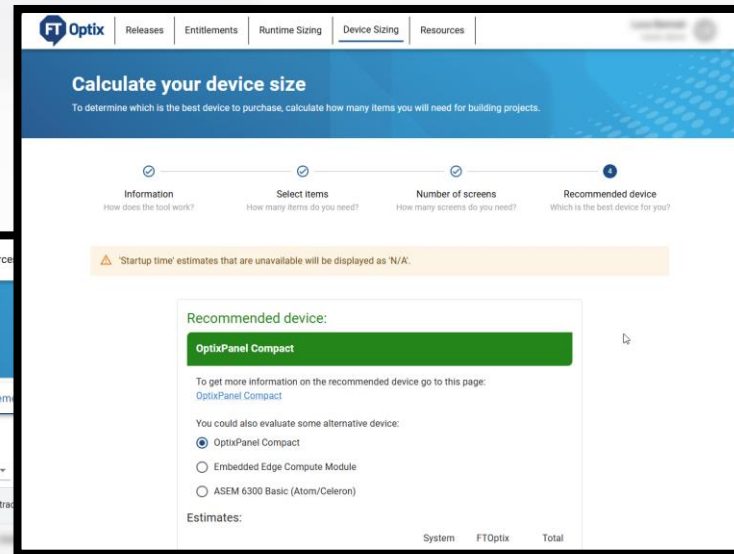
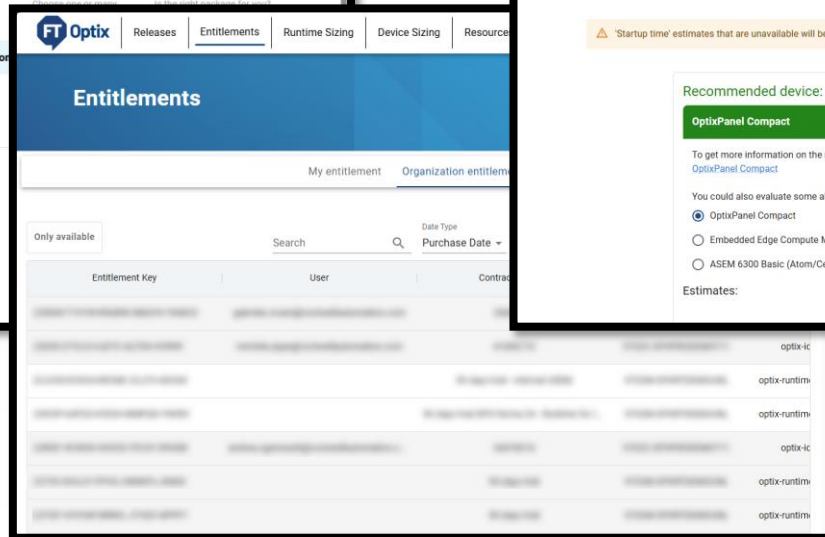
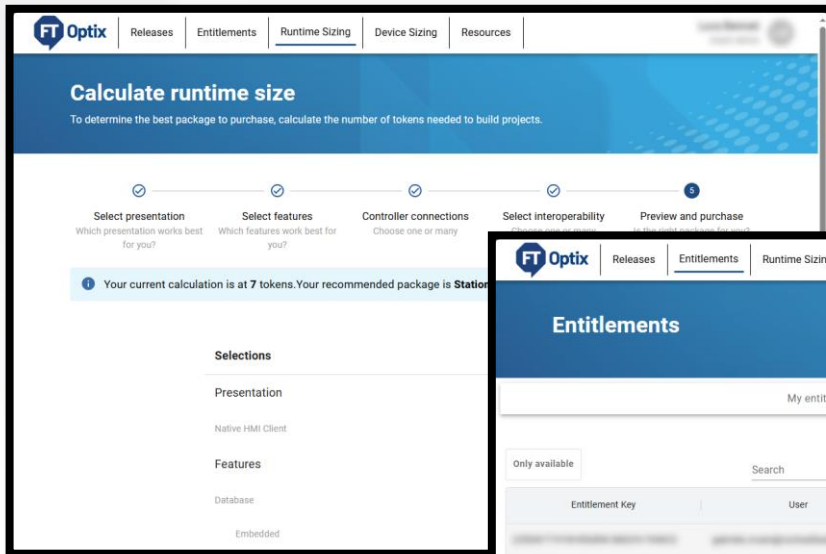
Thumbnail	Title	Type	Description	
	Boller Demo	Demo	This demo was made to demonstrate interoperability between different communication drivers, three aliases in the MainPage are used to interface those drivers, when clicking ...	Open in Browser Download
	Features Demo 2	Demo	This demo contains most of the features exposed by FactoryTalk Optix.	Open in Browser Download
	SMT Demo A	Demo	Demo representing the Asem SMT line based in Artegha (UD) - Italy	Open in Browser Download
	SMT Demo B	Demo	Demo representing the pick and place part of the Asem SMT line based in Artegha (UD) - Italy	Open in Browser Download



[FactoryTalk Hub](#)

FactoryTalk Hub

- Used to view, manage and activate FactoryTalk Optix licenses
- Contains some useful tools
 - Runtime License size calculator (which license is needed to run the project)
 - Device sizing calculator (which device is recommended to run the project)



[FactoryTalk Hub](#)



THANK YOU!